What Follows from What:

Learning Inference Rules from Imitation, Coherence, and the World

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Abstract

Can a learner that acquires inference rules come to reason soundly, and with what justification? We consider a learner that imitates human inferences, is then conditioned on coherence (no good arguments for both P and $\neg P$), and receives occasional feedback from the world. We study this question for formal mathematics, informal mathematics, and the checking of physics solutions. All guarantees are relative to stated assumptions, chiefly that the target is realizable in a structured hypothesis class and that data and designations are truthful.

The answer is a division of labour among the learning signals, each with provable strengths and a provable residue.

- **Search forces worst-case soundness.** Accuracy on the distribution of human steps says nothing about a reasoner that searches: one tonk-like rule of arbitrarily small probability mass makes everything derivable. Under realizability, cautious acceptance over schematic hypotheses is sound against every adaptive prover. Its price is a combinatorial dimension of the hypothesis class.
- **Imitation identifies rules.** For rules cited by name, positive examples identify a schematic calculus exactly at coupon-collector rates, and this generalizes to derivations of any size. But systematic human errors are indistinguishable from rules.
- **Coherence is eliminative negative data.** It pins the target down exactly when the target has no coherent uniform proper extension. Classical logic has none, by Post-completeness, and is pinned by a single coherence datum. Coherence is blind among intermediate logics and Popperian for arithmetic. Carnap's categoricity problem can be read, in part, as Gold's problem in the dual space of valuations.
- **Formal mathematics.** We prove an end-to-end theorem for a two-tier learner: audit, then assert, with bold conjectures in a sandbox. It holds under realizability, a frequency floor, truthful designation and an exact refutation oracle, and each of several ingredients is shown to be necessary.
- **Informal mathematics.** We model it as the shadow of a latent formalization. A bounded-gap verifier is sound against adaptive provers when the formalization is realizable. With complete data, validity is identified up to informal-validity equivalence and no finer. Counterexample objects can be exponentially more informative than paradoxes.

*Claude is an AI system made by Anthropic. Claude did the research and wrote this report in response to questions posed by Kaarel Hänni and under his direction; the design and success criteria in §1.1 are his. Quotations from his public working notes (Hänni, 2026) are verbatim and cited by file path. As the notes' README says, they are working notes rather than claims asserted as in a paper, and the readings of them given here are ours. Every main result was attacked by separate Claude instances acting as referees, and was repaired and re-checked where needed (§1.3); a core of discrete results is machine-checked in Lean 4 (Appendix K). No human has checked all of the mathematics.

- **Physics.** Contexts receive a filter semantics. No truth-functional eternalist reading tracks in-context truth, and exports from idealized contexts need certificates of finite information radius. A structured-solution checker is sound against adversarial solvers relative to a judgment layer, consisting of the reading of the problem and the adequacy of the top model, which we prove ineliminable. We illustrate it on one olympiad problem.

We also answer, in part, two questions raised in the notes that motivated this work: whether a steeper simplicity penalty leads from imitation to norms, and whether there is a "philosophical completeness theorem".

Every main result was attacked by independent referees and repaired where needed. Core discrete results are machine-checked in Lean 4, and small symbolic experiments illustrate the phenomena. We argue that principled justification beyond proof has a definite shape: proof within contexts, certified bridges between them, and an explicit, minimized, ineliminable judgment layer, together with learning signals whose one-sided limits are known.

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1 Introduction

1.1 The question

Consider a learning process that learns *which inferences are valid*, and thereby learns to make valid inferences. The following design, and the three success criteria below, were posed by Kaarel Hänni. Many of the questions behind them are recorded in his public working notes (Hänni, 2026). We call these *the motivating notes* and quote them verbatim, citing them by file path. As their own README says, they are working notes rather than asserted claims, and the readings of them given here are ours. The design has six parts.

- It starts by imitating human inferences. These are the steps of formal or informal mathematical proofs, or of physics olympiad solutions.
- It treats an argument as valid when it is a sequence of valid inferences.
- It conditions the learner on *coherence*: it should not have good arguments for both P and $\neg P$.
- It also conditions the learner on practical success. Occasionally the world reports a truth value.
- On an inferentialist picture of meaning (Brandom, 1994), learning which inferences are valid is roughly learning what the words mean.
- Arguments happen in *contexts*. A physics problem says "assume air pressure is 0", and from background knowledge one can prove that it is not. Arguments for P and for $\neg P$ in such contexts should not be penalized, yet some coherence constraint is clearly needed.

Does anything of this shape work, even with unlimited compute? Does it give rise to a reasoner that derives many correct conclusions? Can the arguments it accepts be trusted when they come from an untrusted, possibly adversarial, generator? Can one *prove* any of this? The bar rises in three steps:

- a setup that provably works for formal mathematics;
- one that would have worked for mathematics *before* we knew how to formalize proofs;
- a principled system for solving physics olympiad problems or for checking their solutions. We take the checking reading (Section 11).

Behind all of these sits a broader question: what does principled justification look like beyond mathematical proof?

The motivating notes put verification first. Formal proof checking is what could make it safe to obtain superhuman mathematical work from an untrusted system. A defeasible checker that faces an untrusted prover, by contrast, "might have lost all the safety" (`ai/verification and generation/formalizing philosophy.pdf`). We therefore treat soundness against an adversarial arguer as the central requirement throughout, and we build checkers, not provers.

This report answers these questions as precisely as we could. The main results are summarized below and collected in Table 1. Each is stated as a theorem in the body and proved there or in the appendix. Most theorems are paired with an impossibility result showing that its hypotheses cannot be dropped.

1.2 The answers in brief

(1) Search makes soundness a worst-case property (Section 3).

- A reasoner searches for derivations whose steps its learned checker accepts. It is sound iff every accepted step is derivable (Lemma 3.1). One accepted bad step is fatal.
- For every distribution over steps and every $\varepsilon > 0$, a single tonk-like schema with mass $< \varepsilon$ makes every formula derivable (Theorem 3.2). Every PAC learner of calculi containing three basic introduction rules can be modified to stay PAC while producing outputs that trivialize the reasoner (Corollary 3.3).
- Majority-vote ensembles of individually coherent verifiers can derive $\bot$ forever without any member being refuted (Theorem 3.12).
- Accuracy on human steps, which is what step-level reward models optimize, is therefore not a soundness certificate. In a small closed algebra domain, a boosted-tree step classifier with in-distribution AUC 0.995 is exploited by search to prove most of a set of false equations (Section 12). The phenomenon is known in practice (Gao et al., 2023); the theorem explains why it is unavoidable.

(2) Cautious acceptance is the canonical sound verifier, and structure makes it affordable (Section 4).

- Accepting exactly what every surviving hypothesis accepts is sound against every adaptive prover. It is also the largest deterministic sound policy. A randomized δ-sound verifier gets anything outside it accepted only with joint probability at most δ (Theorem 4.2).
- Its worst-case cost in human escalations is exactly a positive-elasticity dimension of the class (Theorem 4.4).
- For rules cited by name, learned by anti-unification, this cost is linear in step size. Without citations it is at least $\lfloor (N-1)/k \rfloor^k$ even for one rule, and unstructured classes cost exponentially (Table 4).
- A Bayesian version is time-uniformly sound via Ville's inequality, using the prior–posterior-ratio martingale of Waudby-Smith and Ramdas (2020) (Theorem 4.15).
- This is reliable or KWIK-style learning (Rivest and Sloan, 1988; Li et al., 2008) applied to step checkers; what is specific here is the escalation dimension and its values for rule schemas.

(3) Imitation identifies rules but not norms (Section 5).

- For rules cited by name, positive examples identify a schematic calculus exactly, at coupon-collector rates (Theorem 5.6). Uncited rules need a diversity condition (Theorem 5.8).
- Exact identification generalizes to derivations of any size (Corollary 5.10). This is a precise sense in which finite usage fixes inferential role for unbounded use.
- Systematic human errors are indistinguishable from rules at every rate (Corollary 5.19).

(4) Coherence is eliminative negative data (Section 6).

- A derivation of $\bot$ from a context certified consistent is a *negative bag*: some step in it is invalid. Only bold learners ever receive this signal. A halving-type learner, in the tradition of Barzdin–Freivalds and Littlestone (1988), suffers at most $\log_2(1/w^*)$ detections (Theorem 6.5).
- With $\neg$ and one of $\wedge, \vee, \to$ in the language, every structural extension of classical consequence is classical or trivial. This is Post-completeness in consequence form (Post, 1921). So one coherence datum pins down classical logic with no simplicity bias (Theorems 6.13 and 6.15, machine-checked). In general, coherence pins the target exactly when the target has no coherent uniform proper extension.
- By Glivenko's theorem, coherence is blind among intermediate logics (Theorem 6.16). In arithmetic it is Popperian, detecting Π_1 truth but not Σ_1 truth, and runs out at Σ_2 (Theorem 6.21).
- Carnap's categoricity problem is, in part, Gold's problem in the dual space of valuations (Table 7). Single-conclusion data cannot fix the classical meanings, while, among structural meanings, finitely many bilateral data plus one coherence datum can (Theorems 6.24 and 6.26); the bilateral idea is that of Shoesmith–Smiley, Rumfitt and Restall.
- What survives everything is the set of coherent uniform alternatives: a precise "Kripkensteinian residue" (Theorem 6.34).

(5) An end-to-end theorem for formal mathematics (Section 7).

- The naive two-tier design (cautious assertion plus a bold red team) is provably unsound, because practice contains fallacies (Proposition 7.5).
- The corrected learner audits, then asserts. It assigns blame through the union of minimal conflicts, which is Reiter's duality (Lemma 7.7), and states soundness relative to a refutation depth.
- Assume realizability, with each systematic fallacy used under its own latent rule tag. Assume also a known frequency floor, truthful designation and an exact depth-d refutation oracle. Then, with probability at least $1 - \delta$, the learner is sound against every adaptive prover and red team. Its asserted calculus is the target plus at most the depth-d residue, minus a set of collateral genuine rules when blame is ambiguous (Theorem 7.12). If fallacies are cited under the names of genuine rules, the theorem does not apply as stated.
- Several ingredients are shown to be necessary by matching lower bounds (Table 9).
- For classical propositional logic the residue is empty and convergence is exact. For complete decidable theories this holds conditionally on a realizability hypothesis that is not established for standard axiomatizations (Corollary 7.26).

(6) Informal mathematics before formalization (Section 10).

- We model informal practice as the shadow of an unknown latent formalization. A bounded-gap conservative verifier is sound against adaptive provers. It must check links between contexts as well as steps, since Cauchy's sum theorem is an equivocation (Proposition 10.7).

- Under complete closure-level presentation, validity is identified exactly up to informal-validity equivalence, and no finer. Bounded-gap data identify the short-step relation and only bracket meaning, which is pinned when long inferences decompose into short steps (Theorem 10.14, Example 10.15). Ontology is never identifiable: standard and nonstandard reals agree on the practice (Proposition 10.16).
- Counterexample objects need at most $\log_2 |\mathcal{H}|$ corrections, by the classical halving argument, whereas paradoxes can need $|\mathcal{H}| - 1$ on some classes; a sorites-shaped chain is the tight case (Theorems 10.19 and 10.20).
- The history Frege→Russell→Zermelo comes out as constrained repair selected by coverage of practice (Theorem 10.28).
- A robust-core theorem shows the following. Suppose pre-formal practice used only steps valid under every admissible sharpening of its vague concepts; this is a hypothesis from the motivating notes, untested here. Then its proofs survive every such formalization (Theorem 10.30).
- Selection of steps by counterexample objects, together with Gödel completeness, explains why the surviving informal steps are derivable, though not why they are short (Proposition 10.31).

(7) Contexts and the principled checking of physics (Section 11).

- Contexts denote filters of models from a declared deformation family. Suppositions are exact discharge. Idealizations may contradict the background.
- No truth-functional eternalist reading of in-context claims tracks in-context truth (Theorem 11.8, an observation once set up). Eternalism survives only at the meta level, as McCarthy's $\mathrm{ist}(c, \varphi)$. In the frame-free semantics, coherence is needed exactly on the root and the anchored contexts (Theorem 11.11(a)). In the frame semantics this condition is necessary but not sufficient, and Proposition J.3 gives the exact condition under limit semantics.
- Designating "idealization plus full background" as coherent forces a structural learner to drop some classical schema in every context. If the background states the negated stipulation outright, the learner must also become atomically paraconsistent everywhere (Proposition 11.9). This is a formal version of the air-pressure worry.
- Exporting a conclusion from an idealization needs information of finite radius about the de-idealized family. There is no free export, by the standard adversary argument of information-based complexity (Traub et al., 1988) (Theorem 11.19). Context choice is an attack surface (Theorem 11.22).
- A Structured Physics Solution checker is sound against adversarial solvers relative to a judgment layer (Theorem 11.27). That layer, consisting of the reading of the problem and the adequacy of the top model, is ineliminable (Theorem 11.28).
- "True in the context at hand" becomes supertruth over admissible completions (Theorem 11.29).
- The approach is worked through one EuPhO problem (Section 11.7). Its setup was reconstructed from the motivating notes, not checked against the official text, and the checker run is a problem-specific prototype.

(8) Two questions from the motivating notes.

- A *steeper simplicity penalty* on function induction with private randomness does produce the hoped-for kink at the simple model, proved for parity teachers (Theorem 8.4); kinks beyond parities are open. But selection is a rate threshold. Rare valid rules are lost and cheap fallacies kept, and the threshold cannot be calibrated from imitation data (Section 8).
- The notes ask for a *"philosophical version" of Gödel's completeness theorem*: "a mode of talking makes sense (is coherent) iff there is something it could be talking about". In the generalized sense the question asks for, the answer is positive, for positions, credences and context systems. Coherence cannot certify existence of the *intended* (standard) model, a limitation the notes themselves anticipate. Bounded-arity coherence constraints of the kind used in consistency-based probing do not characterize coherence (Section 9).

Three conclusions sit on top of these results (Section 13):

- every learning signal is *one-sided*;
- principled justification is a *set* of signals that jointly cover every direction of error, together with a declared residue;
- mathematical and physical justification differ in degree, measured by the size of an explicit judgment layer, not in kind.

1.3 How the results were established

The work was carried out by many cooperating instances of a language model, coordinated by one instance. It proceeded in four stages.

- Literature memos on eleven strands.
- Seven theory documents (T1–T7), which developed the mathematics.
- Adversarial verification. Each theory document was divided among two or three independent referees (one literature memo had a single referee). Each referee was instructed to refute the results in its part, with brute-force computation on small cases where possible, so every main result was attacked by one referee. The results were then repaired, and a fresh referee re-checked every item whose statement or proof had changed.
- Drafting, followed by a fidelity check of each section against its verified source, and a review of the whole paper.

In the first verification round the referees rated four issues fatal and twenty major. The fatal issues were false statements, and at least eight of the major ones were also counterexamples to statements as written. Among them were the conditional reading of Theorem 4.2(b) and an optimality clause of Theorem 7.19 that fails at finite depth. The rest were gaps in proofs or hypotheses. Each issue was repaired, weakened or retracted. The second round found two further major problems in Section 7, in Theorem 7.18 and Theorem 7.27(e). These were repaired and then re-checked by a third referee, who found no further major problem. The second round also found two major problems in glosses of the literature memo L3. The paper states the affected result (Proposition E.3) only in the form the referee confirmed, but the memo itself has not been

Success criterion	Status
A setup that provably works for formal mathematics	**Yes for classical propositional logic; conditionally beyond.** The two-tier learner (Theorem 7.12) is sound under its stated assumptions. It converges exactly for classical propositional logic (Corollary 7.23). For complete decidable theories, exactness holds conditionally on a realizability hypothesis that is not established for standard axiomatizations (Corollary 7.26); without it the learner stays sound but loses some rules. It is Popperian for arithmetic, with sharp impossibility results beyond Σ_1 (Theorem 7.27).
A setup that would have worked before formalization	**Conditionally.** Suppose the practice's latent formalization lies in the learner's class. Then a bounded-gap verifier over latent formalizations is sound against adaptive provers. For a finite class under complete presentation, it identifies the practice's step relation and brackets its meaning, which is identified exactly up to informal-validity equivalence when long inferences decompose into short steps (Section 10). History violated the realizability condition at its decisive steps (ε–δ, uniform convergence, ideals). What the criterion really requires, affordable soundness with language invention, remains open (Section 14).
A principled system for checking physics olympiad solutions	**A principled design with a relative soundness theorem, not yet a system.** Solutions written in the Structured Physics Solution format are checked soundly against adversarial solvers, relative to a trusted base (kernel, laws, proved bridge schemas and validated numerics), the reading of the problem and the top model (Theorem 11.27). The reading and top-model layer is provably ineliminable (Theorem 11.28). The design is illustrated on one EuPhO problem by a problem-specific prototype, which also rejects adversarial variants. No general checker has been built. We deliberately built a checker, not a solver.

Table 1: The three success criteria, and what this report establishes for each.

revised. The records are in `research/verification/` and in the Verification log section of each theory document (for T3, only there). The repaired statements are the ones used here.

A core of discrete results is machine-checked in Lean 4 (de Moura and Ullrich, 2021) with Mathlib (The mathlib Community, 2020) (Appendix K). It covers about fifty paper results in 14,600 lines with no `sorry`; about a third of them are formalized as special cases, weaker statements or variants (Table 24). Among them are:

- the Post-completeness and coherence theorems;
- Carnap's problem in learning form, with the bilateral tell-tale;
- tonk beyond the horizon, the doctrinal paradox and the halving bounds;
- Reiter's blame duality, and the separation between paradoxes and objects;
- bilateral completeness and the arity counterexample;
- the rate threshold and the rate-inversion counterexample;
- no free export from samples, and the Lipschitz certifier.

Experiments in a small symbolic toolkit illustrate the phenomena (Section 12). They concern school algebra with simulated derivations containing well-known systematic student errors,

propositional natural deduction, adversarial proof search against learned verifiers, and a prototype physics checker. They are illustrations, not evidence of scalability.

We have tried to mark scope honestly:

- Many results are "trivial once set up". Their contribution is the setup, and we say so.
- Many ingredients are classical, and we credit them where they are used. Among them:
  - learning from positive data and anti-unification: Gold, Angluin, Plotkin and Reynolds, Natarajan and Helmbold–Sloan–Warmuth, Wright and Motoki–Shinohara, Lange–Zeugmann;
  - reliable and selective learning: Rivest–Sloan, El-Yaniv–Wiener, KWIK;
  - martingales and online bounds: Ville and Waudby-Smith–Ramdas, Barzdin–Freivalds and Littlestone;
  - logic: Post and Pogorzelski, Glivenko, Carnap, Shoesmith–Smiley, Rumfitt, Restall and Scott;
  - diagnosis and inductive inference: Reiter and Shapiro; Putnam, Gold and Kelly;
  - contexts and numerical analysis: chunk-and-permeate (Brown–Priest), information-based complexity (Traub–Wasilkowski–Woźniakowski);
  - probability and logic: de Finetti and Gaifman.
- The soundness guarantees are relative to stated assumptions: chiefly realizability of the target in a structured hypothesis class and truthful data and designations, together with the technical hypotheses stated with each theorem. Computational cost is mostly not addressed.

Conventions. Results carry three kinds of marker.

- A superscript tag such as [T1 Thm 3.1] gives a result's provenance. T1–T7 are the theory documents in `research/theory/` of the accompanying repository, and L1–L11 are the literature memos in `research/lit/`. "As repaired" means the statement was changed during adversarial verification, with records in `research/verification/`.
- A superscript Lean: `name` names a Lean declaration (Appendix K).
- Status labels mark results that are known (cited), computed by a script, a proof sketch, or a conjecture.

1.4 Reader's guide

The report is long because it treats three domains and two side questions in full. Readers can take the following routes.

- **Core argument:** Sections 2 to 7, about the formal case, then Section 13.
- **Informal mathematics and the history of rigor:** Section 10, after Section 2.
- **Contexts, idealization and physics checking:** Section 11. It is largely self-contained given Section 2.
- **Learning theory of imitation and simplicity:** Section 8.

- **Logic and the semantics of coherence:** Section 9.
- **Experiments and code:** Section 12.

Section 14 collects open problems. Full proofs, computed evidence and secondary results are in the appendices, one per section.

2 Setting: judgments, steps, rules, provers, and three channels

This section fixes the vocabulary of the paper. Three questions are kept apart: *what* is learned (a step set, a consequence relation, or a set of valuations; Section 2.3), *from what* (three channels of different logical form; Section 2.5), and *against whom* (an adaptive prover that searches for wrongly accepted steps; Section 2.6). Two running examples, propositional natural deduction and school algebra (Section 2.4), recur throughout.

2.1 Judgments, steps and closure

Fm is the set of formulas over atoms $p, q, r, p_0, p_1, \ldots$ with $\neg, \wedge, \vee, \to, \bot, \top$; first-order languages are used for arithmetic and physics. A *context* Γ is a finite set of formulas.

Definition 2.1 (Judgments, positions)**.** [T1 §1.1; T2 §1] A *judgment* is $\Gamma \vdash \varphi$ (single conclusion) or $\Gamma \vdash \Delta$ (multiple conclusion, Δ finite); J denotes a countable set of judgments. Contexts are part of judgments, so hypothetical reasoning, and "assume the air pressure is 0", lives inside the step language. A *position* $[\Gamma : \Delta]$ asserts all of Γ and denies all of Δ. Following Restall (2005), $\Gamma \vdash \Delta$ says that *the position* $[\Gamma : \Delta]$ *is out of bounds*; a position is *coherent* if it is in bounds. When $\bot$ is present it counts as always denied, so $[\Gamma : \emptyset]$ is coherent iff $\Gamma \nvdash \bot$. Without $\bot$, "coherent" means *non-trivial* (Γ does not entail every formula; for a structural relation, equivalently $\Gamma \nvdash q$ for an atom q not occurring in Γ).

Restall's reading gives inference data and coherence data opposite polarity: an inference says a position is *out of* bounds, a coherence datum certifies that one is *in* bounds, so the two inform the learner from opposite sides. This is the bilateral picture of Smiley (1996) and Rumfitt (2000) and the multiple-conclusion logic of Shoesmith and Smiley (1978), anticipated in Carnap's "full formalization" (Carnap, 1943). Section 6.4 makes this precise, recasting Carnap's categoricity problem as Gold's problem in the dual (valuation) space.

Definition 2.2 (Steps, rule sets, closure)**.** [T1 §1.1] A *step* $P \rhd j$ has a finite set $P \subseteq J$ of premises and a conclusion $j \in J$; S is the set of steps, and a *rule set* (step set) is any $R \subseteq S$. An *R-derivation* π of j from $B \subseteq J$ is a finite tree with root j whose nodes are leaves in B or conclusions of steps $P \rhd j' \in R$ whose premises P label their children ($P = \emptyset$ allowed); $\mathrm{Steps}(\pi)$ is the set of steps used. The *closure* $\mathrm{Cl}_R(B)$ is the least $X \supseteq B$ such that $P \rhd j \in R$ and $P \subseteq X$ imply $j \in X$; the *derived steps* are $\mathrm{Sound}(R) = \{P \rhd j : \ j \in \mathrm{Cl}_R(P)\}$. The *target* is an unknown rule set R^*: its members are the *valid* steps, and the members of $\mathrm{Sound}(R^*)$ the *derivable* ones.

Lemma 2.3 (Closure facts; trivial)**.** *[T1 §1.1]* (a) $\mathrm{Cl}_R(B)$ *is the set of judgments with an R-derivation from B;* Cl_R *is extensive, monotone, idempotent and finitary, and monotone in R.* (b) $R \subseteq \mathrm{Sound}(R)$ *and* $\mathrm{Cl}_{\mathrm{Sound}(R)} = \mathrm{Cl}_R$*; hence* $\mathrm{Sound}(R)$ *is the largest step set A with* $\mathrm{Cl}_A = \mathrm{Cl}_R$*.*
Lean: Cl_eq_sInter Lean: Cl_subset_Cl_iff_subset_Sound

Proof in Appendix A.

Two formats recur. In the *formula format* (Hilbert style) $J = \mathrm{Fm}$ and a step is $\varphi_1, \dots, \varphi_k \rhd \psi$. In the *sequent format*, natural deduction (Gentzen, 1935; Prawitz, 1965) written with sequents as in Gentzen (1936), judgments are sequents and discharge is an ordinary step, e.g. $\to$I is $\{\Gamma, \varphi \vdash \psi\} \rhd \Gamma \vdash \varphi \to \psi$. A rule set induces a relation on sets of formulas: $A \vdash_R \varphi$ iff $\varphi \in \mathrm{Cl}_R(A)$ (formula format), or iff $(A_0 \vdash \varphi) \in \mathrm{Cl}_R(\emptyset)$ for a finite $A_0 \subseteq A$ (sequent format). In the formula format $\vdash_R$ is a finitary consequence relation by Lemma 2.3(a): arguments chain, so cut is built in; for the natural-deduction calculi below, weakening and cut are admissible and the same holds. R is *A-coherent* if $A \nvdash_R \bot$.

Reading. A verifier is safe under derivations of every length iff everything it accepts lies in $\mathrm{Sound}(R^*)$ (Lemma 3.1). So "learning which inferences are valid" has a sharp target: a set between R^* and $\mathrm{Sound}(R^*)$.

2.2 Schemas and anti-unification

Steps are encoded as ground terms over a first-order signature Ω, e.g. $\mathsf{s}(\mathsf{and}(p_0, p_1), p_0)$ for an $\wedge$E step; object-language atoms are *constants* of Ω.

Definition 2.4 (Schemas)**.** [T1 §§1.3, 5; T7 §1.1] A *schema* σ is a term with metavariables x, y, A, B, …, and $\mathrm{inst}(\sigma)$ is its set of ground instances. $\sigma \succeq \tau$ (σ is at least as general) iff $\tau = \sigma\theta$ for a substitution θ; $\sigma \equiv \tau$ iff both (equality up to renaming). σ is *pure* if no atom constant occurs in it; its *purification* σ° replaces distinct atoms by distinct fresh metavariables. A *guarded schema* (σ, Ψ) carries guards $\Psi \subseteq \Phi$, a fixed finite family of decidable predicates on ground steps (freshness; "the facts entail $a \neq 0$"), and $\mathrm{inst}(\sigma, \Psi)$ is the set of instances of σ satisfying every guard in Ψ. A *calculus* is a finite set Σ of (guarded) schemas, with $R_\Sigma = \bigcup_{\sigma \in \Sigma} \mathrm{inst}(\sigma)$; a *tagged* step (i, s) records the schema i that it cites.

Anti-unification is the learning algorithm of this paper. For a tuple $\bar{u} = (u_1, \dots, u_n)$ of ground terms, $\mathcal{A}(\bar{u}) = f(\mathcal{A}(\bar{u}^1), \dots, \mathcal{A}(\bar{u}^r))$ if all u_j have root f of arity r ($\bar{u}^k$ is the tuple of k-th arguments); otherwise $\mathcal{A}(\bar{u}) = z_{\bar{u}}$, a metavariable indexed by the tuple, so equal tuples get equal metavariables.

Theorem 2.5 (Least general generalization; Plotkin 1970; Reynolds 1970)**.** [T1 §1.3] *(known) Every nonempty finite set $P = \{t_1, \dots, t_n\}$ of ground terms has a least general generalization $\mathrm{lgg}(P)$, a schema with $P \subseteq \mathrm{inst}(\mathrm{lgg}(P))$ and $\mathrm{lgg}(P) \preceq \tau$ whenever $P \subseteq \mathrm{inst}(\tau)$. It is unique up to renaming and computed by $\mathcal{A}(t_1, \dots, t_n)$. Hence $\mathrm{inst}(\mathrm{lgg}(P)) = \bigcap\{\mathrm{inst}(\tau) : P \subseteq \mathrm{inst}(\tau)\}$: anti-unification is the* closure algorithm *for single-schema instance sets.*

For a schema t let $\mu(t) := |t| - |\mathrm{vars}(t)|$, where $|t|$ counts symbol occurrences (metavariables included) and $\mathrm{vars}(t)$ is the set of distinct metavariables; so $\mu \geq 0$, $\mu(x) = 0$, and $\mu(t) = |t|$ for ground t.

Lemma 2.6 (Rank)**.** [T1 Lemma 1.2] *If $\tau = \sigma\theta$ then $\mu(\tau) \geq \mu(\sigma)$, with equality iff θ restricted to $\mathrm{vars}(\sigma)$ is a renaming. Hence every strictly increasing generalization chain $\tau = \sigma_0 \prec \sigma_1 \prec \cdots \prec \sigma_m$ has $m \leq \mu(\tau) - \mu(\sigma_m) \leq \mu(\tau)$.*

Lemma 2.7 (When the lgg recovers the schema)**.** [T1 Lemma 1.3] *Let $t_j = \sigma\theta_j$ ($1 \leq j \leq n$) be ground instances of σ. Then $\mathrm{lgg}(t_1, \dots, t_n) \preceq \sigma$, with $\equiv$ iff both* witness events *hold:* (R) *for each metavariable x of σ, the roots of $\theta_1 x, \dots, \theta_n x$ are not all equal;* (D) *for each $x \neq y$ there is j with $\theta_j x \neq \theta_j y$.*

Proofs are in Appendix A, which also shows that lggs exist for infinite P (Appendix A.4). The positive-data theory of Sections 4 and 5 runs on counting witness events.

Lemma 2.8 (Guarded closure; trivial)**.** *[T1 §5, side conditions] For nonempty finite $P \subseteq S$ let $\Psi_P = \{\phi \in \Phi : \phi \equiv 1 \text{ on } P\}$. Then $\mathrm{inst}(\mathrm{lgg}(P), \Psi_P)$ is the least set of the form $\mathrm{inst}(\sigma, \Psi)$ containing P.*

So a learned guard is the strongest guard satisfied by *every* datum, and one datum that omits a side condition deletes it (Example 2.12). Proof in Appendix A.

2.3 Hypothesis classes

Definition 2.9 (Hypothesis classes)**.** [T2 §1, §4.1; T1 §1.2, §3] Hypotheses come in three guises. (i) *Step sets* $\mathcal{H} \subseteq 2^S$: e.g. single schemas $H_1 = \{\mathrm{inst}(\sigma)\} \cup \{\emptyset\}$, tagged calculi $H_k^{\mathrm{tag}} = \{\bigcup_{i \le k} \{i\} \times \mathrm{inst}(\sigma_i)\}$, untagged unions $H_k = \{\bigcup_{l \le k} \mathrm{inst}(\tau_l)\}$, finite unstructured classes. A learned step scorer (a process reward model) is a step set, and need not be closed under chaining. (ii) *Consequence relations* $\vdash$ (finitary, reflexive, monotone, cut-closed), equivalently finitary closure operators C; $\vdash$ is *structural* if $\Gamma \vdash \varphi$ implies $s\Gamma \vdash s\varphi$ for every substitution s of formulas for atoms (Łoś and Suszko, 1958; Wójcicki, 1988). Each R induces $\vdash_R$; many R induce the same relation. (iii) *Meanings as sets of valuations*: a set V of admissible valuations $v : \mathrm{Fm} \to \{0,1\}$ (not necessarily compositional), with $\Gamma \models_V^m \Delta$ iff no $v \in V$ has $v[\Gamma] = 1$ and $v[\Delta] = 0$; dually, $\mathrm{Val}(S')$ is the set of valuations violating no sequent in S'. The *version space* of data D is $\mathrm{VS}(D) = \{h \in \mathcal{H} : h \text{ consistent with } D\}$ $(= \{h : D \subseteq h\}$ for positive data). $\mathcal{H}$ is *realizable* if it contains the target; w is a prior and $w^* = w(R^*)$.

Structurality is uniformity. If σ is pure, $\mathrm{inst}(\sigma)$ is closed under substitutions s for atoms ($s(\sigma\theta) = \sigma\theta'$ with $\theta' x = s(\theta x)$). So a learner that generalizes human steps to pure schemas is structural. A schema that mentions an atom is not, and can be unsound without trivializing (Example 3.6); purification restores the link (Lemma 3.7).

2.4 Practice and the running examples

Definition 2.10 (Practice)**.** [T7 Defs 1.1–1.2; T1 §6] The *target* is a finite calculus Σ^* (the *genuine rules*), with $R^* = R_{\Sigma^*}$. The *systematic fallacies* are a finite set F of schemas, $F \cap \Sigma^* = \emptyset$, each with an instance outside $\mathrm{Sound}(R^*)$. The *practice* is $\Sigma^P = \Sigma^* \sqcup F$, with $R^P = R^* \cup R_F$. *Human data* are i.i.d. tagged steps: tag $i \in \Sigma^P$ has probability π_i, and given i the step is an instance of i with probability $1 - a_i$ and otherwise an arbitrary step outside $\mathrm{inst}(i)$ tagged i (*sporadic noise*). Alternatively, the data are a *text*, an enumeration of R^P (Gold, 1967).

Fallacy tags are idealized as *latent but distinct*: a student who affirms the consequent "cites" affirming the consequent. If fallacies are cited under a genuine rule's name, the per-tag class becomes H_k (Sections 5 and 7).

Example 2.11 (Propositional natural deduction)**.** Σ^* is a natural-deduction calculus for CPC ($\wedge$I/E, $\vee$I/E, $\to$I, $\to$E = modus ponens (MP), $\neg$I/E, ex falso, double negation). In the formula format MP is $\varphi,\ \varphi \to \psi \rhd \psi$. A student sometimes *affirms the consequent*, AC : $\psi,\ \varphi \to \psi \rhd \varphi$; so $F = \{\mathrm{AC}\}$. The data are steps such as $p \wedge q \rhd p$ tagged $\wedge$E. The two instances $p \wedge q \rhd p$ and $(r \to p) \wedge p \rhd r \to p$ already satisfy (R) and (D), so their lgg is $A \wedge B \rhd A$ (Lemma 2.7). (In the sequent format contexts are sets, and anti-unification modulo sets is finitary, not unitary; we flag this where it matters.)

Example 2.12 (School algebra)**.** Judgments are equations $s = t$ between terms built from $+$, $-$, $\cdot$, $/$, powers, $\sqrt{}$, numerals and variables, under finite sets of *facts* such as $x \neq 0$. Terms denote partial functions ($a/0$ is undefined), and $s = t$ is true under the facts iff s, t are Kleene-equal (both undefined, or both defined and equal) at every real assignment satisfying them. Steps are instances of rewrite schemas such as $(a+b)^2 \rhd a^2 + 2ab + b^2$ and $a/a \rhd 1$ with guard $a \neq 0$, plus trusted equational steps (symmetry, transitivity, congruence) and trusted numeral evaluation. The student commits the *freshman's dream* $(a+b)^2 \rhd a^2 + b^2$, an invalid schema, and *unguarded cancellation* $a/a \rhd 1$, a genuine rule used without its guard (the instance $x/x \rhd 1$ fails at $x = 0$). If the student cites "cancellation" for both uses, one datum such as $(y+1)/(y+1) \rhd 1$ under no facts drops the guard from the learned rule (Lemma 2.8); with distinct latent tags the unguarded rule is learned as a schema of its own. Either way, positive data learn it as a rule.

These are Experiments A (algebra) and B (natural deduction) of Section 12.

2.5 Three channels

Definition 2.13 (The channels)**.** [T1 §1.2; T2 §1, §2.1–2.2; T7 Defs 1.2–1.4; L3 Lemma A]

(P) *Practice/imitation* delivers human steps $X_1, X_2, \ldots$, possibly tagged with rule names, possibly with sporadic noise (Definition 2.10). A noise-free datum lies in R^P; no datum says that a step is invalid.

(C) *Coherence* supplies a family $\mathcal{A}$ of *designated contexts*, finite premise sets certified coherent for the target ($A \nvdash_{R^*} \bot$), or, bilaterally, *designated positions* $[\Gamma : \Delta]$ certified in bounds ($\bot \notin \mathrm{Cl}_{R^*}(\Gamma)$ and $\Delta \cap \mathrm{Cl}_{R^*}(\Gamma) = \emptyset$). By default $\mathcal{A}$ is fixed and known, independently of the target. A hypothesis h is *$\mathcal{A}$-coherent* if $A \nvdash_h \bot$ for all $A \in \mathcal{A}$. A *negative bag* is $\mathrm{Steps}(\pi) \setminus T_{\mathrm{trust}}$ for an argument π from some $A \in \mathcal{A}$ to $\bot$ (trusted steps T_{trust} as in (W)); it contains an invalid step, since designation excludes $\bot \in \mathrm{Cl}_{R^* \cup T_{\mathrm{trust}}}(A)$ (for $T_{\mathrm{trust}} \neq \emptyset$ via (WS), Lemma A.1).

(W) The *world* is an evaluable fragment $J_W \subseteq J$ with truth function $W : J_W \to \{0, 1\}$, plus optional *trusted steps* T_{trust}. Its kinds: (W1) *truth values on a fragment* (closed propositional formulas, Δ_0 sentences, a decision procedure); (W2) *counterexample objects*, structures in which every line of a given argument can be evaluated; (W3) *computation with fresh randomness*, e.g. testing a polynomial identity at a point drawn after the claim is committed; (W4) *simulation*, a validated numerical solution of a less idealized model (Section 11).

Coherence is enforced *only* on designated contexts: deriving $\bot$ from an undesignated supposition is a reductio and is never penalized. A negative bag is a multiple-instance label (Lemma 6.2). Objects (W2) turn bags into singletons: descent from a false conclusion through false premises ends at one step with true premises and a false conclusion (Lemmas 7.8 and 10.17). Fresh randomness (W3) is sound against adaptive provers under sufficient conditions on commitment order, degree bounds and the field, and fails when the points are drawn before the prover searches (Lemma 3.11).

Assumption 2.14 ((WS), sound evaluation)**.** [T7 §1.3] For each designated $[\Gamma : \Delta]$, the partial evaluation with $e(j) = 1$ if $j \in \Gamma$ or $W(j) = 1$, and $e(j) = 0$ if $j \in \Delta \cup \{\bot\}$ or $W(j) = 0$, is well defined and extends to a total $V : J \to \{0, 1\}$ under which no step of $R^* \cup T_{\mathrm{trust}}$ has true premises and a false conclusion.

	(P) practice	(C) coherence	(W) world
datum	"this step is used"	an argument from a designated context to $\bot$	a truth value; an object; a random test; a simulation
logical form	positive instance	negative bag ("some step is invalid")	singleton blame with objects; derivation-level otherwise
blind to	systematic errors	errors coherent with the target	what lies outside the evaluable fragment

Table 2: The three channels (Definition 2.13).

(WS) does not require $T_{\text{trust}} \subseteq \text{Sound}(R^*)$, and the relevant target closure is then $\text{Cl}_{R^* \cup T_{\text{trust}}}$. Under (WS) every designated position is in bounds for $R^* \cup T_{\text{trust}}$, hence for R^*; without a world the converse holds (Lemma A.1; Appendix A.6 also discusses arithmetic and formal mathematics). When $J_W \neq \emptyset$ we assume the empty position $[\emptyset : \emptyset]$ is designated, so that world-only falsifications count (Definition 2.19).

Example 2.15 (Example 2.11 continued)**.** (C) Designate $A_1 = \{q,\ p \to q,\ p \to \bot\}$: AC gives p, MP then gives $\bot$. The bag $\{q, p \to q \rhd p;\ p, p \to \bot \rhd \bot\}$ does not say which step is at fault, and the rival diagnosis "MP is the fallacy" reads $\to$ as $\leftarrow$ (Section 7). (W2) The valuation $v(p) = 0$, $v(q) = 1$ makes A_1 true and the line p false, so descent blames AC alone. (W1) The closed instance $\top,\ \bot \to \top \rhd \bot$ of AC has true premises and a false conclusion.

Example 2.16 (Example 2.12 continued)**.** (C) Designate the empty fact set, with $\bot$ any equation between distinct numerals. The freshman's dream gives $(1+1)^2 = 1^2 + 1^2$, i.e. $4 = 2$ after trusted evaluation: a singleton bag. Unguarded cancellation gives $2 \cdot (0/0) = 2 \cdot 1 = 2$ and $2 \cdot (0/0) = (2 \cdot 0)/0 = 0/0 = 1$, so $1 = 2$; this bag also contains the genuine rules $a \cdot (b/c) \rhd (a \cdot b)/c$, $a \cdot 0 \rhd 0$ (for defined a) and $a \cdot 1 \rhd a$. (W2) An evaluator that observes "undefined" blames $0/0 \rhd 1$ alone. (W3) Random evaluation exposes the freshman's dream with high probability over $\mathbb{Z}$, but never over $\mathbb{F}_2$ (Lemma 3.11). The instance $x/x \rhd 1$ fails only at $x = 0$, which random evaluation essentially never hits; the object $x = 0$ does. Without (W), coherence in Experiment A converges, up to a few schemas, to a coherent calculus for the total reading $1/0 = 0$, a different meaning of "/" (computed; Section 12.1).

Reading. The channels are the three signals of the design in Section 1.1 (see also the motivating notes, Hänni, 2026): imitation of human inferences, conditioning on coherence, and occasional feedback from the world. They are not three noisy versions of one signal. (P) can only confirm, (C) can only condemn sets, and (W) can condemn single steps when it supplies objects. Each later section says what one of them fixes and what it provably cannot.

2.6 Provers, verifiers, soundness and cost

Definition 2.17 (Protocol)**.** [T1 §1.2; T2 §2.2] In rounds $t = 1, 2, \ldots$ a *prover* submits $q_t \in S$ and the *verifier* answers $a_t \in \{\textsf{ACCEPT}, \textsf{REJECT}, \textsf{ESCALATE}\}$; on ESCALATE an oracle returns $\mathbf{1}[q_t \in R^*]$ (possibly noisy), which joins the history. The *acceptance region* $\hat{R}_t$ is the set of steps accepted outright at time t; a reasoner chains them and derives $\text{Cl}_{\hat{R}_t}(B)$. The prover may be randomized, adaptive and computationally unbounded; it knows the target, the practice, the verifier's code and the history, but not the verifier's future coins, future human data, or the

oracle's future (fresh) noise. It models any proof search, the learner's own included. It is *honest* if all its queries are valid.

Definition 2.18 (Soundness)**.** [T1 Def 1.2; T7 §1.5] Fix a class $\mathcal{H}$ of targets and, per target, a set $S^\dagger$ of admissible steps (default $S^\dagger = R^*$). A verifier is *δ-sound* if for every target and every prover $\mathbb{P}[\exists t:\ a_t = \mathsf{ACCEPT} \wedge q_t \notin S^\dagger] \le \delta$ (a time-uniform guarantee); *deterministically* (or 0-)*sound* if $\delta = 0$; and *uniformly δ-sound* (hence δ-sound) if some event G with $\mathbb{P}(G) \ge 1 - \delta$, depending only on the human data and the verifier's coins, has no prover ever getting a step outside $S^\dagger$ accepted on G. *Target-soundness* takes $S^\dagger = R^*$ for the genuine Σ^*, *practice-soundness* $S^\dagger = R^P$. (*Truth-soundness*, in Section 7, means deriving only truths of the intended structure.)

Asking for $S^\dagger = R^*$ is stronger than the condition $\hat{R}_t \subseteq \mathrm{Sound}(R^*)$ that chaining needs; the version-space verifiers of Section 4 meet it. Soundness is relative to the target calculus, not to truth: improving on R^* is the job of (C) and (W). Because coherence is Π_1-complete (Theorem 6.30), no computable learner is, on every practice, both eventually complete and sound at all times relative to the full-depth residue of fallacies (Theorem 7.20); soundness is therefore stated relative to a refutation budget.

Definition 2.19 (Depth-relative soundness)**.** [T7 Defs 1.5–1.6] Assume (WS). A *d-refutation* of a finite schema set B is a derivation of size $\le d$ (in symbols of distinct judgments; Definition 7.2), relative to a designated position, whose leaves evaluate to 1, whose steps lie in $R_B \cup T_{\mathrm{trust}}$, and whose conclusion evaluates to 0 ($\bot$, a denied judgment, or a W-false one). B is *d-clean* if it has none. The *depth-d residue* $F^{(d)}_{\mathrm{res}} = \{\tau \in F : \Sigma^* \cup \{\tau\} \text{ is } d\text{-clean}\}$ decreases in d. A verifier is *depth-d sound* if it is (uniformly) δ-sound for $S^\dagger = R^* \cup R_{F^{(d)}_{\mathrm{res}}}$.

Definition 2.20 (Cost)**.** [T1 Def 1.3; T2 §2.2; T7 §1.4] (a) *Escalations*: the number of rounds in which a valid query is not accepted outright (in expectation, for randomized verifiers). $\mathrm{Esc}(\mathcal{H} \mid D)$ is the infimum over 0-sound verifiers of the supremum over targets in $\mathcal{H}$ containing the data D, and over honest query sequences, of this number; $\mathrm{Esc}(\mathcal{H}) = \mathrm{Esc}(\mathcal{H} \mid \emptyset)$ is the *escalation dimension*. Invalid queries are not charged. (b) *Detections*: rounds in which an argument π from some $A \in \mathcal{A}$ to $\bot$ with $\mathrm{Steps}(\pi) \subseteq \hat{R}_t$ is exhibited; *incompleteness errors*: rounds presenting a valid step outside $\hat{R}_t$. (c) *Mind changes*: rounds with $\hat{R}_{t+1} \neq \hat{R}_t$; those that remove an accepted step are *retractions* (Kelly, 1996). (d) *Samples* N (human steps) and *calls* to a refutation oracle that returns a d-refutation of a queried set or reports none.

Example 2.21 (One bad step suffices)**.** If AC is accepted, a prover derives any φ from no premises: $\top$ and $\varphi \to \top$ are theorems, and AC concludes φ. Accepted unguarded cancellation derives $1 = 2$ (Example 2.16). A verifier whose errors are rare on the human distribution still accepts such a step whenever a prover asks for it: Prior's *tonk* (Prior, 1960) as an attack on a learned checker, which Section 3 turns into theorems.

Reading. The question asks what works even with unlimited compute (Section 1.1). So the prover is unbounded and adaptive, and soundness is a worst-case property (Section 3); its price is counted separately in escalations, samples, detections and oracle calls (Section 4).

2.7 Contexts: a preview

Contexts let one argue inside a supposition or an idealization without being penalized for contradicting the world. They form a tree with root @, the actual or intended situation, and $\mathrm{ist}(c, \varphi)$

says that φ holds in c (McCarthy, 1993). Suppositions $\mathsf{SUP}(A)$ are exact discharge (Proposition 11.4); idealizations DEF ("air pressure $\to 0$") are filters over declared deformations that may contradict the root without exploding (Definition 11.2 and Proposition 11.3); and coherence is enforced only on the root and on anchored contexts (Theorem 11.11 and Proposition 11.9).

3 Search makes soundness a worst-case property

The architecture studied in this paper has three parts: a learned relation $\hat{V}$ that says which steps are valid, a prover that searches for derivations all of whose steps $\hat{V}$ accepts, and a training signal that starts from human steps. Once a prover searches, the only property of $\hat{V}$ that bears on the soundness of the resulting reasoner is a worst-case one: whether its acceptance region contains a non-derivable step. Accuracy on typical steps, on human steps, or on most instances of a rule does not bear on it, because search can seek out exactly the steps on which $\hat{V}$ is wrong.

The phenomenon parallels two observations in the motivating notes (Hänni, 2026). Solomonoff function induction "doesn't work that well when there is a simple property distinguishing test from train",[1] and a defeasible checker facing an untrusted prover "might have lost all the safety" (Section 1.1). For a learned checker the prover supplies the distinguishing property; in Corollary 3.3 it is mere size beyond the largest training step. This section gives precise versions of both observations for learned step checkers.

None of the results below is deep; what they contribute is the setup and the quantifiers. Reasoner soundness is stepwise soundness (Lemma 3.1). Arbitrarily small average error is compatible with a reasoner that derives everything, and every PAC learner of calculi containing three basic introduction rules has a PAC variant, with essentially the same sample complexity, whose every output does so (Theorem 3.2 and Corollary 3.3). In classical propositional logic every unsound *pure* schema added to a complete calculus trivializes the reasoner, so errors are total but visible to coherence (Proposition 3.5 and Lemma 3.7). Simplicity is the wrong bias for positive data (Proposition 3.9). Average-case validity survives search in two regimes, both with randomness drawn after the rules or claims are fixed and outside the prover's control (rules valid per random scene, with the library fixed first; fresh randomness per check), and fails in a fixed world with prover-chosen instances (Lemma 3.11). Voting ensembles of coherent verifiers can be incoherent in a way that coherence feedback cannot localize (Theorem 3.12). Except in Section 3.5, "sound" means sound relative to the human calculus R^* (or a target consequence relation), not relative to truth; going beyond R^* is the business of coherence and the world (Sections 6, 7 and 11).

3.1 Reasoner soundness is stepwise soundness

Steps, closure and $\mathrm{Sound}(R^*)$ are as in Definition 2.2; we write Π for a premise set. A deterministic verifier $\hat{V}$ has *acceptance region* $A = \{s \in S : \hat{V}(s) = \mathsf{ACCEPT}\}$, and an unbounded prover chaining accepted steps from premises B reaches exactly $\mathrm{Cl}_A(B)$.

Lemma 3.1 (Reasoner soundness is stepwise soundness)**.** *[T1 Lemma 1.1] For every $A \subseteq S$,*

$$\forall B \subseteq J:\ \mathrm{Cl}_A(B) \subseteq \mathrm{Cl}_{R^*}(B) \qquad \Longleftrightarrow \qquad A \subseteq \mathrm{Sound}(R^*).$$

Lean: StepSoundness.lemma_1_1

[1] `ai/learning/solomonoff induction/solomonoff function induction.md`.

Proof. ($\Leftarrow$) If $(\Pi \rhd j) \in A$ and $\Pi \subseteq \mathrm{Cl}_{R^*}(B)$, then $j \in \mathrm{Cl}_{R^*}(\Pi) \subseteq \mathrm{Cl}_{R^*}(B)$ by monotonicity and idempotence. So $\mathrm{Cl}_{R^*}(B) \supseteq B$ is closed under A, and contains $\mathrm{Cl}_A(B)$ by leastness. ($\Rightarrow$) For $(\Pi \rhd j) \in A$ take $B = \Pi$: $j \in \mathrm{Cl}_A(\Pi) \subseteq \mathrm{Cl}_{R^*}(\Pi)$. □

There is no "small error rate" middle ground: one accepted step $\Pi \rhd j \notin \mathrm{Sound}(R^*)$ yields, from Π, a conclusion that R^* does not derive. The cautious verifiers of Section 4 satisfy the stronger $A \subseteq R^*$.

3.2 Average-case accuracy cannot give soundness

Fix a propositional language with atoms $p_0, p_1, \ldots$, the constant $\top$ and connectives $\wedge, \vee$ (others optional); judgments are formulas. Let R^* contain the instances of

$$\top\mathrm{I}: \ \rhd \top, \qquad \wedge\mathrm{I}: \ A, B \rhd A \wedge B, \qquad \wedge\mathrm{E}_{1,2}, \qquad \vee\mathrm{I}_1: \ A \rhd A \vee B, \qquad \vee\mathrm{I}_2: \ B \rhd A \vee B,$$

and possibly of other classically sound rules. Put $\top^{(1)} = \top$, $\top^{(j+1)} = \top \wedge \top^{(j)}$, and consider the pure one-metavariable schema $T_j := \{A \vee \top^{(j)} \rhd A : \ A \in \mathrm{Fm}\}$. Composed with $\vee\mathrm{I}_2$, T_j is the rule "from $\top^{(j)}$ infer anything": a tonk (Prior, 1960; Belnap, 1962) with its trigger hidden beyond a size horizon. For a distribution Q on S and $h \subseteq S$ write $\mathrm{err}_Q(h) := Q(h \triangle R^*)$.

Theorem 3.2 (Tonk beyond the horizon)**.** *[T1 Thm 2.1]*

(i) Every formula C lies in $\mathrm{Cl}_{R^* \cup T_j}(\emptyset)$, by a derivation of $j+2$ steps of which only the last uses T_j. For non-tautologous C that step is unsound, hence not in $\mathrm{Sound}(R^*)$.

(ii) For every probability distribution Q on S (over valid steps, invalid steps, or both), $Q(T_j) \to 0$ as $j \to \infty$.

So for every Q and $\varepsilon > 0$ some $R^* \cup T_j$ has $\mathrm{err}_Q(R^* \cup T_j) \le Q(T_j) < \varepsilon$ and makes every formula derivable. If R^* is given by finitely many pure schemas, so is $R^* \cup T_j$.

Lean: StepSoundness.thm_2_1_error_and_trivial

Proof. (i) Derive $\top$ by $\top$I, then $\top^{(2)}, \ldots, \top^{(j)}$ by $j-1$ uses of $\wedge$I, then $C \vee \top^{(j)}$ by $\vee\mathrm{I}_2$, and finally C by T_j. The premise $C \vee \top^{(j)}$ is a tautology, so for non-tautologous C the last step is unsound. (ii) The T_j are pairwise disjoint ($A \vee \top^{(j)} = A' \vee \top^{(j')}$ forces $A = A'$, $j = j'$), so $\sum_j Q(T_j) \le 1$. Finally $(R^* \cup T_j) \triangle R^* \subseteq T_j$. □

Corollary 3.3 (PAC learners can be maximally unsound)**.** *[T1 Cor 2.2, as repaired] In the setting of Theorem 3.2, let $\mathcal{L}$ be any learner mapping a sample $X_1, \ldots, X_m$ of steps, drawn i.i.d. from Q and labelled by membership in R^*, to a step set. Let $|s|$ count the symbol occurrences in a step s, put $\hat{\jmath} := 1 + \max_i |X_i|$, and*

$$\mathcal{L}^+(X_{1..m}) := \mathcal{L}(X_{1..m}) \ \cup \ T_{\hat{\jmath}} \ \cup \ \mathrm{inst}(\top\mathrm{I}, \wedge\mathrm{I}, \vee\mathrm{I}_2).$$

Then for every Q, $m \ge 1$ and $\varepsilon \in (0,1)$: (a) $\mathrm{err}_Q(\mathcal{L}^+) \le \mathrm{err}_Q(\mathcal{L}) + Q(T_{\hat{\jmath}})$ and $\Pr[Q(T_{\hat{\jmath}}) > \varepsilon] \le (1-\varepsilon)^m$. (b) Hence if $\mathcal{L}$ PAC-learns a class of targets R^*, each containing $\top I$, $\wedge I$ and $\vee I_2$, with sample complexity $m_{\mathcal{L}}(\varepsilon, \delta)$, then $\mathcal{L}^+$ PAC-learns the same class (improperly) with sample complexity at most $\max\{m_{\mathcal{L}}(\varepsilon/2, \delta/2), \lceil (2/\varepsilon)\ln(2/\delta)\rceil\}$. (c) Yet every *output of $\mathcal{L}^+$ makes every formula derivable from $\emptyset$.*

Lean: StepSoundness.cor_2_2_trivializes *((c))*

Proof idea. $T_{\hat{\jmath}}$ lies beyond every sample, in a tail of Q that is light unless all m samples missed a heavy set. The added schemas lie in R^* and, with $T_{\hat{\jmath}}$, carry the derivation of Theorem 3.2(i). Appendix B.1. □

The added schemas are needed for (c) (Remark B.1). The corollary is about what a PAC guarantee can certify, not a prediction that natural learners output tonks. A soundness guarantee must be a $\sup_s$, not an $\mathbb{E}_{s\sim Q}$, because the prover chooses s after seeing the verifier; Section 4 computes the price of this uniform requirement.

Selection effect. A search that derives a judgment that R^* does not derive from the premises used has found an accepted step outside $\mathrm{Sound}(R^*)$ (Lemma 3.1), whatever the verifier's error rate on human steps. This is one mechanism that can produce reward-model overoptimization (Gao et al., 2023), and it limits what process reward models trained on human step labels (Lightman et al., 2023) can certify (Section 3.7).

Reading. Theorem 3.2 separates two things "a learner learns which inferences are valid" might mean. "Valid on typical steps" (PAC accuracy, held-out accuracy on human steps, a process reward model's accuracy) is a statement about a measure, which can be arbitrarily close to perfect while the set it measures licenses every conclusion. "Valid on every step it accepts" is a statement about the set, and only it transfers to a reasoner, and to the justification of what the reasoner concludes. A learned checker must certify an inclusion $A \subseteq \mathrm{Sound}(R^*)$; the rest of the paper asks how imitation, coherence and the world can certify it.

3.3 In classical propositional logic, every unsound pure schema added to a complete calculus trivializes

One bad schema *can* trivialize a reasoner (Theorem 3.2). In classical propositional logic it always does once it is added to a complete calculus, provided it mentions no specific atom. Errors are then total, but also maximally visible: the reasoner derives $\bot$ from no premises. (Without the complete calculus a pure schema can be unsound without trivializing: T_j alone derives nothing from $\emptyset$, Remark B.1.)

Definition 3.4 (Substitution-closed step sets)**.** Recall pure schemas and purification τ° (Definition 2.4). Atoms are constants, so learned schemas may mention them: least general generalization produces $x \vee p_0 \rhd x$ whenever a position of the data is constantly p_0. A step set is *substitution-closed* if it is closed under uniform substitution of formulas for atoms (instance sets of pure schemas are). A step $\Pi \rhd \varphi$ is *sound* if $\Pi \models \varphi$, and a schema or step set is sound if all its instances or steps are.

Proposition 3.5 (Every unsound pure schema is a tonk)**.** *[T1 Prop 2.3, repaired scope] Let R^*_{CPC} be sound and complete for classical propositional consequence, i.e. $\mathrm{Cl}_{R^*_{\mathsf{CPC}}}(\Gamma) = \mathrm{Cn}_{\mathsf{CPC}}(\Gamma)$ for all Γ, over a language with $\top, \bot, \neg$. Let $A = R^*_{\mathsf{CPC}} \cup A_1$ with A_1 substitution-closed. If A contains an unsound step, then $\mathrm{Cl}_A(\emptyset) = \mathrm{Fm}$.*

Proof. Let $\Pi \rhd \varphi \in A$ be unsound, with $v(\Pi) = 1$, $v(\varphi) = 0$; it lies in A_1 since R^*_{CPC} is sound. Let σ_v send each atom true under v to $\top$ and each false one to $\bot$. Then $\sigma_v\Pi \rhd \sigma_v\varphi \in A_1$ has variable-free formulas; each $\sigma_v\pi$ and $\neg\sigma_v\varphi$ is a tautology, so lies in $\mathrm{Cl}_{R^*_{\mathsf{CPC}}}(\emptyset) \subseteq \mathrm{Cl}_A(\emptyset)$ by completeness. The step adds $\sigma_v\varphi$, and $\mathrm{Cl}_A(\emptyset) \supseteq \mathrm{Cn}_{\mathsf{CPC}}(\{\sigma_v\varphi, \neg\sigma_v\varphi\}) = \mathrm{Fm}$. □

This is essentially Post (1921): classical propositional logic is Post-complete. The consequence-level form is Theorem 6.13.

Example 3.6 (Schemas that mention atoms)**.** Without substitution-closure the proposition fails. $\tau = (x \vee p_0 \rhd x)$ is unsound (instance $p_1 \vee p_0 \rhd p_1$). Yet $\mathrm{Cn}_{\mathsf{CPC}}(\neg p_0)$ is closed under $R^*_{\mathsf{CPC}} \cup \mathrm{inst}(\tau)$, since $\neg p_0 \models B \vee p_0$ implies $\neg p_0 \models B$. So the reasoner derives the non-tautology $\neg p_0$ (from $\neg p_0 \vee p_0$) but never p_0: unsound, not trivial. A ground error $p_0 \rhd p_1$ never fires from $\emptyset$ at all (Appendix B.2).

Lemma 3.7 (Purification)**.** *[T1 Prop 2.3, repair] In classical propositional logic, τ is sound if and only if τ° is sound.*

Corollary 3.8 (Coherence decides soundness of schema hypotheses)**.** *[T1 Prop 2.3, repair] For any family of schemas τ_l, let $A = R^*_{\mathsf{CPC}} \cup \bigcup_l \mathrm{inst}(\tau_l)$ and $A^\circ = R^*_{\mathsf{CPC}} \cup \bigcup_l \mathrm{inst}(\tau_l^\circ)$. Then A is sound iff A° is sound iff $\bot \notin \mathrm{Cl}_{A^\circ}(\emptyset)$.*

Proof idea. Every instance of τ° is a uniform-substitution image of an instance of τ, and conversely every instance of τ is one of τ°. For the corollary, apply Proposition 3.5 to the substitution-closed set $\bigcup_l \mathrm{inst}(\tau_l^\circ)$. Appendix B.2. □

Reading. This is the positive face of Theorem 3.2. An unsound pure schema cannot hide: it makes the reasoner incoherent in the empty context, which is exactly the signal of the coherence channel (C). Coherence in the empty context is thus a complete soundness test for purified schema hypotheses over CPC; Section 6 turns it into identification of CPC and Section 7 into an end-to-end learner. Two limits. Purification is used as a *test*: one checks A° to decide A, without accepting the purified steps. And the test presupposes a structural target: under atom-specific principles (an axiom about one constant, a physical theory with named parameters) coherence can miss errors as in Example 3.6. Beyond classical propositional logic the dichotomy fails too; Section 6 determines what coherence detects in intuitionistic logic and arithmetic.

3.4 Simplicity is the wrong bias for positive data

A natural response to Theorem 3.2 is to prefer simple hypotheses. On positive data alone this points the wrong way.

Proposition 3.9 (MDL picks "anything from anything")**.** *[T1 Prop 2.4] Encode steps as ground terms and schemas as terms with metavariables, of size equal to the number of symbol occurrences, and let $\mathcal{H}_1 = \{\mathrm{inst}(\sigma) : \sigma \text{ a schema}\}$. For every nonempty set P of positive data, the single metavariable x is a shortest schema with $P \subseteq \mathrm{inst}(x)$, and $\mathrm{inst}(x) \supseteq S$ accepts every step. It is the unique shortest one up to renaming unless P is a single step of size 1.*

Proof. x has the minimum size 1 and covers everything; any other size-1 schema is a constant c, with $\mathrm{inst}(c) = \{c\}$. □

More generally, an MDL learner without a likelihood term over-generalizes: positive data never refute an over-general hypothesis (Gold, 1967; Angluin, 1980), and over-general hypotheses are typically short. The positive-data remedies are *least* generality (Plotkin, 1970), as in the version-space verifiers of Sections 4 and 5, and a likelihood with the size principle; since MAP selection is unsound whenever an over-general hypothesis is temporarily most probable, Section 4 thresholds the posterior conservatively instead. Simplicity has a role in this paper (Section 8), but not as the acceptance criterion for steps.

3.5 When does average-case validity survive search? The commitment order

Theorem 3.2 seems to contradict a respectable tradition. In PAC-semantics (Valiant, 2000), learned rules that are $(1-\varepsilon_i)$-valid over a distribution of scenes are chained with additive error; learning to reason (Khardon and Roth, 1997) and implicit learning (Juba, 2013) give guarantees of the same average-case kind. There is no contradiction. What decides whether an average-case guarantee survives an adversarial prover is *what* is random and *when* it is drawn.

Definition 3.10 (Sequent-level validity in a scene)**.** Fix a distribution D over *scenes* x (structures for a fixed language). A sequent $\Gamma \vdash \varphi$ is *true in* x if every assignment to its free variables that makes all of Γ true in x makes φ true in x. A rule instance (premise sequents, conclusion sequent) *holds in* x if its conclusion is true in x whenever all its premises are. A rule or schema ρ is *valid in* x if all its instances hold in x, and $(1-\varepsilon)$*-valid per scene* if $\Pr_{x\sim D}[\rho \text{ is valid in } x] \geq 1-\varepsilon$.

The sequent level matters: rules that discharge assumptions ($\to$I, reductio) or introduce eigenvariables ($\forall$I) are not truth-preserving at formula level, but they are at sequent level. So hypothetical contexts are covered.

Lemma 3.11 (Commitment order)**.** *[L3 Lemma A, as repaired]*

(i) Random world, rules fixed first. *Let $R = \{\rho_1, \dots, \rho_m\}$ be a finite library fixed before $x \sim D$ is drawn, each ρ_i $(1-\varepsilon_i)$-valid per scene. Then*

$$\Pr_{x\sim D}\left[\textit{some } R\textit{-derivation from premises true in } x \textit{ has a conclusion false in } x\right] \;\leq\; \textstyle\sum_{i=1}^{m} \varepsilon_i,$$

where the derivation may be chosen after seeing x; the bound holds simultaneously for all provers.

(ii) Fixed world, prover-chosen instances. *Fix a world x_0, a schema ρ and a distribution μ on its instances, and call ρ* $(1-\varepsilon)$-valid on average *if $\mu\{\iota : \iota \text{ holds in } x_0\} \geq 1-\varepsilon$. Suppose ρ has an instance failing in x_0. It must have one if $\mu\{\iota : \iota \text{ fails in } x_0\} > 0$, and it may have one even when $\varepsilon = 0$, if the failing instance lies off the support of μ. Then a prover that submits it obtains, with probability* 1*, a one-step derivation from premises true in x_0 to a conclusion false in x_0. No bound of the form (i) holds in this regime.*

(iii) Fresh randomness per check. *Each claim is a polynomial identity $p \equiv q$ in n variables over a field F, committed by an adaptive prover. The verifier enforces* $\deg(p-q) \leq d$*, computing the degree from an explicit expression; only then does it draw a point uniformly from U_k^n, $U_k \subseteq F$ finite, independently of everything before, and accept iff p and q agree there. Call the claim* false *if $p-q$ is nonzero* as a polynomial over F*. A false k-th claim is accepted with probability at most $d/|U_k|$. So some false claim among the first N (N fixed in advance) is accepted with probability at most $\sum_{k\leq N} d/|U_k|$, uniformly over adaptive provers; for unboundedly many claims, $|U_k| \geq d\,2^k/\delta$ gives at most δ.*

Proof idea. (i) With probability $\geq 1-\sum_i \varepsilon_i$ every ρ_i is valid in x (union bound over the library); this event does not depend on the prover, and on it every derivation preserves truth in x. (ii) Nothing is random. (iii) Schwartz–Zippel per claim, conditional on the past, and a union bound. Appendix B.3. □

Part (i) is the chaining lemma of PAC-semantics (Valiant, 2000) with the quantifier order made explicit; part (iii) is the Schwartz–Zippel lemma (Schwartz, 1980; Zippel, 1979; DeMillo

regime	what is random	drawn when	sound under search?
(i) random scene, fixed library	scene $x \sim D$	after the rules are fixed	yes, $\leq \sum_i \varepsilon_i$
(ii) one world, instance average	instance $\iota \sim \mu$	never: the prover chooses	no
(iii) fresh per-check randomness	evaluation point	after each claim	yes, $\leq \sum_k d/\lvert U_k\rvert$
learned verifier	training sample	before the search	no (Corollary 3.3)

Table 3: The commitment order (Lemma 3.11). In the two sound regimes the randomness is drawn after the rules or claims are fixed and outside the prover's control; in regime (i) the rules must be valid per scene.

and Lipton, 1978); the whole is folklore. The lemma gives two sufficient conditions and one failing regime; it is *not* a characterization. Each hypothesis is load-bearing (Appendix B.3): validity *per scene* in (i), which a guarantee per random (scene, instance) pair does not give; and in (iii) an enforced degree bound and a $p - q$ nonzero over the field used.

Reading. Table 3 sorts the channels of this paper. Mathematics has one world, so a learned checker's in-distribution accuracy is randomness over instances that the prover chooses: regime (ii), of which Theorem 3.2 is an instance. Its training sample is drawn before the prover searches, so a learned checker is never a regime-(iii) oracle. Fresh per-check randomness, by contrast, is a legitimate world channel (W) in mathematics, worst-case sound under the conditions of part (iii); in school algebra, part (iii) covers value errors of polynomial identities such as the freshman's dream, not definedness errors such as $x/x \rhd 1$ (Example 2.16). In physics the uncertainty that matters for an export rule (which carries a result from an idealized context back to the world) is over situations, so regime (i) bounds the probability that some derivation from a fixed library of bridges fails in a random situation by the library's summed validity defects, up to distribution shift: olympiad problems are selected to stress idealizations (Section 11). For justification, the lemma gives sufficient conditions for a statistical warrant to compose with search: the chance event is drawn after the reasoner has committed, and outside its control.

3.6 Voting ensembles: incoherence without information

A common response to an exploitable verifier is an ensemble that accepts a step if most members do. With coherence feedback this looks attractive: a derivation of $\bot$ from premises certified coherent shows that something accepted is invalid. Stepwise voting, however, can produce an incoherence that implicates no member.

We use the online protocol of Definition 6.1: hypotheses are step sets with prior w, designated contexts $\mathcal{A}$ are target-coherent, and a detected incoherence (an argument π from some $A \in \mathcal{A}$ to $\bot$ with $\mathrm{Steps}(\pi) \subseteq \hat{R}_t$) deletes every hypothesis containing its steps, never the target (Lemma 6.2). *Majority aggregation* announces $\hat{R}_t = \{s : w(\{R \in \mathrm{VS}_t : s \in R\}) > \frac{1}{2} w(\mathrm{VS}_t)\}$.

Theorem 3.12 (Doctrinal paradox: uninformative incoherence)**.** *[T2 Thm 2.4, as repaired] Let* $\mathcal{H} = \{h_1, h_2, h_3\}$ *with uniform prior, where* $h_i = \{\Gamma \rhd \varphi : \ T_i \cup \Gamma \models \varphi\}$ *for*

$$T_1 = \{p, q\}, \qquad T_2 = \{p, \neg q\}, \qquad T_3 = \{\neg p, q\},$$

and let $\mathcal{A} = \{\emptyset\}$*. Every* h_i *is coherent, and any of them may be the target. Under majority aggregation, for every target there is an environment under which the prover exhibits the same*

detected incoherence in every round and no hypothesis is ever deleted. The environment presents only that incoherence, or interleaves it with uninformative positive data such as $\triangleright p \vee \neg p$, which lie in every h_i. Lean: CoherenceGames.thm_2_4_environment

Proof idea. While $\mathrm{VS}_t = \mathcal{H}$, the majority set is the set of steps in at least two h_i. The argument

$$\triangleright p, \qquad \triangleright q, \qquad p, q \triangleright p \wedge q, \qquad \triangleright \neg(p \wedge q), \qquad p \wedge q,\ \neg(p \wedge q) \triangleright \bot$$

uses only majority steps ($\triangleright p \in h_1, h_2$; $\triangleright q \in h_1, h_3$; $\triangleright \neg(p \wedge q) \in h_2, h_3$; the last two are in all three). Yet each h_i lacks one of them, so the detection deletes nothing, nor do tautologous data. Appendix B.4. □

The environment must withhold discriminating data: on a text enumerating the target, every wrong hypothesis is eventually refuted and the uninformative detections stop (Appendix B.4). So the theorem concerns what coherence feedback can guarantee round by round against an unhelpful environment, not convergence on complete data. It is the doctrinal paradox of Kornhauser and Sager (1986) and the discursive dilemma of Pettit (2001), transposed from judges to verifiers; its general form is the impossibility theorem of List and Pettit (2002).

Reading. Each h_i is a coherent calculus, yet the vote derives $\bot$ from no premises, so by Lemma 3.1 the majority acceptance region is unsound relative to *every* candidate target: voting produces an accepted set that no member endorses. Worse, the incoherence does not say which member is wrong, so the coherence channel, which should supply the negative data that imitation lacks, is silenced. The repair is oligarchic acceptance (Definition 6.4): announce the intersection of a coalition of at least half the remaining weight. Against a worst-case prover exactly this guarantees that every detection removes at least half of that weight, a bit per detection (Proposition 6.6 and Theorem 6.5).

3.7 Search in practice

Experiment C (Section 12.3) illustrates the gap (computed): a gradient-boosting step classifier with in-distribution AUC 0.995, calibrated to 99% in-distribution balanced precision, lets a black-box prover prove 16.3 of 23 false goal equations, while the learned calculus, after world feedback, proves none. That learned verifiers are exploited by search is well documented (Gao et al., 2023; Cobbe et al., 2021). The theorems explain why in-distribution accuracy, a statement about a measure fixed before the prover moves, cannot rule this out (Lemmas 3.1 and 3.11 and Theorem 3.2). They do not predict the *rate* at which a given search finds errors; for neural verifiers it is graded, as the overoptimization curves show.

4 Cautious verification and its price

Section 3 reduced the soundness of a search-driven reasoner to a set inclusion: everything the verifier accepts must lie in $\mathrm{Sound}(R^*)$, and average-case accuracy does not certify this. This section shows how a learner can certify the inclusion, and what it costs. The answer is caution. Accept a step only if every hypothesis consistent with the evidence accepts it, reject it only if none does, and otherwise escalate it to a human or an oracle. This is reliable learning (Rivest and Sloan, 1988), perfect selective classification (El-Yaniv and Wiener, 2010) and the "accept"

side of KWIK (Li et al., 2008), applied to steps. It is sound against every prover from the first datum on, and it is the most permissive 0-sound verifier (Theorem 4.2).

Its price is paid in escalations, never in soundness. The worst case is exactly a bounded, target-relative form of Wright's elasticity (Theorem 4.4), computed for the main classes in Table 4. A Bayesian version adds time-uniform soundness from noisy data, with no union bound over queries (Section 4.4), but its escalation cost is exponential in description length unless structure intervenes. In one line: *structure, not the prior, makes sound learned verification affordable.*

The results are elementary. As far as we know, three are new: the exact single-schema budget, the tagged/untagged separation and the Bell-number bound for two-sided checking over growing signatures. Proposition C.5 checks that Ville's constant cannot be improved in this protocol, as expected since Ville's inequality is essentially tight for test martingales with small increments (Ramdas et al., 2023).

Protocol, soundness and cost are as in Definitions 2.17, 2.18 and 2.20. Human data $P_0 \subseteq R^*$ are fixed unless said otherwise, and $\mathrm{Esc}(\mathcal{H}; N)$ restricts queries to steps of size at most N. We use the rank lemma and the witness events (R), (D) of Lemmas 2.6 and 2.7.

4.1 The version-space verifier is sound and optimal

Definition 4.1 (Version space; VS verifier)**.** [T1 §3] Let $\mathcal{H} \subseteq 2^S$ be realizable. For sets P (positive) and P^- (negative) of labelled steps, $\mathrm{VS}(P, P^-) = \{R \in \mathcal{H} : \ P \subseteq R, \ R \cap P^- = \emptyset\}$, and $\mathrm{VS}(P) = \mathrm{VS}(P, \emptyset)$. The *VS verifier* starts with $P = P_0$, $P^- = \emptyset$, and answers a query q with ACCEPT if $q \in \bigcap \mathrm{VS}(P, P^-)$, with REJECT if $q \notin \bigcup \mathrm{VS}(P, P^-)$, and with ESCALATE otherwise; an escalated q joins P or P^- according to the oracle's label. For a history h, VS_h is the version space computed from P_0 and the labels in h.

In the terms of El-Yaniv and Wiener (2010), this is the consistent selective strategy: it escalates exactly on the disagreement region of the version space.

Theorem 4.2 (Soundness and optimality of caution)**.** *[T1 Thm 3.1, as repaired]*

(a) The VS verifier is uniformly 0*-sound. For every target $R^* \in \mathcal{H}$ with $P_0 \subseteq R^*$ and every prover, $R^* \in \mathrm{VS}(P, P^-)$ at all times. So everything it accepts lies in R^*, and a reasoner that chains accepted steps is sound (Lemma 3.1).*

(b) Let V be any (possibly randomized) δ-sound verifier for $\mathcal{H}$ with human data P_0. Fix a target $R^* \in \mathcal{H}$ with $P_0 \subseteq R^*$, a history h of t rounds (queries $q_1, \ldots, q_t$, answers, oracle labels) whose labels agree with R^*, and a query $q \notin \bigcap \mathrm{VS}_h$. Let p be the joint *(unconditional) probability, over V's coins, that against R^* the prover issuing $q_1, \ldots, q_t, q$ produces h and then has q accepted. Then $p \leq \delta$.*

(c) In particular, a deterministic δ-sound verifier with $\delta < 1$ accepts no $q \notin \bigcap \mathrm{VS}_h$ at any history h that it produces against a target in $\mathcal{H}$. History by history, its acceptance region lies inside $\bigcap \mathrm{VS}_h$, the set the VS verifier accepts.

Lean: StepSoundness.thm_3_1_a (a) *Lean: StepSoundness.thm_3_1_b (c)*

Proof. (a) All labels are truthful, so the target never leaves the version space, and hence $\bigcap \mathrm{VS} \subseteq R^*$.

(b) Pick $R' \in \mathrm{VS}_h$ with $q \notin R'$; it is an admissible target that agrees with every label in h. Coupling the runs against R^* and R' through the verifier's coins, the event "h, then q accepted" has probability p under both (Appendix C.1). As $q \notin R'$, δ-soundness for R' gives $p \leq \delta$.

(c) A deterministic verifier answering fixed queries has $p \in \{0, 1\}$, so $p \leq \delta < 1$ forces $p = 0$. □

The joint form in (b) is essential. Read conditionally on h, (b) is false: a verifier that accepts everything with probability ε, and otherwise runs the VS verifier, is ε-sound, yet after some histories it accepts steps outside $\bigcap \mathrm{VS}_h$ with conditional probability 1 (Appendix C.1). With random human data the coupling must also transport the data, giving only $\mathbb{P}_{R'}[\mathrm{data}] \cdot p \leq \delta$; this is where Section 4.4 takes over. The theorem shows that the VS verifier is the *largest* 0-sound acceptance rule, so its escalations are forced on every 0-sound verifier.

Reading. A learner can *guess* the calculus (the simplest or most probable hypothesis), which is unsound under search (Section 3). Or it can *certify* steps, accepting only what every calculus consistent with the evidence accepts, which is sound from the first datum without negative data. Gold's problem (Gold, 1967) does not threaten certification. An over-general hypothesis, tonk included, never licenses a step by itself, because the target stays in the version space as a guardian. Over-generality costs only completeness. Since contexts are part of judgments, this covers hypothetical steps verbatim (→I; "assume the air pressure is 0").

4.2 The escalation dimension is a positive elasticity

Definition 4.3 (Elastic chain; positive elasticity)**.** [T1 §3] Let $R \in \mathcal{H}$ with $P_0 \subseteq R$. An *elastic chain in* R (relative to P_0) is a sequence $s_1, \ldots, s_m \in R$ with $s_i \notin \bigcap \mathrm{VS}(P_0 \cup \{s_1, \ldots, s_{i-1}\})$ for every i. Equivalently, there are *witnesses* $R_i \in \mathcal{H}$ with $P_0 \cup \{s_1, \ldots, s_{i-1}\} \subseteq R_i \not\ni s_i$. The *positive elasticity* $\mathrm{el}(\mathcal{H}, R \mid P_0)$ is the supremum of the lengths of such chains, and $\mathrm{el}(\mathcal{H}, R \mid P_0; N)$ restricts the s_i to size at most N.

Wright's finite elasticity (Wright, 1989; Motoki et al., 1991) forbids infinite sequences of this kind and suffices for identification in the limit from text. Positive elasticity is its bounded version, with the points in the target.

Theorem 4.4 (Escalation dimension = positive elasticity)**.** *[T1 Thm 3.2, proof as repaired] For every class $\mathcal{H}$ and human data P_0,*

$$\mathrm{Esc}(\mathcal{H} \mid P_0) \;=\; \sup_{R \in \mathcal{H},\ R \supseteq P_0} \mathrm{el}(\mathcal{H}, R \mid P_0).$$

The VS verifier attains it: against an honest prover for target R^*, it never rejects and escalates at most $\mathrm{el}(\mathcal{H}, R^* \mid P_0)$ times. Conversely, if $s_1, \ldots, s_m$ is an elastic chain in some $R \in \mathcal{H}$, the honest prover for R that queries $s_1, \ldots, s_m$ in order forces every δ-sound (possibly randomized) verifier to expected cost at least $(1 - \delta)m$, and every 0-sound verifier to cost m almost surely. The same holds for steps of size at most N. Lean: Unstructured.thm_3_2 Lean: Unstructured.vsVerifier_attains *(deterministic 0-sound verifiers, no size bound)*

Proof idea. Against an honest prover, the VS verifier escalates q exactly when $q \notin \bigcap \mathrm{VS}(P_0 \cup E)$, with E the earlier escalated queries, so these form an elastic chain in R^*. Conversely, R and the witness R_i label $s_1, \ldots, s_{i-1}$ alike, so the transcripts up to round i have the same law under both; s_i is invalid under R_i, so it is accepted with probability at most δ under both. Appendix C.2. □

So the cost of caution depends only on the class, the target and P_0, not on the prover, the human distribution or a prior. It is KWIK's adversarial abstention count (Li et al., 2008, 2011),

made one-sided; Rivest and Sloan (1988) and El-Yaniv and Wiener (2010) bound instead the distributional mass of abstention.

Proposition 4.5 (Intersection-closed classes). *[T1 Prop 3.3] (known in substance) Suppose $\mathcal{H} \cup \{\emptyset\}$ is closed under arbitrary intersections. For P contained in some member, let $\mathrm{cl}(P)$ be the intersection of all members of $\mathcal{H}$ that contain P. Then:*

(i) $\bigcap \mathrm{VS}(P) = \mathrm{cl}(P)$*, the least member of $\mathcal{H} \cup \{\emptyset\}$ containing P. So, against an honest prover, the VS verifier is the* closure algorithm*: it accepts iff $q \in \mathrm{cl}(P_0 \cup E)$, where E is the set of steps escalated so far.*

(ii) $\mathrm{el}(\mathcal{H}, R \mid P_0)$ *is the largest m for which there is a strict chain $\mathrm{cl}(P_0) = C_0 \subsetneq C_1 \subsetneq \cdots \subsetneq C_m \subseteq R$ of members of $\mathcal{H} \cup \{\emptyset\}$.*

Proof idea. An elastic chain gives the strict chain $C_i = \mathrm{cl}(P_0 \cup \{s_1, \ldots, s_i\})$; conversely, pick $s_i \in C_i \setminus C_{i-1}$ (Appendix C.3). This is the closure algorithm of Helmbold et al. (1990); its use for one-sided learning from positive examples goes back to Natarajan (1987). We claim nothing new beyond reading the chain length as an escalation count.

Lemma 4.6 (Single schemas are intersection-closed). *[T1 §3, after Prop 3.3] $H_1 = \{\mathrm{inst}(\sigma) : \sigma$ a schema$\} \cup \{\emptyset\}$ is closed under arbitrary intersections. For nonempty P, $\mathrm{cl}(P) = \mathrm{inst}(\mathrm{lgg}P)$.*

Proof idea. Two instance sets meet in the instance set of a most general common instance, or not at all, and a ground term has only finitely many generalizations up to renaming (Appendix C.3).

Reading. Caution climbs a lattice from below. Each escalation moves the accepted set one strict step up, and the price is the longest climb from what the human data force to the target. For schemas the lattice is Plotkin's (Plotkin, 1970; Reynolds, 1970), the closure algorithm is anti-unification, and each step up costs a unit of rank. A learner that acquires a connective's inferential role from its uses climbs in this way, and the elasticity bounds how often it must ask before its accepted inferences cover that role.

4.3 What caution costs: from named rules to unstructured classes

Theorem 4.4 reduces the cost of caution to a property of the class. This subsection computes it for the classes that matter here (Table 4). The main contrast is between rules cited by name, where caution costs at most $N+1$ escalations per rule, and uncited rules, where even one rule can force $\lfloor (N-1)/k \rfloor^k$. Unstructured classes cost exponentially in description length, and certifying invalidity too costs a Bell number.

Theorem 4.7 (One schema, learned by anti-unification). *[T1 Thm 3.4, as restated] Let the target be $\mathrm{inst}(\sigma^*) \in H_1$, written σ^* inside* el.

(i) For nonempty $P_0 \subseteq \mathrm{inst}(\sigma^*)$: $\mathrm{el}(H_1, \sigma^* \mid P_0) \le \mu(\mathrm{lgg}P_0) - \mu(\sigma^*)$.

(ii) From no data: $\mathrm{el}(H_1, \sigma^* \mid \emptyset; N) \le \max\{0, 1 + N - \mu(\sigma^*)\}$. *Hence* $\mathrm{Esc}(H_1; N) \le N + 1$.

(iii) This is tight. If the signature contains two constants a, b and a unary symbol g, then for the universal schema $\sigma^* = x$, $\mathrm{el}(H_1, x \mid \emptyset; N) = N + 1$. *So* $\mathrm{Esc}(H_1; N) = N + 1$.

Proof idea. (i) With $g_i = \mathrm{lgg}(P_0 \cup \{s_1, \ldots, s_i\})$, Lemma 4.6 gives $s_i \notin \mathrm{inst}(g_{i-1})$, so $\mathrm{lgg}P_0 = g_0 \prec \cdots \prec g_m \preceq \sigma^*$ strictly, and the rank lemma bounds m. (ii) From no data the first query

escalates, and it has rank at most N. (iii) Query $g^{N-1}(a)$, $g^{N-1}(b)$, $g^{N-2}(a)$, $g^{N-3}(a)$, …, a. The successive lggs are $g^{N-1}(a)$, $g^{N-1}(x)$, $g^{N-2}(x)$, …, x, of ranks $N, N-1, \ldots, 0$, and each query lies outside the previous lgg. Full proof, and an exhaustive check, in Appendix C.4. □

After a corpus P_0, the residual budget is at most the *generality gap* $\mu(\mathrm{lgg}P_0) - \mu(\sigma^*)$. It is zero once the corpus satisfies the witness events (Lemma 2.7 and Section 5). These numbers are small only because the protocol is one-sided.

Proposition 4.8 (Certifying invalidity is expensive)**.** *[T1 Prop 3.5, as revised] Call a verifier* two-sided *if it is* 0*-sound and rejects only steps outside* $\bigcup \mathrm{VS}$*, and count all its escalations, of valid and invalid queries alike. Over a signature with constants* $a, b_1, \ldots, b_n$ *and an* n*-ary symbol* f*, there are a target in* H_1 *and a prover that force every two-sided verifier for* H_1 *to escalate at least* $\mathrm{Bell}(n)$ *times on steps of size* $n+1$*. Over a* fixed *finite signature, the two-sided VS verifier escalates on steps of size at most* N *at most as often as there are such steps,* $2^{O(N)}$*.*

Proof idea. Let target and P_0 be $\{f(a, \ldots, a)\}$. For a partition π of $[n]$, s_π puts b_j and σ_π a metavariable x_j on the j-th block; $s_{\pi'} \in \mathrm{inst}(\sigma_\pi)$ iff π refines π'. Query the s_π finest first: each is invalid, yet σ_π is still consistent when s_π is queried. Full proof, and a brute-force check for $n \le 4$, in Appendix C.5. □

So certifying validity costs at most $N+1$ escalations over any signature, and certifying invalidity at least $\mathrm{Bell}(N-1)$ over a signature with N constants and an $(N-1)$-ary symbol; compare the exponential KWIK cost of conjunctions (Li et al., 2011) and the negative results for two-sided reliable learning (Kivinen, 1995). A step checker never needs it: *one-sidedness is what makes learned verification cheap.*

Theorem 4.9 (k rules cited by name)**.** *[T1 Thm 3.6, as repaired] Steps are tagged (Definition 2.4) with the rule* $i \in [k]$ *they cite, as in formal proofs (Lean lemma names, Metamath labels, natural-deduction rule names). Let*

$$H_k^{\mathrm{tag}} = \Big\{ \bigcup_{i \le k} \{i\} \times \mathrm{inst}(\sigma_i) \Big\},$$

with target schemas $\sigma_1^*, \ldots, \sigma_k^*$ *and human data* P_0 *whose tag-*i *part is* $P_0^{(i)}$*. Then*

$$\mathrm{el}(H_k^{\mathrm{tag}}, R^* \mid P_0; N) = \sum_{i=1}^{k} \mathrm{el}_i, \qquad \mathrm{el}_i \le \begin{cases} \mu(\mathrm{lgg}P_0^{(i)}) - \mu(\sigma_i^*) & \text{if } P_0^{(i)} \neq \emptyset, \\ \max\{0,\, 1 + N - \mu(\sigma_i^*)\} & \text{if } P_0^{(i)} = \emptyset, \end{cases}$$

where el_i *is the positive elasticity of* $\mathrm{inst}(\sigma_i^*)$ *among single-schema instance sets, relative to* $P_0^{(i)}$*, on steps of size at most* N*. In particular* $\mathrm{Esc}(H_k^{\mathrm{tag}}; N) \le k(N+1)$*, with equality under the hypothesis of Theorem 4.7(iii).*

Proof idea. H_k^{tag} is not intersection-closed, but its version space *factors*, $\mathrm{VS}(P) = \prod_i \mathrm{VS}_i(P^{(i)})$ with nonempty single-schema factors, so $\bigcap \mathrm{VS}(P) = \bigcup_i \{i\} \times \bigcap \mathrm{VS}_i(P^{(i)})$. Elastic chains are thus interleavings of per-rule chains, each bounded by Theorem 4.7 (Appendix C.6). □

Without citations the picture changes, even when only one rule is in use.

Theorem 4.10 (k rules without citations)**.** *[T1 Thm 3.7, scope as repaired] Let* $H_k = \{\bigcup_{l \le k} \mathrm{inst}(\tau_l)\}$ *be the class of untagged unions of at most* k *schemas.*

(i) Lower bound, even for one rule. *Suppose the signature contains a k-ary p, a unary g and a constant c. Then an honest sequence for the single-rule target* $\mathrm{inst}(p(x_1, \ldots, x_k)) \in H_1 \subseteq H_k$ *forces* $\lfloor (N-1)/k \rfloor^k$ *escalations on steps of size at most N. So* $\mathrm{Esc}(H_k; N) \geq \lfloor (N-1)/k \rfloor^k$, *whereas* $\mathrm{Esc}(H_1; N) \leq N + 1$.

(ii) Upper bound. *For $k \geq 2$,* $\mathrm{Esc}(H_k; N) \leq (k^{N+1} - 1)/(k-1)$.

(iii) Linear flat schemas. *On the steps $\{a, b\}^n$, $n \geq k$, the unions of at most k linear flat schemas (that is, of at most k subcubes) have* $\mathrm{Esc} = \Theta(n^k)$. *With repeated metavariables,* Esc *lies between $\Omega(n^k)$ and $O(n^{2k})$. Exact bounds are in Proposition C.2.*

Proof idea. (i) Query $p(g^{a_1}c, \ldots, g^{a_k}c)$, $a \in \{0, \ldots, n\}^k$, $n + 1 = \lfloor (N-1)/k \rfloor$, by non-increasing $\sum_j a_j$. Each earlier step lies in the union of the k schemas $p(x_1, \ldots, g^{a_j+1}(x_j), \ldots, x_k)$, which misses the current one. (ii) Chains of indices that the witnesses keep in one block give strictly increasing lggs, so they have length at most $N + 1$; a Ramsey-type recursion finishes. Full proofs, the subcube count behind (iii) and exact small values are in Appendix C.7. □

Conjecture 4.11 (Binomial bound)**.** *[T1 Conj 3.8] (conjecture) Let $\mathcal{C}$ be a family of subsets of U that is closed under intersections and contains U. Let h be its* height*: the number of inclusions in the longest strict chain of members from $\bigcap \mathcal{C}$ to U. Then the positive elasticity of the unions of 1 to k members of $\mathcal{C}$, relative to $P_0 = \emptyset$ and with target U, is at most $\binom{h+k-1}{k}$.*

Evidence. The bound is exact for $k = 1$ (Proposition 4.5) and for $h = 2$. The case $(k, h) = (2, 3)$ is proved, graphic matroids show that the bound would be tight at $k = 2$, and computer searches found no counterexample (Appendix C.8). For schemas $h \leq N + 1$, so the conjecture would give $\mathrm{Esc}(H_k; N) \leq \binom{N+k}{k}$. This would match (i) up to $e^{O(k)}$, making the truth $\Theta_k(N^k)$.

Theorem 4.12 (Unstructured classes)**.** *[T1 Thm 3.9] (known: KWIK enumeration) For finite $\mathcal{H}$,* $\mathrm{Esc}(\mathcal{H}) \leq |\mathcal{H}| - 1$. *The class $\{U \setminus \{u\} : u \in U\}$ attains $|U| - 1$.* Lean: Unstructured.thm_3_9_Esc_le_card Lean: Unstructured.thm_3_9_coSingleton *(deterministic verifiers)*

Proof. Each escalation of an honest q removes from the version space a hypothesis that misses q, never the target. For the class, query $R^* = U \setminus \{u^*\}$ in any order; the witness $U \setminus \{s\}$ contains all queries before s and misses s. □

This is KWIK's enumeration bound (Li et al., 2008): 2^L hypotheses of L bits can need $2^L - 1$ escalations. Single schemas of size $\leq N$ (with a binary symbol) also number $2^{\Omega(N)}$, yet need at most $N + 1$. The price is measured by lattice height, not cardinality.

Reading. Table 4 quantifies one way in which creating "a language in which formal proofs can be written" with named rules, as the motivating notes (Hänni, 2026) describe the formalization of mathematics, makes inference learnable. Formal proofs cite their rules, and then caution costs linearly in step size, additively over rules. Informal proofs and physics solutions cite none. The verifier must then pay at least $\lfloor (N-1)/k \rfloor^k$ even when one rule is in use, since it must rule out that the steps seen so far are covered by k narrower rules. The proved upper bound is exponential, and $\Theta_k(N^k)$ is conjectured. Informal steps are moreover multi-step elements of $\mathrm{Sound}(R^*)$, treated in Section 10.

class (steps of size $\leq N$)	worst-case escalations	result
intersection-closed class	longest strict chain of members from $\mathrm{cl}(P_0)$ into the target	Proposition 4.5
one schema, after a corpus P_0	$\leq \mu(\mathrm{lgg}P_0) - \mu(\sigma^*)$ (generality gap)	Theorem 4.7(i)
one schema, no data	$\leq \max\{0, 1 + N - \mu(\sigma^*)\}$; worst case $N+1$, tight	Theorem 4.7(ii,iii)
k rules cited by name (tagged)	$\sum_i \mathrm{el}_i \leq k(N+1)$, tight	Theorem 4.9
k rules without citations	$\geq \lfloor (N-1)/k \rfloor^k$ even for a one-rule target; $\leq (k^{N+1}-1)/(k-1)$; conjecturally $\leq \binom{N+k}{k}$	Theorem 4.10(i,ii), Conjecture 4.11
k linear flat schemas on $\{a,b\}^n$	$\Theta(n^k)$; between $\Omega(n^k)$ and $O(n^{2k})$ with repeated metavariables	Theorem 4.10(iii)
unstructured finite class	$\lvert\mathcal{H}\rvert - 1$, tight; $2^L - 1$ for L-bit hypotheses	Theorem 4.12
two-sided, one schema	$\geq \mathrm{Bell}(N-1)$ over a growing signature; $\leq 2^{O(N)}$ over a fixed one	Proposition 4.8
Bayes-conservative, VS posterior, $\delta \leq w^*$	$\leq \ln(1/w^*)/\delta$; $\tilde{\Theta}(2^L)$ on unstructured classes with $w^* = 2^{-L}$	Theorem 4.16, Corollary 4.17
structure plus Bayes	$\min\{\mathrm{el}(\mathcal{H}, R^* \mid P_0),\ \ln(1/w^*)/\delta\}$	Corollary 4.18

Table 4: The price of caution: escalations (human interventions on valid steps) needed by a sound verifier against an honest prover. "Tight" means attained over a signature with two constants and a unary symbol. All entries are proved except the conjectured bound.

4.4 Bayes-conservative verification and Ville's inequality

A posterior relaxes the version space's demand of truthful data, and makes very large classes usable, which eases (but does not remove) the demand of realizability. MAP acceptance is unsound whenever an over-general hypothesis is temporarily most probable (Section 3.4). Caution instead thresholds the posterior probability of invalidity.

Definition 4.13 (Bayes-conservative verifier)**.** [T1 §4] Let $\mathcal{H}$ be countable with prior w and $w^* = w(R^*) > 0$. Each $R \in \mathcal{H}$ carries a law p_R on S for human data (which may model errors, or follow the size principle) and an oracle kernel $\ell_R(y \mid q)$, $y \in \{0,1\}$, which is *deterministic* if $\ell_R(y \mid q) = \mathbf{1}[y = \mathbf{1}[q \in R]]$. The model is *well specified*: human data are i.i.d. p_{R^*}, and oracle answers follow $\ell_{R^*}(\cdot \mid q_t)$. After the observations of rounds $\leq t$, the likelihood is $L_t(R)$, the posterior is $w_t(R) \propto w(R)L_t(R)$, and

$$Z_t := \sum_{R \in \mathcal{H}} w(R)\,\frac{L_t(R)}{L_t(R^*)}, \qquad \text{so that} \qquad w_t(R^*) = \frac{w^*}{Z_t}.$$

The verifier V_δ answers q with ACCEPT iff $w_t(\{R : q \notin R\}) < \delta$, optionally with REJECT iff $w_t(\{R : q \in R\}) < \delta_r$, and with ESCALATE otherwise. The *VS posterior* is $w_t^{\mathrm{VS}}(R) \propto w(R)\,\mathbf{1}[R \text{ is consistent with all data}]$.

Theorem 4.14 (VS posterior: deterministic soundness)**.** *[T1 Thm 4.1] Suppose all data are truthful: human positives lie in R^*, and oracle labels are correct. If $\delta \leq w^*$, then V_δ with the VS posterior never accepts an invalid step, for any prover and at any time. The guarantee is per target: it covers every target with $w(R^*) \geq \delta$. Conversely, for every $\delta > w^*$ there is a two-hypothesis class on which V_δ accepts an invalid step at time 0.*

Proof. Truthful data never contradict the target, so $w_t^{\mathrm{VS}}(R^*) = w^*/w(\mathrm{VS}_t) \geq w^*$. For an invalid q, $w_t(\{R : q \notin R\}) \geq w_t(R^*) \geq w^* \geq \delta$, so q is not accepted. For the converse, take $\mathcal{H} = \{R^*, R'\}$ with $R' \supsetneq R^*$, $w(R^*) = w^*$ and $q \in R' \setminus R^*$. Then $w_0(\{R : q \notin R\}) = w^* < \delta$. □

The truth is a *guardian*: an invalid step can be accepted only once the target's own posterior has fallen below δ. With noisy data this can happen, but only on one global event.

Theorem 4.15 (Time-uniform soundness)**.** *[T1 Thm 4.2, as repaired] Assume the model is well specified and $\delta \le w^*\delta'$.*

(a) Suppose that, conditional on $\mathcal{F}_t$, the next observation has its model law: a fresh $X \sim p_{R^*}$, or an answer $y \sim \ell_{R^*}(\cdot \mid q_{t+1})$. Here $\mathcal{F}_t$ contains the history and all of the prover's information*, including its internal randomness and its choice of q_{t+1}. So the prover cannot foresee human data or oracle noise, and the noise is fresh at every query. Then for every such prover*

$$\mathbb{P}\big[\exists t:\ V_\delta \text{ accepts some } q_t \notin R^*\big] \le \delta'.$$

(b) If the oracle is deterministic, the guarantee is uniform*. There is an event of probability at least $1-\delta'$, depending only on the human data, on which no prover strategy ever gets an invalid step accepted. This holds even for a prover that knows the target and all future human data. Only the human data need to be i.i.d. p_{R^*}.*

Proof idea. Under (a), each observation multiplies the R-term of Z_t by a likelihood ratio of conditional mean $\sum_{o:\ell_{R^*}(o)>0} \ell_R(o) \le 1$. So Z is a nonnegative supermartingale with $Z_0 = 1$, and Ville's inequality (Ville, 1939) gives $\mathbb{P}[\sup_t Z_t \ge 1/\delta'] \le \delta'$; off this event $w_t(R^*) = w^*/Z_t > \delta$ for all t. In (b), Z_t is bounded pathwise by a supermartingale of the human data alone. Appendix C.9. □

Credit. Part (a) is the prior–posterior-ratio martingale of Waudby-Smith and Ramdas (2020), applied to validity relations. Part (b) is an easy corollary. See Ville (1939); Shafer et al. (2011); Ramdas et al. (2023) on Ville's inequality and test martingales. The closest existing analogues are conservative Bayesian agents that defer to a mentor (Cohen and Hutter, 2020; Cohen et al., 2022). New here is only the check that the constant is tight in this protocol (Proposition C.5), an instance of the known near-tightness of Ville's inequality; the application to step validity is routine.

No union bound over queries. A prover may try 10^{100} steps, and one event of probability at least $1-\delta'$ covers them all, because accepting *any* invalid step requires the posterior of the truth to fall below δ. A per-step scorer has no such single event, and its errors can be found one at a time (Theorem 3.2).

The hypothesis of (a) is needed: a prover that knows in advance when the oracle's answer will be flipped gets an invalid step accepted with probability 1 (Appendix C.9). The constant $\delta \le w^*\delta'$ cannot be improved (Proposition C.5).

Theorem 4.16 (Escalation bound)**.** *[T1 Thm 4.4] Let the oracle be deterministic, let the human data satisfy the hypothesis of Theorem 4.15(a), and let $\delta_m = \min(\delta, \delta_r)$. Write $\#\mathrm{ESC}_t$ for the number of escalations up to time t.*

(i) With probability at least $1-\delta''$, simultaneously for all t, $\#\mathrm{ESC}_t \le \big(\ln(1/w^*)+\ln(1/\delta'')\big)/\delta_m$.

(ii) With the VS posterior, deterministically, $\#\mathrm{ESC}_t \le \ln(1/w^*)/\delta_m$.

For an honest prover, or when only escalations of valid queries are counted, δ_m may be replaced by δ. Without a reject option ($\delta_r = 0$), the bound on escalations of invalid queries is vacuous.

Proof idea. Z_t evolves multiplicatively. A human datum multiplies it by a factor of conditional mean at most 1. Escalating a valid (invalid) q multiplies it by at most $1-\delta$ ($1-\delta_r$), since q was

not accepted (rejected). Combine with $Z_t = w^*/w_t(R^*) \geq w^*$ and Ville's inequality for the data factors. Appendix C.11. □

Corollary 4.17 (Bayes is essentially optimal on unstructured classes)**.** *[T1 Cor 4.5] On $\mathcal{H} = \{U \setminus \{u\} : u \in U\}$ with the uniform prior ($w^* = 1/|U|$), every δ'-sound verifier has worst-case honest cost at least $(1-\delta')(1/w^* - 1)$, while $V_{w^*\delta'}$ with the VS posterior is 0-sound and pays at most $\ln(1/w^*)/(w^*\delta')$. The ratio of the two bounds is $\ln(1/w^*)/(\delta'(1-\delta')(1-w^*))$, which is $O(\ln(1/w^*)/\delta')$ for $w^*, \delta' \leq 1/2$. With a description-length prior $w^* = 2^{-L}$, the cost is $\tilde{\Theta}(2^L)$.*

Proof. Apply Theorem 4.4 to the chain of Theorem 4.12, of length $1/w^* - 1$, and use Theorem 4.16(ii) and Theorem 4.14. □

The factor $1/\delta'$ is an artifact of the threshold: with $\delta = w^*$ the VS posterior accepts exactly $\bigcap$VS on this class (every consistent hypothesis has posterior at least w^*), so it is the VS verifier itself, which is exactly optimal.

Corollary 4.18 (Structure plus Bayes)**.** *[T1 Cor 4.6, as repaired] With truthful data, the VS-posterior verifier with threshold δ never accepts an invalid step, for any prover and every target with $w(R^*) \geq \delta$. Its escalations of valid queries number at most*

$$\min\left\{\mathrm{el}(\mathcal{H}, R^* \mid P_0),\ \ln(1/w^*)/\delta\right\}.$$

Proof idea. Soundness is Theorem 4.14. Escalated valid queries lie outside the current $\bigcap$VS, so they form an elastic chain in R^*; the second term is Theorem 4.16(ii). Appendix C.11.

The guarantee is not 0-soundness (Definition 2.18) unless every target has prior at least δ, which fails for infinite $\mathcal{H}$.

Structure, not the prior. For single schemas with a description-length prior, $\ln(1/w^*)/\delta \geq \ln(1/w^*)/w^*$ is exponential in $|\sigma^*|$, while el $\leq N + 1$. The prior earns its keep elsewhere. If $\mathcal{H}$ contains every finite step set (calculi with unboundedly many ground rules, so exception lists are hypotheses), the bare version space never generalizes: $\bigcap\mathrm{VS}(P) = P$. Under the hypotheses of Theorem 4.15, the Bayesian verifier is sound with probability at least $1 - \delta'$ for every target with $w(R^*) \geq \delta/\delta'$, i.e. under a description-length prior, every rule system of description length at most $\log_2(\delta'/\delta)$. Hence a three-layer recipe:

- a structured version-space core, for cheap escalations and exact soundness when realizable;
- a Bayesian hierarchy over it, as insurance against unrealizability;
- conservative thresholds, never MAP.

One caveat: Theorem 4.15 needs well-specification. Systematic human errors are exactly a misspecification the model does not anticipate, and the posterior then concentrates on "rule plus fallacy" as a rule (Section 5).

4.5 What caution buys, and what it cannot do

Learning which inferences are valid. A cautious learner learns exactly what its evidence forces, never adopts a rule it cannot certify, and pays for completeness in escalations, of which the positive elasticity is the exact worst case. With rules cited by name and varied human data, positive data alone bring it to exactly the human calculus after coupon-collector many steps, soundly at every moment (Section 5).

Coherence. A cautious verifier gets no signal from coherence. Its accepted steps lie in R^*, so no argument from a designated context to $\bot$ uses only accepted steps (Proposition 6.3). Coherence informs only *bold* learners. Hence the two tiers of Section 7: cautious assertion, and a bold sandbox attacked by a red team.

Limits. Realizability is load-bearing. If the true calculus lies outside the class, for example two untagged rules while $\mathcal{H} = H_1$, the lgg over-generalizes and soundness fails. The Bayesian hierarchy repairs this only probabilistically, at a cost exponential in the complexity of what is missing. Soundness is relative to the *human* calculus, naive comprehension included if humans use it, and improving on it is the job of coherence and the world (Sections 6, 7 and 10). The first-order term model leaves out associative-commutative contexts, binders and variable-arity rules, where the bounds of Table 4 would need re-proof. Which contexts may be designated, and how results are exported, is left to Section 11.

Justification. Certified acceptance models a warrant that survives search. A step is warranted when every hypothesis the evidence leaves open licenses it (relative to $\mathcal{H}$ and the human calculus, not to truth), and no amount of adversarial search erodes such a warrant, unlike a statistical one (Section 3). The Bayesian version (Theorem 4.15) trades certainty for a single chance of error δ', fixed in advance, that does not grow with the number of steps tried or the running time, as long as the searcher cannot foresee the data or the noise. Both are conditional reliabilist vindications, the second also fallibilist, relative to explicit assumptions (realizability; truthful or well-specified data), and later sections show that each assumption is needed (Section 13).

5 Imitation: what positive examples fix

The first channel is imitation. The learner sees human steps, possibly with the names of the rules they cite, and nothing else; no step is ever labelled invalid. This section asks what such data fix when acceptance must be cautious, as search requires (Sections 3 and 4). The answer has two halves.

With the right hypothesis class (schemas, learned by least general generalization), positive examples fix the rules *exactly*. A cautious verifier fed only human steps is sound at every time. It becomes exactly complete after finitely many examples iff the target has a finite *anchor* (Theorem 5.3). For rules that humans cite by name this happens at a coupon-collector rate (Theorem 5.6); without citations it needs a diversity condition (Theorem 5.8). Once exact, the verifier generalizes systematically: it accepts exactly the target calculus, on steps and derivations of any size, whatever the distribution of the human data (Corollary 5.10). This is the formal sense in which finite usage fixes inferential role for unbounded use.

The other half is that positive data cannot tell what humans do from what is valid. One mislabelled step collapses a learned rule into "anything follows" (Proposition 5.12). Sporadic errors can be trimmed when they are rarer than the evidence for each rule's generality, and this threshold is necessary up to a factor of two (Theorems 5.13 and 5.14 and Corollary 5.18). Systematic errors are indistinguishable from rules at every rate (Corollary 5.19). Removing them needs negative information (Sections 6 and 7).

Most results below are elementary or known in substance, and we say which. New here, as far as we know, are three things: anchors read as the completeness condition of *certified* verification; the witness-event sample complexities and their noise versions, with necessity up to a factor of two; and the exact statement that schema-generated errors are rules. None is deep. Together they pin down when imitation works. Throughout, the signature Ω in which steps are encoded

(Section 2.2) has at least two constants. Then a proper specialization of a schema has strictly fewer instances, and the instance sets of all schemas have empty intersection (Appendix D.1).

5.1 Cautious learning from positive data

Let $\mathcal{H} \subseteq 2^S$ be a hypothesis class of step sets (Definition 2.9) containing the target R^*, and let the data D be a finite multiset of human steps. Until Section 5.6 the data are clean, $D \subseteq R^*$.

Lemma 5.1 (The cautious verifier; trivial). *[T1 §5; Thm 3.1, Prop 3.3, positive data] With no oracle, the VS verifier of Definition 4.1 is the* cautious verifier*: it accepts exactly $\hat{R}(D) := \bigcap \mathrm{VS}(D)$, where $\mathrm{VS}(D) = \{R \in \mathcal{H} : D \subseteq R\}$ and $\bigcap \emptyset := S$.* (a) *If $D \subseteq R^*$, then $\hat{R}(D) \subseteq R^*$, so the reasoner that chains accepted steps is sound against every prover at every time.* (b) *Every acceptance set contained in all $R \in \mathrm{VS}(D)$ (i.e. sound for every target compatible with D) is contained in $\hat{R}(D)$.* (c) *$D \subseteq D'$ implies $\hat{R}(D) \subseteq \hat{R}(D')$.* (d) *If $\mathcal{H} \cup \{\emptyset\}$ is closed under intersections and $\mathrm{VS}(D) \neq \emptyset$, $\hat{R}(D)$ is the least member of $\mathcal{H} \cup \{\emptyset\}$ containing D; for H_1 and $D \neq \emptyset$ it is* $\mathrm{inst}(\mathrm{lgg}(D))$.
Lean: thm_3_1_a_static Lean: thm_3_1_b_static

Parts (a) and (b) are Theorem 4.2(a) and its static maximality without negative data, (c) is immediate, and (d) is Proposition 4.5(i) with Lemma 4.6. Soundness is deterministic and distribution-free, so every question below is about *completeness*: when does $\hat{R}(D)$ reach R^*?

5.2 Anchors: when positive data suffice

Definition 5.2 (Anchor). [T1 Thm 5.1] A finite $T \subseteq R^*$ is an *anchor* (for R^* in $\mathcal{H}$) if every $R \in \mathcal{H}$ with $T \subseteq R$ satisfies $R \supseteq R^*$.

Theorem 5.3 (Eventual completeness iff finite anchor). *[T1 Thm 5.1, as repaired]* (a) *For finite $D \subseteq R^*$: $\hat{R}(D) = R^*$ iff D contains an anchor.* (b) *Let the data be a* text *for R^* (finite $D_1 \subseteq D_2 \subseteq \cdots \subseteq R^*$ with $\bigcup_t D_t = R^*$), or i.i.d. draws from a law with support R^*. If R^* has an anchor, $\hat{R}(D_t) = R^*$ for all large t (a.s. in the i.i.d. case); if not, $\hat{R}(D_t) \neq R^*$ for every t.*

Proof. (a) If $D \supseteq T$ for an anchor T, every $R \in \mathrm{VS}(D)$ contains R^*, so $\hat{R}(D) \supseteq R^*$; the reverse inclusion is Lemma 5.1(a). Conversely, if $\hat{R}(D) = R^*$, then D is itself an anchor, since every $R \in \mathcal{H}$ containing D lies in $\mathrm{VS}(D)$. (b) A text eventually contains any finite $T \subseteq R^*$, and then (a) and monotonicity give exactness forever. S is countable, so i.i.d. data with support R^* form a text a.s. □

"Finite" matters in (a): in the class of Proposition 5.4(c), $\hat{R}(2\mathbb{N}) = 2\mathbb{N}$ but $2\mathbb{N}$ has no anchor.

Proposition 5.4 (Anchors versus tell-tales). *[T1 Prop 5.2, as repaired] Call a finite $T \subseteq R^*$ an (Angluin)* tell-tale *if no $R \in \mathcal{H}$ satisfies $T \subseteq R \subsetneq R^*$.* (a) *Every anchor is a tell-tale.* (b) *If $\mathcal{H} \cup \{\emptyset\}$ is closed under intersections, every* nonempty *tell-tale is an anchor; the empty one need not be ($\mathcal{H} = \{\{1\}, \{2\}\}$, $R^* = \{1\}$). As existence conditions the two coincide when $R^* \neq \emptyset$.* (c) *In general the converse fails: $\mathcal{H} = \{2\mathbb{N}\} \cup \{R_n\}_{n \in \mathbb{N}}$ with $R_n = \{0, 2, \ldots, 2n, 2n+1\}$ is identifiable in the limit from text, but $2\mathbb{N}$ has no anchor, so on a text for $2\mathbb{N}$ the cautious verifier never becomes complete.*

Proof in Appendix D.2. *Credit.* Identification in the limit is due to Gold (1967), and tell-tales characterize it for indexed families (Angluin, 1980). Anchors are the "$\subseteq$-tell-tales" of strong-monotonic learning, where every conjecture is contained in the target. Lange and Zeugmann

characterize strong-monotonic learning from text by recursively generable finite sets $T_j \subseteq L_j$ such that $T_j \subseteq L_k$ implies $L_j \subseteq L_k$ (cf. Lange and Zeugmann, 1992; Lange et al., 2008); up to effectivity, this is the existence of anchors. What is new here is only the reading of anchors as the completeness condition of certified verification. For intersection-closed classes, anchors exist iff R^* is the least member of $\mathcal{H}$ containing some finite set (Lemma 5.1(d) and Theorem 5.3(a)).

Anchors exist in the schema classes with a bounded number of rules, by finite elasticity (Wright, 1989; Motoki et al., 1991).

Proposition 5.5 (Finite elasticity gives anchors). *[L1 §2.3, Lemma] Say $\mathcal{H}$ has* infinite elasticity *if there are $w_0, w_1, \cdots \in S$ and $R_1, R_2, \cdots \in \mathcal{H}$ with $\{w_0, \ldots, w_{n-1}\} \subseteq R_n \not\ni w_n$ for all $n \geq 1$, and* finite elasticity *otherwise. If $\mathcal{H}$ has finite elasticity, every $R^* \in \mathcal{H}$ has an anchor.*

Proof in Appendix D.2. A ground step is an instance of only finitely many schemas up to renaming (a generalization is never larger than its instances), so H_1 has finite thickness, hence finite elasticity. Finite elasticity survives unions of at most k members (Wright, 1989, as corrected by Motoki et al. 1991). So H_1, H_k^{tag} and H_k all have anchors, as do richer classes such as Shinohara's length-bounded elementary formal systems with at most k clauses (Shinohara, 1994), which model learning theorem sets rather than steps.

Reading. "Learning which inferences are valid" can mean *guessing* R^* in the limit, as in Gold's paradigm, or *certifying* a step: every hypothesis compatible with the evidence licenses it. Under search only the second protects. A learner that guesses $2\mathbb{N}$ in Proposition 5.4(c) converges, but it is unsound whenever the truth is a large R_n, and a prover can find the difference. An anchor is a finite body of usage that, given $\mathcal{H}$, justifies a rule for all of its uses. This is the kind of justification by imitation that survives adversarial search. It is relative to $\mathcal{H}$, a point we return to after Corollary 5.10.

5.3 Rates: cited rules and the coupon collector

Formal proofs cite their rules (natural-deduction rule names, Lean lemma names, Metamath labels). Take tagged calculi H_k^{tag} (Definition 2.9) with target rules $\sigma_1, \ldots, \sigma_k$, and let $v_i = |\text{vars}(\sigma_i)|$. The version space factors over tags, as in Theorem 4.9 (Appendix D.1). So the cautious verifier is the *per-rule lgg verifier* $\hat{R}(D) = \bigcup_i \{i\} \times \text{inst}(\text{lgg}(D^{(i)}))$, where $D^{(i)}$ is the tag-i data (a tag with $D^{(i)} = \emptyset$ contributes nothing), and rule i is identified once its data satisfy the witness events (R) and (D) of Lemma 2.7. Human data are i.i.d. pairs $(I, \sigma_I\Theta)$ with $\mathbb{P}(I = i) = \pi_i$ and $\Theta \sim \Lambda_i$ given $I = i$. For a metavariable x of σ_i, and distinct metavariables $x \neq y$ of σ_i, let

$$\rho_{i,x} := \max_{G \subseteq \Omega} \min(\mathbb{P}[\text{root}(\Theta x) \in G],\ \mathbb{P}[\text{root}(\Theta x) \notin G]), \qquad r_{i,xy} := \mathbb{P}[\Theta x \neq \Theta y],$$

let ρ_i be the minimum of all these numbers, and $c_i := 2v_i + \binom{v_i}{2}$. For a ground rule ($v_i = 0$) put $\rho_i := c_i := 1$.

Theorem 5.6 (Exact identification at a coupon-collector rate). *[T1 Thm 5.3] For N i.i.d. tagged human steps, $\mathbb{P}[\hat{R}(D) \neq R^*] \leq \sum_{i=1}^{k} c_i\, e^{-N\pi_i\rho_i}$. So $N \geq \max_i \frac{1}{\pi_i\rho_i} \ln \frac{k\,c_i}{\delta}$ suffices for exact identification with probability at least $1 - \delta$. Order $1/(\pi_i\rho_i)$ samples are also necessary: if $\mathbb{P}[\text{root}(\Theta x) = f] = 1 - \rho$ for some rule i, metavariable x, symbol f and $\rho \in (0, 1]$, then $\mathbb{P}[\hat{R}(D) \neq R^*] \geq (1 - \pi_i\rho)^N$.*

N	12	24	48	96	192
empirical exact-identification rate	.003	.41	.98	1.00	1.00
bound $1 - 30e^{-N/12}$ of Theorem 5.6	vacuous	vacuous	.45	.99	.999997

Table 5: Toy natural deduction with six tagged rules ($\wedge$I, $\wedge$E$_{1,2}$, $\vee$I$_{1,2}$, MP), random formulas over four atoms, 400 trials per N; $\pi_i = 1/6$, $\rho_i = 1/2$, $c_i \le 5$ (computed; `research/theory/T1-code/toy_nd.py`).

Proof idea. Given n rule-i samples, (R) fails at x only if all roots fall on one side of the best split (probability at most $2(1-\rho_i)^n$), and (D) fails at x, y with probability at most $(1-\rho_i)^n$. A union bound, averaged over $n \sim \mathrm{Bin}(N, \pi_i)$, gives $c_i(1-\pi_i\rho_i)^N$ (Appendix D.3).

Table 5 compares the bound with simulation. The coupons are the witness events of each rule; in spirit this is stochastic finite learning, as Rossmanith and Zeugmann (2001) studied for pattern languages.

Remark 5.7 (Teaching sets, corpora, and distribution dependence). [L1 §4.2, §9] Given $2v_i$ distinct constants, two instances of σ_i whose metavariables are replaced by two disjoint sets of distinct constants satisfy (R) and (D). So a textbook with two well-chosen examples per rule is an anchor, while a corpus needs about $\ln(kc_i/\delta)/(\pi_i\rho_i)$ steps. No distribution-free completeness rate exists, even for single schemas. For every $m = 2^h$, the complete binary trees of depth $\log_2 m$ with one distinguished leaf carry a law under which every sound learner has completeness error above $1/2$ after fewer than $m/2$ samples (Appendix D.4). Soundness is distribution-free; completeness cannot be, and $\pi_i\rho_i$ is what the distribution must supply.

Reading. A rule used in a fraction π_i of human steps, with variability ρ_i, is pinned down after about $\ln(kc_i/\delta)/(\pi_i\rho_i)$ steps; further examples add nothing. What makes this cheap is the citation of rules, which lets the version space factor. This is the positive-data counterpart of the gap between cited and uncited rules in Table 4. Without names the problem becomes a union problem with a diversity condition (Theorem 5.8).

5.4 Uncited rules: a diversity condition

Now let steps carry no citations: $R^* = \bigcup_{i \le k'} \mathrm{inst}(\sigma_i)$, in the class $H_k = \{\bigcup_{l \le k} \mathrm{inst}(\tau_l)\}$ with $k \ge k'$. Some bound k is unavoidable. With unbounded k every finite set of ground steps is a hypothesis, the class is superfinite in Gold's sense, and $\hat{R}(D) = D$: the cautious verifier never generalizes (Gold, 1967). For rule i let $\mathrm{Sub}_i(D) := \{\theta|_{\mathrm{vars}(\sigma_i)} : \sigma_i\theta \in D\}$, with *failure sets*

$$\mathfrak{F}_i := \{\{\theta : \mathrm{root}(\theta x) = f\} : x \in \mathrm{vars}(\sigma_i),\ f \in \Omega\} \cup \{\{\theta : \theta x = \theta y\} : x \ne y\}.$$

By Lemma 2.7, a nonempty set Q of substitutions has $\mathrm{lgg}\{\sigma_i\theta : \theta \in Q\} \equiv \sigma_i$ (call Q *generic*) iff Q lies in no single failure set.

Theorem 5.8 (Untagged unions: anchors and sample complexity). [T1 Thm 5.4] (a) *If for each $i \le k'$ the set $\mathrm{Sub}_i(D)$ is nonempty and not contained in a union of k failure sets from $\mathfrak{F}_i$, then $\hat{R}_{H_k}(D) = R^*$. In particular, $k+1$ pairwise-generic instances of each rule form an anchor of size $k'(k+1)$.* (b) *Let the rule-i data be i.i.d. $\sigma_i\Theta$, $\Theta \sim \Lambda_i$. For $v_i \ge 1$ let*

$$\zeta_i := \inf_{F_1,\dots,F_k \in \mathfrak{F}_i} \Lambda_i\big(\overline{F_1 \cup \cdots \cup F_k}\big), \qquad d_i := 2k\log_2\big(4k(v_i^2+1)\big).$$

If every $\zeta_i > 0$ and the data contain, for each rule with $v_i \geq 1$, at least

$$n_i \;\geq\; \frac{8d_i}{\zeta_i}\log_2\frac{13}{\zeta_i} + \frac{4}{\zeta_i}\log_2\frac{2k'}{\delta}$$

independent rule-i samples, and one sample of each ground rule, then $\hat{R}_{H_k}(D) = R^*$ with probability at least $1-\delta$.

For (a), a hypothesis in $\mathrm{VS}(D)$ splits $\mathrm{Sub}_i(D)$ among its k schemas; some part is generic, so its schema generalizes σ_i. Part (b) bounds the VC dimension of complements of k-unions of failure sets by d_i and applies the ε-net theorem (Haussler and Welzl, 1987; Blumer et al., 1989) with $\varepsilon = \zeta_i$ (Appendix D.5). The condition $\zeta_i > 0$ asks for more instantiation variety than k rules can absorb, and some such condition is needed:

Proposition 5.9 (When variety is too low)**.** *[T1 Remark after Thm 5.4, as repaired]* (a) *Suppose that for some i the data in $\mathrm{inst}(\sigma_i)$ not covered by the other $k'-1$ rules are covered by $k-k'+1$ specializations of σ_i of the forms $\sigma_i[x \mapsto f(\bar{z})]$ ($\bar{z}$ fresh) or $\sigma_i[y \mapsto x]$, and that some instance of σ_i lies outside these specializations and the other rules. Then $\hat{R}_{H_k}(D) \neq R^*$.* (b) *For $k' \geq 2$, low variety need not block identification: for $k = k' = 2$, $R^* = \mathrm{inst}(p(x)) \cup \mathrm{inst}(q(y))$ and $D = \{p(a), p(b), q(a), q(b)\}$, $\hat{R}_{H_2}(D) = R^*$, although x takes only $2 = k$ roots.*

For $k' = 1$, (a) says that identification fails while a metavariable is instantiated with at most k root symbols (and others are possible). Proofs are in Appendix D.6. For $k' \geq 2$ the exact threshold, between "no cover by k failure sets" and "a cover by $k-k'+1$ plus an uncovered instance", is open. So "allow up to k rules" is a real assumption about the diversity of usage: without rule names, imitation must see each rule used in more ways than the $k-k'+1$ spare schemas of the class can absorb. This is the positive-data face of the escalation lower bound for untagged unions (Theorem 4.10).

5.5 Systematic generalization: finite usage fixes inferential role

Corollary 5.10 (Systematic out-of-distribution generalization; trivial)**.** *[T1 Cor 5.5] If $\hat{R}(D) = R^*$, the verifier accepts exactly R^* and the reasoner derives exactly $\mathrm{Cl}_{R^*}(B)$ for every B, including conclusions that need derivations far longer, and judgments far larger, than any in the data, independently of the distribution that produced the data.*

Proof. $\hat{R}(D) = R^*$ gives $\mathrm{Cl}_{\hat{R}(D)} = \mathrm{Cl}_{R^*}$. □

Theorems 5.6 and 5.8 supply its hypothesis, and Lemma 5.1 keeps the verifier sound before that. The result is a learner of inference rules from imitation that is sound at all times against adaptive search, exact after about $\max_i \ln(kc_i/\delta)/(\pi_i\rho_i)$ cited steps, and *systematic*: correctness on terms of size 40 is the same event as on terms of size 10. A learned step scorer comes with none of these guarantees (Sections 3 and 5.8).

Reading. Under an inferentialist picture of meaning (Brandom, 1994), the meaning of a connective is its inferential role, the steps in which it may figure. Corollary 5.10 says that a finite set of uses (an anchor; for cited natural-deduction rules, two generic instances per rule) fixes that role for unbounded use. Three qualifications make this precise. (i) The role is fixed *relative to* $\mathcal{H}$. An anchor leaves open, within $\mathcal{H}$, only hypotheses that contain R^*, so the cautious verifier

accepts exactly R^*; it says nothing about hypotheses outside $\mathcal{H}$. Quus-like alternatives (Kripke, 1982) are excluded by the choice of hypothesis class, not by the data. Which of simplicity, coherence and convention does that work is the subject of Sections 6, 8 and 13. (ii) On clean data, overgeneralizations such as *tonk* are never adopted, because the lgg is the *least* generalization covering usage. The learner adds to a connective's role only what usage forces. This is a learning-theoretic analogue of Belnap's conservativeness requirement (Belnap, 1962), and it is why simplicity is the wrong bias here (Proposition 3.9). (iii) The role fixed is that of *practice*: a fallacy used systematically becomes part of it (Section 5.6). *Contexts.* Contexts are part of judgments (Definition 2.1), so steps inside suppositions are data like any other, and the theorems apply to them unchanged as long as contexts are encoded as first-order terms (Remark 5.11). But they must *be* in the data. Theorems determine a structural consequence relation only up to the interval between derivable and admissible rules, which is non-degenerate for intuitionistic logic (the Kreisel–Putnam rule is admissible but not derivable). So, unless the hypothesis class enforces a deduction theorem, assertions alone cannot in general teach reasoning under hypotheses.

Remark 5.11 (Side conditions). [T1 §5, extensions] With guards from a finite family Φ (Definition 2.4), the guarded single-schema class stays intersection-closed. Its closure is the lgg with the strongest guard true on all data (Lemma 2.8), anchors exist, and the escalation budget of Section 4 grows by at most $|\Phi|$ (Appendix D.7). But a positive example shows that a guard *held*, never that it was *needed*. A guard is learned only from the absence of violating uses. It is lost as soon as one human use omits it (Example 2.12), and it cannot be learned at all if Φ cannot express it. Set-valued contexts and binders lie outside first-order anti-unification; modulo AC it is finitary rather than unitary, and the bounds above would need re-proof.

5.6 Noise: sporadic and systematic errors

Human data contain errors, and the two kinds behave completely differently.

Proposition 5.12 (One error collapses the lgg)**.** *[T1 Prop 6.1] Let D consist of $\wedge E_1$ instances $\mathsf{s}(\mathsf{and}(A_j, B_j), A_j)$ whose A_j do not all have the same root, plus one invalid step $\mathsf{s}(\varphi, \psi)$ tagged $\wedge E_1$, with φ not a conjunction. Then $\mathrm{lgg}(D) = \mathsf{s}(z, w)$ with $z \neq w$: the learned rule accepts every one-premise step, and from any derivable formula it derives everything.*

Proof. The premise column mixes the root and with the root of φ; the conclusion column has mixed roots because the A_j do. Both become metavariables, distinct because the columns differ ($\mathsf{and}(A_1, B_1) \neq A_1$). □

(computed) A mis-tagged "$p_0 \vee p_1 \rhd p_0$" does exactly this. A 15% rate of affirming the consequent, cited as MP, turns MP's lgg into $\mathsf{s}_2(\mathsf{imp}(v_0, v_1), v_2, v_3)$: from $A \to B$ and anything, infer anything. This is the tonk of Theorem 3.2, produced by learning: the lgg is least among generalizations of *all* the data, errors included.

The remedy for sporadic errors is to let hypotheses miss a few data. Counting with multiplicity, let $\mathrm{VS}_e(D) = \{R \in \mathcal{H} : |D \setminus R| \leq e\}$, and with per-tag budgets $\mathrm{VS}_{(e_i)}(D) = \{R \in \mathcal{H} : |D^{(i)} \setminus R^{(i)}| \leq e_i \text{ for all } i\}$, where $D^{(i)}, R^{(i)}$ are the tag-i parts. The *trimmed verifiers* accept $\bigcap \mathrm{VS}_e(D)$, resp. $\bigcap \mathrm{VS}_{(e_i)}(D)$.

Theorem 5.13 (Trimmed version space)**.** *[T1 Thm 6.2, as repaired]* (a) *For any $\mathcal{H}$: if D contains at most e invalid steps, then $\bigcap \mathrm{VS}_e(D) \subseteq R^*$; likewise $\bigcap \mathrm{VS}_{(e_i)}(D) \subseteq R^*$ if at most e_i invalid steps are tagged i, for each i. The verifier is then sound against every prover.* (b) *For tagged schemas*

and the per-tag verifier, suppose that for each i: (i) *at most e_i invalid steps are tagged i; and* (ii) *the valid rule-i samples are e_i-*robustly generic*: there are more than e_i of them; for every x, f, more than e_i of them have* $\mathrm{root}(\Theta x) \neq f$*; and for every $x \neq y$, more than e_i of them have $\Theta x \neq \Theta y$. Then $\bigcap \mathrm{VS}_{(e_i)}(D) = R^*$.*

Proof idea. (a) R^* lies in the trimmed version space. (b) A hypothesis in $\mathrm{VS}_{(e_i)}$ covers all but at most e_i valid rule-i samples. By (ii) the rest is nonempty and still satisfies (R) and (D), so its tag-i schema generalizes σ_i. Full proof in Appendix D.8.

Two features of (b) are needed, with counterexamples in Appendix D.8. The clause "more than e_i of them" is the only content of (ii) for ground rules, and without it the theorem fails for them. The budgets must be *per tag*: one global budget lets a hypothesis spend everything on one rule. (computed) With $e = 1$ (resp. 2), one (resp. two) mis-tagged errors are removed exactly; with $e = 1$ and two errors the lgg collapses again.

Under i.i.d. noise the budgets can be set from the rates. Suppose each datum independently is a valid rule-i step with probability β_i (with $\Theta \sim \Lambda_i$), or an invalid step tagged i with probability α_i (with arbitrary law); in terms of Definition 2.10, $\beta_i = \pi_i(1 - a_i)$ and $\alpha_i = \pi_i a_i$.

Theorem 5.14 (i.i.d. noise: witness frequency must beat error frequency)**.** *[T1 Thm 6.3, as repaired] Let ρ_i, c_i be as in Theorem 5.6 ($\rho_i = c_i = 1$ for ground rules). If every margin $\Delta_i := (\rho_i\beta_i - \alpha_i)/2$ is positive and $e_i := \lfloor(\alpha_i + \Delta_i)N\rfloor$, then $\mathbb{P}[\bigcap \mathrm{VS}_{(e_i)}(D) = R^*] \geq 1 - \sum_i (1 + c_i)\, e^{-2N\Delta_i^2}$.*

Proof idea. The number of invalid steps tagged i is $\mathrm{Bin}(N, \alpha_i)$ and exceeds e_i with probability at most $e^{-2N\Delta_i^2}$ (Hoeffding). Robust genericity follows from c_i counts (valid rule-i samples with root in G and in G^c for the best split G at each x; with $\Theta x \neq \Theta y$ for each pair), each $\mathrm{Bin}(N, p)$ with $p \geq \rho_i\beta_i = \alpha_i + 2\Delta_i$. Appendix D.9.

The budgets grow linearly with N; a fixed support threshold is not a soundness guarantee under sporadic noise, because coincidences between random errors grow with N (Section 5.8; Appendix D.13). The threshold $\rho_i\beta_i > \alpha_i$ is necessary up to a factor of two, because a noisy law for one target is a clean law for another.

Definition 5.15 (Robust soundness)**.** [T1 Thm 6.4] A *positive-data verifier* maps the data so far (and private coins) to an acceptance set Acc_t. It is (δ, α)*-robustly sound* for $\mathcal{H}$ if for every $R^* \in \mathcal{H}$ and every law Q on S with $Q(S \setminus R^*) \leq \alpha$, i.i.d. data from Q give $\mathbb{P}[\exists t :\ \mathrm{Acc}_t \not\subseteq R^*] \leq \delta$.

Theorem 5.16 (Indistinguishability; trivial)**.** *[T1 Thm 6.4] Let the verifier be (δ, α)-robustly sound for $\mathcal{H}$, let $R^* \subsetneq R'$ both lie in $\mathcal{H}$, and let Q be a law on R' with $Q(R' \setminus R^*) \leq \alpha$. Under i.i.d. data from Q, every $q \in R' \setminus R^*$ has $\mathbb{P}[\exists t :\ q \in \mathrm{Acc}_t] \leq \delta$. So if $\delta < 1$, the verifier is not complete for the target R' under its clean law Q.*

Proof. Q is an admissible noisy law for the target R^*, and $q \in \mathrm{Acc}_t$ implies $\mathrm{Acc}_t \not\subseteq R^*$; the data law is the same in both scenarios. □

The argument is in the spirit of Kearns and Li (1993). For schemas it yields a matching necessity. Let $m_{i,x} := \max_f \mathbb{P}[\mathrm{root}(\Theta x) = f]$.

Lemma 5.17 (Balanced splits)**.** *[T1 §6 after Thm 6.4, repaired] $\rho_{i,x} \leq 1 - m_{i,x} \leq 2\rho_{i,x}$, and the constant 2 is attained by the uniform law on an odd number of symbols.*

Corollary 5.18 (Witness frequencies must exceed the noise rate)**.** *[T1 §6 after Thm 6.4, repaired] Let $\mathcal{H} = H_k^{\mathrm{tag}}$ and let Q be a clean law for R^* under which a datum is a rule-i instance $\sigma_i\Theta$, $\Theta \sim \Lambda_i$,*

with probability β_i. If a (δ, α)-robustly sound verifier with $\delta < 1$ has $\mathbb{P}_Q[R^* \subseteq \mathrm{Acc}_t] > \delta$ for some t, then for all x, f and all $x \neq y$,

$$\beta_i \, \mathbb{P}[\mathrm{root}(\Theta x) \neq f] > \alpha \qquad \text{and} \qquad \beta_i \, \mathbb{P}[\Theta x \neq \Theta y] > \alpha.$$

With per-tag noise budgets, α may be read as α_i.

Proof idea. If a witness frequency is at most α, then Q is an α-noisy law for the target in which σ_i is replaced by $\sigma_i[x \mapsto f(\bar{z})]$ (resp. $\sigma_i[y \mapsto x]$), and Theorem 5.16 applies. Proofs in Appendices D.10 and D.11.

So Theorem 5.14 asks for $\rho_{i,x}\beta_i > \alpha_i$ and $r_{i,xy}\beta_i > \alpha_i$ (plus margins), while necessity asks for $(1 - m_{i,x})\beta_i > \alpha_i$ and $r_{i,xy}\beta_i > \alpha_i$. By Lemma 5.17 the two differ by at most a factor of two in the root frequencies and agree on the equality frequencies; the factor two in the lemma is attained. The trade-off is unavoidable. A positive-data verifier that tolerates error rate α must refuse every generalization supported by less than a fraction α of the data. With human error rates around 1%, the generality of a rule used in less than about 1% of steps needs negative information.

Corollary 5.19 (Systematic errors are rules)**.** *[T1 Cor 6.5] Let human errors be* schema-generated*: the data law is $Q = (1 - \alpha)Q_0 + \alpha Q_{\mathrm{err}}$, with Q_0 supported on R^* and Q_{err} on $\mathrm{inst}(\tau)$ for a schema τ with $\mathrm{inst}(\tau) \not\subseteq R^*$. Suppose $R' := R^* \cup \mathrm{inst}(\tau) \in \mathcal{H}$, as holds whenever $\mathcal{H}$ is closed under adding a schema (tagged calculi with a fresh tag for τ; untagged unions with room for one more schema). Then Q is a* clean *law for R' at every error rate α. Hence a (δ, α)-robustly sound verifier with $\delta < 1$ accepts each $q \in \mathrm{inst}(\tau) \setminus R^*$ with probability at most δ under Q. So with probability at least $1 - \delta$ it never accepts all of R', a calculus in which τ is a genuine rule used at the same rate.*

Proof. $Q(S \setminus R') = 0$; apply Theorem 5.16 to $R^* \subsetneq R'$. □

So no positive-data method, with any amount of data, can both tolerate schema-generated errors at rate α and learn genuine rules used at rate α. This also removes the protection of the Bayes-conservative verifier (Section 4). Its time-uniform guarantee assumes a well-specified likelihood. A model of errors as sporadic is misspecified for Q, and the clean hypothesis R' fits Q exactly.

Remark 5.20 (Imitation identifies practice). [T7 Prop 2.2; T1 §6] With latent but distinct fallacy tags (Definition 2.10), the theorems above apply to the practice $\Sigma^P = \Sigma^* \sqcup F$ tag by tag. On the event of Theorem 5.14 the trimmed verifier accepts exactly $R^P = R^* \cup R_F$, including an instance of every fallacy outside $\mathrm{Sound}(R^*)$, which a prover will query. In classical logic with a complete Σ^* and a pure fallacy schema, one closed instance with true premises and a false conclusion trivializes the reasoner (Proposition 7.5; cf. Proposition 3.5). If fallacies are cited under genuine rules' names, a learner with one schema per tag collapses the rule instead (Proposition 5.12).

Reading. The line between sporadic and systematic errors is exact. Positive data remove an error only if it is rarer than the witness frequencies of the genuine rules. Rule-like errors, at any rate, need *negative* information: escalation (Section 4), coherence (Section 6), or the world (Sections 7 and 11). For the program of the motivating notes (Hänni, 2026), imitation learns which inferences are *used*, and which are *valid* only insofar as use is valid. Getting from the first to the second is what the coherence and world channels are for, and positive data provably

cannot do it. In the language of justification, imitation can certify a step as licensed by the practice, never as licensed by anything beyond it. This is why the cautious tier of Section 7 is sound only relative to practice, and why coherence must act on a separate, bold tier.

5.7 An exactly solvable case: dimensions as inferential roles

The dimension of a physical quantity is its inferential role: which sums and equations it may enter. For dimensions, learning meaning from positive examples is solvable in closed form (Remark D.2; proof sketch in Appendix D.12): the finest grading that makes the observed equations homogeneous is computed by Smith normal form, every target has a finite anchor, positive data never refute "everything dimensionless" but the cautious learner never adopts it, and the choice of base dimensions is a convention that no inference can see.

5.8 Imitation in practice

Experiment A (Section 12.1) illustrates each limit on school algebra: rule shapes are recovered once a rule has a few instances and generalize out of distribution, but no guard is ever displayed, all five systematic fallacies reach support, and spurious schemas from sporadic noise grow with N under a fixed support threshold. Appendix D.13 gives the positive-data numbers.

In short, what positive data fix is: rule shapes, given an anchor (at coupon-collector cost for cited rules, under a diversity condition for uncited ones); their extension to unbounded use; side conditions only as "strongest guard true on all data"; and sporadic errors rarer than every witness frequency. What they cannot fix is: the choice of $\mathcal{H}$, the necessity of a guard, generalizations rarer than the error rate, and any schema-generated error. Sections 6 and 7 show what coherence and world feedback do about the latter.

6 Coherence: what contradictions fix

Imitation cannot tell a systematic human error from a rule (Corollary 5.19). The second ingredient of the motivating design (Section 1.1) is a coherence loss: penalize having good arguments for both P and $\neg P$ (Hänni, 2026). In our setting this is channel (C) of Definition 2.13: a family $\mathcal{A}$ of *designated contexts* is certified coherent, and a derivation of $\bot$ from one of them shows that one of its steps is invalid. In brief: coherence is negative data, but only for *bold* learners, and it controls only one of two error types (Section 6.1); it pins the target exactly when the target sits at the top of its coherent version space, as classical propositional logic does by Post-completeness, and it is blind for intuitionistic logic (Section 6.2); in arithmetic it is Popperian (Section 6.3); Carnap's categoricity problem is, in part, Gold's problem in the dual space; among structural meanings it is solved by data with two denials plus one coherence datum (Section 6.4); coherence catches tonk but not non-conservativity (Section 6.5); what survives is the Kripkensteinian residue of coherent uniform alternatives (Section 6.6); and this use of coherence is eliminative, untouched by the coherentist impossibility theorems (Section 6.7).

Most results are known in substance. Post-completeness (Post, 1921; Wójcicki, 1988), Glivenko's theorem, Gold's limit-point argument, halving and weighted majority (Littlestone, 1988; Littlestone and Warmuth, 1994), the multiple-conclusion answer to Carnap (Carnap, 1943; Shoesmith and Smiley, 1978), Belnap's conservativity (Belnap, 1962) and the Σ_1/Π_1 asymmetry of limiting inquiry (Putnam, 1965; Gold, 1965; Kelly, 1996) are classical. New, as far as we know, are the reduction of contradictions to coalitions, the two-sided trade-off, the denial-rank

trichotomy and twelve-datum tell-tale as learning results, the rule-level analysis of complete theories, and the reading of all of these as statements about what a coherence loss can teach.

Cn_2 is classical propositional consequence ($\varphi \in \mathrm{Cn}_2(X)$ iff every Boolean valuation making X true makes φ true), Taut $= \mathrm{Cn}_2(\emptyset)$, and $C_{\mathrm{Fm}}(X) = \mathrm{Fm}$ is the trivial operator. Hypotheses are step sets (Definition 2.2) or finitary consequence relations h, with $h(A) = \{\varphi : A \vdash_h \varphi\}$; h is *$\mathcal{A}$-coherent* if $A \nvdash_h \bot$ for all $A \in \mathcal{A}$, read as non-triviality $h(A) \neq \mathrm{Fm}$ in languages without $\bot$. Unless stated otherwise $\mathcal{A}$ is *fixed and known*, independently of the target.

6.1 Coherence as negative data

Here hypotheses are step sets R in a countable class $\mathcal{H}$ with prior w, $\sum_R w(R) = 1$, and target R^*; every $A \in \mathcal{A}$ is target-coherent. An *argument* π from A to $\bot$ is a derivation of $\bot$ from A; $\mathrm{Steps}(\pi)$ is its finite set of steps.

Definition 6.1 (Online coherence protocol)**.** [T2 §2.2] In round t the learner holds a version space $\mathrm{VS}_t \subseteq \mathcal{H}$ ($\mathrm{VS}_0 = \mathcal{H}$) and announces an *accepted set* $\hat{R}_t$, whose steps the reasoner may chain. The environment, possibly adversarial, possibly the learner's own proof search, then either (P) presents some $s \in R^*$, and the learner deletes every $R \in \mathrm{VS}_t$ with $s \notin R$; or (N) exhibits an argument π from some $A \in \mathcal{A}$ to $\bot$ with $\mathrm{Steps}(\pi) \subseteq \hat{R}_t$, and the learner deletes every $R \in \mathrm{VS}_t$ with $\mathrm{Steps}(\pi) \subseteq R$; or does nothing. A (P)-round with $s \notin \hat{R}_t$ is an *incompleteness error*; an (N)-round is a *detection*.

Lemma 6.2 (Negative bags; trivial)**.** *[T2 Lemma 2.1, as repaired] Let π be an argument from a designated A to $\bot$.* (a) $\mathrm{Steps}(\pi) \not\subseteq R^*$. (b) *Every $R \supseteq \mathrm{Steps}(\pi)$ has $\bot \in \mathrm{Cl}_R(A)$, witnessed by π.* (c) *The designation alone excludes every h with $A \vdash_h \bot$; deletion by the witness removes only $\{h : \mathrm{Steps}(\pi) \subseteq h\}$, which can be strictly smaller even for consequence relations (target Cn_2, $A = \{p\}$, π with steps $p \rhd q$, $q \rhd \bot$, and h the least consequence relation containing $p \rhd \bot$).* Lean: CoherenceGames.negative_bag *((a),(b))*

Proof. (a) Otherwise chaining gives $\bot \in \mathrm{Cl}_{R^*}(A)$. (b) Chain in R. (c) $h(X) = X \cup \{\bot\}$ if $p \in X$, else $h(X) = X$. □

So a contradiction is a multiple-instance label (Dietterich et al., 1997): "not all of these steps are valid". For a learner that can test $A \rhd \bot \in h$ by membership it collapses to an ordinary negative example and π is only a checkable witness; the bag structure matters for step verifiers that are not cut-closed, such as process reward models and ensembles.

Proposition 6.3 (Caution receives no signal)**.** *[T2 Prop 2.9(a)–(c), as repaired]* (a) *If $\hat{R}_t \subseteq R^*$, no detection can occur; since deletions never remove R^*, a* certified-sound *learner ($\hat{R}_t \subseteq \bigcap \mathrm{VS}_t$) never sees one.* (b) *Certified soundness can be expensive. For $k \geq 1$ and distinct atoms a_i, p_i ($1 \leq i \leq k$), $\ell_i^1 = p_i$, $\ell_i^0 = \neg p_i$, let $h_b = \{\Gamma \rhd \varphi : \{a_i \to \ell_i^{b_i}\}_{i \leq k} \cup \Gamma \models \varphi\}$ for $b \in \{0,1\}^k$, and $\mathcal{A} = \{\emptyset\}$. Every certified-sound learner makes $2^k - 1 = |\mathcal{H}| - 1$ incompleteness errors on some text.* (c) *If $\mathcal{A}$ is fixed and known and membership in and $\mathcal{A}$-coherence of each $h \in \mathcal{H}$ are decidable, coherence is prior knowledge (a restriction of $\mathcal{H}$), not data.*

(a) and (c) are immediate; (b) is proved in Appendix E.3. So coherence feedback is a dividend paid on boldness, and with $\mathcal{A}$ fixed and coherence decidable it is not even data. It is an informative *signal* when consistency is undecidable (Π_1; Theorem 6.30), so that a prover must find contradictions for the learner, and when designations are data about the target (Proposition 6.12).

Oligarchic acceptance. How should a bold learner accept, so that each contradiction teaches it something? Not by voting: stepwise majority over coherent verifiers can derive $\bot$ while implicating no member (Theorem 3.12).

Definition 6.4 (Oligarchic halving)**.** [T2 §2.2] OH chooses a *coalition* $S_t \subseteq \mathrm{VS}_t$ with $w(S_t) \geq \frac{1}{2}w(\mathrm{VS}_t)$ and announces $\hat{R}_t = \bigcap_{R \in S_t} R$.

Theorem 6.5 (Oligarchic halving)**.** *[T2 Thm 2.2] Under OH with $w(R^*) > 0$:* (i) *R^* is never deleted;* (ii) *there are at most $\log_2(1/w(R^*))$ detections (at most $\log_2 |\mathcal{H}|$ for the uniform prior), and the bound says nothing about incompleteness errors;* (iii) *if $R^* \in S_t$, then $\mathrm{Cl}_{\hat{R}_t}(B) \subseteq \mathrm{Cl}_{R^*}(B)$ for all B;* (iv) *always $\mathrm{Cl}_{\hat{R}_t}(B) \subseteq \mathrm{Cl}_R(B)$ for all $R \in S_t$, so chaining never leaves what each coalition member accepts.* Lean: CoherenceGames.thm_2_2 *(finite classes)*

Proof idea. In a detection every member of S_t contains $\hat{R}_t \supseteq \mathrm{Steps}(\pi)$, so the whole coalition is deleted, while R^* never is (Lemma 6.2(a)). Appendix E.1.

The halving argument is that of Barzdin and Freivalds (1972) and Littlestone (1988); new is only the reduction of a bag to a coalition. Oligarchy needs *truth maintenance*: lemmas certified by an old coalition must be re-validated, or the union over coalitions reproduces the doctrinal paradox across time.

Proposition 6.6 (Per-round halving forces oligarchy)**.** *[T2 Prop 2.3, as repaired] Let* VS *be countable with $w(\mathrm{VS}) > 0$, and assume every finite $P \subseteq \hat{R}$ with $P \not\subseteq \bigcap \mathrm{VS}$ (i.e. legal for some target in* VS*) can occur as* $\mathrm{Steps}(\pi)$*. Then every possible detection deletes at least $\frac{1}{2}w(\mathrm{VS})$ iff $\hat{R} \subseteq \bigcap S$ for some $S \subseteq \mathrm{VS}$ with $w(S) \geq \frac{1}{2}w(\mathrm{VS})$.* Lean: CoherenceGames.prop_2_3 *(finite version spaces)*

Proof in Appendix E.2. The scope is per-round halving against a worst-case prover; coalitions of weight $\frac{1}{3}$ still give $\log_{3/2}(1/w(R^*))$. This is the learning-theoretic face of the judgment-aggregation result that, for rich agendas, aggregation yielding deductively closed collective judgments is oligarchic (Gärdenfors, 2006; Dietrich and List, 2008). Designations can be wrong ("air pressure is 0" plus full background knowledge is inconsistent), so we also need a robust version.

Theorem 6.7 (Mis-designated contexts)**.** *[T2 Thm 2.5] On a detection multiply $w(R)$ by $\beta \in [0, 1)$ for every $R \supseteq \mathrm{Steps}(\pi)$; choose S_t with $w_t(S_t) \geq \frac{1}{2}W_t$ (W_t the total weight, $W_0 = 1$) and announce $\bigcap S_t$; positive data are noise-free. If m of the D detections are* false alarms *(their context is target-incoherent), then $D \leq \big(\ln(1/w_0(R^*)) + m\ln(1/\beta)\big)/\ln\big(2/(1+\beta)\big)$, with $0 \cdot \ln(1/0) := 0$.* Lean: CoherenceGames.thm_2_5 *(finite classes)*

This is the bound of Littlestone and Warmuth (1994), with their proof (Appendix E.1). A structural learner that trusts a false alarm fully fares worse: it must become sub-classical everywhere (Proposition 11.9).

Proposition 6.8 (Boldness is a two-sided trade-off)**.** *[T2 Prop 2.9(d),(e), added after verification]* (a) *On the class of Proposition 6.3(b) with uniform prior, every learner with $\hat{R}_t \subseteq R$ for some $R \in \mathrm{VS}_t$ in every round, in particular every OH learner, makes $2^k - 1$ incompleteness errors on some text for some target. The non-oligarchic union learner $\hat{R}_t = \bigcup \mathrm{VS}_t$ makes no error of either type there, and is silently unsound.* (b) *On the class of Theorem 3.12, no accepted set guarantees that every possible incompleteness error and every possible detection deletes half the weight.*

Proof in Appendix E.3. *Reading.* Oligarchy makes each contradiction worth a bit, robustly, but does not make boldness cheap: on this class every oligarchic learner pays exactly the price of caution in missed valid steps, and a learner that avoids this price must at some round accept a set inside no surviving hypothesis, hence a target-invalid step. The two-sided minimax is open (Section 14); Section 7 resolves the trade-off architecturally, with cautious assertion and a bold sandbox under attack.

What coherence cannot see. Now let hypotheses be finitary consequence relations with target $\vdash^*$, let $h \oplus D$ be the least consequence relation containing $h \cup D$, and let the environment be complete. Call h *text-refutable* if $\vdash^* \not\subseteq h$, *coherence-refutable* if it is $\mathcal{A}$-incoherent, and *refutable by coherence with data* if $h \oplus D$ is $\mathcal{A}$-incoherent for some finite $D \subseteq \vdash^*$ (accepted human conclusions used as premises).

Theorem 6.9 (Silent over-generalizations; trivial)**.** *[T2 Thm 2.6]* (i) *h is refutable by coherence with data iff $h \oplus \vdash^*$ is $\mathcal{A}$-incoherent.* (ii) *The hypotheses refuted in none of the three ways are exactly $U(\vdash^*) = \{h : \vdash^* \subseteq h,\ h$ $\mathcal{A}$-coherent$\}$.* (iii) *Each $h \in U(\vdash^*) \setminus \{\vdash^*\}$ is unsound, and if it is a possible target, every finite stage of every data stream for $\vdash^*$ is one for h.*

Proof idea. (i) A finite derivation of $\bot$ in the directed union $h \oplus \vdash^*$ lies in some $h \oplus D$. Appendix E.1.

Proposition 6.10 (Coherent chains have coherent unions)**.** *[T2 Prop 2.7] The union of a chain of $\mathcal{A}$-coherent finitary consequence relations is one: cut and $A \rhd \bot$ involve finitely many sequents, which lie in one member.*

Corollary 6.11 (Fixed designations do not remove limit points)**.** *[T2 Cor 2.8, as repaired] Let $\mathcal{A}$ be fixed and known independently of the target. If the possible targets include a strictly increasing chain $h_1 \subsetneq h_2 \subsetneq \cdots$ of $\mathcal{A}$-coherent hypotheses together with $h_\omega = \bigcup_n h_n$, no learner, computable or not, identifies them in the limit (EX or BC) from text plus coherence information.*

Proof. The coherence information is target-independent, so fold it into the learner. A learner for h_ω has a locking sequence σ (Blum and Blum, 1975), or its BC analogue; its finite content lies in some h_k, and on a text for h_k extending σ the learner conjectures h_ω forever. □

This is Gold's limit-point theorem (Gold, 1967). The hypothesis "$\mathcal{A}$ fixed" is essential: designations that are data about the target, as when a problem setter certifies a setup consistent with the true theory, are genuine negative data.

Proposition 6.12 (Target-dependent designations are negative data; trivial)**.** *[T2 Prop 2.8′, repair] Suppose the environment designates only target-coherent finite contexts, and eventually all of them. If every hypothesis satisfies classical reductio ($\Gamma \vdash_h \varphi$ iff $\Gamma \cup \{\neg\varphi\} \vdash_h \bot$), text plus designations is an informant, and every countable such class is EX-identifiable by enumeration. Example: for $h_n = \mathrm{Cn}_2\{\neg p_i : i < n\}$ and $h_\omega = \mathrm{Cn}_2\{\neg p_i : i \in \omega\}$, $\{p_j\}$ is h_n-coherent iff $j \geq n$, so the limit point is identified.*

Proof in Appendix E.4. So with fixed designations coherence prunes only over-generalizations that are not limit points, and positive-data structure (Section 5) decides the rest; text plus coherence *refutes* every wrong hypothesis exactly when $U(\vdash^*) \cap \mathcal{H} = \{\vdash^*\}$ (a condition for refutation, not for identifiability). *Reading:* a coherence loss removes over-acceptance that *conflicts*; the silent over-generalizations $U(\vdash^*)$ are the precise content of "coherent but wrong", and whether they exist depends on the target, not on the learner.

6.2 Post-completeness: when coherence suffices

A consequence operator C is *structural* if $\sigma C(X) \subseteq C(\sigma X)$ for every substitution σ: rules are uniform schemas, so a learner that generalizes steps to schemas (Section 2.2) is a structural learner.

Theorem 6.13 (Structural Post-completeness; known). *[T2 Thm 3.1] If* Fm *contains* $\neg$ *and one of* $\wedge, \vee, \to$*, all interpreted by their truth functions, every structural closure operator* $C \supseteq \mathrm{Cn}_2$*, finitary or not, is* Cn_2 *or* C_{Fm}*. The same holds for pure* $\{\to\}$*.* *Lean: Post.eq_Cn2_or_eq_trivial*
Lean: Post.frag_postComplete_neg Lean: Post.impFrag_postComplete

Proof idea. Take $\varphi \in C(X) \setminus \mathrm{Cn}_2(X)$ and a falsifying row v. Substituting a tautology or a contradiction for each atom according to v, as in the proof of Proposition 3.5, maps X into $\mathrm{Taut} \subseteq C(\emptyset)$ and φ to a contradiction, so structurality gives $C(\emptyset) = \mathrm{Fm}$. Appendix E.5.

At the level of theorems this is Post (1921); the consequence-level form is standard too (Pogorzelski, 1971; Wójcicki, 1988; Pogorzelski and Wojtylak, 2008). One falsifying row of one bad rule yields a derivation of $\bot$ from no premises. Without a theorem in the language there is one more extension: in the $\{\wedge, \vee\}$ fragment ($\mathrm{Cn}_2(\emptyset) = \emptyset$) let $C_{\mathrm{ai}}(\emptyset) = \emptyset$ and $C_{\mathrm{ai}}(X) = \mathrm{Fm}$ for $X \neq \emptyset$ (*almost inconsistent*).

Proposition 6.14 (The theorem-free fragment; presumably folklore). *[T2 Prop 3.2] In the* $\{\wedge, \vee\}$ *language the structural closure operators extending* Cn_2 *are exactly* Cn_2*,* C_{ai} *and* C_{Fm}*.*

Proof in Appendix E.5. So there coherence must be enforced on a non-empty designated context. For sequents D let $\langle D \rangle$ be the least structural consequence relation containing D, and for a designated Cn_2-consistent A_0 let $\mathcal{S}(D, A_0) = \{h \text{ structural} : D \subseteq h,\ A_0 \nvdash_h \bot\}$.

Theorem 6.15 (Coherence pins classical logic). *[T2 Thm 3.3, as repaired] Let the language be as in Theorem 6.13.* (a) *For* $D \subseteq \mathrm{Cn}_2$*:* $\mathcal{S}(D, A_0) = \{\mathrm{Cn}_2\}$ *iff* $\langle D \rangle = \mathrm{Cn}_2$*, and a finite* D *suffices (with* $\to$ *in the language, Hilbert-style bases with modus ponens as the only rule, plus* $\rhd\neg\bot$ *and* $\rhd\top$ *when these constants are primitive; without* $\to$*, by Rautenberg 1981; Appendix E.5).* (b) *Let* $\mathcal{H} = (h_i)$ *be any uniformly decidable family of structural consequence relations with* $\mathrm{Cn}_2 \in \mathcal{H}$*. The learner outputting the least* i *such that* h_i *contains the data so far and* $A_0 \nvdash_{h_i} \bot$ *identifies* Cn_2 *in the limit from every text, with at most* i^* *mind changes (*i^* *the least index of* Cn_2*), for* every *ordering of* $\mathcal{H}$*: no simplicity or minimality bias is needed.* (c) *Each assumption is needed:* $\mathrm{Cn}_2 + \{\rhd p\}$ *(*p *not in* A_0*) is non-structural,* A_0*-coherent and contains every text;* C_{Fm} *contains every text; in the theorem-free* $\{\wedge, \vee\}$ *fragment,* C_{ai} *survives the empty context.* *(all but the finite-basis clause of (a))*
Lean: Post.versionSpace_eq_singleton_iff Lean: Post.learner_identifies

Proof idea. (a) Members of $\mathcal{S}$ extend $\langle D \rangle = \mathrm{Cn}_2$, so are Cn_2 or C_{Fm}, and A_0 excludes C_{Fm}. (b) For $j < i^*$, either the text refutes h_j, or $h_j \supsetneq \mathrm{Cn}_2$ is C_{Fm} and A_0 rejects it. Details, and why the $\wedge, \vee$ clauses must be $\to$-axioms rather than sequent rules, in Appendix E.5. □

Reading. This is a setup of the requested shape that provably works for formal propositional mathematics. The algorithm is trivial; the content is Theorem 6.13: substituting $\top/\bot$ along a falsifying row turns an invalid rule into "from tautologies infer a contradiction", so a uniform learner cannot hide an invalid schema from coherence. It must generalize schematically, though: closure under *renaming* atoms is not enough (Appendix E.5). In Experiment B (Section 12.2) a bold learner over classical natural deduction accepted all 59 valid and refuted all 118 invalid

structural candidate schemas, each by an explicit derivation of $\vdash \bot$, while the non-structural $\vdash p$ survived every designated context.

Learning theorems is enough here: Cn_2 is the largest structural closure operator with theorems Taut, so a bold learner can learn the theorems and take the structural completion. For IPC, derivability and admissibility differ (Harrop's rule) but have the same theorems and the same coherence profile, so theorems plus coherence data cannot tell them apart (Proposition E.1 in Appendix E.5).

Theorem 6.16 (Intuitionistic logic: coherence is blind)**.** *[T2 Thm 3.6]* (a) *For every intermediate logic L and finite A: $A \vdash_L \bot$ iff $A \vdash_{\mathsf{CPC}} \bot$.* (b) *A learner outputting a maximal coherent structural hypothesis never outputs* IPC. (c) *There is a strictly increasing chain of finitely axiomatizable intermediate logics $L_1 \subsetneq L_2 \subsetneq \cdots$ whose union L_ω is intermediate, and no learner identifies $\{L_k : k \leq \omega\}$ from text plus coherence information, for any $\mathcal{A}$, even with target-dependent designations.* (d) *The conservative learner outputting $\langle D_n \rangle$ BC-identifies every finitely axiomatizable structural target.*

Proof idea. (a) is Glivenko's theorem (Glivenko, 1929). (c) uses Jankov formulas of an infinite antichain of finite Heyting algebras (Jankov, 1968; Chagrov and Zakharyaschev, 1997); by (a) all designations are the same for every member, so Corollary 6.11 applies. Appendix E.6.

So for IPC the work is done by *minimality* (Section 5), not coherence: minimality is right when the target is at the bottom of its coherent version space, coherence plus boldness when it is at the top, and classical logic is at both ends. In Experiment B a bold learner over intuitionistic natural deduction drifts toward CPC, accepting excluded middle, Peirce's law and other IPC-underivable schemas (Section 12.2).

Complete theories. Much of the algebra in a physics calculation is real-closed-field reasoning, and RCF is complete. Is it pinned like CPC?

Proposition 6.17 (Complete theories: theorems, and rules under $\exists$-closure)**.** *[T2 Prop 3.11, repaired]* *Let T be a complete consistent first-order theory.* (a) *The only consistent closed theory in the language of T containing T is T; if T is decidable, the learner of Theorem 6.15(b), run on a uniformly decidable family of theories containing T, identifies T from a text and the coherence datum $\emptyset$.* (b) *Add parameters (fresh constants); the target is $\vdash_T$ ($\Gamma \vdash_T \varphi$ iff $T \cup \Gamma \vdash \varphi$). If $h \supseteq \vdash_T$ is closed under substituting terms for parameters and under $\exists$-elimination (from $\Gamma \vdash_h \exists x\psi$ and $\Gamma, \psi(d) \vdash_h \chi$, d fresh, infer $\Gamma \vdash_h \chi$), and $\emptyset \nvdash_h \bot$, then $h = \vdash_T$.* (c) *Without $\exists$-closure (b) fails, even for a non-empty context. For $T =$ RCF, the least parameter-structural $h_2 \supseteq \vdash_{\mathsf{RCF}}$ containing all instances of $t^3 - 4t + 1 = 0 \rhd t > 1$ has $h_2(\emptyset) = \mathrm{Cn}_{\mathsf{RCF}}(\emptyset)$ and $h_2(\{c^3 - 4c + 1 = 0\}) = \mathrm{Cn}_{\mathsf{RCF}}(\{c^3 - 4c + 1 = 0,\ c > 1\})$: coherent on both, yet unsound (the cubic has three real roots and Galois group S_3). The context $\{c^3 - 4c + 1 = 0,\ c < 0\}$ catches it.*

Proof idea. (b) An invalid $\Gamma(\bar{c}) \vdash_h \varphi(\bar{c})$ gives $T \vdash \exists \bar{x}(\bigwedge \Gamma \wedge \neg\varphi)$; a fresh instance derives $\bot$ in h, and $\exists$-elimination discharges the parameters one at a time. (c) The rule fires only on terms the context forces to be roots. Appendix E.7.

Reading. At the level of theorems, RCF, Presburger arithmetic and Tarski's geometry are pinned as cleanly as CPC. At the level of the motivating question, rules applied in contexts with parameters, a set of uniform rules is *not* pinned by one coherence datum; it is pinned if its hypotheses are closed under $\exists$-elimination, a discharge meta-rule rather than a step. Without it, no fixed finite family of designated contexts suffices: for even d large enough, $t^d = 1 + 1 \rhd t > 0$

never fires in a context of the family, so it is coherent there but unsound (Appendix E.7, which also discusses sin and exp) (proof sketch). Designation is soundness-critical in the other direction too (Proposition 11.9).

6.3 Arithmetic: coherence is Popperian

Arithmetic is where coherence stops being a complete test. Coherence with a Σ_1-complete base already does everything Δ_0 world feedback can (Lemma 6.18); no r.e. hypothesis is maximal, and Turing progressions are not identifiable (Theorems 6.19 and 6.20). The right use of coherence is Popperian: it detects false Π_1 claims but not false Σ_1 claims, and no computable channel decides Σ_2 truth in the limit (Theorem 6.21).

Hypotheses are r.e. theories $T \supseteq \mathsf{PA}$, world feedback is a Δ_0-oracle, and $\mathbb{N}$ is the intended model; Q is Robinson arithmetic.

Lemma 6.18 (Δ_0 feedback is subsumed by coherence; standard)**.** *[T2 Lemma 3.7] A consistent $T \supseteq \mathsf{Q}$ proves every true Δ_0 sentence and no false one, so no Δ_0 feedback refutes it.*

Proof. Q is Σ_1-complete (Boolos et al., 2007); if T proved a false δ it would also prove $\neg\delta$. □

Theorem 6.19 (No maximal r.e. hypothesis; cited)**.** *[T2 Thm 3.8, as repaired] Every consistent r.e. $T \supseteq \mathsf{Q}$ has a ρ with $T + \rho$ and $T + \neg\rho$ consistent (Gödel, 1931; Rosser, 1936; Tarski et al., 1953).*

So a bold learner with r.e. hypotheses and $\mathcal{A} = \{\emptyset\}$ has no maximal choice. (If $\mathcal{A}$ contains all finite sets of true sentences, T is $\mathcal{A}$-coherent iff $T \subseteq \mathrm{Th}(\mathbb{N})$, and at most one refinement survives.)

Theorem 6.20 (Turing-chain impossibility)**.** *[T2 Thm 3.9, as repaired] Let $T_0 = \mathsf{PA}$, $T_{k+1} = T_k + \mathrm{Con}(T_k)$, $T_\omega = \bigcup_k T_k$ (Turing, 1939; Feferman, 1962), and let $\mathcal{A}$ be any fixed family of finite sets of* true *sentences. No learner identifies a class containing $\{T_k : k \leq \omega\}$ from text, $\mathcal{A}$-coherence and a Δ_0-oracle.*

Proof. All T_k are sound, so all side information is target-independent; the chain is strict by Gödel II; apply Corollary 6.11. □

The Σ_1-unsound $T_k + \neg\mathrm{Con}(T_k)$ are natural *hypotheses*: consistent, unrefuted by Δ_0 data, and refuted by coherence only relative to accepted premises proving $\mathrm{Con}(T_k)$. Restricting designations to true contexts is essential: with $\{\neg\mathrm{Con}(T_k)\}$ designated as data about the target, the chain becomes identifiable (Proposition 6.12).

Theorem 6.21 (Coherence as a Π_1-truth detector; the Σ_2 barrier)**.** *[T2 Thm 3.10, as repaired]* (a) *For Π_1 π, $\mathsf{PA} + \pi$ is consistent iff π is true.* (b) *For Σ_1 σ, truth implies consistency of $\mathsf{PA} + \sigma$, not conversely ($\sigma = \neg\mathrm{Con}(\mathsf{PA})$).* (c) *The* Popperian *learner ("true" on a Π_1 sentence until PA refutes it; "false" on a Σ_1 sentence until PA proves it) converges to the truth on $\Sigma_1 \cup \Pi_1$ with at most one mind change per sentence.* (d) *The* naively bold *learner (accept φ_n iff consistent with PA plus the earlier accepted sentences) converges to a Lindenbaum completion that depends on the enumeration, and on every enumeration beginning with $\neg\mathrm{Con}(\mathsf{PA})$ it is Σ_1-unsound.* (e) *Every computable learner using a computable text, coherence searches and a Δ_0-oracle fails to converge to the truth value of some Σ_2 sentence.*

Proof idea. (a) A false Π_1 sentence has a true, hence PA-provable, Σ_1 negation. (e) Convergent computable guesses define a Δ_2 set (Shoenfield, 1959), and the true Σ_2 sentences are Σ_2-complete. Appendix E.8.

This is Popper's asymmetry in its standard learning-theoretic form (Putnam, 1965; Gold, 1965; Kelly, 1996), read for coherence-learners. The same boundary appears for credences: computable, weakly coherent, Π_1-convergent credences are dogmatic on infinitely many true Π_2 sentences (Theorem E.2, a self-contained form of a theorem of Sawin and Demski 2013), and Proposition E.3 records what weak coherence costs (both in Appendix E.9).

Reading. In arithmetic, computation adds nothing beyond coherence with a Σ_1-complete base. What matters is where boldness is applied: on universal claims, which coherence can refute, not on existential ones, which need a witness. Beyond Δ_2 no computable channel decides truth in the limit (Theorem 6.21(e)); only the weaker gradual verification, with credence tending to 1 exactly on truths, reaches further (Proposition E.3). The Gödel–Rosser alternatives are a permanent residue, removed only by accepting stronger principles (reflection, Con(PA), a stronger theory): a justification that is not a PA-proof (Section 13).

6.4 Carnap's problem is Gold's problem

Under inferentialism, learning valid inferences should be learning meanings. Carnap (1943) observed that the classical rules do not fix the truth tables: the all-true valuation, and "true iff tautologous", respect every classical inference.

Definition 6.22 (Meanings). [T2 §4.1] A *valuation* is any $v : \mathrm{Fm} \to \{0,1\}$, identified with its true-set in Cantor space 2^{Fm}; a *meaning* is a set V of valuations; BV is the set of Boolean ones. For finite Γ, Δ, $\Gamma \models^1_V \varphi$ iff every $v \in V$ with $v[\Gamma] = 1$ has $v(\varphi) = 1$, and $\Gamma \models^{\mathrm{m}}_V \Delta$ iff no $v \in V$ has $v[\Gamma] = 1$, $v[\Delta] = 0$. $\mathrm{Val}(S)$ is the set of valuations satisfying the sequents S; $V^\cap$ closes V under intersections of true-sets, including the empty one $v_\top \equiv 1$; v_{Taut} is the characteristic function of Taut. V is *structural* if $v \in V$ implies $v \circ \sigma \in V$ for all substitutions σ.

The sequent $\Gamma \rhd \Delta$ excludes exactly the clopen set of valuations realizing the position $[\Gamma : \Delta]$. So *inference data are negative data about valuations*, while *coherence certifications* ("$[A : D]$ is in bounds") and world feedback are *positive* data: Restall's reading of consequence (Restall, 2005) turned into a statement about data.

Lemma 6.23 (Duality). *[T2 Lemma 4.1]* (a) $\mathrm{Val}(\models^{\mathrm{m}}_V) = \overline{V}$. (b) *For closed V, $C_V(X) = \bigcap\{v \in V : X \subseteq v\}$ is finitary and the valuations satisfying every valid single-conclusion sequent form $V^\cap$.* (c) $\mathrm{BV}^\cap$ *is the set of characteristic functions of classical theories, including* Fm. *Lean: Carnap.Val_mrel*
Lean: Carnap.lemma_4_1_b Lean: Carnap.intClosure_BV_eq

Theorem 6.24 (Carnap's problem in learning form). *[T2 Thm 4.2, as repaired]* $\models^1_{\mathrm{BV}} = \models^1_W$ *whenever* $\mathrm{BV} \subseteq W \subseteq \mathrm{BV}^\cap$. *So no data that are a function of the single-conclusion relation (text, informant, or metarules with discharge read* globally*, as closure conditions) distinguish* BV *from W. The non-Boolean members of* $\mathrm{BV}^\cap$ *are the v_T for theories T that are not maximal consistent (including $T = \mathrm{Fm}$): $v_\top$ violates the table for $\neg$; v_{Taut} and every non-maximal consistent v_T violate those for $\neg, \vee, \to$; all respect $\wedge$.* $\mathrm{BV}^\cap$ *is closed and structural.* *Lean: Carnap.carnap_single*

Proof idea. By Lemma 6.23(b) every $w \in W$ satisfies $\models^1_{\mathrm{BV}}$. Appendix E.10.

Under the *local* reading of rule validity (Garson, 2013), $\to$I excludes v_{Taut}, but only by imposing the two-conclusion $\rhd \varphi, \varphi \to \psi$. The right measure is the number of denials.

exclusion data (sequents)	rank	excludes
non-contradiction, no denial: $p, \neg p \triangleright$ ("not both P and $\neg P$")	0	$v_\top$, inconsistent valuations
imitation of single-conclusion inferences	1	non-closed valuations
exhaustiveness: $\triangleright p, \neg p$ ("not deny both"), $p \vee q \triangleright p, q$	2	v_{Taut}, all gappy theories

Table 6: What each shape of data excludes (Theorem 6.25). Coherence *certifications* are positive data about meanings: certifying $[A : D]$ excludes exactly the meanings that do not realize $[A : D]$ (for the empty position, only $V = \emptyset$), and adding $v_\top$ to a meaning never makes it fail a certification.

Theorem 6.25 (Denial-rank trichotomy)**.** *[T2 Thm 4.3] With $\neg$ in the language, let the* denial rank $d(v)$ *be the least $|\Delta|$ over classically valid $\Gamma \triangleright \Delta$ (finite) that v violates (∞ if none). Then $d(v) = 0$ iff v's true-set is inconsistent; $d(v) = 1$ iff it is consistent but not closed; $d(v) = 2$ iff it is a consistent, closed, non-maximal theory; $d(v) = \infty$ iff $v \in \mathrm{BV}$.* Lean: Carnap.denialRank_eq_two_iff

Proof idea. A consistent closed non-maximal T omits some φ and $\neg\varphi$, violating $\triangleright\varphi, \neg\varphi$. Appendix E.10.

Reading. "Coherence" covers two kinds of data (Table 6): certifications that a position is in bounds (positive about V), and 0-denial constraints such as $p, \neg p \triangleright$ (negative about V), which remove $v_\top$. Neither removes v_{Taut}. A learner trained on imitation plus a coherence loss, in either sense, is provably free to settle on gappy valuations: consistent, inferentially competent, never committing on open questions. These are van Fraassen's supervaluations (van Fraassen, 1966), the honest semantics of "settled by the premises". Pinning classical meaning needs a loss with *two* denials. Bilateral signed sequents (Smiley, 1996; Rumfitt, 2000) carry the same rank-2 information. For language-model training this raises a question the theorems do not settle, since they concern identifiability, not training dynamics: imitation plus a contradiction penalty does not rule out consistent but noncommittal reasoners, and whether training drifts toward them is empirical.

With rank-2 data finitely many data suffice. Let TT be the eleven truth-table schemata: $\varphi, \neg\varphi \triangleright$ and $\triangleright\varphi, \neg\varphi$; $\varphi \wedge \psi \triangleright \varphi$, $\varphi \wedge \psi \triangleright \psi$, $\varphi, \psi \triangleright \varphi \wedge \psi$; $\varphi \triangleright \varphi \vee \psi$, $\psi \triangleright \varphi \vee \psi$, $\varphi \vee \psi \triangleright \varphi, \psi$; $\triangleright\varphi, \varphi \to \psi$, $\psi \triangleright \varphi \to \psi$, $\varphi, \varphi \to \psi \triangleright \psi$. $\mathrm{TT}(p, q)$ are their instances in distinct atoms.

Theorem 6.26 (Bilateral determinacy with a twelve-datum tell-tale)**.** *[T2 Thm 4.4, as repaired] Let the language be $\{\neg, \wedge, \vee, \to\}$ (with $\bot, \top$ add $\bot\triangleright$ and $\triangleright\top$; the Lean version has $13 + 1$ data).* (a) $\mathrm{Val}(\mathrm{TT}) = \mathrm{BV}$*, which is closed.* (b) *If V is structural, every $v \in V$ satisfies the eleven sequents $\mathrm{TT}(p, q)$, and $V \neq \emptyset$ (one coherence datum), then $V = \mathrm{BV}$.* (c) *So among structural meanings* BV *is identified finitely by twelve data; without the coherence datum $V = \emptyset$ survives, the Gold over-generalization.* (d) *Without structurality* BV *is not identifiable in the limit from text plus the coherence datum $[\,:\,]$, or any other target-independent certifications: every finite set of valid sequents is valid in a closed meaning $\mathrm{BV} \cup \{v'_\chi\}$, v'_χ a Boolean valuation flipped at a formula χ the set does not mention. (If every target-certified position is supplied, text plus certifications is an informant, as in Proposition 6.12, and this obstruction disappears; the next one does not.) And the dense non-structural $\mathrm{BV} \setminus \{u_0\}$ has the same valid sequents and in-bounds positions as* BV*.* Lean: Carnap.structural_eq_BV Lean: Carnap.BV_not_BC_learnable

Proof idea. (b) Any Boolean u is $v \circ \sigma$ for σ sending atoms to $p_0 \vee \neg p_0$ or its negation. (d) No tell-tale, then the locking-sequence argument. Appendix E.10.

	Gold (sequent space)	Carnap (valuation space)
hypothesis	consequence relation	set V of admissible valuations
positive data	valid sequents (text)	coherence certifications; world feedback
negative data	coherence certifications	sequents, incl. 0-denial ones
over-generalization	trivial logic	$v_\top$, v_{Taut}, …
failure	*learnability*: finite data never exclude supersets	*identifiability*: complete single-conclusion data fix V only up to $(\overline{V})^\cap$
remedy	one coherence datum plus Post-completeness	two-denial data, or a compositional prior

Table 7: Gold's and Carnap's problems as two instances of one duality [T2 §4.5; L4 §3].

The three two-conclusion schemata in TT are exactly the rank-2 data that v_{Taut} violates. Brute-force checks confirm (a), Theorem 6.24 and the denial ranks on small formula sets, and find no non-Boolean structural survivor of the twelve data among two- and three-element matrices, each datum being necessary (Appendix E.10) (computed). A second remedy restricts the hypothesis class, as in Gold's setting:

Proposition 6.27 (A compositional prior suffices)**.** *[T2 Prop 4.5] Exactly one two-element matrix with designated value* 1 *validates explosion,* $\neg\neg p \rhd p$*,* $p \rhd \neg\neg p$*, the three* $\wedge$ *rules,* $\vee$*-introduction, disjunctive syllogism, modus ponens,* $q \rhd p \to q$ *and* $\neg p \rhd p \to q$*: the standard one.*

Proof in Appendix E.10; compare Bonnay and Westerståhl (2016), whose compositionality is more general. World feedback alone fixes nothing, since the actual valuation is Boolean and never refutes $V \supseteq \mathrm{BV}$; under the compositional prior it identifies the tables once every row has been observed [T2 Prop 4.6].

Reading. Carnap's problem is a semantic twin of Gold's only in part (Table 7). Gold's trivial logic ($V = \emptyset$) is removed by a coherence certification, Carnap's $v_\top$ by a 0-denial sequent, and v_{Taut}, the Horn blindness of single-conclusion data, by neither. So imitation plus coherence fixes inferential role, not classical truth conditions. That does not matter for deriving; it matters for interpreting world feedback ("A is false" licenses $\neg A$ only through the bilateral bridge). Rules can be identifiable when truths are not (Section 13.2).

6.5 Tonk, harmony and conservativity

Prior (1960)'s tonk ($\varphi \rhd \varphi \,\mathrm{tonk}\, \psi$, $\varphi \,\mathrm{tonk}\, \psi \rhd \psi$) is the paradigm of a rule search exploits (Theorem 3.2); Belnap (1962) demanded that new rules be *conservative*.

Proposition 6.28 (Tonk)**.** *[T2 Prop 5.1] Under cut, tonk gives* $\varphi \vdash \psi$ *for all* φ, ψ*, and coherence refutes it in two steps (*$a \rhd a\,\mathrm{tonk}\,\bot \rhd \bot$*) if* $\mathcal{A}$ *has a non-empty context or the logic has a theorem. In a theorem-free base with* $\mathcal{A} = \{\emptyset\}$ *it survives: classical* $\{\wedge, \vee\}$ *plus tonk is* C_{ai}*. Without cut tonk need not trivialize (Cook, 2005; Ripley, 2015).*

Proof. One cut gives the first claim. In the theorem-free case every rule has premises, so nothing is derivable from $\emptyset$, while from any premise tonk yields everything. □

Transitivity lets one bad rule poison everything, and lets coherence catch it. Boldness is order-dependent when individually coherent rules are jointly incoherent: offered tonk's two rules one at a time, a bold learner keeps whichever comes first (Section 12.2).

Proposition 6.29 (Coherence is not conservativity). *[T2 Prop 5.2, as repaired] Intuitionistic $\to$ plus $\neg$-introduction, $\neg$-elimination and $\neg\neg$-elimination is the $\{\to, \neg\}$ fragment of* CPC*: coherent, but not conservative over* $\mathsf{IPC}_\to$ *(it proves Peirce's law). $\neg\neg$-elimination alone would be conservative. Conservativity implies coherence when coherence is non-triviality on the old vocabulary, or $\bot$ with ex falso is in the base, but not in general (*$\mathsf{IPC}_\to + \{\triangleright\bot\}$*, with $\bot$ new and without ex falso).*

Proof in Appendix E.11. So Belnap's learner rejects classical negation over intuitionistic positive logic, and the coherence learner accepts it. Harmony *with normalization* gives conservativity (Prawitz, 1965; Dummett, 1991), but local harmony alone does not: Read's bullet (Read, 2000) has local reductions yet derives $\bot$ from nothing. Over a Post-complete base the two notions coincide [L5 Cor 2.4]: if C' is structural on $L' \supseteq L$ with $C'|_L \supseteq \mathrm{Cn}_2$ (L as in Theorem 6.13), then $C'|_L$ is Cn_2 or C_{Fm}, so C' is conservative iff it is coherent on one classically consistent context. Belnap's existence condition is, over classical logic, non-triviality; harmony matters only over sub-classical bases.

Theorem 6.30 (Coherence is Π_1, conservativity Π_2). *[T2 Thm 5.3] For finite rule systems (elementary formal systems; Post 1943; Smullyan 1961):* (a) *$\{R : R \nvdash \bot\}$ is Π_1-complete;* (b) *for some fixed finite base B, $\{N : B + N$ conservative over $B\}$ is Π_2-complete;* (c) *so coherence is decidable in the limit with one mind change, while conservativity is not decidable in the limit by any computable learner, only refutable in the limit (Kelly, 1996);* (d) *for decidable B (e.g.* CPC*,* IPC*) conservativity is Π_1.*

Proof idea. (a) Simulate $\varphi_e(e)$ by rules and add "halt $\triangleright\bot$". (b) Reduce totality of φ_e to conservativity of one new symbol over a universal base. Appendix E.11.

Reading. A coherence filter can be built into a convergent learner (Theorem 6.15(b)); a conservativity filter cannot in general. Coherence plus decidable local checks such as harmony are the learner-friendly norms. Where non-conservativity is the point, as for theoretical terms in physics, a concept whose rules smuggle in a substantive inference (Dummett's *Boche*) is coherent, and only world feedback can test it.

6.6 Fallacies and the Kripkensteinian residue

A learner with a *generalization operator* $\mathcal{G}$ (e.g. structural closure of least generalizations) turns finitely many fallacious human steps F into rules $\mathcal{G}(F)$. Let h^* be the target consequence relation.

Theorem 6.31 (Elimination criterion; trivial). *[T2 Thm 6.1] Coherence (with data) eventually refutes the generalized fallacy iff $h^* \oplus \mathcal{G}(F)$ is $\mathcal{A}$-incoherent; for OH the refutation is charged to the budget of Theorem 6.5. If h^* is sound for a class of intended structures and each $A \in \mathcal{A}$ holds in one that also validates $\mathcal{G}(F)$, the fallacy survives.*

Proof. $h^* \oplus \mathcal{G}(F)$ contains every positive datum; apply Theorem 6.9. A structure validating h^*, $\mathcal{G}(F)$ and A shows coherence on A. □

Corollary 6.32 (All structural propositional fallacies die, quickly). *[T2 Cor 6.2] If $\mathcal{G}$ is structural closure and $h^* \supseteq \mathrm{Cn}_2$ is structural and coherent on a designated context (so $h^* = \mathrm{Cn}_2$), every classically invalid rule is eliminated: substituting $\top/\bot$ along a falsifying row makes its premises*

fallacy	rule	witness
affirming the consequent	$q,\ p \to q \ /\ p$	$p := \bot,\ q := \top$
denying the antecedent	$\neg p,\ p \to q \ /\ \neg q$	$p := \bot,\ q := \top$
conversion	$p \to q \ /\ q \to p$	$p := \bot,\ q := \top$
illicit contraposition	$p \to q \ /\ \neg p \to \neg q$	$p := \bot,\ q := \top$
"or" read as "xor"	$p \vee q,\ p \ /\ \neg q$	$p := q := \top$

Table 8: Propositional fallacies and their Post witnesses (computed).

tautologies and its conclusion a contradiction (Table 8). Given the row, the $\bot$-derivation has polynomial size in any Frege system (proof sketch for the size bound); finding the row is an NP search.

Proof. $h^* \oplus \mathcal{G}(F)$ is a structural proper extension of Cn_2, hence C_{Fm} (Theorem 6.13). □

Experiment B (Table 19) shows the negative-data role end to end: with systematic fallacies and noise ($N \geq 50$), coherence pruning cut the unsound rules left by imitation from 45.6 to 1.2 on average, and the budgeted adversary no longer derived $\vdash \bot$, at no cost in completeness. It is not complete: 19 of 72 runs still end with an unsound rule, most surviving unsound rules are non-structural, and at $N \leq 10$ even affirming the consequent can survive, because a small learned calculus may lack the rules needed to derive the contradiction. Outside a Post-complete target, fallacies can survive: the quantifier swap is coherent until a context forces two objects, the gambler's fallacy until independence is accepted, and conditional perfection until two exclusive laws share a consequent (Example E.4 in Appendix E.12).

Proposition 6.33 (Without uniformity, coherence is toothless; trivial)**.** *[T2 Prop 6.3] If $h^* \oplus F$ is $\mathcal{A}$-coherent, the non-structural $h^* \oplus F$ survives forever; e.g. $\mathrm{Cn}_2 + \{\triangleright p_{17}\}$, the "quus" of propositional logic, when every designated context is consistent with p_{17}.*

Theorem 6.34 (The Kripkensteinian residue)**.** *[T2 Thm 6.4, as repaired]* $\mathrm{Alt}_{\mathcal{G}}(h^*, \mathcal{A}) = \{h\ \mathcal{G}\text{-closed} : h \supseteq h^*,\ h\ \mathcal{A}\text{-coherent}\}$ *is, in the limit of complete data, the set of $\mathcal{G}$-closed hypotheses never refuted by positive data and coherence (Theorem 6.9).* (i) *$\mathcal{G}$ structural, $h^* = \mathrm{Cn}_2$, $\mathcal{A} \neq \emptyset$:* $\mathrm{Alt} = \{\mathrm{Cn}_2\}$. (ii) *$\mathcal{G}$ structural, $h^* = \mathsf{IPC}$, $\mathcal{A}$ a non-empty family of classically consistent contexts:* $\mathrm{Alt} = \{h\ \text{structural} : \mathsf{IPC} \subseteq h \subseteq \mathsf{CPC}\}$*, all intermediate logics.* (iii) *$\mathcal{G}$ deductive closure, $h^* = \mathsf{PA}$, $\mathcal{A} = \{\emptyset\}$:* Alt *is the set of all consistent closed extensions of* PA *(the r.e. ones, if hypotheses are r.e.), e.g.* $\mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$*, whose completions (not r.e.) have only non-standard models. Designating all finite sets of true sentences removes exactly the unsound ones; the sound T_k of Theorem 6.20 remain, and by that theorem no learner identifies them together with T_ω from text plus true-context coherence.*

Proof idea. (i) is Theorem 6.13. For (ii) $\subseteq$, a classically invalid sequent in a structural $h \supseteq \mathsf{IPC}$ becomes, under σ_v, an IPC-provable premise set with an IPC-refutable conclusion. Appendix E.12.

Division of labour. *Community practice* (positive data) eliminates under-generalizations. *Uniformity* (structurality, least generalization) eliminates gerrymanders such as Proposition 6.33; it is vocabulary-relative (Goodman's point), and it turns an instance-level fallacy into a schema that coherence can refute. *Coherence* eliminates uniform alternatives that conflict with accepted uniform rules: tonk, affirming the consequent, "quus" as a second rule for $+$. *Nothing* in the setup

eliminates coherent uniform alternatives that agree with all data. For classical propositional logic there are none; for arithmetic they are the Gödel–Rosser and non-standard alternatives, a precise mathematical counterpart of Kripkenstein for arithmetic (Kripke, 1982), removable only by accepting reflection or stronger theories on other grounds (Section 13).

6.7 A remark on eliminative versus confirmational coherence

Coherentist epistemology has impossibility results: no informative coherence measure is truth-conducive ceteris paribus (Bovens and Hartmann, 2003b,a; Olsson, 2005a,b). They do not threaten the use of coherence in this section, and they do refute a tempting other use.

Remark 6.35 (Three uses of coherence). [L11 §3.4, TC4] (U1) Eliminative: "h derives $\bot$ from a context certified consistent, so h has an invalid rule." This is modus tollens, not a measure. Conditioning a prior on "h is $\mathcal{A}$-coherent" multiplies every coherent hypothesis by $1/\mathbb{P}(\text{coherent})$: under truthful designation the target's posterior weakly rises and odds among coherent rivals are unchanged. The impossibility theorems, about degrees of coherence of consistent information sets, do not apply; the prices are Theorem 6.9 and the need for truthful designation. (U2) Probabilistic consistency of credences: de Finetti constraints, satisfied by the noncommittal assignment, the credal analogue of gappy valuations. (U3) Agreement as evidence: annotators or sampled derivations agree. This is the witness scenario of Lewis (1946) that the impossibility theorems address, and they bite in full.

Appendix E.13 makes (U3) and (U1) quantitative: unanimity among annotators who share an error source, or among samples of one model, moves the odds that a step is valid by a bounded factor however many agree (Remark E.5); and coherentizing credences by projection is safe exactly when the constraints are sound, and strictly harmful when the actual world violates them (Lemma E.6).

Reading. The setup's use of coherence is eliminative, hence immune to the coherentist impossibility theorems, but weak: it never confirms, it only removes what conflicts. Corroboration must come from channels with independent errors: computation, randomized evaluation, measurement (Section 2.5). Where the target is coherence-pinned, uniqueness of the coherent extension, not truth-conduciveness of coherence, does the work. Section 13 reads this division as vindicated wide reflective equilibrium (Goodman, 1955; Daniels, 1979).

7 An end-to-end theorem for formal mathematics: the two-tier learner

The first success criterion of Section 1.1 is a setup of this shape that provably works for formal mathematics: imitate human inferences, then condition on coherence and on occasional feedback from the world. Sections 4 to 6 supply the parts. The natural way to assemble them is a statement of roughly this form. For every target calculus with finite anchors, human data with sporadic noise below the margin and a systematic fallacy set F, and every adaptive prover: (i) a cautious assertion tier is sound at all times with probability at least $1 - \delta$; (ii) the number of detected incoherences is at most $\log_2(1/w^*)$ plus a noise cost; (iii) after N human steps the tier equals the target plus the fallacies coherent with it; and (iv) that residue is empty for classical propositional logic and for complete decidable theories.

This statement is false as stated, and the way it fails is the main finding of this section. Three corrections are needed, and each is forced by a matching lower bound.

1. *Caution is relative to a class, and fallacies break realizability.* Positive data cannot tell a systematic fallacy from a rule (Corollary 5.19), so the realizable object is the practice, not the target. A tier that is cautious over the positive-data version space asserts every frequent fallacy (Proposition 7.5). A tier that is cautious over hypotheses allowed to declare practice rules fallacious asserts nothing (Proposition 7.6). What works is *presumption of validity applied only after an audit*, with abstention until then, and the waiting is necessary (Theorem 7.18).

2. *Coherence condemns sets, and the blame problem is real.* The practice schemas that can be soundly condemned as schemas are exactly those in the union of the minimal conflicts, which is the hitting-set duality of model-based diagnosis (Lemma 7.7). At stable depth this collateral damage is unavoidable for every sound learner (Theorem 7.19). The extremal witness is almost too clean: in the practice {modus ponens, affirming the consequent} the rival diagnosis is the hypothesis that "$\rightarrow$" means "$\leftarrow$". World feedback with descent, or a single denial, turns bags into singletons (Lemma 7.8).

3. *Soundness must be relativized to refutation size.* No computable learner is both sound at all times relative to full coherence and eventually complete (Theorem 7.20).

With these corrections the theorem holds (Theorem 7.12). We call the resulting learner TTL, the *two-tier learner*: a cautious, audited assertion tier and a bold sandbox that a red team attacks.

Depth, honestly. The ingredients are known or transported from Sections 3 to 6: Lemma 7.7 is Reiter's diagnosis theory, Lemma 7.8 is Shapiro's contradiction backtracing, Theorem 7.18 is a standard two-point rare-event bound, and Theorem 7.20 is a routine reduction. What is new is diagnostic: where the natural statement breaks (Propositions 7.5 and 7.6); the audit principle; the blame theorem with its $\rightarrow/\leftarrow$ witness and its stable-depth form; the depth relativization; the observation that world feedback earns its keep by *localization*, not refutation; and the assembly into one statement with explicit constants. None of it is deep mathematics. Its value is that it pins down, with matching lower bounds, what a setup that provably works for formal mathematics can and cannot deliver. All results of this section passed two rounds of adversarial verification[T7]. The second-round repairs (assumption (OE) in Theorem 7.18, the floor bound in Theorem 7.27(e), the size convention of Definition 7.2, the ground-rule convention $\rho_i = c_i = 1$ behind the N_1 formula of Definition 7.9, and the oracle specification of Corollary 7.23) were then re-checked by a third, independent referee, who found no fatal or major issue; its minor corrections are incorporated below.

7.1 Set-up: practice, evaluation, refutations

We use the practice of Definition 2.10: a target calculus Σ^* of genuine rules, a finite set F of systematic fallacies, the practice $\Sigma^P = \Sigma^* \sqcup F$ with $|\Sigma^P| \leq K$, and i.i.d. human steps tagged by practice schemas, with sporadic noise. Fallacy tags are latent but distinct. For tag i write $\beta_i = \pi_i(1 - a_i)$ and $\alpha_i = \pi_i a_i$, let ρ_i, c_i be the variability and witness count of Theorem 5.6 ($\rho_i = c_i = 1$ for ground rules, so $c_i \geq 1$ for every tag), and let $\Delta_i = (\rho_i\beta_i - \alpha_i)/2$ be the margin of Theorem 5.14.

Assumption 7.1 (Floor)**.** [T7 §1.2, Def 1.2] Constants $\Delta > 0$, $c \geq 1$, K and $\bar{\alpha}_i \geq \alpha_i$ are known, with $|\Sigma^P| \leq K$, $c_i \leq c$ and $(\rho_i\beta_i - \bar{\alpha}_i)/2 \geq \Delta$ for every tag i, fallacy tags included.

The evaluation sources are those of Definition 2.13: a finite family $\mathcal{A}$ of designated positions $[A : D]$, a world $W : J_W \to \{0, 1\}$, and trusted steps T_{trust} (for instance, first-order logic when only non-logical axioms are learned). Each position induces the partial evaluation $e_{[A:D]}$ of (WS) (Assumption 2.14): value 1 on A and on W-true judgments, value 0 on $D \cup \{\bot\}$ and on W-false ones. When $J_W \neq \emptyset$, the empty position $[\emptyset : \emptyset]$ is designated. A step is *falsified* under e if its premises are e-true and its conclusion e-false. (WS) is assumed for every designated position unless stated otherwise.

Definition 7.2 (Derivations, refutations, conflicts, oracle)**.** [T7 Def 1.5, as repaired]

(a) A *derivation* is a finite sequence of *distinct* judgments, each a leaf or the conclusion of one designated step whose premises occur earlier; its last judgment is its conclusion. Its *size* $|\pi|$ is the total number of symbols in its distinct judgments. A tree derivation (Definition 2.2) compresses to such a sequence with the same conclusion, using only its steps and leaves and no larger: keep, for each judgment, the step at an occurrence of least height, order the judgments by that height, and cut the sequence after the tree's conclusion.

(b) For a finite schema set B (or a step set S) a *d-refutation* is a derivation of size at most d, relative to some $[A : D] \in \mathcal{A}$, whose leaves have e-value 1, whose steps lie in $R_B \cup T_{\text{trust}}$ (resp. $S \cup T_{\text{trust}}$), and whose conclusion has e-value 0. This is Definition 2.19 with the size convention made explicit. A single falsified instance is a one-step refutation. B is *d-clean* if it has no d-refutation, and *clean at every depth* if it is d-clean for all d.

(c) $\text{Conf}_d = \{C \subseteq \Sigma^P : C \text{ is not } d\text{-clean}\}$ is upward closed. Its minimal members $\mathcal{C}_d$ are the *minimal d-conflicts*, and $\mathcal{C}_d(B)$ denotes those contained in B. Since Σ^P is finite there is d_0 with $\text{Conf}_d = \text{Conf}_{d_0}$ for $d \geq d_0$; we write $d = \infty$ for these stable values.

(d) The *refutation oracle* $\text{Ref}_d(B)$ returns a d-refutation of B or NONE, deterministically. By default it returns the lexicographically least among the smallest refutations. The *prefer-unblocked* variant returns one whose descent (Lemma 7.8) is unblocked if one exists, then the smallest, then the lexicographically least.

Ref_d is computable by brute force when instance membership and W are decidable and there are finitely many derivations of size at most d up to renaming the symbols that occur in neither $\mathcal{A}$ nor B and on which W does not depend. Its cost is $\exp(O(d))$, and it is accounted separately from samples. As $d \to \infty$ refutation search is only semi-decidable (Theorem 6.30).

Definition 7.3 (Residue, survivors, collateral)**.** [T7 Def 1.6] The *depth-d residue* is $F_{\text{res}}^{(d)} = \{\tau \in F : \Sigma^* \cup \{\tau\} \text{ is } d\text{-clean}\}$ (Definition 2.19), and $F_{\text{res}} = \bigcap_d F_{\text{res}}^{(d)}$; for $\tau \in F$, $d_\tau = \min\{d : \Sigma^* \cup \{\tau\} \text{ is not } d\text{-clean}\}$. The *survivors* are $F_d^{\text{surv}} = F \setminus \bigcup \mathcal{C}_d$, and the *collateral* is $\text{Coll}_d = \Sigma^* \cap \bigcup \mathcal{C}_d$, the genuine rules lying in some minimal conflict; $\text{Coll}_d(B) = \Sigma^* \cap \bigcup \mathcal{C}_d(B)$. *Singleton blame* ($\text{SB}_d$) holds if every member of $\mathcal{C}_d$ is a singleton.

The residue is the finite-depth version of the Kripkensteinian residue of Theorem 6.34: the fallacies that are coherent with the target on the designated positions, and world-adequate, up to size d. Provers are as in Definition 2.17: adaptive, unbounded, and knowing Σ^*, F, the learner's code and the history, but not future data. Uniform δ-soundness relative to a step set $S^\dagger$ is that of Definition 2.18; TTL uses no coins, so its event G depends only on the human data.

Lemma 7.4 (Refutation is one-sided and target-blind; trivial)**.** *[T7 Lemma 2.1 and §1.3, as repaired]* *Assume (WS).* (a) *Every step set $S \subseteq \text{Sound}(R^*)$ is clean at every depth; in particular every*

subset of Σ^* *is.* (b) *Every member of* Conf_d *contains a fallacy.* (c) Conf_d *and every answer of* Ref_d *are functions of* $(\mathcal{A}, W, T_{\mathrm{trust}}, d)$ *and the queried set alone; they do not depend on which schemas are genuine.* (d) *If* $e_{[A:D]}$ *is well defined, then (WS) holds for* $[A : D]$ *iff* Σ^* *is clean at every depth relative to* $[A : D]$*. The same holds with any calculus* Σ *in place of* Σ^**.*

Proof idea. The total valuation V of (WS) extends e and preserves truth along $R^* \cup T_{\mathrm{trust}}$, hence along $\mathrm{Sound}(R^*)$; so every node of a derivation from e-true leaves is V-true, and its conclusion is not e-false. For (d,$\Leftarrow$) take V to be the indicator of $\mathrm{Cl}_{R^* \cup T_{\mathrm{trust}}}(A \cup W^{-1}(1))$. Appendix F.1.

So refutations can never condemn a subset of the target, and two targets with the same practice receive identical refutational evidence on every query. Coherence and the world can say that a *set* of practice rules contains a fallacy; they can never say that a rule is genuine. Everything positive that is ever learned about a rule comes from imitation.

7.2 Why the naive two-tier claim is false

Proposition 7.5 (Caution over practice asserts the fallacies; trivial)**.** *[T7 Prop 2.2, as repaired] Let the positive-data tier be the per-tag trimmed version-space verifier (Theorem 5.13) over the practice tags. On the event of Theorem 5.14 its accepted set is exactly* $R^P = R^* \cup R_F$*. So it accepts an instance outside* $\mathrm{Sound}(R^*)$ *of every* $\tau \in F$*, and an adaptive prover gets that instance accepted by querying it. In* CPC *with a complete* Σ^* *and a pure* τ*, the prover can query the closed* $\top/\bot$ *instance of* τ *with true premises and a false conclusion (Lemma 7.22), and that single step makes every formula derivable from* $\emptyset$*.*

Proof. Theorem 5.14 applies tag by tag, and fallacy tags are tags like any other. In the CPC case, Σ^* derives the closed true premises and the negation of the closed false conclusion j, both tautologies; the step adds j, and then $\bot$ and everything follow. □

Purity matters for the last clause: adding the single invalid step $p \rhd q$ to a complete calculus for CPC leaves $\mathrm{Cl}(\emptyset)$ unchanged, and for an impure τ even all of $\mathrm{inst}(\tau)$ need not trivialize (Example 3.6). The proposition is the precise sense in which "the cautious assertion tier is sound" is *false* without further structure. Caution is soundness relative to the realizable class (Theorem 4.2), and systematic fallacies make the practice, not the target, the realizable object (Remark 5.20).

The obvious repair is to be cautious over hypotheses that may declare practice rules fallacious. Suppose the practice is known, and let $\mathcal{V} = \{\Sigma : \Sigma \subseteq \Sigma^P\}$, where a candidate Σ reads "Σ is the target and the rest of the practice is fallacious". All candidates explain the positive data equally well. Let $\mathcal{V}_d$ be the d-clean candidates and $\mathcal{V}'_t$ those consistent with the refutations found by time t.

Proposition 7.6 (The caution–presumption dilemma)**.** *[T7 Prop 2.4; (c) as repaired]*

(a) Caution without presumption is vacuous. $\emptyset \in \mathcal{V}_d$ *by Lemma 7.4(a), so* $\bigcap_{\Sigma \in \mathcal{V}_d} R_\Sigma = \emptyset$*, and the version-space verifier over* $\mathcal{V}$ *never asserts anything.*

(b) Presumption without audit is unsound. *The verifier asserting* $\bigcap\{R_\Sigma : \Sigma$ *maximal in* $\mathcal{V}'_t\}$ *asserts all of* R^P *until the first refutation is found: an instance outside* $\mathrm{Sound}(R^*)$ *of every fallacy, including those that a d-refutation will condemn when* $F \not\subseteq F^{(d)}_{\mathrm{res}}$*.*

(c) Presumption after audit is sound. *At the level of schemas, asserting the schemas common to all maximal members of* $\mathcal{V}_d$ *asserts*

$$R_{\bigcap\{\Sigma:\ \Sigma \text{ maximal in } \mathcal{V}_d\}} = R_{\Sigma^P \setminus \bigcup \mathcal{C}_d} = R_{(\Sigma^* \setminus \mathrm{Coll}_d) \cup F^{\mathrm{surv}}_d}, \qquad F^{\mathrm{surv}}_d \subseteq F^{(d)}_{\mathrm{res}},$$

which is sound relative to $R^* \cup R_{F_{\mathrm{res}}^{(d)}}$. *At the level of instances,* $\bigcap\{R_\Sigma : \Sigma \text{ maximal in } \mathcal{V}_d\}$ *contains this set, can be strictly larger, and is still contained in* $R^* \cup R_{F_{\mathrm{res}}^{(d)}}$ *and d-clean.*

Proof idea. (a) and (b) are immediate. In (c) the schema-level equality is Lemma 7.7(b) below; a non-residual fallacy τ has $\Sigma^* \cup \{\tau\} \supseteq C'$ for some $C' \in \mathcal{C}_d$, and $\tau \in C'$ because $C' \not\subseteq \Sigma^*$. For the instance level, Σ^* lies in a maximal clean candidate, all of whose fallacies are residual. Appendix F.4.

The strictness is real (computed). In the practice $\{\mathrm{MP}, \mathrm{AC}\}$ on the designated context $A_1 = \{q,\ p \to q,\ p \to \bot\}$ of Example 2.15, the maximal candidates are $\{\mathrm{MP}\}$ and $\{\mathrm{AC}\}$, so the schema-level set is empty; but MP and AC share the valid instances $(A,\ A \to A \rhd A)$, which may be asserted. A Bayesian variant that waits for the audit automatically, but cannot break the symmetry of Theorem 7.19, is sketched in Remark F.1 (Appendix F.4).

Reading. The dilemma is the formal shape of a familiar norm. A practice is presumed valid, but the presumption is applied only after the hearing at which it could have been impeached. A learner that presumes too early asserts the community's fallacies; a learner that never presumes asserts nothing, because refutational evidence cannot certify a rule. "Audit, then assert" is the only stable design, and it is the design of TTL.

7.3 Blame: from bags to culprits

A contradiction from a designated context condemns a set of steps (Lemma 6.2). Which members should a sound learner withhold?

Lemma 7.7 (Hitting-set duality). *[T7 Lemma 2.3] Let* U *be finite, let* $\mathcal{K} \subseteq 2^U$ *be upward closed with* $\emptyset \notin \mathcal{K}$, *and let* $\mathcal{C}$ *be its minimal members; call* $M \subseteq U$ clean *if* $M \notin \mathcal{K}$. (a) *The maximal clean sets are exactly the complements of the minimal transversals (hitting sets) of* $\mathcal{C}$. (b) *An element lies in some minimal transversal iff it lies in some member of* $\mathcal{C}$. *Hence* $\bigcap\{\text{maximal clean sets}\} = U \setminus \bigcup \mathcal{C}$, *and this set is clean.* (c) *The minimal transversal is unique iff every member of* $\mathcal{C}$ *is a singleton.*
Lean: Blame.isMaxClean_iff Lean: Blame.iInter_maxClean_eq Lean: Blame.existsUnique_minTransversal_iff

Proof in Appendix F.3. This is Reiter's theory of diagnosis from first principles (Reiter, 1987; de Kleer and Williams, 1987), and (b) is the standard fact that the vertices of the transversal hypergraph are those of the minimal edges (Berge, 1989). Components are practice schemas, "abnormal" means fallacious, conflicts are the negative bags of Section 6, and diagnoses are hypotheses about which practice rules are fallacies. All three parts are formalized in Lean for a finite universe, and checked on 4000 random families with no failure (computed). With $U = \Sigma^P$ and $\mathcal{K} = \mathrm{Conf}_d$, every maximal candidate keeps $\Sigma^P \setminus \bigcup \mathcal{C}_d$. So $\bigcup \mathcal{C}_d$ is exactly what a learner that presumes validity, and is cautious across the remaining ambiguity *at the level of schemas*, must withhold. At the level of instances it can do better (Proposition 7.6(c)); that the withholding is forced for *every* sound learner is true at stable depth (Theorem 7.19(b)) but can fail at finite depth (Theorem 7.19(c)).

Descent. Bags shrink to culprits when the argument can be evaluated line by line. Fix a derivation π relative to $[A : D]$ and let e_π^+ be the partial map on the judgments of π obtained from $e_{[A:D]}$ by closing under two rules that use only the *trusted steps of* π: *forward*, if all premises of a trusted step of π have value 1 then its conclusion gets 1; *backward*, if a trusted step of π has conclusion value 0 and all but one premise at 1, then the remaining premise gets 0. A worklist

computes this fixed point with $O(|\pi|\varphi)$ evaluations of e, where φ is the maximal fan-in. (Closing under *all* trusted steps globally is not computable in general: in arithmetic, forward closure from the true Δ_0 sentences contains, among the Σ_1 sentences, exactly the true ones. It is also not needed.) To *descend* on a refutation, start at its conclusion; at a node of value 0 produced by a step s, output s if all premises of s have value 1, and otherwise move to a premise of value 0. The descent is *unblocked* if at every visited node either all premises have value 1 or some premise has value 0.

Lemma 7.8 (Descent with trusted steps)**.** *[T7 Lemma 2.5, as repaired] Assume (WS) for $[A : D]$, with total valuation V. Then e^+_π agrees with V wherever it is defined; in particular, a judgment receiving both values certifies that $[A : D]$ violates (WS). Let π be a d-refutation of B relative to $[A : D]$ whose descent is unblocked. Then:* (a) *the descent outputs an untrusted step $s \in R_B$ with V-true premises and a V-false conclusion, so $s \notin R^*$;* (b) *computing e^+_π and descending costs $O(|\pi|\varphi) \le O(d\varphi)$ evaluations of e;* (c) *every $\sigma \in B$ with $s \in \mathrm{inst}(\sigma)$ is a fallacy, so removing all of them removes no genuine rule (*certified singleton blame*);* (d) *for each such σ, $\{\sigma\}$ is a d-conflict: s together with the trusted steps of π that justify the values of its premises and conclusion forms a d-refutation of $\{\sigma\}$.*

Proof idea. Both propagation rules are V-sound because trusted steps preserve V-truth. The descent follows nodes of value 0 strictly downward and leaves have value 1, so it stops at a step with V-true premises and a V-false conclusion, which by (WS) is neither trusted nor in R^*. Part (d) extracts the refutation from the judgments of π; this is where size must count distinct judgments (Definition 7.2(a)). Appendix F.5.

This is Shapiro's *contradiction backtracing* from algorithmic debugging and model inference (Shapiro, 1981, 1983), the counterexample-to-blame step of Lemma 10.17, and Lakatos's conversion of a global counterexample into a local one (Lakatos, 1976). Two features are specific to this setting. Trusted steps carry values *backward* through unevaluable nodes, which arithmetic needs because $\forall x\, \theta$ is not Δ_0 (Theorem 7.27(a)). And by (c) and (d), blame lands on *schemas*, with no collateral, and every blamed schema is a singleton conflict. When the descent is blocked, all we have is the bag $\{\sigma \in B : \sigma \text{ used in } \pi\}$. Only the related Lemma 10.17 is formalized, for a total evaluation of a line-list argument (`Blame.LineArg.descent`, `Blame.LineArg.descent_not_mem`); trusted-step propagation and part (d) are not.

Reading: localization. In classical propositional logic and in arithmetic, world feedback refutes nothing that coherence does not already refute: by Post-completeness an invalid schema is incoherent in the empty context (Theorem 6.13), and a consistent extension of Q is Δ_0-sound (Lemma 6.18). Its whole added value is that descent turns its refutations into *singletons*. World feedback earns its keep by localization, not by refutation. Example 2.15 shows this in miniature: the context A_1 yields the bag $\{\mathrm{AC}, \mathrm{MP}\}$, and the valuation $p = 0, q = 1$ blames AC alone.

7.4 The learner $\mathrm{TTL}(d, \delta)$

Definition 7.9 (The two-tier learner)**.** [T7 §3.1, as repaired] *Inputs:* the floor constants $(\Delta, c, K, \bar{\alpha})$; an oracle Ref_d (for $\mathcal{A}, W, T_{\mathrm{trust}}$) with either tie-breaking rule; a sandbox class $\mathcal{H}_S$ of step sets with prior w. The *burn-in* is

$$N_1 = \left\lceil \frac{1}{2\Delta^2} \ln \frac{(1+c)K}{\delta} \right\rceil, \qquad \text{or, in } \textit{noise-free mode}, \qquad N_1 = \left\lceil \frac{1}{\lambda} \ln \frac{cK}{\delta} \right\rceil,$$

with a known $\lambda \le \min_i \pi_i \rho_i$ in noise-free mode.

0. *Phase 0* ($t < N_1$). The assertion tier *abstains*. A query may be escalated to a human, or, if all its judgments lie in J_W, answered by a separate *world channel* that accepts it iff it is not W-falsified; world-channel answers are not acceptances of the tier. Human steps are recorded; human answers to escalated queries are not added to the sample.

1. *Practice identification* (at $t = N_1$). For each tag i with $n_i > e_i$ samples, where $e_i = \lfloor(\bar{\alpha}_i + \Delta)N_1\rfloor$ ($e_i = 0$ in noise-free mode), let $\hat{\sigma}_i = \mathrm{mgu}\{\mathrm{lgg}(P_i \setminus E) : |E| = e_i\}$, the most general common instance of the trimmed lggs of the tag-i data P_i; its instance set is the accepted set of the tag's trimmed version space (Theorem 5.13). Tags with $n_i \le e_i$ contribute nothing. Let $\hat{P} = \{\hat{\sigma}_i\}$.

2. *Audit.* Set $B := \hat{P}$ and repeat: call $r = \mathrm{Ref}_d(B)$; if $r = \mathrm{NONE}$, return B; descend on r; if the descent outputs s, set $B := B \setminus \{\sigma \in B : s \in \mathrm{inst}(\sigma)\}$ and continue; if it is blocked, return $B \setminus \bigcup \mathcal{C}_d(B)$ (the *fallback*).

3. *Freeze* the returned set A. In *Phase 1* ($t \ge N_1$) the tier accepts q iff $q \in R_A$, decided by matching; other queries may be escalated.

4. *Sandbox* (all times). Oligarchic halving (Definition 6.4) over $\mathcal{H}_S$, for example over candidate generalizations of asserted schemas or candidate new rules: announce $B_t = \bigcap S_t$ with $w(S_t) \ge \frac{1}{2}w(\mathrm{VS}_t)$. A red team, which may be the prover or the learner's own search, exhibits refutations π of B_t, and each deletes every $h \supseteq \mathrm{Steps}(\pi) \setminus T_{\mathrm{trust}}$. *Isolation:* nothing from the sandbox enters R_A as a primitive step. A conjecture κ is promoted only when a schematic R_A-derivation of it is found, and then as a *macro*: a κ-instance is answered by expanding it into that derivation.

5. *Truth maintenance.* If the audit is re-run at a larger depth, every cached lemma or macro that uses a removed schema is invalidated.

Why noise-free mode needs the budget $e_i = 0$, and why a tag with $n_i \le e_i$ must contribute nothing rather than "accept all", is shown in Appendix F.9.

Lemma 7.10 (Audit correctness)**.** *[T7 Lemma 3.1] Assume (WS) and $\hat{P} = \Sigma^P$.* (a) *If no descent is blocked, the audit returns a d-clean A with $\Sigma^* \subseteq A \subseteq \Sigma^* \cup F_{\mathrm{res}}^{(d)}$, using at most $|F|+1$ calls to Ref_d; every successful call removes at least one fallacy and no genuine rule.* (b) *If the fallback triggers at stage B_0, then $\Sigma^* \subseteq B_0$, and $A = B_0 \setminus \bigcup \mathcal{C}_d(B_0)$ is d-clean with $\Sigma^* \setminus \mathrm{Coll}_d(B_0) \subseteq A \subseteq \Sigma^* \cup F_{\mathrm{res}}^{(d)}$. If $B_0 = \Sigma^P$, then $A = (\Sigma^* \setminus \mathrm{Coll}_d) \cup F_d^{\mathrm{surv}}$.*

The proof, and the cost of the fallback, which needs $\bigcup \mathcal{C}_d(B_0)$, are in Appendix F.6.

Mis-designated positions. Theorem 7.12 below assumes (WS) for *every* designated position. When at most m positions may violate (WS), a per-position voting audit keeps Σ^* and removes every fallacy that conflicts with Σ^* at $m+1$ complete positions (Proposition F.2, Appendix F.7). The soft version of this is Theorem 6.7.

Proposition 7.11 (Sandbox isolation and budget)**.** *[T7 Prop 3.3, as repaired; T2 Thms 2.2, 2.5]* (a) *The assertion tier is a function of the first N_1 human steps and of the audit's oracle answers alone. No sandbox state, red-team action or prover action affects it, and promotion by derivation leaves Cl_{R_A} unchanged.* (b) *If $\mathcal{H}_S$ contains some h_0 that is clean at every depth with $w(h_0) > 0$ (for instance $h_0 = R^*$, by Lemma 7.4(a)), then for every red team the number of detections*

against the coalition is at most $\log_2(1/w(h_0))$*, and with* m *false alarms and penalty factor* β *at most* $\left(\ln \frac{1}{w(h_0)} + m \ln \frac{1}{\beta}\right) / \ln \frac{2}{1+\beta}$*. Without such an* h_0 *the bound is vacuous.* (c) *Coalition unanimity is decidable for finite* S_t*:* $s \in \bigcap_{h \in S_t} h$ *iff* s *matches a schema of every* h*, and for schema-valued hypotheses the intersection of instance sets is the instance set of the most general unifier (Robinson, 1965; Plotkin, 1970; Reynolds, 1970).*

Proof idea. (a) holds by construction. For (b), the proofs of Theorems 6.5 and 6.7 use only that the protected hypothesis is never deleted, and a detection by π deletes only hypotheses containing the untrusted steps of π, of which π is then a refutation. Appendix F.8, which also shows what fails if promoted rules were primitive: a rule packaging the first half of a residual fallacy's refutation of size $2d$ yields one of size at most d.

Reading. How does cautious acceptance interact with the bold coalition? *It does not.* The coalition is bold, and its errors are bounded by halving. The assertion tier is cautious, and its soundness is certified by the audit. The only channel from the sandbox to the tier is derivation, which cannot enlarge the closure (Lemma 3.1). Re-deriving sandbox lemmas inside R_A is what makes the separation safe; truth maintenance is the same point made across time. This resolves the tension left by Sections 4 and 6: caution receives no signal from coherence (Proposition 6.3), and boldness is unsound; so both tiers exist, and only the cautious one asserts.

7.5 The main theorem

Theorem 7.12 (Two-tier coherent inferential learning)**.** *[T7 Thm 4.1, as repaired] Assume (Floor) (Assumption 7.1);* realizability*, i.e. the practice consists of tagged single schemas as in Definition 2.10, with fallacy tags latent but distinct; (WS) for every designated position; and* Ref_d *exact for refutations of size at most* d*, with either tie-breaking rule. Run* $\mathrm{TTL}(d, \delta)$*. There is an event* G *with* $\mathbb{P}(G) \geq 1 - \delta$*, depending only on the first* N_1 *human steps, on which the following hold for every adaptive prover and every red team.*

(i) Uniform soundness at all times. *For* $t < N_1$ *the tier accepts nothing. For* $t \geq N_1$ *every accepted step lies in* $R^* \cup R_{F_{\mathrm{res}}^{(d)}}$*, so for every premise set* B

$$\mathrm{Cl}_{R_A}(B) \subseteq \mathrm{Cl}_{R^* \cup R_{F_{\mathrm{res}}^{(d)}}}(B), \qquad \mathrm{Cl}_{R_A \cup T_{\mathrm{trust}}}(B) \subseteq \mathrm{Cl}_{R^* \cup T_{\mathrm{trust}} \cup R_{F_{\mathrm{res}}^{(d)}}}(B).$$

If $F_{\mathrm{res}}^{(d)} = \emptyset$*, the reasoner is target-sound against arbitrary search. The schema set* A *is* d*-clean: no prover can ever exhibit a refutation of* R_A *of size at most* d*.*

(ii) Refutation budget. *On its descent path the audit uses at most* $|F| + 1$ *oracle calls and* $O((|F| + 1)d\varphi)$ *evaluations, and removes no genuine rule. If* $\mathcal{H}_S$ *contains an* h_0 *clean at every depth with* $w(h_0) > 0$*, the sandbox suffers at most* $\log_2(1/w(h_0))$ *detections (with* m *false alarms, the bound of Proposition 7.11(b)).*

(iii) Convergence modulo the residue. *For* $t \geq N_1$ *the tier accepts exactly* R_A*, where* $\Sigma^* \subseteq A \subseteq \Sigma^* \cup F_{\mathrm{res}}^{(d)}$ *if no descent is blocked, and* $\Sigma^* \setminus \mathrm{Coll}_d(B_0) \subseteq A \subseteq \Sigma^* \cup F_{\mathrm{res}}^{(d)}$ *if the fallback triggers at* B_0*, with* $A = (\Sigma^* \setminus \mathrm{Coll}_d) \cup F_d^{\mathrm{surv}}$ *when* $B_0 = \Sigma^P$*. So under* (SB_d)*, or whenever no descent is blocked, the tier is* complete for the target*: it accepts every genuine step, and the only fallacies it accepts are residual.*

(iv) Depth schedule. *Re-run the audit on the frozen $\hat{P}$ at depths $d_1 < d_2 < \cdots$, with truth maintenance. Then A changes only finitely often, and is constant for $d \geq d_1^*$, the largest depth at which some answer $\mathrm{Ref}_d(B)$, $B \subseteq \Sigma^P$, changes. At each stage (i) holds with the current depth, and from d_1^* on the tier is sound relative to $R^* \cup R_{F_{\mathrm{res}}}$. The changes need not be retractions: A is not monotone in d.*

(v) Uniformity. *G is defined by the human data alone, and Ref_d and W are deterministic, so (i)–(iv) hold simultaneously for all provers and red teams, with no union bound over queries.*

Proof idea. Identification: G is the event of Theorem 5.14 at N_1, with α_i replaced by $\bar{\alpha}_i$ and Δ_i by Δ. Hoeffding and a union bound over the $1+c_i$ events per tag give $\mathbb{P}(G^c) \leq (1+c)Ke^{-2N_1\Delta^2} \leq \delta$. On G, Theorem 5.13(b) makes each tag's trimmed verifier accept exactly $\mathrm{inst}(\sigma_i)$, so $\hat{P} = \Sigma^P$. In noise-free mode Theorem 5.6 gives $\mathbb{P}(G^c) \leq \sum_i c_i e^{-N_1\pi_i\rho_i} \leq cKe^{-N_1\lambda} \leq \delta$. *Audit:* Lemma 7.10. *Soundness:* $R_A \subseteq R^* \cup R_{F_{\mathrm{res}}^{(d)}}$, and Lemma 3.1 with that calculus as target. *Sandbox:* Proposition 7.11. *Depth:* each $\mathrm{Ref}_d(B)$ changes at most twice as d grows, and there are $2^{|\Sigma^P|}$ sets B. Full proof, with an MP/AC example of the non-monotonicity in (iv), in Appendix F.9. Given Theorems 5.14 and 6.5, the proof is about a page: the theorem is an assembly, and its interest lies in the hypotheses it needs (Section 7.6), not in its proof.

Corollary 7.13 (Truth-soundness)**.** *[T7 Cor 4.2, as repaired] Suppose R^* is $\mathfrak{M}$-sound, every member of $F_{\mathrm{res}}^{(d)}$ is $\mathfrak{M}$-valid (vacuous if $F_{\mathrm{res}}^{(d)} = \emptyset$), and trusted steps, if used, are $\mathfrak{M}$-sound. Then on G the reasoner derives only $\mathfrak{M}$-consequences, at all times and against all provers. This also holds for a reasoner that chains world-channel steps with R_A-steps, provided W is truth in $\mathfrak{M}$ on J_W.*

World-channel steps are truth-sound but need not be target-sound: for $\Sigma^* = \{\wedge\mathrm{I}, \wedge\mathrm{E}_1, \wedge\mathrm{E}_2\}$ the step $\top \rhd \top \vee \bot$ preserves truth but is not R^*-derivable. That is why the world channel is kept outside the tier, and why Theorem 7.12(i) concerns the tier only.

Remarks. (1) *What is and is not certified.* Genuine rules are never certified valid by anything in the architecture: their assertion rests on positive data (anchoring with margin) plus presumption after audit. Fallacies are certified invalid by descent, or implicated as a set by conflicts. Lemma 7.4 forces this asymmetry. (2) *Realizability is load-bearing.* If the human calculus has a rule outside the class, identification fails, and nothing downstream repairs it; at best the audit removes an over-general identified schema, which restores soundness but not completeness. Quantifier rules and the induction schema are such rules (Corollary 7.26 and Theorem 7.27). If fallacies are cited under genuine rules' names, the per-tag class must be unions of schemas (Theorem 5.8). (3) *Why freeze.* Freezing $\hat{P}$ keeps the accounting to one event. Learning can continue at the cost of a union bound over $N \geq N_1$ (an additive $\approx \ln \frac{1}{2\Delta^2}$ inside the logarithm) and a re-audit whenever $\hat{P}$ changes. (4) *Before N_1.* Abstention can be softened by the world channel or by a trusted base. Both are truth-sound (Corollary 7.13) but outside (i) unless they lie in $\mathrm{Sound}(R^*)$, and Theorem 7.18 shows that no learner of the relevant kind can do much better on the schemas that might be fallacies.

Reading. For the motivating question, the theorem says what "a learner of inference rules, conditioned on coherence and the world" can deliver. It identifies the target calculus exactly when blame is localized (no descent blocked, or (SB_d)), and otherwise up to the collateral $\mathrm{Coll}_d(B_0)$, which every sound learner must give up at stable depth (Theorem 7.19(b)); in both cases modulo

a residue that no refutational channel at depth d can see. It is sound at every moment against any amount of search. Its costs are explicit: N_1 human steps, at most $|F|+1$ oracle calls on the descent path, $\log_2(1/w(h_0))$ detections in the sandbox. And it never asserts a rule because the rule is coherent, only because the community uses it robustly *and* it survived the audit. Coherence is a filter on imitation, not a source of rules.

7.6 Every ingredient is necessary

We now show that each ingredient of TTL is forced. Here a *learner* is any randomized procedure that receives the same kinds of information as TTL, possibly with one channel removed: human samples, the queries of a prover and red team, answers of Ref_d and W, its own coins and, optionally, escalation answers. Lower bounds use a *fixed* prover and red team, i.e. strategies mapping the history and their own coins to queries, the same in every scenario. Escalation answers, if used, *follow the practice*: they are drawn from a law depending only on the query and on the practice (for instance $\mathbf{1}[q \in R^P]$), so they have the same law in scenarios with the same practice; for Theorem 7.20 the law is moreover computable uniformly in the practice and the query. That covers Propositions 7.15 and 7.16 and Theorems 7.19 and 7.20; Theorem 7.18 compares different practices and needs an explicit assumption instead. A target-truthful labeller ($\mathbf{1}[q \in R^*]$, the labelled setting of Section 4) is a different and stronger channel, and defeats Theorems 7.18 and 7.19.

Fix the practice with its frequencies, instance laws and noise law, and fix $\mathcal{A}, W, T_{\mathrm{trust}}$. A target $\Sigma \subseteq \Sigma^P$ is *admissible* if it satisfies (WS), equivalently is clean at every depth (Lemma 7.4(d)), and each member of $\Sigma^P \setminus \Sigma$ has an instance outside $\mathrm{Sound}(R_\Sigma)$. All admissible targets share the data law and the oracle answers (Lemma 7.4(c)) and satisfy (Floor) with the same constants. $F_{\mathrm{res}}^{(d)}(\Sigma)$ denotes the residue when Σ is the target.

Proposition 7.14 (Positive data are needed; trivial)**.** *[T7 Prop 5.1, as repaired] Let $\Sigma_1 \subsetneq \Sigma_2$ be calculi such that every subset of Σ_2 is clean at every depth and $R_{\Sigma_2} \not\subseteq \mathrm{Sound}(R_{\Sigma_1})$. A learner that receives only refutation evidence (oracle answers on schema sets of its choice, but no human steps and no escalations) has the same view under the targets Σ_1 and Σ_2. So it is unsound for Σ_1 or incomplete for Σ_2 on the nonempty set $R_{\Sigma_2} \setminus \mathrm{Sound}(R_{\Sigma_1})$. Example: $\Sigma_1 = \{\wedge\mathrm{I}, \wedge\mathrm{E}_1, \wedge\mathrm{E}_2\}$, $\Sigma_2 = \Sigma_1 \cup \{\vee\mathrm{I}_1\}$ with classical evaluation. The non-derivability hypothesis is needed: for $\Sigma_2 = \Sigma_1 \cup \{A \wedge B \rhd B \wedge A\}$, accepting R_{Σ_2} is both sound for Σ_1 and complete for Σ_2.*

The proof is Lemma 7.4(c). This concerns soundness relative to the *human* calculus. In CPC, structurality and closed evaluation decide schema validity (Lemma 7.22), so a learner wanting only truth-soundness could enumerate schemas and assert the valid ones without human data; positive data then only say *which* valid rules the practice uses. In arithmetic, positive data are needed even for truth-completeness, because no refutational evidence ever certifies a true Π_1 axiom.

Proposition 7.15 (Caution is needed)**.** *[T7 Prop 5.2; (a) as repaired]*

(a) Let $\mathcal{T}$ be the admissible targets. For any learner, a fixed prover, a finite history h and a step q, the joint probability $p(h,q)$ that the run produces h and then accepts q is the same under every $\Sigma \in \mathcal{T}$. Hence if $q \notin \bigcap_{\Sigma\in\mathcal{T}} \mathrm{Sound}(R_\Sigma \cup R_{F_{\mathrm{res}}^{(d)}(\Sigma)})$, the learner is unsound with probability at least $p(h,q)$ under some admissible target, and a learner that is δ-sound under every admissible target has $p(h,q) \le \delta$. TTL passes: on G, $R_A \subseteq R_\Sigma \cup R_{F_{\mathrm{res}}^{(d)}(\Sigma)}$ for every *$\Sigma \in \mathcal{T}$.*

(b) *The following accept a falsified instance of some* $\tau \in F \setminus F_{\mathrm{res}}^{(d)}$ *before the refutation implicating* τ *is found: presumption without audit (Proposition 7.6(b)); the bold coalition used as the assertion tier, under priors for which every coalition member contains such an instance, for instance a prior concentrated on* R^P *(Theorem 6.5(iii) gives soundness only while* $R^* \in S_t$*, while a prior with more than half its mass on* R^* *gives a sound coalition); and any MAP or MDL learner over practice (Proposition 3.9). A prover that knows* F *queries such an instance at once, and in* CPC *with a complete* Σ^* *and a pure* τ *that one step trivializes the reasoner (Proposition 7.5).*

Part (a) is the joint-probability form of Theorem 4.2(b), relativized to the residue. Both features are needed: the conditional probability given h fails for randomized learners, and plain Sound(R_Σ) would make the intersection trivial and convict TTL itself, since typically $\emptyset \in \mathcal{T}$. Proofs and the counterexample are in Appendix F.10.

Proposition 7.16 (Negative evidence is needed)**.** *[T7 Prop 5.3] Let the practice be* $\{\wedge\mathrm{I}, \mathrm{AC}\}$ *with fixed frequencies. In world* W_1 *the connectives have their classical tables, AC is a fallacy and the target is* $\{\wedge\mathrm{I}\}$*. In* W_2 *the tables are classical except that* $\to$ *is read as* $\leftrightarrow$*; then AC is valid (from* B *and* $A \leftrightarrow B$ *infer* A*) and the target is* $\{\wedge\mathrm{I}, \mathrm{AC}\}$*. A learner without refutation evidence sees identically distributed data in both, so it is unsound under* W_1 *or incomplete under* W_2*. The closed instance* $(\top,\ \bot \to \top \rhd \bot)$ *separates them: under* W_1 *its premises are true and its conclusion false, while under* W_2 *the premise* $\bot \leftrightarrow \top$ *is false.*

This is Corollary 5.19 in two-world form.

Proposition 7.17 (Structurality is needed; trivial)**.** *[T7 Prop 5.4]* (a) *If hypotheses are instance sets rather than schemas, each refutation removes finitely many instances, so a fallacy with infinitely many invalid instances keeps some invalid instance asserted at every finite time by any learner that asserts the practice minus the refuted instances.* (b) *Without uniformity coherence is toothless (Proposition 6.33:* $\mathrm{Cn}_2 + \{\rhd p_{17}\}$*).* (c) *Without structure positive data cannot generalize: for classes of finite step sets the cautious verifier accepts exactly the data (Section 5.4).*

Theorem 7.18 (Burn-in is necessary: the rare refuter)**.** *[T7 Thm 5.5, as repaired in two rounds] Let* $0 < \pi < 1$*, and let* σ, τ, μ *be schemas such that* $\{\sigma, \tau\}$ *and* $\{\sigma, \mu\}$ *are clean at every depth and* $\{\mu, \tau\}$ *is a d-conflict. Compare, with the same* $\mathcal{A}, W, T_{\mathrm{trust}}$ *and a fixed prover and red team:* scenario 1*, practice* $\{\sigma, \tau, \mu\}$*, target* $\{\sigma, \mu\}$*,* $F = \{\tau\}$*, tag frequencies* $(1-\pi)q_\sigma$*,* $(1-\pi)q_\tau$*,* π*; and* scenario 2*, practice* $\{\sigma, \tau\}$*, target* $\{\sigma, \tau\}$*,* $F = \emptyset$*, frequencies* q_σ, q_τ*, where* $q_\sigma + q_\tau = 1$ *and the per-tag laws of* σ *and* τ *are the same in both. Assume*

> ***(OE)*** *at each round, every input to the learner other than the human sample (prover and red-team queries, answers of* Ref_d *and* W*, escalation answers if any, the learner's coins) is generated from the history by the same kernel in both scenarios.*

Then some τ*-instance lies outside* Sound($R_{\{\sigma,\mu\}}$)*, and for every learner satisfying (OE), every* t *and every such instance* s*: if in scenario 1 the learner accepts* s *at time* t *with probability at most* δ*, then in scenario 2 it does so with probability at most* $\delta(1-\pi)^{-t}$*. Hence, if* $\delta + \delta' < 1$*, accepting* s *at time* t *in scenario 2 with probability at least* $1 - \delta'$ *requires*

$$t \;\geq\; \frac{\ln \frac{1-\delta'}{\delta}}{\ln \frac{1}{1-\pi}} \;\geq\; \frac{1-\pi}{\pi} \ln \frac{1-\delta'}{\delta}.$$

The first bound is attained.

Proof idea. On samples whose tag is not μ, the scenario-1 law is $(1-\pi)$ times the scenario-2 law, so on the event E_t that no μ-tag appears among the first t samples, $P_1^{\otimes t} = (1-\pi)^t P_2^{\otimes t}$. By (OE) the learner's whole view has law $P_j^{\otimes t}(dx)K_t(x,\cdot)$ with one kernel K_t for both scenarios; integrate the acceptance probability over E_t. The learner that accepts τ-instances at time t iff no μ-tag has occurred has scenario-1 error exactly $(1-\pi)^t$ and scenario-2 completeness 1. Appendix F.11.

(OE) holds for Ref_d and W (Lemma 7.4(c)), for the fixed prover and red team, for learners that do not escalate, and for TTL's tier (Proposition 7.11(a); escalation answers are not added to the sample). It can fail for practice-following escalation, for instance with answers $\mathbf{1}[q \in R^P]$, and then the conclusion is false. Appendix F.11 gives this counterexample, a computed check of tightness, and a concrete instance with $\sigma = \wedge\mathrm{I}$, $\tau = \mathrm{AC}$ and $\mu = \mathrm{MP}$ on the context A_1.

Reading. Suppose modus ponens is rare, with frequency π. No learner that is sound whether or not affirming the consequent is a fallacy, and that learns about the practice only from the stream, can accept AC in the world where it is genuine before about $\pi^{-1}\ln(1/\delta)$ samples: roughly the time by which MP, the rule that would expose AC if it were a fallacy, has been seen with confidence. TTL's noise-free burn-in $\lambda^{-1}\ln(cK/\delta)$, with $\lambda \le \pi\rho_{\mathrm{MP}}$, matches this up to the factor $\ln K$ and the variability ρ_{MP}. This is the standard two-point rare-event bound behind PAC sample-complexity lower bounds; the only new point is that the relevant rare event is the absence of the *refuter*. Waiting is not timidity: a rule can be presumed valid only once the rules that could impeach it have had their chance.

Theorem 7.19 (Blame: completeness needs world or bilateral evidence)**.** *[T7 Thm 5.6, as repaired]*

(a) Two-point core. *Let $\Sigma_0 \cup \{\mu\}$ and $\Sigma_0 \cup \{\tau\}$ be clean at every depth and $\{\mu,\tau\}$ a d-conflict. Compare scenario 1 (target $\Sigma_0 \cup \{\mu\}$, $F = \{\tau\}$) and scenario 2 (target $\Sigma_0 \cup \{\tau\}$, $F = \{\mu\}$), with equal tag frequencies and a fixed prover. The data and all oracle answers have identical laws, and some μ-instance s, genuine in scenario 1, is accepted at any given time in scenario 1 with probability at most δ by every learner that is δ-sound in both scenarios. So μ, which lies in Coll_d in scenario 1, is collateral for* every *sound learner.*

(b) Stable depth: the fallback is optimal among schema-level policies. *Let $d \ge d_0$ and suppose that on G the audit falls back at B_0. Then for every $g \in \bigcup \mathcal{C}_d(B_0)$ there are an admissible target $M \subseteq B_0$ with $g \notin M$ and $F_{\mathrm{res}}^{(d)}(M) = \emptyset$, and an instance $s_g \in \mathrm{inst}(g) \setminus \mathrm{Sound}(R_M)$. Consequently every learner that is δ-sound under every admissible target accepts s_g at any fixed time with probability at most δ, under every admissible target, the true one included; every genuine $g \in \mathrm{Coll}_d(B_0)$ is withheld on some instance by every such learner; and a policy that asserts $R_{A'}$ for a possibly random $A' \subseteq \Sigma^P$, and is δ-sound under every admissible target, includes each schema outside TTL's A with probability at most δ. The same holds on the descent path, with $M := A$.*

(c) At finite depth the fallback can be strictly suboptimal *(computed). For the practice* $\{\mathrm{MP}, \mathrm{AC}\}$ *on two designated contexts there are depths at which TTL falls back to $A = \emptyset$, although asserting R_{MP} is sound under every admissible target (Appendix F.12).*

Proof idea. (a) A μ-instance outside $\mathrm{Sound}(R_{\Sigma_0 \cup \{\tau\}})$ exists, or the refutation of $\{\mu,\tau\}$ would transfer to the clean set $\Sigma_0 \cup \{\tau\}$ (Lemma 3.1); in scenario 2 it is unsound and non-residual, and the scenarios are indistinguishable. (b) Pick, by Lemma 7.7(b), a maximal clean $M \subseteq B_0$ avoiding g; at stable depth it is clean at every depth, hence admissible. Appendix F.12.

The witness: $\to$ versus $\leftarrow$. Take $\mathcal{A} = \{A_1\}$, no world, $\Sigma_0 = \emptyset$, μ = MP and τ = AC. {MP} is classically sound and A_1 satisfiable; {AC} is clean, since the closure of A_1 under AC alone is $\{q,\ p \to q,\ p \to \bot,\ p\}$; and AC followed by MP derives $\bot$. Scenario 2 is literally the hypothesis that "$\to$" denotes converse implication. Under that reading AC *is* modus ponens for $\leftarrow$, MP is affirming the consequent for $\leftarrow$, and A_1 reads $\{q,\ q \supset p,\ \bot \supset p\}$, which is consistent. *Coherence alone cannot tell which way the arrow points.* Either of two things breaks the symmetry: world feedback, since the closed instance $(\bot \to \bot,\ \bot \to (\bot \to \bot) \rhd \bot)$ of AC is falsified; or a single denial, since in the designated position $[A_1 : p]$ AC alone derives the denied p, and the conflict becomes {AC}. Both make scenario 2 violate (WS) (computed). In the Hilbert practice $\{K, S, \mathrm{MP}, \mathrm{DN}, \mathrm{AC}\}$ the same mechanism makes K and MP necessarily collateral at full depth; whether S or DN is collateral too is open (Appendix F.14).

Reading. This is the blame problem in its sharpest form. A community whose practice contains both modus ponens and affirming the consequent has an incoherent practice, and coherence correctly detects it. But coherence does not say which half of the practice is the meaning of "if". A learner that is sound whatever the answer must give up both as rules, keeping at most their shared instances (Proposition 7.6(c)). What settles the direction is the world, or a denial: the bilateralists' rejection (Smiley, 1996; Rumfitt, 2000; Restall, 2005) does syntactically what a counterexample object does semantically.

Theorem 7.20 (Depth relativization is necessary)**.** *[T7 Thm 5.7, as repaired] There is a uniformly computable family of practices $(\Sigma_e^P)_{e\in\mathbb{N}}$, with $\mathcal{A} = \{[\emptyset : \emptyset]\}$, no world, and frequencies and instance laws computable uniformly in e, such that for the fixed prover that queries the step $\rhd\mathrm{start}_e$ in every round no computable learner achieves both of the following for all e, for any $\delta < \frac{1}{2}$:* (a) *δ-soundness at all times relative to the full-depth residue, $\mathbb{P}[\exists t :$ a step outside $R^* \cup R_{F_{\mathrm{res}}}$ is accepted at $t] \le \delta$; and* (b) *eventual completeness, $\mathbb{P}[\exists t\ \forall t' \ge t :\ R^* \subseteq$ accepted at $t'] \ge 1 - \delta$.*

Proof idea. A finite elementary formal system Σ_0 simulates the universal machine on input e from start_e and contains halt $\rhd \bot$ (Theorem 6.30); add the axiom $\tau_e = (\rhd\mathrm{start}_e)$. If $\varphi_e(e)$ diverges the target is $\Sigma_0 \cup \{\tau_e\}$; if it converges, τ_e is a non-residual fallacy. A learner meeting (a) and (b) would make $\{e : \varphi_e(e)$ diverges$\}$ a Σ_1 set. Appendix F.13.

This is the Π_1-completeness of coherence turned into a statement about *anytime* soundness. TTL's form of soundness (relative to $F_{\mathrm{res}}^{(d_t)}$ at time t, and to F_{res} only in the limit, with finitely many changes) cannot be improved by any computable procedure.

Proposition 7.21 (Truthful designation is needed)**.** *[T7 Prop 5.8; T2 Prop 7.1, Thm 2.5] If a designated position violates (WS), descent can blame a genuine rule: with target* {MP} *and the bad position $[a,\ a \to b : b]$, MP is blamed. Under structurality, one classically inconsistent designated context forces the loss of a classical schema everywhere (Proposition 11.9). Proposition F.2 bounds the damage when at most m positions, not known which, violate (WS), and Theorem 6.7 is the soft version. Nothing bounds the damage of an unknown number of mis-designations.*

7.7 Specializations: what provably works for formal mathematics

Theorem 7.12 is stated relative to a residue and to realizability. This subsection evaluates both for three targets. For classical propositional logic the residue is empty, closed-instance evaluation gives singleton blame, and TTL is exact (Corollary 7.23); coherence alone is sound

ingredient dropped	failure mode	witness
positive data	refutational evidence is target-blind; nested sound calculi are indistinguishable	Proposition 7.14
caution (audit, then assert)	an adaptive prover exploits an unaudited fallacy; one step trivializes CPC	Proposition 7.15
negative evidence	fallacy and rule are indistinguishable ($\to$ versus $\leftrightarrow$)	Proposition 7.16
structurality	instance-level refutation never converges; coherence is toothless	Proposition 7.17
burn-in	rare refuter: $t \gtrsim \pi^{-1} \ln(1/\delta)$	Theorem 7.18
world or bilateral blame	MP/AC ($\to$ versus $\leftarrow$) symmetry; collateral for every sound learner	Theorem 7.19
depth relativization	Π_1 barrier for anytime soundness	Theorem 7.20
truthful designation	global loss of classical schemas	Proposition 7.21

Table 9: Each ingredient of the two-tier learner is necessary.

but may withhold genuine rules, and one denial per fallacy repairs this (Proposition 7.24). For complete decidable theories the same holds conditionally on a realizability hypothesis that is not established for the usual axiomatizations (Corollary 7.26). Arithmetic is Popperian, with a permanent residue beyond Σ_1 (Theorem 7.27).

Classical propositional logic. The language has $\neg, \wedge, \vee, \to, \top, \bot$. Judgments are formulas, or sequents whose contexts are *lists* with explicit structural rules (set or multiset contexts would need anti-unification modulo associativity and commutativity, for which the bounds of Section 5 are not proved). All schemas are *pure* (Definition 2.4); by Proposition 7.17 this cannot be dropped. Σ^* is any finite set of classically valid pure schemas that is complete for classical consequence, F any finite set of classically invalid pure schemas, the world is J_W = closed formulas with W = truth value ("computation"), and $\mathcal{A}$ consists of truthful positions and contains $[\emptyset : \emptyset]$.

Lemma 7.22 (Closed-instance refutability; Post's substitution)**.** *[T7 Lemma 6.1] A pure schema τ is classically invalid iff some substitution of $\top/\bot$ for its metavariables yields a closed instance whose premises are W-true and whose conclusion is W-false. As a one-step derivation this instance has size at most $|\tau|$, and there are at most $2^{v(\tau)}$ candidates, $v(\tau)$ the number of metavariables.*
Lean: Post.schema_invalid_iff_closedRefutation Lean: Post.card_candidates_le

Proof. ($\Leftarrow$) A falsified instance is invalid. ($\Rightarrow$) As in the proof of Proposition 3.5: if a valuation v falsifies an instance $\tau\theta$, put $c_x = \top$ if $v(\theta x) = 1$ and $c_x = \bot$ otherwise; by induction on τ, $\tau[c_x/x]$ has the values of $\tau\theta$ under v, and closed formulas have the same value under every valuation. □

This is essentially Post (1921). For sequents with a context metavariable Γ, substitute the one-element list $[\top]$ or $[\bot]$ according to $v(\bigwedge \Gamma\theta)$; depending on how lists are encoded, the instance can then exceed $|\tau|$ by a constant. A brute-force check on 30,000 random pure schemas (16,309 invalid) found no mismatch (computed), and the formula case is formalized in Lean.

Corollary 7.23 (TTL on classical propositional logic is exact)**.** *[T7 Cor 6.2, as repaired] Let $d \geq \max_{\tau \in F} |\tau|$, and let Ref_d return the first closed $\top/\bot$ instance, in a fixed enumeration, of a schema of B with W-true premises and a W-false conclusion, and NONE if there is none. On every $B \subseteq \Sigma^P$, the only sets the audit queries, this oracle is exact for refutations of size at most d and deterministic; its refutations are one-step and fully evaluable, so every descent is unblocked;*

and its answers do not depend on $d \geq \max_\tau |\tau|$*. Then on* G*, with* $\mathbb{P}(G) \geq 1-\delta$*:* (a) *the minimal conflicts are exactly the singletons* $\{\tau\}$*,* $\tau \in F$*, so (SB*$_d$*) holds and* $\mathrm{Coll}_d = F_{\mathrm{res}}^{(d)} = \emptyset$*;* (b) *the tier abstains before* N_1 *and accepts exactly* R^* *afterwards, so for every premise set* B *the reasoner derives exactly* $\mathrm{Cl}_{R^*}(B) = \mathrm{Cn}_{\mathsf{CPC}}(B)$*, for derivations of any length and formulas of any size (Corollary 5.10), against any prover;* (c) *the audit makes at most* $|F|+1$ *oracle calls and at most* $|F|$ *successful descents, and at most* $(|F|+1)\sum_{\sigma\in\Sigma^P} 2^{v(\sigma)}$ *closed-instance checks (*$\sum_{\sigma\in\Sigma^P} 2^{v(\sigma)}$ *with caching);* (d) *the tier is truth-sound at all times (Corollary 7.13).*

Proof. If every schema of B is valid, a classical valuation agreeing with the truthful position and with W falsifies no step of R_B, so there is no refutation at any size; otherwise Lemma 7.22 gives a one-step refutation of size at most $|\tau| \leq d$. Hence each $\{\tau\}$ is a conflict, and since every conflict contains a fallacy (Lemma 7.4(b)) it contains a singleton conflict. So (SB$_d$) holds and $F_{\mathrm{res}}^{(d)} = \emptyset$, and Theorem 7.12(iii) gives $A = \Sigma^*$; completeness of Σ^* gives (b). Each descent removes the schemas of which the falsified step is an instance, all fallacies and at least one. □

The counts are upper bounds: one falsified step can be an instance of several fallacies, as $(\top,\ \bot \to \top \rhd \bot)$ is of both AC and $(\top,\ A \to \top \rhd A)$, and one descent removes both. The instance checks include those of the genuine schemas, which certify the final NONE. The oracle need not return a smallest refutation; exactness and determinism are all that the audit uses, and part (iv) of Theorem 7.12 is trivial here.

Reading. This meets the first success criterion in its strongest form. From imitation of human proofs containing systematic fallacies, each used under its own latent tag, and sporadic slips, plus truth-table evaluation of closed formulas, the learner identifies exactly the classical calculus the community uses, with probability at least $1-\delta$, after N_1 human steps, with at most $|F|$ refutations. On that event it is sound against every prover at every moment, and its reasoner derives exactly classical consequence, out of distribution.

Proposition 7.24 (Coherence alone on CPC is lossy; a denial repairs it)**.** *[T7 Prop 6.3; (b) as repaired] Let* $J_W = \emptyset$ *and* $\mathcal{A}$ *be truthful.* (a) *If* $[\emptyset : \emptyset] \in \mathcal{A}$*, every* $\tau \in F$ *lies in some minimal conflict once* $d \geq d_\tau$*, because* $\Sigma^* \cup \{\tau\}$ *derives* $\bot$ *from* $\emptyset$ *(Proposition 3.5). Hence* $F_{\mathrm{res}}^{(d)} = \emptyset$ *for* $d \geq \max_\tau d_\tau$*, and TTL is truth-sound.* (b) *The collateral can be nonempty, and then TTL's fallback is incomplete, necessarily so at stable depth (Theorem 7.19(b)). For a complete Hilbert system* Σ^* *for the full language containing* K *and MP,* $F = \{\mathrm{AC}\}$ *and* $\mathcal{A} = \{[\emptyset : \emptyset]\}$ *or* $\{[\emptyset : \emptyset], [A_1 : \emptyset]\}$*, every descent is blocked, and* K *(with* A_1*, also MP) lies in* Coll_d *for all large* d*: every sound learner using coherence alone must withhold it as a rule.* (c) Bilateral repair. *If for each* $\tau \in F$*,* $\mathcal{A}$ *contains a counterexample position* $[\Pi' : \varphi']$*, where* $\Pi' \rhd \varphi'$ *is a closed falsified instance of* τ*, and* $d \geq \max_\tau |\tau|$*, then (SB*$_d$*) holds and* $A = \Sigma^*$*.*

Proof in Appendix F.14. *Reading.* In CPC closed evaluation refutes exactly the schemas that coherence refutes; by Post, both refute all invalid ones. The whole difference between the channels is localization: computation names the culprit, a contradiction only a list of suspects. A counterexample position is a syntactic object, a partial valuation, so "bilateral coherence with denials" and "world feedback with descent" are the same resource in two notations.

Complete decidable theories. Let T be complete and decidable, such as real closed fields (Tarski, 1951), Presburger arithmetic (Presburger, 1929), ACF$_p$ or dense linear orders without endpoints, with $\mathfrak{M} \models T$. Judgments are sequents $\Gamma \vdash \varphi$ of first-order formulas, with $W(\Gamma \vdash$

$\varphi) = [\mathfrak{M} \models \forall \bar{x}(\bigwedge \Gamma \to \varphi)]$, decidable by quantifier elimination, so $J_W = J$. Σ^* is a finite schema calculus, sound and complete for T-consequence on sequents, including $\to$-introduction and $\forall$-introduction.

Lemma 7.25 (Validity is truth preservation)**.** *[T7 Lemma 6.4]* (a) Sound(R^*) *is exactly the set of W-truth-preserving steps.* (b) *A schema is a fallacy iff it has a W-falsified instance, so each $\tau \in F$ has a smallest falsified instance, of some size $f_\tau \geq d_\tau$.*

Proof idea. If all premises are true they are derivable, since T is complete, so Cl_{R^*} of them is the set of true sequents; if one is false, Σ^* derives its false universal closure, whose negation T proves, and then everything. Appendix F.15.

Corollary 7.26 (TTL on complete decidable theories, given realizability)**.** *[T7 Cor 6.5, as repaired]* *Suppose in addition that Σ^* and F lie in a schema class for which the identification of Theorem 5.14 holds, for instance first-order patterns, possibly with decidable side conditions such as the eigenvariable condition of $\forall I$. Let $d \geq \max_\tau f_\tau$, $[\emptyset : \emptyset] \in \mathcal{A}$, and let Ref_d search the instances of size at most d of the schemas of B for a W-falsified one. Then everything in Corollary 7.23 holds verbatim: (SB_d), $F_{\mathrm{res}}^{(d)} = \emptyset$, $A = \Sigma^*$, truth-soundness at all times on G, and at most $|F| + 1$ oracle calls.*

The realizability hypothesis is not established for the listed theories. The rule $\forall$E, from $\forall x \varphi$ infer $\varphi[t/x]$, is not a first-order pattern, because its conclusion is a substitution result; with t arbitrary it is not a higher-order pattern in the sense of Miller (1991) either. The lgg of three $\forall$E instances is $(\forall x(z_0 + z_1 = z_2) \rhd z_3 + z_4 = z_5)$, with the invalid instance $(\forall x(0+0 = 0) \rhd 0+0 = S0)$, and more data only generalize it further (computed). The usual axiomatizations of RCF and ACF_p contain degree-indexed families, and Presburger arithmetic an induction or congruence schema; none is a single first-order pattern. Dense linear orders without endpoints are finitely axiomatized, so there only the quantifier rules matter. Without realizability, the identified schema for a non-pattern rule is in general an over-general lgg. If it has an instance outside Sound(R^*), as for $\forall$E and induction, the total oracle W falsifies one (Lemma 7.25(b)), and an audit of sufficient depth removes it with singleton blame. So TTL stays sound, eventually so under the depth schedule, but A lacks the rule and exactness fails. Two repairs are open: an encoding with auxiliary decidable judgments $\mathrm{Sub}(\varphi, x, t, \psi)$ ("ψ is $\varphi[t/x]$") evaluated by W, carried out below for induction; or an identification theorem for higher-order-pattern schemas. Why one should learn at all when W already decides validity, and what changes when the oracle evaluates only quantifier-free sentences, is discussed in Appendix F.15. The real-closed-field core of olympiad algebra and geometry would be covered by Corollary 7.26 once its realizability hypothesis is met; trigonometric steps are not, since $(\mathbb{R}, +, \cdot, \sin)$ defines $\mathbb{Z}$.

Arithmetic. The language is that of PA and $\mathfrak{M} = \mathbb{N}$. T_{trust} is first-order logic: the logic is fixed, and non-logical axiom schemas are learned (its propositional fragment can be learned first by Corollary 7.23; the quantifier rules are not first-order patterns and must be trusted, Corollary 7.26). The world is $J_W = \Delta_0$ sentences with $W =$ computation, and $\mathcal{A}$ consists of truthful positions. A practice axiom schema is genuine iff all its instances are true in $\mathbb{N}$, so Σ^* may contain PA together with true extras such as Con(PA), and (WS) holds with $V =$ truth in $\mathbb{N}$. Since R^* consists of axiom steps, the relevant target closure is $\mathrm{Cl}_{R^* \cup T_{\mathrm{trust}}}$ (Theorem 7.12(i)). Ground axioms (those of Q, each $\mathrm{Con}(T_j)$, $\neg\mathrm{Con}(\mathsf{PA})$) are realizable; the induction schema is not, and part (e) encodes it. Parts (a)–(d) and (f) concern the audit and hold for any $\hat{P} = \Sigma^P$.

Theorem 7.27 (Arithmetic is Popperian). *[T7 Thm 6.6, as repaired in two rounds]*

(a) False Π_1 axioms get singleton blame. *Let τ have a false instance $\forall \bar{x}\, \theta(\bar{x})$ with $\theta \in \Delta_0$ and least counterexample $\bar{n}$. The refutation "$\nvdash \forall \bar{x}\theta$ (by τ); $\vdash \theta(\bar{n})$ (by trusted $\forall E$)", with $W(\theta(\bar{n})) = 0$, has an unblocked descent that blames τ alone, by the backward rule through $\forall E$. With the prefer-unblocked oracle, τ is removed and no* PA *axiom is put at risk. For one quantified variable with m free occurrences in θ, $\tau \notin F_{\mathrm{res}}^{(d)}$ once $d \geq 2|\theta| + m(n+1) + O(1)$; for k variables $(k+1)(|\theta| + m(n+1) + O(k))$ suffices, n the largest component of $\bar{n}$.*

(b) Under presumption of validity, false Σ_1 axioms survive. *For the practice $\Sigma^P = \mathsf{PA} \cup \{\tau\}$ with $\tau = (\triangleright \neg \mathrm{Con}(\mathsf{PA}))$ (so $\Sigma^* = \mathsf{PA}$, $F = \{\tau\}$) and positions with Δ_0 assertions and denials, $\Sigma^* \cup \{\tau\}$ is clean at every depth, so $\tau \in F_{\mathrm{res}}$ is asserted forever.*

(c) Popperian Σ_1-caution removes syntactically Σ_1 residue. *Asserting an instance of a practice axiom with Σ_1 normal form only once W has verified a witness never asserts a false such axiom, $\neg\mathrm{Con}(\mathsf{PA})$ included. The policy is syntactic: an equivalent sentence with a Π_2 normal form escapes it and is residual.*

(d) Beyond Σ_1 there is a permanent residue. *False Π_2 or Σ_2 axioms τ with $\Sigma^* \cup \{\tau\}$ clean at every depth (for example, $\Sigma^* \supseteq \mathsf{Q}$ and $\Sigma^* \cup \{\tau\} \cup A \cup \neg D$ consistent for every designated $[A : D]$) admit no finite refutation and survive, and no computable policy sorts the true from the false ones in the limit (Theorem 6.21(e)). The same holds for randomized learners that are correct in the limit with probability above $\frac{1}{2}$ (proof sketch).*

(e) Finite reflection-extended practices are learned under the floor, given a realizable encoding of induction. *Under (Floor) a practice has at most $\min(K, \lfloor 1/(2\Delta) \rfloor)$ axiom schemas, and, with induction written as a first-order pattern through decidable* Sub *judgments, $T_k = \mathsf{PA} + \mathrm{Con}(\mathsf{PA}) + \cdots + \mathrm{Con}(T_{k-1})$ is identified from positive data for every k within that bound. Without a floor there is no uniform sample bound, by a tight two-point bound that mirrors Theorem 7.18.*

(f) Reflection as designation. *If $\mathcal{A}$ asserts* $\mathrm{Con}(\mathsf{PA})$*, then $\neg\mathrm{Con}(\mathsf{PA})$ becomes a singleton conflict.*

Proof idea. (a) is Lemma 7.8. (b) $\mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$ is consistent by Gödel's second theorem (Gödel, 1931) and proves every true Δ_0 sentence and no false one (Lemma 6.18); so the audit's first call returns NONE. (d) reduces to Shoenfield's limit lemma (Shoenfield, 1959) and Post's hierarchy theorem. (e) is the mirror image of Theorem 7.18. Appendix F.16 gives the proofs, the precise forms of (c) and (e), and computed checks of (a) and (e).

Reading. In arithmetic TTL achieves exact, localized, Popperian falsification of false Π_1 axioms; exclusion of syntactically Σ_1 errors under Σ_1-caution; identification of finite reflection-extended practices, given a realizable induction schema; and soundness at all times relative to the stated residue. It does not achieve, and no computable procedure can achieve, the elimination of false Σ_2 axioms consistent with the practice, or uniform sample bounds without a frequency floor; and under presumption of validity it keeps false Σ_1 axioms such as $\neg\mathrm{Con}(\mathsf{PA})$. The Gödel–Rosser alternatives are the Kripkensteinian residue of arithmetic (Theorem 6.34(iii)), and TTL leaves it exactly where Section 6 located it, now with explicit sample sizes, refutation budgets and soundness against adaptive provers. Reflection enters as a *designation*, a justification beyond PA-proof that is made explicit rather than smuggled in as a rule.

N	positive-only cautious tier unsound	TTL unsound	TTL exact ($A = \Sigma^*$)	untrimmed lgg + audit exact
60	12/20	0/20	0/20	6/20
120	20/20	0/20	10/20	8/20
250	20/20	0/20	20/20	1/20
500	20/20	0/20	20/20	0/20

Table 10: TTL against an exhaustive adversarial prover on a propositional practice with four systematic fallacies and 1% sporadic noise, 20 trials per N (computed). At $N = 500$ one trial had more than e noisy steps, on the fallacy tag "xor", and was still exact.

7.8 Simulation evidence

The two-tier learner on a propositional practice. A simulation runs TTL in the setting of Corollary 7.23. The practice has nine genuine tagged natural-deduction rules ($\wedge$I, $\wedge$E$_{1,2}$, $\vee$I$_{1,2}$, MP, modus tollens, disjunctive syllogism, double-negation elimination) and four fallacies with their own tags (affirming the consequent, denying the antecedent, conversion, "or" as "xor"). Sporadic noise at rate $\alpha = 0.01$ is mis-tagged at random, and the trim budget is $e = 2$. The audit is the per-schema closed-instance oracle. An adversarial prover enumerates every step over a pool of ten formulas and checks it by truth table. Table 10 shows the result (seed 0; the run is deterministic across hash seeds).

The positive-only cautious tier asserts the fallacies as soon as they are anchored (Proposition 7.5): it is unsound in 72 of 80 runs. TTL is unsound in none of the 80, and exact in all 20 runs at $N = 250$ and at $N = 500$. Excess noise on a *fallacy* tag is harmless, since whatever schema the trimmed lgg produces for that tag is world-refuted and dropped anyway. The untrimmed learner is kept sound by the audit, which refutes the collapsed lgg of Proposition 5.12, but it loses whole tags, and loses more of them as N grows. The simulation's scope is limited: it tests the singleton-conflict case of CPC with a fixed budget and a per-schema world audit. It does not exercise descent on multi-step refutations, the fallback, or the N_1 formula; a simulation of the full audit is future work.

The propositional experiment. Experiment B (Section 12.2) runs a related pipeline, not an implementation of TTL, and shows this section's phenomena: coherence pruning closes every route to $\vdash \bot$, Post probes refute all 118 invalid structural candidate schemas, non-structural rules escape (Proposition 7.17), and world feedback mainly speeds up blame (5.0 versus 9.1 pruning rounds): localization, not refutation.

7.9 Summary

Learning which inferences are valid is learning the *audited practice*. Imitation finds the practice's rules, fallacies included, and is the only channel that says anything positive about a rule (Lemma 7.4 and Proposition 7.14). Confrontation with designated positions and the world sorts them, withholding every rule in a minimal conflict and, where descent applies, only the culprit (Lemmas 7.7 and 7.8). Descent makes an algorithm, with a guarantee, out of a lesson recorded in the motivating notes (Hänni, 2026): "one way to resolve a tension/contradiction is to go back to what caused one (or each) of the contradicting views/predictions/claims" (`philosophy/philosophy of thinking/how do i resolve contradictions, tensions?.md`). Assertion

comes last, after the burn-in (Theorem 7.18). Coherence fixes the learned role only up to which member of a conflicting set is given up: in the MP/AC example only the world or a denial decides whether "$\to$" means $\to$ or $\leftarrow$. This gives one precise sense to the notes' question about "the hooking of a model onto the world" (`philosophy/philosophy of language/meaning and truth/individual sentences being meaningful.md`): the hooking is needed not to make the role coherent, but to make it the intended one when practice conflicts with itself. The residue F_{res} is empty for CPC and, given realizability, for complete decidable theories; in arithmetic it contains $\neg\text{Con}(\text{PA})$ under presumption of validity. Section 13 discusses the resulting warrant. Open here are the query complexity of sound blame, the full-depth status of the Hilbert rival $\{S, \text{DN}, \text{AC}\}$ (Appendix F.14), blame with priors, the size of coherence witnesses in first-order logic, and TTL with anchored physical contexts as designated positions (see also Section 14).

8 Simplicity and normativity from imitation

Imitation identifies practice, errors included: for positive data, a systematic error generated by a schema *is* a rule (Corollary 5.19). This section examines a different remedy, proposed in the motivating notes (Hänni, 2026), and answers the question it raises.[2] The note asks how one can understand "what someone means when using a word. as opposed to becoming a predictor of what they would say". It proposes

> "an even stronger simplicity prior than solomonoff. eg if there are explainable mistakes on a simple model, you want the simple model that doesn't predict the mistakes. [...] let's just do a version of the simple hypothesis with noise, and then penalize the likelihood term less."

The note anticipates an objection: in sequence prediction, a steeper penalty favours a universal hypothesis that absorbs everything. It suggests, while "very much uncertain", that this pathology "is not present in the function case, if [...] each input has to sample its own random bits". It asks whether, given the models it lists, the good model "could be a vertex of the convex hull of the set of attainable (hypothesis complexity, expected neg log likelihood) tuples". And it adds: "I consider it very much plausible that there is some other pathology".

Answers in brief. The function-case intuition and the convex-hull argument are both right: private randomness makes every hypothesis pay for mixing per input, the product-model frontier has the hoped-for kink (proved for parity models), and selection is a threshold on *rates* (risk reduction per bit) along the lower hull (Theorems 8.2, 8.4 and G.6 and Lemma 8.6). But there are other pathologies: universality returns at block granularity, selection is by rate rather than validity, and the right steepness is not identifiable from imitation data. Steeper simplicity is a good *third* filter beside coherence and the world; the note's own alternative, "you specify properties of the thing or notion", removes cheap fallacies at every steepness. Section 8.7 gives the verdict in the note's terms, with references.

As far as we know, three results are mildly new: the product-model frontier and its kink, the rate threshold for arbitrary hypotheses, and the realizability of every convex frontier after rescaling.

[2] The source is the public working note `philosophy/philosophy of thinking/beating solomonoff induction at grokking a notion.md`, which lists this proposal as one of several ideas. Quotations are verbatim.

8.1 The proposal, formalized

Inputs $x \in X$ are drawn from μ, and labels lie in a finite set Y. The teacher is a channel $P^*(y \mid x)$. In the note's picture, P^* arises from an intended *norm* h° through sporadic noise and systematic error. Data are i.i.d. from $\mu \times P^*$. *Normativity* asks the learner to output h°, or its deterministic core, rather than P^*. In Section 8.6, x is a proof context and y the next human step; then P^* is the practice R^P of Definition 2.10, viewed as a channel.

Definition 8.1 (Randomness and objective)**.** [T5 Def 1.1, §1.3] A *product hypothesis* (private randomness) is a conditional distribution $h(\cdot \mid x)$, with likelihood $\prod_i h(y_i \mid x_i)$. A *shared-randomness* hypothesis is a distribution $P(y_{1:N} \mid x_{1:N})$; sequence prediction is its causally conditioned case. In a *block* hypothesis, the input at position i of a block (e.g. a document) includes the block's earlier pairs. The note's loss is $\lambda \ell(h) + \sum_{i \le N} -\log_2 h(y_i \mid x_i)$, with $\lambda > 1$ growing with N. Writing $\lambda = cN$ and dividing by N gives $\hat{J}_c = c\ell + \hat{\mathcal{R}}_N$, and its population version $J_c = c\ell + \mathcal{R}$, where $\mathcal{R}(h) = \mathbb{E}_{\mu \times P^*}[-\log_2 h(y \mid x)]$. Let $\Phi(c) = \inf_h J_c(h)$. In the *idealized* version, $\ell = K$ (prefix complexity) over total programs with rational outputs, and p_h is a shortest program for h. In the *computable* version, ℓ is a prefix code on a countable class. A *component* costing k bits that lowers $\mathcal{R}$ by r has *rate* r/k.

The steepness c is measured in bits per sample per bit of description. In the computable version, with a noise floor $h(y \mid x) \ge 2^{-B}$, a standard Occam bound (Lemma G.1) gives $J_c(\hat{h}) \le \Phi(c) + O(B\sqrt{(\ell(h_0) + B/c + \log(1/\delta))/N})$ for every minimizer $\hat{h}$ of $\hat{J}_c$, with probability $1 - \delta$ and for every fixed h_0; so we work with J_c.

Fixed versus growing steepness. With fixed λ, $c \to 0$, so every component of positive rate is eventually absorbed, error or not [T4 Prop 7.1(i); L11 Prop 2.2]: imitation with a fixed simplicity weight converges to the teacher's practice, the learner's version of Cohen's view that normative standards are a systematization of ordinary judgments (Cohen, 1981). With $\lambda \propto N$, as the note proposes, selection is asymptotically stable, so the obstruction found below is *rate inversion*, not sample size.

Relation to the literature. The note's λ is the inverse $1/\eta$ of a *learning rate* $\eta < 1$ in a tempered posterior (Grünwald, 2012; Grünwald and van Ommen, 2017; Zhang, 2006; Bissiri et al., 2016; Bhattacharya et al., 2019), and $c\ell + \mathcal{R}$ is the algorithmic rate–distortion Lagrangian with log-loss distortion (Vereshchagin and Vitányi, 2010). Appendix G.2 gives the details and relates the proposal to SafeBayes and to Kolmogorov's structure function.

8.2 No universal hypothesis in the function case

The note asks whether it is "just failing to construct the right 'universal hypothesis' ". The answer is no as long as randomness is private and inputs carry no other labels, and yes as soon as they do.

Theorem 8.2 (Per-input versus once-only pricing)**.** *[T5 Thm 2.1, Prop 2.2, as repaired] Let $\mathcal{C}$ be a finite class of deterministic labelers, $x_1, \dots, x_N$ inputs, and $\mathcal{C}(x) = \{f(x) : f \in \mathcal{C}\}$.*

(i) For every product h and every distribution Q on $\mathcal{C}$,

$$\max_f \sum_i -\log h(f(x_i) \mid x_i) \ge \sum_i H(Q_{x_i}),$$

where Q_x is the law of $f(x)$. The minimax value over product h is exactly $\max_Q \sum_i H(Q_{x_i}) \le \sum_i \log|\mathcal{C}(x_i)|$. Equality holds iff some Q has uniform marginals on every $\mathcal{C}(x_i)$. Such a Q (the uniform one) exists when a group acts on $\mathcal{C}$ by $(g\cdot f)(x) = \sigma_{g,x}(f(x))$, with permutations $\sigma_{g,x}$ of Y, transitively on each $\mathcal{C}(x)$; parities are an example.

(ii) For shared randomness, $\min_P \max_f -\log P(f(x_{1:N})) = \log|\{f(x_{1:N})\}| \le \log|\mathcal{C}|$ (Shtarkov, 1987).

(iii) A mixture with weights w_q pays $\log(1/w_q)$ at every input *if it mixes per input, and* once *if it mixes sequences. Both bounds are tight.*

Lean: NoAdaptation.thm_2_1_i_ennreal_maxF Lean: NoAdaptation.minimax_of_transitive

The core of the proof is one line: $\max_f \ge \mathbb{E}_{f\sim Q}$, and $\mathbb{E}_Q[-\log h(f(x) \mid x)] = H(Q_x) + \mathrm{KL}(Q_x \| h(\cdot \mid x))$. The minimax claim (via Sion's theorem) and the group condition are proved in Appendix G.3. Lean covers the lower bound in (i) and the value $\sum_i \log|\mathcal{C}(x_i)|$ under the group condition, not the general value $\max_Q \sum_i H(Q_{x_i})$, (ii) or (iii).

For parities on $\{0,1\}^{10}$ with $N = 200$, the per-input mixture pays 200.0 bits and the Bayes mixture 10.0; the uniform-marginal condition is needed, since $\{(a,a),(a,b),(b,a),(c,a)\}$ has value $2.5431 < \log_2 3 + 1$ (computed).

In sequence prediction, by contrast, a universal semimeasure $\mathbf{M}$ beats every model whose complexity exceeds a threshold that tends to $K(\mathbf{M})$ as $\lambda \to \infty$ (Proposition G.2); in the function case even the universal conditional semimeasure is a fixed predictor, with positive risk against every teacher with binary labels (Appendix G.4).

Proposition 8.3 (Per-block pricing: in-context universality)**.** *[T5 Prop 2.6, as repaired] Suppose the input at position i of block b contains the block's earlier pairs. Let (q_θ) be a finite family of computable, rational-valued "persona" models $q_\theta(y \mid x, \text{history})$, with rational weights $\sum_\theta w_\theta \le 1$. The in-context Bayes mixture h_w, with weights $w(\theta \mid \text{history}) \propto w_\theta \prod_{k<i} q_\theta(y_{b,k} \mid \cdot)$, is a single function hypothesis with $K(h_w) \le K(w,(q_\theta)) + O(1)$. On every block,*

$$\sum_{i\in b} -\log h_w(y_{b,i} \mid \cdot) \le \min_\theta \Big[\log \tfrac{1}{w_\theta} + \sum_{i\in b} -\log q_\theta(y_{b,i} \mid \cdot) \Big].$$

The block bound holds for every countable family, but for an infinite one h_w need not be computable.

The block likelihood of h_w telescopes (Appendix G.4). So h_w pays $\log(1/w)$ per block, not per input. Suppose that 20% of humans each have an idiosyncratic error, "in error contexts play wrong move j", with j uniform among 2^{16} moves. Each such rule has rate about 0, yet an $O(1)$-bit copier beats the norm model once a block has two error contexts: it pays 3.03 bits per error context at two and 1.33 at sixteen, against exactly 4.635 for the norm model at any block length (computed).

Reading. An LLM imitating whole proof documents is a block model: it can learn each author's systematic errors in context. A step checker whose input is a single step (Π, j) (Section 2) is a function model. The step is therefore the granularity at which the note's defence holds. This gives a second, independent argument for the step-level architecture of Sections 3 and 4.

8.3 The kink: shared versus private randomness

In sequence prediction, the note says, "it's pretty much just 1 bit paying for 1 bit". In the function case one might instead "see some point at which the derivative changes". We prove this for the note's example, made rigorous.

Setting. Let $X = \{0,1\}^d$, $n = 2^d$ and μ uniform. The simple model is a parity $f_a(x) = \langle a, x\rangle \bmod 2$ with $K(a) \geq d$. The error set E has $|E| = pn$, $p < 1/4$ and $K(p) = O(\log d)$, and the teacher is $y^* = f_a \oplus 1_E$. Let $L_E = \log_2 \binom{n}{pn} \approx nH(p)$, and let $\delta_E = L_E - K(E \mid a, K(a))$ be the randomness deficiency of E, which is $O(\log n)$ for most E. Write $\hat{s}_h(a')$ for the correlation of h's vote $h(0 \mid x) - h(1 \mid x)$ with $(-1)^{\langle a', x\rangle}$. The *good model* is f_a with flip rate p; it has $K \leq d + \kappa'$ with $\kappa' = O(\log d)$, and risk $H(p)$. The *bad model* y^* has risk 0.

Theorem 8.4 (Kink)**.** *[T5 Thm 2.5, as repaired] There is a machine constant κ such that the following hold.*

(a) Shared randomness has (almost) no kink. *Let P_u be the uniform shared-randomness mixture of the 2^u noisy parities that agree with a on its first $d - u$ coordinates. Then $|K(P_u) - (d-u)| \leq \kappa'$ and $-\log P_u(y^*) = u + nH(p) - o(1)$, where the $o(1)$ term is nonnegative and exponentially small in n (so super-exponentially small in d). So the total $K(P_u) - \log P_u(y^*)$ is $d + nH(p)$ up to $\pm(\kappa' + o(1))$ for every u, and every minimizer u^* of $\lambda K(P_u) - \log P_u(y^*)$ has $d - u^* \leq \frac{\lambda}{\lambda - 1}(2\kappa' + o(1))$.*

(b) Private randomness has a kink. *For every product h and $t \in \mathbb{N}$ with $K(h) \leq d - 2t - 2\log_2(t+1) - 2\log_2 d - \kappa$,*

$$\mathcal{R}(h) \geq 1 - \log_2(1 + \beta_t) \geq 1 - \beta_t / \ln 2,$$

where $\beta_t = \min\{2^{-t} + 2p,\ (1-2p)2^{-t} + \varepsilon_E\}$, with $\varepsilon_E = \sqrt{32 \ln 2 \cdot p(\delta_E^+ + d + \kappa_n)/n}$, $\delta_E^+ = \max(\delta_E, 0)$ and $\kappa_n = \log_2 n + 2\log_2(\log_2 n + 2) + \kappa$.

(c) The error segment. *For every product h, $K(h) + n\mathcal{R}(h) \geq d + L_E - \delta_E - 2\log_2 d - \kappa$.*

The proof (Appendix G.5) combines the list-size bound for the Hadamard code (Goldreich and Levin, 1989; Kushilevitz and Mansour, 1993), which makes a short h correlating with f_a encode a in fewer than d bits, with Jensen's inequality and the coding theorem; the slack $2p$ arises because h may be right on E for free, and the randomness of E relative to h replaces it by ε_E.

Reading. Part (a) is the note's "1 bit paying for 1 bit": the shared frontier is the sufficiency line, so a steeper penalty selects a mixture that knows almost nothing of a and learns the rest from the data, as any Bayes mixture does. Part (b) makes the note's hope quantitative. A product hypothesis missing $u = d - K(h)$ bits gains at most about $2^{-u/2 + O(\log d)}$ bits per sample over the coin, plus slack: exponentially little, though not nothing (list mixtures missing $u = 1, 2, 3$ bits gain 0.454, 0.221, 0.110 bits (computed)). So on the scale of d bits the product frontier is flat near the coin and drops at the simple model; that drop is the change of derivative.

Two caveats. The kink is resolved only up to $O(\log d)$ bits next to the simple model. And the slack $2p$ alone gives a kink only for $p < 0.1421$, where $1 - 2p/\ln 2 > H(p)$; the slack ε_E gives one for every $p < 1/4$ once n is large and $\delta_E = O(\log n)$ (at $p = 0.2$, $d = 20$ and $\delta_E = \kappa = 0$, $\varepsilon_E = 0.014$).

8.4 Selection traces the hull: rate thresholds

Let $z_h = (\ell(h), \mathcal{R}(h))$, $Z = \{z_h\}$ and $\hat{C} = \mathrm{conv}(Z + \mathbb{R}^2_{\geq 0})$. Only finitely many h have $\ell \leq L$, and $\mathcal{R} \geq 0$, so each J_c with $c > 0$ attains its minimum.

Definition 8.5 (Lower hull)**.** [T5 §3.1, as repaired] $\mathrm{hull}^-(Z)$ is the set of boundary points of $\hat{C}$ supported by a line of slope $-c$ with $0 < c < \infty$. A *vertex* is an extreme point of $\hat{C}$ that lies in $\mathrm{hull}^-(Z)$; vertices belong to Z.

Lemma 8.6 (Selection traces the lower hull; standard)**.** *[T5 Lemma 3.1, as repaired]* (i) *For $c > 0$, the minimizers of J_c are the h with $z_h \in \mathrm{hull}^-(Z)$ on a supporting line of slope $-c$, and every point of* $\mathrm{hull}^-(Z)$ *in Z arises this way.* (ii) *If $c < c'$, and h and h' minimize J_c and $J_{c'}$, then $\ell(h) \geq \ell(h')$ and $\mathcal{R}(h) \leq \mathcal{R}(h')$.* (iii) *A point $z \in Z$ is the unique minimizing point for all c in some nonempty open interval iff it is a vertex. The interval is then (s_+, s_-), given by the absolute slopes of the adjacent edges. A* hypothesis *is the unique minimizer there iff, in addition, no other hypothesis has the same point.*

Lean: RateThreshold.onLowerHull_iff Lean: RateThreshold.lemma_3_1_ii Lean: RateThreshold.lemma_3_1_iii_hyp

Part (i) holds because $J_c(h) = \langle (c, 1), z_h \rangle$ has the same minimum over Z as over $\hat{C}$; (ii) follows by adding the two optimality inequalities; (iii) is proved in Appendix G.6. The Lean proofs of (i) and (iii) cover finite classes.

Reading. The note's convex-hull argument is correct, with two refinements. The intended model must be a *vertex*, and the only hypothesis at its point. And λ must lie in $(s_+ N, s_- N)$. The note's alternative, to "set abs bound on model complexity", selects every model the penalty selects (bound it at that model's length), as the note observes. The converse fails: a hard bound can also select points of the staircase $\rho^\times(k) = \min\{\mathcal{R}(h) : \ell(h) \leq k\}$ that lie off the hull, though never a Pareto-dominated point such as that of Proposition 8.10.

Theorem 8.7 (Rate threshold, separable classes; trivial once set up)**.** *[T5 Thm 3.2; T4 Prop 7.1] Partition X into finitely many regions X_j, and let $\mathcal{H} = \prod_j \mathcal{H}_j$ with $\ell = \sum_j \ell_j$. Then $J_c = \sum_j [c\ell_j + \mathcal{R}_j]$, and the minimizers are exactly the products of per-region minimizers. In particular, let $\mathcal{H}_j = \{$default, component $j\}$, where the component costs $k_j > 0$ extra bits and lowers $\mathcal{R}_j$ by r_j. Then component j is selected iff $r_j/k_j > c$; at equality both choices are optimal.*

Lean: RateThreshold.prod_isMin_iff Lean: RateThreshold.rate_threshold

Proof. The objective is a sum of functions of disjoint coordinates. This is the standard hard-thresholding fact for separable penalties. □

Corollary 8.8 (Separating rules from fallacies)**.** *[T4 Cor 7.2] Split finitely many separable components into valid ones and fallacies. Some $c > 0$ makes the valid set the unique minimizer iff* $\min_{v \text{ valid}} r_v/k_v > \max_{e \text{ fallacy}} r_e/k_e$, *where* $\max \emptyset := 0$.

Lean: RateThreshold.selectable_iff

Proof. If the inequality holds, take c between the two sides. If instead $r_e/k_e \geq r_v/k_v$, any c that strictly selects v has $c < r_e/k_e$, and so also selects e. □

Arbitrary hypotheses. With $\ell = K$ and all hypotheses allowed, which may share information across regions or learn components partially, independent parities behave as separable components: the rate threshold of Theorem 8.7 survives up to explicit $O(\log)$ slack (Theorem G.6). Every convex, decreasing, piecewise-linear frontier that starts at height 1, has rational slopes and drops by at most one bit is, after rescaling and up to logarithmic bands, the product-model frontier of a teacher made of independent parities and a fair-coin region (Corollary G.7); so whether a good model is a vertex is a property of the data, not a theorem. And in the note's example, for small p and c well inside the window ($\approx 1/n, \approx (1 - H(p))/d$), every near-optimal hypothesis correlates strongly with the parity and memorizes almost none of the errors (Theorem G.8 and Corollary G.9). All four results are stated and proved in Appendix G.

The note's own numbers (computed). These are a 100-bit simple model (plus 10 bits for the flip rate, our stand-in for the note's "something small"), and a 10 000-bit error on a 2^{-10} fraction of inputs, with $H(2^{-10}) = 0.0112$ bits per sample (about 1/100, as the note says). The rates are $8.99 \cdot 10^{-3}$ for the model and $1.12 \cdot 10^{-6}$ for the error. So the good model is selected iff c lies between these, a window spanning a factor of about 8000. At $N = 10^6$ this means $\lambda \in (1.12, 8990)$. Standard MDL ($\lambda = 1$) switches to the error model at $N \approx 8.9 \cdot 10^5$.

Reading. The error's rate is about $1/n$: a K-random error set costs one bit per $1/n$ of risk, the slowest possible rate. So *the note's example is the most favourable case for the proposal.* Normativity is achieved there because the errors are incompressible relative to the norm.

8.5 Pathologies

Theorem 8.9 (Rate blindness; trivial once set up)**.** *[T5 Thm 4.1, as repaired]*

(i) $c \mapsto \arg\min J_c$ *depends on the data only through* (μ, P^*)*; validity enters nowhere.*

(ii) Let v *be valid and* e *an error with* $r_v/k_v \le r_e/k_e$*. In the setting of Theorem 8.7, if* v *is selected then so is* e*, for every* c *other than a common tie value. In the setting of Theorem G.6, the same holds for near-optimal* h *at every* c *at which each of* v *and* e *satisfies the hypothesis of clause (a) or of clause (b) of that theorem; this excludes slack bands around their rates.*

(iii) h° *is the unique minimizer on an open* c*-interval iff* z_{h° *is a vertex and no other hypothesis shares it.*

(iv) Take two worlds with the same (μ, P^*)*, in which some component is an error in one and a rare valid rule in the other. Every imitation-only procedure gives the same output in both, so it is wrong in one of them.*

Lean: RateThreshold.rate_blindness

Proof. Parts (i) and (iv) hold because J_c is a functional of (μ, P^*). Part (iii) is Lemma 8.6.

For (ii) in the separable case: if v is selected and c is not a tie, then $c < r_v/k_v \le r_e/k_e$; and if $c = r_v/k_v < r_e/k_e$, then e is strictly selected. The idealized case is proved in Appendix G.10. □

The Lean proof covers (ii) in the separable case only. Part (iv) is Corollary 5.19 again. What is new is the quantitative criterion: the penalty separates errors from rules *by rate, not validity.*

For instance, a 200-bit valid rule used in 10^{-4} of steps, gaining at most 2.2 bits per use, has a lower rate than the note's 10 000-bit error, so every c that rejects the error discards it; rarely used irregular verbs likewise regularize first (Lieberman et al., 2007).

Cheap fallacies [T5 Prop 4.2]. In step imitation (Section 8.6), a schema used in a fraction π of steps has rate $\pi g/\ell$. Affirming the consequent (AC: $q,\ p \to q \,/\, p$) has the size of modus ponens. The freshman's dream (FD: $(a+b)^2 \to a^2+b^2$) is shorter than the correct expansion.

Take a toy with $M = 64$ candidate steps and slip rate $\varepsilon = 0.05$, so that the gain is $g = 5.42$ bits (computed). AC, at frequency 0.02 and 20 bits, then has rate $5.4 \cdot 10^{-3}$. That beats proof by cases ($1.4 \cdot 10^{-3}$), reductio ($1.6 \cdot 10^{-3}$), ex falso ($7.2 \cdot 10^{-4}$) and induction ($1.2 \cdot 10^{-4}$). So no steepness keeps every valid rule while removing every error.

We conjecture that this is no accident (conjecture; empirical). A fallacy becomes systematic because it is a short, natural variant of a valid pattern.

Proposition 8.10 (Rate inversion: the intended model need not be a vertex)**.** *[T4 Prop 7.3] In Theorem 8.7, take $k_r = \ell(r)$ and $r_r = \pi_r g_r$, with three rules: a common valid rule A, with $\ell = 10$, $\pi = 0.5$, $g = 8$; a rare valid rule B, with $\ell = 40$, $\pi = 0.01$, $g = 8$; and the freshman's dream F, with $\ell = 8$, $\pi = 0.05$, $g = 8$. The rates are 0.4, 0.002 and 0.05. As c grows, the selection passes through $\{A, B, F\}$, $\{A, F\}$, $\{A\}$ and $\emptyset$, so no c selects the valid set $\{A, B\}$. In fact, in (bits, residual bits per datum), $\{A, B\} = (50, 0.4)$ is Pareto-dominated by $\{A, F\} = (18, 0.08)$. So no criterion strictly increasing in both coordinates selects it.*

Lean: RateThreshold.valid_not_minimizer Lean: RateThreshold.pareto_dominated

Proof. Apply Theorem 8.7. The residual of a set R is $\sum_{r \notin R} \pi_r g_r$. □

This answers the note's vertex question in general. The hull mechanism is right, but the intended model is a vertex only if every valid component out-rates every error (Corollary 8.8). That holds for rare, expensive, idiosyncratic errors. It fails for cheap systematic fallacies, which, we expect, include common student-algebra errors and Euler-style manipulations.

Theorem 8.11 (The steepness is not identifiable from imitation data)**.** *[T5 Thm 4.3, as repaired]*

(i) Along the population path, $\mathcal{R}(h_c)$ is non-increasing as c decreases. So, for a fixed finite grid in the computable version with a noise floor, choosing c by held-out log-loss picks, with probability tending to one as $N \to \infty$, a grid value whose selected hypothesis has least risk among the selected ones; the smallest grid value always qualifies. (At a grid value lying exactly at a hull breakpoint, the hypothesis selected there may alternate between co-minimizers of different risk.) At finite N this can fail, through the B/c term of Lemma G.1.

(ii) For every strictly proper scoring rule, the teacher channel strictly maximizes the expected score at every x (Gneiting and Raftery, 2007). So every validation criterion that is an empirical proper score on fresh imitation data targets P^*, errors included.

(iii) No imitation-data statistic picks a good c in both worlds of Theorem 8.9(iv).

The proof is in Appendix G.11. In the note's example, cross-validation picks the error model: held-out log-loss is 1.0 at $c = 10^{-1}$, 0.0112 on $[10^{-5}, 10^{-3}]$, and 0 at $c = 10^{-7}$ (computed). A non-proper "knee" heuristic (the interior vertex with the widest $\log c$-interval) picks the good model in this example; by (iii), no such rule is right in both twin worlds.

Reading. Choosing c is a normative act, not an estimate. The note says that "is this a chair?" is "much closer to 'should the person who taught me the chair notion consider it a chair?'" than to what that person would say. Part (iii) makes literal the claim that this target is not a function of P^*.

Proposition 8.12 (Guard erosion). *[T5 Prop 4.4, as repaired] Let $\sigma|_G$ guard a schema σ (e.g. $x/x \to 1$ if $x \neq 0$). Humans apply σ in G-contexts (frequency π_G, per-step gain g). In contexts where σ applies but G fails (frequency $\pi_{\neg G}$), they play a non-σ step. In the σ-applicable region a hypothesis has three options:* none*: uniform over the M candidates, at cost 0;* unguarded*: mass $1 - \varepsilon + \varepsilon/M$ on the σ-step, at cost k_σ; and* guarded*: the same as unguarded in G-contexts, and uniform on the non-σ steps otherwise, at cost $k_\sigma + k_G$. Define*

$$\rho_\sigma = \frac{\pi_G g - \pi_{\neg G} \log_2(1/\varepsilon)}{k_\sigma}, \qquad \rho_G = \frac{\pi_{\neg G} \Delta_G}{k_G}, \qquad \Delta_G = \log_2(M/\varepsilon) - \log_2(M-1).$$

If $\rho_G < \rho_\sigma$, then for $c \in (\rho_G, \rho_\sigma)$ the unguarded, unsound *σ is selected, although the teacher never errs. For $c < \rho_G$ the guarded rule is selected, and for $c > \rho_\sigma$ neither is. If $\rho_G \geq \rho_\sigma$, then apart from ties the path goes from guarded to none, and the learner stays sound.*

In (bits, risk) relative to *none*, the unguarded option is a vertex of the lower hull of the three options iff $\rho_\sigma > \rho_G$; the proof is in Appendix G.12.

Take $\pi_G = 0.05$, $\pi_{\neg G} = 5 \cdot 10^{-4}$, $k_\sigma = 22$, $k_G = 12$, $M = 64$ and $\varepsilon = 0.05$ (computed). Then $\rho_G = 1.8 \cdot 10^{-4}$ and $\rho_\sigma = 1.22 \cdot 10^{-2}$, so the unguarded rule is selected on $(1.8 \cdot 10^{-4}, 1.22 \cdot 10^{-2})$, a range that includes steepnesses at which ex falso (rate $7.2 \cdot 10^{-4}$) is kept. With $\pi_G = \pi_{\neg G} = 0.002$ instead, $\rho_\sigma < \rho_G$ and there is no unsound phase.

Reading. What a careful teacher exhibits most rarely is what goes first: side conditions, eigenvariable conditions, edge cases. One unsound schema can trivialize a classical calculus (Proposition 3.5). With field rules, $0/0 = 1$ and $0/0 = 0 \cdot 0^{-1} = 0$ give $1 = 0$, a negative bag (Lemma 6.2). So coherence must restore what simplicity erodes.

Further pathologies [T5 §4.4]. Four more are stated precisely in Appendix G.13. *Error envelopes:* errors concentrated in a cheaply describable region are learned at the envelope's own rate, and where their density there exceeds $\frac{1}{2}$ the learner predicts the error label throughout the region. *Parasitic errors:* an error that is cheap given a valid rule ($K(e \mid v) \ll K(e)$, as FD is given $(ab)^n = a^n b^n$) can enter the hull bundled with it, so a valid rule may be kept only because of the error it subsidizes. *Distribution dependence:* a valid rule that is rare in training stays discarded as $N \to \infty$. *Language relativity:* for any $c > 0$, some prefix code makes any given set of positive-gain separable components the unique selection; between universal machines marginal costs move by at most $2c_{UV}$, negligible for 10 000-bit errors but decisive for 20-bit schemas, so the note's "simplicity prior defined in terms of existing understanding" cuts both ways.

8.6 Inference rules: the division of labour

In *situation-typed step imitation*, a context x is a premise set and a goal (Section 2), and y is the next human step among M_τ candidates. Contexts fall into computable situation types τ of frequency π_τ. In type τ the human applies a schema $\sigma_\tau \in \Sigma^P$, genuine or fallacious, with probability $1 - \varepsilon$, and otherwise moves uniformly.

Modelling type τ costs $\ell(\sigma_\tau) + O(1)$ bits and gains $g_\tau = \log_2 M_\tau - H_\tau$ per step. Its rate is therefore $\rho_\tau = \pi_\tau g_\tau / (\ell(\sigma_\tau) + O(1))$, and by Theorem 8.7 it is selected iff $\rho_\tau > c$. The induced verifier accepts the instances of the selected schemas. It is sound (Definition 2.18) iff no error schema is selected and no guard has eroded.

Definition 8.13 (Admissible family, witnesses)**.** [T5 §5.2, as repaired] The components are $V \sqcup F$ (valid, erroneous), with net values $\nu_j = r_j - c\,k_j$. The *admissible family* $\mathfrak{A} \subseteq 2^{V \sqcup F}$ is the intersection of the filters in use: $\mathcal{A}$-coherence (Definition 2.13), W-adequacy, and coherence with designated meaning postulates Ax. Here W-adequacy means that no claim the world refutes is derivable. The family is down-closed and contains 2^V. A *witness* for $e \in F$ is a minimal $W \subseteq V$ with $W \cup \{e\} \notin \mathfrak{A}$. It is the negative bag of Lemma 6.2, seen from the rules. Let $V_+ = \{v : \nu_v > 0\}$ and $F_+ = \{e : \nu_e > 0\}$. At a given c we impose two conditions. *Error-independence*: for $U \subseteq V$, $U \cup F' \in \mathfrak{A}$ iff $U \cup \{e\} \in \mathfrak{A}$ for each $e \in F'$. *Valid dominance*: every $e \in F_+$ with a witness inside V_+ has $\beta(e) > \sum_{F_+} \nu$. Here $\beta(e)$ is the least value of $\sum_B \nu$ over sets $B \subseteq V_+$ meeting all of e's witnesses inside V_+.

Theorem 8.14 (Division of labour)**.** *[T5 Thm 5.2, as repaired] Fix c, and assume separability, error-independence and valid dominance at c. Then the optimal admissible sets are exactly the admissible sets $S^*(c) \cup Z$ with $Z \subseteq \{\nu = 0\}$, where*

$$S^*(c) = V_+ \cup \{e \in F_+ : V_+ \cup \{e\} \in \mathfrak{A}\}.$$

So, up to zero-value ties, simplicity *removes exactly the errors of rate at most c, coherent or not;* coherence*, the* world *and* postulates *remove exactly the errors that clash with the valid rules actually kept (V_+, plus* Ax*), whatever their rate; the residue is the high-rate errors that pass every filter; and valid rules of rate at most c are lost, whatever the other filters do.*

The proof is in Appendix G.14: valid dominance makes it better to restore the valid rules an error's witnesses need than to keep the error. A brute-force check of 17 806 instances found no violation (computed).

Corollary 8.15 (Best steepness given the other filters)**.** *[T5 Cor 5.2a, as repaired] Assume error-independence and valid dominance at every c considered, and $k_j > 0$. Then every optimal admissible set contains V and no error iff*

$$\max\{\rho_e : V \cup \{e\} \in \mathfrak{A}\} < c < \min_{v \in V} \rho_v, \qquad \textit{with } \max \emptyset := 0.$$

The proof, and a counterexample showing that valid dominance is essential, are in Appendix G.14.

So coherence and the world *widen the window*: the rates of incoherent or refutable errors stop mattering. Simplicity is needed only against coherent, adequate, low-rate errors, such as idiosyncratic false lemmas and quus-like patches (Proposition 6.33).

Proposition 8.16 (Repair by sacrifice)**.** *[T5 Prop 5.3, as repaired] Assume error-independence. Let $e \in F_+$ have a witness inside V_+, and let $B \subseteq V_+$ be a cheapest set meeting all of e's witnesses inside V_+, with $\beta(e) = \sum_B \nu < \nu_e$. Then the clean set $S_{\text{clean}} = V_+ \cup \{e' \in F_+ : V_+ \cup \{e'\} \in \mathfrak{A}\}$ is not optimal. Indeed, $(V_+ \setminus B) \cup \{e\} \cup (S_{\text{clean}} \cap F)$ is admissible, with value $\text{val}(S_{\text{clean}}) + \nu_e - \beta(e)$.*

The proof is in Appendix G.14. Here is a computed instance (computed). AC's Post witness (Corollary 6.32), with $p := \bot$ and $q := A \to A$, needs only $\to$I. If $\to$I is rare, with rate $8.7 \cdot 10^{-4}$, then at $c = 5 \cdot 10^{-4}$ the coherent optimum is {AC, MP, $\top$I}. *The learner drops $\to$I to keep the fallacy.*

This instance assumes $\mathcal{A} = \{\emptyset\}$. With a designated context such as $\{p_0 \to q_0, q_0, \neg p_0\}$, MP is itself a witness partner, and AC is dropped instead. The phenomenon is Lakatos's monster-barring (Lakatos, 1976) run backwards: the repair is chosen by description cost, not by validity. It refines Theorem 6.31, whose criterion must be read relative to the rules the learner keeps.

Meaning postulates. The note lists a separate route: "you specify properties of the thing or notion sometimes". For example, "1+1=2 won't be true if you accidentally assign 1->rabbit and 2->chicken from a demonstration (for any reasonable meaning of plus)". Formally, a finite set Ax of sentences endorsed as constitutive of a notion enters as designated premises. These are Carnap's meaning postulates (Carnap, 1952).

Proposition 8.17 (Postulates are rate-independent)**.** *[T5 Prop 5.4, as repaired]*

(a) Assume the hypotheses of Theorem 8.14 at every c, with Ax *designated, and let* $\rho_e > 0$*. Up to zero-value ties,* $e \notin S^*(c)$ *iff* $\rho_e \le c$ *or some witness of e lies inside* $V_+(c) = \{v : \rho_v > c\}$*. Hence e is excluded at* every *c iff some witness W of e has* $\rho_w \ge \rho_e$ *for all* $w \in W$*.*

(b) Let $\mathrm{Ax}_{\mathrm{num}} = \{1+1=2,\ 1^2=1,\ 2^2=4,\ 4\neq 2\}$*, and keep substitution of equals. Then* $(1+1)^2 = 1^2+1^2$ *rewrites to* $4=2$*, so FD is excluded however frequent it is. Without postulates, FD together with the ring rules derives* $2=0$*. That is coherent, since FD holds in characteristic 2.*

(c) The demonstrations ("1", one rabbit) and ("2", two chickens) fit both numeral $\mapsto$ *cardinality and numeral* $\mapsto$ *kind. With + read as disjoint union, the postulate* $1+1=2$ *holds under the first reading only.*

(d) Postulates eliminate exactly the readings outside Mod(Ax)*. For first-order arithmetic, the non-standard models remain (Theorem 6.34).*

Part (a) is proved in Appendix G.14; (b)–(d) are direct. Here "rate-independent" means that exclusion needs only that the witness rules be kept. For FD the witness rule is substitution of equals, of rate $4.07 \cdot 10^{-2}$, against $\rho_{\mathrm{FD}} = 2.46 \cdot 10^{-3}$ (computed). Random numerical evaluation through the world channel also refutes FD (Schwartz, 1980; Zippel, 1979). Part (c) is Quine's gavagai (Quine, 1960), with a cure.

Reading: avowal versus performance. Postulates are the teacher's *avowals* ("1+1=2", "MP is valid, AC is not", "a chair is for sitting"), not the teacher's *performance*. The steeper penalty sees only performance. This is where the debate between Stich and Nisbett (1980) and Cohen (1981) lands [L11 §2].

Cohen grounds norms in a systematization of ordinary judgments; Stich and Nisbett had argued, against reflective equilibrium as a test of justification, that systematic fallacies such as the gambler's fallacy can survive such a systematization. Performance data indeed cannot separate a fallacy from a rule with the same profile (Theorem 8.9(iv)). Avowals are a second channel: few bits that exclude whole high-rate classes of errors. Where an avowal is itself wrong, as with naive comprehension, coherence catches it (Section 10).

The note's abstract/concrete example, "if something is abstract, then it usually helps a lot to study examples to understand it", is a cross-notion postulate: it ties the notion to another notion with independent evidence, turning a labelling error into a W-refutation. This makes "should the person who taught me [...] consider it a chair" operational: the norm is whatever fits the notion's avowed role, checked against the other channels.

Table 11 puts the pieces together. The residue is the alternative meanings of Theorem 6.34 *that humans actually use*, such as "if" read as "iff" in closed worlds. Calling them errors is itself a normative decision outside the data, and it may be wrong: conditional perfection is arguably part of everyday "if".

error	rate	simplicity alone removes it at	removed by
idiosyncratic false lemma (300 bits, freq. $5 \cdot 10^{-4}$)	$9 \cdot 10^{-6}$	$c > 9 \cdot 10^{-6}$	**simplicity** (coherent; outside W)
affirming the consequent	$5.4 \cdot 10^{-3}$	$c > 5.4 \cdot 10^{-3}$ (loses 4 valid rules)	**coherence**, given →I kept
freshman's dream	$2.5 \cdot 10^{-3}$	$c > 2.5 \cdot 10^{-3}$ (loses 4 valid rules)	**postulates** or **world**
quantifier swap	$5.4 \cdot 10^{-4}$	$c > 5.4 \cdot 10^{-4}$ (loses induction)	**postulates**/designation ($0 \neq 1$)
gambler's fallacy	$9.0 \cdot 10^{-4}$	$c > 9 \cdot 10^{-4}$ (loses ex falso, induction)	**world** (frequencies)
conditional perfection	$2.2 \cdot 10^{-3}$	$c > 2.2 \cdot 10^{-3}$ (loses 4 valid rules)	**nothing**: the residue

Table 11: Which channel removes which error, in the situation-typed toy ($M = 64$, $\varepsilon = 0.05$, illustrative frequencies). With all filters, the window that removes the false lemma and keeps every valid rule is $c \in (9 \cdot 10^{-6}, 1.2 \cdot 10^{-4})$. With simplicity alone, no window works (computed).

8.7 Verdict

Where the proposal is vindicated. Function induction with private randomness has no universal-hypothesis pathology, by exactly the mechanism the note names: "you have to pay the likelihood term separately at every x" (Theorem 8.2). Shared randomness pays for partial knowledge once, product models gain exponentially little from it, so the product frontier kinks, as proved for parity teachers (Theorem 8.4). The convex-hull argument is correct; the change of derivative is the vertex condition, plus uniqueness at the vertex (Lemma 8.6). The note's example works, asymptotically, on a window of more than three orders of magnitude (Corollary G.9).

Where it fails. The penalty is validity-blind: it sorts by rate, betting that norms are high-rate structure and errors low-rate structure (Theorem 8.9). The bet is right for idiosyncratic slips and wrong for cheap, frequent fallacies (Proposition 8.10). It costs rare valid rules and, worse, rarely exercised guards, so the learner can be less sound than its teacher (Proposition 8.12). The good c cannot be estimated from imitation data (Theorem 8.11), and private randomness fails whenever inputs carry the same person's earlier labels (Proposition 8.3).

What does the normativity work. Of the note's other ideas, "you specify properties of the thing or notion" is decisive: postulates are avowal rather than performance, and they remove high-rate fallacies at every c, provided the valid rules in a witness out-rate the fallacy (Proposition 8.17). The idea "what is it that this person is trying to teach me" calls for a teacher model with a competence/performance split. Armstrong and Mindermann (2018) show that a policy does not determine its decomposition into planner and reward, and that a simplicity prior over decompositions does not single out the intended one. That is Theorem 8.9(iv) in another guise.

The division of labour comes to this: simplicity removes the *expensive* errors; coherence removes the *incoherent* ones; the world removes the *refutable* ones; postulates remove those that *violate avowed properties*. What survives is the meanings humans actually use. Steeper simplicity still has a job no other channel does: removing idiosyncratic, coherent, unrefutable errors (Table 11, first row). It is a good third filter, not a route to normativity on its own.

In the terms of Section 13, a learner fitted to practice alone is in narrow reflective equilibrium with its teacher, with or without a simplicity penalty. Stich's objection applies to it exactly. What removes coherent systematic fallacies is information that is not a function of performance: avowed postulates, and an independent world channel. That world channel is what makes the equilibrium wide.

Besides the open problems of Section 14, a product-model algorithmic statistics and the best block size for a step imitator remain open. The simulations are in Section 12.

9 Coherence and existence: a "philosophical completeness theorem"

Sections 6 and 7 used coherence as a learning signal. This section asks what coherence *means*: if a learned mode of talking is coherent, is there something it is talking about? The motivating notes (Hänni, 2026) pose exactly this question:[3]

> "a mode of talking makes sense (is coherent) iff there is something it could be talking about"
> as long as the 'mode of talking' is a first-order theory and 'something' is a model of it, then this is literally the completeness theorem
> is a generalization of the completeness thm available that justifies this philosophical proposition more broadly?

The note suggests further instances ("probabilities describe world iff coherent", after Christiano et al. 2013; "maybe something could be said about vnm stuff also?"). Under the heading "from sam"[4] it records a colleague's remark, "completeness is an adjunction between sentences and models", the observation "if you are in image, then round trip is just identity", the question "what is this image?" and its answer, "this is all sets of sentences closed under inference rules. this is by the completeness theorem". Elsewhere the notes add the caveat that "there might only be such a thing in the same sense that there is a 'proof' of the Gödel sentence G".[5]

Answer in brief. There is a whole family of "coherent iff there is something it could be talking about" theorems with one anatomy: completeness holds iff the *points* of a calculus are realized by models (§9.1). The adjunction remark, as the note answers it, is strong completeness; the note's proposition is weak completeness; the two differ. Every calculus is complete for models built from its own syntax, so the proposition has content only for an independently specified semantics. Instances: positions (Stone duality, §9.2), credences (de Finetti with *counting sequents*; no bounded-arity, CCS-style constraint family characterizes coherence, §9.3), and contexts (the somethings are belief states, not worlds, §9.4). A coherent learned calculus has a model, but a syntactic one; the Kripkensteinian residue of Section 6 becomes a set of non-isomorphic things talked about; and a compactness and a computability barrier show that coherence cannot in general deliver the *intended* model (§9.5; verdict in §9.6).

Credit. Almost everything here is a standard result, reorganized; the organization is the contribution. The Galois connection is Birkhoff (1940) and Ore (1944); bilateral completeness is Scott (1974) and Shoesmith and Smiley (1978); probabilistic coherence is de Finetti (1937) and Gaifman (1964); local-models semantics is Ghidini and Giunchiglia (2001). The arity result Theorem 9.14 is an instance of the classical fact that locally consistent marginals need not admit a joint distribution (the marginal problem of Vorob'ev 1962; the correlation polytopes of Pitowsky 1991), and NP-hardness of probabilistic satisfiability is due to Georgakopoulos et al. (1988). New as far as we know, and all elementary: the counting-sequent form of de Finetti's theorem (probably folklore; cf. Paris 1994); the threshold hierarchy for counting-sequent constraints (Theorem H.1); the simplest monotone case of multi-context completeness, fitted to context calculi like that of Section 11 (Theorem 9.19; completeness results for local-models semantics already exist, Ghidini and Giunchiglia 2001; Serafini and Bouquet 2004); and the corollary on

[3] `logic/a 'philosophical version' of gödel's completeness theorem.md`.

[4] The README of the notes acknowledges discussions with Sam Eisenstat.

[5] `philosophy/philosophy of thinking/advent of reason/real attempt 2024/3f math, thinking, and technology are equi-infinite.md`, footnote 5.

logical inductors (Corollary 9.17). Theorems 9.22 and H.9 are standard arguments, newly applied. In Lean (Table 24), module `Bilateral` formalizes Theorems 9.8 and 9.10 and Corollary 9.9 for an arbitrary set Fm and Theorem 9.11 for the fixed propositional term algebra; module `Specker` formalizes Theorem 9.14 and the probabilistic half of Proposition 9.23; nothing else here is formalized. Sanity checks are in `research/theory/T6-checks/`.

9.1 The adjunction, and the anatomy of a completeness theorem

Definition 9.1 (Semantic frames and completeness)**.** [T6 Def 1.1, §1.1] A *semantic frame* $(S, M, \models)$ consists of a set S of "sentences", a set or class M of "models" and a relation $\models \subseteq M \times S$. Put $\mathrm{Mod}(\Sigma) = \{m : m \models \sigma \ \forall \sigma \in \Sigma\}$ and $\mathrm{Th}(K) = \{\sigma : m \models \sigma \ \forall m \in K\}$. A *deductive system* is a closure operator C on $\mathcal{P}(S)$ (extensive, monotone, idempotent), with closed sets (*theories*) $\mathrm{Fix}(C)$; it is *finitary* if $C(X)$ is the union of the $C(X_0)$ over finite $X_0 \subseteq X$. C is *sound* if $C \leq \mathrm{Th} \circ \mathrm{Mod}$ pointwise, *strongly complete* if $C = \mathrm{Th} \circ \mathrm{Mod}$, and *weakly complete* if every *coherent* X (i.e. $C(X) \neq S$) has $\mathrm{Mod}(X) \neq \emptyset$.

Lemma 9.2 (Galois connection)**.** *[T6 Fact 1.2] (known) $K \subseteq \mathrm{Mod}(\Sigma)$ iff $\Sigma \subseteq \mathrm{Th}(K)$. Hence* $\mathrm{Th} \circ \mathrm{Mod}$ *and* $\mathrm{Mod} \circ \mathrm{Th}$ *are closure operators, and* $\mathrm{Mod}, \mathrm{Th}$ *are mutually inverse inclusion-reversing bijections between their fixed points (Birkhoff, 1940; Ore, 1944).*

Read as an adjunction between $(\mathcal{P}(S), \subseteq)$ and $(\mathcal{P}(M), \supseteq)$, this is $\mathrm{Mod} \dashv \mathrm{Th}$: the precise content of "completeness is an adjunction between sentences and models". The adjunction exists for *every* frame; a completeness theorem is a statement about its fixed points.

Proposition 9.3 (The image of the round trip)**.** *[T6 Prop 1.3] For every closure operator C: $\{\mathrm{Th}(K) : K \subseteq M\} = \mathrm{Fix}(C)$ iff C is strongly complete.*

Proof. The image of Th is $\mathrm{Fix}(\mathrm{Th} \circ \mathrm{Mod})$, as $\mathrm{ThModTh} = \mathrm{Th}$; and a closure operator is determined by its fixed points, $C(X) = \bigcap\{T \in \mathrm{Fix}(C) : X \subseteq T\}$. □

This is the note's own answer, for arbitrary closure operators (soundness is not needed): the round trip fixes exactly the sets closed under the rules iff the rules are complete.

Theorem 9.4 (Tautological completeness)**.** *[T6 Thm 1.4] Every closure operator C is strongly complete for its* canonical frame $(S, \mathrm{Fix}(C), \ni)$*, whose models are the theories T, with $T \models \sigma$ iff $\sigma \in T$.*

Proof. $\mathrm{Th}(\mathrm{Mod}(X)) = \bigcap\{T \in \mathrm{Fix}(C) : X \subseteq T\} = C(X)$. □

Reading. If "something it could be talking about" may be *any* object, the note's proposition is trivially true: every mode of talking is about its own theories. Real completeness theorems have content because their models (truth tables, structures, points of $\bar{k}^n$, measures) are specified *independently of* the calculus. For a learner of inference rules, that a learned calculus is "about something" is cheap; what matters is whether it is about something we can access independently.

Theorem 9.5 (Completeness is realization of points)**.** *[T6 Thm 1.5, as repaired] Call a theory T a* point *if $T \neq S$ and, for some $\sigma \notin T$, T is maximal among theories omitting σ; write* $\mathrm{Pt}(C)$*. Let C be finitary.* (a) Lindenbaum. *Every theory is the intersection of the points containing it.* (b) *If C is sound, it is strongly complete iff every point is the theory of a model:* $\mathrm{Pt}(C) \subseteq \{\mathrm{Th}(\{m\}) : m \in M\}$. (c) *If C is sound, some finite F has $C(F) = S$, and $\mathrm{Th}(\{m\}) \neq S$ for all m, then C is weakly complete iff every* maximal *proper theory is the theory of a model. Without soundness, ($\Rightarrow$) fails.*

mode of talking (calculus)	something	realization of points	theorem
finite sequents (Ov, Wk, Cut)	valuations $\mathrm{Fm} \to 2$	trivial	Scott (1974); §9.2
first-order sentences	structures	Henkin witnesses	Gödel (1930); Henkin (1949)
polynomial equations over k (ideal, radical)	points of $\bar{k}^n$	Zariski's lemma	Hilbert (1893); Zariski (1947)
finite linear inequalities	points of $\mathbb{R}^n$	LP duality	Farkas (1902)
real polynomial (in)equalities	points of $\mathbb{R}^n$	real closure	Stengle (1974)
credences, finite agenda (counting sequents)	mixtures of valuations	separating hyperplane	de Finetti (1937); §9.3
first-order credences (Gaifman coherence)	measures on complete theories	Carathéodory	Gaifman (1964); §9.3
finite preference data	utility functions	Motzkin	von Neumann and Morgenstern (1944); §9.3
labelled claims (local rules, bridges)	belief-state families	canonical chain	§9.4

Table 12: One anatomy, many completeness theorems [T6 §1.5]. Typically coherence buys existence only in a *closure* of the naive model class (see text).

Proof idea. (a) Zorn, using finitarity; (b) realize each point in the decomposition (a), and conversely use complete meet-irreducibility. Full proof and the counterexample for (c) in Appendix H.1. So every completeness proof has two steps: (i) points exist, which is Zorn and free; (ii) points are realized by models of the independently given kind, which is where content lives (Henkin witnesses, Zariski's lemma, Carathéodory extension; for coherent logic, Deligne's theorem that coherent toposes have enough points, Makkai and Reyes 1977).

Proposition 9.6 (Weak $\neq$ strong)**.** *[T6 Prop 1.6]* IPC *is sound and weakly complete, but not strongly complete, for the Boolean valuations* BV.

Proof idea. $p \vee \neg p$ holds in all of BV but is not IPC-provable; a maximal IPC-consistent theory contains excluded middle, so Theorem 9.5(c) applies (Appendix H.1; alternatively Glivenko's theorem, Glivenko 1929).

Lemma 9.7 (Classical points are maximal)**.** *[T6 reading of Prop 1.6, as repaired] Every point of* Cn_2 *is a maximal consistent theory. (Proof by cases; Appendix H.1.)*

Reading. Weak completeness needs the *maximal* theories to be realized, strong completeness *all* points; they coincide when every point is maximal, as in classical logic, while IPC has non-maximal points, realized only by Heyting or Kripke models. This mirrors Theorem 6.16: intuitionistic and classical logic have the same coherence data, and every coherent intuitionistic position *is* about a classical world, yet those worlds do not determine intuitionistic consequence.

Table 12 lists the family. The Nullstellensatz row is one of the note's two further pointers (the other is LP duality): the weak Nullstellensatz ("$1 = 0$ is not derivable from $f_1 = \cdots = f_m = 0$ by ideal manipulations iff the f_i have a common zero in $\bar{k}^n$") is weak completeness; the strong one, $I(V(J)) = \sqrt{J}$, is strong completeness, so the image is the radical ideals; the points are the maximal ideals, realized by Zariski's lemma. The rows typically carry a *non-standard caveat*,

existence only in a closure of the naive model class: $x^2+1=0$ is coherent over $\mathbb{Q}$ with no rational zero; $\{x \geq n : n \in \mathbb{N}\}$ has no finite refutation and no real solution; $2x=1$ is LP-coherent with no integer solution; and $\mathbb{R}[x]/(x^2+1)$, Cauchy's construction of the complex numbers, is a coherent way of talking about $\sqrt{-1}$ whose object is the Lindenbaum quotient.

9.2 Positions: bilateral completeness is Stone duality

Fix any set Fm; until Theorem 9.11 connectives play no role. A sequent $\Gamma \rhd \Delta$ has finite $\Gamma, \Delta \subseteq \mathrm{Fm}$; a *valuation* is any $v : \mathrm{Fm} \to \{0,1\}$, and $v \models \Gamma \rhd \Delta$ iff v falsifies some $\gamma \in \Gamma$ or verifies some $\delta \in \Delta$. A *position* $[X : Y]$ asserts all of X and denies all of Y; it is *incoherent* for a set $\vdash$ of sequents if some $\Gamma \rhd \Delta \in \vdash$ has $\Gamma \subseteq X$, $\Delta \subseteq Y$ (Restall 2005; Definition 6.22). A *Scott relation* is a set of sequents closed under (Ov) $\Gamma \rhd \Delta$ if $\Gamma \cap \Delta \neq \emptyset$; (Wk) from $\Gamma \rhd \Delta$ infer $\Gamma \cup \Gamma' \rhd \Delta \cup \Delta'$; (Cut) from $\Gamma, \varphi \rhd \Delta$ and $\Gamma \rhd \varphi, \Delta$ infer $\Gamma \rhd \Delta$. $\langle S \rangle$ is the least Scott relation containing S, and $\mathrm{Val}(\vdash)$ the set of valuations satisfying $\vdash$.

Theorem 9.8 (Bilateral Lindenbaum). *[T6 Thm 2.2] (known) Let $\vdash$ be a Scott relation.* (a) *Every coherent position $[X : Y]$ is realized by some $v \in \mathrm{Val}(\vdash)$: $v[X] = 1$, $v[Y] = 0$.* (b) *The maximal coherent positions are exactly the $[v^{-1}(1) : v^{-1}(0)]$ with $v \in \mathrm{Val}(\vdash)$.* Lean: Bilateral.thm_2_2_a Lean: Bilateral.thm_2_2_b

Corollary 9.9 (Strong bilateral completeness). *[T6 Cor 2.3] For every set S of sequents, the sequents valid in every valuation that satisfies S are exactly those in $\langle S \rangle$: $\mathrm{Th}(\mathrm{Mod}(S)) = \langle S \rangle$.* Lean: Bilateral.cor_2_3

Proof idea (Scott, 1974; Shoesmith and Smiley, 1978). Zorn gives a maximal coherent extension, since sequents are finite; (Ov) makes it disjoint, and (Cut) makes it exhaustive: if φ can be added to neither side, weaken the two witnessing sequents to a common context and cut on φ. Full proofs in Appendix H.2.

Theorem 9.10 (The duality). *[T6 Thm 2.4] (known) Give 2^{Fm} the product topology. The Galois connection between sets of sequents and sets of valuations has syntactic closure $S \mapsto \langle S \rangle$ and semantic closure $V \mapsto \overline{V}$, the topological closure. Hence $\vdash \mapsto \mathrm{Val}(\vdash)$ is a dual lattice isomorphism from Scott relations onto closed subsets of 2^{Fm}, with inverse* Th. Lean: Bilateral.scottClosedIso Lean: Bilateral.Val_Th

The syntactic half is Corollary 9.9, the semantic half is Lemma 6.23(a): each sequent excludes a basic clopen set. This is Stone duality (Stone, 1936) for the free Boolean algebra B on Fm, every formula an independent generator: $\Gamma \rhd \Delta$ is the clause $\bigvee_\Gamma \neg\gamma \vee \bigvee_\Delta \delta$ of B, and Scott relations, closed sets of valuations and filters of B correspond. So *at this level the connectives play no role*; they enter only through *which* closed set the rules carve out. This is Suszko's thesis seen as a duality: a Tarskian consequence is determined by the characteristic functions of its theories, which need not be truth-functional (Suszko 1977 stated it for structural logics).

Theorem 9.11 (Structural calculi and Lindenbaum bundles). *[T6 Thm 2.6] Let* Fm *be a term algebra and call* $\vdash$ structural *if it is closed under substitutions σ.* (a) *A Scott relation $\vdash$ is structural iff $\mathrm{Val}(\vdash)$ is closed under $v \mapsto v \circ \sigma$.* (b) *If $\vdash$ is structural, each matrix $L_v = \langle \mathrm{Fm}, v^{-1}(1) \rangle$, $v \in \mathrm{Val}(\vdash)$, validates $\vdash$ under every assignment; $\vdash$ is exactly the set of sequents valid in all L_v; and every coherent position is realized in some L_v by the identity assignment.* Lean: Bilateral.thm_2_6_a Lean: Bilateral.thm_2_6_b_realize *(fixed propositional signature)*

The proof (Appendix H.2) rests on $v \circ \sigma \models \Gamma \rhd \Delta$ iff $v \models \sigma\Gamma \rhd \sigma\Delta$.

Reading. Theorem 9.8 is the cleanest form of the note's proposition: a position is coherent iff it partially describes an admissible valuation, "some way the sentences could all come out". Theorem 9.11 upgrades the something to a compositional model whose algebra is the language itself, the multiple-conclusion analogue of Łoś and Suszko (1958) and Wójcicki (1988). Whether the *intended* semantics is reached depends on the rules: with the truth-table sequents and one coherence datum, $\mathrm{Val}(\vdash) = \mathrm{BV}$ (Theorem 6.26); with single-conclusion data only, the admissible valuations include the "gappy" $\mathrm{BV}^{\cap}$ (Theorem 6.24), so a single-conclusion coherence loss licenses talk about undecided worlds.

9.3 Credences: de Finetti, counting sequents, and why local constraints fail

Let F be a finite agenda that is a union of *complementary pairs* $\{\varphi, \varphi^*\}$, $\varphi^* \equiv \neg\varphi$ (a finite agenda cannot be closed under $\neg$); BV the Boolean valuations of its atoms; $\bar{v} = (v(\varphi))_{\varphi \in F}$. A credence $P : F \to [0,1]$ is *negation-coherent* if $P(\varphi^*) = 1 - P(\varphi)$, and *coherent* if $P = \sum_v \mu(v)\bar{v}$ for a probability μ on BV, i.e. P lies in the *coherence polytope* $\Pi_F = \mathrm{conv}\{\bar{v}\}$. A *counting sequent* (Φ, m) is a finite multiset Φ of members of F with an integer *threshold* m; it is *valid* if every $v \in \mathrm{BV}$ makes at least m members of Φ true (with multiplicity), and P *satisfies* it if $\sum_{\varphi \in \Phi} P(\varphi) \geq m$. An ordinary sequent $\Gamma \rhd \Delta$ is the case $m = 1$, $\Phi = \{\gamma^* : \gamma \in \Gamma\} \cup \Delta$.

Theorem 9.12 (de Finetti, with counting sequents)**.** *[T6 Thm 3.1, as repaired] For negation-coherent P the following are equivalent:* (i) *P is coherent;* (ii) *there is no Dutch book, i.e. no $\lambda \in \mathbb{R}^F$ with $\lambda \cdot (\bar{v} - P) < 0$ for all $v \in \mathrm{BV}$;* (iii) *for rational λ, c, if $\lambda \cdot \bar{v} \geq c$ for all v then $\lambda \cdot P \geq c$;* (iv) *P satisfies every valid counting sequent. Negation coherence is needed: $P \equiv 1$ satisfies (iv) but is incoherent.*

Proof idea. Π_F is a rational polytope, so a point outside it violates one of finitely many rational defining inequalities, which gives a Dutch book; clearing denominators and moving negative coefficients from φ to φ^* turns the inequality into a counting sequent. Appendix H.3.

Reading. The probabilistic completeness theorem says a credence is coherent iff it is a mixture of worlds. Its proof-theoretic side is not ordinary sequents but *counting sequents*, graded positions such as "at least two of these six claims hold". An ordinary sequent gives only the union bound $\sum_\Gamma (1 - P(\gamma)) + \sum_\Delta P(\delta) \geq 1$, the multiple-conclusion form of Adams's uncertainty-sum bound (Adams, 1966).

Which consistency constraints suffice? The motivating notes on a consistency-based probing project[6] record candidate constraints: negation coherence $P(\neg Q) = 1 - P(Q)$ (the CCS loss), the identity $P(A \wedge B) + P(A \vee B) = P(A) + P(B)$ (listed alongside a collaborator's suggestion), and, tentatively ("one choice would e.g. be"), modus ponens read as the tautology $\neg P \vee \neg(P \to Q) \vee Q$, i.e. the union bound $P(Q) \geq P(P) + P(P \to Q) - 1$. Contrast-consistent search (CCS) trains a probe with exactly a negation-consistency loss plus a confidence loss (Burns et al., 2023).

Example 9.13 (Small agendas)**.** [T6 Prop 3.2, Ex 3.3](computed) On $\{p, q, p \wedge q\}$ plus complements, $P(p) = P(q) = 0.2$, $P(p \wedge q) = 0.9$ is negation-coherent but violates the union bound of $p \wedge q \rhd p$. For $(P(p), P(q), P(p \to q)) = (a, b, c)$ the coherence polytope is the tetrahedron with facets

[6] `ai/DLK/`, in particular `DLK notes.md` and `june meeting notes.md`.

$a+c \geq 1$ (from $\triangleright p, p \to q$), $b \geq a+c-1$ (modus ponens), $b \leq c$ (from $q \triangleright p \to q$) and the trivial $c \leq 1$: on this tiny agenda the MP constraint considered in those notes, two other union bounds and a trivial bound *are* coherence.

This does not scale. For $k \geq 3$ let $F_k = \{A_i, \neg A_i : i \leq k\} \cup \{A_i \wedge A_j, \neg(A_i \wedge A_j) : i < j \leq k\}$, and $P_k(A_i) = \frac{1}{k-1}$, $P_k(A_i \wedge A_j) = 0$, with complements on negations.

Theorem 9.14 (No bounded-arity family of constraints suffices)**.** *[T6 Thm 3.4]* (a) *For every $F' \subseteq F_k$ whose formulas jointly mention at most $k-1$ atoms, $P_k|_{F'}$ extends to a coherent credence on F_k.* (b) *P_k is incoherent. Hence no family of necessary coherence constraints each depending only on sub-agendas over fewer than k atoms characterizes coherence; in particular pairwise (CCS-style) constraints fail, and so does every bounded arity.* *Lean: Specker.thm_3_4*

Proof. (a) With J the mentioned atoms, put mass $\frac{1}{k-1}$ on each world making exactly one A_j, $j \in J$, true, and the rest, $1 - \frac{|J|}{k-1} \geq 0$, on the all-false world. (b) A representing μ would make the A_i pairwise disjoint, so $\mu(\bigvee_i A_i) = \frac{k}{k-1} > 1$. □

This persists if F_k is enlarged to all formulas in at most two atoms, valued by the two-atom marginals of the μ in (a), which do not depend on J: every pair of atoms has a genuine joint distribution, yet P_k is globally incoherent. This is the frustration in Specker's parable of the three boxes (Specker, 1960).

The hierarchy by *threshold* is strict too: P_k violates a repetition-free valid counting sequent of threshold $k-1$ but satisfies every valid one of threshold at most $k-2$, including the union bounds of all valid ordinary sequents (Theorem H.1). For negation-coherent credences any *single* agenda is characterized by finitely many counting sequents, read off its facets; what is ruled out is a bound on arity or threshold that works uniformly across agendas. On agendas forming a finite Boolean algebra up to equivalence, the three-formula identities $P(\varphi) = P(\varphi \wedge \psi) + P(\varphi \wedge \neg\psi)$ with $P \geq 0$, $P(\top) = 1$ suffice (Proposition H.2). Both results are in Appendix H.3.

A caution for consistency-based probing. A CCS-style probe reads credences off a model on an *unclosed* list of statements and enforces negation consistency, perhaps with further local constraints. By Example 9.13 and Theorems 9.14 and H.1, no fixed stock of constraint *types* (pairwise, k-wise for fixed k, or all union bounds of ordinary sequents) guarantees that the probe's credences are a mixture of worlds, i.e. "about" anything at all; that needs the agenda's facets, or an agenda closed under the connectives. And even global coherence buys only *some* distribution over *some* worlds (Theorem 9.15): coherence is eliminative, not confirmational (Section 6.7). An untested experiment: train probes with negation-only, pairwise and counting-sequent losses on agendas with k-way exclusive alternatives, and measure their distance from the coherence polytope.

First-order credences. For a countable first-order language L, $P : \mathrm{Sent}_L \to [0,1]$ is *(Gaifman-)coherent* if (G1) $P(\varphi) = 1$ whenever $\vdash \varphi$, and (G2) $P(\varphi \vee \psi) = P(\varphi) + P(\psi)$ whenever $\vdash \neg(\varphi \wedge \psi)$ (Gaifman, 1964); this is the notion of the draft cited in the note (Christiano et al., 2013).

Theorem 9.15 (Coherent credences are mixtures of complete theories)**.** *[T6 Thm 3.7]* *(known) Let S_L be the Stone space of complete consistent L-theories, with clopens $[\varphi] = \{T : \varphi \in T\}$. Then P is coherent iff $P(\varphi) = \mu([\varphi])$ for a (unique) Borel probability μ on S_L.*

Proof idea. P is a finitely additive probability on the clopen algebra; by compactness this is automatically countably additive there, and Carathéodory extends it (Appendix H.3). By Gödel completeness, a coherent credence is the expected truth value of φ in a random model: the note's proposition holds verbatim. So does the notes' caveat: for complete $T \supseteq \mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$, δ_T is coherent, gives 1 to every true quantifier-free sentence, and "believes" there is a proof of $0 = 1$.

Theorem 9.16 (The Gaifman condition is a probabilistic ω-rule)**.** *[T6 Thm 3.8, as repaired] In the language of arithmetic, let P be coherent, give probability 1 to every true quantifier-free sentence, and satisfy the* Gaifman condition $P(\exists x\, \varphi(x)) = \sup_n P(\bigvee_{i \le n} \varphi(\underline{i}))$. *Then $P = \delta_{\mathrm{Th}(\mathbb{N})}$. No arithmetic axioms are needed.*

Proof idea. Induction on logical symbols: coherence forces $\{0,1\}$-values through $\neg, \wedge$; numerals carry no logical symbols, and every number is a numeral's value (Appendix H.3). This is essentially a consequence of Gaifman (1964): a probability with the Gaifman property is determined by its quantifier-free values, here the true diagram of $\mathbb{N}$.

Corollary 9.17 (Logical inductors' limits are non-standard)**.** *[T6 Cor 3.9, as repaired] Let $\mathbb{P}$ be a logical inductor (Garrabrant et al., 2016) over a consistent r.e. $\Gamma \supseteq \mathsf{Q}$ in the language of arithmetic. Then $\mathbb{P}_\infty$ violates the Gaifman condition. More generally, no limit-computable coherent credence giving 1 to every true quantifier-free sentence satisfies it; the quantifier-free hypothesis cannot be dropped.*

Proof idea. By Convergence and Limit Coherence, $\mathbb{P}_\infty$ exists, is coherent and gives 1 to true quantifier-free sentences, so Theorem 9.16 would make it $\delta_{\mathrm{Th}(\mathbb{N})}$. Non-Dogmatism refutes this (the false member of a Gödel–Rosser pair gets positive probability), and so, independently, does Tarski's theorem, as $\mathrm{Th}(\mathbb{N})$ would be Δ_2. Appendix H.3.

Reading. The probabilistic "something" is always available; a *standard* something is fixed by an infinitary condition that no limit-computable coherent credence of this kind meets. A Gaifman-style probe of a language model's arithmetic would compare $P(\exists x\, \varphi(x))$ with $\max_{i \le n} P(\varphi(\underline{i}))$: a large gap marks belief in a witness that is never named.

Finally, the note's "vnm stuff" has the same anatomy: finitely many strict and weak comparisons of lotteries have an expected-utility representation iff no nonnegative combination of them derives $L \succ L$ by mixing and transitivity (Proposition H.4, by Motzkin's transposition theorem).

9.4 Contexts: a completeness theorem for "true in the context at hand"

What is a coherent *system of contexts* (Section 11) about? We use a monotone multi-context calculus whose labelled formulas $i{:}\varphi$ are McCarthy's $\mathrm{ist}(i, \varphi)$ (McCarthy, 1993), with a simplified local-models semantics (Ghidini and Giunchiglia, 2001).

Definition 9.18 (Multi-context systems, MC, belief states)**.** [T6 Def 4.0, as repaired; §4.1] For each context $i \in I$ fix a language L_i, local models M_i with $C_i = \mathrm{Th}_i \circ \mathrm{Mod}_i$ (e.g. classical logic; for a *learned* logic C_i, the proper theories $M_i = \mathrm{Fix}(C_i) \setminus \{L_i\}$ with $T \models_i \varphi$ iff $\varphi \in T$), local axioms K_i, and a falsum $\bot_i$ with $\mathrm{Mod}_i(\bot_i) = \emptyset$. A *bridge rule* is $j_1{:}\varphi_1, \ldots, j_n{:}\varphi_n \Rightarrow i{:}\psi$. For labelled premises Γ, $\mathrm{Der}(\Gamma) = (T_i)_i$ is the least family with $T_i \supseteq K_i \cup \Gamma_i$, each T_i C_i-closed, and closed under bridges; $\Gamma \vdash_{\mathrm{MC}} i{:}\varphi$ iff $\varphi \in T_i$. A *belief state* is $c = (c_i)$, $c_i \subseteq M_i$, with $c \models i{:}\varphi$ iff every $m \in c_i$ satisfies φ. It is a *model* if it satisfies every K_i and is *bridge-compatible* (if $c \models j_k{:}\varphi_k$ for all premises then $c \models i{:}\psi$); $\Gamma \models_{\mathrm{LMS}} i{:}\varphi$ iff every model of Γ satisfies $i{:}\varphi$.

Theorem 9.19 (Soundness and completeness; the canonical chain)**.** *[T6 Thm 4.1] Let* $c_i^\Gamma = \mathrm{Mod}_i(T_i)$ *for* $(T_i) = \mathrm{Der}(\Gamma)$*.* (a) c^Γ *is a model of* Γ*.* (b) *Every model* c *of* Γ *has* $c_i \subseteq c_i^\Gamma$*:* c^Γ *is the least committal model.* (c) $\Gamma \vdash_{\mathrm{MC}} i{:}\varphi$ *iff* $\Gamma \models_{\mathrm{LMS}} i{:}\varphi$*.*

Proof idea. $c^\Gamma \models i{:}\varphi$ iff $\varphi \in \mathrm{Th}_i\mathrm{Mod}_i(T_i) = T_i$; and the theories $\mathrm{Th}_i(c_i)$ of any model form a closed, bridge-closed family containing $K \cup \Gamma$, hence contain $\mathrm{Der}(\Gamma)$. Appendix H.4.

Corollary 9.20 (Context coherence iff existence)**.** *[T6 Cor 4.2] Let* $D \subseteq I$ *be the* designated *contexts (Definition 2.13).* $\Gamma \nvdash_{\mathrm{MC}} i{:}\bot_i$ *for every* $i \in D$ *iff some model* c *of* Γ *has* $c_i \neq \emptyset$ *for every* $i \in D$*.*

Proof. $\bot_i \in T_i$ iff $c_i^\Gamma = \emptyset$ by Theorem 9.19(a), and c^Γ is the largest model by (b). □

Remark 9.21 (Learned local logics). [T6 remark after Cor 4.2, as repaired] With the semantics $M_i = \mathrm{Fix}(C_i) \setminus \{L_i\}$, $c_i^\Gamma \neq \emptyset$ iff $T_i \neq L_i$, so for learned local logics designated coherence must be read as *non-triviality* of T_i; this coincides with $\bot_i \notin T_i$ exactly when $\bot_i$ is C_i-explosive.

Non-designated contexts, such as suppositions under reductio, may have $c_i = \emptyset$: they are about nothing, which is what a successful reductio shows; this is the designation discipline of Section 6 as an existence theorem. The canonical chain is the least equilibrium in the sense of Brewka and Eiter (2007); with non-monotone (default) bridges equilibria can fail to exist even when every K_i is consistent, so coherence alone does not guarantee existence there (Proposition H.5, Appendix H.4).

Rules of proof versus conditionals between worlds. A *world family* $w = (w_i)$ picks a *single* model $w_i \in M_i$ of K_i for each i and satisfies each bridge as a *material conditional*: if $w_{j_k} \models \varphi_k$ for all k then $w_i \models \psi$. Write $\models_W$ for consequence over world families.

Theorem 9.22 (Belief states, not worlds)**.** *[T6 Thm 4.5, as repaired]* (a) $\models_{\mathrm{LMS}} \subseteq \models_W$*.* (b) *The inclusion is strict: with classical local logics,* $K_j = \{p \vee q\}$*,* $K_i = \{\neg r\}$ *and bridges* $j{:}p \Rightarrow i{:}r$*,* $j{:}q \Rightarrow i{:}r$*, MC does not derive* $i{:}\bot$*, but there is no world family.* (c) *For classical propositional local logics,* $\models_W$ *is classical consequence in the disjoint union of the local languages from the tagged axioms* $K_i^{(i)}$*, the tagged premises and the bridge axioms* $\bigwedge_k \varphi_k^{(j_k)} \to \psi^{(i)}$ *(the "eternalist" translation).*

Proof idea. In (b), $T_j = \mathrm{Cn}_2(p \vee q)$ contains neither p nor q, so no bridge fires; but every w_j satisfies p or q, forcing $w_i \models r$ (Appendix H.4).

This is the context-logic instance of a standard distinction, rule of proof versus conditional (global versus local consequence): $c \models j{:}\varphi$ is a box over c_j, and (b) is the failure of $\Box(p \vee q) \models \Box p \vee \Box q$; it is why Ghidini and Giunchiglia (2001) use *sets* of local models. New is only its use as a criterion for physics bridges.

Reading. MC reads a bridge as "export what context j *establishes*"; the eternalist translation reads it as a conditional between worlds. Only the first is right for idealizations: a physics bridge ("if the idealized model establishes $Q = q$, then in the world $Q \approx q \pm \varepsilon$", Section 11.4) is certified for what the idealization establishes, not for case splits over its undetermined possibilities, none of which is actual. So *a context system makes sense iff there is something it could be about* holds with belief states as the somethings (Corollary 9.20) and fails with worlds (Theorem 9.22(b)).

Local coherence versus a global world. The motivating notes hold that claims always sit in background frames, that "there are many different background contexts, not one, and it is very useful that this is so", and that "these different frames needn't be easily reconcilable".[7] The following makes precise when locally coherent frames nevertheless glue into one world, and when they cannot. On a finite tree of classical propositional contexts with running intersection and mutually conservative edges, every local model extends to a global valuation (Theorem H.6, Appendix H.4; the join-tree argument, cf. Beeri et al. 1983). On a cycle this fails:

Proposition 9.23 (The frustrated triangle)**.** *[T6 Prop 4.7] On atoms* $\{a,b\},\{b,c\},\{c,a\}$ *let* $T_{ab} = \mathrm{Cn}_2(a \leftrightarrow \neg b)$, $T_{bc} = \mathrm{Cn}_2(b \leftrightarrow \neg c)$, $T_{ca} = \mathrm{Cn}_2(c \leftrightarrow \neg a)$. *Each is consistent, each pair is mutually conservative on its shared atom, and the union is inconsistent; likewise pair marginals uniform on* $\{01, 10\}$ *agree on all single marginals but have no joint distribution.* Lean: Specker.frustrated_triangle *(probabilistic part)*

Proof. An odd cycle of negations has no Boolean solution; each theory has only tautologies in a single atom. □

This is Specker's parable in logical form (Specker, 1960) and the simplest instance of *contextuality*: local sections agreeing on overlaps with no global section (Abramsky and Brandenburger, 2011); acyclicity of the cover rules it out (Beeri et al., 1983). So an olympiad solution's contexts need not be reconcilable into one world. Coherence of the designated contexts guarantees local belief states, nonempty where designated (Corollary 9.20); one world for all contexts is strictly stronger and cover-dependent.

Definition 9.24 (True in the context at hand)**.** [T6 Def 4.8, as repaired] φ is *true in context* c iff $\Gamma \vdash_{\mathrm{MC}} c{:}\varphi$, i.e. (Theorem 9.19) φ holds in every local model that c's canonical belief state leaves open. φ is *true enough in* c *for a world claim* ε iff (i) φ is true in c; (ii) a bridge $c{:}\varphi \Rightarrow @{:}\varepsilon$, possibly with side conditions σ, is in the system and σ holds in the actual world (for a checker, $\Gamma \vdash_{\mathrm{MC}} @{:}\sigma$); and (iii) that bridge is *sound* in the sense of Definition 11.13: every instance of it preserves truth at every world $w \in W_\rho$ admitted by the background, i.e. whenever c establishes φ and σ holds at w, ε holds at w. (Quantifying over the actual world alone would make (iii) equivalent to the truth of ε.)

This is the multi-context counterpart of the answer given in Theorem 11.29: truth in the context at hand is supervaluational truth over the context's admissible local models, and a checker accepting exactly MC-derivations accepts exactly what is true in this sense. It does not justify bridges: bridge soundness is world-relative and is certified by stability arguments or world feedback (Section 11.4), never by coherence, which can only demand $c_i^\Gamma \neq \emptyset$ at designated contexts. The context-tree calculus[L7 §8] from which Definition 11.13 is adapted is an MC system, and its own semantics agrees with ours when every step and export instance is sound (proof sketch; Appendix H.4).

9.5 What coherence of a learned calculus buys, and what pins the model down

We return to the learner. Coherence of a learned structural calculus on a designated context always yields a model, but a syntactic one (Theorem 9.25); it reduces to the intended one

[7] `philosophy/philosophy of thinking/a few notes on frames.md`.

exactly when the intended semantics is a surjective image of the language (Theorem 9.26). Compactness and computability bound how far coherence can pin down an intended infinite structure (Theorems H.8 and H.9).

Let R be a learned set of rule schemas, C_R the least structural finitary consequence operation containing it, and $\langle R\rangle$ the least structural Scott relation. The designated contexts $A \in \mathcal{A}$ are certified coherent: $\bot \notin C_R(A)$, or $[A : D]$ is $\langle R\rangle$-coherent. As in Section 6, this is all the learner knows.

Theorem 9.25 (What coherence buys)**.** *[T6 Thm 5.1] For each designated A:* (a) *some C_R-theory $T \supseteq A$ is maximal among theories omitting $\bot$;* (b) *the* Lindenbaum matrix $\langle \mathrm{Fm}, T\rangle$ *validates every rule of C_R under every assignment, and the identity assignment designates all of A and not $\bot$ (Łoś and Suszko, 1958; Wójcicki, 1988);* (c) *bilaterally, each designated coherent $[A : D]$ is realized by some $v \in \mathrm{Val}\langle R\rangle$, and L_v is a matrix model of $\langle R\rangle$.*

Proof idea. (a) is the Zorn step of Theorem 9.5(a), (b) is structurality, and (c) is Theorems 9.8 and 9.11; (b) holds for any C_R-theory $T \supseteq A$ omitting $\bot$ (Appendix H.5).

So *if a learned mode of talking is coherent on a context, there is something it is talking about there.* But the model's universe *is the language*: the formal content of the notes' caveat about the "proof" of the Gödel sentence. When is the syntactic something the intended one? For a matrix $\langle \mathbf{A}, D\rangle$ the *Leibniz congruence* $\Omega_{\mathbf{A}}(D)$ is the largest congruence compatible with D ($a\,\theta\,b$ and $a \in D$ imply $b \in D$), and the *reduction* $\langle \mathbf{A}/\Omega, D/\Omega\rangle$ identifies elements no formula context can tell apart (Blok and Pigozzi, 1989; Font, 2016); write $\Omega(T)$ for $\Omega_{\mathrm{Fm}}(T)$.

Theorem 9.26 (The syntactic model collapses to the intended one)**.** *[T6 Thm 5.2]* (a) *If $h : \mathrm{Fm} \to \mathbf{A}$ is a* surjective *homomorphism and $T = h^{-1}(D)$, then*

$$\langle \mathrm{Fm}/\Omega(T),\, T/\Omega(T)\rangle \;\cong\; \langle \mathbf{A}/\Omega_{\mathbf{A}}(D),\, D/\Omega_{\mathbf{A}}(D)\rangle;$$

if $\langle \mathbf{A}, D\rangle$ is reduced, the reduced Lindenbaum matrix of T is $\langle \mathbf{A}, D\rangle$. (b) *Every maximal consistent theory of* CPC *has reduced Lindenbaum matrix $\cong \langle \mathbf{2}, \{1\}\rangle$.*

Proof idea. $\Omega(T) = h^{-1}(\Omega_{\mathbf{A}} D)$, because a congruence compatible with T, joined with $\ker h$, pushes forward along h to one compatible with D (Appendix H.5). Surjectivity matters: in general one gets the reduction of the *generated* submatrix.

Proposition 9.27 (What the points of a finite-matrix logic talk about)**.** *[T6 Prop 5.3, as repaired] Let C be the consequence of a finite matrix $\langle \mathbf{A}, D\rangle$. Every point of C has reduced Lindenbaum matrix isomorphic to the reduction of a generated submatrix $\langle h(\mathrm{Fm}), D \cap h(\mathrm{Fm})\rangle$ (a strict homomorphic image of a submatrix, not necessarily a submatrix); up to isomorphism there are finitely many.*

Proof idea. Every point is some $h^{-1}(D)$ by Theorem 9.5(b); apply Theorem 9.26(a) to h onto its image (Appendix H.5).

So for classical logic every coherent position is, after identifying indiscernibles, about the two truth values, because every point of CPC is maximal (Lemma 9.7) and maximal theories are Boolean. The reason is *not* Post-completeness: IPC is not Post-complete, yet its maximal consistent theories reduce to $\mathbf{2}$ (Proposition 9.6); calculi differ in their non-maximal points.

Example 9.28 (The residue as a set of things talked about)**.** [T6 Ex 5.4] (i) Map Fm onto the three-element Heyting chain $\mathbf{H}_3 = \{0 < \frac{1}{2} < 1\}$ with $h(p) = \frac{1}{2}$. Then $T = h^{-1}(1)$ is a non-maximal point of IPC (maximal among theories omitting p) whose reduced matrix is $\langle \mathbf{H}_3, \{1\}\rangle$:

one coherent intuitionistic position talks about the three-element chain while the maximal ones talk about **2**. (ii) First-order logic with equality plus the fallacy $\forall x \exists y\, R \rhd \exists y \forall x\, R$ is coherent (Example E.4); its models are exactly the one-element structures, which is what the learned fallacy is "about".

In general, for multiple-conclusion hypotheses, Theorems 9.10 and 9.11 put the residue $\mathrm{Alt}(h^*, \mathcal{A})$ of coherent uniform alternatives (Theorem 6.34) in exact correspondence with the closed, substitution-invariant $V \subseteq \mathrm{Val}(h^*)$ realizing every designated position. *Non-identifiability of meaning is non-uniqueness of the thing talked about.*

What pins the intended model. Two layers of non-standardness remain. Coherent but false theories such as $\mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$ have a prime model containing a definable non-standard "proof of $0 = 1$" (Theorem H.7). True theories need not be categorical, since by compactness no first-order mode of talking pins down an infinite structure (Theorem H.8). For any computable coherence notion, "coherent iff there is something it could be talking about" can hold only if having such a something is Π_1 (Theorem H.9): "having some model" is Π_1, while "having the standard model", "having an integer solution" and "having a finite model" are not. Anchoring a vocabulary pins, uniformly over anchorings, exactly what is definable from it (Proposition H.10). Every known pin of $\mathbb{N}$ (second-order induction, initiality, the ω-rule, and Tennenbaum's theorem, Theorem H.11) adds something beyond finitary computable coherence (Appendix H.6).

9.6 Verdict

Where yes. In each row of Table 12, "coherent iff there is something it could be talking about" is a theorem, with a realization step that is non-trivial except for positions, where points already are valuations; for the modes of talking that matter to the learner, see Theorems 9.8, 9.12, 9.15 and 9.25, Proposition H.4, and Corollary 9.20. The adjunction slogan holds in each strongly complete case (Proposition 9.3 and Theorem 9.10).

Where no. (1) The something may be made of syntax (Theorems 9.4 and 9.25); the proposition is substantive only for an independently specified semantics, and which semantics is right is answered by which models we can independently access: the truth values, $\mathbb{N}$ through computation, the world through observation. (2) Weak $\neq$ strong (Proposition 9.6). (3) Intended existence needs more than coherence: the coherent-but-false layer is defeated only by truth-tracking principles (reflection, stronger theories, Popperian policies, Theorem 6.21), the non-categorical layer only by infinitary, second-order or meta-level ingredients, and both limits are forced (Theorems H.8 and H.9). (4) The something for a context system is a family of belief states, not of worlds (Theorem 9.22), and a global world exists automatically only on tree-like covers with conservative overlaps (Theorem H.6 and Proposition 9.23). (5) Probabilistic coherence is global: local constraints on an unclosed agenda need not make a probe about any mixture of worlds (Theorems 9.14 and H.1).

What this means for the learner. The notes ask about "the hooking of a model onto the world";[8] against coherence this becomes two precise questions. Coherence answers "is there *any* world this way of talking could be about?": yes, canonically, but partly syntactically. Hooking

[8] `philosophy/philosophy of language/meaning and truth/individual sentences being meaningful.md`.

answers "is it *this* world?": world feedback is positive data in valuation space (Section 6.4), and uniformly over anchorings it pins exactly the definable vocabulary (Proposition H.10). Neither does the other's job: as in Sections 6, 7 and 11, coherence removes what is incoherent with the target, and only world channels or extra pins select among the non-isomorphic somethings of its residue Alt. A draft comment in the notes, marked "text below now disendorsed probably", imagines a conception (a C-model) given with "axioms and inference rules" for which no model can be constructed.[9] In our terms such a conception is an incoherent calculus used only through coherent chunks: local belief states (Corollary 9.20) without a global section (Proposition 9.23). And by the computability barrier there is no final formula for $\mathbb{N}$: further pins, such as reflection, must be added on other grounds, as in the open-ended picture of justification of Section 13.

Verdict. The philosophical proposition is *true in a precise generalized sense*: for positions, credences, preferences, context systems and learned structural calculi, coherence is equivalent to the existence of a model of the kind the semantics allows. It does not deliver *intended-model* existence, which the notes' Gödel-sentence footnote already anticipates: standard or intended existence needs categoricity, anchoring or an infinitary condition, and no computable coherence notion can certify it beyond Π_1.

10 Informal mathematics: learning a latent formalization

This section is the paper's answer to the second success criterion: a setup that *would have worked for mathematical proofs before anyone knew how to formalize them.* The motivating notes (Hänni, 2026) ask it in two sharper forms.[10] "How come mathematicians ended up with such a nice notion of proof? ... this nice structure of a formal proof is sorta there in their activities. how did it come to be there?" And, offered as "an attempt at an explanation" for the case of early theorems about functions: mathematicians used only the part of the concept that later survived formalization, because "the 'rest of the concept' ... was untrustworthy/messy ... and for this reason unusable whenever people were proving things". We call its generalization to pre-formal practice the *robust-core hypothesis*.

We model a pre-formal practice as the shadow of an unknown *latent formalization*: a language, a calculus with a library, a partial *reading* of informal sentences into the language, and an intended semantics (Section 10.1). We show that a verifier checking steps *and* re-uses of sentences across contexts against every surviving latent formalization is sound against adaptive provers (Section 10.2); that practice identifies meaning at best up to sameness of informal consequence, and ontology not at all (Section 10.3); that counterexample objects can be exponentially more informative than paradoxes (Section 10.4); that Frege → Russell → Zermelo is a theorem about simplicity, coherence and coverage in a toy (Section 10.5); and that steps valid under every admissible sharpening survive every formalization, while selection by objects plus Gödel completeness explains why surviving steps are derivable, though not why they are short (Section 10.6).

Two caveats govern the section. Every positive result assumes *realizability* (the target lies in the learner's fixed class), whereas the decisive historical steps (ε-δ, uniform convergence,

[9] `philosophy/philosophy of thinking/analogy/logical models as distinct from mental models.md`.

[10] The quotations in this paragraph are from `philosophy/emergence of structures/how come mathematicians ended up with such a nice notion of proof?.md` and `philosophy/metaphilosophy/why is it a good idea to make vague notions precise?.md`.

ideals) were inventions of language; *affordable soundness with language invention* is the hardest open problem we leave (Section 14). And most results are simple once the definitions are in place. The genuine mathematical content is the paradox side of the object–paradox separation (Theorem 10.20 and Proposition 10.22; its novelty relative to the Littlestone-dimension literature is uncertain) and the comprehension-toy facts (Lemma 10.27); the deep ingredients are cited (Gödel completeness, Robinson's transfer, Nelson's conservativity, Incurvati and Murzi's theorem).

10.1 The model

Definition 10.1 (Occurrences, steps, arguments, links)**.** [T4 §1.1] Ξ is a countable set of informal sentence strings and C a set of *contexts* (definitions, conventions, notation and local suppositions in force). An *occurrence* is $x = (c, \xi) \in X := C \times \Xi$, with $\mathrm{str}(x) := \xi$. An *informal step* is $\Gamma \rhd y$ with $\Gamma \subseteq_{\mathrm{fin}} X$, $y \in X$. An *informal argument* α is a list of lines $y_1, \dots, y_m \in X$, each a *premise*, an *inference* $\Gamma_j \rhd y_j$ with Γ_j among earlier lines, or a *link* $y_i \rightsquigarrow y_j$ with $i < j$ and $\mathrm{str}(y_i) = \mathrm{str}(y_j)$ ("the same claim, used in another context"). $\mathrm{Prem}(\alpha)$ is its set of premises, y_m its conclusion, and its depth the length of its longest dependency path.

Links are the formal locus of *equivocation*: a sentence proved in one context is re-used in another, where it may be read differently.

Definition 10.2 (Latent formalization)**.** [T4 Def 1.1, §1.2] A *latent formalization* is a quadruple $h = (L_h, R_h, \rho_h, \mathbb{M}_h)$ consisting of a countable language L_h with a sentence $\bot$; a set R_h of formal steps $\Pi \rhd \varphi$ over $\mathrm{Sent}(L_h)$ (instances of finitely many schemas plus a library of definitions and derived lemmas); a partial *reading* $\rho_h : X \rightharpoonup \mathrm{Sent}(L_h)$ with domain D_h; and a class $\mathbb{M}_h$ of L_h-structures for which R_h is sound and $\bot$ is false.

(i) $\Gamma \rhd y$ is *valid under* h, written $(\Gamma \rhd y) \in \models_h$ or $\Gamma \models_h y$, iff $\Gamma \cup \{y\} \subseteq D_h$ and $\rho_h(y) \in \mathrm{Cl}_{R_h}(\rho_h \Gamma)$.

(ii) It is *g-valid*, $(\Gamma \rhd y) \in \mathrm{St}_h^g$, iff moreover $\rho_h(y)$ has an R_h-derivation from $\rho_h\Gamma$ with at most g rule applications and formulas of size bounded by a fixed function of g and the input. We assume ρ_h computable with decidable domain, R_h decidable, and derivations restricted to the input's symbols plus a finite, computably given part of L_h depending on g; then St_h^g is decidable and $\mathrm{St}_h^g \subseteq \models_h$, which is all that is used.

(iii) A link $y_i \rightsquigarrow y_j$ is *valid under* h iff $(\{y_i\} \rhd y_j) \in \mathrm{St}_h^g$ (typically $\rho_h(y_i) = \rho_h(y_j)$: zero applications).

(iv) The *admissible valuations* are $\mathcal{V}_h := \{v_M \circ \rho_h : M \in \mathbb{M}_h\}$, with $v_M(\varphi) = [M \models \varphi]$.

(v) h is *first-order complete* if L_h is first-order, $\mathbb{M}_h = \mathrm{Mod}(T_h)$, and $\mathrm{Cl}_{R_h}(\Gamma) = \{\varphi : T_h \cup \Gamma \models \varphi\}$.

De Bruijn's ratio of formal to informal proof size is roughly constant (about 4, Wiedijk 2000) only relative to a definitional library, as Mathias's 4.5×10^{12}-symbol term for Bourbaki's "1" shows (Mathias, 2002); so g is measured over R_h with its library.

Assumption 10.3 ((BG), bounded gap)**.** [T4 §1.3] There is a target $h^* = (L^*, R^*, \rho^*, \mathbb{M}^*)$, with $D^* := D_{h^*}$, such that (BG1) $h^* \in \mathcal{H}$ (*realizability*); (BG2) each accepted human step is, with probability $1-\eta$, drawn from a distribution over $\mathrm{St}_{h^*}^g$, and otherwise noise, sporadic or systematic (instances of a fallacy schema); (BG3) human links are h^*-valid up to the same noise.

Steps that are valid but not g-short are routed out of the practice stream: they are escalated with a request to expand them (Proposition 10.11), or count as noise.

Definition 10.4 (The three channels)**.** [T4 §1.4] (P) *Practice*: the stream of (BG2); escalation answers are practice data labelled by $\mathrm{St}^g_{h^*}$. (C) *Coherence*: a family $\mathcal{A}$ of finite *designated contexts* $A \subseteq X$ with the promise $\bot \notin \mathrm{Cl}_{R^*}(\rho^* A)$; a *paradox for* h is an R_h-derivation of $\bot$ from $\rho_h A$, $A \in \mathcal{A}$, and refutes h. Suppositions are not designated, so reductio is never penalized. (O) *Objects*: a finite partial valuation $v : X \rightharpoonup \{0,1\}$ with the promise $v \subseteq u$ for some $u \in \mathcal{V}_{h^*}$; it refutes h iff $v|_{D_h}$ extends to no member of $\mathcal{V}_h$. Interactively, the learner holds an object and queries its values ("checks the step on an example"). *Computation* is the special case of an intended object $M_0 \in \mathbb{M}^*$ queried on a decidable fragment, like Euler's six-decimal value of $\sum 1/n^2$. *Convention*: $\rho_h A := \rho_h(A \cap D_h)$, so a sentence h cannot read asserts nothing under h.

(O) is the world channel (W) of Definition 2.13 inside mathematics: objects (W2), with computation (W1, W3) as a special case. The model fixes ρ^* (Section 10.6 makes it vague instead) and the class of languages.

10.2 Sound bounded-gap verification

At time t, h is *consistent with the data* if it labels every practice step and escalation answer correctly by St^g_h, agrees with every link answer, and has not been refuted by a paradox or an object datum found within the search budget so far. VS_t is the set of such h. Every refutation is a certificate, so budgeted search never refutes an unrefutable hypothesis.

Definition 10.5 (Bounded-gap conservative verifier)**.** [T4 Def 2.1] On a query q (an informal step or a link), V^g_t answers ACCEPT iff q is g-valid, resp. link-valid, under *every* $h \in \mathrm{VS}_t$, with a certificate for each; otherwise ESCALATE (ask the oracle, or ask the prover to expand q).

V^g_t is computable for finite $\mathcal{H}$; for infinite $\mathcal{H}$ unanimity is a Π_1 condition, and the effective version is the Bayesian verifier below, truncated to finitely many hypotheses. Any budgeted proof search that counts "not found" as "not valid" only shrinks the accepted set, so it preserves the soundness results below, though not the escalation bounds.

Lemma 10.6 (Argument soundness)**.** *[T4 Lemma 2.2, repaired] If every line of α lies in D_h and every inference and link of α (a link as the step $\{y_i\} \rhd y_j$) lies in $\models_h$, then $\rho_h(y_j) \in \mathrm{Cl}_{R_h}(\rho_h \mathrm{Prem}(\alpha))$ for every line y_j. Hence every line is true in every $M \in \mathbb{M}_h$ that makes the premises true.*

Induction on lines, by monotonicity and idempotence of Cl_{R_h} (Appendix I.1).

Proposition 10.7 (Equivocation: step-checking alone is unsound; trivial)**.** *[T4 Prop 2.3] There are h and α such that every inference of α is h-valid and every link joins equal strings, yet $\mathrm{Prem}(\alpha) \rhd y_m$ is h-invalid.*

Proof (Cauchy's sum theorem). Let ρ_h read "$\sum f_n$ converges" as uniform convergence in context c_1 (Cauchy's proof) and as pointwise convergence in c_2 (the statement). Lines: (1), (2) premises (c_2, "each f_n is continuous"), (c_2, "$\sum f_n$ converges"); (3), (4) links of (2), (1) into c_1; (5) the valid inference to (c_1, "the sum is continuous"); (6) a link of (5) back into c_2. Links (4) and (6) are valid; link (3), pointwise to uniform, is not; and $\{1,2\} \rhd 6$ fails on Abel's series $\sum_n (-1)^{n+1} \sin(nx)/n$, whose sum is $x/2$ on $(-\pi, \pi)$, extended periodically. □

So the verifier checks links. Lakatos's "hidden lemma" in Cauchy's proof (Lakatos, 1976, App. 1) is, here, an invalid link.

Theorem 10.8 (Deterministic soundness; trivial once set up)**.** *[T4 Thm 2.4] Assume (BG1) and noise-free truthful data: practice and escalation answers lie in and are labelled by* $\mathrm{St}^g_{h^*}$*, link answers agree with* h^**, designated contexts are* h^**-coherent, object data are* h^**-admissible. Then for every prover (randomized, adaptive, unbounded, knowing* h^**, the verifier and the history) and every* t*, everything* V^g_t *accepts lies in* $\mathrm{St}^g_{h^*} \subseteq \models_{h^*}$*; so every argument assembled from accepted items has an* h^**-valid conclusion (unused premises outside* D^* *dropped), true in every* $M \in \mathbb{M}^*$ *satisfying its premises.*

Proof. No datum can refute h^*, so $h^* \in \mathrm{VS}_t$, and unanimity implies g-validity under h^*; then apply Lemma 10.6. □

Incoherent survivors only shrink the intersection: Frege's inconsistent 1903 "way out" (Quine, 1955) would have made a conservative verifier timid, never unsound. With noisy practice h^* can be refuted, so we use a posterior; the threshold then involves the prior mass of the target's observational-equivalence class $[h^*]$ (Definition 10.12).

Theorem 10.9 (Time-uniform soundness with a class prior)**.** *[T4 Thm 2.5; T1 Thm 4.2] Let* $\mathcal{H}$ *be countable with prior* w *and assume (BG1). Each* h *has a practice likelihood* p_h *(including noise); practice data are i.i.d. from* p_{h^*}*, each independent of everything before it (the prover does not foresee them); likelihoods are* shadow-measurable *(*$p_h = p_{h'}$ *whenever* $h \equiv_g h'$*); all other data (escalation and link answers, paradoxes, objects) are truthful and multiply the weight of* h *by* 1 *if consistent with* h*, by* 0 *otherwise. Let* $V^{g,\delta}_t$ *accept* q *iff the posterior gives* $\{h : q \notin \mathrm{St}^g_h\}$ *mass below* δ*, and put* $W^* := w([h^*])$*. If* $\delta \le W^*\delta'$*, there is an event of probability at least* $1-\delta'$*, depending only on the practice data, on which no prover ever gets a query outside* $\mathrm{St}^g_{h^*}$ *accepted.*

The proof (Appendix I.1) is that of Theorem 4.15, with shadow-measurability giving the whole class $[h^*]$ the likelihood of h^*. W^* counts the $\equiv_g$-class (renamings; the ontological variants of Proposition 10.16), not the meaning class: at gap g, soundness is paid for at the level of g-shadows. Its price is escalations. Let $\mathcal{H}^g := \{\mathrm{St}^g_h : h \in \mathcal{H}\}$, with positive elasticity $\mathrm{el}(\mathcal{H}^g, \mathrm{St}^g_{h^*})$ (Definition 4.3; also Definition 10.18).

Theorem 10.10 (Escalation costs)**.** *[T4 Thm 2.7, repaired; T1 §§3–4] Let the prover be honest and count only escalations answered by the oracle (expansion requests yield no label). (a) With noise-free data* V^g_t *escalates at most* $\mathrm{el}(\mathcal{H}^g, \mathrm{St}^g_{h^*}) \le |\mathcal{H}/\equiv_g| - 1$ *times. (b) In the setting of Theorem 10.9, with probability* $\ge 1-\delta''$*,* $V^{g,\delta}_t$ *escalates at most* $(\ln(1/W^*) + \ln(1/\delta''))/\delta$ *times in all. (c) If the* n *hypotheses' g-step relations are* $B \cup (U \setminus \{u\})$*,* $u \in U$*, then (a) equals* $n-1$ *and is attained, and with the uniform prior every* δ'*-sound verifier fails to accept (escalates or rejects), in expectation, at least* $(1-\delta')(n-1)$ *queries of some honest prover, so (b) with* $\delta = W^*\delta'$ *is tight up to a factor* $O((\ln(1/W^*) + \ln(1/\delta''))/\delta')$*.*

Proof in Appendix I.1, by reduction to the escalation theorems of Section 4. Expansion closes the gap. Call h^* *step-expressive* if every $(\Gamma \rhd y) \in \models_{h^*}$ is the conclusion of an informal argument from Γ whose inferences lie in $\mathrm{St}^1_{h^*}$ (one rule application each) and whose links join occurrences with equal ρ^*-readings.

Proposition 10.11 (Expansion makes the verifier complete; trivial)**.** *[T4 Prop 2.8] If* h^* *is step-expressive and* $\bigcap_{h \in \mathrm{VS}_t} \mathrm{St}^1_h \supseteq \mathrm{St}^1_{h^*}$*, then every* h^**-valid step has an argument all of whose steps and links* V^1_t *accepts.*

Proof. Each inference of the expansion lies in $\mathrm{St}^1_{h^*}$, and each link is in $\mathrm{St}^1_{h^*}$ with zero applications; all lie in every survivor's St^1_h. □

Under complete presentation, the hypothesis on VS_t is what Theorem 10.14(d) delivers at $g = 1$.

Reading. A checker that asks the arguer to "explain/justify/derive" what it does not recognize[11] is *sound at all times* and *complete via expansion* once the atomic level is learned. No measure over steps enters the guarantee. For contexts, Proposition 10.7 says that a sentence true in one context is not thereby usable in another: the re-use is an inference to be checked, the zero-length case of the export rules of Section 11.

10.3 What practice can identify

Definition 10.12 (Equivalences). [T4 Def 3.1] $h \approx h'$ iff $\models_h = \models_{h'}$ (this fixes $D_h = \{x : \{x\} \models_h x\}$); $h \approx_D h'$ iff they agree on steps with all occurrences in D. $h \equiv_g h'$ iff $\mathrm{St}^g_h = \mathrm{St}^g_{h'}$, $\mathcal{V}_h = \mathcal{V}_{h'}$, and h, h' are coherent on the same $A \in \mathcal{A}$; $\equiv_\infty$ is the same with $\models$ for St^g.

If all data and queries depend on the target only through its g-shadow, no learner distinguishes $\equiv_g$-equivalent targets (Proposition I.1, stated and proved in Appendix I.2). Which shadow determines which? Call D_h *negation-closed* if an informal operation $x \mapsto {\sim}x$ ("it is not the case that ...") preserves D_h with $T_h \vdash \rho_h({\sim}x) \leftrightarrow \neg\rho_h(x)$, and call $v : D_h \to \{0,1\}$ a *coherent valuation of* $\models_h$ if $v({\sim}x) = 1 - v(x)$ and $v(y) = 1$ whenever $\Gamma \models_h y$ for a finite $\Gamma \subseteq v^{-1}(1)$.

Lemma 10.13 (Informal completeness). *[T4 Lemma 3.3, repaired]* *(standard corollary of compactness and completeness) Let h be first-order complete with negation-closed $D_h \neq \emptyset$. (i) $\mathcal{V}_h$ is exactly the set of coherent valuations of $\models_h$. (ii) $\Gamma \models_h y$ iff $\Gamma \cup \{y\} \subseteq D_h$ and every $v \in \mathcal{V}_h$ with $v(\Gamma) = 1$ has $v(y) = 1$. (iii) A finite $A \subseteq D_h$ is h-coherent iff $v(A) = 1$ for some $v \in \mathcal{V}_h$, iff $A \not\models_h y$ for some $y \in D_h$. So $\models_h$ determines $\mathcal{V}_h$ and coherence; conversely $\mathcal{V}_h$ determines $\models_h$ if $\mathcal{V}_h \neq \emptyset$, and coherence of all finite subsets of D_h determines it, but coherence on a fixed family $\mathcal{A}$ does not.*

Reading. This is the notes' "philosophical version" of the completeness theorem ("a mode of talking makes sense (is coherent) iff there is something it could be talking about") for a practice's informal shadow: *the coherent ways of evaluating the practice's sentences are exactly its admissible objects.* It is a direct corollary of Lindenbaum, compactness and completeness (Gödel, 1930) (proof in Appendix I.2). The notes' caveat ("as long as the 'mode of talking' is a first-order theory and 'something' is a model of it") is visible in $\mathbb{M}_h = \mathrm{Mod}(T_h)$: if only standard objects are presented, strengthenings such as $T + \mathrm{Con}(T)$ survive, over-generalizations of the practice's validity relation though not errors about truth (Section 9).

For a step set S and $D \subseteq X$, let $\mathrm{Ch}_D(S)$ be the steps $\Gamma \rhd y$ derivable from Γ by an argument whose lines lie in D and whose inferences and links lie in S. h^* is *step-expressive on D^* at gap g* if $\mathrm{Ch}_{D^*}(\mathrm{St}^g_{h^*}) = \models_{h^*}$ (implied by step-expressiveness, for $g \geq 1$).

Theorem 10.14 (Identification up to $\approx$ on the practice's domain). *[T4 Thm 3.4, (d) repaired]* *Let every $h \in \mathcal{H}$ be first-order complete with negation-closed nonempty domain (one fixed ${\sim}$), let T^* be consistent, and let $\mathcal{A}$ consist of subsets of D^*.*

[11] The phrase comes from a remark in the notes about questioning a solver who uses primitives one does not already trust: `ai/verification and generation/is verification really easier than solving the problem yourself?.md`.

(a) $h \equiv_\infty h'$ *iff* $h \approx h'$.

(b) *Under the closure-level version of Proposition I.1, no learner identifies the target more finely than* $\approx$.

(c) *Let* $\mathcal{H}$ *be finite with* $h^* \in \mathcal{H}$ *(BG1), and the presentation* complete *(every element of* $\models_{h^*}$ *appears in (P), every finite restriction of every member of* $\mathcal{V}_{h^*}$ *in (O)). After finitely many data the survivors are exactly* $U^* := \{h : h \approx_{D^*} h^*\}$*, and unanimity on* $\models_h$ *accepts on* D^* *exactly* $\models_{h^*}$*. (This concerns the ideal version space: refutation by a practice datum needs a certificate of non-validity, which is co-r.e.)*

(d) *If instead (P) presents* $\mathrm{St}^g_{h^*}$ *completely, the survivors are eventually exactly*

$$S^g := \{h \in \mathcal{H} : \ \mathrm{St}^g_h \supseteq \mathrm{St}^g_{h^*} \text{ and } \models_h|_{D^*} \subseteq \models_{h^*}\},$$

V^g accepts exactly $\mathrm{St}^g_{h^*}$*, and every survivor's meaning is bracketed:* $\mathrm{Ch}_{D^*}(\mathrm{St}^g_{h^*}) \subseteq \models_h|_{D^*} \subseteq \models_{h^*}$*. If* h^* *is step-expressive on* D^* *at gap* g*, then* $S^g = \{h \in U^* : \mathrm{St}^g_h \supseteq \mathrm{St}^g_{h^*}\}$*; without it, survivors can have strictly weaker meaning. Here all refutations are semi-decidable, so the budgeted* VS_t *reaches* S^g*.*

Proof in Appendix I.2. The key is an upper-bracket lemma (Lemma I.2): among hypotheses with $D_h \supseteq D^*$, objects refute exactly those that validate on D^* a step h^* does not, by restricting an h^*-countermodel, and paradoxes refute no others.

Example 10.15 (A weaker rival survives bounded-gap data)**.** [T4 Thm 3.4(d), verification A1] Read D^* as $p_0, \neg p_0, p_3, \neg p_3$; let $T^* = \{p_0 \to p_1, p_1 \to p_2, p_2 \to p_3\}$ with axiom steps plus one-step tautological consequence, and let the rival have $T_h = \emptyset$. For $g \leq 3$ the g-step relations agree on D^*, but $(p_0 \rhd p_3) \in \models_{h^*} \setminus \models_h$, and neither objects nor paradoxes refute the rival ($\mathrm{Mod}(T^*) \subseteq \mathrm{Mod}(\emptyset)$); at $g = 4$ it is refuted. Verified in Appendix I.2, together with a random test of (d).

Reading. Bounded-gap data identify the *g-step relation* and *bracket* meaning: practice bounds it from below, objects from above (they refute a hypothesis for validating too much, never too little). *Coarser* rivals (same inferences, larger steps) survive by Gold's asymmetry (Gold, 1967), now about obviousness; *weaker* rivals miss inferences of h^* that are not chains of short steps through the practice's own sentences. So a practice is guaranteed to fix its meaning when its long inferences so decompose, and otherwise it may not; expansion is the remedy.

The equivalence classes are large: practices fix inferential role and leave ontology and latent language open.

Proposition 10.16 (Ontology is unidentifiable at every gap)**.** *[T4 Prop 3.5, repaired] Let* $h = (L, R, \rho, \mathbb{M})$ *and* $h' = (L, R, \rho, \mathbb{M}')$ *be latent formalizations such that every member of* $\mathbb{M}'$ *agrees with some member of* $\mathbb{M}$ *on all sentences in* $\rho(D_h)$*, and vice versa. Then* $h \equiv_g h'$ *for every* g*, and* $h \equiv_\infty h'$*.*

Proof. R, ρ are shared, so St^g, $\models$ and paradoxes are; and $v_M \circ \rho$ depends only on the truth values of $\rho(D_h)$ in M. □

Instance (modulo the cited transfer principle). For readings into *bounded* sentences over the superstructure $V(\mathbb{R})$, Robinson's transfer principle (Robinson, 1966) makes $\{V(\mathbb{R})\}$ and $\{{}^*V(\mathbb{R})\}$ such a pair, so *whether the intended reals of practice contain infinitesimals* cannot be learned

from any data in this language. Likewise ε-δ analysis and a literal reading in Nelson's internal set theory agree in informal consequence on sentences with standard parameters (Proposition I.3), and von Neumann and Zermelo numerals on arithmetic sentences (Proposition I.4), which turns Benacerraf (1965) into a non-identifiability theorem. Details for all three cases are in Appendix I.2.

Reading: $\approx$ is inferentialist meaning. Inferential role, incompatibility (here incoherence, Lemma 10.13(iii)) and Sellarsian language-entry transitions (here object evaluations, Lemma 10.13(i)) are all functions of $\models_h$ on D^*, and soundness refers only to $\models_{h^*}$. Everything finer (ontology, latent language, junk like "$1 \in 3$") never changes a practice inference and is unidentifiable. Granularity is partly identifiable, but it is pragmatics (which steps are *acceptable*), not semantics (which are *valid*). Whether ε-δ and infinitesimals differ in what they are about the practice cannot settle; the verifier needs only $\approx$.

10.4 Counterexample objects versus paradoxes

Lemma 10.17 (Descent along false lines)**.** *[T4 Lemma 4.1] If v is defined on the lines of α, true on the premises and false on the conclusion, then α contains an inference with true antecedents and false conclusion, or a link with true source and false target, found within* $\mathrm{depth}(\alpha)$ *moves evaluating only antecedents of visited lines. If v is h-admissible, the item is h-invalid.*
Lean: Blame.LineArg.descent_depth Lean: Blame.ArgTree.descent_tree_not_mem

From the false conclusion, repeatedly move to a false antecedent or link source (Appendix I.3). This is Shapiro's contradiction backtracing (Shapiro, 1983) for informal arguments, Lakatos's conversion of a global counterexample into a local one (Lakatos, 1976), and Easwaran's demand that proofs be fine enough to convert rebutting into undercutting defeaters (Easwaran, 2015). Descent can end at a link: the blame then falls on an equivocation, not a rule (Lakatos's "global but not local" case).

Definition 10.18 (Correction game)**.** [T4 Def 4.2] Let $\mathcal{H} \subseteq 2^S$ be finite, S a finite step universe, $h^* \in \mathcal{H}$. Each round the learner announces $A_t \subseteq S$; if $A_t \neq h^*$ the environment returns $(+)$ some $s \in h^* \setminus A_t$, or $(-\mathrm{obj})$ some $s \in A_t \setminus h^*$, or $(-\mathrm{bag}_r)$ some $B \subseteq A_t$ with $|B| \leq r$, $B \not\subseteq h^*$; the learner deletes the hypotheses $h \not\ni s$, $h \ni s$, $h \supseteq B$ respectively. A *correction* is a round with feedback. $M_{\mathrm{obj}}(\mathcal{H})$ (feedback $(+)$ or $(-\mathrm{obj})$) and $M^{(r)}_{\mathrm{bag}}(\mathcal{H})$ (feedback $(+)$ or $(-\mathrm{bag}_r)$) are the optimal worst-case numbers of corrections; $M_{\mathrm{bag}} := M^{(|S|)}_{\mathrm{bag}}$. The positive elasticity $\mathrm{el}(\mathcal{H}, h^*)$ is the supremum of lengths of sequences $s_1, s_2, \ldots \in h^*$ with $s_i \notin \bigcap\{h \in \mathcal{H} : h \supseteq \{s_1, \ldots, s_{i-1}\}\}$, and $\mathrm{el}^*(\mathcal{H}) := \max_{h^*} \mathrm{el}(\mathcal{H}, h^*)$.

In reality a negative channel can be silent (objects need presentable countermodels; coherent-but-wrong hypotheses yield no paradox, Section 6); the bounds below then still bound corrections, but unsound rounds may pass silently.

Theorem 10.19 (Objects: logarithmic)**.** *[T4 Thm 4.3] (classical) (a) $M_{\mathrm{obj}}(\mathcal{H}) \leq \lfloor \log_2 |\mathcal{H}| \rfloor$; with a prior w, weighted halving makes at most $\log_2(1/w(h^*))$ corrections, so with $w = 2^{-\ell}$ the number of counterexamples needed is at most the description length of the target. (b) $M_{\mathrm{obj}}(\mathcal{H}) \leq M^{(r)}_{\mathrm{bag}}(\mathcal{H})$ for all $r \geq 1$, with equality at $r = 1$. (c) $M_{\mathrm{obj}}(\mathcal{H}) = \mathrm{Ldim}(\mathcal{H})$, the Littlestone dimension.*
Lean: (a) CoherenceGames.thm_4_3_a; (b) ParadoxLowerBound.thm_4_3_b

Part (a) is classical halving (Barzdin and Freivalds, 1972; Littlestone, 1988; Angluin, 1988): announce the steps carried by more than half of the surviving weight. The object game is the mistake-bound model, hence (c) (Littlestone, 1988). Proofs in Appendix I.3.

Theorem 10.20 (Paradoxes: linear, tightly). *[T4 Thm 4.4] (a)* $M^{(r)}_{\text{bag}}(\mathcal{H}) \le \text{el}^*(\mathcal{H}) \le |\mathcal{H}| - 1$. *(b)* Single culprit*: for* $S = \{b_1, \dots, b_n\}$ *and* $\mathcal{H}_n = \{S \setminus \{b_j\} : j \le n\}$, $M_{\text{obj}}(\mathcal{H}_n) = 1$ *and* $M^{(r)}_{\text{bag}}(\mathcal{H}_n) = \min(r, n-1)$*; so* $M_{\text{bag}}(\mathcal{H}_n) = |\mathcal{H}_n| - 1$ *against* $\log_2 |\mathcal{H}_n|$*, and (a) is tight. (c)* The chain paradox*: take sentences* $q_0, \dots, q_n$*, suspect steps* $b_i = (q_{i-1} \rhd q_i)$*, and the designated context* $\{q_0, \sim q_n\}$*, e.g.* $q_k =$ *"a pile of* $n-k$ *grains is a heap". Let* h_j *validate every* b_i *with* $i \ne j$ *(axioms* $p_{i-1} \to p_i$*,* $i \ne j$*). Every* h_j *is coherent on the designated context; every paradox uses all the* b_i*, so every bag is* S*; every learner using paradoxes suffers* $n-1$ *corrections; and one admissible object pins down the culprit by descent.* *(deterministic learners)* Lean: (b) ParadoxLowerBound.thm_4_4_b; (c) thm_4_4_c

Proof in Appendix I.3; for (a), the cautious learner $A_t = \bigcap \text{VS}_t$ receives only (+) items, which form an elastic chain. "Paradoxes are worthless" ($M_{\text{bag}} = \text{el}^*$) is false in general: $\mathcal{H} = \{\emptyset, \{0\}, \{0,1\}\}$ has $M_{\text{bag}} = 1 < 2 = \text{el}^*$ (announce $\{0\}$); exact minimax values for small classes are in Appendix I.3. Real paradoxes have few *suspect* steps, since steps every survivor accepts drop out of the bag.

Theorem 10.21 (Bags of size at most r). *[T4 Thm 4.5] The* super-majority learner $A_t = \{s : w(\{h \in \text{VS} : s \in h\}) > \frac{r}{r+1} w(\text{VS})\}$ *makes at most* $\ln(1/w(h^*))/\ln(1+1/r)$ *corrections. With the uniform prior,* $M^{(r)}_{\text{bag}}(\mathcal{H}) \le \ln|\mathcal{H}|/\ln(1+1/r) \le (r+1)\ln|\mathcal{H}|$. Lean: CoherenceGames.thm_4_5

Each correction keeps at most a $\frac{r}{r+1}$ fraction of the surviving weight (proof in Appendix I.3).

Proposition 10.22 (Linear dependence on r is necessary). *[T4 Prop 4.6] For the product of* k *single-culprit blocks of size* $b \ge 2$ *(*$|\mathcal{H}| = b^k$*),* $M_{\text{obj}} = k$ *and* $M^{(r)}_{\text{bag}} \ge k \min(r, b-1)$*. With* $b = r+1$*:* $M^{(r)}_{\text{bag}} \ge r \log|\mathcal{H}|/\log(r+1)$ *while* $M_{\text{obj}} = \log|\mathcal{H}|/\log(r+1)$.

Proof in Appendix I.3. So adversarial multiple-instance feedback costs $\tilde{\Theta}(r \log|\mathcal{H}|)$ over classes of a given size, whereas in the i.i.d. multiple-instance setting (Dietterich et al., 1997) bag size enters far more mildly (Sabato and Tishby, 2012); and the form "Ldim $\times$ polylog(r)" is refuted, as products have Ldim $= k$.

Proposition 10.23 (Bag price when $M_{\text{obj}} = 1$). *[T4 Prop 4.6′] If* $M_{\text{obj}}(\mathcal{H}) = 1$ *then* $M^{(r)}_{\text{bag}}(\mathcal{H}) \le r$.

Conjecture 10.24 (Bag price). *[T4 §4.2]* $M^{(r)}_{\text{bag}}(\mathcal{H}) \le r \cdot \text{Ldim}(\mathcal{H})$ *for every finite* $\mathcal{H}$.

Proposition 10.23 (proof in Appendix I.3) is the case Ldim $= 1$. The conjecture is tight on single-culprit classes ($r \le n-1$), products ($r \le b-1$) and some two-culprit classes, and computer searches found no violation (Appendix I.3).

Objects can also be neutralized. A step blamed by an object can be repaired by changing R (*lemma-incorporation*), changing ρ so that the monster no longer satisfies the premise (*monster-barring* by redefinition), rejecting or re-interpreting the datum (*monster-barring* by exclusion, *monster-adjustment*), or extending L (a *proof-generated concept* naming the hidden lemma) (Lakatos, 1976).

If barring is free, objects constrain nothing; if each barred datum costs $c > 0$ bits, a hypothesis that must bar m truthful data loses to h^* in description length once $m > (\ell(h^*) - \ell(h))/c$

(Proposition I.5, Appendix I.3). So objects are powerful only if certified independently (by the community, or by computation) or if revisions of ρ are paid for, which is Lakatos's own preference ordering.

Reading. Coherence says that *some* step of a paradox is bad; an object says *which*. The sorites is the canonical uninformative paradox. The oligarchic halving of Section 6 bounds *detected incoherences* by $\log_2(1/w(h^*))$, yet total corrections, counting rejections of valid steps, can still be $|\mathcal{H}|-1$: coherence bounds the price of boldness, not of learning; objects bound both. Russell's paradox, by contrast, is a bag of size one; what it could not do is choose the *generalization* (Section 10.5). A learner of inference rules must generate and evaluate objects, not only search for $\bot$.

10.5 Frege → Russell → Zermelo

Definition 10.25 (Repair setting)**.** [T4 §5.1] I is a set of rule instances; $\mathrm{Coh} \subseteq \mathcal{P}(I)$ is a coherence predicate, downward closed and of finite character; $\mathcal{R} \subseteq \mathcal{P}(I)$ is a finite class of *uniformly specified restrictions* with description lengths ℓ. Practice at time t is a finitely supported weight $w_t : I \to [0,\infty)$, with coverage $\mathrm{cov}_t(c) := \sum_{i\in c} w_t(i)$. Budget-$t$ $\bot$-search refutes c iff c has a refutation of size $\le t$; incoherent c has one of size $k(c) < \infty$. $\mathrm{MDL}_t := \arg\min\{\ell(c) : c \in \mathcal{R},\ \mathrm{supp}(w_t) \subseteq c\}$, and REP_t is the lexicographic $\arg\max$ of $(\mathrm{cov}_t, -\ell)$ over the members of $\mathcal{R}$ not refuted with budget t.

Theorem 10.26 (Abstract repair; trivial)**.** *[T4 Thm 5.1, repaired] (a) If $I \in \mathcal{R}$ is strictly shortest, $\mathrm{MDL}_t = I$ for all t. (b) Coherent members are never refuted; incoherent c is refuted for $t \ge k(c)$. (c) For $t \ge t_0 := \max\{k(c) : c \in \mathcal{R}$ incoherent$\}$, REP_t maximizes $(\mathrm{cov}_t, -\ell)$ over the coherent members of $\mathcal{R}$. (d) If $w_t/\|w_t\|_1 \to w_\infty$ in ℓ_1 and the coherent maximizer of cov_∞ is unique, REP_t is eventually equal to it. (e) If $c_1, c_2 \in \mathcal{R}$ are coherent and $c_1 \cup c_2$ is not, no coherent set contains both, and while $\mathrm{supp}(w_t) \subseteq c_1 \cap c_2$ only ℓ chooses between them.*

Proof in Appendix I.4. The content is in an instance that makes every step of the story checkable. In first-order logic with $\in, =$, let $\mathrm{Comp}(\varphi) := \forall a \forall b \exists y \forall x\, (x \in y \leftrightarrow \varphi(x, a, b))$, with φ from the menu Φ of Table 13. A formula is *stratified* (as in Quine's NF, Quine 1937) if it has a typing t with $t(v) = t(u) + 1$ for $u \in v$ and $t(u) = t(v)$ for $u = v$; $x \in x$ has none. $\mathcal{R}$ consists of NC (the whole menu), POS (positive formulas), STRAT (stratified; NF-like), SEP (of the form $x \in a \wedge \psi$), and $\mathrm{Z} := \mathrm{SEP} \cup \{\mathrm{EMP}, \mathrm{PAIR}, \mathrm{UNI}\}$ (Zermelo-like). The description lengths are *stipulated*: $\ell(\mathrm{NC}) < \ell(\mathrm{POS}) < \ell(\mathrm{SEP}) < \ell(\mathrm{STRAT}) < \ell(\mathrm{Z})$, and outcomes depend on this choice.

Lemma 10.27 (Comprehension facts)**.** *[T4 §5.2, F1–F6] (F1) NC is incoherent (*$\mathrm{Comp(R)}$ *with $x := y$ refutes it in four lines). (F2) POS is coherent, even for all positive formulas. (F3)* $\mathrm{STRAT} \cap \Phi$ *is coherent. (F4)* $\mathrm{Z} \cap \Phi$ *is coherent. (F5) POS, STRAT, Z are pairwise jointly incoherent. (F6)* $\mathrm{SEP} \subsetneq \mathrm{Z}$*, and POS, STRAT, Z are pairwise $\subseteq$-incomparable.*

The models are a one-element $u \in u$ (F2), $\mathbb{N}$ coding the finite and cofinite sets (F3) and the hereditarily finite sets (F4); $\{\mathrm{V}, \mathrm{ZR}\}$ and $\{\mathrm{S}, \mathrm{CMP}\}$ each yield Russell's set (F5). Proof in Appendix I.4.

Theorem 10.28 (The Frege–Russell–Zermelo trajectory)**.** *[T4 Thm 5.2, (iv) repaired] (i) For every practice* $\mathrm{MDL} = \mathrm{NC}$*: Frege's Basic Law V (Frege, 1893) as the compression of "every concept has an extension". (ii) Coherence refutes it with a four-line budget. (iii) The maximal coherent*

name	$\varphi(x, a, b)$	role	pos.	strat.	$x \in a \wedge \psi$
V	$x = x$	Frege's extension of "object"	✓	✓	
EMP	$x \neq x$	empty set		✓	
S	$x \in x$		✓		
R	$x \notin x$	Russell's set			
INT	$x \in a \wedge x \in b$	intersection (Dedekind)	✓	✓	✓
UNI	$x \in a \vee x \in b$	union	✓	✓	
PAIR	$x = a \vee x = b$	pairs	✓	✓	
DIFF	$x \in a \wedge x \notin b$	difference (Cantor)		✓	✓
CMP	$x \notin a$	complement		✓	
ZR	$x \in a \wedge x \notin x$	Zermelo's 1908 subset			✓

Table 13: The comprehension menu Φ [T4 §5.2]. ZR is Cantor's diagonal set $\{x \in a : x \notin f(x)\}$ (Cantor, 1891) for $f = \mathrm{id}$; identifying diagonal practice with ZR is a modelling simplification.

members of $\mathcal{R}$ are exactly POS, STRAT and Z, pairwise jointly incoherent: there is no minimal repair. (iv) Coverage selects ($t \geq t_0$; $\mathrm{DC} := \{\mathrm{INT}, \mathrm{UNI}, \mathrm{PAIR}, \mathrm{DIFF}, \mathrm{EMP}\}$*, the Dedekind–Cantor operations, all supports with positive weights):*

- *support* DC*: STRAT and Z cover everything and ℓ selects* ***STRAT****. For supports* $\subseteq \mathrm{DC}$*: STRAT iff the support meets both* $\{\mathrm{DIFF}, \mathrm{EMP}\}$ *and* $\{\mathrm{UNI}, \mathrm{PAIR}, \mathrm{EMP}\}$*; otherwise POS if it avoids DIFF and EMP (a positive practice selects positive set theory); otherwise SEP;*
- *support* $\mathrm{DC} \cup \{\mathrm{ZR}\}$*: only Z covers everything, and* ***Z*** *is selected; for* $\mathrm{ZR} \in$ *support* $\subseteq \mathrm{DC} \cup \{\mathrm{ZR}\}$*, Z iff the support meets* $\{\mathrm{UNI}, \mathrm{PAIR}, \mathrm{EMP}\}$*, else SEP;*
- *support* $\mathrm{DC} \cup \{\mathrm{ZR}, \mathrm{V}\}$*: Z iff $w(\mathrm{ZR}) > w(\mathrm{V})$, ties to STRAT. For* $\{\mathrm{ZR}, \mathrm{V}\} \subseteq$ *support* $\subseteq \mathrm{DC} \cup \{\mathrm{ZR}, \mathrm{V}\}$*: if $w(\mathrm{ZR}) > w(\mathrm{V})$, SEP if the support avoids UNI, PAIR, EMP, else Z; if $w(\mathrm{V}) > w(\mathrm{ZR})$, POS if it avoids DIFF, EMP, else STRAT (e.g.* $\{\mathrm{INT}, \mathrm{UNI}, \mathrm{PAIR}, \mathrm{ZR}, \mathrm{V}\}$ *gives POS); if $w(\mathrm{ZR}) = w(\mathrm{V})$, POS if it avoids DIFF, EMP, else SEP if it avoids UNI, PAIR, EMP, else STRAT.*

Proof in Appendix I.4, from Theorem 10.26(c) and the membership table; the conditions are also checked there against the learner by computation, and experiment G (Section 12) reproduces the dynamics.

Historical reading ("a nice thing shadowed in" practice, not a claim about what Zermelo did). Zermelo (1908) proves from Separation that every set M has a subset $\{x \in M : x \notin x\}$ not in M: ZR at arbitrary a, the diagonal argument as a theorem. So *practice containing Zermelo-style diagonal subsets selects Separation over stratification, and Frege's universal extension is the price.* Without diagonal practice, and under the stipulated ℓ, NF-like stratification (on NF's consistency see Holmes 2015) would have won only for a practice using difference or $\emptyset$ and also union, pairs or $\emptyset$; a purely positive practice selects positive set theory. A second separating datum lies outside the toy: NF refutes choice (Specker, 1953), which Zermelo defended by its uses (Moore, 1982). His stated method, to restrict "sufficiently to exclude all contradictions" yet keep "all that is valuable" (Zermelo, 1908) (trans. van Heijenoort 1967, p. 200), is REP's constrained optimization; the theorem adds that the optimum exists only in a uniform class and is chosen by coverage. Without uniformity it is far worse.

The instances of naive comprehension have exactly $2^{\aleph_0}$ maximal consistent subsets, and even $\mathrm{Comp}(x \neq x)$ alone has $2^{\aleph_0}$ pairwise incompatible maximal consistent extensions (Proposition I.6,

Appendix I.4). Incurvati and Murzi (2017) prove more: under minimal assumptions, multiple incompatible maximal consistent sets of instances of naive comprehension exist and *none is recursively axiomatizable*, generalizing McGee's theorem for the T-schema (McGee, 1992). So "the maximal repair" does not exist and no learner can output a recursive axiomatization of one. *Uniformity of the restriction class is what makes the learning problem well posed.* The toy omits Replacement (coverage of a *conclusion*), the cumulative-hierarchy conception (a choice of ℓ), and undecidable coherence, which by Theorem 10.8 costs escalations, never soundness.

Reading. Imitation plus simplicity produces the over-general rule (Section 3); coherence refutes it, here with sharp blame; but coherence cannot choose the repair, since the maximal coherent generalizations are several. Coverage of what the practice does chooses: Lakatos's "heuristic falsifiers", Zermelo's "all that is valuable". What coverage leaves open is a Kripkensteinian residue (Section 6): coherent meanings of "set" that agree on everything the practice has used.

10.6 The robust core and the emergence of a notion of proof

Definition 10.29 (Sharpenings; robust core)**.** [T4 Def 6.1] Fix a first-order L, a base theory T_0 and vague words W. A *sharpening* σ gives each $w \in W$ an explicit L-definition; Σ is the set of *admissible* joint sharpenings. Each σ gives a first-order complete $h_\sigma = (L, R_\sigma, \rho_\sigma, \mathrm{Mod}(T_\sigma))$, with $T_\sigma = T_0$ plus σ's definitions and ρ_σ reading each w by its definition. *Local supervaluational validity* is $\models_{\mathrm{SV}} := \bigcap_\sigma \models_\sigma$; put $\mathrm{St}^g_{\mathrm{SV}} := \bigcap_\sigma \mathrm{St}^g_\sigma$. A precise object $M \models T_0$ expands uniquely to $M_\sigma \models T_\sigma$; an occurrence is *super-determinate* at M if all M_σ agree on it, and object data are *SV-truthful* if reported only for super-determinate occurrences. The *robust-core hypothesis* RCH_g: every non-noise practice step lies in $\mathrm{St}^g_{\mathrm{SV}}$ and every practice link is valid under every σ.

Local validity (truth-preservation at each sharpening) differs from the *global* validity (preservation of supertruth) usually called supervaluational (Fine, 1975; Williamson, 1994; Keefe, 2000; Varzi, 2007); local validity is the right notion for "survives every formalization".

Theorem 10.30 (Robust core)**.** *[T4 Thm 6.2, repaired]*

(a) Survival. *Under* RCH_g*, an argument whose inferences and links are non-noise practice items has, for every* σ*, its conclusion in* $\mathrm{Cl}_{R_\sigma}(\rho_\sigma\mathrm{Prem})$*; a theorem so proved from premises every* T_σ *proves is a theorem of every admissible sharpening, whichever a later formalization adopts.*

(b) SV-truthful data never refute a sharpening. *Let* $\{h_\sigma\} \subseteq \mathcal{H}$*, and let practice lie in* $\mathrm{St}^g_{\mathrm{SV}}$*, escalation and link answers be given only when all* σ *agree ("yes" on* $\mathrm{St}^g_{\mathrm{SV}}$*, "no" outside* $\bigcup_\sigma \mathrm{St}^g_\sigma$*), designated contexts be coherent under every* σ*, and objects be SV-truthful. Then every* h_σ *survives forever, and* V^g_t *accepts only steps in* $\mathrm{St}^g_{\mathrm{SV}}$*: it is sound for every sharpening at once. If moreover* $\mathcal{H}$ *is finite and every element of* $\mathrm{St}^g_{\mathrm{SV}}$ *eventually appears in (P), then* V^g_t *eventually accepts exactly* $\mathrm{St}^g_{\mathrm{SV}}$*.*

(c) Nothing more is soundly acceptable. *Under these data, (c1) an acceptance set sound for every possible target* h_σ *lies in* $\models_{\mathrm{SV}}$*; (c2) one that is g-sound for every target lies in* $\mathrm{St}^g_{\mathrm{SV}}$*, which* V^g_t *attains when* $\mathcal{H} = \{h_\sigma\}$*. The gap* $\models_{\mathrm{SV}} \setminus \mathrm{St}^g_{\mathrm{SV}}$ *is in general nonempty.*

(d) Permanent abstention. *Steps in* $\bigcup_\sigma \mathrm{St}^g_\sigma \setminus \mathrm{St}^g_{\mathrm{SV}}$ *are never accepted. (d1) Those outside* $\models_{\mathrm{SV}}$ *are invalid under some admissible sharpening, and only a definition (shrinking* Σ*) or lemma-incorporation removes them; (d2) those in* $\models_{\mathrm{SV}}$ *are valid but not short under every*

sharpening; adding the step as a library lemma to every R_σ removes them, and so does an expansion into steps of $\mathrm{St}^g_{\mathrm{SV}}$ when one exists.

Proof in Appendix I.5, which also shows the gap in (c) and that for infinite Σ the robust core can be undecidable.

A step in the domain that is invalid under an admissible σ has a σ-*monster*, a model of T_σ making its premises true and its conclusion false (Theorem I.7, by completeness; Appendix I.5). Steps in $\models_{\mathrm{SV}} \setminus \mathrm{St}^g_{\mathrm{SV}}$ have none, and a monster need not be presentable under SV-truthful reporting. *Cauchy, again*: the sum-theorem step is valid only under the uniform sharpening of "converges", so it lies outside the robust core, and Abel's series is a monster for the pointwise one. Lakatos's responses change Σ or the language: monster-barring drops the pointwise sharpening; lemma-incorporation adds "uniformly"; a proof-generated concept introduces a word whose only sharpening is uniform convergence. The robust-core hypothesis predicts *where* pre-formal mathematics broke: at the non-robust steps.

Only unanimity chains: if each of n steps is valid with probability $\geq 1-\varepsilon$ under a measure on sharpenings, the argument is guaranteed validity only with probability $1-n\varepsilon$, and a sorites valid under no sharpening attains this at $\varepsilon = 1/n$ (Proposition I.8, known; Appendix I.5). The bound is Adams's (Adams, 1966), its use for the sorites is Edgington's, with her verities in place of a measure over precisifications (Edgington, 1997), and the disjoint-failure tightness is the lottery paradox's (Kyburg, 1961). "Valid on most readings" is to vagueness what majority vote is to verifier ensembles (the doctrinal paradox of Section 3): it does not compose, and unanimity is the weakest property that guarantees validity whatever the admissible sharpening turns out to be.

Proposition 10.31 (How the nice notion of proof came to be there)**.** *[T4 Prop 6.5, repaired] Let T be an r.e. first-order theory, ρ^* a reading, and A_0 the practice's initial repertoire of steps (within D^*). Suppose the community drops any step refuted by a presented monster (a model of T), and every step outside $\models_T$ is eventually refuted.*

(a) Derivability*: the survivors are $A_\infty = A_0 \cap \models_T$, and the ρ^*-images of any argument chained from them interpolate into a derivation in any complete calculus for T.*

(b) No gap bound follows*: for A_0 the instances of the single schema "from χ infer θ" (χ, θ arithmetic sentences) and $T = \mathsf{PA}$, A_∞ is undecidable and no g has $A_\infty \subseteq \mathrm{St}^g$ for any complete calculus for PA meeting the effectivity assumptions of Definition 10.2.*

(c) A bounded gap needs two extra hypotheses*: if A_0 is the instance set of finitely many schemas, selection acts on* schemas *(a schema is dropped when any instance is refuted), and each surviving schema is* uniformly derivable *(a derivation template with at most g rule applications, in the schema's metavariables, instantiates to a derivation of each instance), then all surviving steps are in St^g of a complete calculus for T, the shadow of finitely many derived rules. Uniform derivability is automatic for pure propositional schemas and fails in general: in a standard Hilbert calculus for PA (rules closed under substituting a fresh variable for a numeral), "infer $\theta(\bar{n})$" with $\theta(n) =$ "n codes no PA-proof of $0=1$" has all instances provable, but a uniform template would yield a PA-proof of $\mathrm{Con}(\mathsf{PA})$; uniform derivability is relative to the calculus and its library.*

Proof in Appendix I.5: (a) is completeness, and in (b) a gap bound would make A_∞ decidable. In the vague case, (a) applied to each (T_σ, ρ_σ) gives survivors $A_0 \cap \models_{\mathrm{SV}}$, though complete monster presentation is then in tension with SV-truthful reporting.

episode	faulty informal rule	negative evidence	repair
Cauchy 1821	convergent series of continuous functions has continuous sum	object: Fourier series (Abel 1826); hidden lemma (Seidel, Stokes 1847) (Lakatos, 1976, App. 1)	uniform convergence (L)
Ampère 1806	continuous functions differentiable except at isolated points	object: Weierstrass's function (1872)	function concept broadened (ρ)
Riemann 1851/57	Dirichlet's principle	object: Weierstrass's variational problem (1870) (Monna, 1975)	side conditions (Hilbert) (R)
Steiner 1848	7776 conics	excess component of double lines (Eisenbud and Harris, 2016)	3264; genericity
Lamé 1847	unique factorization in cyclotomic integers	failure known (Kummer 1844) (Edwards, 1977)	ideal numbers (L)
Poincaré 1900	homology characterizes S^3	object: homology sphere (1904)	fundamental group (L)
Frege 1893	Basic Law V	paradox: Russell (1902) (van Heijenoort, 1967)	restricted comprehension (R)
Zermelo 1908	no Replacement	coverage: $\aleph_\omega$ not provable (van Heijenoort, 1967)	Replacement, 1922 (R)

Table 14: Negative evidence in informal mathematics, and the component of $(L, R, \rho, \mathbb{M})$ revised; sources in the rows, dates as checked in [L6 §1].

Reading: how the nice notion of proof came to be there. The model's partial answer has three ingredients. (1) *Local checkability*: steps are checked on objects line by line (Lemma 10.17), so monster pressure selects steps that preserve truth in every object. (2) *Completeness*: for r.e. T, Gödel's theorem turns "truth-preserving in every object of an elementary class" into "derivable in a finite calculus", with no bound on length. (3) *Schema-level selection of uniformly derivable schemas*, an extra hypothesis not derived here, which alone yields a bounded gap. So the formal structure is "sorta there" in the activity because the activity was *selected by objects*, and completeness says that being so selected is being shadowed by a calculus. Caveats: complete monster presentation is an idealization (the selection took centuries); for intended-model semantics such as $\{\mathbb{N}\}$, (2) fails and selection heads toward true arithmetic, which no calculus captures; and selection only removes steps, so the surviving calculus need not be complete for T. With Theorem 10.30 this also answers the robust-core question: *if* pre-formal practice used only steps valid under every admissible sharpening, its proofs survive every formalization among them, and the conservative verifier never accepts beyond that core.

The notes suggest looking at Euclid, who "didn't have a full set of axioms"; reading a diagram's sharpenings as nearby configurations, Manders's co-exact properties and the system E of Avigad, Dean and Mumma give an analogue, not an instance, of the robust core (Remark I.9, Appendix I.5).

10.7 Historical reading, and what a learner before formalization needed

How informal mathematics stayed reliable (Table 14). (i) The dominant negative datum was an *object*, not a contradiction; only the set-theoretic antinomies were pure derivations of $\bot$. Objects localize blame, at logarithmic rather than linear cost (Theorems 10.19 and 10.20); mathematicians still check how particular steps apply to specific examples (Weber and Mejía-Ramos, 2011), and Euler accepted $\pi^2/6$ on six-decimal agreement. (ii) The faulty rules were *average-case* sound: right on the tame objects practitioners had in mind, wrong on degenerate ones (Steiner's count fails for *every* input, because of an excess component). Rigour arrived when mathematicians *became* adversarial over objects (Weierstrass, Peano, Russell). This is Section 3 in history: a checker trained on human steps inherits the tameness assumption, and a strong prover finds the monsters. (iii) Lags ran from weeks to decades, so guarantees must be time-uniform (Theorem 10.9). (iv) Labels are noisy both ways; some "known errors" are not errors (Hales, 2007), and whether Cauchy's theorem was false depends on the reading (Proposition 10.7 and Theorem I.7). Modularity (Avigad, 2021), a track record of heavy use, and a social process good at locating weak points (Thurston, 1994) tell the object search where to look. Kreisel's squeezing argument (Kreisel, 1967) pins informal *logical* validity to derivability; the axioms cannot be squeezed, and they are what coherence and coverage leave non-unique.

What a learner before formalization would have needed: (1) a class of latent formalizations with a growing definitional library, so that practice has a bounded gap; (2) conservative acceptance by unanimity, applied to links as well as steps, with a time-uniform Bayesian version (Theorems 10.8 and 10.9); (3) an expansion protocol (Proposition 10.11); (4) an object generator, adversarial over objects, with descent (Lemma 10.17 and Theorem 10.19); (5) paid revisions of readings (Proposition I.5); (6) coverage of practice to choose among coherent repairs within a uniform class (Theorem 10.28 and Proposition I.6); (7) acceptance restricted to the robust core when concepts are vague, with non-robust steps flagged as where monsters live (Theorems 10.30 and I.7); (8) a human corpus treated as noisy and corrigible.

The answer to success criterion 2 is yes, in a precise and conditional sense. A learner with (1)–(8) whose class contains the practice's latent formalization is sound at all times against adaptive provers; for a finite class under complete presentation it identifies the practice's step relation, brackets its meaning and converges to the robust core; and in the comprehension toy it reproduces Frege–Russell–Zermelo, with coverage, not coherence, choosing the repair. But *soundness is relative to realizability of the latent formalization class* (BG1), and history violated realizability at every decisive step: ε-δ, uniform convergence, ideals and the fundamental group were inventions of language. Compare the observation in the notes that in mathematics "we created a language in which formal proofs can be written" rather than translating existing proofs. The hardest open problem, affordable soundness with language invention, is the first problem of Section 14; Appendix I.6 records what this section adds to it.

11 Contexts, idealization, and checking physics

Physics is where the programme of the motivating question meets its hardest test, and where the problem of contexts cannot be postponed. Existing olympiad solutions, the motivating notes observe, often do not reduce the problem to one clear mathematical problem: "there's more of a dance with many clear setups, with some relations to the physical problem of interest" (Hänni, 2026, `ai/verification and generation/verification of physics olympiad solutions/the structure of physics olympiad solutions.md`). Some setups are supposi-

tions made for reductio. Some are idealizations that contradict known facts ("assume the air pressure is 0"). Some complete a problem that leaves something open (the sun's elevation is not given). A coherence loss that penalizes good arguments for both P and $\neg P$ would penalize all three. A checker that asks whether each claim is true fails at once, because good solutions are full of literally false claims; the motivating notes ask what else it could check: "check that a claim is true in the context at hand. but wtf is that???" (`ai/verification and generation/verification from truth.md`).

This section answers with a semantics of contexts (§11.2), a theory of coherence, export and tolerances (§§11.3–11.5), and a checker that is sound against every solver relative to a provably ineliminable judgment layer, with a worked run (§§11.6–11.7).

We build a *checker*, not a solver. Gold-level scores on IPhO 2025 theory under rubric grading have been reported for current AI systems (Yu et al., 2025; Qiu et al., 2025), though in the benchmark's September 2025 data not on the modelling-heavy EuPhO 2025 of our worked example (best model 14.9 of 29, gold threshold 16.5; a later update reports gold on all its olympiads). The unmet target is principled checking. The motivating notes also hold that publishing insightful research on AI theorem provers is gravely dangerous ("in my opinion, publishing such research is omnicidal", `ai/verification and generation/formalizing philosophy.pdf`, a July 2024 draft). We have therefore avoided solver-side contributions.

Most results here are trivial once the definitions are in place, or import standard tools (Łoś's theorem, Gronwall's inequality, split conformal prediction, information-based complexity, Lipschitz covering bounds, interval arithmetic). New, as far as we know, are the frame-free realizability criterion (Theorem 11.11(a)), the two-sided analysis of adversarial stipulation (Theorem 11.22), the table of one-sided signals (Table 15), and a rigorous thin-leg certificate with a complete worked checker run (Section 11.7). Proofs not given inline are in Appendix J.

11.1 The problem: eternalism, contexts, and three import regimes

The simplest coherence constraint is *eternalist*: each proposition is true or false independently of context, and a learner is penalized whenever it can argue for both P and $\neg P$. Physics reasoning violates this constantly and correctly. A hydrostatics problem stipulates $p_{\text{atm}} = 0$, while the background derives $p_{\text{atm}} \approx 1.0 \times 10^5\,\text{Pa}$. A reductio derives $\bot$ under a supposition. One exam treats the same physics in incompatible models: at IPhO 2025, Theory Problem 2 excludes surface tension and lets a gas phase appear exactly at the equilibrium threshold, while Theory Problem 3 makes surface tension the mechanism of bubble growth. The motivating notes record the difficulty with two obvious moves. Conjoining every sentence with the whole background framework "leads to just assigning probability 0 to everything"; and proofs by contradiction inside a framework known to be inadequate "might ... really involve subverting the background framework" (`philosophy/epistemology/probabilities/assigning probabilities in a conception of the world you know to be inadequate.md`).

Three import regimes. "Suppose" covers three operations that differ in what the new context imports from its parent:

Regime	Example	Imports	$\perp$ inside means	What leaves the context
supposition	"suppose $\sqrt{2} = p/q$"	everything	success (reductio)	$\neg A$, or $A \to B$
counterfactual	"if the atmosphere vanished, the water would boil"	a maximal compatible part	the revision was done badly	$A \mathrel{\square\!\!\to} B$
idealization	"neglect drag", "$p_{\text{atm}} = 0$", "massless rope"	a designated minimal fragment	the model is ill-posed	query answers, through a certified bridge

Physics idealizations are *not* counterfactuals. "If the air pressure were 0, the water in the beaker would boil" is true and irrelevant: the problem asks what follows in a designated model that imports incompressible hydrostatics and the data, and nothing else. Similarity semantics (Stalnaker, 1968; Lewis, 1973) or AGM revision (Alchourrón et al., 1985) imports exactly the background (the vapour pressure of water) that makes the context degenerate. Idealization is designated minimal import, as in chunk-and-permeate (Brown and Priest, 2004), McCarthy's ist(c, φ) (McCarthy, 1993), local-models semantics (Ghidini and Giunchiglia, 2001), Cartwright's models (Cartwright, 1983), and compositional modelling (Falkenhainer and Forbus, 1991). We keep their shape, "classical inside, filtered between"; what none of them supplies, as far as we know, is different coherence obligations for suppositions and idealizations, and rigorous certificates on the bridges.

Anatomy of a solution. Three decomposed olympiad solutions (IPhO 2012 Theory Problem 1, part A; EuPhO 2025 Theory Problem 1(a); IPhO 2025 Theory Problem 2) share one structure. A *root* c_P holds the problem text, figures and conventions; for grading, the "world" is c_P, not the actual world. A *reading* ρ maps c_P to an admissible model family, possibly with free completion parameters, and types the query ("express in terms of a, I_0, r, φ"). *Local models* are clear mathematical problems, joined by *bridges*: exact ones (symmetry, decomposition, isomorphism) and approximate ones (a small parameter set to its limit, with an error claim), ending in an *export*. Only approximate bridges introduced by the solver, rather than stipulated or regime-stated by the problem, need a certificate from the solver. The notes' hypothesis that "there's a local setting up of a clear thing" holds in all three: the unclear part concentrates in the reading and in a few approximate bridges.

11.2 Filter semantics for contexts

Definition 11.1 (Frames, worlds, readings)**.** [T3 Def 1.1] Let L be a many-sorted first-order language with a sort $\mathbb{R}$ and parameter constants $p_1, \dots, p_n$ of that sort. (i) A *deformation frame* is a partial map $\mathfrak{M} : \Lambda \rightharpoonup \mathrm{Str}(L)$, $\Lambda \subseteq \mathbb{R}^n$, with $p_i^{\mathfrak{M}(\lambda)} = \lambda_i$; $\operatorname{dom}\mathfrak{M}$ is where the model exists (is well-posed). In every $\mathfrak{M}(\lambda)$ the sort $\mathbb{R}$ is the ordered field of reals with standard numerals; $\mathcal{K}$ is the class of L-structures with this standard real sort. Ideal values at infinity are reparametrized ($m \to \infty$ as $1/m \to 0$). (ii) A *world* is $w = (\mathfrak{M}_w, \lambda_w^*)$ with $\lambda_w^* \in \operatorname{dom}\mathfrak{M}_w$: a top-model family and the actual (or intended) parameter point. (iii) A *reading* ρ yields a set W_ρ of worlds, the *admissible completions* (given parameters fixed, measured ones in intervals, free ones in ranges); a query term Q; and a set Ax_ρ of sentences true at $\mathfrak{M}_w(\lambda_w^*)$ for every $w \in W_\rho$. The *laws* of a frame are the sentences true at every $\mathfrak{M}_w(\lambda)$, $\lambda \in \operatorname{dom}\mathfrak{M}_w$.

For a projectile, $\mathfrak{M}(\lambda)$ solves $\ddot{\mathbf{x}} = -g\hat{z} - k|\dot{\mathbf{x}}|\dot{\mathbf{x}}$ with parameters (g, v_0, θ, k), and "neglect air" deforms k toward 0.

Definition 11.2 (Contexts; filter semantics)**.** [T3 Defs 1.2–1.3] A *context tree* has root @ and parent map π; each other context is $\mathsf{SUP}(A)$, a supposition of a sentence A; or $\mathsf{DEF}(D, \lambda^\circ, \mathcal{G})$, a *declared deformation* with deformed parameters $D \subseteq \{1, \dots, n\}$, ideal values $\lambda^\circ \in \mathbb{R}^D$, and a proper filter $\mathcal{G}$ on $\mathbb{R}^D$ converging to λ° (the *path*); or AUX, a definitional extension that shares its parent's semantics. A DEF context uses *limit semantics* if $\mathcal{G}$ is principal at λ°, and *germ semantics* if $\mathcal{G}$ is the punctured neighbourhood filter, possibly restricted to a curve. Fix a world w, write $\|\varphi\|_w = \{\lambda \in \operatorname{dom}\mathfrak{M}_w : \mathfrak{M}_w(\lambda) \models \varphi\}$ and $\lambda[D{:=}\mu]$ for λ with its D-coordinates replaced by μ. Each context c gets a (possibly improper) filter $\mathcal{F}_c(w)$ on $\operatorname{dom}\mathfrak{M}_w$: $\mathcal{F}_@(w)$ is principal at λ^*_w; for $c = \mathsf{SUP}(A)$ with parent p, $\mathcal{F}_c(w)$ is generated by $\mathcal{F}_p(w) \cup \{\|A\|_w\}$; for $c = \mathsf{DEF}(D, \lambda^\circ, \mathcal{G})$, it is generated by the sets $\{\lambda[D{:=}\mu] : \lambda \in F, \mu \in G\} \cap \operatorname{dom}\mathfrak{M}_w$, $F \in \mathcal{F}_p(w)$, $G \in \mathcal{G}$. *Truth in a context* is $c \models_w \varphi$ iff $\|\varphi\|_w \in \mathcal{F}_c(w)$, and the context-free proposition $\mathrm{ist}(c, \varphi)$ is true at w iff $c \models_w \varphi$. A context is *proper* at w if $\emptyset \notin \mathcal{F}_c(w)$. A claim is *supertrue* under ρ if it is true at every $w \in W_\rho$.

Below a root parent, a germ context makes φ true iff φ holds for all deformed parameters close enough to the ideal along the path, at the actual values of everything else; a limit context is evaluated at the single point $\lambda^*[D{:=}\lambda^\circ]$, and is improper if no model exists there.

Proposition 11.3 (Classicality without explosion; Łoś)**.** [T3 Prop 1.4] *Let* $\mathrm{Th}_w(c) = \{\varphi : c \models_w \varphi\}$. (a) $\mathrm{Th}_w(c)$ *is closed under first-order consequence.* (b) *It is consistent iff* c *is proper at* w. (c) $\mathrm{Th}_w(c) = \bigcap_U \mathrm{Th}(\prod_U \mathfrak{M}_w)$, *where* U *ranges over the ultrafilters on* $\operatorname{dom}\mathfrak{M}_w$ *extending* $\mathcal{F}_c(w)$ *and* $\prod_U \mathfrak{M}_w$ *is the ultraproduct of the structures* $\mathfrak{M}_w(\lambda)$. Lean: Contexts.cn2_closed_ThIn Lean: consistent_ThIn_iff *((a), (b); propositional)*

Proof idea. (a), (b) by compactness and filter closure; (c) because a set lies in a filter iff it lies in every ultrafilter extending it, with Łoś's theorem (Łoś, 1955). Details in Appendix J. □

If c and all its DEF ancestors use limit semantics, $\mathcal{F}_c(w)$ is principal or improper, so the ultraproducts in (c) are just the standard model at the context's point. Under germ semantics with a root parent they are nonstandard models in which the deformed parameters are infinitely close to the ideal along the path and the undeformed ones take *exactly* their actual values: a concrete semantics for McCarthy's ist, and the sense in which Robinson vindicated Leibnizian infinitesimals (Robinson, 1966). $\mathrm{Th}_w(c)$ may be incomplete (germs can oscillate); that is underdetermination the reading must settle.

Proposition 11.4 (Supposition is exact discharge)**.** [T3 Prop 1.5] *For* $c = \mathsf{SUP}(A)$ *with parent* p: $c \models_w \psi$ *iff* $p \models_w A \to \psi$. *In particular,* c *is improper at* w *iff* $p \models_w \neg A$. Lean: Contexts.holdsIn_sup_iff *(propositional)*

Proof. $X \in \mathcal{F}_c(w)$ iff $X \supseteq F \cap \|A\|$ for some $F \in \mathcal{F}_p(w)$, and $\|\psi\| \supseteq F \cap \|A\|$ iff $\|A \to \psi\| \supseteq F$. Take $\psi = \bot$. □

Proposition 11.5 (What idealized contexts may do)**.** [T3 Prop 1.6, rev.] (a) Contradicting the root. *In a hydrostatics frame whose laws do not mention phase change (so the model exists at* $p_{\mathrm{atm}} = 0$*), let* c *deform* p_{atm} *toward* 0 *(limit semantics). Then* $c \models p_{\mathrm{atm}} = 0$ *and* $@ \models p_{\mathrm{atm}} = 1.0 \times 10^5$*, both contexts are proper, and no sentence is both true and false in one context.* (b) Inconsistent limit, consistent germ. *A rope of mass* m_r *is pulled by a force* $\Phi \neq 0$*, with* $\mathrm{dom} = \{m_r > 0\}$*;*

deform m_r toward 0 below the root. The limit context is improper; the germ context is proper and proves $|a| > K$ for every numeral K. (c) Norton's distinction. *Let $c_{\lim}, c_{\mathrm{germ}}$ be the limit and germ children of the root for the same D, λ°, with $\lambda^*[D{:=}\lambda^\circ] \in \mathrm{dom}$, and let $q(\mu)$ be the value of Q at $\lambda^*[D{:=}\mu]$. Then $c_{\mathrm{germ}} \models Q \in I$ for every open interval $I \ni q(\lambda^\circ)$ iff $q(\mu) \to q(\lambda^\circ)$ along $\mathcal{G}$ (where the deformed models exist).*

Part (c) is Norton's "limit property versus property of the limit system" (Norton, 2012): reasoning inside the limit model justifies a value exactly when the query is continuous along the path; d'Alembert's paradox is the failure case. Part (b) is the shape of idealizations that are inconsistent at the limit but consistent along it, such as Painlevé's rigid bodies with Coulomb friction (Stewart, 2000) or a uniform isothermal gas ball in vacuum; *germ semantics* requires well-posedness along the family, not at the limit. Different filters on one family give different contexts, so the path to the limit is part of the reading.

Imports are fixed by the deformation. The import filter of an idealized context is not a third object to learn besides rules and tolerances.

Definition 11.6 (Stability)**.** [T3 Def 1.7] *φ is D-stable* in w if $\mathfrak{M}_w(\lambda) \models \varphi \iff \mathfrak{M}_w(\lambda') \models \varphi$ for all $\lambda, \lambda' \in \mathrm{dom}\,\mathfrak{M}_w$ differing only in coordinates in D. Laws are D-stable, and so are *undeformed data*: arithmetic formulas in the p_i, $i \notin D$.

Lemma 11.7 (Import soundness)**.** *[T3 Lemma 1.8, rev.: (b), (c) added]* (a) *If φ is D-stable in w and $p \models_w \varphi$, then $c \models_w \varphi$ for every* DEF *child c of p deforming D, for any parent filter and path.* (b) Uniform converse. *Fix $\mathfrak{M}_w$ and D. If for every actual point $\lambda^* \in \mathrm{dom}\,\mathfrak{M}_w$ and every ideal value $\lambda^\circ \in \mathbb{R}^D$ the limit child c of the root satisfies $@ \models \varphi \Rightarrow c \models \varphi$, then φ is D-stable.* (c) Scope. *For one fixed deformation the converse of (a) fails (e.g. $\varphi := (k \le 1)$ under "$k \to 0$", limit semantics; Appendix J).* Lean: Contexts.import_sound *((a); propositional)*

Proof idea. (a) The generator built from $F = \|\varphi\|$ lies in $\|\varphi\|$. (b) Put the actual point at a φ-point λ and the ideal value at λ'_D. (c) The child always satisfies $k = 0$. Details in Appendix J. □

Reading. The imports that are sound *uniformly* in the actual point and the ideal value are exactly what the declared top-model family makes stable. "Water's vapour pressure is 2.3 kPa" is a law of a frame with phase equilibrium, where the $p_{\mathrm{atm}} \to 0$ context correctly boils; in incompressible hydrostatics it is not in the language. So learning sound import filters reduces to learning which top-model family the problem intends: part of the reading (§11.6).

11.3 What coherence should constrain

Theorem 11.8 (No truth-functional eternal reading)**.** *[T3 Thm 2.1, rev.] Let c be a* DEF *context, proper at w, with a stipulation S_c such that $c \models_w S_c$ and S_c is false at $\mathfrak{M}_w(\lambda^*_w)$ (e.g. $p_{\mathrm{atm}} = 0$). Let $\tau(c, \varphi)$ translate "φ holds in c" into a world sentence whose truth value at w is $f(S_c^w, \varphi^w)$ for some $f : \{0,1\}^2 \to \{0,1\}$. Then τ does not track in-context truth at w: $c \models_w \top$ and $c \not\models_w \neg S_c$, but $\tau(c, \top)^w = \tau(c, \neg S_c)^w$.* Lean: Contexts.not_exists_truthFunctional_reading *(propositional filters)*

Proof. $\top$ and $\neg S_c$ are both true at w, so both translations have value $f(0,1)$. And $c \models_w S_c$ with $\mathrm{Th}_w(c)$ consistent (Proposition 11.3(b)). □

Stripping ($f = \wedge$) makes every in-context claim false at w, hence incoherent with the root; *conditionalizing* ($f = \rightarrow$) makes every one true, hence vacuous; the other two choices collapse "in c" into "at w" or its negation. Properness is needed: for the improper rope limit of Proposition 11.5(b), conditionalizing tracks in-context truth perfectly.

Consequently an *eternalist* check, which deletes context indices and flags unsatisfiable sets of recorded judgments, flags every recorded sound reductio, every sound idealization contradicting a recorded root judgment, and every pair of sound contexts that record different answers to one query (IPhO 2025 Theory Problems 2 and 3); what is penalized depends on how much of the reasoning is recorded (Corollary J.2, Appendix J).

A learner trained to keep such sets coherent loses classical logic, and not only in the offending context. This is a formal version of the air-pressure worry of Section 1.1.

Proposition 11.9 (Mis-designation forces global sub-classicality)**.** *[T2 Prop 7.1] Let C be a structural consequence relation that is $\mathcal{A}$-coherent ($A \nvdash_C \bot$ for $A \in \mathcal{A}$), where some designated $A \in \mathcal{A}$ is classically inconsistent.* (a) *$C \not\supseteq \mathrm{Cn}_{\mathsf{CPC}}$; moreover, for every classical derivation of $\bot$ from A in a Hilbert or sequent-step format (no discharge metarules), some rule schema used in it is not C-valid: its most general instance is not in C.* (b) *If $A \supseteq \{\alpha, \neg\alpha\}$, then $p, \neg p \nvdash_C \bot$ for every atom p: C is atomically paraconsistent everywhere.* *Lean: Contexts.misdesignation_not_Cn2_le*
Lean: atomic_paraconsistent

Proof idea. (a) Chaining valid instances would give $A \vdash_C \bot$. (b) Substitution and weakening. Details in Appendix J. □

Reading. Designate "$p_{\mathrm{atm}} = 0$ plus the full background" as coherent, and a coherence loss teaches a sub-classical logic for *all* reasoning: atomic paraconsistency if the background states $p_{\mathrm{atm}} \neq 0$ outright, the loss of some classical schema if it only derives it. The fix has three parts. Designate only consistent chunks (Brown and Priest, 2004). Keep the root *phenomenological* (interval claims about measurables, observations, exports), with laws living in model contexts (Cartwright, 1983; Wilson, 2006); a root holding unrestricted laws may itself be inconsistent. And adopt *meta-level eternalism*: $\mathrm{ist}(c, \varphi)$ is an eternal proposition about a finitely specified context (McCarthy, 1993), by Theorem 11.8 necessarily non-truth-functional in the world's facts. Which contexts, then, must be consistent?

11.3.1 The right constraint: realizability

Definition 11.10 (Judgment sets, bridges, realizations)**.** [T3 Def 2.3, rev.] "Model" and "satisfiable" refer to $\mathcal{K}$; for a set M of structures, $M \models \chi$ means $N \models \chi$ for all $N \in M$. A *judgment set* J is a finite set of triples (c, Γ, φ) over a context tree in which SUP contexts have only SUP descendants; a set C_{cert} of DEF contexts is *certified* (claimed well-posed). A *bridge* β has a pattern $\phi_\beta(v)$, a side condition σ_β and an export formula $\varepsilon_\beta(v)$, $v \in V$. It is a *value-export bridge* for Q if $\phi_\beta(v)$ is $Q = \bar{v}$ with standard names (distinct values denote distinct objects in $\mathcal{K}$) and σ_β does not depend on v; it is *informative* if $\{\sigma_\beta\} \cup \{\varepsilon_\beta(v) : v \in V_0\}$ has no model for some *finite* $V_0 \subseteq V$ (e.g. interval exports over unbounded V). A *bridge use* is $(c, \beta, t) \in U$ with c a DEF context and $(c, \emptyset, \phi_\beta(t))$, $(\pi c, \emptyset, \sigma_\beta)$, $(\pi c, \emptyset, \varepsilon_\beta(t)) \in J$. A *(frame-free) realization* assigns each context a set $M_c \subseteq \mathcal{K}$ with (R0) $M_{@} \neq \emptyset$; (R1) $M_c = M_{\pi c} \cap \|A\|$ for $c = \mathsf{SUP}(A)$; (R2) M_c arbitrary, possibly empty, for DEF contexts; (R3) every judgment holds: for $N \in M_c$, $N \models \bigwedge \Gamma$ implies $N \models \varphi$; (R4) every used bridge is sound: for all v, $M_c \models \phi_\beta(v)$ implies $M_{\pi c} \models \sigma_\beta \rightarrow \varepsilon_\beta(v)$; (R5) $M_c \neq \emptyset$ for $c \in C_{\mathrm{cert}}$. The *SUP-collapse* J^* replaces each (c, Γ, φ) with c a SUP context by

$(b, \Gamma \cup \mathrm{Asm}(c), \varphi)$, where b is the nearest non-SUP ancestor and $\mathrm{Asm}(c)$ the suppositions on the path; $\mathrm{Th}^*(c) := \{\bigwedge \Gamma \to \varphi : (c, \Gamma, \varphi) \in J^*\}$. The *anchored set* Anc is the least set of non-SUP contexts containing @ and C_{cert} and containing c whenever $(c, \beta, t) \in U$ with β informative and $\pi c \in \mathrm{Anc}$.

Theorem 11.11 (Realizability criterion)**.** *[T3 Thm 2.4, rev.: (a) frame-free; (b), (c) new] Suppose every used bridge is an informative value-export bridge.* (a) Frame-free. *J has a realization satisfying (R0)–(R5) iff $\mathrm{Th}^*(c)$ is satisfiable for every $c \in \mathrm{Anc}$.* (b) Necessity in the frame semantics. *Call J* frame-realizable *if for some world w, with the filters of Definition 11.2, every judgment holds ($c \models_w \bigwedge \Gamma \to \varphi$), every $c \in C_{\mathrm{cert}}$ is proper, and every bridge use is sound at w ($c \models_w \phi_\beta(v)$ implies $\pi c \models_w \sigma_\beta \to \varepsilon_\beta(v)$). If J is frame-realizable, the criterion of (a) holds, for limit or germ paths.* (c) Non-sufficiency. *The converse of (b) fails. CE1: a certified limit child c of the root deforms $D = \{2\}$ toward 0, and $J = \{(@, \emptyset, p_1 = 1), (c, \emptyset, p_1 = 0)\}$; the criterion holds, but the root forces $\lambda_1^* = 1$, and c, proper because certified, sits at $\lambda^*[2{:=}0]$, so $c \models p_1 = 1$. CE2: two certified siblings with the same declaration have equal filters, so $Q = 1$ in one and $Q = 2$ in the other cannot both hold.*

Proof idea. By (R1), SUP contexts impose on their base exactly the constraints Th^*. An *anchoring lemma* says every anchored context has $M_c \neq \emptyset$: from an empty M_c an informative bridge would export every $\varepsilon_\beta(v)$. Conversely, put one model of $\mathrm{Th}^*(c)$ at each anchored context and $\emptyset$ elsewhere. Part (b) uses properness for nonemptiness. Full proof in Appendix J. □

Brute-force checks agree with (a), show that informativeness is needed, and find no violation of (b) (computed) (Appendix J). Bridge soundness here is ω-free; for judgment sets produced by the calculus of Definition 11.13, add $(\pi c, \emptyset, \omega_c)$ to each bridge use and read (R4) as $M_{\pi c} \models \sigma_\beta \wedge \omega_c \to \varepsilon_\beta(v)$; the anchoring arguments are unchanged. The locality of (a) is a *consequence of (R2)*: in the frame semantics rigid parameters and shared filters force cross-context constraints, which under limit semantics are a pointwise adjunction (Proposition J.3), the structural core of discussive and preservationist logics (Jaśkowski, 1969; Schotch and Jennings, 1980). Germ semantics is open.

Consequences for reductio and ill-posed contexts are collected in Corollary J.4 (Appendix J): a SUP context deriving $\bot$ contributes exactly the negation of its suppositions, an idealized one deriving $\bot$ constrains frame-free realizability only if anchored, and eternalist coherence is strictly stronger than frame-free realizability.

Reading. Arguments for P and $\neg P$ in contradictory contexts should not be penalized, and Theorem 11.11(a) says exactly which contexts *must* be consistent: the root, and idealizations that are certified or export informatively. Ill-posed idealizations must not export, and this is now an iff. The frame-semantic constraints CE1, CE2 are enforced in the checker by a declaration discipline, not by coherence (§11.6).

Remark 11.12 (Belief states, worlds, contextuality). Section 9 gives the multi-context counterpart: coherence on designated contexts is equivalent to a model nonempty there (Theorem 9.19 and Corollary 9.20); bridges must be read as rules of proof over belief states, not conditionals between worlds (Theorem 9.22); and locally coherent contexts need not glue into one world (Proposition 9.23), a logical form of contextuality (Abramsky and Brandenburger, 2011).

11.3.2 A calculus, its soundness, and blame localization

Definition 11.13 (Context calculus)**.** [T3 §2.3, rev.] Judgments are $c \Vdash \varphi$; $\hat{R}$ is the (learned) rule set and Imp_c the declared import set of c. (Ax) $c \Vdash \varphi$ for $\varphi \in \mathrm{Ax}(c)$, where $\mathrm{Ax}(@) = \mathrm{Ax}_\rho$, $\mathrm{Ax}(\mathsf{SUP}(A)) = \{A\}$, and $\mathrm{Ax}(\mathsf{DEF}(D, \lambda^\circ, \mathcal{G}))$ is the set of *canonical stipulations*: $p_i = \lambda_i^\circ$ $(i \in D)$ under limit semantics, $p_D \in U$ for definable $U \in \mathcal{G}$ under germ semantics; stipulations about undeclared parameters are not axioms. (Step) From $c \Vdash \varphi_1, \dots, c \Vdash \varphi_k$ and an instance $\varphi_1, \dots, \varphi_k \rhd \psi$ of $\hat{R}$, infer $c \Vdash \psi$. (Imp) From $\pi c \Vdash \varphi$ with $\varphi \in \mathrm{Imp}_c$, infer $c \Vdash \varphi$. (Dis) For $c = \mathsf{SUP}(A)$: from $c \Vdash \psi$ infer $\pi c \Vdash A \to \psi$. (Exp) For a DEF context c: from $c \Vdash \phi_\beta(t)$, $\pi c \Vdash \sigma_\beta$ and $\pi c \Vdash \omega_c$, infer $\pi c \Vdash \varepsilon_\beta(t)$; here the *well-posedness certificate* ω_c is a sentence such that for every w and $\lambda \in \|\omega_c\|_w$, $\{\mu : \lambda[D{:=}\mu] \in \operatorname{dom} \mathfrak{M}_w\} \in \mathcal{G}$. An instance is *sound at* w if it preserves truth there (for Step: $c \models_w \bigwedge_i \varphi_i \to \psi$), and *sound* if sound at every $w \in W_\rho$.

Step soundness is *context-relative*: time reversal is sound in a drag-free DEF context and unsound at the root.

Theorem 11.14 (Soundness)**.** *[T3 Thm 2.5] If every Ax in a derivation of $c \Vdash \varphi$ is true at w and every Step, Imp and Exp instance in it is sound at w, then $c \models_w \varphi$.*

Proof. Induction: Ax and Step by hypothesis and filter closure (Proposition 11.3(a)); Imp and Exp by soundness of the instance; Dis by Proposition 11.4. □

Corollary 11.15 (Blame localization)**.** *[T3 Cor 2.6, rev.] Let $W_\rho \neq \emptyset$. Call d* designated *if it is the root, or a certified* DEF *context with a* properness witness *$w_d \in W_\rho$ at which d is proper. Let $\mathcal{D}$ derive $d \Vdash \bot$, and let its* cone *be all instances in its derivation tree, including those in sub-contexts whose conclusions reach d by Dis or Exp. (a) At every w at which d is proper (every $w \in W_\rho$ if $d = @$), the cone contains an Ax false at w or an instance unsound at w. (b) Items outside the cone are never implicated: an undischarged* SUP *contradiction, or a pair of contradictory* ist*-claims. (c) If $\pi d \models_w \omega_d$ and πd is proper at w, then d is proper at w; so parent-derived certificates are properness witnesses.*

Proof idea. (a) is the contrapositive of Theorem 11.14; (b) follows. (c) In each parent filter set pick $\lambda \in \|\omega_d\|$. Proposition J.5 adds how $\bot$ in an anchored DEF context propagates toward the root or a certified context, and where the blame then falls. □

The cone is a *negative bag* in the sense of Section 6, and by (a) a correct target never contains it whole. So with the designated contexts of Corollary 11.15 (the root, and certified contexts with properness witnesses) as the *only* designated contexts, an oligarchic-halving learner makes at most $\log_2(1/w^*)$ detections and no false alarms while the reading and the certifications are correct (Theorem 6.5; false alarms are priced by Theorem 6.7). What root coherence cannot see is silent unsoundness: exports that are false but consistent with everything else (§11.5).

11.3.3 Reductio hygiene, and $100 = 99.9$

The motivating notes' second worry (§11.1) continues: "and there's a chance such an argument could equally be provided starting from the statement itself". Let c have an explicit *imported fragment* K_c, a finite set of premises. A *reductio of A in c* is an $\hat{R}$-derivation of $\bot$ from $K_c \cup \{A\}$ in $\mathsf{SUP}(A)$; accepting it yields $c \Vdash \neg A$. An instance is *c-sound at w* if $c \models_w \bigwedge \varphi_i \to \psi$.

Theorem 11.16 (Reductio hygiene)**.** *[T3 Thm 1.9, rev.: (c)–(e)]* (a) Soundness. *If $c \models_w K_c$ and every instance used is c-sound at w, every accepted reductio of A yields $c \models_w \neg A$.* (b) Inconsistent imports refute everything. *If $K_c \vdash_{\hat{R}} \bot$, every sentence, A and $\neg A$ alike, has a reductio in c, whatever the world.* (c) Derivation-local tests fail. *Let $K = \{A \to q, \neg A \to q, \neg q\}$. K is unsatisfiable, but the reductio of A uses only the satisfiable $\{A \to q, \neg q\}$ and that of $\neg A$ only the satisfiable $\{\neg A \to q, \neg q\}$; any test that is a function of the derivation alone and accepts every sound reductio from a consistent premise set accepts both.* (d) A consistency certificate suffices. *If a structure N satisfies K_c and every instance used ($N \models \bigwedge \varphi_i \to \psi$), every reductio-refuted A is false in N, so no A and $\neg A$ are both refuted. The condition on instances is automatic for logically sound $\hat{R}$ and needed otherwise. If c is proper at w, $c \models_w K_c$ and the instances are c-sound at w, a suitable $N = \mathfrak{M}_w(\lambda)$ exists: a properness witness is a certificate.* (e) Learning consequence. *A coherence loss fed with reductios from a context whose fragment is $\hat{R}$-inconsistent receives "A false" and "$\neg A$ false" for every A: jointly unsatisfiable, systematic, world-independent labels.* Lean: Contexts.refutes_of_inconsistent Lean: derivation_local_test_accepts_both Lean: certificate_not_both *((b)–(d))*

Proof idea. (a) $\mathsf{SUP}(A)$ refines c's filter, so induction gives $\mathsf{SUP}(A) \models_w \bot$, and Proposition 11.4 gives $\neg A$. (b) Monotonicity. (c) Truth tables. (d) Induction inside N; the instance condition is needed when $c \models q$ and $K_c = \{\neg q\}$. Details in Appendix J. □

Reading. A reductio in an inadequate framework can indeed be spurious, as the notes suggest; the remedy is not to abandon reductio. Hygiene is a *global* certificate on the imported fragment, not a syntactic relevance condition on the derivation, which (c) shows cannot work. In physics the certificate is a well-posedness witness: an explicit or validated-numerical solution of the local model.

Proposition 11.17 (The continuity asymmetry)**.** *[T3 Prop 1.10] Let premises $x_i = m_i$ hold only to tolerance, $x_i \in [m_i \pm r_i]$, with box B.* (a) *A* direct *derivation of $y = g(x)$, g continuous on B, errs by at most $\sup_{x \in B} |g(x) - g(m)|$, which tends to 0 with the tolerances by continuity of g.* (b) Exact-equality *reasoning derives $\bot$ from approximately true premises read as exact: $x = 100\,m$ (flat Earth) and $x = 99.9\,m$ (sphere), both true to $\pm 0.2\,m$, give $0 = 0.1$ by substitution, while as intervals they give $x \in [99.8, 100.1] \neq \emptyset$.* (c) Sound replacement. *If A with the exact laws implies $f(x) = 0$, and F is an inclusion-monotone interval extension of f with $0 \notin F(B)$, the interval premises refute A soundly; the condition is sufficient, not necessary.*

Proof. (a) Definition. (b) Arithmetic. (c) Inclusion monotonicity (Moore, 1966) gives $f(B) \subseteq F(B) \not\ni 0$. □

Reading. The notes ask: "how come we usually avoid this sort of catastrophic reasoning [100 = 99.9]? i don't think any nice 'formal criterion' is going to be right?" (`philosophy/philosophy of science/the domain of applicability of a scientific mode(l).md`). Here is a partial answer, a sufficient condition rather than a definition of good reasoning: "$x = y$" defines a closed set of data with empty interior, so derivability of $\bot$ is a *discontinuous* function of the data, while derived values are continuous ones. This gives one precise reason behind a tentative suggestion in the notes, that the danger of reductio in inadequate frameworks is "maybe ... some argument in favor of trusting 'constructive' reasoning more than 'non-constructive' reasoning, at least in messy domains" (the note quoted in §11.1). The fix is interval typing with a *margin* (Theorem J.8(c)).

11.4 Export: what can justify leaving a context

Contexts let an idealized model contradict the world; exports carry its conclusions back. No export is free: information about the idealized model alone, however complete (its jet, its germ, finitely many values elsewhere), never justifies a finite-tolerance claim at the actual parameter (Theorem 11.19). What justifies one is information of small radius about the de-idealized family (Proposition 11.20). A solver who may choose contexts can then stipulate any answer unless side conditions are evaluated in the parent and contexts are declared deformations (Theorem 11.22).

Fix a one-parameter deformation (the multi-parameter case is identical) with ideal value 0 and actual value $\lambda^* \in (0, \Lambda]$. A family is represented by its query function $Q : [0, \Lambda] \to \mathbb{R}$; a *family class* is a set $\mathcal{C}$ of them. An *information map* I gives what an export rule may use (the jet at 0, the germ, a restriction $Q|_S$, finitely many values). An *export rule* E maps $(I(Q), \lambda^*)$ to "abstain" or a finite interval $[q \pm \epsilon]$, and is *sound on* $\mathcal{C}$ if $Q(\lambda^*) \in E(I(Q), \lambda^*)$ whenever it does not abstain.

Lemma 11.18 (Invisible perturbation)**.** *[T3 Lemma 3.2; Traub et al. 1988] If some f with $f(\lambda^*) \neq 0$ has $Q + Af \in \mathcal{C}$ and $I(Q + Af) = I(Q)$ for all $Q \in \mathcal{C}$, $A \in \mathbb{R}$, then every export rule sound on $\mathcal{C}$ abstains at λ^* on every $Q \in \mathcal{C}$.* Lean: Export.invisible_perturbation

Proof. An output $[q \pm \epsilon]$ on $I(Q) = I(Q + Af)$ would have to contain $Q(\lambda^*) + Af(\lambda^*)$ for every A. □

Theorem 11.19 (No free export)**.** *[T3 Thm 3.3, rev.: closure hypothesis] Every rule sound on $\mathcal{C}$ abstains at every $\lambda^* > 0$, on every $Q \in \mathcal{C}$, if (a) $\mathcal{C} + C^\infty[0, \Lambda] \subseteq \mathcal{C}$ and I is the full jet at 0; or (b) $\mathcal{C} + C^\infty[0, \Lambda] \subseteq \mathcal{C}$ and I is the germ at 0, or $Q|_S$ for any S whose closure omits λ^*; or (c) $\mathcal{C} + \{\text{polynomials}\} \subseteq \mathcal{C}$ (e.g. $\mathcal{C}$ is the class of all restrictions of entire functions) and I is the jet up to a finite order N; or (d) $\mathcal{C} + C^\infty[0, \Lambda] \subseteq \mathcal{C}$ and I is any finite set of values $Q(\lambda_j)$, $\lambda_j \neq \lambda^*$. With only $\mathcal{C} \supseteq C^\infty$, the conclusion still holds for $Q \in C^\infty$ in (a), (b) and (d), and for every $Q \in \mathcal{C}$ in (d); in (b) it can fail outside C^∞.* Lean: Export.no_free_export_germ Lean: no_free_export_finite_jet Lean: no_free_export_samples Lean: no_free_export_samples_of_poly_subset *((b)–(d); variants for (b), (c))*

Proof idea. Lemma 11.18 with the flat e^{-1/λ^2} in (a), a C^∞ bump at λ^* vanishing near S in (b) and (d), and λ^{N+1} in (c). Under the weaker hypothesis, (d) perturbs the interpolating polynomial, and the germ of $\sqrt{\lambda}$ breaks (b). Details in Appendix J. □

For analytic families the full jet determines $Q(\lambda^*)$, but a sound use of a series needs a remainder bound, which is information *away from* 0. This vindicates McMullin's and Laymon's demand for de-idealization information (McMullin, 1985; Laymon, 1987).

Proposition 11.20 (Minimax export is the radius of information)**.** *[T3 Prop 3.4; Traub et al. 1988] For an information value i let $\mathcal{C}_i = \{Q \in \mathcal{C} : I(Q) = i\}$ and $r(i) = \frac{1}{2}(\sup_{\mathcal{C}_i} Q(\lambda^*) - \inf_{\mathcal{C}_i} Q(\lambda^*))$. A sound rule with tolerance ϵ at i exists iff $r(i) \leq \epsilon$; the optimal one outputs the midrange.* Lean: Export.exists_interval_iff_radius_le

Proof. A sound interval must contain $\{Q(\lambda^*) : Q \in \mathcal{C}_i\}$, and the midrange interval of radius $r(i)$ does. □

So an *export certificate is information of small radius*: a uniform derivative bound, a Gronwall bound, an invariance that pins Q, a monotone sandwich between the ideal and a computed less-ideal value, or data near λ^*.

Theorem 11.21 (Certified-neighbourhood and rate-qualified export)**.** *[T3 Thm 3.5] (a) If the parent proves $\lambda^* \in U$ and a frame theorem gives $\sup_{\lambda\in U} |Q(\mathfrak{M}(\lambda)) - q| \le \epsilon$, the parent may soundly assert $Q \in [q \pm \epsilon]$. (b) A germ claim $Q \sim q$ exports nothing at a fixed λ^*; with a rate $|Q/q - 1| \le C\|\lambda\|^k$ on U it exports $|Q/q - 1| \le C \sup_U \|\lambda\|^k$. (c) If the problem declares a regime, the root is a germ context and asymptotic equivalence is the correctness relation itself; for exp-log expressions in one parameter it is decidable modulo zero-equivalence of constants (Gruntz, 1996; Richardson, 1968).*

Part (a) fixes the logical form of every side condition: a parent-certified set plus a uniform bound over it, not a value at a guessed point. Certificates of this form include a checkable Gronwall bound for finite-horizon exports (Theorem J.6); a rigorous range certificate for projectiles under any drag dominated by quadratic drag (Theorem J.7), which for a steel ball certifies $R \in [9.653, 10.483]$ m against a simulated 9.978 m and for a ping-pong ball is silent; and error budgets for chains of bridges (Theorem J.8), under which "$100 - 99.9$" with tolerances 0.2 certifies not even a sign. All three are stated and proved in Appendix J.5.

Adversarial stipulation. Context choice is an attack surface. Take a schema "neglect p_D": pattern $Q = \bar{v}$ in the child, side condition $s(\vec{p}) \le \eta$ for a dimensionless smallness parameter (e.g. kv_0^2/g), conclusion $Q \in [v \pm \epsilon]$ in the parent.

Theorem 11.22 (Where side conditions live, and what contexts may be)**.** *[T3 Thm 3.9, rev.: (c)] (a)* Child-evaluated side conditions are unsound. *If Exp accepts $c \Vdash s \le \eta$ in place of $\pi c \Vdash s \le \eta$, the export fires on every world (at $p_D = 0$, $s = 0$); on any C^∞-closed class this is unsound for every ϵ. The ping-pong ball gets the drag-free range $\pm 5\%$, while its true range is 49% short. (b)* Free stipulations are unsound even with parent evaluation. *If contexts may set undeclared parameters, a solver stipulates $p_D = 0$ and $g_c = g'$; the parent condition holds for the steel ball, and the child's $Q = R_0(g')$ is any desired value. (c)* Sufficiency. *Suppose contexts are declared deformations with only canonical stipulations as axioms; σ_β and a well-posedness certificate ω_c are derived in the parent; and the schema is a theorem of the frame: for all w, all $\lambda \in \operatorname{dom}\mathfrak{M}_w$ with $\mathfrak{M}_w(\lambda) \models \sigma_\beta$ and all v, if $\{\mu : \lambda[D{:=}\mu] \in \operatorname{dom}\mathfrak{M}_w,\ Q(\mathfrak{M}_w(\lambda[D{:=}\mu])) = v\} \in \mathcal{G}$ then $\mathfrak{M}_w(\lambda) \models \varepsilon_\beta(v)$. Then every accepted export is sound, for every solver. Without ω_c, (c) is false: in the rope frame of Proposition 11.5(b) under limit semantics, the improper child proves $a = 0$, the schema holds vacuously, the parent proves $m_r \le \eta$, and Exp would accept $|a| \le \epsilon$ while $|a| = |\Phi|/m_r \ge |\Phi|/\eta$.*

Proof idea. (a), (b) are the constructions. For (c), the child-filter generator witnessing $Q = \bar{v}$, intersected with the parent sets for σ_β and ω_c, puts the schema's antecedent at every point of a parent filter set. Details in Appendix J. □

Theorem J.7 is such a schema (Appendix J.5). Germ contexts naturally conclude $Q \simeq v$ (standard part v); the analogous asymptotic schema is sound below a principal parent given ω_c (Lemma J.9), and under a nested germ parent it also needs uniform convergence on a parent filter set.

Reading. This is the search problem of Section 3 transplanted to idealizations: a solver chooses contexts as a prover chooses steps. The remedy is the analogue of cautious acceptance: contexts restricted to declared deformations with canonical stipulations (Strevens's "default values" made into a syntactic discipline; Strevens 2008), side conditions evaluated in the parent, and a parent-derived well-posedness certificate supplied by the trusted catalogue (Assumption 11.26).

11.5 Learning tolerances and validity regions

Export certificates need tolerances and validity regions, which are natural targets for learning. This subsection asks which signals learn them soundly against a solver who picks the regime. Coherence among exports catches only tolerances that are too narrow (Theorem 11.23); exchangeable calibration gives average coverage and nothing under selection; and only a *proved* regularity property turns validated simulation of the top model into a sound certifier. A bridge schema s has a dimensionless regime parameter $\pi \in \Pi$ and an error function $e_s(\pi)$, the relative difference between top model and idealization; the task is to learn *where* $e_s \le \tau$.

Theorem 11.23 (Coherence cannot calibrate tolerances)**.** *[T3 Thm 4.1] Schemas $s = 1, \dots, m$ export $[q_{s,i} \pm \epsilon_s]$ for true values V_i; a* coherence-only calibrator *chooses $\hat{\epsilon}$ from the discrepancies $(q_{s,i} - q_{s',i})$ alone. If the environment class is closed under* common shifts *((q, V) admissible implies $(q, V + b)$ admissible for $|b| \le \beta$: all models may share an omitted effect), every $\hat{\epsilon}$ sound on the class has $\hat{\epsilon}_s \ge \beta$; with unbounded shifts, $\hat{\epsilon} = \infty$.* Lean: Export.coherence_cannot_calibrate

Proof. $(q, V \pm \beta)$ give the same data, hence the same $\hat{\epsilon}$, and soundness for both signs gives $2\beta \le 2\hat{\epsilon}_s$. □

Worse, coherent tolerance vectors form an up-set, so a minimal coherent repair only ever widens tolerances and never restricts a schema's regime, and two schemas sharing an idealization never disagree: their agreement is common-mode, not independent evidence (Levins, 1966; Odenbaugh and Alexandrova, 2011). Appendix J.6 gives pendulum numbers for both effects (computed).

The other signals are treated in Appendix J.6: split conformal calibration fails under selection (Theorem J.10); without regularity, finitely many simulations certify only the queried points (Theorem J.11); a *proved* Lipschitz constant or monotonicity gives a sound certifier at a matching query cost (Theorems J.12 and J.13); and monotonicity can fail, for a damped pendulum (Proposition J.14).

Reading. The monotonicity that Laymon (1987) uses to explicate idealized data (better data, better predictions) is a property of a model–query pair, to be proved, not assumed. Once such a property is proved, simulation of the top model is a sound oracle for validity regions; world feedback proper is needed only to calibrate the top model itself, and for olympiad grading, where the root is c_P, not even that.

11.6 The Structured Physics Solution checker

Definition 11.24 (Structured Physics Solution)**.** [T3 Def 5.1; L9 §7.1] An *SPS* is a tuple $(\rho, \mathcal{T}, \mathcal{D}, X)$: a purported reading ρ (frame class and parameter roles: given, measured, free, deformable; the query, its dimension and allowed symbols; conventions), which, if the setter supplies a formal specification, must be that specification or agree with it (W_ρ, Q and its type equal, Ax_ρ its axioms or kernel-derivable from them); a context tree $\mathcal{T}$; a derivation $\mathcal{D}$ in the calculus of Definition 11.13 whose Exp instances cite schemas from a catalogue $\mathcal{B}$; and a final judgment $X = (@ \Vdash Q \in [q \pm \delta])$. A declared-regime problem uses a top-level germ context d whose declaration belongs to the trusted specification, and $X = (d \Vdash \chi)$.

Definition 11.25 (The checker)**.** [T3 Def 5.2, rev.: Ax legality, schema indexing] The checker accepts iff all of the following hold. *Reading:* if a formal specification is supplied, ρ is it or passes the

comparison of Definition 11.24 (check C0 in §11.7). *Ax legality:* every Ax at the root is in Ax_ρ, every Ax in $\mathsf{SUP}(A)$ is A, every Ax in a DEF context is a canonical stipulation of its declaration. *Kernel and laws:* every Step is accepted by the kernel, and every law it cites is in the trusted law catalogue. *Schema indexing:* every Exp cites a $\beta \in \mathcal{B}$ whose catalogue entry is indexed by *exactly* the context's declaration $(D, \lambda^\circ, \mathcal{G})$; σ_β and the well-posedness certificate ω_β listed in the catalogue entry of β (used as ω_c) are derived *in the parent*, so the solver may not choose ω_c; asymptotic schemas are cited only below a principal parent (the root, or a limit-semantics context all of whose DEF ancestors use limit semantics). *Imports:* every Imp into a DEF context is a catalogue law or an undeformed datum for its D (imports into SUP children are unrestricted). *Shape:* every non-root context is SUP, DEF or AUX, every AUX bridge is a kernel-checked theorem, and X is derived at the root (or the declared germ root). All checks are deterministic.

Ax legality and schema indexing are necessary: without them the checker accepts the attack of Theorem 11.22(b). The list of "deformable" parameters need not be trusted: if a solver uses $\mathsf{DEF}(\{k, g\}, (0, g'))$, the "neglect k" schema, indexed by $\mathsf{DEF}(\{k\}, 0)$, cannot be cited.

Assumption 11.26 (Trusted base)**.** [T3 §5, rev.] (TB1) The kernel and the law catalogue are sound: every accepted Step instance and every catalogue law is valid in every $\mathfrak{M}_w(\lambda)$, $\lambda \in \operatorname{dom} \mathfrak{M}_w$, $w \in W_\rho$ (including top-model theorems used as certified-neighbourhood bridges). (TB2) Every $\beta \in \mathcal{B}$ is a theorem of the frame for its indexed declaration, in the pointwise form of Theorem 11.22(c) or the asymptotic form of Lemma J.9; and each catalogue entry carries a well-posedness certificate ω_β for its declaration: for every w and $\lambda \in \|\omega_\beta\|_w$, $\{\mu : \lambda[D{:=}\mu] \in \operatorname{dom} \mathfrak{M}_w\} \in \mathcal{G}$. (TB3) Validated numerics return correct enclosures.

Theorem 11.27 (Relative soundness of the SPS checker)**.** *[T3 Thm 5.3, rev.] Assume (TB1)–(TB3), and that ρ is a reading in the sense of Definition 11.1, i.e. Ax_ρ holds at every $w \in W_\rho$ (guaranteed when the Reading check compares ρ with a formal specification). For every solver whatsoever, if the checker accepts $(\rho, \mathcal{T}, \mathcal{D}, X)$ with a root judgment X, then $Q^{\mathfrak{M}_w(\lambda^*_w)} \in [q \pm \delta]$ for every $w \in W_\rho$ (for a germ root, $d \models_w \chi$). Hence if the intended reading satisfies $W_{\rho^*} \subseteq W_\rho$ the export is supertrue for the intended problem, and if the actual system is in W_{ρ^*} it is true of the world.*

Proof idea. Theorem 11.14 at each $w \in W_\rho$: Step by (TB1); root axioms because ρ is a reading; DEF axioms because canonical stipulations hold in the child filter (the check that blocks Theorem 11.22(b)); imports by Lemma 11.7(a); exports by Theorem 11.22(c) with (TB2), (TB3) and schema indexing (by (TB2), ω_c is a genuine certificate). Nothing depends on how $\mathcal{D}$ was produced. Details in Appendix J. □

The residual *judgment layer* is exactly (J1) *reading adequacy*: ρ is a reading (Ax_ρ holds at every $w \in W_\rho$) and $W_{\rho^*} \subseteq W_\rho$; when ρ is the setter's formal specification only the second half remains; (J2) *top-model adequacy*, that the actual system is a world in W_{ρ^*} (needed only for world claims, not for olympiad grading, where c_P is the root); and (J3) the *trusted base* (TB1)–(TB3). (J3) is the trusted core of any proof checker. (J1) and (J2) cannot be removed:

Theorem 11.28 (The judgment layer is ineliminable)**.** *[T3 Thm 5.4, rev.] (a)* Reading. *Let $\mathcal{R}$ be the readings that the checker's inputs (text, figures, SPS) do not exclude. A checker sound for every intended $\rho^* \in \mathcal{R}$ accepts X only if X holds on $\bigcup_{\rho \in \mathcal{R}} W_\rho$. (b)* Open world. *Extend the frame by an unmodelled effect μ, with $\mathfrak{M}'(\lambda, 0) = \mathfrak{M}(\lambda)$. If, for the given slice $\mathfrak{M}'(\cdot, 0)$ and actual λ^*, some admissible extension has actual $\mu^* \neq 0$, and the admissible extensions are closed under $Q \mapsto Q + Af(\mu)$ for all $A \in \mathbb{R}$ and C^∞ f with $f(0) = 0$, then every finite-tolerance world claim*

accepted by a checker that sees only the slice $\mathfrak{M}'(\cdot, 0)$ and the SPS is false in some admissible extension.

Proof. (a) If X fails at some $w \in W_{\rho_0}$, $\rho_0 \in \mathcal{R}$, the checker is unsound for $\rho^* = \rho_0$. (b) Lemma 11.18 in the μ direction, with $f(\mu^*) = 1$. □

If $\mu^* = 0$, or if laws bound Q, (b) can fail: then the open world has been closed by a modelling assumption, i.e. moved into (J2).

Theorem 11.29 (Supervaluational correctness; completion invariance)**.** *[T3 Thm 5.5, rev.] Write $Q^w = Q^{\mathfrak{M}_w(\lambda_w^*)}$, with W_ρ ranging over all admissible completions (free parameters $\theta_f \in \Theta_f$, measured ones in their intervals). (a) "$Q \in [q \pm \delta]$" is supertrue iff $\sup_{w \in W_\rho} |Q^w - q| \leq \delta$; a supertrue answer at precision δ exists iff $\operatorname{osc}_{W_\rho} Q \leq 2\delta$, and the midrange is one. (b) A* completion-invariance certificate *is a root derivation of $Q \in [q \pm \delta]$ from Ax_ρ in which θ_f enters only through "$\theta_f \in \Theta_f$". By Theorem 11.27 the conclusion of every accepted root derivation holds at every completion, so in the formal checker completion invariance is not a separate obligation but the content of deriving from Ax_ρ rather than from a narrowed axiom set. Informal forms are $\partial Q / \partial \theta_f = 0$ on a connected Θ_f* where Q is differentiable in θ_f*, or a variation bound; in §11.7, E is constant in α only while the point stays lit for every admissible α, and jumps to 0 at $s = L \tan \alpha$. (c) Supertruth over $\bigcup \mathcal{R}$ is the* largest *acceptance criterion sound under reading uncertainty.*

Reading. This answers the question quoted at the opening of this section: a claim is true in the context at hand iff it holds on the context's filter in every admissible completion of the reading. It is supervaluation (Fine, 1975) over completions of an underspecified problem. Checking it decomposes into kernel steps, stable imports, certified bridges with parent-evaluated side conditions, and a reading; only the reading is not mechanizable. Hence our strongest recommendation: *the problem setter should supply a formal specification of the admissible completions*, after which the checker is sound relative to the top model and the trusted base (TB1)–(TB3) alone. An episode in the notes' introspective solution of the problem of §11.7 ("i suspected [the angle doesn't matter] but then i messed up a calculation", `introspection/introspect-solving physics olympiad problems/introspect-solving eupho 2025-T1.md`) is a failed completion-invariance obligation that a checker would flag at once.

Proposition 11.30 (Gricean determinacy is one-sided)**.** *[T3 Prop 5.6] Call ρ* determinate at δ *if $\operatorname{osc}_{W_\rho} Q \leq 2\delta$. If $W_{\rho'} \subseteq W_\rho$ then $\operatorname{osc}_{\rho'} Q \leq \operatorname{osc}_\rho Q$. So "the problem is well-posed" can refute only (some) over-wide readings, never over-narrow ones; and silent narrowing (fixing the sun's angle, choosing a specular mirror, importing an extra law, deciding "the finger is horizontal") never decreases determinacy, while it violates (J1).*

Proof. A supremum over a subset is at most the supremum over the set; dually for infima. □

The free internal signal for readings thus pushes *toward* the unsafe error: silent narrowing is the solver's natural attack. With Section 6 and Theorem 11.23 this completes a table of three one-sided signals: The opposite signal must come from outside: a formal specification by the setter (which moves (J1) into the checked part); intended-completion data (official solutions, marking schemes; the introspection note, on whether a finger in the part-(b) photograph is tilted: "ok i peeked at the solution and it seems to assume that's not the case"); and round-trip checks such as rendering the model's observable and comparing it with the figure, which are also one-sided: passing one does not certify a reading. Section 13 returns to the table.

Table 15: Three objects, three free signals, three blind spots.

Object	Free internal signal	Error it catches	Error it misses
rules	coherence (Section 6)	over-acceptance	under-acceptance
tolerances	coherence across exports (Theorem 11.23)	too narrow	too wide
readings	Gricean determinacy (Proposition 11.30)	(some) too wide	too narrow

11.7 Worked example: EuPhO 2025 Theory 1(a), the polished chair leg

The setup is reconstructed from the introspection note of §11.6 and has not been checked against the official text; the answer $E = \frac{aI_0}{2}|\sin\frac{\varphi}{2}|/r$ is the one reached in that note, which records it as matching the official answer. *Specification:* a vertical specular cylinder of radius a, lit over height L, stands on a flat floor; the sun is a parallel beam of irradiance I at zenith angle α, travelling in $+x$; the floor's direct illuminance is $I_0 = I\cos\alpha$. The query is the reflected illuminance ("surplus") $E(r,\varphi)$, with φ measured from the anti-sun direction $+x$. Given a, I_0, L; free completion $\alpha \in [30°, 60°]$, which the photograph does not fix; conventions: perfect specular reflectance, geometric optics, incoherent addition, sun radius $\delta_s = 4.65\,\text{mrad}$, leg vertical; answer type: a function of (a, I_0, r, φ) with the dimension of I_0. Write $e(\vartheta) = (\cos\vartheta, \sin\vartheta)$.

Proposition 11.31 (Exact reflected illuminance)**.** *[T3 Prop 6.1] (SymPy-verified) Parametrize the lit side by $\vartheta \in (\pi/2, 3\pi/2)$ and the reflection height by $h \in [0, L]$; let $\psi = 2\vartheta - \pi$, $s = h\tan\alpha$ and $c = \sin\frac{\psi}{2}$. (a) The reflected ray lands at $P = a\,e(\vartheta) + s\,e(\psi) = (s + ac)\,e(\psi) + a\cos\frac{\psi}{2}\,e(\psi + \frac{\pi}{2})$. (b) $(\vartheta, s) \mapsto P$ is injective with $|\det \partial P/\partial(\vartheta, s)| = 2s + ac$. (c) $E(P) = I_0\,ac/(2s + ac)$ for $s \le L\tan\alpha$, and $E = 0$ off the image and where the preimage has $s > L\tan\alpha$; the total reflected flux $2aLI\sin\alpha$ equals the intercepted flux. (d) At a fixed floor point E does not depend on α; only the lit domain $s \le L\tan\alpha$ does.*

Part (d) is the completion-invariance certificate, *exact* in the top model. Injectivity (the caustic of a convex cylindrical mirror is virtual) is proved in Appendix J; a Monte Carlo agrees to its ≈0.6% noise (computed). The formula is elementary catoptrics; we have not found it in the sources checked, but the official solution may contain it, so its novelty is uncertain.

Proposition 11.32 (Thin-leg bridge, rigorous)**.** *[T3 Prop 6.2, rev.: lit domain] Let $E_0 = \frac{aI_0}{2}\sigma/r$, $\sigma = |\sin\frac{\varphi}{2}|$, $\epsilon = a/r$, $m = \frac{1}{2}\arcsin\epsilon$. If $\epsilon < 2/\sqrt{5}$, $\sigma > m$ and $r \le L\tan\alpha$ (or $L = \infty$), then $1 - \frac{m}{\sigma} \le \frac{E}{E_0} \le \frac{1+m/\sigma}{\sqrt{1-\epsilon^2}-\epsilon/2}$.*

The bound is rigorous but conservative (computed); its export domain is an annulus $a \ll r < L\tan\alpha$, an instance of intermediate asymptotics (Barenblatt, 1996). A *finite-sun bridge* (Proposition J.15, Appendix J.8) widens it by the sun's azimuth spread, for any radiance profile over the solar disc.

The checker run. A mini-checker (`sps_checker.py`, re-run in Section 12) encodes the specification with $L = 60a$ and runs concrete versions of the checks of Definition 11.25: (C0) reading containment against the formal specification; (C1) answer type: symbols (a, I_0, r, φ) only, and dimension; (C2) declared imports are in the specification; (C3) completion invariance: identity with the certified α-free form; (C4) side conditions recomputed from the root's domain, every hypothesis of Propositions 11.32 and J.15 asserted (the script checks $\alpha_{\max} \le \pi/2$; the stronger $\alpha_{\max} + \delta_s \le \pi/2$ holds trivially here), and the composed error budget evaluated at the worst

corner. All admissible α light the disc $r \le 60\tan(29.73^\circ)a = 34.3a$. SymPy stands in for a proof kernel.

Table 16: Mini-checker run on EuPhO 2025 Theory Problem 1(a). "Syntactic" marks attacks that a purely syntactic version of checks C1, C3 and C4 accepts.

Submitted SPS	Outcome
Honest: $\frac{aI_0}{2}\sin\frac{\varphi}{2}/r$, $r \in [20a, 34a]$, $\sigma \ge 0.5$, tolerance 10%	ACCEPT, certified error $\le 8.88\%$ (exact point-sun error $\le 2.56\%$; sun term $\le 0.93\%$)
V0: same, tolerance 5%	REJECT: budget $0.0888 > 0.05$ (completeness gap: the truth is within 5%)
V1: overclaimed domain $r \ge 5a$	REJECT: budget $0.380 > 0.10$
V2: thin-leg side condition evaluated in the child ($a := 0$), declared	REJECT (Theorem 11.22(a)); a silent one cannot change the budget, which C4 recomputes
V3: $\frac{aI_0}{2\cos\alpha}\sin\frac{\varphi}{2}/r$ (a natural slip, made and later corrected in the introspection note: I for I_0)	REJECT by C1 (α in answer), C3 (half-oscillation 0.42)
V4: reading narrowed to $\alpha = 45^\circ$	REJECT by C0, only because the specification is formal; else (J1)
V5: declared extra import "10% Lambertian"	REJECT: import not in the specification
V6: $2E_{\text{thin}}\cos(\alpha)^{1/1000}$ (true error $\approx$0.5)	REJECT by C1, C3; syntactic version accepts ("≤ 0.0888")
V7: $\frac{23}{20}E_{\text{thin}}\cos(\alpha)^{1/50}$ (true error 0.150)	REJECT by C1, C3; syntactic version accepts
V8: $1.5E_{\text{thin}}e^{\alpha/1000}$	REJECT by C1, C3; syntactic version accepts
V9: $r \ge 1.05a$, $\sigma \ge 1$, tolerance 0.7 (true error 0.909)	REJECT by C4 ($\epsilon = 0.952 > 2/\sqrt{5}$); syntactic version accepts ("≤ 0.635")
V10: $1.09E_{\text{thin}}$ (control)	REJECT by C3: differs from the certified form
V11: undeclared 10% diffuse correction $1.1E_{\text{thin}}$	REJECT by C3: differs from the certified form

Plausible syntactic versions of C1, C3 and C4 accept four simple attacks (V6–V9), which are rejected only once every check is made semantic: a small instance of the lesson of Section 3. The prototype is still a toy; its residual defects are listed in Appendix J.8.

What remains judgment. The photo-to-model reading (a *specular* cylinder, lit over its full height, its unsupplied reflectance, the meaning of "illuminance surplus", the α range) and the open world (taper, tilt, floor roughness, the seat's shadow). Each effect could be added as a declared deformation with its own bridge, moving it from judgment to checking; what cannot be moved is the claim that *these* are all the effects the setters intend (Theorem 11.28); the reading can only be tested by round trips such as rendering the predicted floor pattern. A second, germ-root example from IPhO 2025 is in Appendix J.8.

11.8 The four logical roles of physicists' checks

Dimensional analysis, limiting cases, symmetry and conservation are often grouped together as "coherence checks". They play at least four logical roles, and only the first two are coherence in the sense of Section 6:

Dimensions are a type system (Kennedy, 1994), and the Buckingham Π theorem (Buckingham, 1914) turns "express in terms of" into a type signature; but dropping a small Π group is a bridge (role 4), not a type fact. *Symmetry* needs uniqueness, as the notes observe (`physics/symmetry.md`): a symmetric problem implies only a symmetric *set* of solutions, and

Table 17: Logical roles of the physicist's checks.[L9 §3]

Role	Checks	What it certifies	Soundness status
1. invariance of the rule set	dimensions (unit group); declared context symmetries	homogeneity, equivariance of derived equations	worst case, decidable; necessary only
2. in-model theorem	conservation laws	a computation inside one local model	worst case within the model; silent on exports
3. regression against another context	limiting cases	agreement with solved sub-models	sound only for regular limits (Norton)
4. export side condition	order of magnitude; "the neglected term is small"	a bridge at a stated tolerance	sound with explicit budgets; heuristic otherwise

a declared symmetry gives free negative data. *Limiting cases* are valid only under Norton's continuity (Proposition 11.5(c)), and *order-of-magnitude* checks are export side conditions (Theorem J.8). Appendix J.9 gives the details and the closest precedent, the Andes tutor (VanLehn et al., 2005).

What is and is not established. Contradictory contexts are fine, and realizability says which must be consistent. Reductio in inadequate frameworks needs a consistency certificate, and with approximate premises a margin. Untrusted solvers are handled for everything except the reading, whose characteristic attack, silent narrowing, no internal signal sees. Not established (Section 14): any theorem about *learning* readings (the deep part, J1); frame realizability under germ semantics; idealizations inside suppositions and non-value exports; automated proofs of the regularity constants of Theorems J.12 and J.13; a checker beyond one worked problem. The mini-checker illustrates the checks on one problem and twelve variants (ten attacks, one control and one completeness test); it is not a system, and its own soundness is argued, not proved.

12 Experiments

This section reports three small computational experiments (A–C) and re-runs of four check scripts of the theory (D–G). They *illustrate* theorems of the earlier sections in settings where the target calculus is known, validity can be decided or tested, and adversaries can be run. They are not evidence that the methods scale, whether to real human mathematics, to neural verifiers, or to domains where world feedback is scarce. Experiments A–C use the Python package `cil` in `code/`. All runs are deterministic given the seeds. A–C use seeds $0, 1, 2$ throughout, and "$a \pm b$" is the mean and the population standard deviation over these three seeds, not a confidence interval; rows that pool several values of N give the standard deviation over the pooled runs. Full tables are in the reports `code/results/*.md`; reproduction instructions and run times are in `code/README.md`. An independent review of the code re-ran everything and fixed two reporting errors in B and a latent one in A. The fallacy-fate counts in B's report changed (for example, for affirming the consequent at $N = 10$); no other number changed (computed).

12.1 A: algebra, from imitation to a sound calculus, and the meadow

Setup. The target has 41 oriented rewrite schemas for field algebra over *partial* terms (Example 2.12). Eleven carry guards built from defined, nonzero and nonneg atoms on schematic variables, such as $x/x \to 1$ if nonzero(x); numeral arithmetic is trusted, not learned. A simulator writes N human derivations in which every valid step instantiates a target schema and cites its name. Noise is `sporadic` (5% random mutations under random rule names), `fallacies` (each opportunity for one of five systematic fallacies taken with probability 0.3: the freshman's dream, unguarded $x/x \to 1$, unguarded $\sqrt{x^2} \to x$, $(a+b)/(c+d) \to a/c + b/d$, $-(a+b) \to -a+b$), `both`, or none. The learner clusters rewrite cores by MDL and takes Plotkin's least general generalization (Plotkin, 1970) of each cluster, bucketing by the cited rule (`tagged`) or by shape (`untagged`). The verifier is conservative: it accepts a numeral evaluation or an instance of an active schema with human support $m \geq 2$ whose guard follows from the context. A Lakatos-style loop (Lakatos, 1976) of at most six rounds probes the schemas, derives with them, detects contradictions, blames and repairs (minimal guard, else split the cluster, else delete). In mode `numeral` the only signal is coherence (C): two derivations of distinct numerals from one ground term, or a derivation contradicting arithmetic. Modes `bag` and `step` add a world oracle (W), on whole derivations with blame by spectrum fault localization (Abreu et al., 2007), or on single steps. The oracle compares partial values with Kleene equality at boundary and random rational points (Schwartz, 1980; Zippel, 1979). Held-out valid and invalid steps in distribution (ID, about 10-node terms) and out of it (OOD, about 40 nodes) are labelled by an independent 60-point oracle, and a white-box attack targets every active schema.

verifier	exact /41	unsound	FA ID / OOD	C ID / OOD	exploits	oracle queries
imitation only	30.0 ± 0.0	15.7 ± 0.5	0.506 / 0.507	1.00 / 1.00	65 ± 2	0
+ coherence only (`numeral`)	33.0 ± 0.0	8.3 ± 0.5	0.166 / 0.156	1.00 / 1.00	27.3 ± 5.6	0
+ world, derivations (`bag`)	41.0 ± 0.0	0.0 ± 0.0	0.000 / 0.000	1.00 / 1.00	0.0 ± 0.0	11524 ± 1461
+ world, steps (`step`)	41.0 ± 0.0	0.0 ± 0.0	0.000 / 0.000	1.00 / 1.00	0.0 ± 0.0	13329 ± 450
boosted trees, with labels	n/a	n/a	0.533 / 0.503	0.888 / 0.728	n/a	labels

Table 18: Experiment A at $N = 500$, noise `both`, tagged learner, three seeds (computed). *Exact*: schema and guard recovered; *unsound*: active schemas failing the soundness test; FA and C: acceptance of held-out invalid and valid steps; *exploits*: accepted invalid steps found by the white-box attack. Of the 11 guarded rules, 0, 3, 11 and 11 are recovered exactly. The gradient-boosting baseline (34 features, threshold 0.5) is trained on the same corpus with validity labels.

Imitation and world feedback. Positive data alone (Section 5.8; numbers in Appendix D.13) recover shapes once a rule has a few instances (in 67% of rule–run pairs with two, 100% with five or more), and these generalize out of distribution; but no guard is ever displayed, all five fallacies reach support at $N = 500$ (Corollary 5.19), and spurious schemas from sporadic noise grow with N (Theorem 5.14). The first row of Table 18 is the outcome. With step-level world feedback, 0 of the 90 runs of the coherence grid (N from 20 to 500, three noise settings, both learners, three seeds) end with an unsound schema, the largest false-accept rate on any held-out set is 0.000, and for $N \geq 200$ the loop recovers 40.6 ± 0.8 of the 41 schemas exactly, guards included, at ID/OOD completeness 0.993 ± 0.015 / 0.992 ± 0.018. Bag-level feedback leaves an unsound schema in 2/90 runs (for example $0 \cdot (x - a) \to 0$). The guarantee is only as good as

the red-teaming of one's own rules: with four probes per schema and round, bag feedback leaves 2.0 ± 0.8 unsound schemas; with thirteen, none.

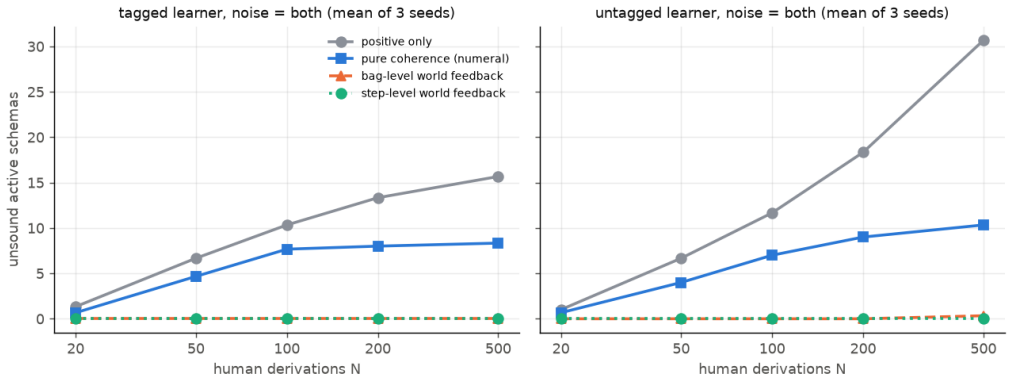


Figure 1: Experiment A: unsound active schemas against the number of human derivations N (noise `both`, mean of three seeds), tagged learner (left) and untagged learner (right) (computed). Imitation alone (grey) gets worse with N; coherence without a world (blue) removes about half; world feedback (orange, green) leaves one schema in one run.

Coherence without a world: the meadow. Pure coherence removes the errors that change the *value* of a numeral expression. In paired runs of the tagged learner it deleted the freshman's dream in all 11 runs in which imitation had produced it, $(a+b)/(c+d) \to a/c + b/d$ in 17/17 and $-(a+b) \to -a+b$ in 20/21, and repaired unguarded $x/x \to 1$ into the guarded rule in 31/35. It almost never learned a guard that concerns only definedness: $a \cdot 0 \to 0$ ended with a guard weaker than $\mathsf{defined}(a)$ in all 54 runs with $N \geq 100$. Of the 90 pure-coherence runs, 88 end with an unsound schema (Figure 1).

What survives has a meaning. Every unsound learned schema was re-tested in a *total* semantics: complex values, $x/0 := 0$, the principal square root, non-integer powers set to 0. This is a version of the *meadows* of Bergstra and Tucker (2007); its division convention $x/0 = 0$ is also that of Lean's Mathlib (Section 13). All 41 target rules hold in it, all five fallacies fail in it, and seven of the 11 guards are unnecessary in it. For $N \geq 200$, 304 of the 315 schemas that survive pure coherence (96.5%) are sound in it. Of the other eleven, ten mention an object constant (as in $x \cdot z \to x$; pure coherence compares only numeral expressions) and one is a value error. For $N < 200$ the share is 86.3%, since a smaller calculus derives fewer contradictions. Conversely, apart from one run, the guards that pure coherence *did* learn are guards the total semantics also needs: $x/x \to 1$ and $\sqrt{x^2} \to x$ came out exact whenever learned in general form, and in Experiment C the coherence-only calculus guards $\sqrt{ab} \to \sqrt{a}\sqrt{b}$ in every seed by exactly the one atom, $\mathsf{nonneg}(a)$, that the total semantics requires. So for $N \geq 200$, up to those eleven of 315 survivors, pure coherence converged not to the intended partial calculus but to a coherent calculus for another meaning of "/" and "$\sqrt{\ }$". This is the algebraic counterpart of the Kripkensteinian residue of Theorem 6.34: a coherent uniform alternative that agrees with everything positive examples and contradictions can supply. Only a channel that observes "undefined", here the world oracle at the corner point $x = 0$, separates the two meanings (Example 2.16).

Checks. Soundness of schemas is a random test with 250 assignments that include undefined and irrational values. The review re-tested every active schema of five final calculi with 20,000 random assignments plus a grid of special values and found no unsound one. Fresh seeds $3, 4, 5$ ($N = 200$) reproduce the pattern: 14 to 15 unsound schemas after imitation, 8 to 9 after pure coherence, none after world feedback.

12.2 B: natural deduction, Post probes and the bold extension of IPC

Setup. Formulas use atoms p, q, r (s only out of distribution) and $\to, \wedge, \vee, \neg, \bot$; steps are sequent-style natural-deduction inferences (Example 2.11). The target is classical natural deduction with 13 rules, the assumption rule carrying the side condition $A \in \Gamma$ as a mem guard; IPC drops reductio. Simulated humans prove random tautologies backwards. Their fallacies, each taken with probability 0.3 and cited as →E, are affirming the consequent (AC), denying the antecedent (DA) and an illegitimate discharge (ID); `noise` corrupts 5% of steps. The learner `vs` buckets steps by cited rule, anti-unifies them under an MDL criterion, adds most-specific mem guards and verifies with support $m = 2$. Coherence pruning uses five designated contexts: $\emptyset$ and four satisfiable premise sets. Negative bags (Lemma 6.2) come from blind search for $\Gamma_d \vdash \bot$ and from *Post probes*, which substitute $\bot$ or $\neg\bot$ for a rule's metavariables, prove its premises and refute its conclusion (Proposition 3.5 and Lemma 7.22). Blame is a minimum-weight hitting set, weighted by the human steps the cheapest repair loses. The `world` phase adds two observed valuations of the 16. Rule soundness is decided exactly by truth tables. An adversary searches with the learned rules for derivations of 34 invalid goals, $\vdash \bot$ among them; *sound completeness* is the share of 25 random tautologies the sound active rules prove within budget (target calculus: 0.88 ± 0.03).

phase	unsound	tonk-like	exploits /34	$\vdash \bot$	rules /13	sound compl.	world queries
imitation only	45.6 ± 19.3	30.2 ± 12.2	34.0 ± 0.0	9/9	13.0	0.87 ± 0.04	0
+ coherence, no world	1.2 ± 0.9	0.0	1.7 ± 4.1	0/9	13.0	0.87 ± 0.04	0
+ sparse world	0.2 ± 0.4	0.0	0.0	0/9	13.0	0.87 ± 0.04	910 ± 309

Table 19: Experiment B, learner `vs`, noise `both`, $N \in \{50, 100, 200\}$: 9 runs, sd over the 9 runs (computed). *Tonk-like*: unsound rules whose conclusion is unconstrained by the premises; $\vdash \bot$: runs in which the adversary derives $\vdash \bot$; *rules*: target rules recovered exactly or with a stronger guard. Held-out invalid steps accepted: 0.86 ± 0.02, 0.00 ± 0.01, 0.00.

Imitation, coherence, world. Imitation recovers all 13 rules, but on noisy data MDL anti-unification merges corrupted steps that share a skeleton into tonk-like rules (Prior, 1960), and the adversary proves all 34 invalid goals (Table 19). Coherence alone cuts the unsound rules from 45.6 to 1.2 and closes every route to $\vdash \bot$, at no cost in completeness. Over all 72 runs of `vs` (N from 5 to 200, four noise settings), 47 runs have an unsound rule after imitation, 19 after coherence and 6 after world feedback, and $\vdash \bot$ is derivable in 32, 0 and 0. On clean data no phase has an unsound rule for $N \geq 50$, and the assumption rule's side condition is learned from positives, as most specific guard. A fallacy counts as surviving if an active unsound rule licenses it or a special case of it. For $N \geq 50$, AC survives imitation in all 18 runs with fallacies, DA in 14 and ID in 10, and coherence removes all of them except one closed instance of DA that mentions atoms, outside the structural hypothesis of Corollary 6.32; world feedback removes that

one too. At $N \leq 10$ coherence leaves unsound rules in 4 of 18 runs, AC itself among them: a small learned calculus may lack the rules needed to *derive* the contradiction. Plotkin's learner without clustering (`aggressive`, 18 runs) produces 4.8 ± 2.3 tonk-like rules per run and lets the adversary derive $\vdash \bot$ in every run; after pruning, in none. Blame needs a prior (Theorem 7.19): for this learner at $N = 50$ on noise `both`, blaming the least-supported rule recovers 0.3 ± 0.5 target rules and no completeness, while blaming by cost of the cheapest repair recovers all 13 with completeness 0.87 ± 0.04.

Post-completeness probes. A *bold* learner over classical natural deduction adds a candidate schema iff no derivation of $\vdash \bot$ is found with it. Of 177 structural candidates (19 hand-picked, 8 pairwise LGGs of target rules, 150 random), the 59 classically valid ones were all accepted, and the 118 invalid ones were all refuted by an explicit derivation of $\vdash \bot$, as Theorems 6.13 and 6.15 predict. The derivations are short: median 2.5 and maximum 6 steps for the LGGs of target rules, median 6 and maximum 17 for the random schemas. For AC, the probe $A := \bot$, $B := \neg\bot$ gives

$$\bot \vdash \bot \text{ (ax)}, \quad \bot \vdash \neg\bot \text{ (}\neg\text{I)}, \quad \vdash \bot \to \neg\bot \text{ (}\to\text{I)}, \quad \vdash \neg\bot \text{ (}\neg\text{I)}, \quad \vdash \bot \text{ (AC)}.$$

Non-structural rules escape (Proposition 7.17): $\vdash p$ is coherent with every designated context and falls only to an observed valuation, and $p \vdash q$ falls to the designated context $\{p, \neg q\}$. Of the 45 unsound rules that survive pruning in any run, 38 mention specific atoms. Offered tonk's two rules one at a time, the bold learner accepts whichever comes first and rejects the other (Proposition 6.28). One observed valuation is a complete test for a schematic rule with k metavariables only if all 2^k literal instantiations are probed. The code probes at most eight, and the three schematic rules that survive the world phase in some run each have $k \geq 4$; probing all instantiations removed them. What world feedback adds is exact blame, since a refuted step names its rule: 5.0 ± 2.1 pruning rounds against 9.1 ± 5.9 without it ($N \geq 20$, 36 runs).

The bold extension of intuitionistic logic. Over intuitionistic natural deduction the bold learner accepted excluded middle, double-negation elimination, reductio, Peirce's law, weak excluded middle and Dummett's linearity, each underivable in IPC (a Kripke countermodel was found for each), and rejected the classically invalid candidates by derivations of $\vdash \bot$. Completeness on classical tautologies rose from 0.72 to 0.80 (classical natural deduction: 0.84). The same happens when IPC is first *learned*: from 60 simulated intuitionistic proofs, `vs` with coherence recovers the 12 rules (10 exactly, 2 with a stronger guard), never learns reductio, has no classically or Kripke-unsound rule, and its bold phase then accepts the same six principles. This is Theorem 6.16: coherence cannot separate the intermediate logics, so a maximal coherent structural learner never outputs IPC, and the residue of Theorem 6.34(ii) shows up as a drift toward CPC. Only the cautious, imitation-based tier keeps IPC.

Caveats. The designated contexts are certified satisfiable with ground-truth satisfiability, a modelling assumption (Assumption 2.14). Most-specific guards make $\wedge$E and $\vee$E stronger than the target, since the simulated humans only eliminate from open assumptions: sound, but incomplete out of distribution. Fresh seeds 3 and 4 ($N = 50$, `both`) reproduce the pattern.

12.3 C: an adversarial prover against statistical and rule-learned verifiers

Setup. All verifiers are trained on the same $N = 500$ noisy algebra derivations of Experiment A, except one trained on a clean corpus. The statistical verifiers, akin to process reward models, are logistic regression, gradient-boosted trees and a random forest from scikit-learn (Pedregosa et al., 2011) on hashed subterm-shape n-grams, rule signatures and tree-edit features. They receive oracle labels for positives and for negatives (refuted human steps, perturbations, fallacy instances), so they get *more* supervision than the positive-only rule learner. Thresholds are calibrated in distribution to a balanced precision $\mathrm{TPR}/(\mathrm{TPR}+\mathrm{FPR})$; "@p$X$" means precision X. The rule-learned verifiers are A's learner without guards (`lgg_pos`), with most-specific guards (`lgg_ms`; `lgg_ms_clean` on the clean corpus), after pure coherence, and after world feedback; the ground-truth calculus is a control. The prover sees only accept or reject. It runs a bidirectional best-first search whose proposals are target rules, fallacies and near-miss schemas with guards ignored, arithmetic, random mutations, and goal-directed leaps and grafts, with 3000 verifier queries for each of 23 false goal equations and 1500 for each of 33 true identities. A derived equation counts as false when exact evaluation at 5 fixed points refutes it, so these counts are lower bounds.

verifier	ID bal. prec.	ID recall	false goals /23	true goals /33	false eq./10^3 q.
boosted trees, threshold 0.5	0.9752 ± 0.0048	0.958 ± 0.003	22.3 ± 0.9	31.0 ± 0.0	281.0 ± 46.9
boosted trees @p0.99	0.9925 ± 0.0049	0.885 ± 0.030	16.3 ± 2.4	28.3 ± 1.7	259.2 ± 45.7
boosted trees @p1.0	1.0000 ± 0.0000	0.450 ± 0.048	1.7 ± 0.5	4.7 ± 0.5	43.4 ± 14.5
random forest @p0.99	0.9918 ± 0.0014	0.796 ± 0.018	13.0 ± 0.8	24.0 ± 2.2	168.0 ± 30.2
logistic regr. @p0.99	0.9749 ± 0.0209	0.061 ± 0.060	3.3 ± 1.9	1.7 ± 1.7	104.4 ± 35.8
`lgg_pos` (no guards)	0.9064 ± 0.0057	1.000 ± 0.000	20.7 ± 0.9	32.3 ± 0.5	269.0 ± 41.3
`lgg_ms` (most-specific guards)	0.9158 ± 0.0052	0.995 ± 0.002	14.0 ± 0.8	32.3 ± 0.5	141.2 ± 7.6
`lgg_ms_clean` (clean corpus)	1.0000 ± 0.0000	0.988 ± 0.005	0.0 ± 0.0	32.3 ± 0.5	0.0 ± 0.0
LGG + coherence, no world	1.0000 ± 0.0000	0.997 ± 0.001	5.0 ± 0.0	32.3 ± 0.5	98.7 ± 8.4
LGG + world (bag or step)	1.0000 ± 0.0000	0.997 ± 0.001	0.0 ± 0.0	32.3 ± 0.5	0.0 ± 0.0
ground-truth calculus	1.0000 ± 0.0000	0.997 ± 0.001	0.0 ± 0.0	32.3 ± 0.5	0.0 ± 0.0

Table 20: Experiment C: in-distribution metrics on held-out human-like steps, and outcomes under black-box adversarial search, three seeds (computed). Bag- and step-level world feedback give identical rows. Six further statistical configurations are in `code/results/adversarial_prover.md`.

Accuracy in distribution does not transfer to the search distribution. The boosted-tree verifier reaches test AUC 0.995 on held-out human-like steps. Calibrated to in-distribution balanced precision 0.9925, it still lets the prover prove 16.3 ± 2.4 of the 23 false goals, with a median of 514 queries to a false proof (Table 20 and Figure 2). At the strictest threshold, with no false accept on the calibration set, recall falls to 0.45; it then proves 4.7 of 33 true identities, and still 1.7 false goals. All 11 statistical configurations derive false equations under search. Of their 1082 invalid proof edges, 81% come from goal-directed leaps, grafts and random mutations, and only 1% from the known fallacies: steps that look like human steps and are not. This is Theorem 3.2 and Lemma 3.11(ii) in a closed domain: in-distribution accuracy is a property of a measure fixed before the prover moves, soundness a property of a set. That learned reward models are exploited by optimization is documented at scale, notably by the overoptimization scaling laws of Gao et al. (2023); the experiment adds a domain in which every exploited step can be certified invalid.

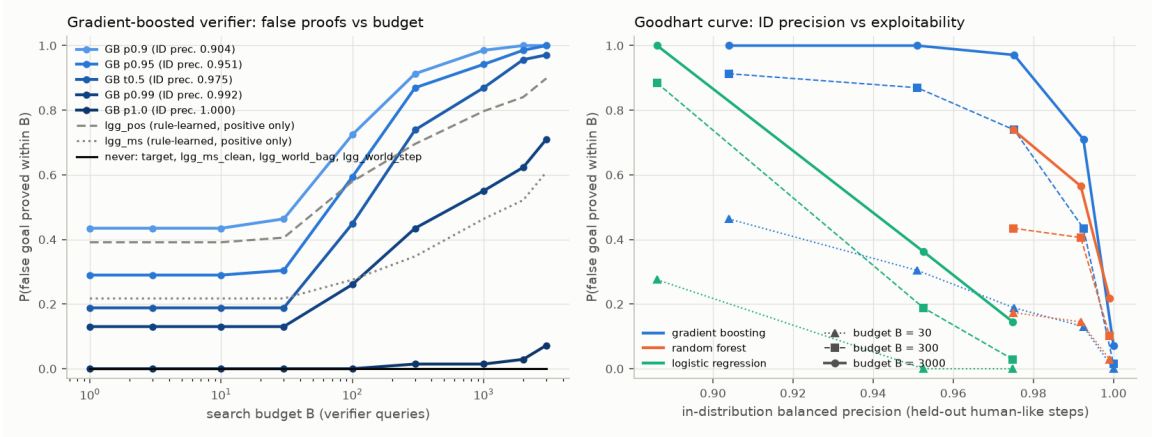


Figure 2: Experiment C (computed). Left: fraction of the 3×23 (seed, false goal) searches that found a false proof within B verifier queries, for boosted trees at five thresholds and the positive-only rule learners; the ground-truth calculus, `lgg_ms_clean` and the world-repaired calculi never do. Right: the same probability at budgets 30, 300 and 3000 against in-distribution balanced precision.

Rule-learned verifiers fail only through named schemas. All 154 invalid proof edges of the rule-learned verifiers are licensed by a schema that the soundness test flags; the most used is the noise-born projection $x + y \to y$, which alone collapses arithmetic ($1 = 1 + 1 = 2$). A counterexample deletes or guards such a schema and removes its whole family of exploits; a classifier has no such unit of repair. After world feedback the learned calculus proves no false goal and 32.3 of 33 true identities, like the ground-truth calculus. With clean data and an adequate guard language positive data suffice: `lgg_ms_clean` proves no false goal, as the version-space verifier of Theorem 4.2 should, although in one seed an over-specialized schema $0/(y + 3) \to 0$, whose true guard the guard language cannot express, is active and was not reached. Together, the four rule-based verifiers with no false goal derive no false equation in 900,305 queries. Coherence without a world leaves A's residue: in-distribution balanced precision 1.0000, yet 5 false goals per seed, all definedness errors such as $0/x = 0$ and $(x - x)/y = 0$, each true when $1/0 = 0$. In-distribution precision does not rank verifiers by soundness, even among rule-based ones.

Patching the classifier with its own exploits. The natural remedy is adversarial data collection, as for PRM800K (Lightman et al., 2023): harvest the certainly invalid accepted steps from the searches on the 23 false goals, add them as negatives, retrain and recalibrate. Over four rounds (boosted trees at p0.99, seeds 0 and 1 only), false goals proved on the harvested set fell from 15.5 to 0.0 and false equations per thousand queries from 195.7 to 14.3. The patched verifier still proved 1.5 of 12 fresh false goals and 2.0 of 35 goals against a second, differently configured prover, while true identities proved fell from 28.0 to 24.5. Retraining is an effective practical mitigation; its soundness is relative to the attacks it was patched against.

12.4 D–G: re-runs of the check scripts

Four check scripts were re-run for this section and reproduce the numbers quoted where the results are stated (computed): `ttl_sim.py`, the two-tier learner, in the singleton-conflict case of CPC only

(D, Table 10); `sps_checker.py`, the physics mini-checker on the chair-leg solution (E, Table 16); the simplicity scripts `c1`–`c5` (F, the `computed` figures of Section 8); and `comprehension_toy.py`, the Frege–Russell–Zermelo trajectory (G, Theorem 10.28(iv)). Three outputs are not quoted elsewhere. In F, `c2` confirms the kink of Theorem 8.4 on parities over $\{0,1\}^{10}$ with $N = 200$ and 16 errors: a product model one bit short of the simple model has risk 0.546 against 0.116, while shared-randomness code lengths stay in $[0.116, 0.126]$; and `c3` finds no violation of the rate threshold of Theorem 8.7 in 6000 (instance, c) pairs. In G, on the Dedekind–Cantor operations alone, stratification (STRAT) and the Zermelo-like class (Z: separation plus elementary sets) tie on coverage, and description length selects STRAT; once Zermelo's diagonal subsets outweigh Frege's universal extension, Z is selected; with the weights reversed, STRAT is.

12.5 What the experiments do and do not show

They show the mechanisms of the theorems at work in small, fully checkable domains: positive data fix shapes but not guards (A, B); systematic errors reach support (A, B, D); coherence refutes uniform errors through Post's substitution and leaves a residue of coherent alternative meanings (the meadow in A and C, intermediate logics in B); a world channel that can see the residue removes it (A–C); and accuracy in distribution says nothing about soundness under search (C). They show no more than that. *Realizability is assumed*: valid human steps are exact instances of the target rules, the guard language contains the true guards, and the simulated humans are the target plus known fallacies and noise; real practice satisfies none of this (Section 10). *Soundness is measured, not proved*: in A and C it is random testing, and "0" means that no counterexample was found; B decides rule soundness exactly, but its coherence verdicts and adversaries are budgeted searches. *World feedback is cheap in algebra*: random evaluation is an almost perfect oracle for polynomial identities, and a run of A uses about 10^4 queries; in physics, and in mathematics with undecidable fragments, B's two observed valuations or the certified bridges of Section 11 are the more realistic picture. *Few seeds*: three per configuration (two for C's retraining), although fresh seeds reproduce the patterns of A and B. *Toy scale*: closed small domains, heuristic and informed black-box adversaries (for which "not proved" is not "unprovable"), small statistical baselines on hand-built features. Nothing here measures how often larger learned verifiers are exploited, and the theorems do not predict that rate; they explain only why accuracy cannot rule it out.

13 Justification beyond proof

The investigation began with a question about principled justification. What does it look like, once we go beyond mathematical proof? This section steps back from the theorems and says what they show about that question. It also says what they do not show. Throughout, "we" means the authors of this report. The view argued for is deliberately modest. The theorems do not deliver a theory of justification. They make precise one shape that justification can take, and they locate exactly where that shape runs out.

13.1 The setup is Goodman's circle, made into a learning problem

The setup we studied has three parts:

- learn inference rules from accepted inferences;

- revise them when they generate unacceptable conclusions;
- revise accepted inferences when they conflict with rules one is unwilling to give up.

This is, almost verbatim, Goodman's account of the justification of deduction and induction:

> "Rules and particular inferences alike are justified by being brought into agreement with each other. *A rule is amended if it yields an inference we are unwilling to accept; an inference is rejected if it violates a rule we are unwilling to amend.*" (Goodman, 1955, ch. III)

In our vocabulary, the channels correspond to Goodman's account as follows:

- Imitation (P) supplies the accepted particular inferences.
- Rule schemas, learned by least generalization, are the rules.
- Coherence (C) is the mechanism by which a rule "yields an inference we are unwilling to accept". A derivation of $\bot$ from a context certified consistent is a negative bag, and some rule in it must go.
- World feedback (W) adds a channel whose errors are independent of human errors.

Without (W), the process is *narrow* reflective equilibrium (Rawls, 1971, 1974). With (W) it is *wide* reflective equilibrium in the sense of Daniels (1979). Daniels' "independence constraint" requires that the background theories not rest on the same considered judgments as the principles. It can be read as requiring that the corroborating channel's errors be independent of the human ones. Without such independence, agreement moves the odds by at most a bounded factor (Section 6.7). Even with it, no informative coherence measure is truth-conducive in general.

What our theorems add to Goodman's picture is a set of *vindications with explicit conditions*. Under stated channel assumptions, mutual adjustment is *sound against every adaptive arguer*, with explicit costs:

- realizability of the target in a structured hypothesis class;
- truthful designation of the contexts that are certified consistent;
- a world channel of the right kind.

For several of these assumptions there is a matching impossibility result (Table 9). Realizability is shown to be load-bearing by example rather than by a theorem. World feedback or bilateral evidence (a designated denial) is needed for completeness under ambiguous blame, not for soundness (Theorem 7.19 and Proposition 7.24). This is the sense in which the theorems are *conditional reliabilist meta-justifications* of reflective equilibrium. They are not a non-circular foundation for logic. As Dummett (1973) observed, any justification of deduction is explanatory rather than suasive: it uses deduction. What they provide is an explicit list of what the stated guarantees rely on, with proofs that the guarantees hold under them. The list consists of the three items above, together with the technical hypotheses of Theorem 7.12: a known frequency floor, fallacies used under their own latent tags, and an exact depth-d refutation oracle.

Two features of the guarantees matter philosophically.

- First, they are *anytime*, not merely in the limit. Limit theorems are Reichenbachian vindications (Reichenbach, 1938). They fall to the objection of Salmon (1967) that infinitely many methods share the same limit while disagreeing at every finite stage. Our soundness guarantees hold at every time, with probability over the data (Theorem 4.15). Our costs

are explicit finite quantities: escalations, detections, samples. The motivating notes are suspicious of limit framings, and hold that thinking and mathematics are "infinite endeavors" with no final formula. They also propose "trajectories of moral-reflective flight" as an alternative to reflective equilibrium, which they find "pretty flat/dead" as a picture of a mind's development (`philosophy/ethics/Trajectories of moral-reflective flight - an alternative to reflective equilibrium.md`). Our results bear on that view only in a limited way. They concern a *fixed* target calculus, for which convergence is the right notion. They say nothing about the open-ended development of concepts and calculi that the notes have in mind (language invention, Section 14).

- Second, they are *worst case over arguers*. A step checker that is accurate on the distribution of human steps can be exploited by search (Theorem 3.2). Justification that is to survive an untrusted or adversarial generator must be uniform over what the generator may propose. This is the case that motivates the question in the first place.

13.2 What each signal can do, and the residue it leaves

The main results combine into a division of labour. Table 21 summarizes it.

Two observations about this table seem to us to be the real philosophical content of the investigation.

Every signal is one-sided. Each learning signal we studied can refute errors in only one direction:

- Positive data never refute an over-generalization (Gold).
- Coherence never refutes over-caution: a cautious learner never sees a contradiction (Proposition 6.3). Nor does it refute a coherent over-generalization (Theorem 6.9).
- Coherence among exported tolerances never refutes a tolerance that is too wide. Coherent tolerance vectors form an up-set, so minimal coherent repairs only widen (Table 15). And when all models may share an omitted effect, no coherence-only calibrator can certify a tolerance below the size of that effect (Theorem 11.23).
- The demand that a problem be well posed refutes readings that are too wide, but it rewards readings that are too narrow (Proposition 11.30).

Each one-sided signal is an attack surface for an adversarial arguer, and it calls for a complementary signal that refutes errors in the other direction. Principled justification, on this picture, is not one master criterion. It is a set of signals that *together* cover every direction of error, plus an explicit statement of the directions that remain uncovered. Table 15 lists the pairs for physics. The same structure appears in mathematics. Imitation proposes and caution guards. Coherence condemns, but only bold hypotheses. Objects localize.

Each residue has a name, and different philosophies disagree only about the residue. Outside the residue of coherent, uniform, world-adequate alternatives, every position we know of counts the learner as correct:

- the realist;
- the Peircean, for whom truth is the limit of inquiry;

Signal	What it provably fixes	What it provably cannot fix (residue)
Imitation (P)	The *rules* actually followed. Exact identification of a schematic calculus of cited rules at coupon-collector rates (Theorem 5.6), with systematic out-of-distribution generalization (Corollary 5.10).	Systematic errors, which are indistinguishable from rules at every rate (Corollary 5.19). Imitation converges to "competence" in the sense of Cohen (1981), not to validity (Stich and Nisbett, 1980).
Cautious acceptance	Soundness against every adaptive arguer (Theorems 4.2 and 4.15).	Nothing new. It receives no coherence signal at all (Proposition 6.3), and its price is set by the structure of the class (Table 4).
Coherence (C)	Every error that is incoherent with the target on a designated context (Theorem 6.31). Exactly the target, when it has no coherent uniform proper extension: classical logic (Theorem 6.15), and complete theories at the level of theorems. The classical *meanings* of the connectives, among structural meanings, once denials are allowed (Theorem 6.26).	Coherent uniform alternatives: the Kripkensteinian residue (Theorem 6.34). Intermediate logics (Theorem 6.16). Σ_1-unsound extensions of arithmetic (Theorem 6.21). Which valuation is actual.
Simplicity	Expensive, idiosyncratic errors, via a rate threshold (Section 8).	Cheap, frequent fallacies. Rare valid rules are lost. Its strength cannot be calibrated from imitation data.
World (W)	Objects give step-level blame by descent, so $\log_2 \lvert\mathcal{H}\rvert$ corrections suffice, against up to $\lvert\mathcal{H}\rvert - 1$ from paradoxes alone on some classes (Theorems 10.19 and 10.20). Computation with fresh randomness gives worst-case-valid stochastic checks (Lemma 3.11). Calibration of top models (Section 11).	Errors invisible on every presentable object. Effects outside the top model (Theorem 11.28).
Contexts	Reasoning under contradictory idealizations without explosion. Coherence where it belongs (Theorem 11.11).	The reading of a problem, and the adequacy of the top model. Both are ineliminable judgments (Theorem 11.28).

Table 21: The division of labour among the learning signals, with the residue each one provably leaves.

- the Dummettian, for whom truth is warranted assertibility;
- Carnap's tolerance;
- Quine's pragmatism.

Inside the residue, they disagree. The realist sees undetectable error. The Peircean or Dummettian sees no fact of the matter. Carnap sees a free choice of language. Quine sees a choice made on pragmatic grounds. The theorems make the residue precise in several cases:

- It is empty for classical propositional logic.
- It contains all intermediate logics for an intuitionistic practice.
- It contains $\mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$ for arithmetic.
- In the algebra experiment it contains the total semantics with $1/0 = 0$, which pure coherence converged to for $N \geq 200$, up to 11 of 315 surviving schemas (Section 12). Its division convention $x/0 = 0$ is, as it happens, the one adopted by Lean's Mathlib and by Isabelle/HOL. Its other total conventions, such as complex square roots, differ from Mathlib's.

A theorem can say exactly where these philosophies part ways. It cannot settle who is right inside the residue. We think that is the correct division of labour between mathematics and philosophy here.

One consequence deserves to be stated on its own. *Rules can be identifiable when truths are not.* Among structural meanings, the bilateral tell-tale identifies the classical ones from finitely many data (Theorem 6.26). Yet in arithmetic, no computable learner using a computable text, coherence searches and a Δ_0 oracle decides Σ_2 truth in the limit (Theorem 6.21). This bears on Dummett's acquisition argument but does not refute it. That argument targets truth-conditional meaning: a meaning one could acquire must be manifestable in use, so meaning cannot outrun decidability. What the learner acquires is the rules, for the propositional connectives (Theorem 6.26), and Dummett agrees that rules are manifestable. The theorems show only that acquiring rules exactly is compatible with truth values, here of arithmetic Σ_2 sentences, staying beyond computable reach.

13.3 Rules are not premises

The tortoise of Carroll (1895) refused to infer Z from A, B and "if A and B then Z" until that inference was itself added as a premise, and so on forever. For a learner the lesson has a precise form:

- A learned *rule* is a schema. Its validity is a universally quantified claim over all instances, of Π form.
- No finite amount of world feedback verifies such a claim. Feedback can only refute it.
- The justification for *using* a learned rule therefore cannot be further premises. It has to be a guarantee about the process that accepted it: cautious acceptance, and an anchor that has been observed.

This is why the right acceptance policy for learned rules is Popperian. Accept tentatively. Assert only what every surviving hypothesis licenses. Treat each refutation as permanent. It is also why self-certification must never feed back into acceptance:

- a learned verifier that "proves its own soundness" has provided no evidence that distinguishes it from an unsound verifier sharing the certifying rule;
- in arithmetic, by Löb's theorem a theory that proves the reflection schema $\mathrm{Prov}_T(\ulcorner\varphi\urcorner) \to \varphi$ for its *own* provability predicate, for all φ, proves everything. By contrast, the corresponding reflection *rule* is admissible for Σ_1-sound theories.

Reflection is a principled source of genuinely new inferences, beyond imitation, that a learner who has accepted a calculus is committed to. Where it is used, it must enter as a stratified rule, never as a self-referential premise.

13.4 What principled justification beyond proof looks like

Putting the pieces together, the results suggest the following shape for a principled justification of a claim about something other than pure mathematics. A physics olympiad answer is the running case.

1. **Proof inside contexts.** Each step inside a context is checked against a calculus. The calculus is formal where possible. Where it is learned, acceptance is cautious, so the check is sound against an arguer who searches for weaknesses (Sections 4 and 10).
2. **Certified bridges between contexts.** Moving a conclusion from an idealized context to the claim of interest is not a logical step. It needs a certificate carrying information of finite radius about how the quantity behaves away from the idealization. Facts about the idealized model alone never suffice (no free export, Theorem 11.19). Side conditions are evaluated where they are true, in the parent context, and contexts are declared deformations. Otherwise an arguer can "stipulate" its way to anything (Theorem 11.22).
3. **An explicit judgment layer.** What remains is a judgment layer of three parts. The first two are provably ineliminable (Theorem 11.28). The third is the trusted core of any proof checker.

   - that the problem was read correctly, i.e. the admissible worlds were not narrowed;
   - that the top model is adequate for the world claims;
   - that the trusted base is trustworthy.

   A principled justification does not hide this layer. It minimizes it and declares it.
4. **Correctness as supertruth.** For underspecified questions, "true in the context at hand" means true in every admissible completion of the intended model, to the demanded precision. This is the largest acceptance criterion that stays sound under uncertainty about the reading (Theorem 11.29). Answer-invariance under unspecified features, which experienced solvers notice and rely on, is exactly a certificate of supertruth.
5. **Learning signals with known limits.** Every learned component comes with a statement of the signal that trained it and the residue that signal leaves. The learned components are the calculus, the bridge validity regions and the tolerances.

Measured against this shape, mathematical proof is the limiting case in which:

- the judgment layer has shrunk to the reading of the formal statement and trust in the axioms;

- every bridge is exact;
- the calculus is fixed rather than learned.

Informal mathematics before formalization sits in between. In our model it is a bounded-gap practice over a latent calculus. If the robust-core hypothesis holds, its proofs survive every admissible formalization. In the historical episodes we examined, counterexample objects did most of the corrective work, and the few paradoxes were the set-theoretic antinomies (Section 10). Physics adds a nonempty, explicit judgment layer and non-logical bridges, but the shape is the same. On this view the difference between mathematical and physical justification is one of degree, measured by the size of the declared residue, not one of kind.

13.5 What this says about meaning

The inferentialist premise of the question was that learning which inferences are valid is roughly learning meanings. The results support a qualified version:

- **Inferential roles are learnable, exactly, from finite use.** A finite anchor fixes the inferential role of a vocabulary for unbounded use (Corollary 5.10). This is a precise, if modest, answer to the rule-following question of how finite training fixes an infinite norm. The answer is: through structure (schemas) plus a small set of telling instances. The skeptic's "quus" survives only as a non-uniform patch, which uniformity rules out, or as a coherent uniform alternative, which is a named residue. This answer is relative to the hypothesis class and its vocabulary, which the data do not fix. That is Goodman's point, and it is exactly where Kripke's skeptic presses (Section 5.5).
- **More than role is not identifiable.** Under complete closure-level data, practice, coherence and object data identify a latent formalization exactly up to informal-validity equivalence, and no further. Bounded-gap data only bracket it (Theorem 10.14). Ontology is not identifiable: standard and nonstandard reals are observationally equivalent on the practice (Proposition 10.16). Neither is the choice of set-theoretic reduction (Proposition I.4).
- **Coherence buys existence, but not intended existence.** Every coherent mode of talking has *something* it could be talking about. That is a generalized completeness theorem, true for positions, credences and context systems alike (Section 9). But that something may be a syntactic or nonstandard model. Intended existence needs one of three additional things:
  - categoricity;
  - anchoring of the vocabulary to the world (Beth definability);
  - an infinitary condition, such as the probabilistic ω-rule.

  No computable coherence notion can certify intended existence beyond Π_1.

Inferential role is one aspect of meaning, and the project captures that aspect precisely. Other aspects are not captured by anything studied here:

- the "hooking" of a conception onto the world;
- the activities a concept supports;
- the questions it lets one ask.

We do not claim otherwise. All three aspects are emphasized in the motivating notes. These treat inferential role ("compositional inference warranting") as one aspect of meaning among several (`philosophy/philosophy of language/meaning and truth/`; `philosophy/philosophy of thinking/beating solomonoff induction at grokking a notion.md`).

13.6 The motivating questions

The questions that prompted this investigation are recorded in Hänni's public working notes (Hänni, 2026). Section 1.2 and Table 1 list them, with pointers to where each is answered or partly answered. The readings of the notes given there are ours.

14 Limits and open problems

We collect the main limitations of the results and the open problems they leave. The hardest of them is the first.

1. Affordable soundness when the learner must invent language. Every soundness guarantee here is relative to *realizability*: the target calculus, or the latent formalization, must lie in the hypothesis class. History did not respect any fixed class. It invented uniform convergence, ideals, ε–δ and the fundamental group as proof-generated concepts.

- A universal, language-inventing class restores realizability, but sound verification then costs on the order of $2^{K(h^*)}$ escalations (Theorems 4.12 and 10.10).
- What is needed is a class that is closed under the definitional extensions mathematics actually made, and that still has small escalation dimension. Equivalently, we need a theory of when concept invention, for example by anti-unification over definitions or by naming the hidden lemma found by descent, is cheap.

2. Computational cost. Our bounds count samples, escalations, detections and refutation-oracle calls, not computation.

- Membership in the version-space intersection for untagged unions is likely NP-hard.
- Consistency with negative bags is a multiple-instance problem. Its hardness is the formal shadow of the obfuscated-arguments problem in debate.
- Refutation search is only semi-decidable. Our depth-relative soundness (Theorem 7.20) is the honest form of what can be had.

A complexity-theoretic version of the two-tier theorem would be valuable. It would identify which audits are polynomial and for which classes.

3. The reading layer. In physics the irreducible judgment is the *reading*: which admissible worlds the problem text determines (Theorem 11.28).

- Determinacy refutes only over-wide readings (Proposition 11.30).
- What positive data identify the intended admissible class? We conjecture that official solutions together with marking schemes form a tell-tale within a class of "minimal completions consistent with the intended imports".

- Learning readings soundly is, we think, the central open problem for principled checking beyond mathematics.

4. **Sharp combinatorics.** Several bounds have gaps:
   - The binomial conjecture for k-unions of intersection-closed classes (Conjecture 4.11) would close the gap for untagged schema learning.
   - The bag-price conjecture $M_{\text{bag}}^{(r)} \leq r \cdot \text{Ldim}$ (Conjecture 10.24) would give the exact exchange rate between paradoxes and counterexample objects.
   - The query complexity of sound blame, i.e. computing the union of the minimal conflicts with a conflict oracle, is open.

5. **Coherence beyond Post-completeness.**
   - For which first-order or modal targets does every invalid uniform rule have a short coherence witness in a natural designated context?
   - Is there a hierarchy of coherence-witness complexity?
   - Which logics and theories are *coherence-pinned*, meaning that the residue is empty for natural families of designated contexts?
   - For arithmetic: is there a principled learner over reflection progressions that converges to "the right amount of reflection" under a natural feedback notion?

6. **Simplicity and normativity.**
   - For which classes of simple models does the product-model frontier kink, beyond parities?
   - Is there a principled, data-free choice of the simplicity coefficient, for example one tied to the cheapest postulate-violating error?
   - Which assumptions on a performance model, such as errors being time-bounded shortcuts, make joint description length identify competence despite the known non-identifiability of preferences from behaviour?

7. **Probabilistic and contextual coherence.**
   - Find a finitely presented proof system for valid counting sequents that is complete for de Finetti coherence.
   - Prove a probabilistic local-to-global theorem for context systems on tree covers.
   - Decide whether contextual frustration between idealized contexts should be penalized or tolerated.

8. **Scale.**
   - Our experiments are illustrations in small symbolic domains with realizable targets, cheap world feedback and few seeds.

- Whether learned calculi with cautious acceptance remain affordable at the scale of real mathematical corpora (for example Lean or Metamath step data with injected fallacies at measured human rates) is an empirical question we have not addressed.
- We have deliberately not built a strong prover. The motivating notes regard publishing insightful research on AI theorem provers as gravely dangerous. The results here concern checking, and we make no recommendations for prover development.

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A Proofs for Section 2

Theorem 2.5 is due to Plotkin (1970) and Reynolds (1970), and the rank lemma is folklore in their lattice of schemas. All other statements of Section 2 are proved here.

A.1 Proof of Lemma 2.3 (closure facts)

(a) Let D be the set of judgments with an R-derivation from B. D contains B (one-node derivations) and is closed under R: if $P \rhd j \in R$ and $P \subseteq D$, graft derivations of the finitely many members of P under a new root j. So $\mathrm{Cl}_R(B) \subseteq D$. Conversely, if $X \supseteq B$ is R-closed, induction on the height of derivations shows that every node of every derivation from B lies in X; so $D \subseteq \mathrm{Cl}_R(B)$. Extensivity and monotonicity in B and in R hold because a derivation from B using steps of R is also one from any $B' \supseteq B$ using steps of any $R' \supseteq R$. Idempotence: a derivation from $\mathrm{Cl}_R(B)$ becomes a derivation from B after grafting a derivation from B above

each leaf. Finitarity: a derivation has finitely many leaves, so it is a derivation from a finite $B_0 \subseteq B$.

(b) A single step $P \rhd j \in R$ is a derivation of j from P, so $R \subseteq \mathrm{Sound}(R)$, and monotonicity in the rule set gives $\mathrm{Cl}_R(B) \subseteq \mathrm{Cl}_{\mathrm{Sound}(R)}(B)$. Conversely, $\mathrm{Cl}_R(B)$ is closed under $\mathrm{Sound}(R)$: if $P \subseteq \mathrm{Cl}_R(B)$ and $j \in \mathrm{Cl}_R(P)$, then by (a) $j \in \mathrm{Cl}_R(\mathrm{Cl}_R(B)) = \mathrm{Cl}_R(B)$; it contains B, so it contains $\mathrm{Cl}_{\mathrm{Sound}(R)}(B)$. Finally, $\mathrm{Sound}(A) = \{P \rhd j : j \in \mathrm{Cl}_A(P)\}$ depends only on Cl_A, so $\mathrm{Cl}_A = \mathrm{Cl}_R$ gives $A \subseteq \mathrm{Sound}(A) = \mathrm{Sound}(R)$; and $\mathrm{Sound}(R)$ itself has the closure Cl_R. □

A.2 Proof of Lemma 2.6 (rank)

Throughout, $|t|$ is the number of symbol occurrences in t, metavariable occurrences included, and $\mathrm{vars}(t)$ is the set of distinct metavariables of t. A *renaming* of a set X of metavariables is a substitution whose restriction to X is an injective map into metavariables. If $\tau = \sigma\theta$ with θ a renaming of $\mathrm{vars}(\sigma)$, then $\sigma = \tau\theta^{-1}$, so $\sigma \equiv \tau$; conversely, $\sigma \equiv \tau$ means equality up to such a renaming, and then $\mu(\sigma) = \mu(\tau)$.

Let $\tau = \sigma\theta$. Let $x_1, \ldots, x_k$ be the distinct metavariables of σ, and let x_i occur $m_i \geq 1$ times in σ.

Size. Replacing each of the m_i occurrences of x_i (one symbol) by θx_i (with $|\theta x_i|$ symbols) gives

$$|\tau| = |\sigma| + \sum_{i=1}^{k} m_i(|\theta x_i| - 1).$$

Metavariables. Every metavariable of τ occurs in some θx_i, so

$$|\mathrm{vars}(\tau)| \leq \sum_{i=1}^{k} |\mathrm{vars}(\theta x_i)|,$$

with equality iff the sets $\mathrm{vars}(\theta x_i)$ are pairwise disjoint.

Difference. Since $|\mathrm{vars}(\sigma)| = k$,

$$\mu(\tau) - \mu(\sigma) \;\geq\; \sum_{i=1}^{k} \Big[m_i(|\theta x_i| - 1) + 1 - |\mathrm{vars}(\theta x_i)| \Big]. \tag{1}$$

We bound each summand.

- If θx_i is a metavariable, then $|\theta x_i| = 1 = |\mathrm{vars}(\theta x_i)|$ and the summand is 0.
- Otherwise θx_i contains at least one function or constant symbol and at least one occurrence of each of its metavariables, so $|\theta x_i| \geq |\mathrm{vars}(\theta x_i)| + 1$. Using $m_i \geq 1$, the summand is at least $(|\theta x_i| - 1) + 1 - |\mathrm{vars}(\theta x_i)| \geq 1$.

Hence $\mu(\tau) \geq \mu(\sigma)$.

Equality. If $\mu(\tau) = \mu(\sigma)$, then every summand in (1) is 0 and the inequality there is tight. The first forces every θx_i to be a metavariable; the second forces the sets $\mathrm{vars}(\theta x_i) = \{\theta x_i\}$ to be pairwise disjoint, i.e. the θx_i to be pairwise distinct. So θ restricted to $\mathrm{vars}(\sigma)$ is a renaming. Conversely, if it is a renaming, then $|\tau| = |\sigma|$ and $|\mathrm{vars}(\tau)| = |\mathrm{vars}(\sigma)|$, so $\mu(\tau) = \mu(\sigma)$.

Chains. Suppose $\sigma_{l+1} \succ \sigma_l$ strictly, i.e. $\sigma_l = \sigma_{l+1}\theta_l$ and $\sigma_l \not\equiv \sigma_{l+1}$. Then θ_l is not a renaming of $\mathrm{vars}(\sigma_{l+1})$ (otherwise the two would be equivalent), so $\mu(\sigma_l) > \mu(\sigma_{l+1})$, i.e. $\mu(\sigma_l) \geq \mu(\sigma_{l+1}) + 1$. Summing along $\tau = \sigma_0 \prec \sigma_1 \prec \cdots \prec \sigma_m$ gives $\mu(\tau) \geq \mu(\sigma_m) + m$, and $\mu(\sigma_m) \geq 0$. □

A.3 Proof of Lemma 2.7 (when the lgg recovers the schema)

Idea. Substituting θx for the occurrences of x adds more symbols than distinct metavariables unless θx is a metavariable, which is the rank lemma. Anti-unifying the t_j copies σ and puts $\mathcal{A}(\theta_1 x, \ldots, \theta_n x)$ at x; this is a metavariable iff (R) holds, and two of them coincide iff (D) fails. In detail:

Let σ have metavariables $x_1, \ldots, x_v$, let $\theta_1, \ldots, \theta_n$ be ground substitutions ($n \geq 1$), and let $t_j = \sigma\theta_j$. For each metavariable x of σ write $c_x = (\theta_1 x, \ldots, \theta_n x)$ for its *column*, and let η be the substitution $\eta(x) := \mathcal{A}(c_x)$.

Claim. For every subterm σ' of σ, $\mathcal{A}(\sigma'\theta_1, \ldots, \sigma'\theta_n) = \sigma'\eta$.

By induction on σ'. If $\sigma' = x$ is a metavariable, both sides are $\mathcal{A}(c_x)$. If $\sigma' = f(\sigma'_1, \ldots, \sigma'_r)$ (with $r = 0$ for a constant), then every $\sigma'\theta_j = f(\sigma'_1\theta_j, \ldots, \sigma'_r\theta_j)$ has root f, so by definition $\mathcal{A}$ outputs f applied to $\mathcal{A}$ of the argument columns $(\sigma'_k\theta_1, \ldots, \sigma'_k\theta_n)$, which by the induction hypothesis are $\sigma'_k\eta$. Hence the output is $f(\sigma'_1\eta, \ldots, \sigma'_r\eta) = \sigma'\eta$.

Taking $\sigma' = \sigma$ and using Theorem 2.5,

$$\mathrm{lgg}(t_1, \ldots, t_n) = \mathcal{A}(t_1, \ldots, t_n) = \sigma\eta \preceq \sigma.$$

When η is a renaming. By the definition of $\mathcal{A}$:

- $\eta(x) = \mathcal{A}(c_x)$ is a metavariable iff the entries of c_x do not all have the same root, i.e. iff (R) holds at x. If (R) fails at x, $\eta(x)$ is a compound term or a constant.
- If (R) holds at x and at y, then $\eta(x) = z_{c_x}$ and $\eta(y) = z_{c_y}$, and these coincide iff $c_x = c_y$, i.e. iff (D) fails for the pair x, y.

So η restricted to $\mathrm{vars}(\sigma)$ is a renaming iff (R) holds for every x and (D) holds for every pair $x \neq y$.

Conclusion. If (R) and (D) hold, η is a renaming and $\mathrm{lgg}(t_1, \ldots, t_n) = \sigma\eta \equiv \sigma$. If either fails, η is not a renaming of $\mathrm{vars}(\sigma)$, so by Lemma 2.6 $\mu(\sigma\eta) > \mu(\sigma)$; since equivalent schemas have equal rank, $\mathrm{lgg}(t_1, \ldots, t_n) \not\equiv \sigma$, and it is strictly more specific. □

A.4 Least general generalizations of infinite sets

Every nonempty set P of ground terms, finite or not, has a least general generalization, unique up to renaming. To see this, choose a nonempty finite $Q \subseteq P$ for which $\mu(\mathrm{lgg}(Q))$ is minimal (μ takes values in $\mathbb{N}$). Let $p \in P$. Since $Q \subseteq Q \cup \{p\} \subseteq \mathrm{inst}(\mathrm{lgg}(Q \cup \{p\}))$, Theorem 2.5 gives $\mathrm{lgg}(Q) \preceq \mathrm{lgg}(Q \cup \{p\})$, so by Lemma 2.6 $\mu(\mathrm{lgg}(Q \cup \{p\})) \leq \mu(\mathrm{lgg}(Q))$. By minimality the two ranks are equal, so by the equality case of Lemma 2.6 $\mathrm{lgg}(Q \cup \{p\}) \equiv \mathrm{lgg}(Q)$ and $p \in \mathrm{inst}(\mathrm{lgg}(Q))$. Hence $P \subseteq \mathrm{inst}(\mathrm{lgg}(Q))$. If $P \subseteq \mathrm{inst}(\tau)$, then $Q \subseteq \mathrm{inst}(\tau)$ and $\mathrm{lgg}(Q) \preceq \tau$. So $\mathrm{lgg}(Q)$ is a least general generalization of P; any two are $\preceq$ each other, hence equivalent. □

A.5 Proof of Lemma 2.8 (guarded closure)

$P \subseteq \mathrm{inst}(\mathrm{lgg}(P))$ by Theorem 2.5, and every $\phi \in \Psi_P$ holds on P, so $P \subseteq \mathrm{inst}(\mathrm{lgg}(P), \Psi_P)$. Let $P \subseteq \mathrm{inst}(\sigma, \Psi)$. Then $P \subseteq \mathrm{inst}(\sigma)$, so $\sigma \succeq \mathrm{lgg}(P)$ and $\mathrm{inst}(\mathrm{lgg}(P)) \subseteq \mathrm{inst}(\sigma)$. Every $\phi \in \Psi$ holds on all of P, so $\Psi \subseteq \Psi_P$ and $\bigcap_{\phi \in \Psi_P} \phi^{-1}(1) \subseteq \bigcap_{\phi \in \Psi} \phi^{-1}(1)$. Intersecting, $\mathrm{inst}(\mathrm{lgg}(P), \Psi_P) \subseteq \mathrm{inst}(\sigma, \Psi)$. □

A.6 Sound evaluation and coherence

Lemma A.1 ((WS) and coherence; trivial)**.** *[T7 §1.3] Under (WS) every designated position is in bounds for $R^* \cup T_{\text{trust}}$, hence for R^*. Conversely, if $J_W = \emptyset$ and every designated position is in bounds for $R^* \cup T_{\text{trust}}$ (for instance, in bounds for R^* with $T_{\text{trust}} \subseteq \text{Sound}(R^*)$), then (WS) holds.*

Remarks. (WS) does not require $T_{\text{trust}} \subseteq \text{Sound}(R^*)$: in arithmetic the target may consist of non-logical axioms with first-order logic trusted (Section 7), and the relevant target closure is then $\text{Cl}_{R^* \cup T_{\text{trust}}}$. In formal mathematics V is truth in the intended structure $\mathfrak{M}$: (WS) holds when designated positions are true of $\mathfrak{M}$, W is truth in $\mathfrak{M}$ on its fragment, and target and trusted steps are $\mathfrak{M}$-sound.

Proof. Write $R^+ = R^* \cup T_{\text{trust}}$. Let $[\Gamma : \Delta]$ be designated and V as in (WS). V makes every member of Γ true, and no step of R^+ has true premises and a false conclusion, so the set of V-true judgments is an R^+-closed superset of Γ and contains $\text{Cl}_{R^+}(\Gamma)$. Since V makes $\bot$ and every member of Δ false, $\bot \notin \text{Cl}_{R^+}(\Gamma)$ and $\Delta \cap \text{Cl}_{R^+}(\Gamma) = \emptyset$. As $\text{Cl}_{R^*}(\Gamma) \subseteq \text{Cl}_{R^+}(\Gamma)$ (Lemma 2.3(a), monotonicity in the rule set), the position is also in bounds for R^*.

Conversely, let $J_W = \emptyset$ and let $[\Gamma : \Delta]$ be a designated position that is in bounds for R^+. Then e is well defined, because $\Gamma \subseteq \text{Cl}_{R^+}(\Gamma)$ is disjoint from $\Delta \cup \{\bot\}$. Let V be the indicator function of $\text{Cl}_{R^+}(\Gamma)$. It extends e: it is 1 on Γ and 0 on $\Delta \cup \{\bot\}$. The set $\text{Cl}_{R^+}(\Gamma)$ is closed under R^+, so no step of R^+ has V-true premises and a V-false conclusion, and (WS) holds. For the parenthetical instance: if $T_{\text{trust}} \subseteq \text{Sound}(R^*)$, then $R^* \subseteq R^+ \subseteq \text{Sound}(R^*)$, so by Lemma 2.3(a),(b) $\text{Cl}_{R^+} = \text{Cl}_{R^*}$, and being in bounds for R^* is the same as being in bounds for R^+. □

B Proofs for Section 3

Lemma 3.1, Theorem 3.2, Proposition 3.5 and Proposition 3.9 are proved in full in the main text. This appendix proves the remaining results of Section 3, verifies the claims of Example 3.6, and records the examples showing that the hypotheses of Lemma 3.11 and Theorem 3.12 are needed. Notation is as in Section 3; in particular S is the set of all steps (Definition 2.2). Throughout, $\models$ is classical propositional consequence, and a step $\Pi \rhd \varphi$ over formulas is *sound* if $\Pi \models \varphi$.

B.1 PAC learners can be maximally unsound

Proof of Corollary 3.3. For $N \geq 0$ let $S_{\geq N} := \{s \in S : |s| \geq N\}$.

Step 1: $T_j \subseteq S_{\geq j}$. Every step of T_j has the premise $A \vee \top^{(j)}$, and $\top^{(j)}$ contains j occurrences of $\top$. So every step of T_j has size at least j.

Step 2: the tail bound. The sets $S_{\geq N}$ decrease in N and have empty intersection, so $Q(S_{\geq N}) \downarrow 0$ by continuity from above. Fix $\varepsilon \in (0,1)$ and let $N_\varepsilon := \min\{N \geq 0 : Q(S_{\geq N}) \leq \varepsilon\}$.

- If $\hat{\jmath} \geq N_\varepsilon$, then $Q(T_{\hat{\jmath}}) \leq Q(S_{\geq \hat{\jmath}}) \leq Q(S_{\geq N_\varepsilon}) \leq \varepsilon$, by Step 1 and monotonicity.
- If $\hat{\jmath} < N_\varepsilon$, then $N_\varepsilon \geq 2$, because $\hat{\jmath} \geq 1$. By minimality of N_ε, $Q(S_{\geq N_\varepsilon - 1}) > \varepsilon$. Moreover $\hat{\jmath} < N_\varepsilon$ means $|X_i| \leq N_\varepsilon - 2$ for every i, i.e. no sample lies in $S_{\geq N_\varepsilon - 1}$. Since the X_i are i.i.d. from Q, this event has probability $(1 - Q(S_{\geq N_\varepsilon - 1}))^m < (1-\varepsilon)^m$.

Hence $\Pr[Q(T_{\hat{\jmath}}) > \varepsilon] \leq \Pr[\hat{\jmath} < N_\varepsilon] \leq (1-\varepsilon)^m$.

Step 3: the error decomposition. Write $h = \mathcal{L}(X_{1..m})$ and $Y = \text{inst}(\top\text{I}, \wedge\text{I}, \vee\text{I}_2) \subseteq R^*$. An element of $(h \cup T_{\hat{\jmath}} \cup Y) \setminus R^*$ lies in $h \setminus R^*$ or in $T_{\hat{\jmath}}$, because $Y \subseteq R^*$. An element of $R^* \setminus (h \cup T_{\hat{\jmath}} \cup Y)$

lies in $R^* \setminus h$. So $(h \cup T_{\hat{\jmath}} \cup Y) \triangle R^* \subseteq (h \triangle R^*) \cup T_{\hat{\jmath}}$, and $\mathrm{err}_Q(\mathcal{L}^+) \le \mathrm{err}_Q(\mathcal{L}) + Q(T_{\hat{\jmath}})$ pointwise in the sample. This proves (a).

Step 4: PAC. Let $m \ge \max\{m_{\mathcal{L}}(\varepsilon/2, \delta/2), \lceil (2/\varepsilon)\ln(2/\delta)\rceil\}$. Then $\Pr[\mathrm{err}_Q(\mathcal{L}) > \varepsilon/2] \le \delta/2$, since as usual the guarantee of $\mathcal{L}$ holds for every sample size $m \ge m_{\mathcal{L}}(\varepsilon, \delta)$. By Step 2 applied with $\varepsilon/2$, $\Pr[Q(T_{\hat{\jmath}}) > \varepsilon/2] \le (1 - \varepsilon/2)^m \le e^{-m\varepsilon/2} \le \delta/2$. By Step 3 and a union bound, $\Pr[\mathrm{err}_Q(\mathcal{L}^+) > \varepsilon] \le \delta$. Both events are computed from the same sample, so no extra samples beyond the maximum are needed. The outputs of $\mathcal{L}^+$ need not lie in the hypothesis class of $\mathcal{L}$, so the guarantee is improper. This proves (b).

Step 5: trivialization. Every output of $\mathcal{L}^+$ contains $Y \cup T_{\hat{\jmath}}$. The derivation in the proof of Theorem 3.2(i), with $j = \hat{\jmath}$, uses only steps of Y (namely $\top$I, $\wedge$I and $\vee$I$_2$) and one step of $T_{\hat{\jmath}}$. So $\mathrm{Cl}_{\mathcal{L}^+(X_{1..m})}(\emptyset) = \mathrm{Fm}$ for every sample. This proves (c). □

Remark B.1. Step 5 is where the three added schemas are needed. If $\mathcal{L} \equiv \emptyset$ and they are omitted, the output is $T_{\hat{\jmath}}$ alone. Every step of $T_{\hat{\jmath}}$ has a premise, so $\emptyset$ is closed under it and $\mathrm{Cl}_{T_{\hat{\jmath}}}(\emptyset) = \emptyset$.

B.2 Schemas that mention atoms, and purification

Verification of Example 3.6. Let $\tau = (x \vee p_0 \rhd x)$ and $A = R^*_{\mathsf{CPC}} \cup \mathrm{inst}(\tau)$.

(a) *τ is unsound.* Its instance $p_1 \vee p_0 \rhd p_1$ fails under $v(p_0) = 1$, $v(p_1) = 0$.

(b) *A is not trivial.* Let $X := \mathrm{Cn}_{\mathsf{CPC}}(\neg p_0)$. Then $X \supseteq \emptyset$. X is closed under R^*_{CPC}, because $\mathrm{Cl}_{R^*_{\mathsf{CPC}}}(X) = \mathrm{Cn}_{\mathsf{CPC}}(X) = X$. X is also closed under $\mathrm{inst}(\tau)$: if $B \vee p_0 \in X$, every valuation with $v(p_0) = 0$ satisfies $B \vee p_0$ and hence B, so $\neg p_0 \models B$ and $B \in X$. By leastness, $\mathrm{Cl}_A(\emptyset) \subseteq X$. Since $p_0 \notin X$ (take $v(p_0) = 0$), $\mathrm{Cl}_A(\emptyset) \neq \mathrm{Fm}$.

(c) *A is unsound.* The tautology $\neg p_0 \vee p_0$ lies in $\mathrm{Cl}_{R^*_{\mathsf{CPC}}}(\emptyset)$, and the instance of τ with $x := \neg p_0$ yields $\neg p_0$, which is not a tautology.

(d) *$\mathrm{inst}(\tau)$ is not substitution-closed.* The substitution $p_0 \mapsto p_1$ maps the instance $p_2 \vee p_0 \rhd p_2$ to $p_2 \vee p_1 \rhd p_2$, which is not an instance of τ.

(e) *A ground error does not fire.* For the single step $p_0 \rhd p_1$, the set $\mathrm{Cn}_{\mathsf{CPC}}(\emptyset)$ is closed under $R^*_{\mathsf{CPC}} \cup \{p_0 \rhd p_1\}$, because $p_0 \notin \mathrm{Cn}_{\mathsf{CPC}}(\emptyset)$. So the closure from $\emptyset$ is just the set of tautologies.

Proof of Lemma 3.7. Let τ have metavariables $\bar{x} = x_1, \ldots, x_k$, and let $p_{i_1}, \ldots, p_{i_l}$ be the distinct atoms occurring in τ. Let $y_1, \ldots, y_l$ be fresh metavariables and $\tau^\circ := \tau[p_{i_t} := y_t]_{t \le l}$. Write $\tau[\bar{x} := \bar{A}]$ for the instance of τ obtained by substituting the formulas $\bar{A}$ for $\bar{x}$, and similarly for τ°.

($\Leftarrow$) Every instance of τ is an instance of τ°: $\tau[\bar{x} := \bar{A}] = \tau^\circ[\bar{x} := \bar{A},\ \bar{y} := (p_{i_1}, \ldots, p_{i_l})]$. So if τ° is sound, so is τ.

($\Rightarrow$) Assume τ is sound, and let $\iota = \tau^\circ[\bar{x} := \bar{A},\ \bar{y} := \bar{B}]$ be an instance of τ°. Then ι is τ with each occurrence of p_{i_t} *in τ itself* replaced by B_t, and each x_s replaced by A_s. Occurrences of p_{i_t} inside the A_s are left alone. There are infinitely many atoms, so we can choose distinct atoms $r_1, \ldots, r_l$ that occur nowhere in $\bar{A}$, $\bar{B}$ or τ. Let ρ be the renaming $p_{i_t} \mapsto r_t$ ($t \le l$), identity on all other atoms, and put $A'_s := \rho(A_s)$. Then $\iota' := \tau[\bar{x} := \bar{A}']$ is an instance of τ, hence sound.

Let σ be the simultaneous uniform substitution with $\sigma(p_{i_t}) = B_t$, $\sigma(r_t) = p_{i_t}$ ($t \le l$), and identity on all other atoms. We claim $\sigma(\iota') = \iota$. The formulas of ι' are built from the skeleton of τ, whose atoms are among the p_{i_t} and include no r_t, with the fillers A'_s plugged in for the metavariables.

- On the skeleton, σ replaces each p_{i_t} by B_t, exactly as the substitution $\bar{y} := \bar{B}$ does in τ°.

- Each filler $A'_s = \rho(A_s)$ contains no p_{i_t}, since ρ renamed them all. It contains r_t exactly where A_s contains p_{i_t}, because A_s contains no r_t. All its other atoms are fixed by both ρ and σ. Hence $\sigma(A'_s) = A_s$.

So $\sigma(\iota') = \tau^\circ[\bar{x} := \bar{A},\ \bar{y} := \bar{B}] = \iota$.

Finally, uniform substitution preserves classical validity. Suppose $\Pi' \models \varphi'$ and let v be any valuation. Define $v'(p) := v(\sigma(p))$. By induction on formulas, $v'(\psi) = v(\sigma\psi)$ for every ψ. So if v satisfies $\sigma\Pi'$, then v' satisfies Π', hence φ', and v satisfies $\sigma\varphi'$. Applied to ι', this shows that $\iota = \sigma(\iota')$ is sound. □

Proof of Corollary 3.8. R^*_{CPC} is sound, so A is sound iff every τ_l is sound. By Lemma 3.7, this holds iff every τ_l° is sound, i.e. iff A° is sound.

Next, $A_1^\circ := \bigcup_l \mathrm{inst}(\tau_l^\circ)$ is substitution-closed. Let $\tau^\circ\theta$ be an instance, where θ maps metavariables to formulas, and let σ be a uniform substitution of formulas for atoms. Since τ° contains no atom, $\sigma(\tau^\circ\theta) = \tau^\circ(\sigma \circ \theta)$, which is again an instance.

If A° is unsound, Proposition 3.5 gives $\mathrm{Cl}_{A^\circ}(\emptyset) = \mathrm{Fm} \ni \bot$. If A° is sound, then every step of A° is in $\mathrm{Sound}(R^*_{\mathsf{CPC}})$ by completeness: $\Pi \models \varphi$ implies $\varphi \in \mathrm{Cn}_{\mathsf{CPC}}(\Pi) = \mathrm{Cl}_{R^*_{\mathsf{CPC}}}(\Pi)$. By Lemma 3.1, $\mathrm{Cl}_{A^\circ}(\emptyset) \subseteq \mathrm{Cl}_{R^*_{\mathsf{CPC}}}(\emptyset) = \mathrm{Cn}_{\mathsf{CPC}}(\emptyset)$, which does not contain $\bot$. □

B.3 The commitment-order lemma

Recall Definition 3.10. An *R-derivation* is a finite tree of sequents in which each node is either a leaf marked as a *premise* (chosen freely by the prover) or, together with its children, forms an instance of some rule of R: the children are the instance's premise sequents (none, for a premise-free rule) and the node is its conclusion sequent. The *conclusion* of the derivation is the sequent at the root.

Proof of Lemma 3.11(i). Let $G := \{x : \rho_i \text{ is valid in } x \text{ for every } i \le m\}$. By the union bound over the whole library,

$$\Pr_{x\sim D}[x \notin G] \ \le\ \sum_{i=1}^{m} \Pr_{x\sim D}[\rho_i \text{ is not valid in } x] \ \le\ \sum_{i=1}^{m} \varepsilon_i.$$

Fix $x \in G$ and any R-derivation whose premises are all true in x. We show by induction on height that every sequent in the tree is true in x. Premise nodes are true by assumption. Every other node is the conclusion of an instance ι of some ρ_i, whose premise sequents are its children (possibly none) and are true in x by the induction hypothesis. Since $x \in G$, ρ_i is valid in x, so ι holds in x and its conclusion is true in x. In particular the conclusion of the derivation is true in x.

So the event "some R-derivation from premises true in x has a conclusion false in x" is contained in the complement of G. Its probability (its outer probability, if it is not measurable) is therefore at most $\sum_i \varepsilon_i$. The event G is determined by x and the library R alone. Because R is fixed before x is drawn, G does not depend on the prover, and the bound holds simultaneously for every prover, including one that sees x and chooses any number of derivations of any length. The sum runs over the whole library, not over the rules a particular derivation uses. □

Remark B.2 (Per-pair guarantees do not give per-scene validity). Suppose each scene in the support of D falsifies exactly one of M instances of a rule ρ, and the instance is drawn uniformly from these M. Then $\Pr_{(x,\iota)}[\iota \text{ holds in } x] = 1 - 1/M$, but ρ is valid in no scene, so its per-scene

validity is 0. A prover that sees x submits the falsified instance; this is regime (ii) inside each scene.

Proof of Lemma 3.11(ii). Let ι_0 be an instance of ρ that fails in x_0: its premise sequents are true in x_0 and its conclusion sequent is false in x_0. The tree consisting of ι_0 alone (its conclusion at the root, its premises as leaves) is a $\{\rho\}$-derivation from premises true in x_0 with a conclusion false in x_0. A prover that knows x_0 submits it, and nothing in this regime is random, so this happens with probability 1. If $\mu\{\iota : \iota \text{ fails in } x_0\} > 0$, such an ι_0 exists. If μ is supported on instances that hold in x_0, the average validity is 1, i.e. $\varepsilon = 0$, but a failing instance off the support of μ may still exist, and the same prover uses it. So no bound in terms of ε holds. □

For part (iii) we use the Schwartz–Zippel lemma (Schwartz, 1980; Zippel, 1979; DeMillo and Lipton, 1978): if $f \in F[z_1, \ldots, z_n]$ is nonzero of total degree at most d and $U \subseteq F$ is finite, then $\Pr_{a \sim \mathrm{Unif}(U^n)}[f(a) = 0] \le d/|U|$.

Proof of Lemma 3.11(iii). Formalize the protocol as follows. In round k the prover, which may be randomized, adaptive and computationally unbounded, commits to explicit expressions for $p_k, q_k \in F[z_1, \ldots, z_n]$. The verifier computes an upper bound on $\deg(p_k - q_k)$ from the expressions and rejects if it exceeds d. Otherwise it draws a_k uniformly from U_k^n, independently of everything so far, and accepts iff $p_k(a_k) = q_k(a_k)$.

Let $\mathcal{H}_k$ be the σ-algebra generated by everything that happens before a_k is drawn: earlier claims, points and answers, and the prover's coins. The claim (p_k, q_k) is $\mathcal{H}_k$-measurable, and a_k is independent of $\mathcal{H}_k$. Let E_k be the event that claim k passes the degree check and $p_k - q_k$ is nonzero in $F[z_1, \ldots, z_n]$, and Acc_k the event that it is accepted. By the Schwartz–Zippel lemma, applied conditionally on $\mathcal{H}_k$,

$$\Pr[E_k \cap \mathrm{Acc}_k \mid \mathcal{H}_k] \;=\; \mathbf{1}_{E_k} \cdot \Pr_{a \sim \mathrm{Unif}(U_k^n)} \big[(p_k - q_k)(a) = 0\big] \;\le\; d/|U_k|.$$

Taking expectations, $\Pr[E_k \cap \mathrm{Acc}_k] \le d/|U_k|$. A false claim (one with $p_k - q_k \neq 0$ over F) that fails the degree check is rejected, so a false claim is accepted only on $E_k \cap \mathrm{Acc}_k$. By the union bound,

$$\Pr\big[\exists k \le N : \text{ claim } k \text{ is false and accepted}\big] \;\le\; \sum_{k \le N} d/|U_k|,$$

which is $Nd/|U|$ when $U_k = U$ for all k. If $|U_k| \ge d\, 2^k/\delta$ for all $k \ge 1$, then $\sum_{k \ge 1} d/|U_k| \le \sum_{k \ge 1} \delta 2^{-k} = \delta$, which bounds the probability that any false claim in an unbounded sequence is accepted. The prover's strategy enters only through the $\mathcal{H}_k$, so the bounds hold uniformly over provers. □

Why the conditions of (iii) are needed.

- *Degree enforcement.* Over $F = \mathbb{F}_{2^k}$ with $U = F$, the claim $z^{2^k} \equiv z$ can be written with k squarings. The polynomial $z^{2^k} - z$ is nonzero of degree 2^k, but it vanishes on all of F, so random evaluation always accepts. The bound $d/|U|$ is vacuous here, since $d = 2^k = |U|$. A verifier that read the degree off the length of the expression, roughly k, would wrongly believe its error to be about $k/2^k$.
- *The field.* For prime p, $f = (a+b)^p - a^p - b^p = \sum_{0<i<p} \binom{p}{i} a^i b^{p-i}$ is nonzero over $\mathbb{Z}$, but p divides every $\binom{p}{i}$ with $0 < i < p$, so $f = 0$ in $\mathbb{F}_p[a, b]$. Evaluation mod p accepts this false integer identity with probability 1. Over $\mathbb{Q}$ with $U \subseteq \mathbb{Z}$, the lemma applies with $d = p$.

- *Order of commitment.* If the evaluation points were drawn before the prover committed to its claims, and visible to it, the prover could choose a nonzero $p - q$ that vanishes at those points. This is regime (ii) again.

B.4 The doctrinal paradox

We use the online protocol of Definition 6.1: (P)- and (N)-data, arguments π and their step sets $\mathrm{Steps}(\pi)$ are as defined there.

Proof of Theorem 3.12. Step 1: each h_i is a coherent, cut-closed calculus. The sets T_1, T_2, T_3 are satisfied by the valuations $(v(p), v(q)) = (1,1), (1,0), (0,1)$ respectively. For every set Γ, $\mathrm{Cl}_{h_i}(\Gamma) = \mathrm{Cn}_{\mathsf{CPC}}(T_i \cup \Gamma)$. For $\supseteq$: if $T_i \cup \Gamma \models \varphi$, compactness gives a finite $\Gamma_0 \subseteq \Gamma$ with $T_i \cup \Gamma_0 \models \varphi$, so $\Gamma_0 \rhd \varphi \in h_i$ and $\varphi \in \mathrm{Cl}_{h_i}(\Gamma)$. For $\subseteq$: $\mathrm{Cn}_{\mathsf{CPC}}(T_i \cup \Gamma)$ contains Γ and is closed under h_i, since $\Pi \subseteq \mathrm{Cn}_{\mathsf{CPC}}(T_i \cup \Gamma)$ and $T_i \cup \Pi \models \varphi$ imply $T_i \cup \Gamma \models \varphi$. In particular $\mathrm{Cl}_{h_i}(\emptyset) = \mathrm{Cn}_{\mathsf{CPC}}(T_i)$, which does not contain $\bot$ because T_i is satisfiable. So every h_i is $\mathcal{A}$-coherent and is a possible target.

Step 2: memberships. The following table records which h_i contain each step of π.

step	h_1 (p, q)	h_2 $(p, \neg q)$	h_3 $(\neg p, q)$
$\rhd p$	✓	✓	–
$\rhd q$	✓	–	✓
$p, q \rhd p \wedge q$	✓	✓	✓
$\rhd \neg(p \wedge q)$	–	✓	✓
$p \wedge q,\ \neg(p \wedge q) \rhd \bot$	✓	✓	✓

The checkmarks are immediate: $p \in T_1, T_2$; $q \in T_1, T_3$; $\neg q \models \neg(p \wedge q)$ and $\neg p \models \neg(p \wedge q)$; and the third and fifth steps are valid outright. For the dashes: $T_3 \models \neg p$, $T_2 \models \neg q$ and $T_1 \models p \wedge q$, and each T_i is satisfiable, so $T_3 \not\models p$, $T_2 \not\models q$ and $T_1 \not\models \neg(p \wedge q)$.

Step 3: π is a legal detection that deletes nothing. The five steps form an argument π from $\emptyset$ to $\bot$: the first two yield p and q, the third yields $p \wedge q$, the fourth yields $\neg(p \wedge q)$, and the fifth yields $\bot$. With uniform weights, s is in the majority set over $\mathcal{H}$ iff s lies in at least two of the h_i, since $2/3 > 1/2 > 1/3$. By the table, every step of π lies in at least two h_i, so $\mathrm{Steps}(\pi) \subseteq \hat{R}_{\mathrm{maj}}$, the majority set over $\mathcal{H}$. Also by the table, $\mathrm{Steps}(\pi) \not\subseteq h_i$ for each i. So the deletion set $\{R \in \mathcal{H} : \mathrm{Steps}(\pi) \subseteq R\}$ is empty.

Step 4: induction on rounds. Fix any target h_{i^*}. Consider the environment that, in each round, exhibits π or presents the positive datum $\rhd p \vee \neg p$, in any order. We show $\mathrm{VS}_t = \mathcal{H}$ for all t. Initially $\mathrm{VS}_1 = \mathcal{H}$. If $\mathrm{VS}_t = \mathcal{H}$, then $\hat{R}_t = \hat{R}_{\mathrm{maj}} \supseteq \mathrm{Steps}(\pi)$. So π is a legal (N)-datum, since $\emptyset \in \mathcal{A}$, and by Step 3 it deletes nothing. The datum $\rhd p \vee \neg p$ lies in every h_i, so it is a legal (P)-datum for the target and deletes $\{R : \rhd p \vee \neg p \notin R\} = \emptyset$. Hence $\mathrm{VS}_{t+1} = \mathcal{H}$. So $\mathrm{VS}_t = \mathcal{H}$ and $\hat{R}_t = \hat{R}_{\mathrm{maj}}$ for all t, and π is available in every round. □

Consequences recorded in the main text.

- *The vote is unsound relative to every target.* By Step 3, $\bot \in \mathrm{Cl}_{\hat{R}_{\mathrm{maj}}}(\emptyset)$. By Step 1, $\bot \notin \mathrm{Cl}_{h_i}(\emptyset)$ for each i. By Lemma 3.1 with $B = \emptyset$, $\hat{R}_{\mathrm{maj}} \not\subseteq \mathrm{Sound}(h_i)$ for each i.
- *Discriminating data break the paradox.* Let the target be h_1 and present $\rhd p \in h_1$. This deletes h_3, since $\rhd p \notin h_3$. Over $\mathrm{VS} = \{h_1, h_2\}$, with weight $1/3$ each, a step is a strict majority iff it lies in both, so $\hat{R} = h_1 \cap h_2$. Since $\rhd q \notin h_2$, π is no longer available. The h_i are pairwise incomparable: $\rhd p \wedge q$ lies only in h_1, $\rhd p \wedge \neg q$ only in h_2, and $\rhd \neg p \wedge q$

only in h_3. So on a text enumerating all of the target, each wrong hypothesis is eventually deleted by a datum it lacks, and from then on the uninformative detections stop.

Relation to judgment aggregation. Theorem 3.12 is the doctrinal paradox of Kornhauser and Sager (1986) and the discursive dilemma of Pettit (2001), with verifiers in place of judges. Its general form is the theorem of List and Pettit (2002): no aggregation function satisfies universal domain, anonymity and systematicity while producing consistent and complete collective judgments, for agendas with at least two atoms and their conjunction (or disjunction, or material conditional). For oligarchic aggregation under mere deductive closure, see Gärdenfors (2006) and Dietrich and List (2008); Proposition 6.6 is its learning-theoretic counterpart.

C Proofs for Section 4

The following are proved in full in the main text: Theorem 4.2(a) and (c), Theorem 4.12, Theorem 4.14 and Corollary 4.17. This appendix proves the remaining results, states two results that the main text only summarizes (Propositions C.2 and C.5), checks the examples of Section 4, and records the evidence for Conjecture 4.11. All statements are the versions repaired during adversarial verification; the repairs are listed at the end of this appendix (Appendix C.12). The scripts cited below are in `research/theory/T1-code/` of the accompanying repository. We re-ran them for this paper, and all values reproduce; the exceptions are the hill-climbing run with $h \le 5$ and the independent search of Appendix C.8, which we report as recorded.

Throughout, a verifier's internal randomness is a single draw $\omega \sim \Lambda$ on a space Ω_V, made before the first round. Nothing is lost by this, since coins drawn later can be drawn up front. Steps are ground terms, $|t|$ is the number of symbol occurrences of t, and $\mu(t) = |t| - |\mathrm{vars}(t)|$. We use the rank lemma (Lemma 2.6) and the characterization of the lgg (Theorem 2.5).

C.1 Optimality of the VS verifier

Fix human data P_0 and a deterministic, non-adaptive prover issuing $q_1, q_2, \ldots$. For a step set T used as the target, the transcript after i rounds is $\tau_i^T(\omega) = (e_1, \ldots, e_i)$ with $e_j = (q_j, a_j, y_j)$. It is defined recursively by $a_j = V(P_0, \tau_{j-1}^T(\omega), q_j, \omega)$, and $y_j = \mathbf{1}[q_j \in T]$ if $a_j = \mathsf{ESCALATE}$, while $y_j = *$ (no label) otherwise. A history $h = (e_1, \ldots, e_t)$ is *consistent* with a step set R if $y_j = \mathbf{1}[q_j \in R]$ whenever $y_j \neq *$.

Lemma C.1 (Coupling through the coins)**.** *Let h be a history of t rounds that is consistent with two step sets T and T', with queries $q_1, \ldots, q_t$. Then for every ω, $\tau_t^T(\omega) = h$ iff $\tau_t^{T'}(\omega) = h$.*

Proof. We show by induction on $i \le t$ that $\tau_i^T(\omega) = h_{\le i}$ iff $\tau_i^{T'}(\omega) = h_{\le i}$. The case $i = 0$ is trivial. Suppose the claim holds for $i-1$. If neither transcript equals $h_{\le i-1}$, neither extends to $h_{\le i}$. If both equal $h_{\le i-1}$, the verifier's answer $a_i = V(P_0, h_{\le i-1}, q_i, \omega)$ is the same under both targets. If a_i differs from the answer recorded in h, neither transcript equals $h_{\le i}$. If it agrees and is not $\mathsf{ESCALATE}$, both transcripts equal $h_{\le i}$. If it agrees and is $\mathsf{ESCALATE}$, the labels are $\mathbf{1}[q_i \in T]$ and $\mathbf{1}[q_i \in T']$. Both equal the label recorded in h, by consistency, so both transcripts equal $h_{\le i}$. □

Proof of Theorem 4.2(b). Since $q \notin \bigcap \mathrm{VS}_h$, there is $R' \in \mathrm{VS}_h$ with $q \notin R'$. Then $R' \in \mathcal{H}$ and $P_0 \subseteq R'$, and R' contains every step labelled positive in h and no step labelled negative,

so h is consistent with R'. By hypothesis, h is consistent with R^*. Let the prover issue $q_1, \ldots, q_t, q_{t+1} := q$. For $T \in \{R^*, R'\}$ put

$$E_T := \{\omega : \ \tau_t^T(\omega) = h \text{ and } V(P_0, h, q, \omega) = \mathsf{ACCEPT}\}.$$

By Lemma C.1, $E_{R^*} = E_{R'}$ as subsets of Ω_V, so $p = \Lambda(E_{R^*}) = \Lambda(E_{R'})$. On $E_{R'}$, with target R', the invalid query $q \notin R'$ is accepted in round $t+1$. V is δ-sound, and R' is an admissible target, so $\Lambda(E_{R'}) \le \mathbb{P}_{R'}[\exists s : \ a_s = \mathsf{ACCEPT} \wedge q_s \notin R'] \le \delta$. □

The bound is on the joint probability; read conditionally on h it is false, as the next example shows. The deterministic form (c) follows because $\Lambda(E) \in \{0, 1\}$ when Λ is a point mass. Any history that a deterministic verifier produces against a target is consistent with that target, since the oracle is truthful.

The reckless verifier. Let $\mathcal{H} = \{R_1 = \{a, c\}, R_2 = \{a, b, c\}, R_3 = \{a, b\}\}$ and $P_0 = \emptyset$. The verifier draws $B \sim \text{Bernoulli}(\varepsilon)$. If $B = 1$ it accepts every query; otherwise it runs the VS verifier.

- It is ε-sound. When $B = 0$ no invalid query is ever accepted, by Theorem 4.2(a), so the probability of an invalid acceptance is at most $\mathbb{P}[B = 1] = \varepsilon$.
- Take target R_2, and let the prover query c and then b. Initially $\bigcap \text{VS} = \{a\}$, so when $B = 0$ the query c is escalated. The history $h = ((c, \mathsf{ACCEPT}, *))$ therefore has probability exactly ε. It carries no label, so it is consistent with every hypothesis, and $\text{VS}_h = \mathcal{H}$ with $\bigcap \text{VS}_h = \{a\} \not\ni b$.
- Given h (that is, $B = 1$), b is accepted with probability 1. The joint probability is $\varepsilon \le \delta = \varepsilon$, as (b) says.

A simulation reproduces this (`thm31b_counterexample.py`).

Random human data. Suppose instead that P_0 is random, with a countable range and a target-dependent law, and independent of ω. Let d be an observed value with $d \subseteq R'$, where R' is as above with VS_h computed from $P_0 = d$. The coupling, applied conditionally on $P_0 = d$, gives

$$p(d) := \mathbb{P}_{R^*}[h, \text{ then } q \text{ accepted} \mid P_0 = d] = \mathbb{P}_{R'}[h, \text{ then } q \text{ accepted} \mid P_0 = d].$$

Hence $\mathbb{P}_{R'}[P_0 = d] \cdot p(d) \le \delta$, which is the weaker bound stated in the main text.

C.2 The escalation dimension

Proof of Theorem 4.4. Upper bound. Let V be the VS verifier, and let $R^* \in \mathcal{H}$ with $P_0 \subseteq R^*$. Consider any honest prover, which may be adaptive.

(1) V never rejects. By Theorem 4.2(a), $R^* \in \text{VS}$, so every query $q \in R^*$ lies in $\bigcup \text{VS}$.

(2) Every label is 1. So $P^- = \emptyset$ throughout, and $P = P_0 \cup E$, where E is the set of queries escalated so far.

(3) Let $s_1, s_2, \ldots$ be the escalated queries in order. When s_i is escalated, $s_i \notin \bigcap \text{VS}(P_0 \cup \{s_1, \ldots, s_{i-1}\})$, and $s_i \in R^*$.

So the escalated queries form an elastic chain in R^*, and since nothing is rejected, the cost equals their number, which is at most $\text{el}(\mathcal{H}, R^* \mid P_0)$. As V is 0-sound, $\text{Esc}(\mathcal{H} \mid P_0) \le \sup_R \text{el}(\mathcal{H}, R \mid P_0)$.

Lower bound. Let V be δ-sound. Let $s_1, \ldots, s_m$ be an elastic chain in some $R \in \mathcal{H}$ with $P_0 \subseteq R$, with witnesses R_i. The prover queries $s_1, \ldots, s_m$ in this order, which is honest for

target R. Fix i. For $j < i$ we have $s_j \in R$ and $s_j \in R_i$, so the oracle's label on s_j, if requested, is 1 under both targets. Induction on rounds, as in Lemma C.1, gives $\tau_{i-1}^{R}(\omega) = \tau_{i-1}^{R_i}(\omega)$ for every ω. Hence the answers in round i coincide, $a_i^R(\omega) = a_i^{R_i}(\omega)$. Under target R_i, which is admissible since $R_i \in \mathcal{H}$ and $P_0 \subseteq R_i$, the query s_i is invalid. So $\Lambda(a_i^{R_i} = \mathsf{ACCEPT}) \leq \delta$, and therefore $\Lambda(a_i^R = \mathsf{ACCEPT}) \leq \delta$. Every s_i is valid for R, so the expected cost under R is at least $\sum_{i=1}^{m} \Lambda(a_i^R \neq \mathsf{ACCEPT}) \geq (1-\delta)m$. If $\delta = 0$, each $a_i^R \neq \mathsf{ACCEPT}$ almost surely, so the cost is m almost surely. Every 0-sound verifier thus pays at least m for every finite $m \leq \sup_R \mathrm{el}(\mathcal{H}, R \mid P_0)$, so $\mathrm{Esc}(\mathcal{H} \mid P_0) \geq \sup_R \mathrm{el}(\mathcal{H}, R \mid P_0)$.

Size restriction. The same argument applies verbatim when all queries and chain elements have size at most N. □

The lower bound uses the coupling directly; the conditional reading of Theorem 4.2(b) would not suffice, since it is false. The cost of a randomized verifier is its expected cost (Definition 2.20).

C.3 Intersection-closed classes; single schemas

Proof of Proposition 4.5. (i) By definition, $\bigcap \mathrm{VS}(P)$ is the intersection of the members of $\mathcal{H}$ that contain P, which is $\mathrm{cl}(P)$. Intersection-closure puts it in $\mathcal{H} \cup \{\emptyset\}$. It contains P, and every member containing P contains it. Against an honest prover there are no negative labels, and $P = P_0 \cup E$, so the VS verifier accepts iff $q \in \mathrm{cl}(P_0 \cup E)$.

(ii) Let $s_1, \ldots, s_m$ be an elastic chain in R, and put $C_i = \mathrm{cl}(P_0 \cup \{s_1, \ldots, s_i\})$. Then $C_0 = \mathrm{cl}(P_0)$, and $C_{i-1} \subseteq C_i$ because cl is monotone. The inclusion is strict: $s_i \in C_i$, while $s_i \notin \bigcap \mathrm{VS}(P_0 \cup \{s_1, \ldots, s_{i-1}\}) = C_{i-1}$ by (i). Finally $C_m \subseteq R$, because R is a member containing $P_0 \cup \{s_1, \ldots, s_m\}$.

Conversely, let $\mathrm{cl}(P_0) = C_0 \subsetneq \cdots \subsetneq C_m \subseteq R$ be a chain of members of $\mathcal{H} \cup \{\emptyset\}$, and pick $s_i \in C_i \setminus C_{i-1}$. Then $s_i \in R$. For $i = 1$, $\mathrm{cl}(P_0) = C_0 \not\ni s_1$. For $i \geq 2$, $C_{i-1} \supsetneq C_0$ is nonempty, hence a member of $\mathcal{H}$. It contains $P_0 \subseteq C_0$ and each $s_j \in C_j \subseteq C_{i-1}$ for $j < i$, so $\mathrm{cl}(P_0 \cup \{s_1, \ldots, s_{i-1}\}) \subseteq C_{i-1} \not\ni s_i$. By (i), $s_i \notin \bigcap \mathrm{VS}(P_0 \cup \{s_1, \ldots, s_{i-1}\})$, so the s_i form an elastic chain in R. □

Proof of Lemma 4.6. (1) Two schemas. Let σ, τ be schemas, renamed apart. A ground term t lies in $\mathrm{inst}(\sigma) \cap \mathrm{inst}(\tau)$ iff $t = \sigma\theta = \tau\theta'$ for ground substitutions θ, θ'. Since the variables are disjoint, $\theta \cup \theta'$ then unifies σ and τ. If σ and τ are not unifiable, the intersection is empty. Otherwise let η be a most general unifier. Every unifier factors through η, so $t = \sigma\eta\rho$ for some ρ, and $t \in \mathrm{inst}(\sigma\eta)$. Conversely, every ground instance $\sigma\eta\rho = \tau\eta\rho$ lies in both sets. So $\mathrm{inst}(\sigma) \cap \mathrm{inst}(\tau) = \mathrm{inst}(\sigma\eta) \in H_1$.

(2) Arbitrary families. Let $\mathcal{F} \subseteq H_1$. The empty family has intersection $S = \mathrm{inst}(x) \in H_1$. If $\bigcap \mathcal{F} = \emptyset$, it lies in H_1. Otherwise $\bigcap \mathcal{F}$ contains a ground term t, and every member of $\mathcal{F}$ is $\mathrm{inst}(\sigma)$ with $\sigma\theta = t$ for some θ. Then $|\sigma| \leq |t|$, and every function or constant symbol of σ occurs in t. So, up to renaming, there are finitely many such σ, and $\mathcal{F}$ has finitely many distinct members. Its intersection is a finite intersection, which lies in H_1 by (1).

(3) Lggs of arbitrary nonempty sets. Let $P \neq \emptyset$. For finite $F \subseteq F' \subseteq P$ we have $\mathrm{lgg}F \preceq \mathrm{lgg}F'$, since $F \subseteq F' \subseteq \mathrm{inst}(\mathrm{lgg}F')$. By the rank lemma, a strictly increasing chain of lggs has length at most $\mu(\mathrm{lgg}\{t\}) = |t|$ for any $t \in P$. So there is a finite $F_0 \subseteq P$ such that $\mathrm{lgg}F \equiv \mathrm{lgg}F_0$ for every finite F with $F_0 \subseteq F \subseteq P$. Put $g := \mathrm{lgg}F_0$. For each $s \in P$, $\mathrm{lgg}(F_0 \cup \{s\}) \equiv g$, so $s \in \mathrm{inst}(g)$. If $P \subseteq \mathrm{inst}(\tau)$, then $F_0 \subseteq \mathrm{inst}(\tau)$, so $g \preceq \tau$. Thus g is the lgg of P, and $\mathrm{inst}(g) = \bigcap\{\mathrm{inst}(\tau) : P \subseteq \mathrm{inst}(\tau)\}$.

(4) The closure. For nonempty P, the members of H_1 that contain P are the sets $\mathrm{inst}(\tau)$ with $P \subseteq \mathrm{inst}(\tau)$ (the empty set does not contain P). So $\mathrm{cl}(P) = \mathrm{inst}(\mathrm{lgg}P)$ by (3), or by Theorem 2.5 when P is finite. □

C.4 One schema

Proof of Theorem 4.7. (i) Let $s_1, \ldots, s_m$ be an elastic chain in $\mathrm{inst}(\sigma^*)$ relative to a nonempty $P_0 \subseteq \mathrm{inst}(\sigma^*)$. Put $g_i := \mathrm{lgg}(P_0 \cup \{s_1, \ldots, s_i\})$ for $0 \le i \le m$. By Lemma 4.6, $\bigcap \mathrm{VS}(Q) = \mathrm{inst}(\mathrm{lgg}Q)$ for nonempty Q. So the elasticity condition says $s_i \notin \mathrm{inst}(g_{i-1})$, while $s_i \in \mathrm{inst}(g_i)$. Now $P_0 \cup \{s_1, \ldots, s_{i-1}\} \subseteq \mathrm{inst}(g_i)$, so $g_{i-1} \preceq g_i$ by the lgg property, and the instance sets differ, so $g_{i-1} \prec g_i$ strictly. This uses the lgg property and no assumption on the signature; mere inclusion of instance sets would not give generality over a poor signature. Also $g_m \preceq \sigma^*$, since $P_0 \cup \{s_1, \ldots, s_m\} \subseteq \mathrm{inst}(\sigma^*)$. By the rank lemma $\mu(g_{i-1}) \ge \mu(g_i) + 1$ for each i, and $\mu(g_m) \ge \mu(\sigma^*)$. Summing, $\mu(\mathrm{lgg}P_0) = \mu(g_0) \ge \mu(\sigma^*) + m$.

(ii) If no instance of σ^* has size at most N, every chain is empty and $\mathrm{el} = 0$. Otherwise let $s_1, \ldots, s_m$, $m \ge 1$, be an elastic chain in $\mathrm{inst}(\sigma^*)$ relative to $\emptyset$, with $|s_i| \le N$. Then $s_2, \ldots, s_m$ is an elastic chain relative to $\{s_1\}$, because the version spaces involved are the same. By (i), $m - 1 \le \mu(s_1) - \mu(\sigma^*)$. Since s_1 is ground, $\mu(s_1) = |s_1| \le N$. So $m \le 1 + N - \mu(\sigma^*) \le N + 1$, as $\mu \ge 0$. (Without the maximum the bound can fail: a ground σ^* with $|\sigma^*| = N + 5$ has no instance of size at most N, so $\mathrm{el} = 0 > 1 + N - \mu(\sigma^*) = -4$.) Taking the supremum over targets and applying Theorem 4.4 with the size restriction gives $\mathrm{Esc}(H_1; N) \le N + 1$.

(iii) Let $N \ge 1$, and consider the steps

$$t_0 = g^{N-1}(a), \quad t_1 = g^{N-1}(b), \quad t_j = g^{N-j}(a) \ \ (2 \le j \le N),$$

all of size at most N and all in $\mathrm{inst}(x)$. Let $L_j = \mathrm{lgg}(t_0, \ldots, t_j)$. Then $L_0 = t_0$. For $j \ge 1$, every t_i with $i \le j$ has the form $g^{N-j}(u_i)$. The u_i have mixed roots: they are a, b when $j = 1$, and include $u_0 = g^{j-1}(a)$ and $u_j = a$ when $j \ge 2$. So anti-unification copies the prefix g^{N-j} and places a metavariable at the bottom, giving $L_j \equiv g^{N-j}(z)$ and $\mu(L_j) = N - j$. Each t_j escapes:

- t_0 escapes because $\emptyset \in H_1$, so $\bigcap \mathrm{VS}(\emptyset) = \emptyset$.
- $t_1 \notin \mathrm{inst}(L_0) = \{g^{N-1}(a)\}$.
- For $j \ge 2$, $t_j = g^{N-j}(a) \notin \mathrm{inst}(g^{N-j+1}(z))$, whose instances begin with $N - j + 1$ copies of g.

By Lemma 4.6, $t_0, \ldots, t_N$ is therefore an elastic chain of length $N + 1$ in $\mathrm{inst}(x)$. Together with (ii) and $\mu(x) = 0$, $\mathrm{el}(H_1, x \mid \emptyset; N) = N + 1$, and $\mathrm{Esc}(H_1; N) = N + 1$ by Theorem 4.4. □

Computation. Exhaustive search over all ground terms of size at most N (`elast.py`) gives a maximal elasticity of $4, 5$ at $N = 3, 4$ over $\{a, b, g, p\}$, which is $N + 1$. With a single constant, over $\{c, g, p\}$ at $N = 3, 4, 5$, it gives $3, 4, 5 = N$.

C.5 Two-sided checking

Proof of Proposition 4.8. Let the signature have constants $a, b_1, \ldots, b_n$ and an n-ary symbol f. Take the target $R^* = \mathrm{inst}(f(a, \ldots, a)) = \{f(a, \ldots, a)\}$ and $P_0 = R^*$. For a partition π of $[n]$ with blocks indexed $1, \ldots, |\pi|$, let $\pi(j)$ be the index of the block containing j, and put

$$s_\pi := f(b_{\pi(1)}, \ldots, b_{\pi(n)}), \qquad \sigma_\pi := f(x_{\pi(1)}, \ldots, x_{\pi(n)}).$$

Three facts:

(1) $f(a,\dots,a) \in \mathrm{inst}(\sigma_\pi)$.

(2) $s_{\pi'} \in \mathrm{inst}(\sigma_\pi)$ iff π refines π'. Indeed, $s_{\pi'}$ is an instance iff the assignment $x_{\pi(j)} \mapsto b_{\pi'(j)}$ is well defined, i.e. iff $\pi(j) = \pi(k)$ implies $\pi'(j) = \pi'(k)$.

(3) Each s_π has size $n+1$ and is invalid.

List the Bell(n) partitions as $\pi_1, \pi_2, \dots$ along a linear extension of refinement with the finest first, so that if π_u properly refines π_v then $u < v$. The prover queries $s_{\pi_1}, s_{\pi_2}, \dots$ in this order.

Let V be two-sided, and consider the query s_{π_v}. Every label received so far is negative, except for the positives in P_0, and it concerns some s_{π_u} with $u < v$. Such a step is not an instance of σ_{π_v}: by (2) that would require π_v to refine π_u, which for $u < v$ and $\pi_u \neq \pi_v$ contradicts the order. With (1), $\mathrm{inst}(\sigma_{\pi_v})$ is in the current version space and contains s_{π_v}. So $s_{\pi_v} \in \bigcup \mathrm{VS}$, and V cannot reject it. V is 0-sound and s_{π_v} is invalid, so V cannot accept it. Hence V escalates every one of the Bell(n) queries.

Fixed signature. The two-sided VS verifier escalates q only if $q \in \bigcup \mathrm{VS} \setminus \bigcap \mathrm{VS}$. Once escalated, q is labelled and joins P or P^-. From then on $q \in \bigcap \mathrm{VS}$ (accepted) or $q \notin \bigcup \mathrm{VS}$ (rejected), so no step is escalated twice. A ground term of size n is a string of n symbols in Polish notation, so over a finite signature Ω there are at most $\sum_{n \le N} |\Omega|^n \le (|\Omega|+1)^N$ steps of size at most N. □

The super-exponential lower bound needs a signature that grows with N; over a fixed signature the count of steps caps it. *Computation.* A brute-force search over all depth-one schemas $f(t_1,\dots,t_n)$ with $t_i \in \{a, b_j, \text{metavariables}\}$, together with x, queried in order of decreasing number of blocks, gives 2, 5 and 15 forced escalations for $n = 2,3,4$. These are Bell(2), Bell(3), Bell(4) (`checks37.py`).

C.6 Tagged rules

Proof of Theorem 4.9. Let $G = \{\mathrm{inst}(\sigma) : \sigma \text{ a schema}\}$ be the class of nonempty single-schema instance sets. For a set Q of steps let $\mathrm{VS}^G(Q) = \{C \in G : Q \subseteq C\}$. For a set P of tagged steps let $P^{(i)} = \{s : (i,s) \in P\}$.

(1) Factorization. A hypothesis $\bigcup_i \{i\} \times \mathrm{inst}(\sigma_i)$ contains P iff $P^{(i)} \subseteq \mathrm{inst}(\sigma_i)$ for every i. So $\mathrm{VS}(P)$ corresponds bijectively to $\prod_i \mathrm{VS}^G(P^{(i)})$, via $(\sigma_i)_i \mapsto (\mathrm{inst}\sigma_i)_i$. If $P \subseteq R^*$, each factor contains $\mathrm{inst}(\sigma_i^*)$ and is nonempty.

(2) Intersection. $(i,s) \in \bigcap \mathrm{VS}(P)$ iff $s \in C_i$ for every tuple $(C_1,\dots,C_k)$ in the product. Because the other factors are nonempty, this holds iff $s \in \bigcap \mathrm{VS}^G(P^{(i)})$. So $\bigcap \mathrm{VS}(P) = \bigcup_i \{i\} \times \bigcap \mathrm{VS}^G(P^{(i)})$.

(3) Chains. Let $(i_1,s_1),\dots,(i_m,s_m) \in R^*$. By (2), (i_u,s_u) escapes relative to P_0 and the earlier steps iff $s_u \notin \bigcap \mathrm{VS}^G(P_0^{(i_u)} \cup \{s_v : v < u,\ i_v = i_u\})$. So the sequence is elastic iff, for each i, its tag-i subsequence is an elastic chain for $\mathrm{inst}(\sigma_i^*)$ in G relative to $P_0^{(i)}$. Splitting an elastic sequence by tags gives $\le$, and concatenating maximal per-tag chains gives $\ge$. Hence $\mathrm{el}(H_k^{\mathrm{tag}}, R^* \mid P_0; N) = \sum_i \mathrm{el}_i$.

(4) Per-rule bounds. The classes G and H_1 differ only by $\emptyset$, which contains no nonempty set. So for nonempty data they have the same version spaces. If $P_0^{(i)} \neq \emptyset$, Theorem 4.7(i) gives $\mathrm{el}_i \le \mu(\mathrm{lgg} P_0^{(i)}) - \mu(\sigma_i^*)$. If $P_0^{(i)} = \emptyset$, drop the first element of a chain, and the rest is a chain relative to a singleton. The argument of Theorem 4.7(ii) then gives $\mathrm{el}_i \le \max\{0, 1 + N - \mu(\sigma_i^*)\}$,

the maximum covering the case that no instance of σ_i^* has size at most N. Hence $\mathrm{Esc}(H_k^{\mathrm{tag}}; N) = \sup \mathrm{el} \le k(N+1)$ by Theorem 4.4.

(5) Tightness. Assume constants a, b and a unary g, and take $\sigma_i^* = x$ for all i and $P_0 = \emptyset$. In G, $\bigcap \mathrm{VS}^G(\emptyset) \subseteq \mathrm{inst}(a) \cap \mathrm{inst}(b) = \emptyset$, so the chain $t_0, \dots, t_N$ of Theorem 4.7(iii) is elastic in G. Concatenate its k tagged copies $(i, t_0), \dots, (i, t_N)$ for $i = 1, \dots, k$. By (3) this is an elastic chain of length $k(N+1)$, so $\mathrm{Esc}(H_k^{\mathrm{tag}}; N) = k(N+1)$. □

The proof does not use intersection-closure, and H_k^{tag} does not have it: for example, $(\{1\} \times \mathrm{inst}(a) \cup \{2\} \times \mathrm{inst}(c)) \cap (\{1\} \times \mathrm{inst}(b) \cup \{2\} \times \mathrm{inst}(c)) = \{2\} \times \mathrm{inst}(c)$ is neither a member nor empty. The factorization (1)–(2) is what the proof uses.

C.7 Untagged rules

Proof of Theorem 4.10(i). Assume $N \ge k+1$, and let $n+1 := \lfloor (N-1)/k \rfloor \ge 1$. For $a \in \{0, \dots, n\}^k$ put $s_a := p(g^{a_1}c, \dots, g^{a_k}c)$, which has size $1 + \sum_j (a_j + 1) \le 1 + k(n+1) \le N$. All s_a lie in the target $\mathrm{inst}(p(x_1, \dots, x_k))$. Order the $(n+1)^k$ vectors by non-increasing $\sum_j a_j$, breaking ties arbitrarily, and query the s_a in this order. Let a be the current vector and $a' \ne a$ an earlier one. Then $\sum a' \ge \sum a$, so $a'_j > a_j$ for some j; otherwise $a' \le a$ componentwise, and with $\sum a' \ge \sum a$ this would give $a' = a$. Put

$$R'_a := \bigcup_{j=1}^{k} \mathrm{inst}(p(x_1, \dots, x_{j-1}, g^{a_j+1}(x_j), x_{j+1}, \dots, x_k)) \in H_k.$$

Every earlier $s_{a'}$ lies in the j-th schema for a j with $a'_j \ge a_j + 1$. The current s_a lies in none, since its j-th argument has only a_j copies of g. So R'_a witnesses that s_a escapes, and the sequence is an elastic chain of length $(n+1)^k$ relative to $P_0 = \emptyset$. By Theorem 4.4, $\mathrm{Esc}(H_k; N) \ge \lfloor (N-1)/k \rfloor^k$. □

Proof of Theorem 4.10(ii). Let $k \ge 2$, and let $s_1, \dots, s_m$ be an elastic chain relative to $\emptyset$, with $|s_i| \le N$ and witnesses $R_i = \bigcup_{l \le k} \mathrm{inst}(\tau_{i,l})$. Unions of fewer than k schemas are padded by repetition (only the witness of s_1 can be empty, and it plays no role, since π_1 partitions $\emptyset$). Let π_i be the partition of $[i-1]$ that assigns j to block l when l is the least index with $s_j \in \mathrm{inst}(\tau_{i,l})$. It has at most k blocks.

Chains of indices. Call $i_0 < i_1 < \cdots < i_r$ a *chain* if, for each $b \ge 1$, the indices $i_0, \dots, i_{b-1}$ all lie in one block of π_{i_b}, say block l. Then $\tau_{i_b,l}$ covers $s_{i_0}, \dots, s_{i_{b-1}}$, and it misses s_{i_b} because $s_{i_b} \notin R_{i_b}$. So $s_{i_b} \notin \mathrm{instlgg}(s_{i_0}, \dots, s_{i_{b-1}}) \subseteq \mathrm{inst}(\tau_{i_b,l})$, and hence

$$\mathrm{lgg}(s_{i_0}) \prec \mathrm{lgg}(s_{i_0}, s_{i_1}) \prec \cdots \prec \mathrm{lgg}(s_{i_0}, \dots, s_{i_r})$$

strictly. The first term is ground, with $\mu \le N$. By the rank lemma, μ drops by at least one at each step and stays nonnegative. So $r + 1 \le N + 1$: every chain has length at most $N+1$. This is the argument of Theorem 4.7, and it needs no common target schema.

The recursion. Let $g(d)$ be the largest m for which there is a sequence $\pi_1, \dots, \pi_m$, with π_i a partition of $[i-1]$ into at most k blocks, all of whose chains have length at most d. A two-element index set $\{i_0 < i_1\}$ is always a chain, so $g(1) = 1$. For $d \ge 2$, split $[m-1]$ into the blocks of π_m. Fix a block B, and restrict each π_j with $j \in B$ to $B \cap [j-1]$. This gives a sequence of the same kind indexed by B. Its chains are chains of the original sequence, and appending m to any of them gives a chain again, because B is one block of π_m. So its chains have length at most $d-1$, and $|B| \le g(d-1)$. Hence $m - 1 \le k\, g(d-1)$. So $g(d) \le 1 + k\, g(d-1)$, and

$g(d) \le \sum_{u<d} k^u = (k^d - 1)/(k-1)$. With $d = N+1$, $m \le (k^{N+1}-1)/(k-1)$, and Theorem 4.4 gives the bound on $\mathrm{Esc}(H_k; N)$. □

Theorem 4.10(iii) summarizes the following proposition.

Proposition C.2 (Linear flat schemas)**.** *[T1 Thm 3.7(iii), scope as repaired] Let the steps be the points of $\{a,b\}^n$, $n \ge k$, encoded as $f(c_1,\dots,c_n)$. Let $\mathcal{H}$ be the unions of at most k* linear *flat schemas $f(t_1,\dots,t_n)$: each t_j is a, b or a metavariable, and no metavariable repeats. These are the unions of at most k subcubes. Then*

$$\sum_{w=0}^{k} \binom{n}{w} \;\le\; \mathrm{Esc}(\mathcal{H}) \;\le\; 1 + (2^k - 1)\binom{n}{k},$$

which is $\Theta(n^k)$. If metavariables may repeat, the lower bound persists, and Esc $\le 1 + (2^{2k}-1)\binom{n}{2k}$ *for $n \ge 2k$. So* Esc *then lies between $\Omega(n^k)$ and $O(n^{2k})$; only the linear case is pinned down to $\Theta(n^k)$.*

Proof idea. A subcube avoiding p lies in a half-cube $\{x_j \ne p_j\}$, so p escapes iff it shows a new pattern on some k coordinates; count patterns, and order points of weight $\le k$ by weight.

Proof of Proposition C.2. Identify a, b with $0, 1$, so that steps are points $x \in \{0,1\}^n$. A linear flat schema denotes a subcube, which fixes some coordinates and leaves the others free. For points y, p and a set J of coordinates, say y *agrees with* p *on* J if $y_j = p_j$ for all $j \in J$. The pair $(J, p|_J)$ with $|J| = k$ is a *pattern*, and a set P of points *realizes* it if some $y \in P$ agrees with p on J.

Escape criterion. Let P be the set of earlier chain elements. A subcube Q with $p \notin Q$ fixes some coordinate j to $1 - p_j$, since otherwise $p \in Q$. So Q lies in the half-cube $\{x_j = 1 - p_j\}$, which is itself a subcube avoiding p. Hence p escapes iff some union of at most k half-cubes $\{x_j \ne p_j\}$, $j \in J$, covers P. That holds iff no $y \in P$ agrees with p on J. Padding J to size k (possible since $n \ge k$) preserves this. So p escapes iff some pattern of p is not realized by P.

Upper bound. There are $2^k\binom{n}{k}$ patterns. The first chain element realizes $\binom{n}{k}$ of them. Each later element escapes, so it realizes at least one pattern not realized before, and realized patterns stay realized. So the chain has length at most $1 + (2^k - 1)\binom{n}{k}$.

Lower bound. The target is the full cube $f(x_1,\dots,x_n)$. Query all points of weight at most k in order of non-decreasing weight. Let x have weight $w \le k$, and choose $J \supseteq \mathrm{supp}(x)$ with $|J| = k$. An earlier point y that agrees with x on J has $y_j = 1$ on $\mathrm{supp}(x)$, so $\mathrm{supp}(y) \supseteq \mathrm{supp}(x)$. Its weight is at most w, so $y = x$, which is impossible for an earlier, distinct point. So every such point escapes, and the chain has length $\sum_{w=0}^{k}\binom{n}{w}$.

Repeated metavariables. A flat schema with repeated metavariables denotes the points that satisfy its constants and the equalities $x_i = x_j$ for positions sharing a metavariable. If such a schema avoids p, then either a constant position j has $p_j \ne$ the constant, so the schema lies in $\{x_j \ne p_j\}$, or all constants agree with p and some pair i, j sharing a metavariable has $p_i \ne p_j$, so the schema lies in $\{x_i = x_j\}$. Each such set is determined by at most two coordinates of p, and each of its points disagrees with p on one of them. If p escapes, there is therefore a set J of at most $2k$ coordinates on which no earlier point agrees with p. Padding J to size $2k$ (for $n \ge 2k$) and counting patterns as above gives a chain length of at most $1 + (2^{2k}-1)\binom{n}{2k}$. The class contains the linear one, and elastic chains stay elastic in a larger class, so the lower bound persists. □

Exact values (computed). For subcubes with $k = 2$ and $n = 2, 3, 4$, the escalation dimension is $4, 7, 11 = \sum_{w \le 2} \binom{n}{w}$, so the lower bound of Proposition C.2 is attained (`subcube.py`). For deep schemas over $\{c, g/1, p/2\}$ with $k = 2$ and $N = 3, 4, 5$ it is $4, 8, 13$ (`elast.py`). The partition abstraction used in the proof of Theorem 4.10(ii) attains $2^d - 1$ ($k = 2$, $d = 3, 4$; `abstr2.py`), so a polynomial bound must use more than this abstraction, such as intersection-closure. Separately, explicit elastic sequences of length $\lfloor (N-1)/k \rfloor^k$ were checked for $(k, n) \in \{(2,1), (2,2), (2,3), (3,1), (3,2)\}$ (`checks37.py`).

C.8 The binomial conjecture: the case $k = 2$, $h = 3$, and the evidence

The case $h = 2$. After removing $\bigcap \mathcal{C}$ (as in the reduction below), the members other than U are $\emptyset$ and pairwise disjoint atoms. Each atom A holds at most one chain element: if $s_i, s_j \in A$ with $i < j$, some member Y of the witness of s_j contains s_i but not s_j, so $A \cap Y$ is a member strictly inside A, hence empty, yet it contains s_i. The witness of the last element is a union of at most k members other than U, so it covers at most k earlier elements. Hence chains have length at most $k + 1 = \binom{k+1}{k}$, and the family of singletons attains it.

Proposition C.3 (Conjecture 4.11 for $(k, h) = (2, 3)$)**.** *[T1 Conj 3.8, case proved during verification] Let $\mathcal{C}$ be a family of subsets of U, closed under arbitrary intersections, with $U \in \mathcal{C}$ and of height 3. Then every elastic sequence for the class of unions of one or two members of $\mathcal{C}$, relative to $\emptyset$, has length at most $6 = \binom{4}{2}$.*

Proof. Reduction. Let $C_0 = \bigcap \mathcal{C}$. Its points lie in every member, so they never escape and never occur in an elastic sequence. Removing C_0 from U and from every member preserves intersection-closure, height and elastic sequences. So we may assume $C_0 = \emptyset$.

Ranks. Let the rank $r(X)$ of a member be the length of the longest strict chain of members from $\emptyset$ to X. If $X \subsetneq Y$ are members, then $r(X) < r(Y)$. Since $r(U) = 3$, every member $X \neq U$ has $r(X) \le 2$. The only member of rank 0 is $\emptyset$. Members of rank 1 are *atoms* and members of rank 2 are *planes*.

Fact 1. If X, Y are members with $X \not\subseteq Y$, then $X \cap Y$ is a member with $r(X \cap Y) < r(X)$. So the intersection of a plane with a member not containing it is $\emptyset$ or an atom, and the intersection of an atom with a member not containing it is $\emptyset$.

Let $s_1, \ldots, s_m$ be elastic, with witnesses $Y_i^1 \cup Y_i^2$ (taking $Y_i^1 = Y_i^2$ for a single member) that cover $s_1, \ldots, s_{i-1}$ and miss s_i.

Claim A: each atom contains at most one s_i. Suppose $s_i, s_j \in A$ with $i < j$. Then $s_i \in Y_j^l$ for some l, and $s_j \notin Y_j^l$, so $A \not\subseteq Y_j^l$. By Fact 1, $A \cap Y_j^l = \emptyset$, contradicting $s_i \in A \cap Y_j^l$.

Claim B: each plane contains at most three s_i. Let X be a plane and s_j the last sequence element in X. The earlier elements of X are covered by $Y_j^1 \cup Y_j^2$. Since $s_j \notin Y_j^l$, $X \not\subseteq Y_j^l$, so each $X \cap Y_j^l$ is $\emptyset$ or an atom. By Claim A, each contains at most one sequence element. So X has at most two elements before s_j.

By Claims A and B, every member other than U contains at most three sequence elements.

The contradiction. Suppose $m \ge 7$. The members Y_7^1, Y_7^2 miss s_7, so they differ from U, and together they cover $s_1, \ldots, s_6$. Each covers at most three, so each covers exactly three, disjointly. Atoms cover at most one, so both are planes; call them X and Y. Moreover, X contains no element covered only by Y, since it would then contain four sequence elements, contrary to Claim B; and likewise for Y. Say $s_6 \in Y$. The witness $Z_1 \cup Z_2$ of s_6 misses s_6 and covers the three X-elements, which all lie among $s_1, \ldots, s_5$.

Let Z be a member with $s_6 \notin Z$. If $Z \not\supseteq X$, then $Z \cap X$ is $\emptyset$ or an atom, so Z covers at most one X-element. If $Z \supsetneq X$, then $r(Z) \geq 3$, so $Z = U \ni s_6$, which is impossible. Hence a member that covers two or more X-elements equals X. One of Z_1, Z_2 covers at least two X-elements; say $Z_1 = X$. Then X contains neither of the two Y-elements among $s_1, \ldots, s_5$, so Z_2 must cover both. But $s_6 \in Y \setminus Z_2$, so $Z_2 \cap Y$ is $\emptyset$ or an atom, which covers at most one of them by Claim A. This is a contradiction, so $m \leq 6$. □

Computational evidence (computed).

- *Graphic matroids: the bound would be tight at $k = 2$.* The ground set of $M(K_{h+1})$ is the edge set of K_{h+1}, with $\binom{h+1}{2}$ points. Its flats are the edge sets that are unions of complete graphs on the blocks of a partition of the vertices. Flats are closed under intersection, and the rank of the matroid is h, so the family has height h. `graphic.py` computes the 2-union elasticity as $3, 6, 10, 15 = \binom{h+1}{2}$ for $h = 2, 3, 4, 5$, so every edge can be placed in an elastic sequence. For $k = 3$ it computes 6 and 10 for K_4 and K_5, which is the number of edges and below $\binom{h+2}{3} = 10, 20$; here the edge count caps the elasticity.
- *Search.* Randomized hill-climbing with $k = 2$ (`icclimb.py`) found exactly the conjectured maxima 6 and 10 for $h \leq 3$ and $h \leq 4$, on universes of 9 and 12 points. A run with $h \leq 5$ used a universe of only 14 points. Elasticity never exceeds the number of points, so its maximum of 14 is a ceiling artifact: it is not evidence of slack below 15, and it does not test the conjecture, since a counterexample needs at least 16 points. An independent search on 11–12 points with $h \leq 4$ found no counterexample.

C.9 Time-uniform soundness

Lemma C.4 (Ville's inequality (Ville, 1939))**.** *If $(M_t)_{t \geq 0}$ is a nonnegative supermartingale with $\mathbb{E}M_0 \leq 1$, then $\mathbb{P}[\exists t : \ M_t \geq 1/\delta'] \leq \delta'$ for every $\delta' \in (0, 1]$.*

Proof of Theorem 4.15. Observations and likelihoods. In round u there is at most one observation O_u: a human datum $X \in S$, an oracle answer $y \in \{0, 1\}$ to an escalated query q_u, or nothing. For each R let $\lambda_R(O_u)$ be $p_R(X)$, $\ell_R(y \mid q_u)$ or 1 accordingly, and let $L_t(R) = \prod_{u \leq t} \lambda_R(O_u)$. Under the hypothesis of (a), conditional on $\mathcal{F}_u$ the law of O_{u+1} is λ_{R^*} restricted to the type of round $u + 1$. That type is $\mathcal{F}_u$-measurable: V_δ is deterministic, and q_{u+1} and the schedule of human data are $\mathcal{F}_u$-measurable. In particular $\lambda_{R^*}(O_{u+1}) > 0$ almost surely, so $L_t(R^*) > 0$ almost surely.

(a) Supermartingale. Observation spaces are countable. For each R,

$$\mathbb{E}\Big[\frac{\lambda_R(O_{u+1})}{\lambda_{R^*}(O_{u+1})} \,\Big|\, \mathcal{F}_u\Big] = \sum_{o:\ \lambda_{R^*}(o) > 0} \lambda_R(o) \leq 1.$$

The quantity $L_u(R)/L_u(R^*)$ is $\mathcal{F}_u$-measurable, and the sum defining Z_{u+1} has nonnegative terms. So conditional monotone convergence gives

$$\mathbb{E}[Z_{u+1} \mid \mathcal{F}_u] = \sum_R w(R) \frac{L_u(R)}{L_u(R^*)} \mathbb{E}\Big[\frac{\lambda_R(O_{u+1})}{\lambda_{R^*}(O_{u+1})} \,\Big|\, \mathcal{F}_u\Big] \leq Z_u,$$

and $Z_0 = \sum_R w(R) = 1$. By Lemma C.4, the event $B = \{\exists t : \ Z_t \geq 1/\delta'\}$ has probability at most δ'. Off B, for every t,

$$w_t(R^*) = w^*/Z_t > w^* \delta' \geq \delta.$$

If the query in round $t+1$ is some $q \notin R^*$, then $w_t(\{R : q \notin R\}) \geq w_t(R^*) > \delta$, so V_δ does not accept it. Hence $\mathbb{P}[\exists t : \ V_\delta \text{ accepts an invalid } q_t] \leq \mathbb{P}(B) \leq \delta'$.

(b) Deterministic oracle. The answer to an escalated q is $y = \mathbf{1}[q \in R^*]$, and $\lambda_R(y)/\lambda_{R^*}(y) = \mathbf{1}[\mathbf{1}[q \in R] = \mathbf{1}[q \in R^*]] \leq 1$. Let $X_1, X_2, \ldots$ be the human data in order of arrival, and let $n(t)$ be the number that have arrived by round t. Then, pathwise,

$$Z_t \leq Z^H_{n(t)}, \qquad Z^H_n := \sum_R w(R) \prod_{j \leq n} \frac{p_R(X_j)}{p_{R^*}(X_j)}.$$

Since the X_j are i.i.d. p_{R^*}, the computation of (a), in the filtration $\mathcal{G}_n = \sigma(X_1, \ldots, X_n)$, shows that Z^H is a nonnegative supermartingale with $Z^H_0 = 1$. Let $G = \{\sup_n Z^H_n < 1/\delta'\}$. It depends only on the human data, and $\mathbb{P}(G) \geq 1 - \delta'$ by Lemma C.4. On G, for every prover strategy and every arrival schedule, $Z_t < 1/\delta'$ for all t, and the argument of (a) shows that no invalid step is ever accepted. The prover may know R^*, the verifier and all of $X_1, X_2, \ldots$ in advance. It can choose only the timing of the data, and the bound holds for every timing. □

A prover that foresees the noise. Let $\mathcal{H} = 2^{\{0,1\}}$ with the uniform prior, so $w^* = 1/4$, and let $R^* = \{0\}$. There are no human data. The oracle flips the true label with probability η, independently at each query, so $\ell_R(y \mid q) = 1 - \eta$ if $y = \mathbf{1}[q \in R]$ and η otherwise. Set $\delta' = 0.05$ and $\delta = w^*\delta'$. The prover knows in advance whether the next answer will be flipped. If it will be, the prover queries the invalid step 1, and the answer reads "valid". If not, it queries the valid step 0, and the answer again reads "valid". So every answer is "valid", and each answer to a query of 1 multiplies the likelihood ratio of $\{0,1\}$ to $\{0\}$ by $(1-\eta)/\eta > 1$. No answer ever lowers that ratio. Flips occur infinitely often, so the posterior mass of the hypotheses that exclude 1 eventually falls below δ, and the next query of 1 is accepted. The simulation `prescient.py` gives acceptance probability 1.0 at $\eta = 0.1$ and at $\eta = 0.3$, against the bound 0.05. This prover violates the hypothesis of (a), because $\mathcal{F}_t$ would have to contain the future noise. In (b) no such prover exists, because there is no noise to foresee.

C.10 Tightness of the constant

The main text states that the constant $\delta \leq w^*\delta'$ of Theorem 4.15 cannot be improved. Precisely:

Proposition C.5 (The constant is tight)**.** *[T1 Prop 4.3] (proved; computed) Let $\mathcal{H} = \{R^*, R'\}$ with $R^* = \{a, b\}$, $p_{R^*}(a) = 1 - u$, $p_{R^*}(b) = u$, $R' = \{a, c\}$, $p_{R'}(a) = 1$, prior $w(R^*) = w^*$, a deterministic oracle, and $\delta = w^*\delta'$. Let t^* be the least t with $(1-u)^{-t} > \theta := (1-w^*\delta')/((1-w^*)\delta')$. A prover that waits for t^* human data and then queries the invalid step c gets c accepted with probability*

$$(1-u)^{t^*} \in [(1-u)/\theta,\ 1/\theta), \qquad \text{where } 1/\theta = \frac{(1-w^*)\delta'}{1-w^*\delta'} \leq \delta'.$$

This tends to δ' as $u, w^* \to 0$. The exact values are $0.989\,\delta'$ at $u = 0.05$, $w^* = \delta' = 0.01$, and $0.991\,\delta'$ at $u = 0.01$, $w^* = 10^{-3}$, $\delta' = 0.05$.

Proof idea. After t copies of a, the likelihood ratio of R' to R^* is $(1-u)^{-t}$, and R' dies at the first b. If c is queried earlier, it is escalated, and the answer "invalid" kills R'.

Proof of Proposition C.5. Let $\mathcal{H} = \{R^*, R'\}$ with $w(R^*) = w^*$ and $w(R') = 1 - w^*$. While all human data seen are a, after t of them $L_t(R^*) = (1-u)^t$ and $L_t(R') = 1$. Write $r_t = (1-u)^{-t}$.

Then

$$w_t(\{R : c \notin R\}) = w_t(R^*) = \frac{w^*}{w^* + (1-w^*)r_t}.$$

V_δ with $\delta = w^*\delta'$ accepts c iff this is less than $w^*\delta'$, i.e. iff $w^* + (1-w^*)r_t > 1/\delta'$, i.e. iff $r_t > \theta = (1 - w^*\delta')/((1-w^*)\delta')$. Note that $\theta \geq 1$ since $\delta' \leq 1$, so $t^* \geq 1$.

- Once a b arrives, $L(R') = 0$, and c is never accepted.
- If c is queried at some $t < t^*$ while only a's have arrived, it is not accepted. If it is escalated, the deterministic answer "invalid" sets $L(R') = 0$, and again c is never accepted.

So the prover's best strategy is to query c exactly once, after the t^*-th datum. It succeeds iff the first t^* data are all a, which has probability $(1-u)^{t^*}$.

By the choice of t^*, $r_{t^*} > \theta$, so $(1-u)^{t^*} < 1/\theta$. By minimality, $r_{t^*-1} \leq \theta$, so $(1-u)^{t^*} \geq (1-u)/\theta$. Also

$$1/\theta = (1-w^*)\delta'/(1-w^*\delta') \leq \delta',$$

and $(1-u)/\theta \to \delta'$ as $u, w^* \to 0$. The exact values $0.989\,\delta'$ and $0.991\,\delta'$ quoted in the statement come from `ville2.py`, which computes $(1-u)^{t^*}$ exactly. It also gives, for example, $0.781\,\delta'$ at $u = 0.5$, $w^* = \delta' = 0.01$. □

C.11 The escalation bound

Proof of Theorem 4.16. Let $\mathcal{F}_u$ be as in Theorem 4.15(a). Z changes only in data rounds and in escalation rounds.

Data round. The update is $Z_u = Z_{u-1}F_u$, where $F_u = \sum_R w_{u-1}(R)\, p_R(X_u)/p_{R^*}(X_u)$. This uses $w_{u-1}(R) = w(R)L_{u-1}(R)/(Z_{u-1}L_{u-1}(R^*))$. Since w_{u-1} is $\mathcal{F}_{u-1}$-measurable and X_u is fresh,

$$\mathbb{E}[F_u \mid \mathcal{F}_{u-1}] = \sum_R w_{u-1}(R) \sum_{x:\, p_{R^*}(x)>0} p_R(x) \leq 1.$$

Escalation round. The deterministic answer is $y = \mathbf{1}[q \in R^*]$, and the update is $Z_u = Z_{u-1}\sum_R w_{u-1}(R)\,\mathbf{1}[\mathbf{1}[q \in R] = y]$.

- If q is valid, the factor is $w_{u-1}(\{R : q \in R\}) = 1 - w_{u-1}(\{R : q \notin R\}) \leq 1 - \delta$, because q was not accepted.
- If q is invalid, the factor is $w_{u-1}(\{R : q \notin R\}) = 1 - w_{u-1}(\{R : q \in R\}) \leq 1 - \delta_r$, because q was not rejected. In every case it is at most 1.

Counting. Let $M_t = \prod_{\text{data rounds } u \leq t} F_u$. This is a nonnegative supermartingale with $M_0 = 1$. Let E_t^+ and E_t^- be the numbers of escalations of valid and of invalid queries up to time t. Since $w_t(R^*) \leq 1$,

$$w^* \leq \frac{w^*}{w_t(R^*)} = Z_t \leq M_t\,(1-\delta)^{E_t^+}(1-\delta_r)^{E_t^-}.$$

By Lemma C.4, with probability at least $1 - \delta''$ we have $M_t < 1/\delta''$ for all t. On that event, $(1-\delta_m)^{E_t^+ + E_t^-} > w^*\delta''$, and using $\ln \frac{1}{1-x} \geq x$,

$$\#\mathrm{ESC}_t = E_t^+ + E_t^- \leq \frac{\ln(1/w^*) + \ln(1/\delta'')}{\delta_m}.$$

Dropping the factor $(1-\delta_r)^{E_t^-} \leq 1$ gives $E_t^+ \leq (\ln(1/w^*) + \ln(1/\delta''))/\delta$ against every prover. For an honest prover $E_t^- = 0$.

VS posterior. Here a datum's likelihood is $\mathbf{1}[X_u \in R]$, and $X_u \in R^*$. So $F_u = w_{u-1}(\{R : X_u \in R\}) \leq 1$ and $M_t \leq 1$ deterministically. The same computation then gives $\#\mathrm{ESC}_t \leq \ln(1/w^*)/\delta_m$, and $E_t^+ \leq \ln(1/w^*)/\delta$. □

The bound on E_t^+ against every prover, with δ in place of δ_m, is read off the same computation, because invalid escalations multiply Z by at most 1. Corollary 4.18 uses it in this form.

Proof of Corollary 4.18. Soundness is Theorem 4.14. An escalated valid q had $w(\{R : q \notin R\}) \geq \delta > 0$, so $q \notin \bigcap \mathrm{VS}(P, P^-) \supseteq \bigcap \mathrm{VS}(P)$. Here P is P_0 together with the earlier escalated valid queries; human data that arrive later only enlarge $\bigcap \mathrm{VS}$ further. So these queries form an elastic chain in R^*, and there are at most $\mathrm{el}(\mathcal{H}, R^* \mid P_0)$ of them. The second term is Theorem 4.16(ii), counting valid queries only. □

C.12 Repairs made during verification

The results of Section 4 went through adversarial verification; the referee reports and the repair log are in `research/verification/T1-soundness-under-search-verification.md`, and a second round is in `research/verification/reverification-round2.md`. The repairs that changed a statement or a proof are these.

- Theorem 4.2(b) first bounded the acceptance probability conditionally on the history. That reading is false (the reckless verifier of Appendix C.1); the statement now bounds the joint probability, and the deterministic case (c) was added.
- The lower bound of Theorem 4.4 first cited the conditional reading; it now uses the coupling of Lemma C.1 directly.
- Theorem 4.7: the displays were restated with the target-relative quantity $\mathrm{el}(H_1, \sigma^* \mid \emptyset; N)$, and the proof of (i) now uses the lgg representatives, so that inclusion gives strict generality without assumptions on the signature. The second round noted the degenerate case of (ii) in which no instance of σ^* has size at most N; hence the maximum with 0 there and in the corresponding case of Theorem 4.9.
- Proposition 4.8 first claimed super-exponential cost without qualification; the growing signature and the fixed-signature bound were made explicit.
- The justification of Theorem 4.9 first asserted that H_k^{tag} is intersection-closed, which is false (Appendix C.6); it now uses the factorization of the version space.
- Theorem 4.10(iii) first said "$\Theta(n^k)$ for flat schemas". This is proved only for linear flat schemas and needs $n \geq k$ (Proposition C.2).
- The evidence for Conjecture 4.11 was corrected. A claim that "a natural one-step decomposition proof fails on some extremal sequences" had no surviving script and was withdrawn; the $h \leq 5$ search was recognized as a ceiling artifact; the graphic-matroid data were added. The case $(k, h) = (2, 3)$ (Proposition C.3) was sketched by a referee and written out in full.
- The hypothesis of Theorem 4.15(a), that the prover cannot foresee the data or the noise, was implicit; the prescient prover of Appendix C.9 shows that it is needed. Part (a) is credited to Waudby-Smith and Ramdas (2020).
- Corollary 4.18 first called the verifier 0-sound; the guarantee covers only targets with $w(R^*) \geq \delta$.

D Proofs for Section 5

Theorem 5.3, Corollary 5.10, Proposition 5.12, Theorem 5.16 and Corollary 5.19 are proved in full in the main text, and Lemma 5.1 is the positive-data case of results of Section 4. This appendix proves the remaining results of Section 5. It also states the remark on dimension inference with a proof sketch (Appendix D.12), reports the positive-data numbers of Experiment A (Appendix D.13), and lists the repairs made to these statements during verification (Appendix D.14).

Notation is as in Sections 2 and 5. Steps are ground terms over a signature Ω with at least two constants $a \neq b$. For a schema σ, $\mathrm{vars}(\sigma)$ is its set of metavariables and $v = |\mathrm{vars}(\sigma)|$. A substitution θ for σ maps $\mathrm{vars}(\sigma)$ to ground terms. Anti-unification is the map $\mathcal{A}$ of Section 2.2, and lgg is computed by it (Theorem 2.5).

D.1 Three elementary facts

Lemma D.1. (F1) *If $\tau = \sigma\theta$ and θ restricted to $\mathrm{vars}(\sigma)$ is not a renaming, then $\mathrm{inst}(\tau) \subsetneq \mathrm{inst}(\sigma)$.* (F2) *The intersection of the instance sets of all schemas is empty.* (F3) *For tagged calculi H_k^{tag} and clean data D, $\bigcap \mathrm{VS}(D) = \bigcup_{i \leq k} \{i\} \times \hat{R}_i$, where $\hat{R}_i = \mathrm{inst}(\mathrm{lgg}(D^{(i)}))$ if $D^{(i)} \neq \emptyset$ and $\hat{R}_i = \emptyset$ otherwise.*

Proof. (F1) Inclusion holds because instances of τ are instances of σ. For strictness there are two cases. First, suppose some θx is not a metavariable, say with root g. Choose a constant $c \neq g$; it exists because there are two constants. Let $t = \sigma\theta'$ with $\theta' x = c$ and θ' ground elsewhere. At a position of σ where x occurs, t carries c, while every instance of τ carries a term with root g. So $t \notin \mathrm{inst}(\tau)$. Second, suppose every θx is a metavariable. Then θ is not injective on $\mathrm{vars}(\sigma)$, say $\theta x = \theta y$ with $x \neq y$. Let $\theta' x = a$ and $\theta' y = b$. Every instance of τ has equal subterms at the positions of x and y, and $\sigma\theta'$ does not. So $\sigma\theta' \notin \mathrm{inst}(\tau)$.

(F2) The ground schemas a and b have disjoint instance sets $\{a\}$ and $\{b\}$.

(F3) Write I for the family of single-schema instance sets. Then $H_k^{\mathrm{tag}} = \{\bigcup_i \{i\} \times A_i : A_i \in I\}$, and $D \subseteq \bigcup_i \{i\} \times A_i$ iff $D^{(i)} \subseteq A_i$ for every i. So $\mathrm{VS}(D)$ is the set of all such unions with $(A_1, \ldots, A_k) \in \prod_i \mathrm{VS}_i$, where $\mathrm{VS}_i := \{A \in I : D^{(i)} \subseteq A\}$. Each VS_i is nonempty, since it contains $\mathrm{inst}(\sigma_i)$. Hence $(i, s) \in \bigcap \mathrm{VS}(D)$ iff $s \in A_i$ for every choice in the product, iff $s \in \bigcap \mathrm{VS}_i$. Finally, $\bigcap \mathrm{VS}_i = \mathrm{inst}(\mathrm{lgg}(D^{(i)}))$ for nonempty $D^{(i)}$ by Theorem 2.5, and $\bigcap \mathrm{VS}_i = \emptyset$ for empty $D^{(i)}$ by (F2). □

(F3) is the positive-data case of the factorization used for Theorem 4.9. It is needed because H_k^{tag} is not literally intersection-closed.

D.2 Proofs of Propositions 5.4 and 5.5

Proof of Proposition 5.4. (a) If T is an anchor and $T \subseteq R \in \mathcal{H}$, then $R \supseteq R^*$, so $R \not\subseteq R^*$.

(b) Let $T \neq \emptyset$ be a tell-tale, and suppose $T \subseteq R \in \mathcal{H}$ with $R \not\supseteq R^*$. Then $R \cap R^* \supseteq T$ is nonempty, so it is a member of $\mathcal{H}$ by intersection-closure of $\mathcal{H} \cup \{\emptyset\}$. And $T \subseteq R \cap R^* \subsetneq R^*$, which contradicts the tell-tale property. For $\mathcal{H} = \{\{1\}, \{2\}\}$ and $R^* = \{1\}$: no member lies strictly inside $\{1\}$, so $\emptyset$ is a tell-tale; $\{2\} \supseteq \emptyset$ does not contain R^*, so $\emptyset$ is not an anchor. Here $\mathcal{H} \cup \{\emptyset\}$ is intersection-closed. For the existence claim, if T is a tell-tale and $r \in R^*$, then $T \cup \{r\}$ is a nonempty tell-tale (fewer members contain it), hence an anchor; anchors are tell-tales by (a).

(c) The class is an indexed family of uniformly recursive sets. $\{0\}$ is a tell-tale for $2\mathbb{N}$, since every R_n contains the odd number $2n+1$ and so no member lies strictly inside $2\mathbb{N}$. R_n is a tell-tale for itself. By Angluin's theorem (Angluin, 1980) the class is identifiable in the limit from text. Every finite $T \subseteq 2\mathbb{N}$ lies in R_n for any n with $2n \geq \max T$, and $R_n \not\supseteq 2\mathbb{N}$. So $2\mathbb{N}$ has no anchor, and by Theorem 5.3(b) the cautious verifier is never complete on a text for $2\mathbb{N}$. The same example shows that "finite" is needed in Theorem 5.3(a): every member containing $2\mathbb{N}$ is $2\mathbb{N}$, so $\hat{R}(2\mathbb{N}) = 2\mathbb{N}$. □

Proof of Proposition 5.5. If $R^* = \emptyset$, then $\emptyset$ is an anchor. Otherwise suppose R^* has no anchor, and pick $w_0 \in R^*$. Given $w_0, \ldots, w_{n-1} \in R^*$, the finite set $\{w_0, \ldots, w_{n-1}\} \subseteq R^*$ is not an anchor. So some $R_n \in \mathcal{H}$ contains it with $R_n \not\supseteq R^*$; pick $w_n \in R^* \setminus R_n$. Then $\{w_0, \ldots, w_{n-1}\} \subseteq R_n \not\ni w_n$ for all $n \geq 1$, which is infinite elasticity. □

This is the standard argument in direct form. The facts used after the proposition in the main text are standard: finite thickness implies finite elasticity, and the union theorem of Wright (1989) with the corrected definition of Motoki et al. (1991).

D.3 Proof of Theorem 5.6

By (F3), $\hat{R}(D) = R^*$ iff every rule i is *identified*, i.e. $\hat{R}_i = \mathrm{inst}(\sigma_i)$. Let E_i be the event that rule i is not identified; we bound each $\mathbb{P}(E_i)$ and take a union bound over i.

Rules with $v_i \geq 1$. Let $N_i = |D^{(i)}|$, so $N_i \sim \mathrm{Bin}(N, \pi_i)$. Given $N_i = n$, the rule-i substitutions $\Theta_1, \ldots, \Theta_n$ are i.i.d. Λ_i. Say that (R) *fails at x* if all n roots $\mathrm{root}(\Theta_j x)$ equal one symbol (vacuously so if $n = 0$), and that (D) *fails at $\{x, y\}$* if $\Theta_j x = \Theta_j y$ for all j. If $n \geq 1$ and no witness event fails, Lemma 2.7 gives $\mathrm{lgg}(D^{(i)}) \equiv \sigma_i$, so $\hat{R}_i = \mathrm{inst}(\sigma_i)$. Hence E_i is contained in the union of the $v_i + \binom{v_i}{2}$ failure events.

Fix x. The function $G \mapsto \min(F(G), 1 - F(G))$, with F the law of $\mathrm{root}(\Theta x)$, attains its maximum $\rho_{i,x}$. Indeed $G \mapsto F(G) = \sum_f \mathbf{1}_G(f) F(f)$ is continuous on the compact space $\{0,1\}^\Omega$, because the series converges uniformly. Let G attain the maximum. If all roots equal f, then all lie in G (if $f \in G$) or all lie in G^c. So

$$\mathbb{P}[\text{(R) fails at } x \mid N_i = n] \leq F(G)^n + (1 - F(G))^n \leq 2(1 - \rho_{i,x})^n \leq 2(1 - \rho_i)^n,$$

since $\min(F(G), 1 - F(G)) = \rho_{i,x}$ gives $\max(F(G), 1 - F(G)) \leq 1 - \rho_{i,x}$. For a pair,

$$\mathbb{P}[\text{(D) fails at } \{x, y\} \mid N_i = n] = (1 - r_{i,xy})^n \leq (1 - \rho_i)^n.$$

By the union bound, $\mathbb{P}(E_i \mid N_i = n) \leq c_i(1 - \rho_i)^n$. For $n = 0$ this is trivially true, since $c_i \geq 2$. Averaging over N_i,

$$\mathbb{P}(E_i) \leq c_i \, \mathbb{E}[(1 - \rho_i)^{N_i}] = c_i \sum_{n=0}^{N} \binom{N}{n} \pi_i^n (1 - \pi_i)^{N-n} (1 - \rho_i)^n = c_i (1 - \pi_i \rho_i)^N \leq c_i e^{-N \pi_i \rho_i}.$$

Ground rules. If $v_i = 0$ and $N_i \geq 1$, then $D^{(i)}$ consists of copies of σ_i and $\hat{R}_i = \{\sigma_i\}$. So $E_i = \{N_i = 0\}$, and $\mathbb{P}(E_i) = (1 - \pi_i)^N \leq e^{-N\pi_i} = c_i e^{-N\pi_i\rho_i}$ under the conventions $\rho_i = c_i = 1$. The general formula would give $c_i = 0$ for ground rules; hence the separate convention.

Sample size. If $N \geq \frac{1}{\pi_i \rho_i} \ln \frac{k c_i}{\delta}$ for every i, each term is at most δ/k, and the sum is at most δ.

Necessity. Fix i, x, f and $\rho \in (0,1]$ with $\mathbb{P}[\mathrm{root}(\Theta x) = f] = 1-\rho$. Let A be the event that no datum is a rule-i instance with $\mathrm{root}(\Theta x) \neq f$. The data are i.i.d., and each datum is such an instance with probability $\pi_i\rho$, so $\mathbb{P}(A) = (1-\pi_i\rho)^N$. On A, either $N_i = 0$, and then $\hat{R}_i = \emptyset \neq \mathrm{inst}(\sigma_i)$; or $N_i \geq 1$, and then every column that anti-unification meets at an occurrence of x has root f throughout. In the second case $\mathrm{lgg}(D^{(i)})$ carries a term with root f there. Since $\rho > 0$, some ground term t with root $\neq f$ is in the support. The instance of σ_i with $x \mapsto t$ lies in $\mathrm{inst}(\sigma_i)$ but not in $\hat{R}_i$. So $\mathbb{P}[\hat{R}(D) \neq R^*] \geq (1-\pi_i\rho)^N$. □

D.4 Proof of the claims in Remark 5.7

Teaching sets. Let θ_1, θ_2 map the v_i metavariables of σ_i to $2v_i$ pairwise distinct constants. For each x, $\theta_1 x \neq \theta_2 x$ are distinct constants, so their roots differ and (R) holds. For $x \neq y$, $\theta_1 x \neq \theta_1 y$, so (D) holds. By Lemma 2.7, $\mathrm{lgg}(\sigma_i\theta_1, \sigma_i\theta_2) \equiv \sigma_i$. In a tagged class this set (one per rule) is an anchor, by (F3).

No distribution-free completeness rate. Let g be binary, $h \geq 1$, $m = 2^h$, and let u_j $(1 \leq j \leq m)$ be the complete g-tree of depth h with leaf j equal to a and all other leaves equal to b. Let $J \subseteq [m]$ with $|J| \geq 2$, and anti-unify $\{u_j : j \in J\}$. All internal positions carry g and are copied. At leaf $\ell \in J$ the column contains a (from u_ℓ) and b (from any other u_j, $j \in J$), so it becomes a metavariable z_ℓ. Columns at distinct leaves $\ell, \ell' \in J$ differ (at index ℓ one has a, the other b), so $z_\ell \neq z_{\ell'}$. At leaves $\ell \notin J$ the column is constantly b. Hence the lgg has distinct metavariables exactly at the leaves in J and b elsewhere. Its instances among the u's are exactly $\{u_j : j \in J\}$, because u_k with $k \notin J$ has a at a leaf where the lgg has b. For $|J| = 1$ the lgg is u_j itself.

Take the target $\mathrm{inst}(x) \in H_1$ and the uniform law Q on $\{u_1, \ldots, u_m\}$. A sample of $n < m/2$ draws hits a set J of at most n indices, and by the above $\hat{R}(D) \cap \{u_1, \ldots, u_m\} = \{u_j : j \in J\}$. So $Q(R^* \setminus \hat{R}(D)) \geq (m-n)/m > 1/2$ for every such sample. Any learner whose acceptance set is sound for every target in H_1 compatible with its data accepts a subset of $\hat{R}(D)$ (Lemma 5.1(b)), so its completeness error is at least as large. Since h is arbitrary, no bound on the number of samples that is independent of Q can guarantee completeness error at most $1/2$. □

D.5 Proof of Theorem 5.8

Part (a). Let $R = \bigcup_{l \leq k} \mathrm{inst}(\tau_l) \in \mathrm{VS}(D)$. Partition $\mathrm{Sub}_i(D)$ by the least l with $\sigma_i\theta \in \mathrm{inst}(\tau_l)$. If no class were generic, each would lie in a failure set, and k failure sets would cover $\mathrm{Sub}_i(D)$. So some class Q is generic, and its $\tau_l \succeq \mathrm{lgg}\{\sigma_i\theta : \theta \in Q\} \equiv \sigma_i$. Hence $R \supseteq R^*$. A failure set contains at most one member of a pairwise-generic set, so $k+1$ such members cannot be covered. □

Part (b). Fix a rule i with $v_i \geq 1$. Write $M := v_i + \binom{v_i}{2}$ and $M' := v_i^2 + 1 \geq M$. Consider the range space on substitutions (a countable set) whose ranges are

$$\mathcal{C}_i := \{\, \overline{F_1 \cup \cdots \cup F_k} : \; F_1, \ldots, F_k \in \mathfrak{F}_i \,\}.$$

Step 1: traces of $\mathfrak{F}_i$. Let $\theta_1, \ldots, \theta_n$ be substitutions. For fixed x, the sets $\{\theta : \mathrm{root}(\theta x) = f\}$ for different f are pairwise disjoint. So their traces on $\{\theta_1, \ldots, \theta_n\}$ are pairwise disjoint subsets, of which at most n are nonempty, giving at most $n+1$ distinct traces. Each pair $x \neq y$ contributes one set, hence one trace. So $\mathfrak{F}_i$ has at most $v_i(n+1) + \binom{v_i}{2} \leq M(n+1)$ distinct traces on n points.

Step 2: traces of $\mathcal{C}_i$. The trace of $\overline{F_1 \cup \cdots \cup F_k}$ is determined by the traces of $F_1, \ldots, F_k$. So $\mathcal{C}_i$ has at most $(M(n+1))^k \leq (M'(n+1))^k$ traces on n points.

Step 3: VC dimension. If $\mathcal{C}_i$ shatters d points, then $2^d \le (M'(d+1))^k$, i.e. $\phi(d) := d - k\log_2(M'(d+1)) \le 0$. Let $L := \log_2(4kM') \ge 2$ and $d_i = 2kL$. Then $\phi(d_i) > 0$ iff $(4kM')^2 > M'(2kL+1)$, iff $16k^2M' > 2kL+1$. This holds because $L < 4kM'$ gives $2kL+1 < 8k^2M'+1 \le 16k^2M'$. Moreover $\phi'(d) = 1 - k/((d+1)\ln 2) > 0$ for $d \ge d_i \ge 4k$. So $\phi(d) > 0$ for all $d \ge d_i$, and the VC dimension of $\mathcal{C}_i$ is less than d_i.

Step 4: the ε-net theorem (Haussler and Welzl, 1987; Blumer et al., 1989). Let a range space have VC dimension at most d, let μ be a probability measure, and let $0 < \varepsilon, \delta' < 1$. If

$$m \ge \max\Big\{\frac{4}{\varepsilon}\log_2\frac{2}{\delta'},\ \frac{8d}{\varepsilon}\log_2\frac{13}{\varepsilon}\Big\},$$

then with probability at least $1-\delta'$ an i.i.d. μ-sample of size m meets every range of μ-measure at least ε. We use the theorem in this "measure $\ge \varepsilon$" form. The symmetrization proof gives it with the same constants. Alternatively, apply the standard "measure $> \varepsilon$" form with $\varepsilon = \zeta_i/2$, at the cost of a constant factor.

Step 5: conclusion. Apply Step 4 with $\mu = \Lambda_i$, $\varepsilon = \zeta_i$, $d = d_i$ and $\delta' = \delta/k'$. The sum in the statement exceeds the maximum. By definition of ζ_i, every range in $\mathcal{C}_i$ has Λ_i-measure at least ζ_i. So with probability at least $1-\delta/k'$ the n_i rule-i substitutions meet every range in $\mathcal{C}_i$, i.e. they are not contained in any union of k failure sets. They are a subset of $\mathrm{Sub}_i(D)$, so neither is $\mathrm{Sub}_i(D)$. For a ground rule, one sample makes $\mathrm{Sub}_i(D)$ nonempty, and $\mathfrak{F}_i = \emptyset$. A union bound over the k' rules and part (a) finish the proof. □

D.6 Proof of Proposition 5.9

Part (b) shows that the stronger claim, that at most k roots at a metavariable make identification fail for good, is false for $k' \ge 2$; part (a) recovers it for $k' = 1$.

Failure sets and specializations. For a metavariable x of σ_i and $f \in \Omega$ of arity r, let $\bar{z} = z_1, \ldots, z_r$ be fresh distinct metavariables. Then $\mathrm{inst}(\sigma_i[x \mapsto f(\bar{z})]) = \{\sigma_i\theta : \mathrm{root}(\theta x) = f\}$ and $\mathrm{inst}(\sigma_i[y \mapsto x]) = \{\sigma_i\theta : \theta x = \theta y\}$. So covering substitutions by failure sets is the same as covering the corresponding instances by these specializations.

(a) Let $\sigma'_1, \ldots, \sigma'_{k-k'+1}$ be the specializations in the hypothesis, and let

$$R := \bigcup_{j \ne i} \mathrm{inst}(\sigma_j) \ \cup \bigcup_{l \le k-k'+1} \mathrm{inst}(\sigma'_l).$$

This is a union of at most k schema instance sets, so $R \in H_k$. Every datum lies in R^*. A datum outside $\mathrm{inst}(\sigma_i)$ lies in some $\mathrm{inst}(\sigma_j)$ with $j \ne i$. A datum in $\mathrm{inst}(\sigma_i)$ lies in another rule or, by hypothesis, in some $\mathrm{inst}(\sigma'_l)$. So $D \subseteq R$, $R \in \mathrm{VS}(D)$ and $\hat{R}_{H_k}(D) \subseteq R$. The instance of σ_i assumed to lie outside R is in $R^* \setminus \hat{R}_{H_k}(D)$.

The case $k' = 1$. If x is instantiated in the data with roots among $f_1, \ldots, f_k$ only, the k specializations $\sigma_1[x \mapsto f_j(\bar{z})]$ cover the data. Any instance whose x has another root lies outside them. So identification fails for as long as this persists.

(b) Let $R = \mathrm{inst}(\tau_1) \cup \mathrm{inst}(\tau_2) \in \mathrm{VS}_{H_2}(D)$, with $D = \{p(a), p(b), q(a), q(b)\}$; possibly $\tau_1 = \tau_2$. If some τ_l covers a p-term and a q-term, then $\tau_l \succeq \mathrm{lgg}(p(\cdot), q(\cdot))$, which is a metavariable because the roots differ. So τ_l is a metavariable and R is everything. Otherwise each τ_l covers only p-terms or only q-terms of D. All four terms are covered, so one schema covers both p-terms and the other both q-terms. Then $\tau_1 \succeq \mathrm{lgg}(p(a), p(b)) = p(x)$ and $\tau_2 \succeq q(y)$ (up to the order of the two schemas). In both cases $R \supseteq R^*$. Since $R^* \in \mathrm{VS}_{H_2}(D)$, $\hat{R}_{H_2}(D) = R^*$. If the substitution law of x is supported on $\{a, b\}$, the two failure sets $\{\mathrm{root}(\theta x) = a\}$ and $\{\mathrm{root}(\theta x) = b\}$ have

full measure, so $\zeta_1 = 0$. A brute-force enumeration confirms (b) (computed; `research/theory/T1-code/union_remark.py`). □

D.7 Proofs for Remark 5.11

Let Φ be finite and let the target be $G^* := \mathrm{inst}(\sigma^*, \Psi^*) \neq \emptyset$. Write $\mathrm{cl}(P) := \mathrm{inst}(\mathrm{lgg}(P), \Psi_P)$ for nonempty P, the least guarded instance set containing P (Lemma 2.8).

Anchors exist. Ascending generalization chains are finite (Lemma 2.6), so the lgg of the (possibly infinite) set G^* exists and equals $\mathrm{lgg}(T_0)$ for some finite $T_0 \subseteq G^*$. For each $\phi \in \Phi$ that is not true on all of G^*, pick $t_\phi \in G^*$ with $\phi(t_\phi) = 0$. Let $T := T_0 \cup \{t_\phi\}$. Then $\mathrm{lgg}(T) = \mathrm{lgg}(G^*)$, because $\mathrm{lgg}(T_0) \preceq \mathrm{lgg}(T) \preceq \mathrm{lgg}(G^*) = \mathrm{lgg}(T_0)$. Also Ψ_T is exactly the set of predicates true on all of G^*. Hence $G^* \subseteq \mathrm{inst}(\mathrm{lgg}(T), \Psi_T) = \mathrm{cl}(T)$. Conversely, $\mathrm{cl}(T) \subseteq G^*$, since G^* is a guarded instance set containing T. So $\mathrm{cl}(T) = G^*$, and every guarded instance set containing T contains G^*: T is an anchor.

Chains. Let $\mathrm{cl}(P_0) = C_0 \subsetneq C_1 \subsetneq \cdots \subsetneq C_m \subseteq G^*$ be closures (the chain of closures of the data plus escalated steps, for nonempty initial data $P_0 \subseteq G^*$), $C_j = \mathrm{inst}(g_j, \Psi_j)$ with $g_j = \mathrm{lgg}(C_j)$ and $\Psi_j = \Psi_{C_j}$. Inclusion gives $g_{j-1} \preceq g_j$ and $\Psi_{j-1} \supseteq \Psi_j$. If both were equalities, then $C_{j-1} = C_j$. So each step either strictly generalizes the schema (at most $\mu(g_0) - \mu(\sigma^*)$ times, by Lemma 2.6, since $g_m \preceq \mathrm{lgg}(G^*) \preceq \sigma^*$) or strictly shrinks the guard set (at most $|\Phi|$ times). By Proposition 4.5(ii) and Theorem 4.4 this bounds the escalations by $\mu(\mathrm{lgg}(P_0)) - \mu(\sigma^*) + |\Phi|$. □

D.8 Proof of Theorem 5.13

(a) If D has at most e invalid steps, then $|D \setminus R^*| \leq e$, so $R^* \in \mathrm{VS}_e(D)$ and $\bigcap \mathrm{VS}_e(D) \subseteq R^*$. The per-tag case is identical. Soundness against every prover follows by Lemma 3.1, as in Lemma 5.1.

(b) By (i), $R^* \in \mathrm{VS}_{(e_i)}(D)$, so $\bigcap \mathrm{VS}_{(e_i)}(D) \subseteq R^*$. Conversely, let $R \in \mathrm{VS}_{(e_i)}(D)$ and fix i. Its tag-i component is $\mathrm{inst}(\tau)$ for a schema τ, or empty. Let V_i be the multiset of valid rule-i samples and $V_i' := V_i \cap \mathrm{inst}(\tau)$. Then $|V_i \setminus V_i'| \leq |D^{(i)} \setminus R^{(i)}| \leq e_i$.

- $V_i' \neq \emptyset$, because $|V_i| > e_i$. In particular the tag-i component is not empty.
- For each x and f, more than e_i members of V_i have $\mathrm{root}(\Theta x) \neq f$, and at most e_i were removed. So some member of V_i' has $\mathrm{root}(\Theta x) \neq f$. Since this holds for every f, the roots at x in V_i' are not all equal, and (R) holds for V_i'.
- Likewise, for each $x \neq y$ some member of V_i' has $\Theta x \neq \Theta y$, and (D) holds.

By Lemma 2.7, $\mathrm{lgg}(V_i') \equiv \sigma_i$. Since $V_i' \subseteq \mathrm{inst}(\tau)$, $\tau \succeq \mathrm{lgg}(V_i') \equiv \sigma_i$, so $\mathrm{inst}(\tau) \supseteq \mathrm{inst}(\sigma_i)$. For a ground rule, V_i' is a nonempty multiset of copies of σ_i, so $\tau \succeq \sigma_i$ directly. Hence $R \supseteq R^*$ for every $R \in \mathrm{VS}_{(e_i)}(D)$, and $\bigcap \mathrm{VS}_{(e_i)}(D) = R^*$. □

Why the clause "more than e_i valid samples". Take a ground rule $\sigma_i = c$, one valid sample (i, c), no invalid samples and $e_i = 1$. The rest of (ii) holds vacuously. But a hypothesis whose tag-i component is $\emptyset$ or $\mathrm{inst}(d)$ with $d \neq c$ misses one tag-i datum and lies in $\mathrm{VS}_{(e_i)}$. So $c \notin \bigcap \mathrm{VS}_{(e_i)}$.

Why per-tag budgets. Take two tags with $e_1 = e_2 = 1$ and the global budget $e = 2$. Let rule 1 be $\mathsf{s}(x)$ with valid samples $\mathsf{s}(a), \mathsf{s}(b), \mathsf{s}(c)$. These are 1-robustly generic: for every f, at least two of them have a root $\neq f$. Let rule 2 be a ground t with samples t, t. The hypothesis

with components inst($\mathsf{s}(a)$) and inst(t) misses exactly two data, so it lies in $\mathrm{VS}_2(D)$. Hence $\mathsf{s}(d) \notin \bigcap \mathrm{VS}_2(D)$ for any constant $d \neq a$, and $\bigcap \mathrm{VS}_2(D) \neq R^*$. Under per-tag budgets, the tag-1 component must contain at least two of the three samples. These satisfy (R), so the component generalizes $\mathsf{s}(x)$.

D.9 Proof of Theorem 5.14

We use Hoeffding's inequality for $X \sim \mathrm{Bin}(N, p)$ and $s > 0$: $\mathbb{P}[X - Np \geq sN] \leq e^{-2Ns^2}$ and $\mathbb{P}[X - Np \leq -sN] \leq e^{-2Ns^2}$. Fix a rule i and write $\alpha = \alpha_i$, $\beta = \beta_i$, $\rho = \rho_i$ and $\Delta = \Delta_i$, so that $\rho\beta = \alpha + 2\Delta$ and $e_i = \lfloor (\alpha + \Delta)N \rfloor$.

Noise count. The number X of invalid steps tagged i is $\mathrm{Bin}(N, \alpha)$. Since $e_i + 1 > (\alpha + \Delta)N$,

$$\mathbb{P}[X > e_i] = \mathbb{P}[X \geq e_i + 1] \leq \mathbb{P}[X - \alpha N \geq \Delta N] \leq e^{-2N\Delta^2}.$$

Witness counts, $v_i \geq 1$. For each metavariable x, let G_x attain $\rho_{i,x}$ (Appendix D.3). Let C_x^+ and C_x^- count the valid rule-i data with $\mathrm{root}(\Theta x) \in G_x$ and $\notin G_x$, respectively. For each pair $x \neq y$, let C_{xy} count those with $\Theta x \neq \Theta y$. These are $2v_i + \binom{v_i}{2} = c_i$ counts. Each is $\mathrm{Bin}(N, p)$ with $p \geq \beta\rho_{i,x} \geq \beta\rho$, resp. $p = \beta r_{i,xy} \geq \beta\rho$. So $pN \geq (\alpha + 2\Delta)N$ and $(\alpha + \Delta)N \leq pN - \Delta N$. Hence, for each count C,

$$\mathbb{P}[C \leq e_i] \leq \mathbb{P}[C \leq (\alpha + \Delta)N] \leq \mathbb{P}[C - pN \leq -\Delta N] \leq e^{-2N\Delta^2}.$$

Suppose all c_i counts exceed e_i. For any f, the side of the split G_x that does not contain f consists of samples with root $\neq f$. So $\#\{\mathrm{root}(\Theta x) \neq f\} \geq \min(C_x^+, C_x^-) > e_i$. The pair conditions hold directly, and the number of valid rule-i samples is at least $C_x^+ > e_i$. So the valid rule-i samples are e_i-robustly generic.

Ground rules. With $\rho = 1$, the single required event is that the number of valid rule-i samples, which is $\mathrm{Bin}(N, \beta)$ with $\beta = \rho\beta$, exceeds e_i. Its failure probability is at most $e^{-2N\Delta^2}$ by the same computation; here $c_i = 1$.

Conclusion. Outside an event of probability at most $(1 + c_i)e^{-2N\Delta_i^2}$ for each i, hypotheses (i) and (ii) of Theorem 5.13(b) hold for all rules, and the trimmed verifier accepts exactly R^*. Take a union bound over i. □

D.10 Proof of Lemma 5.17

Let F be a probability law on the countable set Ω, $m = \max_f F(f)$ (attained, because only finitely many atoms exceed any positive mass), $f_{\max}$ a maximizer, and $\rho = \max_G \min(F(G), 1 - F(G))$.

$\rho \leq 1 - m$. For any G, the side of the split that does not contain $f_{\max}$ has mass at most $1 - m$, so the minimum is at most $1 - m$.

$1 - m \leq 2\rho$. If $m \geq 1/2$, the split $G = \{f_{\max}\}$ gives $\min(m, 1 - m) = 1 - m$, so $\rho \geq 1 - m$. If $m < 1/2$, enumerate Ω and add symbols to G one at a time until $F(G) \geq (1 - m)/2$. This happens at a finite stage, since the partial sums tend to 1. Before the last addition $F(G) < (1 - m)/2$, and the last symbol has mass at most m. So finally $(1 - m)/2 \leq F(G) < (1 + m)/2$, hence $1 - F(G) > (1 - m)/2$, and $\rho \geq (1 - m)/2$.

Tightness. For the uniform law on $2r + 1$ symbols, $m = 1/(2r + 1)$, and the best split has r against $r + 1$ symbols, so $\rho = r/(2r + 1)$. Then $1 - m = 2r/(2r + 1) = 2\rho$. □

Over random laws the ratio $(1 - m)/\rho$ reaches 1.974 (computed; `research/theory/T1-code/rho.py`).

D.11 Proof of Corollary 5.18

Suppose $\beta_i \mathbb{P}[\mathrm{root}(\Theta x) \neq f] \leq \alpha$ for some x, f. Let R'' be R^* with its tag-i component replaced by $\mathrm{inst}(\sigma_i[x \mapsto f(\bar{z})])$, which lies in H_k^{tag}. By (F1), $R'' \subsetneq R^*$. Q is a law on R^*, and the Q-mass of $R^* \setminus R''$ is the probability that a datum is a tag-i instance with $\mathrm{root}(\Theta x) \neq f$, namely $\beta_i \mathbb{P}[\mathrm{root}(\Theta x) \neq f] \leq \alpha$. Apply Theorem 5.16 to $R'' \subsetneq R^*$. Every $q \in R^* \setminus R'' \neq \emptyset$ has $\mathbb{P}_Q[\exists t : \ q \in \mathrm{Acc}_t] \leq \delta$. So $\mathbb{P}_Q[R^* \subseteq \mathrm{Acc}_t] \leq \delta$ for all t, which contradicts the hypothesis. The pair case uses $\sigma_i[y \mapsto x]$ in the same way. All of the alternative's "noise" sits on tag i, so robust soundness is only needed for laws whose noise on tag i is at most α; this is the per-tag reading. □

The factor β_i in the displayed conditions is needed because α is a fraction of all data, while the probabilities are under Λ_i.

D.12 Dimension inference

This subsection states the remark summarized in Section 5.7 and sketches its proof.

Remark D.2 (Dimension inference). [L9 §3.1, TC1] (proof sketch; not adversarially verified) The dimension of a physical quantity is its inferential role: which sums and equations it may enter. Here learning meaning from positive examples is solvable in closed form. Let $q_1, \ldots, q_n$ be quantity symbols, and let observed equations be polynomial (or Laurent) equations among them; each monomial has an exponent vector in $\mathbb{Z}^n$. A *grading* is a homomorphism $g : \mathbb{Z}^n \to A$ into an abelian group ($\deg q_i = g(e_i)$), and an equation is *g-homogeneous* if all its monomials have the same degree. For observed equations E, let $K(E) \leq \mathbb{Z}^n$ be generated by the differences of exponent vectors of monomials in a common equation. (i) The gradings making E homogeneous are the homomorphisms vanishing on $K(E)$, so $\mathbb{Z}^n \to \mathbb{Z}^n/K(E)$ is the *finest* one: every consistent grading factors through it. It is computed by Smith normal form (modulo torsion for rational exponents). The cautious verifier for "dimensionally admissible" accepts exactly the equations whose differences lie in $K(E)$ (in its saturation, if gradings must be torsion-free). (ii) The admissible sets form an intersection-closed class indexed by subgroups of $\mathbb{Z}^n$. Since these are finitely generated, every target has a finite anchor, and Theorem 5.3 gives exact identification once the data exhibit generators of the target's relations. This is an instance of learning algebraic closure systems from text, where Noetherianity is finite generation (Stephan and Ventsov, 2001). (iii) Gold's limitation appears in miniature: positive data never refute a coarser grading ("everything dimensionless" is always consistent), but the cautious learner never adopts one. The collapse of Proposition 5.12 appears too: one wrong equation adds a spurious generator and merges dimensions. (iv) The residue is conventional. The finest grading is fixed only up to isomorphism, i.e. up to the choice of base dimensions, which no inference can see. Whether a constant such as c is dimensional is a choice between data sets: natural units add $c = 1$ and coarsen the grading, and Gaussian units absorb a dimension that SI keeps. These are coherent, empirically equivalent alternative meanings, settled by convention (Section 13). Dimension types implement the type-checking side (Kennedy, 1994); Buckingham's Π theorem is the invariant-theoretic face (Buckingham, 1914).

Sketch of (i)–(ii). The claims are elementary; we record only the argument.

(i) An equation whose monomials have exponent vectors $a_1, \ldots, a_r$ is g-homogeneous iff $g(a_j - a_1) = 0$ for all j. So all of E is homogeneous iff g vanishes on the generators of $K(E)$, iff it vanishes on $K(E)$. The projection $\pi : \mathbb{Z}^n \to \mathbb{Z}^n/K(E)$ is such a grading. By the homomorphism

theorem every such g factors uniquely as $\bar{g} \circ \pi$. An equation is homogeneous under every consistent grading iff its differences lie in $\bigcap_g \ker g = \ker \pi = K(E)$.

Smith normal form. If M is an integer matrix whose rows generate $K(E)$ and $UMV = \mathrm{diag}(d_1, \ldots, d_r, 0, \ldots)$ is its Smith normal form, with U, V unimodular, then $\mathbb{Z}^n/K(E) \cong \bigoplus_j \mathbb{Z}/d_j\mathbb{Z} \oplus \mathbb{Z}^{n-r}$. If the target group is torsion-free, $g(cv) = 0$ with $c \neq 0$ forces $g(v) = 0$. So g vanishes on the saturation $\{v : \ cv \in K(E) \text{ for some } c \neq 0\}$, and the finest torsion-free grading is $(\mathbb{Z}^n/K(E))/\text{torsion}$.

(ii) Let $\mathrm{Adm}(L)$ be the set of equations whose differences lie in $L \leq \mathbb{Z}^n$. Then $\mathrm{Adm}(L) \cap \mathrm{Adm}(L') = \mathrm{Adm}(L \cap L')$, and Adm determines L: the equation $1 = q^v$ (Laurent monomials) is in $\mathrm{Adm}(L)$ iff $v \in L$. The version space of E is $\{\mathrm{Adm}(L) : K(E) \leq L\}$, so $\hat{R}(E) = \mathrm{Adm}(K(E))$. For the target $L^* = \ker g^*$, choose generators $v_1, \ldots, v_s$ (subgroups of $\mathbb{Z}^n$ are finitely generated). The equations $1 = q^{v_j}$ form an anchor. More generally, a finite $E \subseteq \mathrm{Adm}(L^*)$ is an anchor iff $K(E) = L^*$: if $K(E) \subsetneq L^*$, then $\mathrm{Adm}(K(E))$ is consistent with E and misses $\mathrm{Adm}(L^*)$. Theorem 5.3 then gives the stated identification. Claims (iii)–(iv) of the remark are direct readings of these facts.

D.13 Experiment A: the positive-data numbers

(computed) These are the numbers behind the summary in Section 5.8, taken from `code/results/algebra_learning.md`. Experiment A (Section 12.1) learns school algebra over partial terms (Example 2.12) from simulated human derivations. The target has 41 rewrite schemas, 11 of them guarded. The corpus has sporadic noise (5% random corruptions), five systematic fallacies (each committed with probability 0.3 per opportunity), or both. The learner is a heuristic relative of the cautious verifier (MDL-clustered anti-unification within rule-name buckets, support threshold two); by default it infers no guards, and a variant takes the most specific guard true on all instances (Lemma 2.8). Means are over three seeds; soundness is measured, not proved. Positive data alone gave the following.

- *Shapes are recovered where Lemma 2.7 predicts.* A rule's shape was recovered in 0% of (rule, run) pairs with one human instance, 67% with two, 92% with three or four and 100% with five or more; on clean data, all 41 shapes for $N \geq 200$ derivations.
- *Generalization is systematic.* On clean data at $N = 500$, valid out-of-distribution steps (terms of about 40 nodes, against about 10 in human steps) are accepted as often as in-distribution ones (1.00 and 1.00). A gradient-boosted step classifier trained on the noisy corpus *with* validity labels accepts 89% of valid in-distribution steps, 73% of valid out-of-distribution ones, and about half of the invalid ones.
- *Guards are not displayed.* On clean data at $N = 500$, the default learner ends with 11.7 ± 0.5 unsound active schemas (essentially the 11 guarded rules without guards) and accepts 34% of invalid out-of-distribution steps. Most-specific guards reduce this to 0.3 ± 0.5 unsound schemas (those whose guard the guard language cannot express), but with fallacies present the count rises to 5.3 ± 0.5, with 22% false accepts.
- *Fallacies reach support.* At $N = 500$ each of the five fallacies is learned as an active schema in 6/6 runs, with and without guard inference (Corollary 5.19).
- *A fixed support threshold is no soundness guarantee.* Under sporadic noise, spurious unsound schemas grow with N, reaching 102 ± 11 at $N = 500$ with threshold one and 18.5 ± 3.6 for

the untagged learner with threshold two (3.0 on clean data); hence the linear budgets of Theorem 5.14.

- With noise and fallacies together at $N = 500$, the cited-rule learner recovers all 41 shapes, yet 15.7 ± 0.5 active schemas are unsound, 51% of invalid out-of-distribution steps are accepted, and a white-box prover finds 65 ± 2 accepted invalid steps.

Experiment B shows the same pattern for propositional natural deduction (Section 12.2).

D.14 Repairs made during verification

The statements of Section 5 are the ones repaired after adversarial verification (record in `research/verification/T1-soundness-under-search-verification.md`). The repairs were these. The converse in Proposition 5.4(b) was restricted to nonempty tell-tales. Proposition 5.9 replaces an earlier claim that at most k roots at a metavariable make identification fail for good, which is false for $k' \geq 2$. Theorem 5.8(b) uses the ε-net theorem in its "measure $\geq \varepsilon$" form. The ground-rule conventions $\rho_i = c_i = 1$ of Theorems 5.6 and 5.14 were made explicit, with the missing ground-rule event added to the proof of Theorem 5.14. Theorem 5.13(b) gained the clause "more than e_i valid samples" and per-tag budgets. In Lemma 5.17 the constant 3 was sharpened to the tight 2, and Corollary 5.18 gained the factor β_i. Finally, the tagged case rests on the factorization (F3) rather than on intersection-closure, which H_k^{tag} lacks.

E Proofs for Section 6

Lemma 6.2, Proposition 6.3(a),(c), Proposition 6.10, Corollary 6.11, Lemma 6.18, Theorem 6.20, Proposition 6.28, Theorem 6.31, Proposition 6.33 (immediate) and Corollary 6.32 are proved in the main text, and Theorem 6.19 is cited. This appendix proves the remaining results. It also contains, with their proofs, Proposition E.1, the credence results Theorem E.2 and Proposition E.3, Example E.4, Remark E.5 and Lemma E.6, which the main text only summarizes. The proofs follow the sources named in the src tags, in the form repaired after adversarial verification; where a statement or proof was changed by the repair, a repair note says so. Notation is as in Section 6. For formulas, $\equiv_2$ is classical equivalence.

E.1 Proofs of Theorems 6.5, 6.7 and 6.9

Theorem 6.5. (i) (P)-data lie in R^*, and (N)-deletions spare R^* by Lemma 6.2(a). (ii) In a detection every $R \in S_t$ contains $\hat{R}_t \supseteq \mathrm{Steps}(\pi)$, so all of S_t is deleted and $w(\mathrm{VS}_{t+1}) \leq \frac{1}{2} w(\mathrm{VS}_t)$; (P)-rounds never add weight. Since R^* is never deleted, $w(R^*) \leq 2^{-D}$ after D detections. (iii), (iv) $\hat{R}_t \subseteq R$ for every $R \in S_t$, and Cl is monotone in the step set. □

Theorem 6.7. A detection has $\mathrm{Steps}(\pi) \subseteq \hat{R}_t = \bigcap S_t$, so it penalizes a superset of S_t, whose weight is at least $\frac{1}{2} W_t$; hence $W_{t+1} \leq W_t - (1-\beta)\frac{1}{2} W_t = \frac{1+\beta}{2} W_t$. (P)-rounds never raise W_t and never delete R^*. R^* is penalized only in false alarms, so $\beta^m w_0(R^*) \leq w_T(R^*) \leq W_T \leq (\frac{1+\beta}{2})^D$, and taking logarithms gives the bound. This is the argument of Littlestone and Warmuth (1994). □

Theorem 6.9. (i) If $h \oplus D$ is $\mathcal{A}$-incoherent for a finite $D \subseteq \vdash^*$, so is $h \oplus \vdash^* \supseteq h \oplus D$. Conversely, $h \oplus \vdash^*$ is the directed union of the $h \oplus D$ over finite $D \subseteq \vdash^*$, and a finite derivation of $\bot$ from some $A \in \mathcal{A}$ uses finitely many sequents, which lie in one of them. (ii) If $\vdash^* \not\subseteq h$, h is text-refutable. Otherwise $h \oplus D = h$ for all $D \subseteq \vdash^*$, so h is refutable by coherence with data iff it is $\mathcal{A}$-incoherent. (iii) $h \supsetneq \vdash^*$ accepts a sequent outside $\vdash^*$, that is, an invalid one; data for $\vdash^*$ are data for h, since $\vdash^* \subseteq h$ and $\mathcal{A}$ is the same. □

E.2 Proof of Proposition 6.6

($\Leftarrow$) Let $\hat{R} \subseteq \bigcap S$ with $w(S) \geq \frac{1}{2} w(\mathrm{VS})$. A detection has $\mathrm{Steps}(\pi) = P \subseteq \hat{R}$, so every $R \in S$ contains P and is deleted; the deleted weight is at least $w(S)$.

($\Rightarrow$) The set of steps is countable, because Fm is. If $\hat{R} = \emptyset$, take $S = \mathrm{VS}$. Otherwise enumerate $\hat{R} = \{s_1, s_2, \dots\}$ (finite or infinite), and let $P_k = \{s_1, \dots, s_k\}$ and $S_k = \{R \in \mathrm{VS} : P_k \subseteq R\}$. If $P_k \not\subseteq \bigcap \mathrm{VS}$, then by the worst-case assumption P_k is a possible detection, and it deletes exactly S_k; so $w(S_k) \geq \frac{1}{2} w(\mathrm{VS})$. If $P_k \subseteq \bigcap \mathrm{VS}$, then $S_k = \mathrm{VS}$. In both cases $w(S_k) \geq \frac{1}{2} w(\mathrm{VS})$. The sets S_k decrease, and $\bigcap_k S_k = \{R \in \mathrm{VS} : \hat{R} \subseteq R\} =: S$. Since w is a finite measure on the countable set VS, continuity from above gives $w(S) = \lim_k w(S_k) \geq \frac{1}{2} w(\mathrm{VS}) > 0$, and $\hat{R} \subseteq \bigcap S$ by definition. (For finite VS one may instead take one step $s_R \in \hat{R} \setminus R$ for each R with $\hat{R} \not\subseteq R$; this is the version formalized in Lean.) □

Repair note. The first version assumed VS finite and allowed every finite $P \subseteq \hat{R}$ as a detection, including $P \subseteq \bigcap \mathrm{VS}$, which would delete the target and contradicts Lemma 6.2; the hypothesis is now restricted to legal detections and VS may be countable.

E.3 Proofs of Proposition 6.3(b) and Proposition 6.8

Proposition 6.3(b). Let $s_c := \bigvee_i (a_i \to \ell_i^{1-c_i})$ for $c \in \{0,1\}^k$. If $b_i \neq c_i$ for some i, then $\ell_i^{1-c_i} = \ell_i^{b_i}$, so the disjunct $a_i \to \ell_i^{b_i}$ of s_c is an axiom of h_b and $\rhd s_c \in h_b$. If $b = c$, the valuation with every a_i true and $p_i = c_i$ (so $\ell_i^{c_i}$ true and $\ell_i^{1-c_i}$ false) satisfies the axioms of h_c and falsifies s_c. So $\rhd s_c \in h_b$ iff $b \neq c$. Every h_b is consistent, so $\mathcal{A} = \{\emptyset\}$ is coherent for all of them. A certified-sound learner accepts only steps in $\bigcap \mathrm{VS}_t$, so it rejects $\rhd s_c$ while $h_c \in \mathrm{VS}_t$. For target h_{b^*}, present $\rhd s_c$ for each $c \neq b^*$ in any order: each is valid in the target, each deletes exactly h_c, and at the moment it is presented h_c has not yet been deleted, so each is an incompleteness error. That makes $2^k - 1$ errors, although $\log_2 |\mathcal{H}| = k$. □

Proposition 6.8. *(a)* As just shown, $\rhd s_c \in h_b$ iff $b \neq c$. Fix a learner that in every round announces $\hat{R}_t \subseteq R$ for some $R \in \mathrm{VS}_t$; we may assume it is deterministic, and the environment sees $\hat{R}_t$ before it moves. While $|\mathrm{VS}_t| \geq 2$, the adversary picks $h_c \in \mathrm{VS}_t$ with $\hat{R}_t \subseteq h_c$ and presents $\rhd s_c$. Since $\rhd s_c \notin h_c \supseteq \hat{R}_t$, this is an incompleteness error (for any target other than h_c). Since $\rhd s_c \in h_b$ for every $b \neq c$, it deletes exactly h_c. After $2^k - 1$ rounds a single hypothesis h_{b^*} remains; take it as the target. Every presented datum $\rhd s_c$ has $c \neq b^*$, hence lies in h_{b^*}, so the run is a prefix of a text for h_{b^*}; extend it arbitrarily. The target must be chosen at the end: if $\mathrm{VS}_t = \{h_c, h_{b^*}\}$ and the learner's coalition is $\{h_{b^*}\}$, a target fixed in advance as h_{b^*} would not be charged the last error.

The union learner. If $R^* \in \mathrm{VS}_t$ then $R^* \subseteq \bigcup \mathrm{VS}_t = \hat{R}_t$, so no incompleteness error occurs; and R^* is never deleted, by Lemma 6.2. Let $h_\neg = \{\Gamma \rhd \varphi : \Gamma \cup \{\neg a_1, \dots, \neg a_k\} \models \varphi\}$. Since $\neg a_i \models a_i \to \ell$, every $h_b \subseteq h_\neg$, so $\hat{R}_t \subseteq h_\neg$ and $\mathrm{Cl}_{\hat{R}_t}(\emptyset) \subseteq \mathrm{Cn}_2(\{\neg a_1, \dots, \neg a_k\})$, which does not contain $\bot$. So no argument from $\emptyset$ to $\bot$ uses only accepted steps: there are no detections. For

target h_{b^*}, as long as VS_t contains some $h_b \neq h_{b^*}$ the union learner accepts $\rhd a_i \to \ell_i^{b_i}$ for an i with $b_i \neq b_i^*$, which is not in h_{b^*} (make a_i and $\ell_i^{b_i^*}$ true); so it is unsound, and it is not oligarchic, so Theorem 6.5 does not apply to it.

(b) Let $\mathcal{H} = \{h_1, h_2, h_3\}$ be the class of Theorem 3.12, each with weight $\frac{1}{3}$. Any step $s \in \bigcup \mathrm{VS} \setminus \hat{R}$ can be presented as a positive datum for a target containing it, and then it is an incompleteness error that deletes $\{h_i : s \notin h_i\}$. This weighs at least $\frac{1}{2}$ only if at least two h_i omit s, i.e. only if s lies in at most one h_i. So if every possible incompleteness error deletes at least half the weight, $\hat{R} \supseteq M := \{s : \ s \in h_i \text{ for at least two } i\}$. But the argument π of Theorem 3.12 has $\mathrm{Steps}(\pi) \subseteq M \subseteq \hat{R}$ and $\mathrm{Steps}(\pi) \not\subseteq h_i$ for each i. So it is a legal detection for every target, and it deletes nothing. (Under the worst-case hypothesis of Proposition 6.6, that every finite $P \subseteq \hat{R}$ with $P \not\subseteq \bigcap \mathrm{VS}$ is a possible detection, the same conclusion follows from that proposition: detection-halving needs $\hat{R} \subseteq h_i \cap h_j$ for some $i \neq j$, and $M \not\subseteq h_i \cap h_j$ for every pair: $h_1 \cap h_2$ lacks $\rhd q$, while $h_1 \cap h_3$ and $h_2 \cap h_3$ lack $\rhd p$.) □

Repair note. Both parts were added after verification. The first version concluded from Theorem 6.5 that "coherence bounds the price of boldness while nothing bounds the price of caution"; part (a) shows that on Proposition 6.3(b)'s own class every oligarchic learner pays the price of caution. Membership facts were checked by truth tables for $k \leq 4$.

E.4 Proof of Proposition 6.12

(a) For the target, $\Gamma \nvdash^* \varphi$ iff $\Gamma \cup \{\neg\varphi\}$ is target-coherent, by classical reductio. So the designations eventually enumerate (codes for) every finite-premise sequent not in $\vdash^*$, and the text enumerates $\vdash^*$: together they form an informant. Let (h_i) enumerate $\mathcal{H}$, and let the learner output the least i such that h_i contains every sequent seen in the text and is coherent on every designated context seen. Let i^* be the least index of the target and $j < i^*$, so $h_j \neq \vdash^*$. If $\vdash^* \not\subseteq h_j$, the text eventually shows a sequent of $\vdash^* \setminus h_j$, and j is rejected from then on. Otherwise pick $\Gamma \rhd \varphi \in h_j \setminus \vdash^*$. Then $\Gamma \cup \{\neg\varphi\}$ is target-coherent, hence eventually designated, and it is h_j-incoherent by reductio in h_j; so j is rejected from then on. The target's index i^* is never rejected. As there are finitely many $j < i^*$, the output is eventually i^* forever. If membership in the h_i is uniformly decidable, both tests are decidable (coherence of h_i on A is non-membership of $A \rhd \bot$), so the learner is computable.

(b) $\{p_j\}$ is h_n-coherent iff $\{\neg p_i : i < n\} \cup \{p_j\}$ is consistent iff $j \geq n$, and it is h_ω-incoherent for every j. The chain is strictly increasing with union h_ω, and all members are $\emptyset$-coherent. Consider the computable learner that outputs h_j for the least j such that $\{p_j\}$ has been designated, and h_ω if there is none (the enumeration learner of (a) also works). On target h_n, exactly the singletons $\{p_j\}$ with $j \geq n$ can be designated, and $\{p_n\}$ eventually is; from then on the least designated j is n, and the learner outputs h_n. On target h_ω no singleton $\{p_j\}$ is ever designated, and the learner outputs h_ω throughout. □

Repair note. This proposition was added after verification; the referee's counterexample (b) showed that the first version of Corollary 6.11, stated without "$\mathcal{A}$ fixed", was false.

E.5 Proofs for Section 6.2: Post-completeness

Theorem 6.13, main case. Let $\varphi \in C(X) \setminus \mathrm{Cn}_2(X)$, with $v[X] = 1$, $v(\varphi) = 0$. Fix a tautology $\top_0$ in one atom and $\bot_0 = \neg\top_0$, and let $\sigma_v(p) = \top_0$ or $\bot_0$ as $v(p) = 1$ or 0. Every Boolean u has $u(\sigma_v\psi) = v(\psi)$, so $\sigma_v X \subseteq \mathrm{Taut} \subseteq C(\emptyset)$ and $\sigma_v\varphi$ is a contradiction. By structurality

$\sigma_v\varphi \in C(\sigma_v X) \subseteq C(C(\emptyset)) = C(\emptyset)$, hence $\mathrm{Fm} = \mathrm{Cn}_2(\{\sigma_v\varphi\}) \subseteq C(\emptyset)$, and $C = C_{\mathrm{Fm}}$. This is the substitution of the proof of Proposition 3.5, at the level of consequence operators. □

Theorem 6.13, pure $\{\to\}$. Let $\varphi \in C(X) \setminus \mathrm{Cn}_2(X)$ and v a Boolean valuation with $v[X] = 1$, $v(\varphi) = 0$. Put $\sigma_v(p) = p_0 \to p_0$ if $v(p) = 1$ and $\sigma_v(p) = p_0$ otherwise. By induction on ψ, $\sigma_v\psi \equiv_2 p_0 \to p_0$ if $v(\psi) = 1$ and $\sigma_v\psi \equiv_2 p_0$ if $v(\psi) = 0$. For atoms this is the definition. For $\psi = \alpha \to \beta$: if $v(\alpha) = 0$, then $\sigma_v\psi \equiv_2 p_0 \to (\cdot)$, and both $p_0 \to (p_0 \to p_0)$ and $p_0 \to p_0$ are tautologies; if $v(\alpha) = v(\beta) = 1$, then $\sigma_v\psi \equiv_2 \top \to \top \equiv_2 \top$; if $v(\alpha) = 1$, $v(\beta) = 0$, then $\sigma_v\psi \equiv_2 \top \to p_0 \equiv_2 p_0$. Hence $\sigma_v X \subseteq \mathrm{Taut} \subseteq C(\emptyset)$ and, by structurality, $\sigma_v\varphi \in C(\sigma_v X) \subseteq C(\emptyset)$, with $\sigma_v\varphi \equiv_2 p_0$. So $p_0 \in \mathrm{Cn}_2(\{\sigma_v\varphi\}) \subseteq C(C(\emptyset)) = C(\emptyset)$. Substituting any ψ for p_0 and using $\sigma C(\emptyset) \subseteq C(\emptyset)$ gives $C(\emptyset) = \mathrm{Fm}$. □

Proposition 6.14. C_{ai} is a closure operator, and it is structural because σX is non-empty iff X is. In the $\{\wedge, \vee\}$ language $\mathrm{Cn}_2(\emptyset) = \emptyset$, since the all-false valuation falsifies every formula; so $C_{\mathrm{ai}} \supseteq \mathrm{Cn}_2$. Now let $C \supsetneq \mathrm{Cn}_2$ be structural. *Case* $C(\emptyset) \neq \emptyset$. Pick $\varphi \in C(\emptyset)$ and substitute p_0 for every atom. Every one-variable lattice term is $\equiv_2 p_0$, so $p_0 \in \mathrm{Cn}_2(\{\sigma\varphi\}) \subseteq C(C(\emptyset)) = C(\emptyset)$, and by structurality $C(\emptyset) = \mathrm{Fm}$: $C = C_{\mathrm{Fm}}$. *Case* $C(\emptyset) = \emptyset$. Take $\varphi \in C(X) \setminus \mathrm{Cn}_2(X)$ (so $X \neq \emptyset$) and a Boolean v with $v[X] = 1$, $v(\varphi) = 0$. Let σ send the atoms true under v to q and the false ones to r (distinct atoms). Each $\sigma\psi$ is a lattice term in q, r, and its value at $(q, r) = (1, 0)$ is $v(\psi)$. Up to $\equiv_2$, the lattice terms in q, r are q, r, $q \wedge r$ and $q \vee r$. So $\sigma\psi \in \{q,\ q \vee r\}$ up to $\equiv_2$ for $\psi \in X$, and $\sigma\varphi \in \{r,\ q \wedge r\}$. Hence $\sigma X \subseteq \mathrm{Cn}_2(\{q\})$ and $r \in \mathrm{Cn}_2(\{\sigma\varphi\})$, so

$$r \in C(\{\sigma\varphi\}) \subseteq C(C(\sigma X)) = C(\sigma X) \subseteq C(\mathrm{Cn}_2(\{q\})) \subseteq C(C(\{q\})) = C(\{q\}).$$

Substituting χ for q and ψ for r gives $\psi \in C(\{\chi\})$ for all χ, ψ, so $C(Y) = \mathrm{Fm}$ for every non-empty Y: $C = C_{\mathrm{ai}}$. □

Theorem 6.15. *(a)* Suppose $\langle D\rangle = \mathrm{Cn}_2$. Every $h \in \mathcal{S}(D, A_0)$ is structural and contains D, hence contains $\langle D\rangle = \mathrm{Cn}_2$; by Theorem 6.13, $h \in \{\mathrm{Cn}_2, C_{\mathrm{Fm}}\}$, and A_0 excludes C_{Fm}. Conversely, suppose $\langle D\rangle \subsetneq \mathrm{Cn}_2$ (note $\langle D\rangle \subseteq \mathrm{Cn}_2$ because Cn_2 is structural and contains D). Then $\langle D\rangle$ is structural, contains D, and is A_0-coherent because Cn_2 is; so $\langle D\rangle$ and Cn_2 are distinct members of $\mathcal{S}(D, A_0)$.

Explicit finite bases. For a language containing $\neg$ and $\to$ (possibly also $\wedge, \vee$), with distinct atoms p, q, r, take the axioms $\rhd p \to (q \to p)$, $\rhd(p \to (q \to r)) \to ((p \to q) \to (p \to r))$, $\rhd(\neg p \to \neg q) \to (q \to p)$ and modus ponens $p,\ p \to q \rhd q$; if $\wedge, \vee$ are present, add their clauses *as $\to$-axioms*: $\rhd p \wedge q \to p$, $\rhd p \wedge q \to q$, $\rhd p \to (q \to p \wedge q)$, $\rhd p \to p \vee q$, $\rhd q \to p \vee q$, $\rhd(p \to r) \to ((q \to r) \to (p \vee q \to r))$. If $\bot$ or $\top$ is a primitive constant, add the axioms $\rhd\neg\bot$, resp. $\rhd\top$; with the third axiom and the first these give $\rhd\bot \to p$ (from $\neg\bot$ and $\neg\bot \to (\neg p \to \neg\bot)$ infer $\neg p \to \neg\bot$, then $\bot \to p$). Without these axioms the constants behave like atoms that substitution cannot reach, and $\rhd\neg\bot$ is underivable. For pure $\{\to\}$ take the first two axioms, Peirce's law $\rhd((p \to q) \to p) \to p$ and modus ponens. For languages without $\to$, such as $\{\neg, \wedge\}$ and $\{\neg, \vee\}$, finite bases exist because every two-element matrix logic is finitely based (Rautenberg, 1981).

The least structural consequence relation generated by a set of sequents is given by derivations from substitution instances of those sequents. With the axioms and modus ponens above, these are derivations in a Hilbert system with modus ponens as its only rule: Łukasiewicz's axioms for $\{\neg, \to\}$, extended by the usual axioms for $\wedge$ and $\vee$, and for $\{\to\}$ the axioms K, S and Peirce's

law, a variant of the Tarski–Bernays axiomatization (K and S give intuitionistic implication, and Peirce's law makes it classical). These systems are complete for tautologies (Kleene, 1952; Mendelson, 1964); for the $\wedge, \vee$ axioms Kalmár's lemma goes through because the system proves $p, q \vdash p \wedge q$, $\neg p \vdash \neg(p \wedge q)$, $\neg q \vdash \neg(p \wedge q)$, $p \vdash p \vee q$, $q \vdash p \vee q$ and $\neg p, \neg q \vdash \neg(p \vee q)$, and $\vdash \neg\bot$, $\vdash \top$ (base cases for the constants). Since modus ponens is the only rule and the first two axioms are present, the deduction theorem holds. So if $X \models \varphi$, compactness gives a finite $\{x_1, \dots, x_n\} \subseteq X$ with $\models x_1 \to (\cdots \to (x_n \to \varphi))$, which is provable, and n applications of modus ponens derive φ from X. Hence $\langle D \rangle = \mathrm{Cn}_2$.

(b) Let $j < i^*$, so $h_j \neq \mathrm{Cn}_2$. If $h_j \not\supseteq \mathrm{Cn}_2$, some sequent of $\mathrm{Cn}_2 \setminus h_j$ eventually appears in the text, and j is ineligible from then on. If $h_j \supseteq \mathrm{Cn}_2$, then $h_j = C_{\mathrm{Fm}}$ by Theorem 6.13, and $A_0 \vdash_{h_j} \bot$ makes j ineligible from the start. i^* is always eligible. So the learner's output is non-decreasing, never exceeds i^*, and is eventually i^*; in particular it changes its mind at most i^* times. Uniform decidability makes the eligibility tests computable; no property of the ordering was used.

(c) $\mathrm{Cn}_2 \oplus \{\rhd p\}$ with p not in A_0: its consequences of A_0 are $\mathrm{Cn}_2(A_0 \cup \{p\})$, which is consistent, and it contains every sequent of Cn_2. C_{Fm} contains every sequent. C_{ai} contains every sequent of Cn_2 in the $\{\wedge, \vee\}$ fragment and is non-trivial on $\emptyset$. □

The sequent-rule basis fails (countermodel). The form of the basis matters: if the $\wedge$ clauses are sequent rules ($p \wedge q \rhd p$, …) while modus ponens is the only rule for $\to$, then $\rhd p \to (q \to p \wedge q)$ is underivable; giving them as $\to$-axioms, as above, repairs this. Take the language $\{\neg, \to, \wedge\}$ and let D' consist of the three $\{\neg, \to\}$ axioms, modus ponens, and the sequent rules $p \wedge q \rhd p$, $p \wedge q \rhd q$, $p, q \rhd p \wedge q$. We show $\rhd p \to (q \to p \wedge q) \notin \langle D' \rangle$; this countermodel is due to a referee and was re-checked by script. For every assignment s of truth values to atoms define v_s on all formulas, simultaneously for all s, by recursion on complexity: $v_s(p) = s(p)$; v_s treats $\neg$ and $\to$ classically; and $v_s(\alpha \wedge \beta) = 1$ iff $\alpha \in \Theta^*$ and $\beta \in \Theta^*$, where $\Theta^* = \{\varphi : \ v_s(\varphi) = 1 \text{ for all } s\}$. This is well defined, since membership of α, β in Θ^* depends only on values at lower complexity. Every substitution instance of a $\{\neg, \to\}$ tautology lies in Θ^*, because each v_s evaluates $\neg, \to$ classically and treats the substituted subformulas as atoms. Θ^* is closed under modus ponens (if $v_s(\alpha) = v_s(\alpha \to \beta) = 1$ for all s, then $v_s(\beta) = 1$ for all s), under $\wedge$-introduction (if $\alpha, \beta \in \Theta^*$ then $v_s(\alpha \wedge \beta) = 1$ for all s) and under $\wedge$-elimination (if $\alpha \wedge \beta \in \Theta^*$ then $\alpha, \beta \in \Theta^*$ by definition). So $\langle D' \rangle(\emptyset) \subseteq \Theta^*$. But $p \notin \Theta^*$, so $v_s(p \wedge q) = 0$ for every s; at $s(p) = s(q) = 1$ this gives $v_s(p \to (q \to p \wedge q)) = 0$. Hence $p \to (q \to p \wedge q) \notin \Theta^*$, and $\langle D' \rangle \neq \mathrm{Cn}_2$. □

Renaming is not enough. [L5 Prop 2.5] Let $C(X) = \mathrm{Cn}_2(X)$ if $\mathrm{Cn}_2(X)$ contains no atom, and $C(X) = \mathrm{Cn}_2(X \cup \mathrm{At})$ otherwise, where At is the set of atoms. C is extensive; it is monotone because $\mathrm{Cn}_2(X) \subseteq \mathrm{Cn}_2(Y)$ for $X \subseteq Y$; and it is idempotent, since in the first case $\mathrm{Cn}_2(\mathrm{Cn}_2(X))$ has no atom and in the second $C(X)$ already contains every atom. For a renaming ρ (atoms to atoms) we have $\rho C(X) \subseteq C(\rho X)$: in the first case because Cn_2 is structural, and in the second because $\mathrm{Cn}_2(\rho X)$ contains ρp for an atom $p \in \mathrm{Cn}_2(X)$, so $C(\rho X) = \mathrm{Cn}_2(\rho X \cup \mathrm{At}) \supseteq \rho \mathrm{Cn}_2(X \cup \mathrm{At})$. C contains $p \rhd q$, and $C(\emptyset) = \mathrm{Taut}$; so C is $\emptyset$-coherent, has the classical theorems, and differs from Cn_2. It is the least renaming-invariant closure operator above Cn_2 containing $p \rhd q$: any such operator puts every atom into $C'(X)$ as soon as $\mathrm{Cn}_2(X)$ contains one. So a learner that generalizes only by renaming atoms can settle on C. □

Theorems versus consequence. The main text summarizes the following proposition.

Proposition E.1 (Theorems versus consequence)**.** *[T2 Props 3.4, 3.5, as repaired] For a substitution-closed theorem set* Th *let* $C_{\mathrm{adm}}(X) = \{\varphi : \forall\sigma(\sigma X \subseteq \mathrm{Th} \Rightarrow \sigma\varphi \in \mathrm{Th})\}$. (a) C_{adm} *is the largest structural closure operator with theorems* Th*; on finite premise sets it is the structural completion (admissible rules).* (b) *For* $\mathrm{Th} = \mathrm{Taut}$ *in a language as in Theorem 6.13,* $C_{\mathrm{adm}} = \mathrm{Cn}_2$ *(structural completeness); in the* $\{\wedge, \vee\}$ *fragment* $C_{\mathrm{adm}} = C_{\mathrm{ai}}$. *Lean: Post.le_Cn2_of_empty_eq_Taut (special case of (b))* (c) *For* IPC, $C_{\mathrm{adm}} \supsetneq \vdash_{\mathsf{IPC}}$ *(Harrop's rule* $\neg p \to q \vee r \,/\, (\neg p \to q) \vee (\neg p \to r)$ *is admissible, not derivable; Harrop 1960; Rybakov 1997; Iemhoff 2001), and the two have the same theorems and the same coherence profile,* $A \vdash^{\mathsf{IPC}}_{\mathrm{adm}} \bot$ *iff* $A \vdash_{\mathsf{CPC}} \bot$.

So a learner can learn the theorems and then take the structural completion; a learner that is not bold here may settle on $X \mapsto X \cup \mathrm{Taut}$, which has the right theorems but cannot chain inferences from premises. And no learner fed theorems and coherence data can tell IPC-derivability from IPC-admissibility.

Proposition E.1(a). *Extensive and monotone:* clear. *Idempotent:* if $\sigma X \subseteq \mathrm{Th}$, then $\sigma C_{\mathrm{adm}}(X) \subseteq \mathrm{Th}$ by definition, so every $\varphi \in C_{\mathrm{adm}}(C_{\mathrm{adm}}(X))$ has $\sigma\varphi \in \mathrm{Th}$. *Structural:* if $\varphi \in C_{\mathrm{adm}}(X)$ and $\sigma(\tau X) \subseteq \mathrm{Th}$, then applying the definition to $\sigma \circ \tau$ gives $\sigma(\tau\varphi) \in \mathrm{Th}$; so $\tau\varphi \in C_{\mathrm{adm}}(\tau X)$. $C_{\mathrm{adm}}(\emptyset) = \mathrm{Th}$ because Th is substitution-closed. *Maximal:* if C is structural with $C(\emptyset) = \mathrm{Th}$, $\varphi \in C(X)$ and $\sigma X \subseteq \mathrm{Th}$, then $\sigma\varphi \in C(\sigma X) \subseteq C(C(\emptyset)) = \mathrm{Th}$. For finite X, the defining condition is admissibility of X/φ. □

Proposition E.1(b),(c). *(b)* Cn_2 is structural with theorems Taut, so $\mathrm{Cn}_2 \subseteq C_{\mathrm{adm}}$ by (a). If $\varphi \notin \mathrm{Cn}_2(X)$, the substitution σ_v of the proof of Theorem 6.13 maps X into Taut and φ to a contradiction (to a formula $\equiv_2 p_0$ in the pure $\{\to\}$ case), which is not in Taut; so $\varphi \notin C_{\mathrm{adm}}(X)$. In the $\{\wedge, \vee\}$ fragment $\mathrm{Th} = \emptyset$, so the condition "$\sigma X \subseteq \mathrm{Th}$" never holds for non-empty X: $C_{\mathrm{adm}}(X) = \mathrm{Fm}$ for $X \neq \emptyset$ and $C_{\mathrm{adm}}(\emptyset) = \emptyset$, i.e. $C_{\mathrm{adm}} = C_{\mathrm{ai}}$.

(c) The admissibility and non-derivability of Harrop's rule are cited (Harrop, 1960; Rybakov, 1997; Iemhoff, 2001). The two relations have the same theorems by (a). For the coherence profile, first note that IPC decides every variable-free formula according to its classical value: by induction, $\vdash_{\mathsf{IPC}} \neg\bot$; and if α, β are each provable or refutable according to their classical values, so are $\alpha \wedge \beta$, $\alpha \vee \beta$, $\alpha \to \beta$ and $\neg\alpha$. Now if A is classically consistent, pick a Boolean $v \models A$ and substitute $\top := \bot \to \bot$ or $\bot$ for each atom according to v. Then $\sigma_v A$ consists of classically true variable-free formulas, all IPC-theorems, while $\sigma_v \bot = \bot$ is not one; so the rule $A/\bot$ is not admissible, i.e. $A \nvdash^{\mathsf{IPC}}_{\mathrm{adm}} \bot$. If A is classically inconsistent, then $A \vdash_{\mathsf{IPC}} \bot$ by Glivenko's theorem (Theorem 6.16(a)), and a fortiori $A \vdash^{\mathsf{IPC}}_{\mathrm{adm}} \bot$. □

E.6 Proof of Theorem 6.16

(a) Glivenko's theorem (Glivenko, 1929) gives $\vdash_{\mathsf{CPC}} \neg\psi$ iff $\vdash_{\mathsf{IPC}} \neg\psi$. With $\psi = \bigwedge A$ and the deduction theorem (in both logics), $A \vdash_{\mathsf{CPC}} \bot$ iff $\vdash_{\mathsf{CPC}} \neg\bigwedge A$ iff $\vdash_{\mathsf{IPC}} \neg\bigwedge A$ iff $A \vdash_{\mathsf{IPC}} \bot$ (for $A = \emptyset$ neither side holds). An intermediate L is sandwiched between IPC and CPC.

(b) CPC is a structural proper extension of IPC, coherent on every classically consistent context; by (a) it has the same coherence data. So IPC is never a maximal coherent structural hypothesis.

(c) We use the following facts about Jankov formulas (Jankov, 1968); see Chagrov and Zakharyaschev (1997, ch. 9). There is an infinite family $B_1, B_2, \ldots$ of finite subdirectly irreducible Heyting algebras with $B_i \notin \mathbf{SH}(B_j)$ for $i \neq j$, and each B_i has a formula $\chi(B_i)$ such that a

Heyting algebra B refutes $\chi(B_i)$ iff $B_i \in \mathbf{SH}(B)$. At most one B_i is the two-element algebra $\mathbf{2}$ (two copies would violate the antichain condition); drop it, so $|B_i| > 2$ for all i. Let L_k be the least structural consequence relation containing $\vdash_{\mathsf{IPC}}$ and the axioms $\rhd\chi(B_i)$ for $i \leq k$, and $L_\omega = \bigcup_k L_k$.

- *Strictly increasing.* The matrix consequence $\models_{B_{k+1}}$ of $(B_{k+1}, \{1\})$ is structural and contains $\vdash_{\mathsf{IPC}}$, by soundness of IPC for Heyting algebras. B_{k+1} validates $\chi(B_i)$ for $i \leq k$, since $B_i \notin \mathbf{SH}(B_{k+1})$. So $L_k \subseteq \models_{B_{k+1}}$, whereas $B_{k+1} \in \mathbf{SH}(B_{k+1})$ refutes $\chi(B_{k+1}) \in L_{k+1}(\emptyset)$.
- *Intermediate.* $\mathbf{SH}(\mathbf{2})$ consists of $\mathbf{2}$ and the trivial algebra, and B_i is neither, so $\mathbf{2}$ validates every $\chi(B_i)$. Hence every $L_k \subseteq \models_{\mathbf{2}} = \mathsf{CPC}$. By Proposition 6.10, L_ω is a consequence relation; it is structural as a union of structural relations, and $\mathsf{IPC} \subseteq L_\omega \subseteq \mathsf{CPC}$.
- *Impossibility.* By (a), for every $k \leq \omega$ the set of L_k-coherent finite contexts is the set of classically consistent ones. So whether $\mathcal{A}$ is fixed, or designations are target-dependent data, the environment can present one and the same designation stream δ for every target in $\{L_k : k \leq \omega\}$. Folding δ into the learner turns any learner for this class into a text learner for the chain $L_1 \subsetneq L_2 \subsetneq \cdots$ together with its union, which is impossible by the locking-sequence argument of Corollary 6.11.

(d) The target is $\langle G \rangle$ for a finite set G of sequents. Since D_n is contained in the structural target, so is $\langle D_n \rangle$. Once the text has shown every sequent of G, $\langle D_n \rangle \supseteq \langle G \rangle$. From then on every conjecture is correct (its syntactic index, e.g. D_n, may keep changing), which is BC-identification. □

E.7 Proof of Proposition 6.17

(a) If a consistent deductively closed $T' \supsetneq T$ contains $\varphi \notin T$, then $\neg\varphi \in T \subseteq T'$ by completeness of T, so T' is inconsistent. For learning, run the learner of Theorem 6.15(b): for $j < i^*$, either $h_j \not\supseteq T$ and the text refutes it, or $h_j \supsetneq T$ is inconsistent and the datum $\emptyset$ rejects it at once.

(b) Let h be parameter-structural, $\exists$-closed, $h \supseteq \vdash_T$ and $\emptyset \nvdash_h \bot$. Suppose $\Gamma \vdash_h \varphi$ but $T \cup \Gamma \nvdash \varphi$, with parameters $\bar{c} = c_1, \ldots, c_n$. Since the parameters do not occur in T, $T \nvdash \forall\bar{x}(\bigwedge\Gamma(\bar{x}) \to \varphi(\bar{x}))$, so by completeness $T \vdash \exists\bar{x}\,\theta(\bar{x})$ for $\theta(\bar{x}) := \bigwedge\Gamma(\bar{x}) \wedge \neg\varphi(\bar{x})$. Take fresh parameters $\bar{d}$. By parameter-structurality $\Gamma(\bar{d}) \vdash_h \varphi(\bar{d})$. Since $h \supseteq \vdash_T$ is closed under cut and $\theta(\bar{d}) \vdash_T \gamma$ for each $\gamma \in \Gamma(\bar{d})$, we get $\theta(\bar{d}) \vdash_h \varphi(\bar{d})$; also $\theta(\bar{d}) \vdash_T \neg\varphi(\bar{d})$ and $\varphi(\bar{d}), \neg\varphi(\bar{d}) \vdash_T \bot$, so $\theta(\bar{d}) \vdash_h \bot$. For $0 \leq i \leq n$ let $\theta_i := \exists x_{i+1} \cdots \exists x_n\, \theta(d_1, \ldots, d_i, x_{i+1}, \ldots, x_n)$, so $\theta_n = \theta(\bar{d})$ and $\theta_{i-1} = \exists x_i\, \psi(x_i)$ with $\psi(d_i) = \theta_i$. If $\theta_i \vdash_h \bot$, then $\exists$-elimination applied to $\theta_{i-1} \vdash_h \theta_{i-1}$ and $\theta_{i-1}, \psi(d_i) \vdash_h \bot$ (monotonicity), with d_i not in θ_{i-1} or $\bot$, gives $\theta_{i-1} \vdash_h \bot$. By induction $\theta_0 \vdash_h \bot$, and since $\emptyset \vdash_T \theta_0$, cut gives $\emptyset \vdash_h \bot$, a contradiction. (With no parameters, $T \vdash \neg(\bigwedge\Gamma \to \varphi)$ and the last step is immediate.) So $h \subseteq \vdash_T$, i.e. $h = \vdash_T$. Learning is as in (a).

(c) Terms are integer polynomials in the parameters. For a set X of sentences, let $\mathrm{cl}_i(X)$ be the least set containing X, closed under $\mathrm{Cn}_{\mathsf{RCF}}$ and under the instances of rule i ($t \cdot t = 1 + 1 \rhd t > 0$ for $i = 1$; $t^3 - 4t + 1 = 0 \rhd t > 1$ for $i = 2$). Substituting terms for parameters maps $\mathrm{Cn}_{\mathsf{RCF}}$-inferences to $\mathrm{Cn}_{\mathsf{RCF}}$-inferences and rule instances to rule instances, so cl_i is parameter-structural, and it is the least parameter-structural consequence relation h_i containing $\vdash_{\mathsf{RCF}}$ and the rule. Because the parameters do not occur in RCF and RCF is complete with model $\mathbb{R}$, $\mathsf{RCF} \cup Y \vdash \psi$ for parameter sentences iff $\mathbb{R}$ satisfies the universal closure of $\bigwedge Y_0 \to \psi$ for some finite $Y_0 \subseteq Y$.

h_1. An instance fires from $\mathrm{Cn}_{\mathsf{RCF}}(\emptyset)$ only if $t(\bar{x})^2 = 2$ identically on $\mathbb{R}$, which no integer polynomial satisfies (a constant integer n has $n^2 \neq 2$, and a non-constant t has non-constant t^2).

So $h_1(\emptyset) = \mathrm{Cn}_{\mathsf{RCF}}(\emptyset)$, which is consistent. The instance with $t = c$ gives $c \cdot c = 1 + 1 \vdash_{h_1} c > 0$, which RCF does not prove ($c = -\sqrt{2}$). In the context $\{c \cdot c = 1+1\}$, RCF proves $(-c)\cdot(-c) = 1+1$, the rule gives $-c > 0$ and $c > 0$, and RCF derives $\bot$.

h_2. The cubic $f(x) = x^3 - 4x + 1$ has no rational root ($f(\pm 1) \neq 0$), so it is irreducible over $\mathbb{Q}$. Its discriminant is $-4(-4)^3 - 27 = 229$, not a square, so its Galois group is S_3. Its real roots are $r_1 \approx -2.115$, $r_2 \approx 0.254$, $r_3 \approx 1.861$. For a root r, $\mathbb{Q}(r)$ has degree 3 and is not normal (else the Galois group would have order 3), so it contains no other root (if it contained two roots it would contain the third, since the roots sum to 0, and be the splitting field). Now:

- From $\emptyset$, an instance fires only if $f(t(\bar{x})) = 0$ for all real $\bar{x}$. Then t is a continuous map from the connected $\mathbb{R}^m$ into the finite set of roots, hence constant, hence an integer root of f, which does not exist. So $h_2(\emptyset) = \mathrm{Cn}_{\mathsf{RCF}}(\emptyset)$.
- Let $X = \{f(c) = 0\}$ and $Y = \mathrm{Cn}_{\mathsf{RCF}}(X \cup \{c > 1\})$. The instance with $t = c$ shows $Y \subseteq h_2(X)$. Conversely, Y is closed under the rule: if $f(t(c, \bar{d})) = 0 \in Y$, then $f(t(r_3, \bar{d})) = 0$ for all real $\bar{d}$ (the only root > 1 is r_3); by continuity and connectedness $t(r_3, \bar{d})$ is constant in $\bar{d}$, equal to $t(r_3, \bar{0}) \in \mathbb{Z}[r_3] \subseteq \mathbb{Q}(r_3)$, hence equal to r_3; so $\mathbb{R}$ satisfies $\forall c \forall \bar{d}\,(f(c) = 0 \wedge c > 1 \to t(c, \bar{d}) > 1)$ and $t > 1 \in Y$. Hence $h_2(X) = Y$, which is consistent, while $X \vdash_{h_2} c > 1$ is not RCF-valid ($c = r_2$). So h_2 is unsound, and coherent on both $\emptyset$ and X.
- In the context $\{f(c) = 0,\ c < 0\}$ the rule gives $c > 1$, and RCF derives $\bot$. □

No finite family of contexts suffices without $\exists$-closure. (proof sketch) Let $\mathcal{F}$ be a finite family of RCF-consistent finite contexts, d even, and h the least parameter-structural consequence relation containing $\vdash_{\mathsf{RCF}}$ and all instances of $t^d = 1 + 1 \rhd t > 0$. Since $x^d - 2$ is irreducible over $\mathbb{Q}$ (Eisenstein), $2^{1/d}$ has degree d, and h is unsound: $c^d = 1 + 1 \vdash_h c > 0$ fails at $c = -2^{1/d}$. For $A \in \mathcal{F}$ with parameters among $\bar{c}$, the set $S_A = \{\bar{x} : \mathbb{R} \models \bigwedge A(\bar{x})\}$ is a non-empty semialgebraic set defined over $\mathbb{Q}$; it has finitely many connected components, and each contains a point with real algebraic coordinates (a standard fact of real algebraic geometry, via Tarski transfer). Choose one such point in every component of every S_A, $A \in \mathcal{F}$, and let D bound the degrees over $\mathbb{Q}$ of the fields their coordinates generate. As for h_1, an instance fires in A only if $t(\bar{x})^d = 2$ on all of S_A (times $\mathbb{R}^k$ for parameters of t not in A). Then t takes values in $\{\pm 2^{1/d}\}$, so it is constant on each component, and its value at the chosen point (extended by zeros) lies in a field of degree at most D. So for $d > D$ no instance fires in any $A \in \mathcal{F}$, and $h(A) = \mathrm{Cn}_{\mathsf{RCF}}(A)$ is consistent: h is $\mathcal{F}$-coherent but unsound. The observation is due to a referee; we give it as a sketch.

Beyond RCF. With sin on $\mathbb{R}$, which defines $\mathbb{Z}$, incompleteness enters. For exp the question is open: $\mathrm{Th}(\mathbb{R}, \exp)$ is model complete (Wilkie, 1996), and decidable if Schanuel's conjecture holds (Macintyre and Wilkie, 1996).

Repair note. The first version claimed, from (a), that complete theories are pinned "as cleanly as CPC" at the rule level, in contexts with parameters. A referee's counterexamples (c) refuted that; (b) is the repair. The algebraic facts about f were also checked with a computer algebra system.

E.8 Proof of Theorem 6.21

(a) If π is true, $\mathsf{PA} + \pi \subseteq \mathrm{Th}(\mathbb{N})$ is consistent. If π is false, $\neg\pi$ is a true Σ_1 sentence, so $\mathsf{PA} \vdash \neg\pi$ by Σ_1-completeness, and $\mathsf{PA} + \pi$ is inconsistent.

(b) A true σ gives $\mathsf{PA}+\sigma \subseteq \mathrm{Th}(\mathbb{N})$. By Gödel's second theorem $\mathsf{PA}+\neg\mathrm{Con}(\mathsf{PA})$ is consistent, although $\neg\mathrm{Con}(\mathsf{PA})$ is a false Σ_1 sentence.

(c) On a Π_1 sentence π the learner says "true" at stage t unless a PA-proof of $\neg\pi$ with code $\leq t$ exists. By (a) and soundness of PA, such a proof exists iff π is false; so the verdict changes at most once and converges to the truth. On a Σ_1 sentence σ it says "false" unless a PA-refutation of $\neg\sigma$ (equivalently a proof of σ) with code $\leq t$ exists; apply (a) to the Π_1 sentence $\neg\sigma$.

(d) At stage t the learner accepts φ_n iff no PA-proof with code $\leq t$ refutes the conjunction of φ_n with the sentences among $\varphi_0,\dots,\varphi_{n-1}$ accepted at stage t. By induction on n: once the statuses of $\varphi_0,\dots,\varphi_{n-1}$ have stabilized, the status of φ_n is a Π_1 question (consistency of a fixed finite extension of PA), answered by proof search with at most one change. So all statuses converge. In the limit φ_n is accepted iff it is consistent with PA plus the accepted earlier sentences: this is Lindenbaum's construction along the enumeration, so the limit is a complete consistent extension of PA, and it depends on the enumeration. If $\varphi_0=\neg\mathrm{Con}(\mathsf{PA})$, no refutation of $\mathsf{PA}+\neg\mathrm{Con}(\mathsf{PA})$ exists (Gödel II), so φ_0 is accepted forever and the limit contains a false Σ_1 sentence. On the enumeration $\mathrm{Con}(\mathsf{PA})\wedge 0{=}0,\ \neg\mathrm{Con}(\mathsf{PA}),\ \mathrm{Con}(\mathsf{PA}),\dots$, the first sentence is consistent with PA (it is true) and stays accepted; the second is inconsistent with it, which proof search finds, so it is rejected.

(e) On a computable presentation, the learner's guesses about each sentence form a computable function $g(\varphi,n)$. If they converged to the truth value on every Σ_2 sentence, then by Shoenfield's limit lemma (Shoenfield, 1959) the set of true Σ_2 sentences would be Δ_2; but it is Σ_2-complete, hence not Δ_2, by the arithmetical hierarchy theorem (Kleene, 1952, §57). □

Repair note. The first version of (d) said that the naive learner accepts whichever of $\mathrm{Con}(\mathsf{PA})$ and $\neg\mathrm{Con}(\mathsf{PA})$ comes first; a referee's enumeration above refutes that, and (d) now states the Lindenbaum form.

E.9 Credences: the trilemma and what weak coherence costs

The arithmetic boundary of Theorem 6.21 has a counterpart for credences. Quantifier blocks may be empty, so Π_1 sentences also count as Π_2 and Σ_2.

Theorem E.2 (The coherence/Π_1/Π_2 trilemma; cf. Sawin and Demski 2013)**.** *[L3 Thm D, as repaired]* *Let $P_t(\varphi)\in\mathbb{Q}\cap[0,1]$ be computable uniformly in t,φ, with* (i) $\lim_t P_t(\pi)=1$ *for every true* Π_1 *π, and* (ii) $\limsup_t(P_t(\pi)+P_t(\varphi))\leq 1$ *whenever π is* Π_1 *and* $\mathsf{PA}\vdash\pi\to\neg\varphi$. *Then infinitely many true* Π_2 *sentences have* $\liminf_t P_t(\varphi)=0$.

Proof. A false Π_2 $\varphi=\forall x\exists y R$ has a true Π_1 refuter $\forall y\neg R(\bar{x}_0,y)$, so $\limsup_t P_t(\varphi)=0$ by (i), (ii). Thus $\{\varphi\in\Pi_2:\liminf_t P_t(\varphi)>0\}$, a Σ_2 set, contains no false Π_2 sentence. If it contained all but finitely many true ones, the Π_2-complete set of true Π_2 sentences would be Σ_2 (a finite modification of a Σ_2 set is Σ_2), contradicting the arithmetical hierarchy theorem (Kleene, 1952, §57). □

So one can have at most three of: computability, weak coherence, Π_1-convergence, and non-dogmatism on true Π_2 sentences; logical induction drops Π_1-convergence. Coherence does *not*, however, confine credences to Δ_2:

Proposition E.3 (What weak coherence costs)**.** *[L3 Thm D′, as repaired]* *Say computable credences* gradually verify *C if $P_t(\varphi)\to 1$ iff φ is true, for $\varphi\in C$.* (a) *Convergent credences, coherent or not, have a* Π_2 *success set, so never gradually verify the* Σ_2 *sentences.* (b) *Credences as in*

Theorem E.2(ii) never gradually verify all Π_2 sentences. (c) *Some computable credences, weakly coherent for all pairs, gradually verify the Σ_2 sentences (necessarily without converging on all of them).*

Proof of Proposition E.3. *(a)* If limits exist on C, then for $\varphi \in C$: $\lim_t P_t(\varphi) = 1$ iff $\forall q \in \mathbb{Q}^+ \, \forall s \, \exists t \geq s \; P_t(\varphi) > 1 - q$, a Π_2 condition. The set of true Σ_2 sentences is Σ_2-complete, so it is not Π_2 (else it would be Δ_2), and no convergent computable credences have it as their success set.

(b) By the convention on empty quantifier blocks every Π_1 sentence is Π_2. If weakly coherent credences gradually verified all Π_2 sentences, they would be Π_1-convergent, and Theorem E.2 would give a true Π_2 sentence with $\liminf_t P_t = 0$, contradicting $P_t \to 1$ on true Π_2 sentences.

(c) Enumerate the sentences of Σ_2 form, $\sigma = \exists x \forall y \, R(x, y)$ with R in Δ_0 (quantifier blocks coded as single variables; an empty block is a dummy variable), with indices $\mathrm{ind}(\sigma)$. At stage t:

- $x_t(\sigma)$ is the least $x \leq t$ with $R(x, y)$ for all $y \leq t$, or $t + 1$ if there is none;
- $c_t(\sigma)$ is the last stage $s \leq t$ with $x_s(\sigma) \neq x_{s-1}(\sigma)$, or -1 if there is none;
- $a_t(\sigma) := t - \max(\mathrm{ind}(\sigma), c_t(\sigma))$, the age of the current witness, for $t \geq \mathrm{ind}(\sigma)$;
- I_t is the symmetric set of pairs (σ, σ') of Σ_2 sentences with indices $\leq t$ that have a PA-proof of $\neg(\sigma \wedge \sigma')$ with code $\leq t$ (including $\sigma' = \sigma$);
- $P_t(\sigma) := \min(1 - 2^{-a_t(\sigma)}, \; \min\{2^{-a_t(\sigma')} : (\sigma, \sigma') \in I_t, \; a_t(\sigma') \geq a_t(\sigma)\})$.

Set $P_t := 0$ on sentences not of Σ_2 form, and on σ before stage $\mathrm{ind}(\sigma)$.

Weak coherence. If $\mathsf{PA} \vdash \psi \to \neg\varphi$ and one of them is not of Σ_2 form, it has credence 0 and the sum is at most 1. Otherwise $(\psi, \varphi) \in I_t$ from some stage on. At such a stage, say $a_t(\psi) \leq a_t(\varphi)$; then $P_t(\psi) \leq 2^{-a_t(\varphi)}$ and $P_t(\varphi) \leq 1 - 2^{-a_t(\varphi)}$, so the sum is at most 1. *False σ.* Every x is eventually refuted, so $x_t(\sigma) \to \infty$ and changes infinitely often. At each change $a_t(\sigma) = 0$ and $P_t(\sigma) \leq 1 - 2^0 = 0$. So $P_t(\sigma) \not\to 1$. *True σ.* Let x^* be its least witness. Once every $x < x^*$ is refuted and $t \geq x^*$, $x_t(\sigma) = x^*$ forever. Let $T_0 = \max(\mathrm{ind}(\sigma), \text{last change})$; for $t > T_0$, $a_t(\sigma) = t - T_0 \to \infty$. A competitor σ' with $(\sigma, \sigma') \in I_t$ is PA-incompatible with the true σ, hence false since PA is sound. If $\mathrm{ind}(\sigma') > T_0$ then $a_t(\sigma') \leq t - \mathrm{ind}(\sigma') < a_t(\sigma)$. The finitely many competitors with index $\leq T_0$ are false, so each changes its witness at some stage $c > T_0$, after which $a_t(\sigma') \leq t - c < a_t(\sigma)$. So from some stage on no competitor enters the inner minimum, and $P_t(\sigma) = 1 - 2^{-a_t(\sigma)} \to 1$. *Non-convergence.* By (a), these credences cannot converge on every Σ_2 sentence. □

The construction is due to a referee and was sanity-checked by script (`L3-scripts/sigma2_weak_coherence.py`). It assigns 0 to every sentence not of Σ_2 form, so it is dogmatic on the strictly Π_2 truths, consistent with Theorem E.2.

Incoherent credences reach the Π_2 class convergently. For a sentence $\varphi = \forall x \exists y \, R(x, y)$ of Π_2 form (R in Δ_0, blocks coded as single variables) let $k_t(\varphi)$ be the largest $k \leq t$ such that every $x < k$ has some $y \leq t$ with $R(x, y)$, put $P_t(\varphi) = 1 - 2^{-k_t(\varphi)}$, and $P_t = 0$ on other sentences. These credences are computable, and $k_t(\varphi)$ is non-decreasing in t, so $P_t(\varphi)$ converges. If φ is true, $k_t(\varphi) \to \infty$ and $P_t(\varphi) \to 1$; if x_0 is the least x without a witness, $k_t(\varphi) \leq x_0$ for all t, so $\lim_t P_t(\varphi) \leq 1 - 2^{-x_0} < 1$. So these credences gradually verify the Π_2 sentences and converge on all of them; by Proposition E.3(b) they are not weakly coherent. On the other side, the sentences of Σ_1 form, of Π_1 form, and Boolean combinations of Σ_1 sentences are decidable in the limit

(Theorem 6.21(c), applied componentwise), and the 0/1 output of such a decider, with 0 on other sentences, is computable, convergent and weakly coherent, since its limits are truth values and PA is sound.

Reading. Take the classes Σ_1, Π_1, $B(\Sigma_1)$, Σ_2, Π_2 and Π_3 as full syntactic classes. Among convergent credences, the first three are gradually verifiable with or without weak coherence, Σ_2 and Π_3 by neither (a convergent success set is Π_2, as in part (a), and neither set of true sentences is Π_2), and Π_2 only incoherently. So among convergent credences weak coherence costs exactly the Π_2 class. Without convergence, weakly coherent credences reach the Σ_2 class (part (c)), which is not Δ_2; so what a learner gains beyond Δ_2 by giving up convergence is not all lost to coherence. These are statements about full classes; which proper subclasses of Π_2 weakly coherent credences can gradually verify we leave open.

E.10 Proofs for Section 6.4: Carnap's problem

For finite Γ, Δ write $U(\Gamma, \Delta) = \{v : v[\Gamma] = 1,\ v[\Delta] = 0\}$; these sets form a clopen basis of 2^{Fm}, and a valuation satisfies the sequent $\Gamma \rhd \Delta$ iff it lies outside $U(\Gamma, \Delta)$. So $\mathrm{Val}(S)$ is closed for every set S of sequents.

Lemma 6.23. *(a)* If $v \notin \overline{V}$, some basic neighbourhood $U(\Gamma, \Delta) \ni v$ misses V; then $\Gamma \models^{\mathrm{m}}_{V} \Delta$ and v violates it. If $v \in \overline{V}$ violates a valid $\Gamma \rhd \Delta$, then $U(\Gamma, \Delta)$ is a neighbourhood of v, so it contains some $u \in V$, which violates the sequent too; contradiction.

(b) Finitary: if $\varphi \in C_V(X)$, the closed sets $V \cap \{v : v(\varphi) = 0\}$ and $\{v : v(x) = 1\}$, $x \in X$, have empty intersection; by compactness a finite subfamily already does, so $\varphi \in C_V(X_0)$ for a finite $X_0 \subseteq X$. Hence a valuation respects every valid finite-premise single-conclusion sequent iff $C_V(v) = v$ (identifying v with its true-set). $\mathrm{Fix}(C_V) = V^{\cap}$*:* a fixed point X equals $\bigcap\{v \in V : X \subseteq v\}$, an intersection of members of V (the empty intersection being Fm); conversely, for $X = \bigcap W$ with $W \subseteq V$, $C_V(X) \subseteq \bigcap W = X$.

(c) An intersection of classical theories is a theory, and Fm is the empty intersection. Conversely, a consistent theory T is the intersection of the Boolean valuations extending it: if $\varphi \notin T$, completeness gives a Boolean $v \supseteq T$ with $v(\varphi) = 0$. □

Theorem 6.24. BV is closed (Theorem 6.26(a)), so by Lemma 6.23(b) every $w \in \mathrm{BV}^{\cap}$, hence every $w \in W$, satisfies every sequent valid in $\models^{1}_{\mathrm{BV}}$; so $\models^{1}_{\mathrm{BV}} \subseteq \models^{1}_{W}$, and $W \supseteq \mathrm{BV}$ gives the converse. By Lemma 6.23(c) the members of $\mathrm{BV}^{\cap}$ are the v_T for classical theories T (including Fm), and v_T is Boolean iff T is maximal consistent. $v_{\top} = v_{\mathrm{Fm}}$ makes φ and $\neg\varphi$ both true. For consistent non-maximal T, some φ has $\varphi, \neg\varphi \notin T$ while $\varphi \vee \neg\varphi \in T$, which violates the tables for $\neg$ and $\vee$; and $\varphi \to \neg\varphi \notin T$ (it entails $\neg\varphi$), which violates the table for $\to$ at $v_T(\varphi) = 0$. Theories are closed under the $\wedge$ rules, so every v_T respects the table for $\wedge$. $\mathrm{BV}^{\cap}$ is closed, because "$\Gamma \subseteq v$ implies $\varphi \in v$" for each finite $\Gamma \models \varphi$ is a closed condition; and it is structural, because $\{\psi : \sigma\psi \in T\}$ is a classical theory whenever T is, Cn_2 being structural. □

Theorem 6.25. $d(v) = 0$ means that v violates some valid $\Gamma\rhd$, i.e. some finite subset of its true-set is inconsistent. For consistent v, $d(v) \leq 1$ means that some finite $\Gamma \subseteq v$ classically entails some $\varphi \notin v$, i.e. the true-set is not closed. A consistent closed non-maximal T omits some φ and $\neg\varphi$, violating the valid $\rhd\varphi, \neg\varphi$, so $d(v_T) = 2$. Maximal consistent theories are Boolean, and Boolean valuations satisfy every valid sequent, so $d = \infty$ exactly on BV. □

Theorem 6.26. *(a)* Each schema enforces one row condition. For $\neg$: $\varphi, \neg\varphi \rhd$ excludes both true and $\rhd\varphi, \neg\varphi$ excludes both false, so $v(\neg\varphi) = 1 - v(\varphi)$. For $\wedge$: the two eliminations give $v(\varphi \wedge \psi) \leq \min$, introduction gives $\geq$. For $\vee$: the introductions give $v(\varphi \vee \psi) \geq \max$, and $\varphi \vee \psi \rhd \varphi, \psi$ gives $\leq$. For $\to$: $\rhd\varphi, \varphi \to \psi$ gives $v(\varphi \to \psi) = 1$ when $v(\varphi) = 0$; $\psi \rhd \varphi \to \psi$ gives it when $v(\psi) = 1$; modus ponens gives $v(\varphi \to \psi) = 0$ when $v(\varphi) = 1, v(\psi) = 0$. So v satisfies all instances iff it is a homomorphism into the two-element Boolean algebra. (With $\bot, \top$ in the language, $\bot \rhd$ and $\rhd \top$ fix their values.) $\mathrm{BV} = \mathrm{Val}(\mathrm{TT})$ is closed, and $\models^{\mathrm{m}}_{\mathrm{BV}}$ determines BV by Lemma 6.23(a).

(b) For any substitution σ and sequent $s = \Gamma \rhd \Delta$, v satisfies σs iff $v \circ \sigma$ satisfies s. Let $v \in V$. Every instance of a schema in TT is $\sigma(\mathrm{TT}_j(p, q))$ for some σ and j, and $v \circ \sigma \in V$ satisfies $\mathrm{TT}_j(p, q)$; so v satisfies all instances and $V \subseteq \mathrm{Val}(\mathrm{TT}) = \mathrm{BV}$. For equality, take any $v \in V$ (there is one, as $V \neq \emptyset$) and any $u \in \mathrm{BV}$. Let $\sigma(p) = p_0 \vee \neg p_0$ if $u(p) = 1$ and $\sigma(p) = \neg(p_0 \vee \neg p_0)$ otherwise. Then $v \circ \sigma \in V \subseteq \mathrm{BV}$ is Boolean and agrees with u on atoms (v is Boolean, so $v(p_0 \vee \neg p_0) = 1$), hence $v \circ \sigma = u$ and $u \in V$.

(c) The learner that outputs BV once the eleven sequents $\mathrm{TT}(p, q)$ and the coherence datum have appeared is correct by (b), and it never needs to change its mind. $V = \emptyset$ satisfies every sequent vacuously, so without the coherence datum no finite set of positive data excludes it.

(d) Pointwise refutation. A text for $L_{\mathrm{BV}} := \models^{\mathrm{m}}_{\mathrm{BV}}$ eventually presents every instance of TT, and by (a) every non-Boolean valuation violates one. *No tell-tale.* Fix a Boolean u_0, and for a formula χ let v'_χ agree with u_0 except at χ. Then $v'_\chi(\neg\chi) = u_0(\neg\chi) = 1 - u_0(\chi) = v'_\chi(\chi)$, so $v'_\chi \notin \mathrm{BV} = \overline{\mathrm{BV}}$. $\mathrm{BV} \cup \{v'_\chi\}$ is closed, and by Lemma 6.23(a) its relation L_χ satisfies $\mathrm{Val}(L_\chi) = \mathrm{BV} \cup \{v'_\chi\} \neq \mathrm{BV}$, so $L_\chi \subsetneq L_{\mathrm{BV}}$. Given a finite $T \subseteq L_{\mathrm{BV}}$, choose χ occurring in no sequent of T; then v'_χ agrees with u_0 on every formula T mentions and satisfies T, so $T \subseteq L_\chi$. *Non-learnability.* Suppose a learner, computable or not, BC-identifies (a fortiori if it EX-identifies) every meaning in a class containing BV and all $\mathrm{BV} \cup \{v'_\chi\}$ from texts of their relations, plus the coherence datum $[\,:\,]$ or any other fixed, target-independent certifications; these can be folded into the learner, as in Corollary 6.11. (The datum $[\,:\,]$ is in bounds for every member and carries no information.) By the locking-sequence argument (Blum and Blum, 1975; Angluin, 1980), which needs no effectivity, there is a finite σ of L_{BV}-data such that the learner conjectures BV on every extension of σ by L_{BV}-data. Choose χ not mentioned in σ; then σ consists of L_χ-data, and on a text for L_χ beginning with σ the learner conjectures $\mathrm{BV} \neq \mathrm{BV} \cup \{v'_\chi\}$ forever. *Scope.* The argument needs the certifications to be target-independent. If every position in bounds for the target is eventually certified, text plus certifications is an informant (a sequent $\Gamma \rhd \Delta$ is valid iff $[\Gamma : \Delta]$ is not in bounds), as in Proposition 6.12; then a certification of $[\chi, \neg\chi :\,]$ or of $[\,: \chi, \neg\chi]$, whichever v'_χ realizes, separates $\mathrm{BV} \cup \{v'_\chi\}$ from BV, and this obstruction disappears. *Density.* A Boolean valuation is determined by its values on atoms, and the map from atom assignments to BV is a homeomorphism from 2^ω, which has no isolated points. So $\mathrm{BV} \setminus \{u_0\}$ is dense in BV; by Lemma 6.23(a) it has the same multiple-conclusion relation, and a basic open set meets BV iff it meets $\mathrm{BV} \setminus \{u_0\}$, so the two meanings have the same in-bounds positions. Composing any Boolean v with the constant substitution along u_0 yields u_0, so $\mathrm{BV} \setminus \{u_0\}$ is not structural. □

Repair note. The first version of (d) ended "identification in the limit still holds via the text". That was false and has been retracted; (d) now proves non-identifiability, and its scope (target-independent certifications) was made explicit in a second round. The Lean formalization proves (d) for BC-identification from texts with pauses and arbitrary learners.

Computational checks. (computed) Brute force over the sixteen formulas of depth ≤ 1 in two atoms confirms Theorem 6.26(a), Theorem 6.24 and all 2^{16} denial ranks of Theorem 6.25 (`T2-checks/carnap_check.py`). A referee's exhaustive search over all structural meanings induced by two- and three-element matrices found no non-Boolean survivor of the twelve data of Theorem 6.26, and found each datum individually necessary; that search was not re-run.

Proposition 6.27. Row by row, with designated set $\{1\}$. $\neg$: explosion $p, \neg p \rhd q$ at $p = 1$, $q = 0$ forces $f_\neg(1) = 0$; then $p \rhd \neg\neg p$ at $p = 1$ forces $f_\neg(0) = 1$. $\wedge$: $p \wedge q \rhd p$ forces $f_\wedge(0, b) = 0$, $p \wedge q \rhd q$ forces $f_\wedge(a, 0) = 0$, and $p, q \rhd p \wedge q$ forces $f_\wedge(1, 1) = 1$. $\vee$: the introductions force $f_\vee(a, 1) = f_\vee(1, b) = 1$, and disjunctive syllogism at $p = q = 0$ (where $\neg p = 1$) forces $f_\vee(0, 0) = 0$. $\to$: modus ponens forces $f_\to(1, 0) = 0$; $q \rhd p \to q$ forces $f_\to(a, 1) = 1$; $\neg p \rhd p \to q$ at $p = 0$ forces $f_\to(0, b) = 1$. So only the standard matrix can validate all twelve rules, and it does. A brute-force check over all $4 \cdot 16^3$ matrices confirms that exactly one matrix validates them (`T2-checks/matrix_check.py`). □

E.11 Proofs for Section 6.5: tonk and conservativity

Proposition 6.29. Intuitionistic implication with $\neg$-introduction, $\neg$-elimination and $\neg\neg$-elimination is a standard natural-deduction system for the $\{\to, \neg\}$ fragment of classical logic (Prawitz, 1965), so it is coherent and proves Peirce's law $((p \to q) \to p) \to p$. Peirce's law is not intuitionistically valid: in the two-world Kripke model $w \leq w'$ with p true only at w' and q true nowhere, $p \to q$ fails at both worlds, so $(p \to q) \to p$ holds at w vacuously, while p fails at w. So the extension is not conservative over $\mathsf{IPC}_\to$. *$\neg\neg$-elimination alone is conservative.* Let τ erase negations ($\tau(\neg\varphi) = \tau(\varphi)$, τ commuting with $\to$, identity on atoms). It maps each instance of $\neg\neg\varphi \rhd \varphi$ to the identity step $\tau\varphi \rhd \tau\varphi$ and each instance of an $\mathsf{IPC}_\to$ rule to an instance of the same rule, and it fixes $\neg$-free formulas. So a derivation of a $\neg$-free sequent in the extension maps to an $\mathsf{IPC}_\to$-derivation of the same sequent. *Conservativity and coherence.* If the base is non-trivial and coherence means non-triviality on the old vocabulary, a conservative extension proves no new old-vocabulary sequent, so it is non-trivial; likewise if $\bot$ with ex falso is in the base. If $\bot$ is new and has no rules, $\mathsf{IPC}_\to + \{\rhd\bot\}$ adds no $\bot$-free theorem and so is conservative, yet derives $\bot$. □

Theorem 6.30. *(a)* "$R \nvdash \bot$" says that no finite derivation in R ends in $\bot$, a Π_1 condition. For hardness, given e build a finite system R_e that derives the successive configurations of the computation of $\varphi_e(e)$ (elementary formal systems simulate Turing machines; Post 1943; Smullyan 1961), and add the rule "halting configuration $\rhd\bot$". Then $R_e \nvdash \bot$ iff $\varphi_e(e)$ diverges, which reduces the Π_1-complete complement of the halting set.

(b) Membership: "for every old-vocabulary s and every derivation of s in $B + N$ there is a derivation of s in B" is Π_2. *Hardness*, from the Π_2-complete set $\mathrm{Tot} = \{e : \varphi_e \text{ total}\}$. Let B be a universal elementary formal system that derives $T(\bar{x}, \bar{e})$ iff $\varphi_e(x)\!\downarrow$ (unary numerals $\bar{x}, \bar{e}$) and derives $\mathrm{Num}(\bar{n})$ for exactly the unary numerals, designed so that T-strings never occur as premises of its rules. Its productions must also match only predicate-tagged strings over the old alphabet; in pure Post-string format a projection such as $xy \rhd y$ could otherwise strip a new symbol, while in an elementary formal system with predicates this is automatic. Let N_e add a new symbol c with the rules $\rhd c\,\bar{e}$ and $c\,y$, $\mathrm{Num}(x) \rhd T(x, y)$ (schematic in x, y). The only derivable c-string is $c\,\bar{e}$, so the new old-vocabulary theorems are exactly $T(\bar{n}, \bar{e})$ for $n \in \mathbb{N}$, and since T-strings are never premises nothing further follows. So the old-vocabulary theorems of

$B + N_e$ are $\mathrm{Thm}(B) \cup \{T(\bar{n}, \bar{e}) : n \in \mathbb{N}\}$, and $B + N_e$ is conservative over B iff $\varphi_e(n)\!\downarrow$ for every n, iff $e \in \mathrm{Tot}$.

(c) A set is decidable in the limit by a computable learner iff it is Δ_2 (Shoenfield, 1959; Putnam, 1965; Gold, 1965); a Π_1 set is decided in the limit with at most one mind change; a Π_2-complete set is not Δ_2. Refutability in the limit corresponds to Π_2 (Kelly, 1996).

(d) If B is decidable, "every old-vocabulary theorem of $B + N$ is a theorem of B" quantifies universally over derivations and then checks a decidable property: Π_1. □

E.12 Proof of Theorem 6.34, and survivors

The characterization of Alt as the never-refuted $\mathcal{G}$-closed hypotheses is Theorem 6.9(ii), restricted to $\mathcal{G}$-closed h.

(i) A structural $h \supseteq \mathrm{Cn}_2$ is Cn_2 or C_{Fm} by Theorem 6.13, and C_{Fm} is incoherent on every context, so on the non-empty $\mathcal{A}$.

(ii) $\supseteq$: if h is structural with $\mathsf{IPC} \subseteq h \subseteq \mathsf{CPC}$ and A is classically consistent, then $A \nvdash_{\mathsf{CPC}} \bot$, so $A \nvdash_h \bot$. $\subseteq$: let $h \supseteq \mathsf{IPC}$ be structural and suppose $\Gamma \vdash_h \varphi$ with Γ finite and $\Gamma \not\models \varphi$. Take a Boolean v with $v[\Gamma] = 1$, $v(\varphi) = 0$, and let σ_v map each atom to $\bot \to \bot$ or $\bot$ according to v. As shown in the proof of Proposition E.1(c), IPC proves every classically true variable-free formula and refutes every classically false one. So IPC proves each member of $\sigma_v\Gamma$ and proves $\neg\sigma_v\varphi$. By structurality $\sigma_v\Gamma \vdash_h \sigma_v\varphi$; by cut with the IPC-theorems $\sigma_v\Gamma$, $\emptyset \vdash_h \sigma_v\varphi$; with $\emptyset \vdash_h \neg\sigma_v\varphi$ and $\sigma_v\varphi, \neg\sigma_v\varphi \vdash_{\mathsf{IPC}} \bot$, $\emptyset \vdash_h \bot$. By monotonicity h is incoherent on every $A \in \mathcal{A}$, which is non-empty. So every coherent structural $h \supseteq \mathsf{IPC}$ lies inside CPC.

(iii) With $\mathcal{G}$ deductive closure, $h^* = \mathsf{PA}$ and $\mathcal{A} = \{\emptyset\}$, the hypotheses above the target are the deductively closed theories containing PA (the r.e. ones, if hypotheses are restricted to r.e. theories as in Section 6.3), and $\emptyset$-coherence is consistency. $\mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$ is consistent by Gödel II; its completions are not r.e., and they have only non-standard models, since they contain the false $\neg\mathrm{Con}(\mathsf{PA})$. Designating $\{\mathrm{Con}(\mathsf{PA})\}$ removes $\mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$. If every finite set of true sentences is designated, a theory containing a false φ is incoherent on $\{\neg\varphi\}$, and a theory $T \subseteq \mathrm{Th}(\mathbb{N})$ is coherent on every true context; so exactly the unsound ones are removed. The theories T_k ($k \leq \omega$) of Theorem 6.20 contain PA and are sound, so they remain, and by that theorem no learner identifies them all from text plus true-context coherence. Any two of them have different theorems (for $j < k$, T_k proves $\mathrm{Con}(T_j)$ and T_j does not), and any finite subclass is identifiable from text; what fails is identification of the whole chain together with T_ω. □

Repair note. Part (ii) was strengthened from "$\supseteq$" to equality after verification, with the σ_v argument above; part (iii) was made precise ($\mathcal{G}$, h^* and $\mathcal{A}$ specified, and the claim about the T_k restricted to what Theorem 6.20 proves).

Survivors. Outside a Post-complete target, generalized fallacies can be coherent. The main text summarizes the following examples.

Example E.4 (Survivors). [T2 §6, as repaired] (1) *Quantifier swap* $\forall x \exists y R \,/\, \exists y \forall x R$ is sound in one-element structures, hence $\emptyset$-coherent; its alternative meaning is "the domain is a singleton", and any designated context with $0 \neq 1$ kills it. (2) The *gambler's fallacy* is incoherent with independence, but coherent without it: an anti-persistent stationary Markov chain with switching probability 0.6 has fair marginals and validates it. Only frequencies or an accepted independence premise remove it. (3) *Base-rate neglect* as a uniform schema $P(E|H) = a \,/\, P(H|E) = a$ yields $P(H|E) + P(\neg H|E) = 1.8$ when $P(E|H) = P(E|\neg H) = 0.9$, and dies with additivity. (4)

Conditional perfection (from q and a law $p \to q \in K$, infer p; Geis and Zwicky 1971) is coherent whenever the per-law converse $\{q \to p : (p \to q) \in K\}$ is consistent with $K \cup A$; this is sufficient, not necessary ($K = \{p_1 \to q_1, p_2 \to q_2\}$, $A = \{q_1 \vee q_2, \neg p_1, \neg p_2\}$: the converse is inconsistent with $K \cup A$, yet neither q_i is derivable, so the step rule never fires and the fallacy survives, for step-set hypotheses without proof by cases). It dies when two exclusive laws share a consequent (*rain* $\to$ *wet*, *sprinkler* $\to$ *wet*, $\neg$(*rain* $\wedge$ *sprinkler*), $A = \{wet\}$). The per-law converse is not Clark's completion (Clark, 1978): completing only the head *wet* (with *rain*, *sprinkler* abducible) gives *wet* $\leftrightarrow$ *rain* $\vee$ *sprinkler*, which is consistent there.

E.13 Consensus and coherentizing

These two results make the uses (U3) and (U1) of Remark 6.35 quantitative.

Remark E.5 (Consensus does not amplify under a common cause). [L11 TC3] Let m annotators judge a step. With probability $\pi > 0$, independently of validity, they all copy one shared judgment that says "valid" with probability a for valid and $b > 0$ for invalid steps; otherwise they judge independently with rates $p > q$. Then

$$\frac{\mathbb{P}(\text{all say valid} \mid \text{valid})}{\mathbb{P}(\text{all say valid} \mid \text{invalid})} = \frac{\pi a + (1-\pi)p^m}{\pi b + (1-\pi)q^m} \leq \frac{1}{\pi b}$$

for every m, against $(p/q)^m$ when $\pi = 0$. So unanimity among annotators sharing an error source, or self-consistency across samples of one model, moves the odds by a bounded factor. Human errors in mathematics are often systematic in this sense, which is why corroboration must come from channels whose errors are independent of the human ones, Daniels' independence constraint (Daniels, 1979).

The eliminative use (U1) has a quantitative safety guarantee whose converse shows what it depends on. Let $\mathcal{W} \subseteq \{0,1\}^S$ be the worlds allowed by constraints on a finite set S of sentences, K their convex hull, and score credences x in world w by $\|x - w\|^2$.

Lemma E.6 (Coherentizing is safe iff the constraints are sound). *[L3 Lemma E′, as repaired] For $x \notin K$ with Euclidean projection x^* onto K, every $w \in K$ has $\|x - w\|^2 \geq \|x - x^*\|^2 + \|x^* - w\|^2$. Conversely, if the actual world's indicator v is not in $\mathcal{W}$, then $v \notin K$, and projecting $x = v$ raises the actual loss from 0 to $\|v - v^*\|^2 > 0$.*

Proof. $\langle x - x^*, w - x^* \rangle \leq 0$. For the converse, a 0/1 vector is an extreme point of the cube, so it lies in the convex hull of 0/1 vectors only if it is one of them. □

This is the accuracy-dominance argument of de Finetti (1974) and Joyce (1998) (Bregman projections for other proper scores; Predd et al. 2009). It belongs to the projection, not to arbitrary arbitrage removal: for $x = (0.6, 0.6)$ on $\{\varphi, \neg\varphi\}$, moving to $(1, 0)$ costs 2.0 in world $(0, 1)$, against 0.52 before and 0.5 for the projection. The converse is the probabilistic form of the air-pressure example of Section 1.1: coherence pressure from an idealized context the world violates, or from untrusted learned rules, can make credences strictly worse.

F Proofs for Section 7

The proofs of Proposition 7.5, Lemma 7.22 and Corollary 7.23 are complete in the main text; this appendix adds the details they cite, proves the other results, and contains the examples,

computations and secondary results of Section 7 that the main text only points to: the Bayesian remark (Remark F.1), the cost of the audit, the voting audit (Proposition F.2), the details of the learner's definition, the finite-depth counterexample of Theorem 7.19(c), a Hilbert illustration of collateral, the role of learning when W is a complete checker, and the precise forms of Theorem 7.27(c),(e). Figures marked (computed) come from the check scripts listed in Table 22. Throughout, (WS) is assumed for every designated position unless a statement says otherwise, V (or $V_{[A:D]}$) denotes a total valuation as in (WS), and "refutation" means a refutation in the sense of Definition 7.2, with size counting the symbols of distinct judgments. Two facts are used repeatedly.

(F1) *Cleanness is closed downward.* If $B' \subseteq B$, every refutation of B' is a refutation of B, because $R_{B'} \subseteq R_B$. Likewise for step sets.

(F2) *Transfer.* If every step of a set S' lies in $\mathrm{Sound}(R)$ and π is a refutation with steps in $S' \cup R \cup T_{\mathrm{trust}}$, then replacing each S'-step of π by an R-derivation of its conclusion from its premises, and compressing (Definition 7.2(a)), gives a refutation with steps in $R \cup T_{\mathrm{trust}}$, from the same leaves to the same conclusion, of some finite size. In particular, if R is clean at every depth, then every refutation using S' transfers to a refutation of R, which is impossible.

F.1 Proof of Lemma 7.4

Fix a designated position $[A : D]$ and its valuation V, and let $X = V^{-1}(1)$. Since V falsifies no step of R^*, X is closed under R^*, so $\mathrm{Cl}_{R^*}(X) = X$. Hence if $P \rhd j \in \mathrm{Sound}(R^*)$ and $P \subseteq X$, then $j \in \mathrm{Cl}_{R^*}(P) \subseteq \mathrm{Cl}_{R^*}(X) = X$: *$V$ falsifies no step of* $\mathrm{Sound}(R^*)$.

(a) Let $S \subseteq \mathrm{Sound}(R^*)$, and let π be a derivation relative to $[A : D]$ whose leaves have e-value 1 and whose steps lie in $S \cup T_{\mathrm{trust}}$. By induction along the sequence π, every judgment of π is V-true. A leaf is e-true, hence V-true because V extends e. A judgment produced by a step of $S \cup T_{\mathrm{trust}}$ from earlier, V-true judgments is V-true, by the claim for S and by (WS) for T_{trust}. So the conclusion of π is V-true, and since V extends e it is not e-false. Hence π is not a refutation; as π, its size and the position were arbitrary, S is clean at every depth. For $B \subseteq \Sigma^*$, $R_B \subseteq R^* \subseteq \mathrm{Sound}(R^*)$ (Lemma 2.3(b)).

(b) A member of Conf_d has a d-refutation, so by (a) it is not contained in Σ^*; as it is contained in $\Sigma^P = \Sigma^* \sqcup F$, it meets F.

(c) The definition of a d-refutation of B mentions only e (which is determined by $\mathcal{A}$ and W), T_{trust}, d and B. So Conf_d is a function of these data. Both tie-breaking rules choose among the d-refutations of B using only their sizes, a fixed lexicographic order, and whether their descent is blocked; the last depends only on π and e, because e^+_π is computed from e and the trusted steps of π.

(d) ($\Rightarrow$) The proof of (a) used (WS) only for the position of π. ($\Leftarrow$) Suppose $e = e_{[A:D]}$ is well defined and Σ^* is clean at every depth relative to $[A : D]$. Let $Y = \mathrm{Cl}_{R^* \cup T_{\mathrm{trust}}}(A \cup W^{-1}(1))$ and let V be the indicator of Y. Since Y is closed under $R^* \cup T_{\mathrm{trust}}$, V falsifies no step of $R^* \cup T_{\mathrm{trust}}$, and V is 1 on every e-true judgment. Let j be e-false and suppose $j \in Y$. By Lemma 2.3(a), j has a finite $(R^* \cup T_{\mathrm{trust}})$-derivation tree from $A \cup W^{-1}(1)$, that is, from e-true leaves. Compressing it gives a refutation of Σ^* relative to $[A : D]$ of some finite size, contradicting cleanness. So V is 0 on every e-false judgment, V extends e, and (WS) holds for $[A : D]$. Nothing in the argument used that Σ^* is the target. □

F.2 Details for Proposition 7.5

The first two claims are Theorems 5.13 and 5.14 applied to the tagged practice, together with Definition 2.10, which gives every fallacy an instance outside Sound(R^*). For the CPC claim, let $\Pi_c \rhd j_c$ be the closed falsified instance of the pure fallacy τ (Lemma 7.22). Each closed formula has the same value under every valuation, so every member of Π_c and $\neg j_c$ are tautologies, and a complete Σ^* derives them from $\emptyset$. The step adds j_c, and $\{j_c, \neg j_c\}$ classically entails every formula, which Σ^* derives. For the contrast: if Σ^* is complete and $s = (p \rhd q)$ for atoms p, q, the set of tautologies is closed under s because p is not a tautology, so $\mathrm{Cl}_{R^* \cup \{s\}}(\emptyset) = \mathrm{Cl}_{R^*}(\emptyset)$. □

F.3 Proof of Lemma 7.7

(a) $M \subseteq U$ is clean iff it contains no member of $\mathcal{K}$, iff (by upward closure and finiteness) it contains no member of $\mathcal{C}$, iff $U \setminus M$ meets every member of $\mathcal{C}$, i.e. is a transversal. Complementation reverses inclusion, so maximal clean sets correspond to minimal transversals.

(b, $\Leftarrow$) Let $x \in C \in \mathcal{C}$ and $T = (U \setminus C) \cup \{x\}$. Let $E \in \mathcal{C}$. If $E \not\subseteq C$, then E meets $U \setminus C$. If $E \subseteq C$, then $E = C$ by minimality of C, so $x \in E$. Hence T is a transversal, and it contains a minimal transversal T'. Since T' meets C and $T' \cap C \subseteq T \cap C = \{x\}$, we get $x \in T'$.

(b, $\Rightarrow$) If $x \in T$ for a minimal transversal T, then $T \setminus \{x\}$ is not a transversal, so some $E \in \mathcal{C}$ has $E \cap T = \{x\}$, and $x \in E$.

(b, second claim) By (a), x lies outside some maximal clean set iff x lies in some minimal transversal, iff $x \in \bigcup \mathcal{C}$. Every clean set is contained in a maximal one, and $\emptyset$ is clean, so there are maximal clean sets and their intersection is $U \setminus \bigcup \mathcal{C}$. It is clean because cleanness is closed downward.

(c, $\Leftarrow$) If $\mathcal{C} = \{\{x_1\}, \ldots, \{x_r\}\}$, every transversal contains $\{x_1, \ldots, x_r\}$, which is itself a transversal.

(c, $\Rightarrow$) Let T be the unique minimal transversal. By (b), $T = \bigcup \mathcal{C}$. For each $x \in T$, minimality gives some $E \in \mathcal{C}$ with $E \cap T = \{x\}$; since $E \subseteq \bigcup \mathcal{C} = T$, $E = \{x\}$. A member $C \in \mathcal{C}$ with $|C| \geq 2$ would then strictly contain the singleton member $\{x\}$ for any $x \in C$, contradicting minimality. □

The Lean formalization (`InfLearn.Blame`) proves (a) as `isMaxClean_iff` and `setOf_isMaxClean_eq_image`, (b) as `exists_minTransversal_mem_iff`, `iInter_maxClean_eq` and `isClean_sdiff_blameSet`, and (c) as `existsUnique_minTransversal_iff`, for a finite universe; upward closure is needed only inside U, and $\emptyset \notin \mathcal{K}$ only for the intersection claims.

F.4 Proof of Proposition 7.6

(a) $\emptyset \subseteq \Sigma^*$ is d-clean by Lemma 7.4(a), so $\emptyset \in \mathcal{V}_d$ and $\bigcap_{\Sigma \in \mathcal{V}_d} R_\Sigma \subseteq R_\emptyset = \emptyset$. The version-space verifier (Definition 4.1) accepts exactly this intersection.

(b) Before any refutation is found, $\mathcal{V}'_t = \mathcal{V}$, whose unique maximal member is Σ^P; so the verifier accepts R^P, which contains an instance outside Sound(R^*) of every $\tau \in F$.

(c) *Schema level.* $\mathcal{V}_d$ is the family of clean sets for $U = \Sigma^P$ and $\mathcal{K} = \mathrm{Conf}_d$, which is upward closed and does not contain $\emptyset$. By Lemma 7.7(b) the intersection of its maximal members is $\Sigma^P \setminus \bigcup \mathcal{C}_d$, and since $\Sigma^P = \Sigma^* \sqcup F$ this equals $(\Sigma^* \setminus \mathrm{Coll}_d) \cup F_d^{\mathrm{surv}}$. For the inclusion $F_d^{\mathrm{surv}} \subseteq F_{\mathrm{res}}^{(d)}$, let $\tau \in F \setminus F_{\mathrm{res}}^{(d)}$. Then $\Sigma^* \cup \{\tau\} \in \mathrm{Conf}_d$ contains some $C' \in \mathcal{C}_d$, and $C' \not\subseteq \Sigma^*$ by Lemma 7.4(a), so $\tau \in C'$ and $\tau \notin F_d^{\mathrm{surv}}$. Hence the asserted set lies in $R^* \cup R_{F_{\mathrm{res}}^{(d)}}$.

Instance level. Let $\mathcal{M}$ be the maximal members of $\mathcal{V}_d$. For each $\Sigma \in \mathcal{M}$, $\bigcap \mathcal{M} \subseteq \Sigma$ gives $R_{\bigcap \mathcal{M}} \subseteq R_\Sigma$; so the instance-level set contains the schema-level one. Since $\Sigma^* \in \mathcal{V}_d$ and Σ^P is finite, $\Sigma^* \subseteq \Sigma_{\max}$ for some $\Sigma_{\max} \in \mathcal{M}$. For $\tau \in \Sigma_{\max} \cap F$, $\Sigma^* \cup \{\tau\} \subseteq \Sigma_{\max}$ is d-clean by (F1), so $\tau \in F^{(d)}_{\mathrm{res}}$. Hence $\bigcap_{\Sigma \in \mathcal{M}} R_\Sigma \subseteq R_{\Sigma_{\max}} \subseteq R^* \cup R_{F^{(d)}_{\mathrm{res}}}$. As a step set it is d-clean, since a d-refutation of it would be a d-refutation of the schema set $\Sigma_{\max}$. The MP/AC example of the main text shows that the inclusion can be strict: $(A,\ A \to A \rhd A)$ is the instance of MP with $B := A$ and the instance of AC with $B := A$, so its instances lie in $R_{\mathrm{MP}} \cap R_{\mathrm{AC}}$, while $R_{\{\mathrm{MP}\}\cap\{\mathrm{AC}\}} = \emptyset$. □

Remark F.1 (A Bayesian middle way). [T7 Remark 2.4′] (proof sketch) Put a prior w on $\mathcal{V}$ and accept s iff $w\{\Sigma \in \mathcal{V}'_t : s \notin R_\Sigma\} < w_0$. By the argument of Theorem 4.14 this is sound at all times for every target with $w(\Sigma^* \cup F^{(d)}_{\mathrm{res}}) \geq w_0$, provided that set is clean, and it waits for the audit automatically. It cannot break the symmetry of Theorem 7.19, where the rival diagnosis has the same prior mass. We do not develop it; the audit-then-assert design of TTL is simpler and has explicit constants.

F.5 Proof of Lemma 7.8

Agreement. We show by induction on the order in which the fixed-point computation assigns values that every value of e^+_π agrees with V. Initial values come from e, which V extends. If the forward rule assigns 1 to the conclusion of a trusted step whose premises have value 1, those premises are V-true by induction, and since V falsifies no trusted step the conclusion is V-true. If the backward rule assigns 0 to a premise π_0 of a trusted step whose conclusion has value 0 and whose other premises have value 1, then by induction the conclusion is V-false and the other premises are V-true; if π_0 were V-true the step would be falsified by V, so π_0 is V-false. Hence e^+_π never assigns both values to a judgment, and if it does, no total V as in (WS) exists for $[A : D]$.

(a) Each move of the descent goes from a node of value 0 to one of its premises, which occurs strictly earlier in the sequence π; so the descent terminates. Leaves have e-value 1, hence e^+_π-value 1, while every visited node has value 0; so every visited node is the conclusion of some step of π. Since the descent is unblocked, it ends by outputting a step s all of whose premises have value 1 and whose conclusion has value 0 (a step without premises is output at once). By agreement, the premises of s are V-true and its conclusion V-false. By (WS), $s \notin T_{\mathrm{trust}}$ and $s \notin R^*$; as π is a refutation of B, $s \in R_B$.

(b) The worklist assigns each judgment of π a value at most once and re-examines a trusted step only when one of its judgments receives a value, so the fixed point costs $O(|\pi|\varphi)$ operations, with one evaluation of e per judgment. The descent looks at the premises of each visited node once, and visits at most $|\pi|$ nodes. The number of judgments is at most $|\pi| \leq d$.

(c) If some genuine σ had $s \in \mathrm{inst}(\sigma)$, then $s \in R^*$, contradicting (a).

(d) Let $\sigma \in B$ with $s \in \mathrm{inst}(\sigma)$. Let J_1 be the list of judgments of π with e^+_π-value 1, in the order in which they received the value (those with initial e-value 1 first). Each member of J_1 is e-true, and then is treated as a leaf, or received value 1 by the forward rule as the conclusion of a trusted step of π whose premises precede it in J_1. The premises of s lie in J_1. Let c_0 be the conclusion of s. Define $c_0, c_1, \ldots$: if c_i is e-false, stop; otherwise c_i received value 0 by the backward rule from a trusted step t_i of π of which c_i is a premise, whose conclusion c_{i+1} had received value 0 strictly earlier, and whose other premises lie in J_1. Since assignment times strictly decrease along the chain, it terminates at some e-false c_k, and the c_i are pairwise distinct. They are not in J_1, because e^+_π is consistent.

Let π_σ be the list J_1, followed by c_0 (justified by s), c_1 (justified by t_0), …, c_k (justified by t_{k-1}). Its judgments are distinct; each non-leaf judgment is the conclusion of a step whose premises occur earlier; its leaves are e-true; its steps are $s \in \mathrm{inst}(\sigma)$ and trusted steps; and its last judgment c_k is e-false. So π_σ is a refutation of $\{\sigma\}$ relative to $[A : D]$. All its judgments are judgments of π, and size counts distinct judgments, so $|\pi_\sigma| \le |\pi| \le d$. Hence $\{\sigma\}$ is a d-conflict. (Under a tree-size convention this bound could fail, because the premise derivations of s and the side premises of the chain may share judgments.) □

The Lean theorem `InfLearn.Blame.LineArg.descent` proves the descent for a line-list argument under a total evaluation, with the walk length bounded by any height function that decreases along dependencies, and `LineArg.descent_not_mem` proves that the step found lies outside every rule set whose steps preserve truth. Trusted-step propagation and part (d) are not formalized.

F.6 Proof of Lemma 7.10, and the cost of the audit

By Lemma 7.8(a),(c) each successful descent removes at least one schema, all of them fallacies, so $\Sigma^* \subseteq B$ throughout, and there are at most $|F|$ removals. (a) The returned B is d-clean; for $\tau \in B \setminus \Sigma^*$, $\Sigma^* \cup \{\tau\} \subseteq B$ is clean by (F1), so $\tau \in F^{(d)}_{\mathrm{res}}$. (b) By Lemma 7.7 with $U = B_0$, A is the intersection of the maximal clean subsets of B_0, hence clean. The left inclusion holds because $\Sigma^* \subseteq B_0$ and $\mathrm{Coll}_d(B_0) = \Sigma^* \cap \bigcup \mathcal{C}_d(B_0)$. If $\tau \in A \cap F$ were not residual, $\Sigma^* \cup \{\tau\} \subseteq B_0$ would contain a minimal conflict, which contains τ by Lemma 7.4(a), contradicting $\tau \in A$. The last claim is Proposition 7.6(c). □

Cost. The descent path costs at most $|F| + 1$ oracle calls and $O((|F| + 1)\, d\, \varphi)$ evaluations of e. The fallback needs $\bigcup \mathcal{C}_d(B_0)$: at most $2^{|B_0|}$ oracle calls by brute force. From a cleanness (membership) oracle, the minimal conflicts and the maximal clean sets of a monotone property can be generated jointly in time quasi-polynomial in their total number (Bioch and Ibaraki, 1995; Gurvich and Khachiyan, 1999), with the dualization algorithm of Fredman and Khachiyan (1996) as the engine; the second family can be exponentially larger than the first, and both can be exponential in $|F|$. At stable depth the fallback's schema-level incompleteness is unavoidable (Theorem 7.19(b)); its cost is the price of bags.

F.7 The voting audit

Theorem 7.12 assumes (WS) for *every* designated position. When some positions may be wrong, as when an idealization is designated together with background it contradicts, the audit can vote.

Proposition F.2 (Voting audit)**.** *[T7 Prop 3.2, as repaired] Suppose at most m designated positions violate (WS), it is not known which, and $\hat{P} = \Sigma^P$. For each position p run the descent loop of the audit relative to p alone, from $B_p = \Sigma^P$ and* $\mathrm{Blamed}_p = \emptyset$*, with the prefer-unblocked oracle restricted to p. Stop when the oracle returns NONE (p is* finished*), when the descent is blocked, or when e_p is ill defined or e^+_π gives a judgment both values (then p is certified bad and its blame is discarded); otherwise move every $\sigma \in B_p$ with $s \in \mathrm{inst}(\sigma)$ into* Blamed_p*. Let* $\mathrm{votes}(\sigma) = \#\{p : \sigma \in \mathrm{Blamed}_p\}$*, counting blame from $[\emptyset : \emptyset]$ as $m+1$ votes if the world is known to be truthful, and return $A = \Sigma^P \setminus \{\sigma : \mathrm{votes}(\sigma) \ge m+1\}$. Then* (a) $\Sigma^* \subseteq A$*;* (b) *at most $|\mathcal{A}|(|F|+1) + m|\Sigma^*|$ oracle calls are made;* (c) *call p* complete *if it is good and finished; then $A \cap F \subseteq \{\tau \in F : \#\{p \text{ complete} : \Sigma^* \cup \{\tau\} \text{ is not } d\text{-clean relative to } p\} \le m\}$. For $m = 0$ and all positions complete this is Lemma 7.10(a).*

Proof. (a) At a good position p, every schema moved into Blamed_p is a fallacy, by Lemma 7.8(c) applied relative to p. So a genuine schema collects votes only from bad positions, of which there are at most m. The bonus for $[\emptyset : \emptyset]$ is used only when that position is known to be good, and then it blames only fallacies. So every genuine schema has at most m votes and lies in A.

(b) Every call at p either ends p's loop or moves at least one schema out of B_p. Indeed, if the call returns a refutation r, e_p is well defined, $e^+_{p,r}$ is consistent and the descent is unblocked, then the descent outputs a step s whose premises have value 1 and whose conclusion has value 0, by the argument of Lemma 7.8(a), which used only consistency and unblockedness. If s were trusted, the forward rule would have given its conclusion value 1 as well, contradicting consistency. So s is untrusted and lies in R_{B_p}, and some $\sigma \in B_p$ with $s \in \text{inst}(\sigma)$ is moved. Hence the loop at p makes at most $|B_p| + 1 \leq |\Sigma^P| + 1$ calls, and at most $|F| + 1$ at a good position, where only fallacies move. With $m' \leq m$ bad positions the total is at most $(|\mathcal{A}| - m')(|F| + 1) + m'(|F| + |\Sigma^*| + 1) \leq |\mathcal{A}|(|F| + 1) + m|\Sigma^*|$.

(c) Let p be complete. Its loop ended with NONE, so its final B_p is d-clean relative to p; and since p is good, only fallacies were moved, so $\Sigma^* \subseteq B_p$. If $\tau \in F \setminus \text{Blamed}_p$, then $\tau \in B_p$, and $\Sigma^* \cup \{\tau\} \subseteq B_p$ is d-clean relative to p by (F1). Contrapositively, each complete p relative to which $\Sigma^* \cup \{\tau\}$ is not d-clean gives τ a vote. A $\tau \in A$ has at most m votes, which gives the claim. For $m = 0$ with all positions complete, $A = \Sigma^P \setminus \bigcup_p \text{Blamed}_p$ lies inside every final B_p, so it is d-clean relative to every position; it contains Σ^*, and each of its fallacies is residual. This is the conclusion of Lemma 7.10(a). □

Why per-position loops (computed). One smallest-refutation query per position, followed by voting, does not suffice. With one bad position $[a,\ a \to b : b]$, target $\{\text{MP}\}$ and $F = \emptyset$, a single query already makes a successful descent, which blames the genuine MP; so the number of successful descents is not bounded by $(m + 1)|F|$, and votes from bad positions must be tolerated, as in (a) and (b). With $m = 1$, the truthful positions $p_1 = [q,\ p \to q,\ \neg p : p, \neg q]$ and $p_2 = [\neg r,\ r \to t : \neg t]$ and the practice $\{\text{MP}, \text{AC}, \text{DA}\}$, the smallest refutation at p_1 always blames AC and hides denying the antecedent, so DA gets one vote and survives although it is refutable at two positions. The loops above exclude already-blamed schemas, so DA is blamed at both positions and removed, while AC, refutable at one position only, may survive.

F.8 Proof of Proposition 7.11

(a) By Definition 7.9, A is computed at $t = N_1$ from $\hat{P}$, a function of the first N_1 human steps, and from answers of Ref_d, which are deterministic functions of $(B, d, \mathcal{A}, W, T_{\text{trust}})$ (Lemma 7.4(c)). Phase-1 acceptance is membership in R_A. The sandbox, the red team and the prover interact with the tier only through queries, which do not change A, and escalation answers are not added to the sample. A promoted κ is a macro, answered by expanding it into an R_A-derivation, so the primitive accepted steps remain R_A and Cl_{R_A} is unchanged. If promoted rules were accepted as primitive steps instead, then $\kappa \subseteq \text{Sound}(R_A) \subseteq \text{Sound}(R^* \cup R_{F^{(d)}_{\text{res}}})$, so the accepted set would satisfy only this weaker inclusion; the closure would be unchanged (Lemma 2.3(b)). But d-cleanness of the enlarged step set can fail. If some $\tau \in A$ is residual at depth d with a shortest refutation of size $2d$, a promoted κ whose instances package the first half of that refutation into single steps removes its intermediate judgments and yields a refutation of size at most d. Primitive promoted rules would also need truth maintenance under the depth schedule.

(b) Let $h_0 \in \mathcal{H}_S$ be clean at every depth with $w(h_0) > 0$. A detection exhibits a refutation π of the step set $B_t = \bigcap S_t$ and deletes every $h \supseteq \text{Steps}(\pi) \setminus T_{\text{trust}}$. If h_0 were deleted, the steps

of π would lie in $h_0 \cup T_{\text{trust}}$ and π would be a refutation of h_0, which is impossible. So h_0 is never deleted. In a detection, every $h \in S_t$ contains $B_t \supseteq \text{Steps}(\pi) \setminus T_{\text{trust}}$, so all of S_t is deleted and $w(\text{VS}_{t+1}) \leq \frac{1}{2} w(\text{VS}_t)$. After D detections $w(h_0) \leq w(\text{VS}) \leq 2^{-D}$, so $D \leq \log_2(1/w(h_0))$. For the robust bound, run the proof of Theorem 6.7 with h_0 in place of R^*: every detection multiplies by β the weight of a superset of S_t, so the total weight shrinks by a factor at most $\frac{1+\beta}{2}$; and h_0 is penalized only in false alarms, since a refutation of h_0 can only come from a position violating (WS). Hence $\beta^m w(h_0) \leq (\frac{1+\beta}{2})^D$. Deletion must test only the *untrusted* steps of π: if it tested all steps against hypotheses that do not contain T_{trust}, coalition members whose refutations use trusted steps would never be deleted, and halving would fail.

(c) For finite S_t, $s \in \bigcap_{h \in S_t} h$ iff for each h (a finite set of schemas) s matches some schema of h, which is decidable. For schema-valued hypotheses, $\bigcap_h \text{inst}(\sigma_h)$ is empty or the instance set of the most general common instance, obtained by unifying renamed-apart copies (Robinson, 1965). □

F.9 Proof of Theorem 7.12 and Corollary 7.13

Step 1: identification (noisy mode). For each tag i consider the following events on the first N_1 samples: (I_i) at most e_i invalid steps are tagged i; and (II_i) the valid tag-i samples are e_i-robustly generic (Theorem 5.13(b)). As in the proof of Theorem 5.14, (II_i) follows from c_i count events: for each metavariable x with best split $G_{i,x}$ (achieving $\rho_{i,x}$), more than e_i valid tag-i samples have root in $G_{i,x}$ and more than e_i have root outside it; and for each pair $x \neq y$, more than e_i have $\Theta x \neq \Theta y$. (For any f, the samples on the side of $G_{i,x}$ not containing f all have root $\neq f$.) For a ground tag the single event is that more than e_i valid samples occur; this is the convention $\rho_i = c_i = 1$. Let G be the intersection of all these events.

The number of invalid tag-i steps is $\text{Bin}(N_1, \alpha_i)$ with $\alpha_i \leq \bar{\alpha}_i$. Since $e_i = \lfloor (\bar{\alpha}_i + \Delta) N_1 \rfloor$, exceeding e_i means exceeding the mean by more than ΔN_1, which by Hoeffding's inequality (Hoeffding, 1963) has probability at most $e^{-2N_1\Delta^2}$. Each count in (II_i) is $\text{Bin}(N_1, p)$ with $p \geq \rho_i \beta_i \geq \bar{\alpha}_i + 2\Delta$ by (Floor); being at most $e_i \leq (\bar{\alpha}_i + \Delta) N_1$ means falling short of the mean by at least ΔN_1, again with probability at most $e^{-2N_1\Delta^2}$. A union bound over the $1 + c_i$ events per tag gives

$$\mathbb{P}(G^c) \leq \sum_i (1 + c_i) e^{-2N_1\Delta^2} \leq (1 + c) K e^{-2N_1\Delta^2} \leq \delta$$

by the choice of N_1. On G, Theorem 5.13(b), applied to the realizable tagged practice, shows that the trimmed verifier of tag i accepts exactly $\text{inst}(\sigma_i)$.

It remains to see that $\hat{\sigma}_i$ computes this set. On G, $n_i > e_i$. The trimmed version space of tag i consists of the hypotheses $\text{inst}(\tau)$ (and $\emptyset$) with $|P_i \setminus \text{inst}(\tau)| \leq e_i$; $\emptyset$ is excluded because $n_i > e_i$. Each $\text{inst}(\text{lgg}(P_i \setminus E))$ with $|E| = e_i$ belongs to it. Conversely, if $|P_i \setminus \text{inst}(\tau)| \leq e_i$, then $P_i \setminus E \subseteq \text{inst}(\tau)$ for some E of size e_i, so $\tau \succeq \text{lgg}(P_i \setminus E)$ (Theorem 2.5). Hence the accepted set is $\bigcap_{|E|=e_i} \text{inst}(\text{lgg}(P_i \setminus E))$, a finite intersection of instance sets. It equals $\text{inst}(\sigma_i) \neq \emptyset$, and a finite intersection of instance sets with a common instance is the instance set of the most general common instance. So $\text{inst}(\hat{\sigma}_i) = \text{inst}(\sigma_i)$, hence $\hat{\sigma}_i \equiv \sigma_i$, and $\hat{P} = \Sigma^P$ up to renaming.

Step 1 (noise-free mode). Now $e_i = 0$ and $\hat{\sigma}_i = \text{lgg}(P_i)$. Let G be the event that every tag's samples satisfy the witness events (R) and (D) of Lemma 2.7 (for a ground tag: that it is sampled). By Theorem 5.6, with its convention $\rho_i = c_i = 1$ for ground rules, $\mathbb{P}(G^c) \leq \sum_i c_i e^{-N_1 \pi_i \rho_i} \leq cK e^{-N_1 \lambda} \leq \delta$, and on G, $\text{lgg}(P_i) \equiv \sigma_i$. In both modes G depends only on the first N_1 human samples; human answers to escalated queries are not part of the sample.

Two details of the definition (computed). In noise-free mode the budget must be $e_i = 0$: with $\bar{\alpha}_i = 0$ the noisy formula still gives $e_i = \lfloor \Delta N_1 \rfloor > 0$, and then the event of Theorem 5.6 does not imply that trimmed lggs identify the rule. With one tag, one metavariable, two equiprobable roots ($\lambda = \frac{1}{2}$) and $\delta = 0.001$, $N_1 = 16$; identification fails with probability 0.077 for $e = 4$ and $3 \cdot 10^{-5}$ for $e = 0$. And with $n_i \leq e_i$ samples the trimmed version space contains the empty rule, so its intersection is $\emptyset$, not "everything"; coding it as "accept all" made TTL unsound in 12 of 20 simulated runs at $N = 60$.

Step 2: audit. On G, $\hat{P} = \Sigma^P$, and Lemma 7.10 applies. It gives the two cases of (iii), the d-cleanness of A in (i), and the audit part of (ii), with the evaluation count from Lemma 7.8(b). For the completeness claim in (iii) under (SB_d): every member of $\mathcal{C}_d$ is a singleton $\{\tau\}$, and $\tau \in F$ by Lemma 7.4(b), so $\mathrm{Coll}_d = \emptyset$; and $\mathrm{Coll}_d(B_0) \subseteq \mathrm{Coll}_d$, because a minimal member of $\mathrm{Conf}_d \cap 2^{B_0}$ is a minimal member of Conf_d (its proper subsets lie in 2^{B_0}).

Step 3: soundness. Before N_1 the tier accepts nothing. From N_1 on it accepts exactly R_A, and $A \subseteq \Sigma^* \cup F_{\mathrm{res}}^{(d)}$ gives $R_A \subseteq R^* \cup R_{F_{\mathrm{res}}^{(d)}} =: R'$. Since $R' \subseteq \mathrm{Sound}(R')$, Lemma 3.1 with R' as target gives $\mathrm{Cl}_{R_A}(B) \subseteq \mathrm{Cl}_{R'}(B)$ for all B; the version with trusted steps follows from monotonicity of Cl in the rule set (Lemma 2.3(a)). If $F_{\mathrm{res}}^{(d)} = \emptyset$ then $R' = R^*$. The accepted set does not depend on the prover's queries, so the conclusions hold for every prover.

Step 4: sandbox. Proposition 7.11(b) gives the sandbox clause of (ii), and Proposition 7.11(a) shows that sandbox activity cannot affect (i).

Step 5: depth schedule. For $B \subseteq \Sigma^P$ let m_B be the least size of a refutation of B (∞ if none) and, for the prefer-unblocked rule, u_B the least size of a refutation with an unblocked descent. Under the default rule $\mathrm{Ref}_d(B)$ is NONE for $d < m_B$ and the lexicographically least refutation of size m_B for $d \geq m_B$: it changes at most once. Under the prefer-unblocked rule it is NONE for $d < m_B$, the least smallest refutation for $m_B \leq d < u_B$ (all of them blocked), and the least smallest unblocked refutation for $d \geq u_B$: it changes at most twice. The audit at depth d is determined by the answers $\mathrm{Ref}_d(B)$, $B \subseteq \Sigma^P$, and by $\bigcup \mathcal{C}_d(B_0)$, which is determined by $\mathrm{Conf}_d = \{B : \mathrm{Ref}_d(B) \neq \mathrm{NONE}\}$. There are $2^{|\Sigma^P|}$ sets B, each with at most two change points, so the audit's output is piecewise constant in d with finitely many changes and constant for $d \geq d_1^*$. At each stage, Steps 2 and 3 apply with the current d; truth maintenance ensures that cached lemmas and macros are derivations in the current R_A. For $d \geq d_1^*$ the cleanness of each $\Sigma^* \cup \{\tau\}$ no longer changes, so $F_{\mathrm{res}}^{(d)} = F_{\mathrm{res}}$, and the tier is sound relative to $R^* \cup R_{F_{\mathrm{res}}}$.

Non-monotonicity (computed). Take the practice $\{\mathrm{MP}, \mathrm{AC}\}$, no world, and the positions $[A_1 : \emptyset]$ and $[r,\ s \to r : s]$ with r, s formulas of size 7. Then $A = \{\mathrm{MP}, \mathrm{AC}\}$ for $d < 9$; $A = \emptyset$ for $9 \leq d < 29$, by fallback on the conflict $\{\mathrm{MP}, \mathrm{AC}\}$; and $A = \{\mathrm{MP}\}$ for $d \geq 29$, where the new conflict $\{\mathrm{AC}\}$ (of size $7 + 15 + 7$ on the second position) makes the old one non-minimal and restores MP. Restored schemas are genuine or residual at the larger depth, so soundness is unaffected.

Uniformity (v). G is an event about the human samples. On G, the accepted set at each time is fixed ($\emptyset$ before N_1, R_A after, at each depth of the schedule), whatever the prover and the red team do; so (i)–(iv) hold simultaneously for all of them. □

Proof of Corollary 7.13. On G, every step the reasoner chains lies in $R_A \subseteq R^* \cup R_{F_{\mathrm{res}}^{(d)}}$, in T_{trust}, or (for the extended reasoner) is a world-channel step. The first two kinds preserve truth in $\mathfrak{M}$ by hypothesis. A world-channel step has all its judgments in J_W and is accepted only if it is not W-falsified; if its premises are true in $\mathfrak{M}$, they are W-true, so its conclusion is not W-false, hence W-true, hence true in $\mathfrak{M}$. By induction along derivations, everything derived from $\mathfrak{M}$-true

premises is $\mathfrak{M}$-true. □

F.10 Proofs of Propositions 7.14 to 7.17

Proof of Proposition 7.14. Every subset of Σ_2 is clean at every depth, so Σ_1 and Σ_2 both satisfy (WS) (Lemma 7.4(d)), and the oracle answers are the same under both (Lemma 7.4(c)). The learner's view consists of these answers, the fixed prover's queries and its own coins, so it has the same law under both targets. Let the prover query some $q \in R_{\Sigma_2} \setminus \mathrm{Sound}(R_{\Sigma_1})$ in every round. If the learner ever accepts q with probability above δ, it is not δ-sound for Σ_1; otherwise, under Σ_2, it fails with probability at least $1-\delta$ to accept the genuine step q. In the example, $\vee\mathrm{I}_1$ is not derivable from the conjunction rules, while $A \wedge B \rhd B \wedge A$ is derivable by $\wedge\mathrm{E}_2$, $\wedge\mathrm{E}_1$ and $\wedge\mathrm{I}$. □

Proof of Proposition 7.15(a). The prover is fixed. The learner's inputs (human data, oracle answers, W, escalation answers) have the same law under every $\Sigma \in \mathcal{T}$: the data law and the practice are shared, oracle answers by Lemma 7.4(c), and escalation answers follow the shared practice. Coupling the learner's coins, the whole run up to and including the answer to q has the same law, so $p(h, q)$ does not depend on Σ. If $q \notin \mathrm{Sound}(R_\Sigma \cup R_{F^{(d)}_{\mathrm{res}}(\Sigma)})$ for some admissible Σ, the event "h, then q accepted" is an unsound acceptance under Σ.

TTL passes. On G the audit runs on $\hat{P} = \Sigma^P$, identically in all admissible scenarios. Fix $\Sigma \in \mathcal{T}$, with valuations V_Σ from (WS). If a descent on π outputs s, then e^+_π agrees with V_Σ, since the agreement argument of Lemma 7.8 needs only a total valuation that extends e and falsifies no trusted step. So s has V_Σ-true premises and a V_Σ-false conclusion, and $s \notin R_\Sigma$; hence no schema of Σ is removed by descent, and Σ is contained in the returned set (descent path) or in B_0 (fallback). Let $x \in A \setminus \Sigma$. On the descent path $\Sigma \cup \{x\} \subseteq A$ is d-clean by (F1). In the fallback, if $\Sigma \cup \{x\} \subseteq B_0$ were not d-clean it would contain some $C \in \mathcal{C}_d(B_0)$, with $C \not\subseteq \Sigma$ because Σ is clean, so $x \in C$, contradicting $x \in A$. Either way $x \in F^{(d)}_{\mathrm{res}}(\Sigma)$, so $R_A \subseteq R_\Sigma \cup R_{F^{(d)}_{\mathrm{res}}(\Sigma)}$. □

Why joint probabilities and the residue. The conditional probability given h would be false for randomized learners: a learner that with probability ε accepts everything from time 1 on, and otherwise runs TTL, is $(\delta+\varepsilon)$-sound, yet given the history "q_1 accepted at time 1" it accepts every fallacy instance. And plain $\mathrm{Sound}(R_\Sigma)$ would make the intersection trivial, since typically $\emptyset \in \mathcal{T}$, convicting TTL itself.

Part (b) of Proposition 7.15 collects facts proved elsewhere: Proposition 7.6(b); Theorem 6.5(iii) for the coalition, whose members all contain a falsified fallacy instance under the stated priors (so the coalition-unanimous set does too), while a prior with more than half its mass on R^* forces $R^* \in S_t$ and hence soundness; and Proposition 3.9.

Proof of Proposition 7.16. Both worlds have the practice $\{\wedge\mathrm{I}, \mathrm{AC}\}$ with the same frequencies and instance laws, so a learner without refutation evidence has a view with the same law in both. Let the prover query an AC instance outside $\mathrm{Sound}(R_{\wedge\mathrm{I}})$ under W_1, such as $q,\ p \to q \rhd p$; it is genuine under W_2. Accepting it with probability above δ is unsound under W_1; otherwise the learner is incomplete under W_2. For the separation, under W_1 the instance $(\top,\ \bot \to \top \rhd \bot)$ has true premises and a false conclusion, while under W_2 its premise $\bot \leftrightarrow \top$ is false. □

Proof of Proposition 7.17. (a) By time t at most t refutations have been exhibited, each a finite derivation implicating finitely many instances; a fallacy with infinitely many invalid instances keeps one that is not refuted. (b) is Proposition 6.33. (c) With every finite set of steps a hypothesis, the intersection of the version space of data D is D itself (Section 5.4). □

F.11 Proof of Theorem 7.18

Existence of s, and admissibility. If every τ-instance were in $\mathrm{Sound}(R_{\{\sigma,\mu\}})$, the refutation of $\{\mu,\tau\}$ would transfer, by (F2), to a refutation of $\{\sigma,\mu\}$, which is clean at every depth. So some τ-instance s lies outside $\mathrm{Sound}(R_{\{\sigma,\mu\}})$. The target $\{\sigma,\mu\}$ of scenario 1 is clean at every depth, hence satisfies (WS), and its fallacy τ has the instance s outside $\mathrm{Sound}(R_{\{\sigma,\mu\}})$; the target $\{\sigma,\tau\}$ of scenario 2 is clean at every depth and has $F=\emptyset$. In scenario 2, s is a genuine instance.

The inequality. Let P_j be the law of one human sample (a tag and a step) in scenario j. On samples whose tag is not μ, $P_1=(1-\pi)P_2$ as measures: the frequencies are $(1-\pi)q_\sigma$ and $(1-\pi)q_\tau$ against q_σ and q_τ, and the per-tag laws coincide. Let E_t be the event that none of the first t samples has tag μ. Then $P_1^{\otimes t}$ restricted to E_t equals $(1-\pi)^t P_2^{\otimes t}$, and $P_2^{\otimes t}(E_t)=1$.

Human samples are i.i.d. and independent of everything generated before them. By (OE), every other input of round r is drawn from a kernel applied to the history, the same kernel in both scenarios. By induction over rounds, the joint law of the samples $x=(x_1,\dots,x_t)$ and the other inputs y through time t is $P_j^{\otimes t}(dx)\,K_t(x,dy)$, with one kernel K_t for both j. Whether s is accepted at time t is a function of (x,y); let $\phi(x)=K_t(x,\{\text{accept } s \text{ at } t\})\in[0,1]$. Then

$$\mathbb{P}_1(\text{accept } s \text{ at } t) \;\geq\; \int_{E_t}\phi\,dP_1^{\otimes t} \;=\; (1-\pi)^t\int_{E_t}\phi\,dP_2^{\otimes t} \;=\; (1-\pi)^t\,\mathbb{P}_2(\text{accept } s \text{ at } t),$$

so $\mathbb{P}_2\leq\delta(1-\pi)^{-t}$ whenever $\mathbb{P}_1\leq\delta$.

The bounds. $\mathbb{P}_2\geq 1-\delta'$ requires $(1-\pi)^t\leq\delta/(1-\delta')$, i.e. $t\ln\frac{1}{1-\pi}\geq\ln\frac{1-\delta'}{\delta}$, which is positive when $\delta+\delta'<1$. Since $\frac{1}{1-\pi}=1+\frac{\pi}{1-\pi}$, $\ln\frac{1}{1-\pi}\leq\frac{\pi}{1-\pi}$, which gives the second inequality. It fails without $\delta+\delta'<1$: for $\delta=\delta'=0.6$ and $\pi=\frac{1}{2}$ the two sides are -0.585 and -0.405.

Tightness. Let the learner accept τ-instances at time t iff no μ-tag occurs among the first t samples and an independent coin with bias $\min(1,\delta(1-\pi)^{-t})$ comes up heads. It does not escalate, so it satisfies (OE). In scenario 1 it accepts s at time t with probability $(1-\pi)^t\min(1,\delta(1-\pi)^{-t})\leq\delta$; in scenario 2, where μ never occurs, with probability $\min(1,\delta(1-\pi)^{-t})$. Linear programming over all learners that depend only on the tag sequence, for $t\leq 6$, confirms that this is optimal in all 108 cases checked (computed).

Practice-following escalation. By (F2) with τ and μ exchanged, some μ-instance s_μ lies outside $\mathrm{Sound}(R_{\{\sigma,\tau\}})$, hence outside the scenario-2 practice $R_{\{\sigma,\tau\}}$, while it lies in the scenario-1 practice. A learner that escalates s_μ in round 1 and accepts s in every later round iff the answer was "reject" accepts s with probability 0 in scenario 1 and 1 in scenario 2. So without (OE) the conclusion fails; the same linear program, with one practice-following escalation, raises the cap 0 to 1 (computed). A lower bound for learners with membership queries to the human would need the refuter to be unknown among many candidates; that is open. □

A concrete instance. A_1 designated, no world, $\sigma=\wedge\mathrm{I}$, $\tau=\mathrm{AC}$, $\mu=\mathrm{MP}$. The closure of A_1 under AC alone is $\{q,\ p\to q,\ p\to\bot,\ p\}$; under $\{\wedge\mathrm{I},\mathrm{AC}\}$ it is infinite but contains no $\bot$, because $\wedge$I creates no implications and AC needs an implication with consequent q or $\bot$; $\{\wedge\mathrm{I},\mathrm{MP}\}$ is classically sound; and AC followed by MP derives $\bot$.

F.12 Proof of Theorem 7.19

(a) The practice $\Sigma_0\cup\{\mu,\tau\}$ and its frequencies are the same in both scenarios, so the data and practice-following escalation answers have the same laws, and oracle answers are target-blind (Lemma 7.4(c)). Both targets are clean at every depth, hence satisfy (WS), and by (F2) each target's fallacy has an instance outside the closure of the other's calculus; in particular some

μ-instance s lies outside $\mathrm{Sound}(R_{\Sigma_0\cup\{\tau\}})$. In scenario 2, μ is not residual, because $\Sigma_0\cup\{\tau,\mu\}$ contains the conflict $\{\mu,\tau\}$; so $F^{(d)}_{\mathrm{res}}=\emptyset$ there, and accepting s is unsound, which δ-soundness caps at probability δ. The fixed prover queries s, and the acceptance probability at any given time is the same in scenario 1, where s is genuine. Finally $\{\mu\}$ and $\{\tau\}$ are clean by (F1), so $\{\mu,\tau\}\in\mathcal{C}_d$ and $\mu\in\mathrm{Coll}_d$ in scenario 1.

(b) Since $d\geq d_0$, a subset of Σ^P is d-clean iff it is clean at every depth.

Choice of M. Apply Lemma 7.7(b) with $U=B_0$ and $\mathcal{K}=\mathrm{Conf}_d\cap 2^{B_0}$, whose minimal members are $\mathcal{C}_d(B_0)$: since $g\in\bigcup\mathcal{C}_d(B_0)$, some maximal d-clean $M\subseteq B_0$ omits g. M is clean at every depth, so it satisfies (WS) by Lemma 7.4(d) (e is well defined because (WS) holds for the true target).

Admissibility of M. Let $x\in\Sigma^P\setminus M$. If $x\in B_0\setminus M$, then $M\cup\{x\}$ is not d-clean by maximality; if every x-instance were in $\mathrm{Sound}(R_M)$, (F2) would turn a refutation of $M\cup\{x\}$ into one of M, which is impossible. If $x\in\Sigma^P\setminus B_0$, then x was removed by a descent on some refutation π, with output $s\in\mathrm{inst}(x)$. The valuation V_M extends e and falsifies no trusted step, so e^+_π agrees with it (agreement in Lemma 7.8); hence s has V_M-true premises and a V_M-false conclusion, and since V_M preserves truth along R_M-derivations, $s\notin\mathrm{Sound}(R_M)$. Either way x has an instance outside $\mathrm{Sound}(R_M)$.

$F^{(d)}_{\mathrm{res}}(M)=\emptyset$. For $x\in B_0\setminus M$ this is maximality. For $x\notin B_0$, $\{x\}$ is a d-conflict by Lemma 7.8(d), applied in the true scenario, so $M\cup\{x\}$ is not d-clean by (F1).

Witness and conclusion. $g\in B_0\setminus M$, so the first case gives $s_g\in\mathrm{inst}(g)\setminus\mathrm{Sound}(R_M)$. All admissible scenarios share the data law and the oracle answers, the prover is fixed, and escalation answers follow the shared practice; so the probability that s_g is accepted at a given time is the same under every admissible target. Under M, δ-soundness relative to $R_M\cup R_{F^{(d)}_{\mathrm{res}}(M)}=R_M$ bounds it by δ. For the schema-level statement, a schema x outside $A=B_0\setminus\bigcup\mathcal{C}_d(B_0)$ lies in $\bigcup\mathcal{C}_d(B_0)$ (handled above) or outside B_0, and then its descent output is a witness outside $\mathrm{Sound}(R_M)$ for any maximal clean $M\subseteq B_0$, which is admissible as shown. A policy that includes x in A' accepts that witness when the prover queries it, so includes x with probability at most δ. On the descent path take $M=A$: at stable depth A is clean at every depth, every schema outside A has a descent witness outside $\mathrm{Sound}(R_A)$ by the second case, and $F^{(d)}_{\mathrm{res}}(A)=\emptyset$ by Lemma 7.8(d); so A is admissible and the same argument applies.

(c) *The example.* Take no world, the practice $\{\mathrm{MP},\mathrm{AC}\}$, and $\mathcal{A}=\{[A_1:\emptyset],[A_2:\emptyset]\}$ with $A_2=\{\varphi,\ \bot\to\varphi\}$, $\varphi=q\wedge(q\wedge q)$. The minimal refutation sizes are 9 for $\{\mathrm{MP},\mathrm{AC}\}$ (on A_1) and 13 for $\{\mathrm{AC}\}$ (on A_2, where AC infers $\bot$ from φ and $\bot\to\varphi$), and $\{\mathrm{MP}\}$ is clean at every depth. For $9\leq d\leq 12$ the only minimal conflict is $\{\mathrm{MP},\mathrm{AC}\}$, the descent is blocked, and TTL returns $A=\emptyset$. Yet the admissible targets are only $\{\mathrm{MP}\}$ and $\emptyset$, and asserting R_{MP} is sound under both (under $\emptyset$, MP is residual). For $d\geq 13=d_0$, TTL returns $\{\mathrm{MP}\}$.

Verification. These claims were computed with well-founded derivations and size counting distinct judgments (computed), and can be checked by hand. On A_1, AC infers p from q and $p\to q$, and MP infers $\bot$ from p and $p\to\bot$; the distinct judgments q, $p\to q$, p, $p\to\bot$, $\bot$ have $1+3+1+3+1=9$ symbols. On A_2, AC infers $\bot$ from φ and $\bot\to\varphi$, with $5+7+1=13$ symbols. The closure of A_1 under AC alone is $\{q,\ p\to q,\ p\to\bot,\ p\}$, so $\{\mathrm{AC}\}$ has no refutation on A_1; and $\{\mathrm{MP}\}$ is classically sound while both contexts are satisfied by $q=1$, $p=0$. So for $9\leq d\leq 12$, $\mathcal{C}_d=\{\{\mathrm{MP},\mathrm{AC}\}\}$. With no world and no trusted steps e is defined only on the assertions and on $\bot$; at $\bot$ the producing MP step has the unevaluated premise p and no premise of value 0, so the descent is blocked, and the fallback returns $\emptyset$. The subsets of the practice that are clean at every depth are $\emptyset$ and $\{\mathrm{MP}\}$; both are admissible, since AC is classically invalid and

MP has instances outside Sound($\emptyset$). Under $\emptyset$, MP is residual at every depth, so asserting R_{MP} is sound under both. For $d \geq 13$, $\{\mathrm{AC}\}$ is a conflict, so $\{\mathrm{MP}, \mathrm{AC}\}$ is no longer minimal; with the default oracle the blocked refutation of size 9 triggers the fallback with $\mathcal{C}_d = \{\{\mathrm{AC}\}\}$, and with the prefer-unblocked oracle the refutation on A_2 has an unblocked descent that removes AC; either way $A = \{\mathrm{MP}\}$. □

F.13 Proof of Theorem 7.20

Construction. Let Σ_0 be a finite elementary formal system (Post, 1943; Smullyan, 1961) with judgments start_e that simulates the universal machine on input e from start_e and contains the rule halt $\triangleright \bot$, as in Theorem 6.30(a). Σ_0 has no rules without premises, so without a start judgment nothing is derivable; in particular Σ_0 is clean on $[\emptyset : \emptyset]$, and the step $\triangleright\mathrm{start}_e$ is not in Sound(R_{Σ_0}). Let $\tau_e = (\triangleright\mathrm{start}_e)$ and $\Sigma_e^P = \Sigma_0 \cup \{\tau_e\}$, with fixed computable frequencies and instance laws. If $\varphi_e(e)$ diverges, Σ_e^P is clean and is the target, with $F = \emptyset$. If $\varphi_e(e)$ converges, $\Sigma_0 \cup \{\tau_e\}$ derives $\bot$ from $\emptyset$, the target is Σ_0, $F = \{\tau_e\}$ and $\tau_e \notin F_{\mathrm{res}}$; accepting $\triangleright\mathrm{start}_e$ is then a step outside $R^* \cup R_{F_{\mathrm{res}}} = R_{\Sigma_0}$.

Uniformity. For each e only one case occurs, so this is not a two-scenario argument; what is used is that the data laws and the oracle answers Ref_d (for every d, by brute force) are computable uniformly in e.

Reduction. Let $S = \{e : \exists t\ \mathbb{P}(\text{the learner accepts } \triangleright\,\mathrm{start}_e \text{ by time } t) > \frac{1}{2}\}$. If $\varphi_e(e)$ diverges, the prover queries the genuine step $\triangleright\mathrm{start}_e$ in every round, and (b) and continuity of measure give $e \in S$. If $\varphi_e(e)$ converges, (a) gives $\mathbb{P}(\text{ever accepts}) \leq \delta < \frac{1}{2}$, so $e \notin S$. Hence S is the complement of the halting set $\{e : \varphi_e(e) \text{ converges}\}$. But for a computable learner, with computable data laws, oracle and prover, the probability of acceptance by time t is lower semicomputable uniformly in (e, t), so S is Σ_1. This also holds if the learner escalates, because escalation answers follow the practice by a law computable uniformly in (e, q) (Section 7.6), such as $\mathbf{1}[q \in R_e^P]$, which is decidable by matching. Computability of the law is needed: e can be read off τ_e, so the law $\mathbf{1}[\varphi_e(e) \text{ diverges}]$ depends only on the practice, and a learner that escalates $\triangleright\mathrm{start}_e$ and accepts iff the answer is 1 would meet (a) and (b). The complement of the halting set is not Σ_1: a contradiction. □

F.14 Proofs for classical propositional logic

Proof of Proposition 7.24. (a) inst(τ) is substitution-closed and contains an unsound step, and R^* is complete for CPC, so by Proposition 3.5 $\mathrm{Cl}_{R^* \cup \mathrm{inst}(\tau)}(\emptyset)$ contains $\bot$. A derivation of $\bot$ from $\emptyset$ is a refutation of $\Sigma^* \cup \{\tau\}$ relative to $[\emptyset : \emptyset]$, so d_τ is finite. For $d \geq d_\tau$, $\Sigma^* \cup \{\tau\} \in \mathrm{Conf}_d$ contains some $C' \in \mathcal{C}_d$, and $\tau \in C'$ by Lemma 7.4(a). Hence $F_{\mathrm{res}}^{(d)} = \emptyset$ for $d \geq \max_\tau d_\tau$, and by Theorem 7.12(iii) and Corollary 7.13 TTL asserts only instances of Σ^*, which are classically valid.

(b) $\{\mathrm{AC}, K\}$ *is a conflict.* $K = A \to (B \to A)$ gives $\bot \to (p \to \bot)$ (with $A = \bot$, $B = p$) and $(p \to \bot) \to (\bot \to (p \to \bot))$ (with $A = p \to \bot$, $B = \bot$). AC, $B,\ A \to B \triangleright A$, applied to these two gives $p \to \bot$; applied to $p \to \bot$ and $\bot \to (p \to \bot)$ it gives $\bot$. This is a refutation relative to $[\emptyset : \emptyset]$. *It is minimal at every depth beyond its size.* $\{K\} \subseteq \Sigma^*$ is clean (Lemma 7.4(a)). $\{\mathrm{AC}\}$ is clean at every depth on these positions: from $\emptyset$ it derives nothing, since AC has premises, and the closure of A_1 under AC alone is $\{q,\ p \to q,\ p \to \bot,\ p\}$. Likewise, with A_1 designated, AC followed by MP derives $\bot$, and $\{\mathrm{MP}\}$ is clean, so $\{\mathrm{AC}, \mathrm{MP}\}$ is a minimal conflict. *Every descent is blocked.* With no world and no trusted steps, e is defined only on the assertions of the position

and on $\bot$. An output step would need premises among the assertions (or none) and conclusion $\bot$. No genuine step is such a step, because the positions are satisfiable and Σ^* is sound; and no AC step is, because AC with premises in A_1 yields only p, and AC needs premises. So the fallback runs at $B_0 = \Sigma^P$, K (and with A_1 also MP) lies in Coll_d for all large d, and Theorem 7.19(b) applies.

(c) τ derives the denied φ' from the asserted Π' in one step, a refutation of size at most $|\tau| \leq d$ (a closed $\top/\bot$ instance has at most the symbols of τ). So $\{\tau\} \in \mathrm{Conf}_d$ for each $\tau \in F$, every conflict contains a fallacy and hence a singleton conflict, and the minimal conflicts are exactly these singletons. Then $\mathrm{Coll}_d = \emptyset$, $F_{\mathrm{res}}^{(d)} = \emptyset$, and Lemma 7.10 gives $A = \Sigma^*$ on the descent path, and also in the fallback, where $\Sigma^* \setminus \mathrm{Coll}_d(B_0) \subseteq A \subseteq \Sigma^*$ and $\mathrm{Coll}_d(B_0) \subseteq \mathrm{Coll}_d = \emptyset$. □

A Hilbert illustration (computed; formulas with at most seven leaves over $\{p, q, \bot\}$, the least bound at which S has instances; with six leaves the conflicts are the same). Take the practice $\{K, S, \mathrm{MP}, \mathrm{DN}, \mathrm{AC}\}$ with $\mathcal{A} = \{[\emptyset : \emptyset], [A_1 : \emptyset]\}$. The minimal conflicts are $\{\mathrm{AC}, K\}$ and $\{\mathrm{AC}, \mathrm{MP}\}$, the diagnoses are $\{\mathrm{AC}\}$ and $\{K, \mathrm{MP}\}$, and the collateral is $\{K, \mathrm{MP}\}$. The conflict $\{\mathrm{AC}, K\}$ is the four-step derivation from no premises in the proof of (b) above. Both conflicts are minimal at every depth beyond their sizes, and with no world and no trusted steps every descent is blocked; so by Theorem 7.19(b), K and MP are *necessarily* collateral at full depth. With the bilateral position $[A_1 : p]$ the only minimal conflict is $\{\mathrm{AC}\}$, and the closed instance $(\bot \to \bot,\ \bot \to (\bot \to \bot) \rhd \bot)$ of AC is falsified. Whether $\{S, \mathrm{DN}, \mathrm{AC}\}$ is clean at every depth, i.e. whether the collateral also contains S or DN, is open. No matrix with at most four values witnesses its cleanness, that is, validates S and DN, with AC designation-preserving, $\bot$ undesignated and A_1 satisfiable (exhaustive backtracking, 5.0M nodes for four values with three designated). A prior that prefers fewer fallacies, Quine's maxim of minimum mutilation, picks $\{\mathrm{AC}\}$ over $\{K, \mathrm{MP}\}$; in the core $\{\mathrm{MP}, \mathrm{AC}\}$ case it is silent unless it uses frequency, which does not track validity (Section 8).

F.15 Proofs for complete decidable theories

We take Σ^* to be a standard sequent calculus: besides being sound and complete for T-consequence, it has $\to$-introduction, $\forall$-introduction, $\neg$-elimination (deriving $\vdash \bot$ from $\vdash \psi$ and $\vdash \neg\psi$), ex falso and weakening, as derived rules at least.

Proof of Lemma 7.25. (a) Let $\Pi \rhd j$ be a step. *All premises W-true.* Each premise $\Gamma \vdash \varphi$ has a universal closure true in $\mathfrak{M}$; since T is complete, T proves it, and Σ^* derives the premise from $\emptyset$. Hence $\mathrm{Cl}_{R^*}(\Pi) = \mathrm{Cl}_{R^*}(\emptyset)$, which by soundness and completeness of Σ^* and completeness of T is the set of W-true sequents; so $j \in \mathrm{Cl}_{R^*}(\Pi)$ iff $W(j) = 1$. *Some premise $\Gamma \vdash \varphi$ W-false.* By $\to$I and $\forall$I, Σ^* derives from it the sentence $\vdash \forall\bar{x}(\gamma_1 \to \cdots \to \gamma_n \to \varphi)$, which is false in $\mathfrak{M}$. T refutes it, so Σ^* derives its negation; then $\neg$E, ex falso and weakening derive every sequent, and $\mathrm{Cl}_{R^*}(\Pi) = J$. So $\Pi \rhd j \in \mathrm{Sound}(R^*)$ iff some premise is W-false or j is W-true, i.e. iff the step preserves W-truth.

(b) A schema is a fallacy iff it has an instance outside $\mathrm{Sound}(R^*)$, iff by (a) it has an instance with W-true premises and a W-false conclusion. The smallest such instance is a one-step refutation of $\{\tau\} \subseteq \Sigma^* \cup \{\tau\}$ relative to $[\emptyset : \emptyset]$, so $d_\tau \leq f_\tau$. □

Proof of Corollary 7.26. Under the realizability hypothesis Theorem 7.12 applies; (WS) holds with $V =$ truth in $\mathfrak{M}$ (of universal closures), the positions being truthful. The oracle is exact: if B is not d-clean it contains a fallacy (Lemma 7.4(b)), whose smallest falsified instance has

size $f_\tau \leq d$ and is found; if B is d-clean the search cannot find a falsified instance of size at most d, which would be a d-refutation. Each $\{\tau\}$, $\tau \in F$, is then a conflict, and the proof of Corollary 7.23 applies verbatim; truth-soundness follows from Corollary 7.13, since R^* preserves truth in $\mathfrak{M}$ by Lemma 7.25(a). □

Why learn at all? Here W is already a sound and complete step checker, so learning is not needed for epistemic access. TTL adds two things. It identifies *which* calculus the practice uses. And it makes deployed checking fast: schema matching takes linear time, while real quantifier elimination is doubly exponential in the worst case (Davenport and Heintz, 1988), deciding RCF sentences is possible in exponential space (Ben-Or et al., 1986), the first-order theory of $(\mathbb{R}, +)$ already needs exponential time, and Presburger arithmetic needs $2^{2^{\Omega(n)}}$ time (Fischer and Rabin, 1974). The audit calls W only on the few instances needed to falsify each fallacy. This dividend is real but modest.

A restricted oracle. A restricted oracle shows the division of labour more sharply. Suppose W evaluates only quantifier-free sentences with rational constants, and first-order logic is trusted. The fallacy $\tau = (\triangleright \neg \exists x\, (x \cdot x = 1 + 1))$ is a step without premises whose conclusion is not quantifier-free, so it has no instance that W can evaluate: the world alone never refutes it, and Lemma 7.25(b) fails for this oracle. The position $[\exists x\, (x \cdot x = 1 + 1) : \emptyset]$ is truthful in $\mathbb{R}$. In the refutation "$\vdash \neg\exists x(x \cdot x = 1+1)$ (by τ); $\vdash \bot$ (by trusted $\neg$E, with the asserted $\exists x(x \cdot x = 1+1)$)", the backward rule through $\neg$E gives $\neg\exists x(\dots)$ the value 0, and the descent outputs τ's step: $\{\tau\}$ is a singleton conflict. Without trusted logic the conflict is the bag $\{\tau, \neg\mathrm{E}\}$; with the bilateral position $[\emptyset :\vdash \neg\exists x(x \cdot x = 1 + 1)]$ it is $\{\tau\}$, by a one-step refutation. The position must assert the closed sentence: under the universal-closure semantics, $[x \cdot x = 1 + 1 : \emptyset]$ would assert the false $\forall x\, (x \cdot x = 2)$ and violate (WS). Over $\mathbb{R}$, then, coherence with a designated truth catches what this computation misses. Over $\mathbb{N}$ there is no converse: Δ_0 computation refutes nothing that coherence from Q does not already refute (Lemma 6.18); what it adds there is localization (Theorem 7.27(a)).

F.16 Proof of Theorem 7.27

(a) In the refutation, τ's step has no premises and $\forall$E is trusted. The conclusion $\theta(\bar{n})$ is a false Δ_0 sentence, so it has value 0; the backward rule through $\forall$E (whose only premise is $\forall\bar{x}\theta$; for k variables, through each of the k instantiation steps) gives $\forall\bar{x}\theta$ the value 0; and the descent outputs τ's step, whose (empty) premise set has value 1 vacuously. The descent is unblocked, so by Lemma 7.8(c) only schemas having $\vdash \forall\bar{x}\theta$ as an instance are removed, and these are fallacies; no axiom of PA is one, since its instances are true. With the prefer-unblocked oracle, as long as τ remains in B there is an unblocked refutation of size at most d, so the oracle returns an unblocked one, every call removes at least one fallacy and no genuine schema, and the loop does not fall back before τ is removed. *Size.* For one variable, $|\forall x\theta| = |\theta| + 2$, and $\theta(\underline{n})$ replaces each of the m occurrences of x (one symbol) by the numeral S^n0 ($n+1$ symbols), so $|\theta(\underline{n})| = |\theta| + mn$. The total is $2|\theta| + mn + 2 \leq 2|\theta| + m(n + 1) + O(1)$. For k variables the refutation has $k + 1$ judgments, each of size at most $|\theta| + m(n + 1) + O(k)$. For $\theta(x) = x < S^50$ ($|\theta| = 8$, $m = 1$, $n = 5$) the sizes are 10 and 13, total 23. Under the default tie-breaking rule a smaller *blocked* refutation through learned axioms could trigger the fallback, and those axioms could become collateral; and if the refutation ran through Q's axioms by coherence instead of through W, the

bag would contain them. So computation is subsumed by coherence as a source of refutations (Lemma 6.18), but not as a source of blame.

(b) $\mathsf{PA} \nvdash \mathrm{Con}(\mathsf{PA})$ by Gödel's second incompleteness theorem (Gödel, 1931), so $\mathsf{PA}+\neg\mathrm{Con}(\mathsf{PA})$ is consistent. A refutation of $\Sigma^* \cup \{\tau\}$ relative to a position with Δ_0 assertions and denials would derive, in first-order logic, from $\mathsf{PA}+\neg\mathrm{Con}(\mathsf{PA})$ and true Δ_0 leaves, either $\bot$ or a false Δ_0 sentence. $\mathsf{Q} \subseteq \mathsf{PA}$ proves every true Δ_0 sentence and refutes every false one (Lemma 6.18), so in either case $\mathsf{PA}+\neg\mathrm{Con}(\mathsf{PA})$ would be inconsistent. Hence $\Sigma^* \cup \{\tau\}$ is clean at every depth and $\tau \in F_{\mathrm{res}}$. Since $\Sigma^P = \Sigma^* \cup \{\tau\}$ is clean at every depth, the first call $\mathrm{Ref}_d(\Sigma^P)$ returns NONE at every d, so $A = \Sigma^P \ni \tau$ at every depth (given $\hat{P} = \Sigma^P$).

(c) *Precise form.* Fix a computable prenex normal form with negations pushed into the Δ_0 matrix, and assert an instance of a practice axiom with Σ_1 normal form $\exists \bar{x}\, \theta$ only once W has verified a witness. Then false axioms with Σ_1 normal form, $\neg\mathrm{Con}(\mathsf{PA})$ included, are never asserted, and true ones are asserted after a witness search, with no uniform time bound; for schematic axioms, acceptance becomes a semi-decision. The sentence $\forall y\, \exists p\, (\mathrm{Prf}_{\mathsf{PA}}(p, \ulcorner \bot \urcorner) \vee y \neq y)$, equivalent to $\neg\mathrm{Con}(\mathsf{PA})$ but with a Π_2 normal form, escapes the policy and is residual.

Proof. A false sentence with Σ_1 normal form $\exists \bar{x}\, \theta$ has no witness, so W never verifies one and the instance is never asserted. A true one has a witness, which enumeration finds after finitely many W-evaluations, with no computable bound on its size. The disguise $\forall y\, \exists p\, (\mathrm{Prf}_{\mathsf{PA}}(p, \ulcorner \bot \urcorner) \vee y \neq y)$ is logically equivalent to $\exists p\, \mathrm{Prf}_{\mathsf{PA}}(p, \ulcorner \bot \urcorner)$, i.e. to $\neg\mathrm{Con}(\mathsf{PA})$, because $y \neq y$ is refutable; its prenex normal form is Π_2, so the policy does not apply; and PA plus it is consistent and Δ_0-sound, so by the argument of (b) it is residual.

(d) *Cleanness.* Suppose $\Sigma^* \supseteq \mathsf{Q}$ and $\Sigma^* \cup \{\tau\} \cup A \cup \neg D$ is consistent for every designated $[A : D]$. A refutation relative to $[A : D]$ would derive, from $\Sigma^* \cup \{\tau\} \cup A$ and true Δ_0 leaves, either $\bot$, or some $\delta \in D$, or a false Δ_0 sentence. Q proves the true Δ_0 leaves and refutes false Δ_0 sentences, so in each case $\Sigma^* \cup \{\tau\} \cup A \cup \neg D$ would be inconsistent. *Impossibility.* Let (φ_e) enumerate the Σ_2 sentences, let $\Sigma^* = \mathsf{PA}$, and consider the practices $\mathsf{PA} \cup \{\vdash \varphi_e\}$, with the practice data given as a computable text. Suppose a computable learner, using the data, Ref_d at every depth and the Δ_0 oracle, eventually and permanently asserts $\vdash \varphi_e$ if φ_e is true, and eventually and permanently withholds it if φ_e is false and $\mathsf{PA} \cup \{\vdash \varphi_e\}$ is clean at every depth. Define $g(e, s)$: if a refutation of $\mathsf{PA} \cup \{\vdash \varphi_e\}$ has been found by stage s of a systematic search (computable, since each Ref_d is), output "false"; otherwise output the learner's current verdict. A true φ_e makes the practice sound, hence clean (Lemma 7.4(a)), so the search never succeeds and g converges to "true". A false φ_e either has a refutation, which the search eventually finds, or is clean, and then the learner converges to "false". So $\lim_s g(e, s)$ exists and is the truth value of φ_e, which makes Σ_2-truth Δ_2 by Shoenfield's limit lemma (Shoenfield, 1959). But the set of true Σ_2 sentences is Σ_2-complete and therefore not Π_2, let alone Δ_2, by Post's hierarchy theorem (Theorem 6.21(e)). *Randomized learners* (proof sketch). If the learner is randomized and correct in the limit with probability above $\frac{1}{2}$, the sets $\{e : \mathbb{P}(\text{eventually and permanently withholds } \vdash \varphi_e) > \frac{1}{2}\}$ and the corresponding set for assertion are Σ_2, and the same combination would make Σ_2-truth Δ_2.

(e) *Precise form.* The induction schema is not a first-order pattern: its lgg has false instances, and the audit refutes it with singleton blame, so the asserted calculus would lack induction. Add auxiliary decidable judgments $\mathrm{Sub}(\varphi, x, t, \psi)$ to J_W and write induction as $\mathrm{Sub}(\varphi, x, 0, a)$, $\mathrm{Sub}(\varphi, x, Sx, b) \rhd \ \vdash a \wedge \forall x(\varphi \to b) \to \forall x \varphi$, with the human data recording the Sub premises. This is a first-order pattern, recovered by the lgg of generic instances. Then under (Floor) a practice has at most $\min(K, \lfloor 1/(2\Delta) \rfloor)$ axiom schemas, and $T_k = \mathsf{PA} + \mathrm{Con}(\mathsf{PA}) + \cdots + \mathrm{Con}(T_{k-1})$ is identified from positive data for every k within that bound. Without a floor there

are no uniform sample bounds: comparing the targets T_{k-1} and T_k, where the tag $\mathrm{Con}(T_{k-1})$ has frequency π, under (OE), a learner that accepts $\vdash \mathrm{Con}(T_{k-1})$ at time t with probability at most δ under T_{k-1} accepts it under T_k with probability at most $1-(1-\delta)(1-\pi)^t$; so if $\delta+\delta'<1$, acceptance with probability at least $1-\delta'$ needs $t \geq \ln\frac{1-\delta}{\delta'}/\ln\frac{1}{1-\pi}$, unbounded as $\pi \to 0$. This bound is tight.

Tag count. For every tag, $\Delta \leq (\rho_i\beta_i - \bar{\alpha}_i)/2 \leq \rho_i\beta_i/2 \leq \pi_i/2$, since $\rho_i \leq 1$ and $\beta_i \leq \pi_i$. As $\sum_i \pi_i = 1$, there are at most $1/(2\Delta)$ tags. *Identification.* The axioms of Q and each $\mathrm{Con}(T_j)$ are ground, and fall under the convention $\rho_i = c_i = 1$; the Sub-encoded induction rule is a first-order pattern whose lgg on generic instances recovers it: the lgg of five generic Sub-encoded instances equals the pattern up to renaming (computed). So Theorem 5.14 applies and Theorem 7.12 gives $\hat{P} = \Sigma^P$. All practice schemas are genuine (every $\mathrm{Con}(T_j)$ is true), so $F = \emptyset$ and the audit, which never removes genuine schemas, returns $A = \Sigma^P$. Instances of the encoded rule with a false Sub premise are harmless: Sub judgments are concluded by no practice schema, so they enter derivations only as leaves, which must be W-true. *Without the induction encoding*, the lgg of induction instances is over-general and has false instances: the lgg of four instances has the false instance $(0+0=0) \wedge \forall x(0+0=x \to 0+S0=Sx) \to \forall x(0+0=x)$ (computed). For this one, $\forall x(0+0=x \to 0+S0=Sx)$ is derivable by trusted equality logic from the true Δ_0 leaf $0+S0=S(0+0)$, so backward propagation through the trusted $\to$E and $\forall$E gives an unblocked descent, and the audit removes the identified schema with singleton blame.

The floor bound. Accepting $\vdash \mathrm{Con}(T_{k-1})$ is unsound under T_{k-1}: by Gödel's second theorem $\mathrm{Con}(T_{k-1})$ is not a theorem of the consistent r.e. theory T_{k-1}, so the step is not derivable in the target calculus with trusted logic, even with all true Sub judgments as premises, and it lies outside $R^* \cup R_{F_{\mathrm{res}}} = R_{T_{k-1}}$. (It is true, so this is soundness for the target calculus, Theorem 7.12(i), not truth-soundness.) Let E_t be the event that the tag $\mathrm{Con}(T_{k-1})$ does not occur among the first t samples. As in Appendix F.11, $P_k^{\otimes t}$ restricted to E_t is $(1-\pi)^t P_{k-1}^{\otimes t}$, $P_{k-1}^{\otimes t}(E_t) = 1$, and under (OE) the acceptance probability given the samples is the same function ϕ in both scenarios. So

$$\mathbb{P}_k(\mathrm{accept}) = \int_{E_t} \phi\, dP_k^{\otimes t} + \int_{E_t^c} \phi\, dP_k^{\otimes t} \;\leq\; (1-\pi)^t \mathbb{P}_{k-1}(\mathrm{accept}) + 1 - (1-\pi)^t \;\leq\; 1-(1-\delta)(1-\pi)^t.$$

Requiring $\mathbb{P}_k \geq 1-\delta'$ gives $(1-\pi)^t \leq \delta'/(1-\delta)$, i.e. $t \geq \ln\frac{1-\delta}{\delta'}/\ln\frac{1}{1-\pi}$, positive when $\delta+\delta'<1$. *Tightness:* the learner that, by one coin at time 0, accepts the step from the start with probability δ and otherwise accepts it once its tag has occurred is uniformly δ-sound under T_{k-1}, where the tag never occurs, and under T_k accepts at time t with probability $\delta+(1-\delta)(1-(1-\pi)^t) = 1-(1-\delta)(1-\pi)^t$. Linear programming over all tag sequences confirms that the bound is attained, in all 108 cases checked (computed). The bound of Theorem 7.18 does not transfer here: for $\pi = \frac{1}{2}$, $\delta = 10^{-6}$ and $\delta' = 0.1$ it would demand $t \geq 19.8$, while the learner "accept iff the tag has occurred" is 0-sound under T_{k-1} and accepts under T_k at $t=4$ with probability 0.9375 (computed). The reason is that the rare tag now lives in the scenario where acceptance is *sound*: the learner must see the tag at least once, so the bound scales with $\ln(1/\delta')$, not $\ln(1/\delta)$. Over classes containing T_ω, identification in the limit fails altogether (Theorem 6.20); T_ω has infinitely many axioms, so it lies outside the finite practices considered here.

(f) With $[\mathrm{Con}(\mathsf{PA}) : \emptyset]$ designated, $\tau = (\triangleright\neg\mathrm{Con}(\mathsf{PA}))$ and the trusted $\neg$E derive $\bot$ from the asserted $\mathrm{Con}(\mathsf{PA})$. The backward rule gives $\neg\mathrm{Con}(\mathsf{PA})$ the value 0, and the descent outputs τ's step, so $\{\tau\}$ is a singleton conflict (compare Theorem 6.34(iii): designating true sentences removes the unsound alternatives). □

F.17 Check scripts

The figures marked (computed) in Section 7 and in this appendix come from the scripts in `research/theory/T7-checks/` of the accompanying repository, all re-run for this paper. Table 22 maps each script to the claims it checks.

script	claims checked
`duality.py`	Lemma 7.7 on 4000 random families
`overlap.py`	strictness of the instance-level inclusion in Proposition 7.6(c) (MP/AC common instance)
`noisefree.py`	the noise-free budget $e_i = 0$ (Appendix F.9, Step 1)
`prop32_voting.py`	Proposition F.2: per-position loops versus one query per position
`depth_nonmono.py`	non-monotonicity in Theorem 7.12(iv)
`two_point_bounds.py`	tightness of Theorem 7.18, its failure under practice-following escalation, and the floor bound of Theorem 7.27(e)
`thm56_finite_d.py`	the finite-depth counterexample of Theorem 7.19(c)
`hilbert_blame.py`	the $\rightarrow/\leftarrow$ witness and the Hilbert illustration (Appendix F.14)
`matrix_witness.py`, `matrix4.py`	search for a matrix witnessing that $\{S, \mathrm{DN}, \mathrm{AC}\}$ is clean at every depth
`lemma61.py`	Lemma 7.22 on 30,000 random pure schemas
`ind_lgg.py`	over-general lggs of ∀E and of induction (Corollary 7.26, Theorem 7.27(e))
`ind_sub_lgg.py`	the Sub-encoded induction rule is recovered by lgg (Theorem 7.27(e))
`ttl_sim.py`	Table 10

Table 22: Check scripts for Section 7.

G Further results and proofs for Section 8

This appendix states the secondary results that Section 8 only summarizes: Lemma G.1, Proposition G.2, Theorems G.6 and G.8, and Corollaries G.7 and G.9, the relation to the literature on algorithmic statistics, and the further pathologies. It also proves the results of the main text whose proofs are not given there. Proved in full in the main text are Theorem 8.7, Corollary 8.8, and Proposition 8.10, Lemma 8.6(i)–(ii), and Theorem 8.9 outside its idealized case. The kink rests on three lemmas, stated and proved in Appendix G.5. Appendix G.15 lists the scripts behind the computed figures and the statements changed during verification.

Conventions:

- Logarithms are base 2 unless written ln.
- κ denotes a machine constant. It absorbs $O(1)$ terms and may grow from one result to the next, but it is fixed before d, n or m.
- p_h is a shortest program for h, and $K(p_h) = K(h) + O(1)$ (Li and Vitányi, 2008).
- Prefix complexities of concatenated self-delimiting descriptions add, up to $O(1)$.
- Where a result compares the class with a semimeasure or an infinite mixture, the class is enlarged to lower-semicomputable semimeasures and computable real-valued predictors.

G.1 Empirical selection

Lemma G.1 (Empirical selection tracks population selection). *[T5 Lemma 1.2] In the computable version, suppose $h(y \mid x) \geq 2^{-B}$ for all h. With probability at least $1-\delta$, simultaneously for all h, $|\hat{\mathcal{R}}_N(h) - \mathcal{R}(h)| \leq B\sqrt{(\ell(h)\ln 2 + \ln(2/\delta))/(2N)}$. On this event, each minimizer $\hat{h}$ of $\hat{J}_c$ satisfies, for every fixed h_0,*

$$J_c(\hat{h}) \leq \Phi(c) + 2B\sqrt{((\ell(h_0) + B/c)\ln 2 + \ln(2/\delta))/(2N)}.$$

In particular, a unique population minimizer with gap γ is recovered once this bound is below γ.

This is a standard Occam-type bound: Hoeffding's inequality with a Kraft union bound.

Proof. Both minimizers exist. By Kraft, $\{h : \ell(h) \leq L\}$ has at most 2^L elements. Fix h_0. Losses lie in $[0, B]$, so $\mathcal{R}(h_0), \hat{\mathcal{R}}_N(h_0) \leq B$. Any h with $\ell(h) > \ell(h_0) + B/c$ therefore has

$$J_c(h),\ \hat{J}_c(h)\ \geq c\ell(h) > c\ell(h_0) + B \geq J_c(h_0),\ \hat{J}_c(h_0).$$

So both infima are minima over a finite set. In particular a population minimizer h_c exists, and both $\ell(h_c)$ and $\ell(\hat{h})$ are at most $\ell(h_0) + B/c$.

Uniform deviation. For a fixed h, $-\log h(y_i \mid x_i)$ are i.i.d. in $[0, B]$. By Hoeffding's inequality,

$$\mathbb{P}\big[|\hat{\mathcal{R}}_N(h) - \mathcal{R}(h)| \geq \varepsilon\big] \leq 2e^{-2N\varepsilon^2/B^2}.$$

Setting the right side equal to $\delta 2^{-\ell(h)}$ gives $\varepsilon = \varepsilon_N(h) := B\sqrt{(\ell(h)\ln 2 + \ln(2/\delta))/(2N)}$. A union bound over h, using $\sum_h 2^{-\ell(h)} \leq 1$, shows that with probability at least $1-\delta$ all deviations are at most $\varepsilon_N(h)$ simultaneously.

Excess risk. On this event,

$$J_c(\hat{h}) \leq \hat{J}_c(\hat{h}) + \varepsilon_N(\hat{h}) \leq \hat{J}_c(h_c) + \varepsilon_N(\hat{h}) \leq \Phi(c) + \varepsilon_N(h_c) + \varepsilon_N(\hat{h}).$$

Both ε_N terms are at most $B\sqrt{((\ell(h_0) + B/c)\ln 2 + \ln(2/\delta))/(2N)}$.

Uniqueness. If $J_c(h) \geq \Phi(c) + \gamma$ for every $h \neq h_c$, then a bound below γ forces $\hat{h} = h_c$. □

G.2 Relation to the literature

Tempered posteriors. The tempered posterior $\propto 2^{-\ell}\prod_i h(y_i \mid x_i)^\eta$ has its MAP at the minimizer of $\ell/\eta + N\hat{\mathcal{R}}_N$, so the note's λ is $1/\eta$: a learning rate $\eta < 1$ (Grünwald, 2012; Grünwald and van Ommen, 2017; Zhang, 2006; Bissiri et al., 2016; Bhattacharya et al., 2019). With $\lambda = cN$ the posterior tends to the non-concentrating Gibbs measure $\propto 2^{-(\ell+\mathcal{R}/c)}$ (Catoni, 2007), whose mode the proposal selects. SafeBayes learns η from data to repair misspecification; Theorem 8.11 suggests why no such criterion finds the note's η, since an imitation model is well specified *for the teacher* (our reading, not checked against the papers).

Algorithmic statistics. Kolmogorov's structure function $h_x(\alpha) = \min\{\log|S| : x \in S, K(S) \leq \alpha\}$ lies, to logarithmic precision, above the sufficiency line $K(x) - \alpha$, and, within these constraints and to logarithmic precision, every shape is realized by the structure function of some data (Vereshchagin and Vitányi, 2004); see also Gács et al. (2001); Vereshchagin and Shen (2017). Set and probability models share randomness. The *product-model structure function* $\rho^\times(k) = \min\{\mathcal{R}(h) : \ell(h) \leq k\}$ need not have slope -1 and can kink (Theorem 8.4): that is the note's insight in this language. All sufficient statistics tie at $\lambda = 1$, so $\lambda = 1 + \epsilon$ selects roughly the

minimal one and larger λ selects *insufficient* models (proof sketch), as the note wants, since the errors are structure. The objective $c\ell + \mathcal{R}$ is the algorithmic rate–distortion Lagrangian with log-loss distortion (Vereshchagin and Vitányi, 2010); classical two-part MDL (Rissanen, 1978; Barron and Cover, 1991; Grünwald, 2007) has $\lambda = 1$ (whether Barron and Cover also use $\lambda > 1$ we have not checked).

G.3 Proof of Theorem 8.2

The minimax value in (i). Let the distinct inputs be x, with multiplicities n_x. Write $L(h, f) = \sum_x n_x[-\log h(f(x) \mid x)]$. For a distribution Q on $\mathcal{C}$,

$$\mathbb{E}_{f\sim Q} L(h, f) = \sum_x n_x \big[H(Q_x) + \mathrm{KL}(Q_x \,\|\, h(\cdot \mid x))\big] \;\geq\; \sum_x n_x H(Q_x) = \sum_i H(Q_{x_i}),$$

with equality iff $h(\cdot \mid x) = Q_x$ for every x. Hence $\min_h \max_f L \geq \max_Q \sum_i H(Q_{x_i})$, and the maximum over the compact simplex is attained, since the objective is continuous.

For the reverse inequality:

- *Restricting h.* If $h(y \mid x) = 0$ for some $y \in \mathcal{C}(x)$, then $\max_f L(h, f) = \infty$. Mass outside $\mathcal{C}(x)$ only hurts. The function $h \mapsto \max_f L(h, f)$ is lower semicontinuous, with compact sublevel sets on the product of simplices. So a minimizer h° exists, and it puts mass at least some $\tau_0 > 0$ on each $y \in \mathcal{C}(x)$.
- *Sion's theorem.* For $\tau \leq \tau_0$, let H_τ be the convex set of product hypotheses with $h(y \mid x) \geq \tau$ for $y \in \mathcal{C}(x)$. On $H_\tau \times \Delta(\mathcal{C})$ the function $(h, Q) \mapsto \mathbb{E}_Q L(h, f)$ is continuous, convex in h and linear in Q, and $\Delta(\mathcal{C})$ is compact. By Sion's minimax theorem (Sion, 1958),

$$\min_h \max_f L = \inf_{h\in H_\tau} \max_Q \mathbb{E}_Q L = \max_Q \inf_{h\in H_\tau} \mathbb{E}_Q L.$$

- *Letting $\tau \to 0$.* For each Q, the hypothesis $h(\cdot \mid x) = (1-\tau')Q_x + \tau' \mathrm{Unif}(\mathcal{C}(x))$ with $\tau' = \tau|Y|$ lies in H_τ. Since $\mathrm{KL}(Q_x \,\|\, (1-\tau')Q_x + \tau' U) \leq -\log(1-\tau')$, we get $\inf_{H_\tau} \mathbb{E}_Q L \leq \sum_i H(Q_{x_i}) - N \log(1-\tau')$, uniformly in Q. Letting $\tau \to 0$ gives $\min_h \max_f L \leq \max_Q \sum_i H(Q_{x_i})$.

When the bound $\sum_i \log|\mathcal{C}(x_i)|$ is attained. We have $H(Q_x) \leq \log|\mathcal{C}(x)|$, with equality iff Q_x is uniform on $\mathcal{C}(x)$. Since the maximum is attained, the value equals $\sum_i \log|\mathcal{C}(x_i)|$ iff some Q has uniform marginals on every $\mathcal{C}(x_i)$.

The group condition. Suppose G acts on $\mathcal{C}$ with $(g \cdot f)(x) = \sigma_{g,x}(f(x))$, transitively on each $\mathcal{C}(x)$. Let $Q = \mathrm{Unif}(\mathcal{C})$, which is G-invariant. Then

$$\begin{aligned} Q_x(\sigma_{g,x} y) &= Q\{f : f(x) = \sigma_{g,x} y\} = Q\{g \cdot f' : \sigma_{g,x}(f'(x)) = \sigma_{g,x}(y)\} \\ &= Q\{g \cdot f' : f'(x) = y\} = Q\{f' : f'(x) = y\} = Q_x(y), \end{aligned}$$

where the second equality substitutes $f = g \cdot f'$ and the fourth uses the G-invariance of Q. By transitivity Q_x is uniform on $\mathcal{C}(x)$. For parities, $g_b \cdot f_a = f_{a\oplus b}$ and $\sigma_{g_b,x}(y) = y \oplus f_b(x)$. This is transitive on $\mathcal{C}(x) = \{0, 1\}$ for $x \neq 0$, and trivially so on $\mathcal{C}(0) = \{0\}$. A group acting transitively on $\mathcal{C}$ alone does not suffice: in the example of the main text, $\mathrm{Sym}(\mathcal{C})$ acts transitively on $\mathcal{C}$, but not compatibly with the $\mathcal{C}(x)$.

(ii) The uniform distribution over the distinct labelings $\mathcal{L} = \{f(x_{1:N})\}$ pays $\log|\mathcal{L}|$ against every f. Conversely, the labelings are disjoint events, so every P gives some labeling probability at most $1/|\mathcal{L}|$ (Shtarkov, 1987).

(iii) For models q with weights w_q, $\sum_q w_q \le 1$, the per-input mixture is $\xi_{\mathrm{fun}}(y \mid x) = \sum_q w_q q(y \mid x)$ and the sequence mixture is $\xi_{\mathrm{seq}}(y_{1:N} \mid x_{1:N}) = \sum_q w_q \prod_i q(y_i \mid x_i)$. We have $\xi_{\mathrm{fun}}(y \mid x) \ge w_q q(y \mid x)$ and $\xi_{\mathrm{seq}} \ge w_q \prod_i q(y_i \mid x_i)$; take logarithms. For tightness, take the models const 0 and const 1 with $w = \frac{1}{2}$, on N ones. Then $\xi_{\mathrm{fun}}(1 \mid x) = \frac{1}{2}$ at every input, so it pays N bits. And $\xi_{\mathrm{seq}}(1^N) = \frac{1}{2}$, so it pays one bit. The chain-rule conditionals of ξ_{seq} depend on past labels; that dependence *is* the shared randomness. □

G.4 Universality: sequences and blocks

Proposition G.2 (The sequence pathology, exactly)**.** *[T5 Prop 2.3, as repaired] Let $\mathbf{M}$ be a universal lower-semicomputable conditional semimeasure on label sequences, with $\mathbf{M} \ge 2^{-K(q)-c_0} q$ for every computable q (Li and Vitányi, 2008), and let $c_{\mathbf{M}} = K(\mathbf{M})$. For every $\lambda > 1$,*

$$\lambda K(\mathbf{M}) - \log \mathbf{M}(y_{1:N} \mid x_{1:N}) \le \lambda c_{\mathbf{M}} + K(q) + c_0 - \log q(y_{1:N} \mid x_{1:N}).$$

This beats q's own objective whenever $K(q) > c_{\mathbf{M}} + (c_{\mathbf{M}} + c_0)/(\lambda - 1)$.

Proof. Take logarithms in the dominance inequality and add $\lambda c_{\mathbf{M}}$. □

As $\lambda \to \infty$ the threshold tends to $c_{\mathbf{M}}$, not 0. In the function case, even the universal *conditional* semimeasure $\mathbf{m}(y \mid x) \asymp 2^{-K(y|x)}$ is a fixed predictor [T5 Thm 2.4]. For binary Y we have $K(y \mid x) = O(1)$, so $\mathcal{R}(\mathbf{m}) \ge \log_2 \frac{1}{1-2^{-\kappa_U}} > 0$ for every teacher, where κ_U is a machine constant.

Proof of Proposition 8.3. Let $Z_i = \sum_\theta w_\theta \prod_{k<i} q_\theta(y_{b,k} \mid \cdot)$. Then $h_w(y_{b,i} \mid \cdot) = Z_{i+1}/Z_i$, so the block likelihood telescopes to $Z_{|b|+1}/Z_1 \ge Z_{|b|+1} \ge w_\theta \prod_i q_\theta(y_{b,i} \mid \cdot)$, using $Z_1 = \sum_\theta w_\theta \le 1$. This holds for history-dependent personas and inputs too, because everything is conditioned on the realized history. For a finite rational family, h_w has rational outputs and is computed by one program from w and (q_θ). □

G.5 The kink

The setting is that of Section 8.3. For a product h put $s_h(x) = h(0 \mid x) - h(1 \mid x)$ and $\chi_{a'}(x) = (-1)^{\langle a', x\rangle}$, so that $\hat{s}_h(a') = \mathbb{E}_x[s_h \chi_{a'}]$. Put $\sigma_h := s_h \chi_a \in [-1, 1]$, so that $\hat{s}_h(a) = \mathbb{E}_x \sigma_h$.

Overview. In (a), the true component of P_u has weight 2^{-u} and likelihood exactly $2^{-nH(p)}$, and every other component disagrees with y^* on at least $n/2 - pn$ points. In (b), Parseval allows at most 2^{2t} parities to correlate at least 2^{-t} with h, so a short h that correlated with f_a would encode a in fewer than d bits; Jensen's inequality turns small correlation into risk near one bit; and Hoeffding's inequality for sampling without replacement (Hoeffding, 1963), with a counting argument, uses the randomness of E relative to h to remove the slack $2p$. Part (c) is the coding theorem.

Lemma G.3 (List decoding; standard)**.** *[T5 Lemma 2.5a] For $t \in \mathbb{N}$, $|\{a' : \hat{s}_h(a') \ge 2^{-t}\}| \le 2^{2t}$. Hence, if $\hat{s}_h(a) \ge 2^{-t}$, then*

$$K(a \mid p_h) \le 2t + 2\log_2(t+1) + 2\log_2 d + O(1).$$

Proof. Parseval gives $\sum_{a'} \hat{s}_h(a')^2 = \mathbb{E}_x[s_h^2] \le 1$, so at most 2^{2t} coefficients are at least 2^{-t}. Given p_h, compute every $\hat{s}_h(a')$ exactly (finite rational sums), and list those that are at least 2^{-t}. Then specify d and t self-delimitingly, followed by the $2t$-bit index of a in the list. The $2\log_2 d$ term is needed because a program for h need not determine d. For real t, apply the lemma to $\lceil t \rceil$. □

The list-size bound is the standard Parseval (Johnson-type) bound for the Hadamard code, as used by Goldreich and Levin (1989) and Kushilevitz and Mansour (1993). Only its combination with the coding theorem is used here.

Lemma G.4 (Risk versus correlation)**.** *[T5 Lemma 2.5b] For every product h,*

$$\mathcal{R}(h) \geq -\log_2\Big(\frac{1+\hat{s}_h(a)}{2}+p\Big).$$

Proof. By Jensen, $\mathcal{R}(h) \geq -\log_2 \mathbb{E}_x h(y^*(x) \mid x)$. Since $h(0 \mid x)+h(1 \mid x) \leq 1$, we have $h(y \mid x) \leq \frac{1+s_h(x)(-1)^y}{2}$. So off E, $h(y^* \mid x) = h(f_a(x) \mid x) \leq \frac{1+\sigma_h(x)}{2}$; on E it is at most 1. As $\frac{1+\sigma_h}{2} \geq 0$, we get $\mathbb{E}_x h(y^* \mid x) \leq \frac{1+\hat{s}_h(a)}{2}+p$. □

As a function of $\hat{s}_h(a)$ alone the bound is tight: an h that is right on E and has $\hat{s}_h(a)=0$ attains $\mathbb{E}_x h(y^* \mid x) = \frac{1}{2}+p$. So removing the additive p requires that E be random relative to h, which the next lemma exploits.

Lemma G.5 (Randomness of E removes the $2p$ slack)**.** *[T5 Lemma 2.5c] For every product h,*

$$\mathcal{R}(h) \geq 1-\log_2\big(1+(1-2p)\,\hat{s}_h(a)+\varepsilon_h\big), \qquad \varepsilon_h := \sqrt{32\ln 2\cdot p\,(\delta_E^+ + K(h)+\kappa_n)/n},$$

where $\delta_E^+ = \max(\delta_E, 0)$ and $\kappa_n = \log_2 n + 2\log_2(\log_2 n+2)+\kappa$.

Proof. An exact identity. Always $h(y \mid x) \leq \frac{1+s_h(x)(-1)^y}{2}$, with equality when $h(0 \mid x)+h(1 \mid x) = 1$; this is because $h(0)+h(1) \leq 1$. Also $(-1)^{y^*(x)} = \chi_a(x)(1-2\cdot 1_E(x))$. By Jensen, $\mathcal{R}(h) \geq -\log_2 \mathbb{E}_x h(y^* \mid x)$, and

$$\mathbb{E}_x h(y^* \mid x) \leq \tfrac{1}{2}+\tfrac{1}{2}\mathbb{E}_x\big[\sigma_h(1-2\cdot 1_E)\big] = \tfrac{1}{2}+\tfrac{1}{2}\hat{s}_h(a)-\tfrac{1}{n}\textstyle\sum_{x\in E}\sigma_h(x).$$

Reduction. Let $\eta_0 := \big|\frac{1}{pn}\sum_{x\in E}\sigma_h(x)-\hat{s}_h(a)\big|$, the deviation of E's average of σ_h from the population average. Then $\frac{1}{n}\sum_E \sigma_h \geq p(\hat{s}_h(a)-\eta_0)$, so

$$\mathbb{E}_x h(y^* \mid x) \leq \tfrac{1}{2}+\tfrac{1-2p}{2}\hat{s}_h(a)+p\eta_0 \quad \text{and} \quad \mathcal{R}(h) \geq 1-\log_2\big(1+(1-2p)\hat{s}_h(a)+2p\eta_0\big).$$

It suffices to show $2p\eta_0 \leq \varepsilon_h$.

Small deviations. If $\eta_0 < 1/n$, then $2p\eta_0 < 2p/n$. And $(2p/n)^2 \leq 32\ln 2\cdot p\kappa_n/n \leq \varepsilon_h^2$, because $p<1$ and $\kappa_n \geq 1$.

Counting. For $\eta>0$, let B_η be the set of pn-subsets $E' \subseteq X$ with $\big|\frac{1}{pn}\sum_{E'}\sigma_h-\hat{s}_h(a)\big| \geq \eta$. A uniformly random pn-subset is a sample without replacement from the population $\{\sigma_h(x)\}_{x\in X}$, whose mean is $\hat{s}_h(a)$ and whose range has length at most 2. Hoeffding's inequality for sampling without replacement (Hoeffding, 1963, §6) gives $|B_\eta| \leq 2e^{-pn\eta^2/2}\binom{n}{pn}$.

Coding. Otherwise $\eta_0 \geq 1/n$. Since $\eta_0 \leq 2$, we may pick $\eta = 2^{-k}$ with $\eta \leq \eta_0 < 2\eta$ and $-1 \leq k \leq d$; then $E \in B_\eta$.

- B_η is computable from p_h, from a (whose length gives d), from pn (written in $d+1$ bits) and from k (written in $\log_2(d+2)+1$ bits). Here $\sigma_h = s_h\chi_a$ is computed by running p_h on all of X. So

$$K(E \mid a, p_h) \leq \log_2|B_\eta|+\kappa_n \leq L_E - \frac{pn\eta^2}{2\ln 2}+\kappa_n.$$

- Also $L_E - \delta_E = K(E \mid a, K(a)) \le K(E \mid a) + O(1) \le K(p_h) + K(E \mid a, p_h) + O(1)$, and $K(p_h) = K(h) + O(1)$.
- Combining these, with κ absorbing the constants, gives $\frac{pn\eta^2}{2\ln 2} \le \delta_E + K(h) + \kappa_n$.

Hence $\eta_0^2 < 4\eta^2 \le 8 \ln 2\,(\delta_E^+ + K(h) + \kappa_n)/(pn)$, i.e. $(2p\eta_0)^2 < \varepsilon_h^2$. □

A numerical check (computed). The identity holds to machine precision. For random E with $d = 12$ and $p \in \{1/64, 0.15, 0.2, 0.24\}$, the observed η_0 is at most 0.08, and the bound $\frac{1}{2} + \frac{1-2p}{2}\hat{s} + p\eta_0$ is never violated.

Proof of Theorem 8.4. (a), upper bounds.

- P_u is computable from the $d-u$ known bits of a together with d, u and p. So $K(P_u) \le d - u + \kappa'$, using $K(p) = O(\log d)$.
- The true parity is a component of weight 2^{-u}. Its likelihood is exactly $(1-p)^{n-pn}p^{pn} = 2^{-nH(p)}$, because $|E| = pn$. Hence $-\log P_u(y^*) \le u + nH(p)$.

(a), lower bound on $-\log P_u(y^*)$. Every other component $f_{a'}$ disagrees with y^* on $w \ge n/2 - pn$ points. This is because $f_a \oplus f_{a'}$ has weight $n/2$, and flipping E changes at most pn of them. So the likelihood of $f_{a'}$ is

$$2^{-nH(p)}(p/(1-p))^{w-pn} \le 2^{-nH(p)}(p/(1-p))^{n(1/2-2p)},$$

and therefore $-\log P_u(y^*) \ge u + nH(p) - \log_2\left(1 + 2^u(p/(1-p))^{n(1/2-2p)}\right)$. The exponent is positive because $p < 1/4$. The last term is below 2^{-2800} already at $d = 10$, $p = 1/64$ (computed).

(a), lower bound on $K(P_u)$. Given a program for P_u and (d, u), compute the a' that maximizes $P_u(f_{a'})$, the probability of the label vector $f_{a'}$.

- Every a' with the true $(d-u)$-prefix has $P_u(f_{a'}) \ge 2^{-u}(1-p)^n$, from its own component.
- Every other a' is at Hamming distance exactly $n/2$ from every component, so $P_u(f_{a'}) = (p(1-p))^{n/2}$.
- The first is larger, because $\frac{n}{2}\log_2\frac{1-p}{p} > \frac{n}{2} \ge d \ge u$.

So the argmax has the true prefix, and $d \le K(a) \le K(P_u) + u + O(\log d)$.

(a), selection. For every u,

$$\lambda K(P_u) - \log P_u(y^*) \ge \lambda(d - u - \kappa') + u + nH(p) - o(1),$$

while the value at $u = d$ is at most $\lambda\kappa' + d + nH(p)$. A minimizer u^* therefore satisfies $(\lambda - 1)(d - u^*) \le 2\lambda\kappa' + o(1)$. In the computable version the code length is exactly $d - u$. The objective at u then exceeds the objective at $u = d$ by $(\lambda - 1)(d - u) + (o_d - o_u)$, where $o_u := u + nH(p) + \log P_u(y^*) = \log_2\left(1 + \sum_{a' \ne a} q_{a'}/q_a\right) \ge 0$, the sum running over the other components of P_u with likelihoods $q_{a'}$, is non-decreasing in u because the families are nested. So for $\lambda > 1$ it is minimized only at $u = d$.

(b). Suppose $\hat{s}_h(a) \ge 2^{-t}$. Then, by Lemma G.3,

$$d \le K(a) \le K(h) + K(a \mid p_h) + O(1) \le K(h) + 2t + 2\log_2(t+1) + 2\log_2 d + O(1).$$

This contradicts the hypothesis on $K(h)$ once κ exceeds the $O(1)$. So $\hat{s}_h(a) < 2^{-t}$.

- Lemma G.4 gives $\mathcal{R}(h) \ge -\log_2(\frac{1}{2} + 2^{-t-1} + p) = 1 - \log_2(1 + 2^{-t} + 2p)$.

- Lemma G.5, with $K(h) \le d$ (so $\varepsilon_h \le \varepsilon_E$) and $1 - 2p > 0$, gives $\mathcal{R}(h) \ge 1 - \log_2(1 + (1 - 2p)2^{-t} + \varepsilon_E)$.
- Finally, $\log_2(1 + z) \le z/\ln 2$.

(c).

- $P_h(y) := \prod_x h(y_x \mid x)$ is a semi-distribution on $\{0,1\}^n$, computable from p_h *and* d. A program for h need not determine d, which is why (c) carries the term $2\log_2 d$.
- By the conditional coding theorem, $K(y^* \mid p_h, d) \le -\log P_h(y^*) + O(1) = n\mathcal{R}(h) + O(1)$. So $K(y^*) \le K(h) + n\mathcal{R}(h) + 2\log_2 d + O(1)$.
- From y^* one computes a. The correlation of $(-1)^{y^*}$ with χ_a is $1 - 2p$, and with any other $\chi_{a'}$ it is $|-2\mathbb{E}[\chi_{a\oplus a'} 1_E]| \le 2p < 1 - 2p$. Then $E = \{y^* \ne f_a\}$. So $K(a, E) \le K(y^*) + O(1)$.
- Symmetry of information gives $K(a, E) = K(a) + K(E \mid a, K(a)) + O(1) \ge d + L_E - \delta_E - O(1)$.

Combining the last three items proves (c). □

G.6 Proof of Lemma 8.6(iii)

Fix $z = (\ell_z, r_z) \in Z$ and $c > 0$, and write $\varphi_c(y) = \langle (c, 1), y\rangle$. Compare z with the other points of Z:

- *Points to the left*, $Z_< = \{z' : \ell' < \ell_z\}$, a finite set. Here $\varphi_c(z) \le \varphi_c(z')$ iff $c \le \sigma(z') := (r' - r_z)/(\ell_z - \ell')$, strictly iff $c < \sigma(z')$.
- *Points to the right*, $Z_> = \{z' : \ell' > \ell_z\}$. Here $\varphi_c(z) \le \varphi_c(z')$ iff $c \ge \tau(z') := (r_z - r')/(\ell' - \ell_z)$, strictly iff $c > \tau(z')$.
- *Other points with the same* ℓ have a different r, and z beats them iff $r_z < r'$, for every c.

Let $s_- := \min_{Z_<} \sigma$, with $s_- = \infty$ if $Z_< = \emptyset$, and $s_+ := \max(0, \sup_{Z_>} \tau)$. If $\sup_{Z_>} \tau > 0$, the supremum is attained, because $\tau(z') \le r_z/(\ell' - \ell_z)$ and only finitely many points have $\ell' \le \ell_z + r_z/s$ for any $s > 0$.

So the set of $c > 0$ at which z is the *unique* minimizing point is the open interval (s_+, s_-) when $r_z < r'$ for all same-ℓ points, and is empty otherwise.

If $(s_+, s_-) \ne \emptyset$, then z is a vertex. Take c in the interval.

- On $Z + \mathbb{R}^2_{\ge 0}$, φ_c is uniquely minimized at z, because $\varphi_c(z' + v) > \varphi_c(z')$ for $v \ge 0$, $v \ne 0$.
- A point of $\hat{C}$ is a finite convex combination of points of $Z + \mathbb{R}^2_{\ge 0}$. Its φ_c-value equals $\varphi_c(z)$ only if every point in the combination is z. So z is the unique minimizer of φ_c over $\hat{C}$.
- Hence z is an extreme point of $\hat{C}$, supported by a line of slope $-c$, i.e. a vertex of $\mathrm{hull}^-(Z)$.

If z is a vertex, then $(s_+, s_-) \ne \emptyset$. First, extreme points of $\mathrm{conv}(S)$ lie in S, and a point $z' + v$ with $v \ne 0$ is the midpoint of $z' + v/2$ and $z' + 3v/2$. So vertices lie in Z.

- Since $z \in \mathrm{hull}^-(Z)$, some $c_0 > 0$ has z minimizing φ_{c_0}. So $s_+ \le c_0 \le s_-$, and same-ℓ points have $r' > r_z$.
- Suppose $s_+ = s_- = c_0$. Then some $z_L \in Z_<$ has $\sigma(z_L) = c_0$, since $Z_<$ is finite, and some $z_R \in Z_>$ has $\tau(z_R) = c_0$, since the supremum is attained when positive.

- Then z_L, z and z_R lie on one line of slope $-c_0$, with $\ell_L < \ell_z < \ell_R$. So z is a proper convex combination of z_L and z_R, and not extreme. This is a contradiction.

The interval and the hypothesis-level claim. The interval is (s_+, s_-), where $s_\mp$ are the absolute slopes of the edges to the adjacent vertices; at a risk-minimal vertex $s_+ = 0$. Hypotheses with the same point have equal J_c for every c, so a hypothesis is the unique minimizer on the interval iff no other hypothesis has its point. □

A computed check (computed): 300 random point sets and 400 values of c each gave no minimizer off the hull, no monotonicity violation, and every vertex selectable.

G.7 The rate threshold for arbitrary hypotheses

With $\ell = K$ and all hypotheses allowed, a hypothesis might share information across regions, or learn components only partially. For independent parities neither helps, up to explicit slack.

Take regions $X_j = \{j\} \times \{0,1\}^{d_j}$, $j \le m$, with $\mu(j,z) = \pi_j 2^{-d_j}$, and teacher $y^*(j,z) = \langle a_j, z\rangle \bmod 2$, with $K(a_1,\dots,a_m) \ge \sum_j d_j$. The rate of component j is π_j/d_j, and $\hat{s}_{h,j}$ is the correlation of h with a_j on region j. The function $\omega(t) = 1 - \log_2(1+2^{1-t})$ is the exact per-region risk floor (Lemma G.4 with $p = 0$). It is increasing and concave, with $\omega(2) = 0.415$.

Theorem G.6 (Rate threshold, idealized). *[T5 Thm 3.3, as repaired] For all $c > 0$,*

$$\Big|\Phi(c) - \sum_j \min(c\,d_j, \pi_j)\Big| \le \Delta_0(c) := c\Big(\sum_j (2\log_2 d_j + 2\log_2 m + 4 + \kappa) + \kappa\Big) + \sum_j \pi_j 2^{1-d_j/4}.$$

Moreover, suppose every $d_j \ge 8$. Let $J_c(h) \le \Phi(c) + \xi$ and $\Delta = 2\Delta_0(c) + \xi$. Then for each j:

(a) if $\pi_j < c\,d_j - \Delta - 4c$, then $\hat{s}_{h,j} < \frac{1}{2}$, so h has not learned component j;

(b) if $\pi_j\,\omega(d_j/4) > c\,d_j + \Delta$ and $\pi_j\,\omega(2) > \Delta + 8c$, then $\hat{s}_{h,j} \ge \frac{1}{2}$, so h has learned it.

The two conditions in (b) are implied by $\pi_j > c\,d_j + \Delta + \pi_j 2^{1-d_j/4}/\ln 2$ and $\pi_j > 2.41(\Delta + 8c)$.

The constant $1/\omega(2) \approx 2.41$ is the best this method gives. The second condition in (b) cannot be weakened to $\pi_j > 2\Delta + 16c$, even together with the first: with $m = 1$, $\pi_1 = 1$, $d_1 = 40$, $c \approx 10^{-8}$, $h(f_a(z) \mid z) = 0.745$ and $\xi := J_c(h)$, both of these hold, yet $\hat{s} = 0.49$ (computed).

Proof idea. The upper bound is witnessed by a hypothesis that knows a_j when $c\,d_j < \pi_j$, and flips coins elsewhere. For the lower bound, on each region we either list-decode a_j from p_h or write it out literally. Joint randomness then gives

$$K(h) \ge \sum_j (d_j - s(t_j))^+ - O\Big(\sum_j \log d_j + m\log m\Big),$$

where $s(t) = 2t + 2\log_2(t+1)$ and 2^{-t_j} is h's correlation scale on region j. The risk on region j is at least $\omega(t_j)$. Since ω is concave, each per-region minimization reduces to its endpoints.

Proof. Write $R_j(h) = \mathbb{E}_z[-\log h(y^*(j,z) \mid j,z)]$ for the normalized risk on region j, so that $\mathcal{R}(h) = \sum_j \pi_j R_j(h)$. Let $s(t) = 2t + 2\log_2(t+1)$, $u_j = 2^{1-d_j/4}$, and $C := \sum_j (2\log_2 d_j + 2\log_2 m + \kappa) + \kappa$.

Upper bound. Let h_S know a_j for $j \in S$ and flip a fair coin elsewhere. Describing j, d_j and a_j for each $j \in S$ gives $K(h_S) \le \sum_{j\in S}(d_j + 2\log_2 d_j + 2\log_2 m) + \kappa$, and $\mathcal{R}(h_S) = \sum_{j\notin S}\pi_j$. With $S = \{j : c\,d_j < \pi_j\}$,

$$J_c(h_S) \le \textstyle\sum_j \min(c\,d_j, \pi_j) + cC \le \sum_j \min(c\,d_j, \pi_j) + \Delta_0(c).$$

Lower bound: complexity. Fix h. Let t_j be the least $t \in \mathbb{N}$ with $\hat{s}_{h,j}(a_j) \geq 2^{-t}$, or ∞ if there is none. For $t_j \geq 1$ we have $\hat{s}_{h,j}(a_j) < 2^{1-t_j}$.

- Given p_h, j and d_j, we can either list-decode a_j (Lemma G.3, applied on region j) or write it literally. So $K(a_j \mid p_h) \leq \min\{s(t_j), d_j\} + 2\log_2 d_j + 2\log_2 m + \kappa$.
- Then $\sum_j d_j \leq K(a_1, \ldots, a_m) \leq K(h) + \sum_j K(a_j \mid p_h) + \kappa$.
- Since $d - \min\{s, d\} = (d - s)^+$, this gives $K(h) \geq \sum_j (d_j - s(t_j))^+ - C$.

Lower bound: risk. Lemma G.4 with $p = 0$, applied on region j, gives

$$R_j(h) \geq -\log_2 \tfrac{1+\hat{s}_{h,j}(a_j)}{2} > \omega(t_j) \quad \text{for } t_j \geq 1,$$

and $R_j \geq 0$ always. For $t_j = \infty$ we have $\hat{s} \leq 0$, so $R_j \geq 1$. Hence $J_c(h) \geq \sum_j \varphi_j(t_j) - cC$, where

$$\varphi_j(0) := c\, d_j, \qquad \varphi_j(t) := c(d_j - s(t))^+ + \pi_j \omega(t) \ (t \geq 1), \qquad \varphi_j(\infty) := \pi_j.$$

Since $s(t) \leq 4t$ for $t \geq 1$, we have $\varphi_j(t) \geq \psi_j(t) := c(d_j - 4t) + \pi_j \omega(t)$.

- The exact floor ω matters here: the cruder $1 - 2^{1-t}$ is not a lower bound for $t \geq 2$, since $\log_2(1 + u) > u$ on $(0, 1)$.
- ω is increasing and concave: with $u = 2^{1-t}$, $\omega'(t) = u/(1 + u)$, which is positive and decreasing in t.

Per-region bound. We claim $\min_t \varphi_j(t) \geq \min(c\, d_j, \pi_j) - 4c - \pi_j u_j$. The values at $t = 0$ and $t = \infty$ already satisfy it.

- If $d_j < 4$, the right side is negative and the claim is trivial.
- If $d_j \geq 4$: on $t \in [1, d_j/4]$ the bound ψ_j is concave, so it is at least $\min(\psi_j(1), \psi_j(d_j/4)) = \min(c(d_j - 4),\ \pi_j \omega(d_j/4))$. For $t > d_j/4$ we have $\varphi_j(t) \geq \pi_j \omega(t) \geq \pi_j \omega(d_j/4)$, since ω is increasing. Hence $\min_t \varphi_j \geq \min\left(c(d_j - 4),\ \pi_j(1 - \log_2(1 + u_j))\right)$.

Absorption. The first entry is at least $\min(c\, d_j, \pi_j) - 4c$. For the second, use $\log_2(1+u) - u \leq (\frac{1}{\ln 2} - 1)u \leq 0.443u$, together with

$$0.886\, d\, 2^{-d/4} \leq 4 \quad \text{for all } d,$$

whose left side has maximum 1.88.

- If $c\, d_j \geq \pi_j$, we need $\pi_j(\log_2(1+u_j) - u_j) \leq 4c$. This follows from $0.443\pi_j u_j \leq 4\pi_j/d_j \leq 4c$.
- If $c\, d_j < \pi_j$, we need $c(d_j - 4) \leq \pi_j(1 - \log_2(1 + u_j) + u_j)$. This follows from $c(d_j - 4) < \pi_j(1 - 4/d_j)$ and $\log_2(1 + u_j) - u_j \leq 0.443u_j \leq 4/d_j$.

Summing over j gives $\Phi(c) \geq \sum_j \min(c\, d_j, \pi_j) - c(C + 4m) - \sum_j \pi_j u_j = \sum_j \min(c\, d_j, \pi_j) - \Delta_0(c)$.

The per-region bound was also checked numerically, with no violations (computed), for $d \in [1, 63] \cup \{80, 100, 200, 400\}$, for $c = 2^{e/4}$ with $e \in [-60, 9]$ and for seven values of π, with both $s(t)$ and $4t$.

Identification. Let $J_c(h) \leq \Phi(c) + \xi$. For each j,

$$\varphi_j(t_j) \leq J_c(h) + cC - \sum_{i \neq j} \varphi_i(t_i) \leq \Phi(c) + \xi + cC - \sum_{i \neq j} \left[\min(c\, d_i, \pi_i) - 4c - \pi_i u_i\right].$$

Using the upper bound on Φ, this gives $\varphi_j(t_j) \leq \min(c\, d_j, \pi_j) + 2\Delta_0(c) + \xi = \min(c\, d_j, \pi_j) + \Delta$.

- *Clause (a).* If $\hat{s}_{h,j}(a_j) \geq \frac{1}{2}$, then $t_j \leq 1$, and $\varphi_j(t_j) \geq c(d_j - 4)$, since $s(1) = 4$. Under the hypothesis of (a) this exceeds $\pi_j + \Delta \geq \min(c\,d_j, \pi_j) + \Delta$, a contradiction.
- *Clause (b).* If $\hat{s}_{h,j}(a_j) < \frac{1}{2}$, then $t_j \geq 2$. On $[2, d_j/4]$, which is nonempty because $d_j \geq 8$, the bound ψ_j is concave. For $t \geq d_j/4$, including $t = \infty$, the bound is at least $\pi_j\omega(d_j/4)$. So $\varphi_j(t_j) \geq \min\left(c(d_j - 8) + \pi_j\omega(2),\ \pi_j\omega(d_j/4)\right)$. Under the two hypotheses of (b), both entries exceed $c\,d_j + \Delta \geq \min(c\,d_j, \pi_j) + \Delta$, a contradiction.
- *The implied forms* use $\omega(d/4) = 1 - \log_2(1 + u) \geq 1 - u/\ln 2$ and $1/\omega(2) = 2.4094$.

A numerical check (computed): under the conditions of (b), $\varphi_j(t) > c\,d_j + \Delta$ for every integer $t \geq 2$ in 8474 random admissible instances. The counterexample after the theorem violates the second condition of (b) by a factor of 1.01–1.06. □

G.8 Every convex frontier is realizable

Corollary G.7 (Every convex frontier is realizable, after rescaling)**.** *[T5 Cor 3.4, as repaired] Let $s_1 > \cdots > s_m > 0$ be rational and $d_j \geq 1$ integers, with $\sum_j s_j d_j \leq 1$. Set $\pi_j = s_j d_j$, and add a fair-coin region of mass $\pi_0 = 1 - \sum_j \pi_j$. Let P be the convex polygon from $(0, 1)$ whose successive edges have slope $-s_j$ and horizontal length d_j, and which is flat at height π_0 afterwards. Then the lower boundary $\breve{\rho}$ of the closure of $\hat{C}$ satisfies*

$$P(k + C') - \epsilon \leq \breve{\rho}(k) \leq P(k - C') + \epsilon,$$

where $C' = O(\sum_j \log d_j + m \log m + m\kappa)$ and $\epsilon = \sum_j \pi_j 2^{1-d_j/4}$.

Scaling every d_j up by L and every s_j down by L makes the bands negligible. So every convex, decreasing, piecewise-linear frontier that starts at height 1, has rational slopes and drops by at most one bit is realized after rescaling, up to the bands. This is the product-model analogue of the "all shapes" theorem of Vereshchagin and Vitányi (2004), with the same moral: whether a good model is a vertex is a property of the data, not a theorem. The proof is Legendre duality.

Proof. The coin region. On the coin region every hypothesis has cross-entropy at least one bit per unit mass, with equality iff it is uniform there. So $\mathcal{R}(h) \geq \pi_0 + \sum_{j \geq 1} \pi_j R_j(h)$, and the region plays no role in the complexity argument. The lower bound of Theorem G.6 therefore holds with π_0 added. The witnesses h_S are uniform on the coin region, at an extra $O(1)$ bits to recognize it. Hence $|\Phi(c) - \Psi(c)| \leq cC' + \epsilon$ with $\Psi(c) := \pi_0 + \sum_j \min(c\,d_j, \pi_j)$, for all $c \geq 0$. At $c = 0$ both sides equal π_0.

Legendre duality. $\Phi(c) = \inf_{y \in \hat{C}} \langle (c, 1), y \rangle$ is the support function of $\hat{C}$. The lower boundary $\breve{\rho}$ of its closure is convex and lower semicontinuous, so $\breve{\rho}(k) = \sup_{c \geq 0}[\Phi(c) - ck]$. Likewise $P(k) := \sup_{c \geq 0}[\Psi(c) - ck]$.

- Ψ is concave and piecewise linear, with breakpoints at $c = s_j$ and slope $\sum_{j : s_j > c} d_j$ between them.
- Its conjugate P is the polygon described in the statement: it starts at $P(0) = \Psi(\infty) = 1$, has an edge of slope $-s_j$ and length d_j for each j in decreasing order of s_j, and is flat at π_0 afterwards. For $k < 0$, $P(k) = +\infty$.

The sandwich. Then

$$\breve{\rho}(k) \geq \sup_c [\Psi(c) - c(k + C')] - \epsilon = P(k + C') - \epsilon, \qquad \breve{\rho}(k) \leq \sup_c [\Psi(c) - c(k - C')] + \epsilon = P(k - C') + \epsilon.$$

Rescaling. Replacing d_j by Ld_j and s_j by s_j/L keeps every π_j. It raises C' only logarithmically in L, so the band is $o(L)$, and $\epsilon = \sum_j \pi_j 2^{1-Ld_j/4} \to 0$. □

G.9 The note's example, verified

Return to the setting of Theorem 8.4. Let $\tau_t = 2t + 2\log_2(t+1) + 2\log_2 d + \kappa$ and $K_{\mathrm{tot}} = d + L_E - \delta_E - 2\log_2 d - \kappa$.

Theorem G.8 (Parity version of the note's example)**.** *[T5 Thm 3.5, as repaired] For every $c > 0$ and $t \in \mathbb{N}$,*

$$\Phi(c) \le \min\{1 + c\kappa,\ c(d+\kappa') + H(p),\ c(d + L_E + \kappa\log n)\}$$

and

$$\Phi(c) \ge \min\{1 - \beta_t/\ln 2,\ c(d-\tau_t) + (L_E - \delta_E - \kappa)/n,\ c\,K_{\mathrm{tot}}\}.$$

The terms correspond to the coin, the good model and the bad model, which switch near $c = (1-H(p))/d$ and $c = H(p)/L_E \approx 1/n$.

Corollary G.9 (Identification in the parity example)**.** *[T5 Cor 3.5a] If $J_c(h) \le \Phi(c) + \xi$, then $\hat{s}_h(a) \ge 2^{1-c(d+\kappa')-H(p)-\xi} - 1 - 2p$. If moreover $cn > 1$, then $\mathcal{R}(h) \ge H(p) - (\xi + c\Lambda)/(cn-1)$, where $\Lambda = \log_2(n+1) + \delta_E + 2\log_2 d + \kappa + \kappa'$.*

So for c well inside the window ($\approx 1/n, \approx (1-H(p))/d$), i.e. with $cn \gg 1$, $c\,d \ll 1$ and p small (the bound on $\hat{s}_h(a)$ is vacuous once $2^{1-H(p)} - 1 \le 2p$, e.g. at $p = 0.2$), every near-optimal hypothesis (small ξ) correlates strongly with f_a and memorizes almost none of the errors. Near either end of the window the bounds become vacuous. For example, at $d = 20$, $p = 2^{-10}$ and $\kappa = \kappa' = \xi = 0$, $c = 10^{-4}$ gives $\hat{s} \ge 0.980$ and $\mathcal{R} \ge H(p) - 2.8 \cdot 10^{-5}$, and $c = 10^{-3}$ gives $\hat{s} \ge 0.955$ (computed).

Proof of Theorem G.8. Upper bound. We list the three models.

- The coin has $K \le \kappa$ and $\mathcal{R} = 1$.
- The good model has $K \le d + \kappa'$ and $\mathcal{R} = H(p)$ exactly, since $|E| = pn$.
- The error-memorizing model y^* has $K \le d + L_E + \kappa\log n$ (index E among the pn-subsets, and give pn) and $\mathcal{R} = 0$.

Lower bound. Fix h and t.

- If $K(h) \le d - \tau_t$, then Theorem 8.4(b) gives $J_c(h) \ge \mathcal{R}(h) \ge 1 - \beta_t/\ln 2$.
- Otherwise, Theorem 8.4(c) and $\mathcal{R} \ge 0$ give $J_c(h) \ge cK + \max(0, (K_{\mathrm{tot}} - K)/n)$ with $K = K(h) > d - \tau_t$. As a function of K this is piecewise linear: its slope is $c - 1/n$ below K_{tot} and c above. Note that $K_{\mathrm{tot}} - d + \tau_t = L_E - \delta_E + s(t) \ge 0$, because $L_E - \delta_E = K(E \mid a, K(a)) \ge 0$.
  - If $c \ge 1/n$, the minimum over $K \ge d - \tau_t$ is at $K = d - \tau_t$, with value $c(d-\tau_t) + (L_E - \delta_E + s(t))/n \ge c(d-\tau_t) + (L_E - \delta_E - \kappa)/n$.
  - If $c < 1/n$, the minimum is at $K = K_{\mathrm{tot}}$, with value cK_{tot}.

In every case $J_c(h)$ is at least the minimum of the three lower-bound terms. The $2\log_2 d$ in K_{tot} and the slack in β_t move the switch points only within their $O(\log)$ slack. □

Proof of Corollary G.9. Correlation. By the upper bound of Theorem G.8, $\mathcal{R}(h) \le J_c(h) \le c(d+\kappa') + H(p) + \xi$. Lemma G.4 gives $2^{-\mathcal{R}(h)} \le \frac{1+\hat{s}_h(a)}{2} + p$, i.e. $\hat{s}_h(a) \ge 2^{1-\mathcal{R}(h)} - 1 - 2p$. Substitute the bound on $\mathcal{R}(h)$.

Memorization. By Theorem 8.4(c), $K(h) \ge K_{\text{tot}} - n\mathcal{R}(h)$, so

$$J_c(h) \ge cK_{\text{tot}} - (cn-1)\mathcal{R}(h).$$

Combine this with $J_c(h) \le c(d+\kappa') + H(p) + \xi$ and with $L_E \ge nH(p) - \log_2(n+1)$. This gives

$$(cn-1)\mathcal{R}(h) \ge c(L_E - \delta_E - 2\log_2 d - \kappa - \kappa') - H(p) - \xi \ge (cn-1)H(p) - c\Lambda - \xi.$$

Divide by $cn - 1 > 0$. □

G.10 Proof of Theorem 8.9(ii) in the idealized setting

Let h be near-optimal, and let c be such that each of v and e satisfies the hypothesis of clause (a) or that of clause (b) of Theorem G.6. Suppose v is learned, i.e. $\hat{s}_{h,v} \ge \frac{1}{2}$.

- Then v does not satisfy the hypothesis of (a), so it satisfies that of (b). In particular $\pi_v \ge \pi_v \omega(d_v/4) > c\, d_v$, i.e. $\pi_v/d_v > c$.
- If e satisfied the hypothesis of (a), then $\pi_e < c\, d_e$. That would give $\pi_e/d_e < c < \pi_v/d_v$, contradicting $\pi_v/d_v \le \pi_e/d_e$.
- So e satisfies the hypothesis of (b), and is learned.

□

G.11 Proof of Theorem 8.11

(i).

- *Monotone risk.* By Lemma 8.6(ii), for grid values $c_1 < c_i$ and any population minimizers h at c_1 and h' at c_i, $\mathcal{R}(h) \le \mathcal{R}(h')$.
- *Eventually exact.* For each grid value, Lemma G.1 gives $J_c(\hat{h}_c) \to \Phi(c)$ in probability. Only finitely many hypotheses have $\ell \le \ell(h_0) + B/c$, so the gap between $\Phi(c)$ and the best non-minimizer is positive. Hence, with probability tending to one, $\hat{h}_c$ is a population minimizer for every grid value simultaneously. Without a unique population minimizer, which minimizer is selected may change with N; with one, $\hat{h}_c = h_c$ eventually.
- *Uniform validation.* The selected hypotheses lie, with probability tending to one, in the finite set of population minimizers at the grid values. Held-out log-losses converge uniformly to risks over that set, since losses lie in $[0, B]$, and its distinct risk values are separated by a positive gap.

So, with probability tending to one, selection by held-out log-loss picks a grid value whose selected hypothesis has least risk among the selected ones. By the first item, which holds for any choice of minimizers (including ties at a hull breakpoint), the smallest grid value always attains this least risk, and it is the unique choice when its risk is strictly smallest. At finite N the second item may fail at small c, where the bound of Lemma G.1 is weakest. That is the caveat in the statement.

(ii). Strict propriety holds pointwise in x (Gneiting and Raftery, 2007). Integrating over μ, the teacher channel strictly maximizes the expected score among channels that differ from it

on a set of positive μ-mass. With a growing held-out set, empirical scores converge uniformly over finitely many candidates. So the criterion selects, among candidates, one with the largest expected score, i.e. the smallest divergence from P^* in the score's divergence.

(iii). Any statistic of imitation data has the same law in the two worlds of Theorem 8.9(iv), while the good c differs between them. □

G.12 Proof of Proposition 8.12

Measure log-loss relative to the option *none*, which gives every candidate probability $1/M$.

- *G-contexts, unguarded or guarded.* The σ-model gains g per step, by the definition of the per-step gain. Mass π_G.
- *$\neg G$-contexts, unguarded.* The teacher's step is a non-σ step. It receives probability ε/M instead of $1/M$, a loss of $\log_2(1/\varepsilon)$ per step. Mass $\pi_{\neg G}$.
- *$\neg G$-contexts, guarded.* The teacher's step receives $1/(M-1)$ instead of the unguarded ε/M, a gain of $\Delta_G = \log_2(M/\varepsilon) - \log_2(M-1)$ per step over the unguarded option. Note $\Delta_G > \log_2(1/\varepsilon)$, by $\log_2 \frac{M}{M-1}$.

So the unguarded option lowers risk by $\pi_G g - \pi_{\neg G}\log_2(1/\varepsilon) = k_\sigma\rho_\sigma$ at cost k_σ. The guard lowers it by a further $\pi_{\neg G}\Delta_G = k_G\rho_G$ at cost k_G. If the guarded option also models the teacher's alternative step in $\neg G$-contexts, its gain there only increases, and the argument is unchanged with the larger Δ_G.

In (bits, risk) relative to *none*, the three options are therefore the points $(0,0)$, $(k_\sigma, -k_\sigma\rho_\sigma)$ and $(k_\sigma + k_G, -k_\sigma\rho_\sigma - k_G\rho_G)$. The middle point is a vertex of their lower hull iff $\rho_\sigma > \rho_G$. Within the σ-applicable region apply Lemma 8.6, and across regions Theorem 8.7: if $\rho_G < \rho_\sigma$, the guarded option is selected for $c < \rho_G$, the unguarded one for $c \in (\rho_G, \rho_\sigma)$, and none for $c > \rho_\sigma$.

For the case $\rho_G \geq \rho_\sigma$: the middle point lies on or above the chord from $(0,0)$ to the guarded point. The lower hull of the three points is then that chord, with slope $-(k_\sigma\rho_\sigma + k_G\rho_G)/(k_\sigma + k_G)$, and apart from ties the selection jumps from guarded to none at that c. In the second numerical instance of the main text the jump is at $c = (0.0022 + 0.0087)/34 \approx 3.2 \cdot 10^{-4}$. □

G.13 Further pathologies

These four complement Proposition 8.12 [T5 §4.4].

Error envelopes [T5 Prop 4.5]. Suppose the error set E lies inside a region S with $K(S \mid f) = k_S$, $\mu(S) = \sigma$ and error density $\theta = \mu(E)/\sigma$ in S, and let $p = \mu(E)$. The hypothesis "f, with flip rate θ on S and 0 off S" costs $k_S + O(1)$ bits more than the good model and lowers its risk from $H(p)$ to $\sigma H(\theta)$ (a direct computation). So it is learned at its *own* rate, which can far exceed the rate of the full error component; and if $\theta > \frac{1}{2}$, the learner predicts the *error label* on all of S, imitating a coarse version of the error. For example, $p = 2^{-10}$, $\sigma = 2^{-8}$, $\theta = \frac{1}{4}$ and $k_S = 30$ give a risk reduction of $0.0112 - 0.0032 = 0.0080$ bits, a rate of $2.7 \cdot 10^{-4}$, against $1.1 \cdot 10^{-6}$ for the full error. In the note's example E is K-random, so no such envelope exists (Theorem 8.4(c)). Real systematic errors are the opposite case: low-complexity patterns of exactly this kind.

Parasitic errors [T5 §4.4(d3), as repaired]. The separability of Theorem 8.7 fails when an error is cheap given a valid rule, $K(e \mid v) \ll K(e)$: once v is included, the marginal rate of e is $r_e/K(e \mid v)$. The freshman's dream is a one-symbol mutation of $(ab)^n = a^n b^n$; the unguarded rule is the guarded one minus its guard; "or as xor" is $\vee$ plus one clause. Valid rules thus subsidize

their own mis-generalizations. The selection path is the lower hull of the subset points under the *joint* code, and bundles can enter it together. For example (computed), take a valid v with $k_v = 10$ and $r_v = 0.01$ (rate 10^{-3}), and an error e with $K(e) = 100$, $K(e \mid v) = 1$ and $r_e = 0.01$. At $c = 1.5 \cdot 10^{-3}$, relative to the empty set, $\{v\}$ changes J_c by $+0.005$, $\{e\}$ by $+0.14$, and $\{v, e\}$ by $11c - 0.02 = -0.0035$. So $\{v, e\}$ is optimal, although v's own rate and e's unconditional rate (10^{-4}) are both below c: the bundle's rate $0.02/11 \approx 1.8 \cdot 10^{-3}$ exceeds c, and v survives only because it buys e cheaply.

Distribution dependence [T5 §4.4(d7)]. Rates scale with frequency under μ. A valid rule that is rare in the training distribution, e.g. used mainly in hard problems, is discarded, and stays discarded as $N \to \infty$ with $\lambda = cN$. Under a shift to a distribution where it matters, the learned norm is wrong. The verifier of Section 8.6 then stays sound but becomes incomplete; with guard erosion it can become unsound.

Language relativity [T5 §4.4(d6), as repaired]. Take separable components $j \le m$ with risk reductions r_j, a target set S° with $r_j > 0$ for $j \in S^\circ$, and $c > 0$. Choose integers $k_j \ge 0$ with $k_j < r_j/c$ on S° and $k_j > r_j/c$ off it, and give h_S length $m + \sum_{j \in S} k_j$. The Kraft sum is $2^{-m} \prod_j (1 + 2^{-k_j}) \le 1$, so a prefix code with these lengths exists. Then $J_c(h_S)$ is a constant plus $\sum_{j \in S}(c\,k_j - r_j)$, whose terms are negative exactly on S°, so S° is the unique selection (no failures in 3000 random instances (computed)). For universal machines U and V the invariance theorem bounds the damage: $|K_U - K_V| \le c_{UV}$, so a marginal cost $K(h_{S \cup \{j\}}) - K(h_S)$ changes by at most $2c_{UV}$, and rates by the factor $k/(k \pm 2c_{UV})$. That is negligible for 10 000-bit errors and decisive for 20-bit schemas, which is the inference-rule regime. A learner whose description language is shaped by human concepts, for instance by pretraining on human text, therefore gives human-natural errors short codes, and raises exactly the rates of the fallacies humans find natural.

G.14 Proofs for the division of labour

Throughout, the components are finite, $\mathrm{val}(S) = \sum_{j \in S} \nu_j$, and by separability minimizing J_c over admissible sets is maximizing val over $\mathfrak{A}$. Two facts are used repeatedly.

- If $V_+ \cup \{e\} \notin \mathfrak{A}$, then e has a witness inside V_+. Indeed, a minimal $W \subseteq V_+$ with $W \cup \{e\} \notin \mathfrak{A}$ is minimal among all subsets of V.
- Conversely, a witness inside V_+ makes $V_+ \cup \{e\}$ inadmissible, by down-closure.

Proof of Theorem 8.14. Overview. An optimal set has no negative-value component. If it contained an error that is inadmissible together with V_+, it would have to omit some $B \subseteq V_+$ meeting that error's witnesses; valid dominance makes restoring B and dropping the error better, and error-independence keeps the result admissible.

Step 0: $S^*(c)$ is admissible. Apply error-independence with $U = V_+$.

Step 1: optimal sets have no negative components. Let S be admissible and optimal. Removing the components with $\nu < 0$ keeps admissibility (down-closure) and raises the value. So $S \subseteq \{\nu \ge 0\}$.

Step 2: optimal sets contain no error inadmissible with V_+. Let $E_1 := (S \cap F) \setminus \{e : V_+ \cup \{e\} \in \mathfrak{A}\}$, and suppose $E_1 \neq \emptyset$.

- Each $e \in E_1$ has a witness inside V_+. Every such witness W is nonempty, because $\{e\} \subseteq S$ is admissible.

- Since $(S \cap V) \cup \{e\} \subseteq S$ is admissible, $S \cap V$ contains no witness of e. So $B := V_+ \setminus S$ meets every witness of every $e \in E_1$ inside V_+. In particular $B \neq \emptyset$, and $\sum_B \nu > 0$.
- If E_1 contains some $e \in F_+$, then $\sum_B \nu \geq \beta(e) > \sum_{F_+} \nu \geq \sum_{E_1} \nu$, using valid dominance and $E_1 \subseteq \{\nu \geq 0\}$. Otherwise $E_1 \subseteq \{\nu = 0\}$, and $\sum_{E_1} \nu = 0 < \sum_B \nu$.
- Let $S' := V_+ \cup ((S \cap F) \setminus E_1)$. It is admissible by error-independence with $U = V_+$. Since $S \cap V \subseteq V_+ \cup \{\nu = 0\}$, we have $\sum_{S \cap V} \nu = \sum_{V_+} \nu - \sum_B \nu$, and therefore $\mathrm{val}(S') = \mathrm{val}(S) + \sum_B \nu - \sum_{E_1} \nu > \mathrm{val}(S)$.

This contradicts optimality, so $E_1 = \emptyset$.

Step 3: comparison with $S^*(c)$. Hence $S \cap V \subseteq V_+ \cup \{\nu = 0\}$ and $S \cap F \subseteq \{e : V_+ \cup \{e\} \in \mathfrak{A},\ \nu_e \geq 0\}$. So $\mathrm{val}(S) \leq \mathrm{val}(S^*(c))$. Since $S^*(c)$ is admissible, it is optimal, and $\mathrm{val}(S) = \mathrm{val}(S^*(c))$. Every element of $S^*(c)$ has positive value, and S is contained in $S^*(c) \cup \{\nu = 0\}$, so equality forces $S \supseteq S^*(c)$ and $S \setminus S^*(c) \subseteq \{\nu = 0\}$. Conversely, every admissible $S^*(c) \cup Z$ with $Z \subseteq \{\nu = 0\}$ has the optimal value.

Why Step 3 compares values directly. Adding $V_+ \setminus S$ to an optimal S need not keep admissibility when S contains zero-valued valid rules. For example, take $V = \{v_+, v_0\}$ with $\nu(v_0) = 0$, and an error e whose only witness is $\{v_+, v_0\}$. Then $S = \{v_0, e\}$ is admissible, but $S \cup \{v_+\}$ is not.

The attributions in the statement. When $\mathfrak{A}$ is the intersection of the coherence, world and postulate families, $V_+ \cup \{e\} \notin \mathfrak{A}$ iff it fails at least one of them. An error with $\nu_e \leq 0$ (rate at most c) is excluded by simplicity, up to zero-value ties. A valid rule with $\nu_v \leq 0$ is not in V_+, whatever the filters. □

Proof of Corollary 8.15. Recall that $\max \emptyset = 0$ and $k_j > 0$.

If. Suppose $\max\{\rho_e : V \cup \{e\} \in \mathfrak{A}\} < c < \min_v \rho_v$.

- Then $V_+ = V$, and no valid rule has value 0.
- Every e with $V \cup \{e\} \in \mathfrak{A}$ has $\rho_e < c$, so $\nu_e < 0$. Hence $S^*(c) = V$.
- By Theorem 8.14, the optimal sets are the admissible $V \cup Z$ with $Z \subseteq \{\nu = 0\} \cap F$. A zero-valued error has $\rho_e = c$, so $V \cup \{e\} \notin \mathfrak{A}$. By down-closure $V \cup Z$ is then inadmissible unless $Z = \emptyset$.

So V is the unique optimum, and separation holds.

Only if. We show that separation fails if either inequality fails.

- *If $c \geq \rho_v$ for some v,* then $\nu_v \leq 0$ and $v \notin S^*(c)$. Since $S^*(c)$ is optimal, some optimal set omits v.
- *If $c < \min_v \rho_v$, but some e with $V \cup \{e\} \in \mathfrak{A}$ has $\rho_e \geq c$:*
  - if $\rho_e > c$, then $e \in F_+$ and $e \in S^*(c)$, which is optimal;
  - if $\rho_e = c$, then $\nu_e = 0$, and $S^*(c) \cup \{e\}$ is admissible by error-independence with $U = V$, and optimal.

  Either way some optimal set contains an error.

□

Valid dominance is essential in Theorem 8.14 and Corollary 8.15. Take $V = \{v\}$ with rate 10^{-3}, and an error e of rate $2 \cdot 10^{-3}$, both with $k = 10$, whose witness is $\{v\}$. No error survives relative to V, yet for $c < 10^{-3}$ the optimum is $\{e\}$ (computed).

Proof of Proposition 8.17(a). The postulates enter only through $\mathfrak{A}$, so Theorem 8.14 applies at each c.

The characterization at each c. If $c \geq \rho_e$, then $\nu_e \leq 0$, and e is excluded up to zero-value ties. If $c < \rho_e$, then $e \in F_+$. In that case $e \in S^*(c)$ iff $V_+(c) \cup \{e\} \in \mathfrak{A}$, iff no witness of e lies inside $V_+(c)$, by the two facts at the head of this subsection. Moreover $\nu_e > 0$, so e lies in every optimal set iff it lies in $S^*(c)$.

Exclusion at every c.

- *If.* Suppose some witness W has $\min_{w \in W} \rho_w \geq \rho_e$. Then $W \subseteq V_+(c)$ for every $c < \rho_e$, so e is excluded at every such c, and also at every $c \geq \rho_e$. This covers the case $W = \emptyset$, i.e. $\{e\}$ already inadmissible.
- *Only if.* Suppose every witness has a member of rate below ρ_e. There are finitely many witnesses, so we can choose $c \in \big(\max_W \min_{w \in W} \rho_w,\ \rho_e\big)$, with $\max \emptyset = 0$; this uses $\rho_e > 0$. At that c no witness lies inside $V_+(c)$, so $e \in S^*(c)$, and e survives in every optimal set.

□

Proof of Proposition 8.16. The repaired set is admissible. First, $(V_+ \setminus B) \cup \{e\} \in \mathfrak{A}$. Otherwise a minimal $W \subseteq V_+ \setminus B$ with $W \cup \{e\} \notin \mathfrak{A}$ would be a witness of e inside V_+ that B misses. Second, for $e' \in S_{\text{clean}} \cap F$, $(V_+ \setminus B) \cup \{e'\} \subseteq V_+ \cup \{e'\} \in \mathfrak{A}$, so it is admissible by down-closure. By error-independence with $U = V_+ \setminus B$, the set $S' = (V_+ \setminus B) \cup \{e\} \cup (S_{\text{clean}} \cap F)$ is admissible.

Its value. Since e has a witness inside V_+, $V_+ \cup \{e\} \notin \mathfrak{A}$, so $e \notin S_{\text{clean}}$. Hence $\text{val}(S') = \text{val}(S_{\text{clean}}) - \sum_B \nu + \nu_e = \text{val}(S_{\text{clean}}) + \nu_e - \beta(e) > \text{val}(S_{\text{clean}})$. □

In 1 882 random instances with $\beta(e) < \nu_e$, the clean set was strictly suboptimal every time (computed).

G.15 Computations and repairs

The figures marked `computed` in Section 8 and in this appendix come from the scripts in `research/theory/T5-checks/` of the accompanying repository and from the brute-force checks of the adversarial verification (Table 23). Section 12 re-runs `c1`–`c5`.

Statements changed during verification. Results tagged "as repaired" were changed during adversarial verification; the records are in `research/verification/`. The substantive changes are: clause (b) of Theorem G.6, strengthened (the counterexample after the theorem refutes the earlier form); the term $2 \log_2 d$ in Theorem 8.4(c) and the assumption $K(p) = O(\log d)$; Lemma G.5, added to remove the slack $2p$ from Theorem 8.4(b); the uniqueness condition in Lemma 8.6(iii); the exact minimax value and the group condition in Theorem 8.2(i); the threshold in Proposition G.2; the guard gain Δ_G in Proposition 8.12, which an earlier bound understated; the zero-value case in the proof of Theorem 8.14; the fixed-grid form of Theorem 8.11(i); the Kraft construction for language relativity; and, for parasitic errors, the withdrawal of a claim that conditional rates fix the order in which components enter.

H Proofs for Section 9

The following results are proved completely in the main text: Propositions 9.3 and 9.23, Theorems 9.4 and 9.14, and Corollary 9.20. In Example 9.13 the first claim is a one-line check

script	what it computes	used in
`c1`	the note's own numbers: rates of the model and of the error, the window of c, the MDL switch point	Section 8.4
`c2`	per-input versus Bayes mixture for parities; list mixtures missing u bits; shared versus product code lengths	Theorems 8.2 and 8.4
`c3`	threshold rule and hull: random instances and point sets, values of c	Theorem 8.7 and Lemma 8.6
`c4`	rules toy: gains and rates of fallacies and valid rules, guard erosion, sacrifice, witness rates of postulates, held-out log-loss	Propositions 8.12, 8.16 and 8.17 and Theorem 8.11, Table 11
`c5`	in-context copier versus norm model	Proposition 8.3
`repair_checks`	the minimax example; the p-crossover and ε_E; the identity and concentration of Lemma G.5; the $o(1)$ term; the counterexample and per-region checks for Theorem G.6; the numbers for Corollary G.9; the guard's second instance; the parasitic bundle; the Kraft construction; the counterexamples in Appendix G.14	Theorems 8.2, 8.4, 8.14 and G.6, Corollary G.9, and Proposition 8.12, Appendix G.13
verifier	brute force: 17 806 instances of the division theorem, 1 882 of repair by sacrifice	Theorem 8.14 and Proposition 8.16

Table 23: Scripts behind the `computed` figures of Section 8 (`research/theory/T5-checks/`; the verifier's checks are recorded in `research/verification/T5-verification.md`).

(the union bound of $p \wedge q \rhd p$ is $(1 - 0.9) + 0.2 = 0.3 < 1$), and the facet list is computed (`research/theory/T6-checks/prob_coherence.py`); it is also easily verified by hand from the four vertices $(0,0,1), (0,1,1), (1,0,0), (1,1,1)$. Several secondary results are stated only here: Theorem H.1 and Propositions H.2 and H.4 (§H.3), Theorem H.6 (§H.4), and the material on term models and pins (§H.6). There, Theorems H.7, H.8 and H.11 and Beth's theorem in Proposition H.10 are cited, not reproved; the non-standardness clause of Theorem H.7 and the examples after Proposition H.10 are argued where they are stated. Everything else is proved below. Throughout, "theory" means closed set.

H.1 The abstract frame

Proof of Lemma 9.2. Both sides of $K \subseteq \mathrm{Mod}(\Sigma) \iff \Sigma \subseteq \mathrm{Th}(K)$ say that $m \models \sigma$ for all $m \in K$, $\sigma \in \Sigma$. Both maps are inclusion-reversing. Taking $K = \mathrm{Mod}(\Sigma)$ gives $\Sigma \subseteq \mathrm{ThMod}(\Sigma)$, and taking $\Sigma = \mathrm{Th}(K)$ gives $K \subseteq \mathrm{ModTh}(K)$; so both composites are extensive and monotone. Applying the inclusion-reversing Mod to $\Sigma \subseteq \mathrm{ThMod}\Sigma$ gives $\mathrm{ModThMod}\Sigma \subseteq \mathrm{Mod}\Sigma$, and the converse is extensivity of ModTh at $\mathrm{Mod}\Sigma$; so $\mathrm{ModThMod} = \mathrm{Mod}$, and dually $\mathrm{ThModTh} = \mathrm{Th}$. Idempotence of both composites follows. The fixed points of $\mathrm{Th} \circ \mathrm{Mod}$ are exactly the sets $\mathrm{Th}(K)$, those of $\mathrm{Mod} \circ \mathrm{Th}$ the sets $\mathrm{Mod}(\Sigma)$, and the two identities show that Mod and Th are mutually inverse on them. Fixed-point sets of closure operators on power sets are closed under arbitrary intersections, so they are complete lattices. □

Proof of Theorem 9.5. We first record a consequence of finitarity: *the union U of a nonempty chain of closed sets is closed.* Indeed $C(U) = \bigcup\{C(U_0) : U_0 \subseteq U \text{ finite}\}$, each finite U_0 lies in a single member T of the chain, and $C(U_0) \subseteq C(T) = T \subseteq U$.

(a) Let T be closed and $\sigma \notin T$. The family of closed $T' \supseteq T$ with $\sigma \notin T'$ is nonempty (it contains T) and, by the observation, closed under unions of nonempty chains. By Zorn's lemma it has a maximal member T_σ. Then $T_\sigma \neq S$, and every closed $T' \supsetneq T_\sigma$ contains σ by maximality;

so T_σ is a point containing T. Hence $\bigcap\{P \in \mathrm{Pt}(C) : P \supseteq T\} \subseteq \bigcap_{\sigma \notin T} T_\sigma \subseteq T$, since $\tau \notin T$ implies $\tau \notin T_\tau$. The reverse inclusion is trivial. If $T = S$ there is no point above T and the empty intersection is S.

(b) ($\Leftarrow$) Soundness gives $C(X) \subseteq \mathrm{Th}(\mathrm{Mod}(X))$. Conversely, by (a), $C(X) = \bigcap_i P_i$ for the points $P_i \supseteq C(X)$. By hypothesis $P_i = \mathrm{Th}(\{m_i\})$, and $m_i \in \mathrm{Mod}(X)$ because $X \subseteq P_i$. So $\mathrm{Th}(\mathrm{Mod}(X)) \subseteq \bigcap_i \mathrm{Th}(\{m_i\}) = C(X)$.

($\Rightarrow$) Let P be a point, with witness σ. By strong completeness, $P = C(P) = \mathrm{Th}(\mathrm{Mod}(P)) = \bigcap_{m \in \mathrm{Mod}(P)} \mathrm{Th}(\{m\})$. If $\mathrm{Mod}(P) = \emptyset$ the right-hand side is $S \neq P$; so $\mathrm{Mod}(P) \neq \emptyset$. Each $\mathrm{Th}(\{m\})$ with $m \in \mathrm{Mod}(P)$ is closed, by soundness, and contains P. If each of them strictly contained P, each would contain σ, and so would their intersection P, which is impossible. So $P = \mathrm{Th}(\{m\})$ for some m.

(c) Proper closed sets are closed under unions of nonempty chains: the union U is closed by the observation, and if $U = S$ then the finite set F lies in a single member T of the chain, so $T \supseteq C(F) = S$, a contradiction. By Zorn, every proper closed set extends to a maximal proper closed set. ($\Leftarrow$) If X is coherent, $C(X)$ is proper and extends to a maximal proper closed T; by hypothesis $T = \mathrm{Th}(\{m\})$, and $m \in \mathrm{Mod}(X)$. ($\Rightarrow$) Let T be maximal proper closed. Since $C(T) = T \neq S$, weak completeness gives $m \in \mathrm{Mod}(T)$. Then $T \subseteq \mathrm{Th}(\{m\}) \neq S$, and $\mathrm{Th}(\{m\})$ is closed *by soundness*; maximality gives $T = \mathrm{Th}(\{m\})$. *Soundness is needed for ($\Rightarrow$).* Let $S = \{a, b, \bot\}$ with closed sets $\emptyset$, $\{a\}$, S (so C is finitary and $C(\{\bot\}) = S$), and let $M = \{m\}$ with $\mathrm{Th}(\{m\}) = \{a, b\} \neq S$. The C-coherent sets are $\emptyset$ and $\{a\}$, both satisfied by m, so C is weakly complete; $\{a\}$ is maximal proper closed but differs from $\mathrm{Th}(\{m\}) = \{a, b\}$, which is not closed, so C is unsound ($C(\{b\}) = S \not\subseteq \mathrm{ThMod}(\{b\}) = \{a, b\}$). `repair_checks.py` (A) reproduces it and finds no failure of (b) or (c) on 1,602 random sound finite frames. □

Details for Proposition 9.6. Soundness. Every Boolean valuation is a model of IPC, so each $\mathrm{Th}(\{v\})$ is IPC-closed. *Failure of strong completeness.* $p \vee \neg p$ is true in every Boolean valuation, and refuted in the three-element Heyting algebra by $h(p) = \frac{1}{2}$, so $\mathsf{IPC} \nvdash p \vee \neg p$. *Weak completeness.* The hypotheses of Theorem 9.5(c) hold with $F = \{\bot\}$, and $\mathrm{Th}(\{v\}) \not\ni \bot$. Let T be a maximal proper IPC-closed set; proper means consistent, as $\{\bot\}$ is explosive. If $\varphi \notin T$, then $C_{\mathsf{IPC}}(T \cup \{\varphi\})$ is closed and strictly larger than T, hence equals Fm; so $T, \varphi \vdash_{\mathsf{IPC}} \bot$ and, by the deduction theorem, $\neg\varphi = \varphi \to \bot \in T$. So $\varphi \in T$ or $\neg\varphi \in T$, whence $\varphi \vee \neg\varphi \in T$ for every φ. As consequence relations $\mathsf{CPC} = \mathsf{IPC} + \mathrm{EM}$, where EM is the set of instances of excluded middle; so $C_{\mathsf{CPC}}(T) = C_{\mathsf{IPC}}(T \cup \mathrm{EM}) = C_{\mathsf{IPC}}(T) = T \not\ni \bot$. Thus T is CPC-consistent, and maximal CPC-consistent (any CPC-consistent superset is IPC-consistent), hence $T = v^{-1}(1)$ for a Boolean v. Theorem 9.5(c) concludes. Alternatively, weak completeness follows from Glivenko's theorem (Glivenko, 1929). □

Proof of Lemma 9.7. Let T be maximal among Cn_2-theories omitting φ, and suppose neither ψ nor $\neg\psi$ is in T. By maximality $\varphi \in \mathrm{Cn}_2(T \cup \{\psi\})$ and $\varphi \in \mathrm{Cn}_2(T \cup \{\neg\psi\})$. By proof by cases, $\mathrm{Cn}_2(T \cup \{\psi\}) \cap \mathrm{Cn}_2(T \cup \{\neg\psi\}) = \mathrm{Cn}_2(T) = T$, so $\varphi \in T$, a contradiction. Hence T is complete (it contains ψ or $\neg\psi$ for every ψ). Since $T \neq \mathrm{Fm}$ it is consistent, and a complete consistent theory is maximal consistent. □

H.2 Positions

Proof of Theorem 9.8. (a) Order the coherent positions extending $[X : Y]$ componentwise. A union of a nonempty chain $[X_k : Y_k]$ is coherent: an incoherence witness $\Gamma \rhd \Delta$ with $\Gamma \subseteq \bigcup X_k$, $\Delta \subseteq \bigcup Y_k$ involves finitely many formulas, so it lies in a single $[X_k : Y_k]$. Zorn gives a maximal

coherent $[X^* : Y^*]$. *Disjoint:* if $\varphi \in X^* \cap Y^*$, then $\{\varphi\} \rhd \{\varphi\} \in \vdash$ by (Ov) witnesses incoherence. *Exhaustive:* suppose $\varphi \notin X^* \cup Y^*$. By maximality both $[X^* \cup \{\varphi\} : Y^*]$ and $[X^* : Y^* \cup \{\varphi\}]$ are incoherent, so there are $\Gamma_1, \varphi \rhd \Delta_1$ and $\Gamma_2 \rhd \varphi, \Delta_2$ in $\vdash$ with $\Gamma_i \subseteq X^*$, $\Delta_i \subseteq Y^*$. (If φ does not occur in a witness, that witness already shows $[X^* : Y^*]$ incoherent.) By (Wk) both hold with context $\Gamma_1 \cup \Gamma_2$, $\Delta_1 \cup \Delta_2$; (Cut) on φ gives $\Gamma_1 \cup \Gamma_2 \rhd \Delta_1 \cup \Delta_2 \in \vdash$, which witnesses incoherence of $[X^* : Y^*]$: a contradiction. So $v = \chi_{X^*}$ is a valuation with $v^{-1}(0) = Y^*$. It satisfies every $\Gamma \rhd \Delta \in \vdash$, since otherwise $\Gamma \subseteq X^*$ and $\Delta \subseteq Y^*$. And $v[X] = 1$, $v[Y] = 0$.

(b) For $v \in \mathrm{Val}(\vdash)$, the position $[v^{-1}(1) : v^{-1}(0)]$ is coherent, because v satisfies every sequent of $\vdash$. It is maximal: a proper extension puts some φ on both sides, which (Ov) makes incoherent. Conversely, a maximal coherent position is, by the proof of (a), of this form. □

Proof of Corollary 9.9. Soundness, $\langle S\rangle \subseteq \mathrm{Th}(\mathrm{Mod}(S))$: the sequents valid in every $v \in \mathrm{Mod}(S)$ contain S and form a Scott relation. Every valuation satisfies the (Ov) instances and preserves (Wk); for (Cut), if $v[\Gamma] = 1$ and $v[\Delta] = 0$, the premises force $v(\varphi) = 0$ and $v(\varphi) = 1$. *Completeness.* If $\Gamma_0 \rhd \Delta_0 \notin \langle S\rangle$, then $[\Gamma_0 : \Delta_0]$ is $\langle S\rangle$-coherent, since a witness inside it would yield $\Gamma_0 \rhd \Delta_0$ by (Wk). Theorem 9.8 gives $v \in \mathrm{Val}\langle S\rangle \subseteq \mathrm{Mod}(S)$ realizing $[\Gamma_0 : \Delta_0]$, i.e. violating $\Gamma_0 \rhd \Delta_0$. □

Proof of Theorem 9.10. Syntactic closure. By Corollary 9.9, $\mathrm{Th} \circ \mathrm{Mod}$ on sets of sequents is $S \mapsto \langle S\rangle$, so its fixed points are exactly the Scott relations. *Semantic closure.* For $V \subseteq 2^{\mathrm{Fm}}$, $\mathrm{Mod}(\mathrm{Th}(V)) = \overline{V}$. The set $\mathrm{Mod}(\mathrm{Th}(V))$ is an intersection of the sets $\{v : v \models \Gamma \rhd \Delta\}$, each of which is clopen (it depends on finitely many coordinates); so it is closed, and it contains V; hence it contains $\overline{V}$. Conversely, let $w \notin \overline{V}$. Some basic open neighbourhood of w, determined by finitely many coordinates, misses V; it has the form $U(\Gamma, \Delta) = \{v : v[\Gamma] = 1,\ v[\Delta] = 0\}$ with $\Gamma = w^{-1}(1) \cap E$, $\Delta = w^{-1}(0) \cap E$ for a finite $E \subseteq \mathrm{Fm}$. Since $U(\Gamma, \Delta) \cap V = \emptyset$, every $v \in V$ satisfies $\Gamma \rhd \Delta$, so $\Gamma \rhd \Delta \in \mathrm{Th}(V)$; and $w \in U(\Gamma, \Delta)$ violates it. So $w \notin \mathrm{Mod}(\mathrm{Th}(V))$. This is Lemma 6.23(a); it is formalized as `Bilateral.Val_Th` for sequents over an arbitrary set Fm (and as `Carnap.Val_mrel` for the propositional language). *Duality.* By Lemma 9.2, Mod = Val and Th are mutually inverse inclusion-reversing bijections between the fixed points on the two sides, i.e. between Scott relations and closed subsets of 2^{Fm}. □

The Stone-duality reading after Theorem 9.10 needs only Stone's theorem (Stone, 1936), the observation that the clopen set of the clause $\bigvee_{\gamma\in\Gamma} \neg\gamma \vee \bigvee_{\delta\in\Delta} \delta$ is $\{v : v \models \Gamma \rhd \Delta\}$, and conjunctive normal form in B. `scott_duality.py` checks Corollary 9.9 on 300 random sequent sets over four formulas, and Theorem 9.10 exhaustively over three formulas (all 256 sets of valuations are Galois-closed, and $V \mapsto \mathrm{Th}(V)$ is injective).

Proof of Theorem 9.11. For every substitution σ and valuation v, $(v \circ \sigma)(\varphi) = v(\sigma\varphi)$, so

$$v \circ \sigma \models \Gamma \rhd \Delta \quad \text{iff} \quad v \models \sigma\Gamma \rhd \sigma\Delta. \tag{2}$$

(a) ($\Rightarrow$) Let $\vdash$ be structural and $v \in \mathrm{Val}(\vdash)$. For $\Gamma \rhd \Delta \in \vdash$ we have $\sigma\Gamma \rhd \sigma\Delta \in \vdash$, so v satisfies it, so $v \circ \sigma \models \Gamma \rhd \Delta$ by (2). Hence $v \circ \sigma \in \mathrm{Val}(\vdash)$. ($\Leftarrow$) Let $\mathrm{Val}(\vdash)$ be substitution-invariant and $\Gamma \rhd \Delta \in \vdash$. For every $v \in \mathrm{Val}(\vdash)$, $v \circ \sigma \in \mathrm{Val}(\vdash)$ satisfies $\Gamma \rhd \Delta$, so $v \models \sigma\Gamma \rhd \sigma\Delta$. Thus $\sigma\Gamma \rhd \sigma\Delta \in \mathrm{Th}(\mathrm{Val}(\vdash)) = \vdash$, the last equality by Corollary 9.9 since $\vdash$ is a Scott relation. The duality statement is the restriction of Theorem 9.10: a closed substitution-invariant V equals $\mathrm{Val}(\mathrm{Th}(V))$, so $\mathrm{Th}(V)$ is structural by ($\Leftarrow$).

(b) An assignment into $L_v = \langle \mathrm{Fm}, v^{-1}(1)\rangle$ is a homomorphism $\mathrm{Fm} \to \mathrm{Fm}$, i.e. a substitution σ, and it satisfies $\Gamma \rhd \Delta$ in the matrix iff not ($\sigma\Gamma \subseteq v^{-1}(1)$ and $\sigma\Delta \cap v^{-1}(1) = \emptyset$), iff $v \models \sigma\Gamma \rhd \sigma\Delta$,

iff $v \circ \sigma \models \Gamma \rhd \Delta$. So L_v validates $\Gamma \rhd \Delta$ iff $v \circ \sigma \models \Gamma \rhd \Delta$ for all σ. If $\vdash$ is structural and $\Gamma \rhd \Delta \in \vdash$, this holds by (a). If $\Gamma \rhd \Delta \notin \vdash$, the position $[\Gamma : \Delta]$ is coherent (a witness inside it would give $\Gamma \rhd \Delta$ by (Wk)), so Theorem 9.8 gives $v \in \mathrm{Val}(\vdash)$ with $v[\Gamma] = 1$, $v[\Delta] = 0$, and the identity assignment in L_v violates $\Gamma \rhd \Delta$. Likewise, a coherent position $[X : Y]$ is realized by some $v \in \mathrm{Val}(\vdash)$, and the identity assignment in L_v designates X and no member of Y. □

H.3 Credences

Proof of Theorem 9.12. (i)⇒(iii): if $P = \sum_v \mu(v)\bar{v}$ and $\lambda \cdot \bar{v} \geq c$ for all v, then $\lambda \cdot P = \sum_v \mu(v)\, \lambda \cdot \bar{v} \geq c$. (i)⇒(ii): $\sum_v \mu(v)\, \lambda \cdot (\bar{v} - P) = \lambda \cdot P - \lambda \cdot P = 0$, so $\lambda \cdot (\bar{v} - P)$ cannot be negative for every v. (iii)⇒(i) and (ii)⇒(i): Π_F is the convex hull of finitely many 0/1 points, hence a polytope with rational vertices, hence (Weyl–Minkowski) $\Pi_F = \{x : a_i \cdot x \geq b_i,\ i \leq r\}$ with rational a_i, b_i. If $P \notin \Pi_F$, one of these inequalities, say $a \cdot x \geq b$ (a facet inequality, or half of an affine-hull equation), has $a \cdot P < b \leq a \cdot \bar{v}$ for all $v \in \mathrm{BV}$. This violates (iii) with $\lambda = a$, $c = b$. And $\lambda = -a$ is a Dutch book: $\lambda \cdot (\bar{v} - P) = a \cdot P - a \cdot \bar{v} < 0$ for every v. (iii)⇒(iv): for a valid (Φ, m) let λ_φ be the multiplicity of φ in Φ and $c = m$. Then $\lambda \cdot \bar{v}$ is the number of members of Φ true in v, which is $\geq m$, so (iii) gives $\sum_\Phi P = \lambda \cdot P \geq m$. (iv)⇒(iii): let λ, c be rational with $\lambda \cdot \bar{v} \geq c$ for all v. Multiplying both by a positive common denominator preserves both inequalities, so assume $\lambda \in \mathbb{Z}^F$, $c \in \mathbb{Z}$. For each coordinate φ with $\lambda_\varphi < 0$ use $\lambda_\varphi v(\varphi) = |\lambda_\varphi|\, v(\varphi^*) - |\lambda_\varphi|$, valid for every $v \in \mathrm{BV}$ since $v(\varphi^*) = 1 - v(\varphi)$, and the same identity for P, valid by negation coherence. Let Φ be the multiset containing λ_φ copies of φ for each $\lambda_\varphi > 0$ and $|\lambda_\varphi|$ copies of φ^* for each $\lambda_\varphi < 0$ (multiplicities add if φ^* also has a positive coefficient), and let $m = c + \sum_{\lambda_\varphi < 0} |\lambda_\varphi|$. Then for every v, $\lambda \cdot \bar{v} = \#_\Phi(v) - \sum_{\lambda_\varphi < 0} |\lambda_\varphi|$, where $\#_\Phi(v)$ counts the members of Φ true in v; so (Φ, m) is valid. Likewise $\lambda \cdot P = \sum_\Phi P - \sum_{\lambda_\varphi < 0} |\lambda_\varphi|$, so (iv) gives $\lambda \cdot P \geq c$. Finally, $P \equiv 1$ satisfies every valid counting sequent, since validity at any world forces $m \leq |\Phi|$; but it is incoherent, as $\bar{v}$ never has $v(\varphi) = v(\varphi^*) = 1$. □

The threshold hierarchy. With F_k and P_k as in Theorem 9.14:

Theorem H.1 (The counting-sequent hierarchy by threshold is strict)**.** *[T6 Thm 3.5, as repaired]* (a) *P_k violates the valid counting sequent $\Phi = \{\neg A_1, \ldots, \neg A_k\} \cup \{A_i \wedge A_j : i < j\}$, $m = k - 1$, which repeats no formula.* (b) *P_k satisfies every valid counting sequent over F_k with $m \leq k - 2$, with arbitrary repetitions; in particular ($m = 1$) the union bound of every valid ordinary multiple-conclusion sequent.*

Proof. (a) In a world with t true atoms, $(k-t)+\binom{t}{2} \geq k-1$ members of Φ hold, as $(t-1)(t-2) \geq 0$; but $\sum_\Phi P_k = \frac{k(k-2)}{k-1} < k - 1$.

(b) Let Φ contain a_t copies of A_t, b_t of $\neg A_t$, c_{ij} of $A_i \wedge A_j$ and n_{ij} of $\neg(A_i \wedge A_j)$. Put $N = \sum n_{ij}$ and $x_t = b_t - a_t$, and write $\#(S)$ for the number of members of Φ (with multiplicity) true in the world whose true atoms are S. *Two kinds of worlds.* $\#(\emptyset) = \sum_t b_t + N$. In the world $\{t\}$ all conjunctions are still false, A_t becomes true and $\neg A_t$ false, so $\#(\{t\}) = \#(\emptyset) - x_t$. *The credence sum.* Directly,

$$\sum_\Phi P_k = \frac{\sum_t a_t}{k-1} + \frac{(k-2)\sum_t b_t}{k-1} + N = \#(\emptyset) - \frac{\sum_t x_t}{k-1}.$$

Validity. Let $s = \#(\emptyset) - m$. Validity at $\emptyset$ gives $s \geq 0$, and validity at $\{t\}$ gives $x_t \leq s$. *Violation forces $m \geq k - 1$.* Suppose (Φ, m) is valid and $\sum_\Phi P_k < m$, i.e. $\sum_t x_t > (k-1)s$. If $s = 0$ this contradicts $x_t \leq 0$; so $s \geq 1$. Since $a_t \geq 0$, $x_t \leq b_t$, so $\#(\emptyset) \geq \sum_t b_t \geq \sum_t x_t \geq (k-1)s + 1$

by integrality. Therefore $m = \#(\emptyset) - s \geq (k-2)s + 1 \geq k-1$. Contrapositively, every valid counting sequent with $m \leq k-2$ is satisfied. *The $m = 1$ case.* An ordinary sequent $\Gamma \rhd \Delta$ over F_k corresponds to $\Phi = \{\gamma^* : \gamma \in \Gamma\} \cup \Delta$, $m = 1$; it is valid iff the counting sequent is, and by negation coherence its union bound is $\sum_\Phi P_k \geq 1$. Since $1 \leq k-2$ for $k \geq 3$, (b) covers it. □

An exhaustive enumeration for $k = 3$ (503,270 valid ordinary sequents, none violated) agrees. For $k = 3, 4, 5$, `repair_checks.py` (C) minimizes $\sum_\Phi P_k - m$ over valid counting sequents with multiplicities in $[0, 8]$ by exact MILP: the minimum is 0 for $m \leq k-2$ and $-\frac{1}{k-1}$ for $m \leq k-1$, attained by the repetition-free sequent of (a).

Scope. For any *single* finite agenda (a union of complementary pairs) and negation-coherent P, the finitely many rational inequalities defining Π_F (its facets, plus its affine hull when it is not full-dimensional) give finitely many counting sequents that characterize coherence (Theorem 9.12; negation coherence is needed, since counting sequents alone cannot impose $P(\varphi) + P(\varphi^*) \leq 1$). What Theorems 9.14 and H.1 rule out is a bound on arity or threshold that works *uniformly across agendas* (F_k needs arity k and threshold $k-1$). Moreover, deciding coherence on arbitrary agendas is NP-hard (Georgakopoulos et al., 1988) ($P(\varphi) = 1$, $P(\varphi^*) = 0$ is coherent iff φ is satisfiable), so no efficiently checkable constraint family characterizes it unless $\mathsf{P} = \mathsf{NP}$. Closing the agenda changes the picture:

Proposition H.2 (On logically closed agendas, local additivity suffices)**.** *[T6 Prop 3.6] If F represents a finite Boolean subalgebra of the Lindenbaum algebra and P respects equivalence, then P is coherent iff $P \geq 0$, $P(\top) = 1$ and $P(\varphi) = P(\varphi \wedge \psi) + P(\varphi \wedge \neg\psi)$ for all $\varphi, \psi \in F$; each constraint mentions at most three formulas, and the additivity identity $P(A \wedge B) + P(A \vee B) = P(A) + P(B)$ from the probing notes follows from them.*

Proof. ($\Rightarrow$) If $P = \sum_v \mu(v)\bar{v}$, the three conditions hold valuation by valuation: $v(\varphi) = v(\varphi \wedge \psi) + v(\varphi \wedge \neg\psi)$ for every Boolean v. ($\Leftarrow$) Taking $\varphi = \psi = \top$ gives $P(\top) = P(\top) + P(\bot)$, so $P(\bot) = 0$. Let $a_1, \ldots, a_N$ be the atoms (minimal nonzero elements) of the finite Boolean algebra F; distinct atoms are disjoint, and every nonzero element lies above some atom. Splitting successively,

$$P(\varphi) = P(\varphi \wedge a_1) + P(\varphi \wedge \neg a_1) = \cdots = \sum_{j \leq N} P(\varphi \wedge a_j) + P\Big(\varphi \wedge \bigwedge_{j \leq N} \neg a_j\Big),$$

using $\varphi \wedge \neg a_1 \wedge \cdots \wedge \neg a_{j-1} \wedge a_j \equiv \varphi \wedge a_j$; all formulas involved lie in F up to equivalence. The last term is $P(\bot) = 0$, and $\varphi \wedge a_j$ is a_j or $\bot$; so $P(\varphi) = \sum_{a_j \leq \varphi} P(a_j)$. Each a_j is consistent; pick $v_j \in \mathrm{BV}$ with $v_j(a_j) = 1$. For $\varphi \in F$, $v_j(\varphi) = 1$ iff $a_j \leq \varphi$: if $a_j \not\leq \varphi$ then $a_j \wedge \varphi \equiv \bot$, so $a_j \leq \neg\varphi$. Put $\mu = \sum_j P(a_j)\delta_{v_j}$. It is nonnegative with total mass $\sum_j P(a_j) = P(\top) = 1$, and $\mu(\varphi) = \sum_{a_j \leq \varphi} P(a_j) = P(\varphi)$. The additivity identity follows: $P(A \vee B) = P((A \vee B) \wedge A) + P((A \vee B) \wedge \neg A) = P(A) + P(B \wedge \neg A)$ and $P(B) = P(B \wedge A) + P(B \wedge \neg A)$. □

Lemma H.3 (Elementary consequences of Gaifman coherence)**.** *Let P satisfy (G1), (G2). Then:* (i) $P(\neg\varphi) = 1 - P(\varphi)$*;* (ii) *if* $\vdash \varphi \leftrightarrow \psi$ *then* $P(\varphi) = P(\psi)$*;* (iii) *if* $\vdash \varphi \to \psi$ *then* $P(\varphi) \leq P(\psi)$*;* (iv) $\max(P(\varphi), P(\psi)) \leq P(\varphi \vee \psi) \leq P(\varphi) + P(\psi)$*;* (v) $P(\varphi) + P(\psi) - 1 \leq P(\varphi \wedge \psi) \leq \min(P(\varphi), P(\psi))$.

Proof. (i) Apply (G2) and (G1) to $\varphi \vee \neg\varphi$. (ii) $\varphi \vee \neg\varphi$ and $\psi \vee \neg\varphi$ are both provable, and both pairs are provably exclusive; so $P(\varphi) + P(\neg\varphi) = 1 = P(\psi) + P(\neg\varphi)$. (iii) ψ is provably equivalent to $\varphi \vee (\psi \wedge \neg\varphi)$, a provably exclusive disjunction, so $P(\psi) = P(\varphi) + P(\psi \wedge \neg\varphi) \geq P(\varphi)$ by (ii) and (G2). (iv) The lower bound is (iii). For the upper bound, $P(\varphi \vee \psi) = P(\varphi) + P(\psi \wedge \neg\varphi) \leq$

$P(\varphi) + P(\psi)$ by (ii), (G2) and (iii). (v) The upper bound is (iii). For the lower bound, $1 - P(\varphi \wedge \psi) = P(\neg\varphi \vee \neg\psi) \leq P(\neg\varphi) + P(\neg\psi)$ by (i), (ii), (iv). □

Proof of Theorem 9.15. ($\Leftarrow$) If $P(\varphi) = \mu([\varphi])$: a provable φ lies in every complete consistent theory, so $[\varphi] = S_L$; and if $\vdash \neg(\varphi \wedge \psi)$, then $[\varphi] \cap [\psi] = \emptyset$ and $[\varphi \vee \psi] = [\varphi] \cup [\psi]$. ($\Rightarrow$) By the completeness theorem, $[\varphi] = [\psi]$ iff $\vdash \varphi \leftrightarrow \psi$, and $[\varphi] \cap [\psi] = \emptyset$ iff $\vdash \neg(\varphi \wedge \psi)$. Every clopen subset of S_L is compact, hence a finite union of basic clopens, hence of the form $[\varphi]$. So $\mu_0([\varphi]) := P(\varphi)$ is a well-defined function on the clopen algebra (Lemma H.3(ii)), with $\mu_0(S_L) = P(\top) = 1$, nonnegative, and finitely additive by (G2). It is countably additive on the clopen algebra: if $[\varphi] = \bigsqcup_n [\varphi_n]$, compactness of $[\varphi]$ and openness of the $[\varphi_n]$ give a finite subcover, and disjointness makes all other $[\varphi_n]$ empty. By Carathéodory's extension theorem μ_0 extends to a probability on the σ-algebra generated by the clopens. Since L is countable there are countably many clopens, and they form a basis, so every open set is a countable union of clopens and this σ-algebra is the Borel σ-algebra. Uniqueness: the clopens form a π-system generating it (π–λ theorem). □

Proof of Theorem 9.16. Every sentence is provably equivalent to one built from atomic sentences by $\neg, \wedge, \exists$, and both P (Lemma H.3(ii)) and truth in $\mathbb{N}$ respect provable equivalence; so it suffices to treat such sentences. We show $P(\varphi) = [\mathbb{N} \models \varphi]$ by induction on the number of logical symbols, simultaneously for all such sentences. *Atomic sentences* are quantifier-free. A true one has $P = 1$ by hypothesis. If σ is false, $\neg\sigma$ is a true quantifier-free sentence, so $P(\sigma) = 1 - P(\neg\sigma) = 0$. *Negation:* $P(\neg\varphi) = 1 - P(\varphi)$ (Lemma H.3(i)). *Conjunction:* if $P(\varphi), P(\psi) \in \{0, 1\}$, the bounds of Lemma H.3(v) coincide and give $P(\varphi \wedge \psi) = \min(P(\varphi), P(\psi))$. *Existential:* numerals $\underline{i} = S \cdots S0$ contain no logical symbols, so each instance $\varphi(\underline{i})$ has fewer logical symbols than $\exists x\, \varphi(x)$, and by induction $P(\varphi(\underline{i})) = [\mathbb{N} \models \varphi(\underline{i})]$. By Lemma H.3(iv), applied repeatedly, $\max_{i \leq n} P(\varphi(\underline{i})) \leq P(\bigvee_{i \leq n} \varphi(\underline{i})) \leq \sum_{i \leq n} P(\varphi(\underline{i}))$, so $P(\bigvee_{i \leq n} \varphi(\underline{i})) = \max_{i \leq n} P(\varphi(\underline{i})) \in \{0, 1\}$. By the Gaifman condition, $P(\exists x\, \varphi(x))$ is 1 iff some $\varphi(\underline{i})$ is true in $\mathbb{N}$, iff $\mathbb{N} \models \exists x\, \varphi(x)$, because every natural number is the value of a numeral. □

Proof of Corollary 9.17. We use the following from Garrabrant et al. (2016): a market is a *computable* sequence of rational pricings $(\mathbb{P}_n)$; by Convergence and Limit Coherence (Garrabrant et al. 2016, Thms 4.1.1, 4.1.2) the limit $\mathbb{P}_\infty(\varphi) = \lim_n \mathbb{P}_n(\varphi)$ exists and is a coherent probability relative to Γ; and by Non-Dogmatism (Garrabrant et al. 2016, Thm 4.6.2) $\Gamma \nvdash \neg\varphi$ implies $\mathbb{P}_\infty(\varphi) > 0$. By Limit Coherence, $\mathbb{P}_\infty$ is Gaifman-coherent and gives 1 to every theorem of Γ, in particular to every true quantifier-free sentence, which $\mathsf{Q} \subseteq \Gamma$ proves. Suppose it satisfied the Gaifman condition. Then $\mathbb{P}_\infty = \delta_{\mathrm{Th}(\mathbb{N})}$ by Theorem 9.16. Two arguments refute this. *Via Non-Dogmatism.* By the Gödel–Rosser theorem (Gödel, 1931; Rosser, 1936), the consistent r.e. $\Gamma \supseteq \mathsf{Q}$ is incomplete: some σ has $\Gamma \nvdash \sigma$ and $\Gamma \nvdash \neg\sigma$. Let ψ be whichever of $\sigma, \neg\sigma$ is false in $\mathbb{N}$. Then $\Gamma \nvdash \neg\psi$ (as $\neg\neg\sigma$ is equivalent to σ), so $\mathbb{P}_\infty(\psi) > 0 = \delta_{\mathrm{Th}(\mathbb{N})}(\psi)$. *Via* Δ_2. Since $(n, \varphi) \mapsto \mathbb{P}_n(\varphi)$ is computable, "$\mathbb{P}_n(\varphi) > \frac{1}{2}$" is decidable. If the limit were $\delta_{\mathrm{Th}(\mathbb{N})}$, then $\varphi \in \mathrm{Th}(\mathbb{N})$ iff $\exists N\, \forall n \geq N\ \mathbb{P}_n(\varphi) > \frac{1}{2}$ (a Σ_2 condition) iff $\forall N\, \exists n \geq N\ \mathbb{P}_n(\varphi) > \frac{1}{2}$ (a Π_2 condition): if $\mathbb{P}_\infty(\varphi) = 1$ both hold, and if $\mathbb{P}_\infty(\varphi) = 0$ neither does. So $\mathrm{Th}(\mathbb{N})$ would be Δ_2, contradicting Tarski's theorem that $\mathrm{Th}(\mathbb{N})$ is not arithmetical. *The general statement.* The Δ_2 argument used only that the credence is the pointwise limit of a computable sequence of rational-valued functions, that it is coherent, and that it gives 1 to every true quantifier-free sentence; so no such credence satisfies the Gaifman condition. *The quantifier-free hypothesis is needed.* Let T be the theory of the one-element structure, in which $S0 = 0 = 0 + 0 = 0 \cdot 0$. T is decidable, so δ_T is computable and coherent; and it satisfies the Gaifman condition with respect to the

numerals, because the only element is the value of a numeral. It fails the hypothesis only by giving $P(0 = S0) = 1$. □

Preferences. The note's "vnm stuff" has the same anatomy.

Proposition H.4 (Finite preference data)**.** *[T6 Prop 3.10] Finitely many strict comparisons $p_i \succ q_i$ and weak comparisons $r_j \succsim s_j$ of lotteries over a finite set X have an expected-utility representation $u \in \mathbb{R}^X$ ($u \cdot (p_i - q_i) > 0$, $u \cdot (r_j - s_j) \geq 0$) iff no $\lambda, \kappa \geq 0$ with $\lambda \neq 0$ satisfy $\sum_i \lambda_i(p_i - q_i) + \sum_j \kappa_j(r_j - s_j) = 0$.*

Proof. Motzkin's transposition theorem (Motzkin, 1936). □

The certificate of incoherence is a derivation of $L \succ L$ by mixing (independence) and transitivity (von Neumann and Morgenstern, 1944), money-pump-like (an interpretive gloss; no trading is formalized), and the something is a real-valued utility.

H.4 Contexts

Proof of Theorem 9.19. Note first that, since $C_i = \mathrm{Th}_i \circ \mathrm{Mod}_i$, every set of the form $\mathrm{Th}_i(K)$ is C_i-closed (Lemma 9.2: $\mathrm{Th}_i\mathrm{Mod}_i\mathrm{Th}_i = \mathrm{Th}_i$), and $\mathrm{Th}_i\mathrm{Mod}_i(T) = T$ for every C_i-closed T. (a) For every i and φ: $c^\Gamma \models i{:}\varphi$ iff $\mathrm{Mod}_i(T_i) \subseteq \mathrm{Mod}_i(\varphi)$ iff $\varphi \in \mathrm{Th}_i\mathrm{Mod}_i(T_i) = T_i$. Hence c^Γ satisfies $K_i \cup \Gamma_i \subseteq T_i$. It is bridge-compatible: if $c^\Gamma \models j_k{:}\varphi_k$ for all premises of a bridge rule, then $\varphi_k \in T_{j_k}$, so $\psi \in T_i$ by (Br), so $c^\Gamma \models i{:}\psi$. (b) Let $c \models \Gamma$ be a model and $U_i = \mathrm{Th}_i(c_i)$. Each U_i is C_i-closed and contains $K_i \cup \Gamma_i$. The family (U_i) is closed under bridges: if $\varphi_k \in U_{j_k}$ for all k, then $c \models j_k{:}\varphi_k$, so $c \models i{:}\psi$ by bridge-compatibility, i.e. $\psi \in U_i$. By leastness of $\mathrm{Der}(\Gamma)$, $T_i \subseteq U_i$, so every $m \in c_i$ satisfies T_i: $c_i \subseteq \mathrm{Mod}_i(T_i) = c_i^\Gamma$. (c) ($\Rightarrow$) If $\varphi \in T_i$ and $c \models \Gamma$ is a model, then by (b) every $m \in c_i \subseteq c_i^\Gamma$ satisfies φ. ($\Leftarrow$) If $\varphi \notin T_i$, then $c^\Gamma \not\models i{:}\varphi$ by (a), and c^Γ is a model of Γ. □

`mcs_completeness.py` confirms Theorem 9.19(c) on 400 random three-context systems and finds 2,348 pairs where $\models_W$ strictly exceeds $\models_{\mathrm{LMS}}$ (Theorem 9.22).

Proof of Theorem 9.22. (a) A world family is a model chain of singletons. (b) $T_j = \mathrm{Cn}_2(p \vee q)$ contains neither p nor q, so no bridge fires and $c_i^\Gamma = \mathrm{Mod}(\neg r) \neq \emptyset$; but any w_j satisfies p or q, forcing $w_i \models r$. (c) World families are the Boolean valuations of the tagged language satisfying the tagged axioms. □

Proof of Remark 9.21. With $M_i = \mathrm{Fix}(C_i) \setminus \{L_i\}$ we have $\mathrm{Th}_i \circ \mathrm{Mod}_i = C_i$: if $C_i(X) \neq L_i$, then $C_i(X)$ is itself a model and $\mathrm{Th}_i\mathrm{Mod}_i(X) = \bigcap\{T \in M_i : X \subseteq T\} = C_i(X)$; otherwise $\mathrm{Mod}_i(X) = \emptyset$ and $\mathrm{Th}_i(\emptyset) = L_i$. So Theorem 9.19 applies, and $c_i^\Gamma = \mathrm{Mod}_i(T_i)$ is nonempty iff T_i is proper, since then $T_i \in c_i^\Gamma$, while no proper closed theory contains L_i. A falsum with $\mathrm{Mod}_i(\bot_i) = \emptyset$ is exactly a C_i-explosive $\bot_i$, and then $T_i \neq L_i$ iff $\bot_i \notin T_i$. Without explosiveness the two readings come apart: let C_i be the identity closure (no learned rules), $K_i = \{\bot_i\}$ and no bridges into i; then $\bot_i \in T_i = \{\bot_i\} \neq L_i$, yet $\{\bot_i\} \in c_i^\Gamma$. `repair_checks.py` (B) checks the equivalence on 3,000 random finite closure systems with an explosive falsum. □

Proposition H.5 (Equilibria; non-monotone bridges)**.** *[T6 Props 4.3, 4.4, as repaired] Call $S = (S_i)$ an* equilibrium *if $S_i = C_i(K_i \cup \Gamma_i \cup \{\psi : (\cdots \Rightarrow i{:}\psi)$ has all premises in $S\})$ for all i (the monotone case of Brewka and Eiter 2007, with each context accepting exactly the closure of its*

knowledge base). (a) $\mathrm{Der}(\Gamma)$ *is the least equilibrium, and there is an equilibrium with* $\bot_i \notin S_i$ *for every* $i \in D$ *iff the system is* D*-coherent.* (b) *Let context* i *have consistent* K_i *with* $K_i \nvdash p$ *and no premises* ($\Gamma_i = \emptyset$)*, and add the single non-monotone bridge "*i*:*p *if not* i*:*p*" (negation as failure). Then* K_i *is consistent, so the context is coherent without its bridge, yet there is no equilibrium. So existence for default-style bridges needs an extra condition, such as stratification or the absence of odd loops through negation as failure; stratification is sufficient, not necessary, and coherence alone does not supply it.*

Proof of Proposition H.5. Write $R(S)_i$ for the right-hand side of the equilibrium equation, and $T = \mathrm{Der}(\Gamma)$. (a) $T \supseteq R(T)$: each T_i is C_i-closed and contains K_i, Γ_i and, by (Br), the head of every bridge rule whose premises lie in T. Conversely the family $R(T)$ contains $K \cup \Gamma$, is locally closed, and is bridge-closed: a rule whose premises lie in $R(T) \subseteq T$ has its premises in T, so its head lies in $R(T)$ by definition. By leastness of T, $T \subseteq R(T)$. So $T = R(T)$ is an equilibrium. Any equilibrium $S = R(S)$ is locally closed, contains $K \cup \Gamma$ and is bridge-closed, so $S \supseteq T$. Hence a D-consistent equilibrium forces T to be D-consistent, and conversely T itself is an equilibrium. (b) If $p \in S_i$, the rule is inapplicable, so $S_i = C_i(K_i) \not\ni p$; if $p \notin S_i$, the rule fires and $p \in S_i$. Either way $S_i \neq R(S)_i$. □

Theorem H.6 (Local-to-global on trees)**.** *[T6 Thm 4.6] Let the contexts be the vertices of a finite tree, context* i *classical propositional over atoms* A_i *with a consistent closed theory* T_i*. Assume running intersection (*$\{i : a \in A_i\}$ *is connected for each atom* a*) and mutual conservativity on edges (*$T_i \cap L(A_i \cap A_j) = T_j \cap L(A_i \cap A_j)$*). Then every model of every* T_i *extends to a valuation satisfying all* T_j*.*

This is the join-tree, or iterated Robinson-consistency, argument (cf. Beeri et al. 1983): conservativity makes model sets agree on overlaps, and models glue breadth-first.

Proof. Projection lemma. For a consistent closed theory T over atoms A and $B \subseteq A$: $\mathrm{Mod}(T)|_B = \mathrm{Mod}_B(T \cap L(B))$. The inclusion $\subseteq$ is clear. For $\supseteq$, let a B-valuation w satisfy $T \cap L(B)$, and suppose T together with the literals over B true in w were unsatisfiable. By compactness, $T \vdash \neg(l_1 \wedge \cdots \wedge l_r)$ for some literals l_k true in w; this sentence lies in $T \cap L(B)$ and is false in w, a contradiction. By the lemma, mutual conservativity on an edge $\{i, j\}$ says $\mathrm{Mod}(T_i)|_{A_i \cap A_j} = \mathrm{Mod}(T_j)|_{A_i \cap A_j}$. *Gluing.* Root the tree at i_0 and let $m_{i_0} \models T_{i_0}$ be given; assign the atoms of A_{i_0} accordingly. Visit the remaining vertices in breadth-first order. When c is visited, its parent p has been visited and its children have not, so p is the only visited neighbour of c and the tree path from c to any visited vertex passes through p. If an atom $a \in A_c$ is already assigned, it lies in A_d for some visited d; by running intersection it lies in every A_e on the path from c to d, in particular in A_p. So the assigned atoms of A_c lie in $A_c \cap A_p$. After p was visited the global assignment restricted to A_p was a model $m_p \models T_p$, and assigned atoms never change. So the current assignment on $A_c \cap A_p$ is $m_p|_{A_c \cap A_p} \in \mathrm{Mod}(T_p)|_{A_c \cap A_p} = \mathrm{Mod}(T_c)|_{A_c \cap A_p}$, and some $m_c \models T_c$ agrees with it; extend the assignment by m_c. Atoms in no A_i are assigned arbitrarily. At the end the assignment restricted to each A_c is $m_c \models T_c$. □

`contexts_local_global.py` checks the theorem on one fixed three-node path cover with 7,131 random local-theory systems and on 2,000 random tree covers with 2–5 nodes, with no failure; a negative control without running intersection fails in 437 of 2,000 cases.

Relation to the context-tree calculus (proof sketch). The context-tree calculus[L7 §8] from which Definition 11.13 is adapted has stipulations (local axioms), imports $\pi(c){:}\varphi \Rightarrow c{:}\varphi$ for φ in the import filter F_c, discharge $c{:}\psi \Rightarrow \pi(c){:}A \to \psi$, and side-conditioned export $c{:}\varphi_\beta(t),\ \pi(c){:}\sigma_\beta(t) \Rightarrow \pi(c){:}\varepsilon_\beta(t)$. All are monotone bridge rules, so it is an MC system, and Theorem 9.19 is its completeness theorem for local-models semantics with bridges read as compatibility constraints; there an exported ε shrinks the parent's belief state. The calculus's own semantics differs: the models of the root are fixed by background and observations and do not depend on exports, and bridges are required to be sound; soundness for that semantics holds when every step and export instance is sound. The two semantics coincide when the local consequence is the complete base logic and every step and export instance is sound: then $T_c = \mathrm{Th}(\mathrm{Mod}_{\mathrm{ct}}(c))$ for every c, where $\mathrm{Mod}_{\mathrm{ct}}$ is the model class of the calculus's own semantics, $\subseteq$ is its soundness proposition, and $\supseteq$ goes by induction down the tree (at the root by local completeness; an idealized child imports exactly $F_c \cap T_{\pi(c)}$; a supposition child imports all of $T_{\pi(c)}$ and its assumption). Without that hypothesis they differ: if an unsound export yields $@{:}Q = 5$ while background and observations prove $Q = 3$, MC derives $@{:}\bot$ and $c^\Gamma_@ = \emptyset$, as Theorem 9.19 requires, while the root's model class in the calculus's own semantics stays nonempty. For supposition contexts with full import, the canonical chain always satisfies $c^\Gamma_c = c^\Gamma_{\pi(c)} \cap \mathrm{Mod}(A)$.

H.5 Learned calculi

Proof of Theorem 9.25. (a) is the Zorn step in the proof of Theorem 9.5(a), applied to the closed set $C_R(A)$ and $\sigma = \bot$ (§H.1). (b) If $\varphi \in C_R(\Gamma)$ and $\sigma\Gamma \subseteq T$, structurality gives $\sigma\varphi \in C_R(\sigma\Gamma) \subseteq C_R(T) = T$; this works for any C_R-theory $T \supseteq A$ omitting $\bot$. (c) is Theorems 9.8 and 9.11. □

Existence of the Leibniz congruence. The join of a family of congruences of $\mathbf{A}$ is the transitive closure of their union. If each congruence in the family is compatible with D, then a and b related by the join are connected by a chain $a = c_0\,\theta_1\,c_1 \cdots \theta_r\,c_r = b$ with each θ_k in the family; each step preserves membership in D, so the join is compatible. The join of all compatible congruences is therefore the largest one.

Proof of Theorem 9.26. (a) *Claim:* $\Omega(T) = h^{-1}(\Omega_{\mathbf{A}} D)$, where $h^{-1}(\theta) = \{(a,b) : (ha, hb) \in \theta\}$. The preimage of a congruence under a homomorphism is a congruence. It is compatible with T: if $(ha, hb) \in \Omega_{\mathbf{A}} D$ and $a \in T$, then $ha \in D$, so $hb \in D$ and $b \in T$. Maximality: let θ' be any congruence of Fm compatible with T. The kernel $\ker h$ is compatible with T, as $T = h^{-1}(D)$ is a union of $\ker h$-classes, so $\theta'' = \theta' \vee \ker h$ is compatible with T. Let $\theta_{\mathbf{A}} = \{(ha, hb) : (a,b) \in \theta''\}$. It is reflexive because h is onto, and symmetric. It is transitive: if $(ha, hb), (hb', hc) \in \theta_{\mathbf{A}}$ come from pairs in θ'' with $hb = hb'$, then $b\,\theta''\,b'$ because $\ker h \subseteq \theta''$, so $a\,\theta''\,c$. It is compatible with the operations: if $(x_l, y_l) \in \theta_{\mathbf{A}}$ come from $(a_l, b_l) \in \theta''$, then for an operation f, $(f(\bar{x}), f(\bar{y})) = (hf(\bar{a}), hf(\bar{b}))$ with $f(\bar{a})\,\theta''\,f(\bar{b})$. And it is compatible with D: if $(ha, hb) \in \theta_{\mathbf{A}}$ comes from $(a,b) \in \theta''$ and $ha \in D$, then $a \in T$, so $b \in T$ and $hb \in D$. Hence $\theta_{\mathbf{A}} \subseteq \Omega_{\mathbf{A}} D$, and $\theta' \subseteq \theta'' \subseteq h^{-1}(\theta_{\mathbf{A}}) \subseteq h^{-1}(\Omega_{\mathbf{A}} D)$. *Isomorphism.* Define $[a]_{\Omega(T)} \mapsto [ha]_{\Omega_{\mathbf{A}} D}$. By the claim it is well defined and injective; it is onto because h is; it is a homomorphism because h is; and it maps $T/\Omega(T)$ onto $D/\Omega_{\mathbf{A}} D$ because $a \in T$ iff $ha \in D$. If $\langle \mathbf{A}, D\rangle$ is reduced, $\Omega_{\mathbf{A}} D$ is the identity and the right-hand side is $\langle \mathbf{A}, D\rangle$. (b) A maximal consistent theory of CPC is $v^{-1}(1)$ for a Boolean homomorphism $v : \mathrm{Fm} \to \mathbf{2}$. This v is onto, since $v(p \vee \neg p) = 1$ and $v(p \wedge \neg p) = 0$.

The matrix $\langle \mathbf{2}, \{1\}\rangle$ is reduced: the only congruence of $\mathbf{2}$ besides the identity identifies 0 with 1, which is not compatible with $\{1\}$. Apply (a). □

`repair_checks.py` (D) checks a non-surjective example for the remark after Theorem 9.26: in the four-element Heyting chain $0 < a < b < 1$, $h(p) = a$ generates $\{0, a, 1\} \cong \mathbf{H}_3$, which is reduced, so the reduced Lindenbaum matrix of $h^{-1}(1)$ is $\langle \mathbf{H}_3, \{1\}\rangle$, not the intended four-element matrix.

Proof of Proposition 9.27. C is sound and strongly complete for the homomorphisms $h : \mathrm{Fm} \to \mathbf{A}$, and finitary by compactness of $\mathbf{A}^{\mathrm{Var}}$ (the standard argument: each condition $h(\gamma) \in D$ is clopen there, as $\mathbf{A}$ is finite). So every point is some $h^{-1}(D)$ by Theorem 9.5(b); apply Theorem 9.26(a) to h onto its image. Since $\mathbf{A}$ is finite, there are finitely many generated submatrices, hence finitely many reductions up to isomorphism. □

Verification of Example 9.28. (i) $h : \mathrm{Fm} \to \mathbf{H}_3$ is a Heyting homomorphism with $h(p) = \frac{1}{2}$, $h(\bot) = 0$, $h(\top) = 1$, so it is onto. $T = h^{-1}(1)$ is an IPC-theory by soundness of IPC for Heyting algebras, and $p \notin T$. *T is maximal among theories omitting p:* let $\psi \notin T$, so $h(\psi) \in \{0, \frac{1}{2}\}$. If $h(\psi) = \frac{1}{2}$ then $h(\psi \leftrightarrow p) = 1$, so $\psi \leftrightarrow p \in T$ and any theory containing T and ψ contains p. If $h(\psi) = 0$ then $h(\neg\psi) = 1$, so $\neg\psi \in T$ and any theory containing T and ψ contains $\bot$, hence p. So every theory strictly above T contains p, and T is a point. It is not maximal consistent, since $h(p \vee \neg p) = \frac{1}{2} \vee 0 = \frac{1}{2}$ puts neither p nor $\neg p$ in T. *$\langle \mathbf{H}_3, \{1\}\rangle$ is reduced:* a congruence identifying $\frac{1}{2}$ with 1 is not compatible with $\{1\}$; one identifying 0 with $\frac{1}{2}$ also identifies $0 \to 0 = 1$ with $\frac{1}{2} \to 0 = 0$, hence everything, and is not compatible either. By Theorem 9.26(a) the reduced Lindenbaum matrix of T is $\langle \mathbf{H}_3, \{1\}\rangle$. (ii) In a one-element structure $\forall x \exists y\, \varphi$ and $\exists y \forall x\, \varphi$ are equivalent for every formula φ, so every instance of the schema is sound there; with equality, a structure in which every instance of the schema holds must have one element (take $R(x, y)$ to be $x = y$: $\forall x \exists y\, x = y$ holds in every structure, and $\exists y \forall x\, x = y$ says there is one element). So the models of first-order logic plus the schema are exactly the one-element structures, and a designated context asserting two distinct objects has none. □

H.6 Term models, non-standardness and pins

This subsection expands the paragraph "What pins the intended model" of §9.5.

First-order term models. For a consistent first-order theory, the Lindenbaum construction is Henkin's (Henkin, 1949): witnesses, a complete Henkin theory, closed terms modulo provable equality (the Leibniz reduction). A note on consistency-based probing tentatively argues ("I think this is an argument that") that "any assignment of truth-values to sentences which does not violate inference rules can be extended to a full model (this is cool because a model also needs to assign actual functions and objects . . .)".[12] The argument goes through for complete consistent assignments: the objects are classes of witness terms. If the assignment makes $\exists x\, \varphi(x)$ true and every named instance false, the model needs unnamed objects, the Gaifman failure of Corollary 9.17.

Theorem H.7 (Prime models)**.** *[T6 Thm 5.5] (known) For complete consistent $T \supseteq \mathsf{PA}$ the definable elements of any model of T form the prime model $\mathcal{K}(T)$, unique up to isomorphism and*

[12] `ai/DLK/logic.md`.

elementarily embeddable in every model of T (Kaye, 1991); $\mathcal{K}(\mathrm{Th}(\mathbb{N})) = \mathbb{N}$; if $\neg\mathrm{Con}(\mathsf{PA}) \in T$, $\mathcal{K}(T)$ is non-standard.

For the last clause: "the least code of a PA-proof of $0 = 1$" is definable, while T refutes each standard instance $\mathrm{Prf}(\underline{n}, \ulcorner 0{=}1 \urcorner)$, whose negation is a true Δ_0 sentence (PA being consistent) and so provable in $\mathsf{Q} \subseteq T$. This is the precise form of the notes' Gödel-sentence footnote: a complete coherent theory has a canonical thing it talks about, the objects it can define, and the "proof" is one of them, a definable non-standard number.

Two layers of non-standardness. *(i) Coherent but false*: $\mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$ is consistent, false, and even Π_1-sound (if it proves a Π_1 sentence π, then $\mathsf{PA} \vdash \mathrm{Con}(\mathsf{PA}) \vee \pi$; were π false, PA would prove the true Σ_1 sentence $\neg\pi$ and hence Con(PA), against Gödel's second incompleteness theorem). This is finitary and comes from incompleteness (coherence is not truth, Theorem 6.21(b)). *(ii) True but not categorical*: $\mathrm{Th}(\mathbb{N})$ itself has non-standard models, a limit phenomenon from compactness. Pinning comes in levels: P0 coherent (a model exists); P1 complete (all models elementarily equivalent; for logics, Post-complete, Theorem 6.13); P2 κ-categorical; P3 categorical (unique model up to isomorphism; for calculi, all *points* reduce to one matrix, Proposition 9.27).

Theorem H.8 (Compactness barrier)**.** *[T6 Thm 5.6] (known) A first-order theory with an infinite model has models of every infinite cardinality $\geq |L|$ (Löwenheim–Skolem–Tarski), so no first-order mode of talking reaches P3 for an infinite structure.*

Every finite structure in a finite language, by contrast, is characterized up to isomorphism by one sentence. In the icosahedron example of the notes,[13] after the physical icosahedron explodes the mode of talking stays coherent and its object is still pinned up to isomorphism; the caveat bites only for infinite objects.

Theorem H.9 (Computability barrier)**.** *[T6 Thm 5.7, as repaired] Let S be a decidable set of finite syntactic objects and K an intended class of models. If some calculus has an r.e. set of incoherent finite sets, and a finite $X \subseteq S$ is incoherent iff it has no model in K, then $\{X : X \text{ has a } K\text{-model}\}$ is co-r.e. (Π_1). Consequently no such calculus exists for* (a) *arithmetic with $K = \{\mathbb{N}\}$ ($\mathrm{Th}(\mathbb{N})$ is not arithmetical);* (b) *Diophantine equations with $K = \{\mathbb{Z}\}$ (solvability is Σ_1-complete, Matiyasevich 1970);* (c) *first-order sentences in a vocabulary with a binary relation symbol, with K the finite structures (finite satisfiability is Σ_1 and undecidable, Trakhtenbrot 1950; for monadic vocabularies it is decidable).* (d) *For K all structures, satisfiability is Π_1, as Gödel's theorem requires; for polynomial equations over $\bar{k}$ it is decidable.*

Proof. "No K-model" is "incoherent", which is r.e. In (b), (c) the set is also Σ_1, hence would be decidable; in (a) $\mathrm{Th}(\mathbb{N})$ would be Π_1, contradicting Tarski's undefinability theorem (e.g. Boolos et al. 2007). □

This is the textbook argument "a complete r.e. calculus makes semantic consequence r.e."; what is new is the framing. *For any computable notion of coherence, "coherent iff there is something it could be talking about" can hold only if having such a something is a Π_1 property.* "Having some model" is Π_1; "having the standard model", "having an integer solution" and "having a finite model" (in a vocabulary with a binary relation symbol) are not. The proposition is true exactly to the extent that "something" is construed liberally.

[13] `philosophy/philosophy of language/math is a mere string game iff everything is.md`.

Proposition H.10 (Anchoring pins exactly the definable vocabulary)**.** *[T6 Prop 5.8, reading as repaired] Let $L_o \subseteq L$ be the anchored (world-hooked) vocabulary and T an L-theory. Any two models of T with the same universe and the same interpretation of L_o coincide iff every symbol of L is explicitly T-definable from L_o (Beth, 1953).*

Reading. Beth's theorem quantifies over all models of T, i.e. over every possible anchoring structure: anchoring determines the rest of the language *uniformly, whatever the anchoring structure* iff the rest is definable. Over the *actual* structure pinning can hold without definability: if T says that R is a nonempty initial segment of $<$ without a largest element, then over $(\mathbb{N}, <)$ the only expansion is $R = \mathbb{N}$, while over $\omega + \omega$ both $R = \omega$ and $R = \omega + \omega$ work. Undefinable theoretical terms are multiply realizable over *some* anchoring, though perhaps not over the actual one (the Ramsey–Lewis situation, Lewis 1970).

How are the natural numbers pinned down? The notes ask this of numbers as pinned down "in a brain, to a thinker, or something", adding "is there a way they are the 'lowest-complexity-structure' which satisfy some axioms??".[14] We answer only the model-theoretic version. Coherence and first-order completeness cannot do it (Theorems H.8 and H.9). The known pins each add something: *second-order induction* (Dedekind, 1888) adds quantification over all subsets, and no complete r.e. calculus exists for its consequence, as Theorem H.9(a) predicts; internal categoricity (Parsons, 1990; McGee, 1997; Button and Walsh, 2018) pins the structure relative to the available language, not the theory. *Initiality* ($\mathbb{N}$ is the initial $(0, S)$-algebra, the least Herbrand model of $\{N(0), N(x) \to N(Sx)\}$, van Emden and Kowalski 1976, and an initial segment of every model of PA) adds a universal property, "nothing else is a number", which is not first-order. The *ω-rule*, or Gaifman condition, adds an infinitary rule that no limit-computable credence of the relevant kind obeys (Corollary 9.17). World anchoring by computation adds nothing beyond coherence with Q (Lemma 6.18). And:

Theorem H.11 (Tennenbaum)**.** *[T6 §5.5] (known) In no countable non-standard model of* PA *is addition computable, and in none is multiplication computable (Tennenbaum 1959; the separate statements for $+$ and $\times$ are due to Kreisel and McAloon, see Kaye 1991). Hence $\mathbb{N}$ is the unique computable model of* PA *up to isomorphism.*

This is a theorem-level "yes" to the "lowest-complexity structure" question, with two caveats. First, "computable" is defined using the meta-language's natural numbers, a point the notes make about the meta-syntax's $\mathbb{N}$ in general.[15] Second, the axioms matter: weaker theories such as open induction have computable non-standard models (Shepherdson, 1964). Every pin uses something beyond finitary computable coherence: second-order quantification, a universal property, a complexity bound measured in $\mathbb{N}$, or an infinitary rule. Theorems H.8 and H.9 say this is forced.

I Proofs for Section 10

Propositions 10.7, 10.11 and 10.16 and Theorem 10.8 are proved in full in the main text. This appendix proves the remaining results of Section 10, verifies Example 10.15, and states in full the secondary results that the main text only summarizes (Propositions I.1, I.3 to I.6 and I.8,

[14] `philosophy/questions/how are the natural numbers pinned down?.md`.
[15] `logic/confusions/soundness.md`.

Theorem I.7, and Remark I.9). Results marked (computed) were checked by the scripts in `research/theory/T4-checks/` of the accompanying repository. Notation is as in Section 10. Throughout, Cl_R is a closure operator (extensive, monotone, idempotent; Lemma 2.3), and for a first-order complete h and finite $\Gamma \cup \{y\} \subseteq D_h$ we use freely that $\Gamma \models_h y$ iff $T_h \cup \rho_h\Gamma \models \rho_h y$ iff $T_h \vdash \bigwedge \rho_h\Gamma \to \rho_h y$ (Gödel completeness).

I.1 Sound bounded-gap verification

Proof of Lemma 10.6. Induction on j, by monotonicity and idempotence (Lemma 2.3):

$$\rho_h y_j \in \mathrm{Cl}_{R_h}(\rho_h\Gamma_j) \subseteq \mathrm{Cl}_{R_h}(\mathrm{Cl}_{R_h}(\rho_h\mathrm{Prem})) = \mathrm{Cl}_{R_h}(\rho_h\mathrm{Prem});$$

links are the case $\Gamma_j = \{y_i\}$. Then use soundness of R_h. □

Proof of Theorem 10.9. Practice data are steps, so they range over a countable set and each p_h is a probability mass function. Let $X_1, X_2, \ldots$ be the practice data, i.i.d. from p_{h^*} and independent of everything the prover and the verifier do. Every other datum is a truthful constraint $c : \mathcal{H} \to \{0,1\}$ with $c(h^*) = 1$. Moreover $c(h') = 1$ for every $h' \equiv_g h^*$: an escalation or link answer is a $\mathrm{St}^g_{h^*}$-label, and $\mathrm{St}^g_{h'} = \mathrm{St}^g_{h^*}$; a paradox exists only for hypotheses incoherent on some $A \in \mathcal{A}$, and h' is coherent on every A on which h^* is; an object datum is a restriction of a member of $\mathcal{V}_{h^*} = \mathcal{V}_{h'}$.

At time t, let $n(t)$ be the number of practice data seen and $c_1, \ldots, c_{k(t)}$ the constraints received. Put

$$L_t(h) := \prod_{i \le n(t)} p_h(X_i) \cdot \prod_{j \le k(t)} c_j(h), \qquad w_t(h) := \frac{w(h)L_t(h)}{\sum_{h'} w(h')L_t(h')}, \qquad Z_t := \sum_h w(h)\frac{L_t(h)}{L_t(h^*)}.$$

Almost surely $L_t(h^*) > 0$, since $p_{h^*}(X_i) > 0$ and $c_j(h^*) = 1$, so $w_t(h) = w(h)L_t(h)/(L_t(h^*)Z_t)$. For $h \in [h^*]$, shadow-measurability gives $p_h = p_{h^*}$, and $c_j(h) = 1$ by the previous paragraph; so $L_t(h) = L_t(h^*)$ and

$$w_t([h^*]) = W^*/Z_t.$$

Since every $c_j(h) \le 1$, pathwise $Z_t \le Z^H_{n(t)}$, where $Z^H_n := \sum_h w(h) \prod_{i \le n} p_h(X_i)/p_{h^*}(X_i)$ depends only on the practice data. Let $\mathcal{F}_n := \sigma(X_1, \ldots, X_n)$. For every h,

$$\mathbb{E}\Big[\frac{p_h(X_{n+1})}{p_{h^*}(X_{n+1})} \,\Big|\, \mathcal{F}_n\Big] = \sum_{x\,:\,p_{h^*}(x)>0} p_h(x) \le 1,$$

so by Tonelli (all terms are nonnegative) $\mathbb{E}[Z^H_{n+1} \mid \mathcal{F}_n] \le Z^H_n$: Z^H is a nonnegative supermartingale with $Z^H_0 = \sum_h w(h) = 1$. Ville's inequality (Ville, 1939) gives $\mathbb{P}[\sup_n Z^H_n \ge 1/\delta'] \le \delta'$. Let $G := \{\sup_n Z^H_n < 1/\delta'\}$; it depends only on the practice data, and $\mathbb{P}(G) \ge 1 - \delta'$. On G, for every prover and every t, $Z_t < 1/\delta'$, so $w_t([h^*]) > W^*\delta' \ge \delta$. If $q \notin \mathrm{St}^g_{h^*}$, then $q \notin \mathrm{St}^g_h$ for every $h \in [h^*]$, so the posterior mass of $\{h : q \notin \mathrm{St}^g_h\}$ is at least $w_t([h^*]) > \delta$, and q is not accepted. This is the prior–posterior-ratio martingale of Waudby-Smith and Ramdas (2020), and the argument is that of Theorem 4.15. □

Proof of Theorem 10.10. (a) Let the oracle-answered escalations of an honest prover be $s_1, s_2, \ldots$, in order; each s_i lies in $\mathrm{St}^g_{h^*}$ and, when escalated, outside $\bigcap_{h \in \mathrm{VS}} \mathrm{St}^g_h$. After the oracle answers "yes" to s_j, every surviving h has $s_j \in \mathrm{St}^g_h$. So when s_i is escalated, $\mathcal{F}_1 := \{\mathrm{St}^g_h : h \in \mathrm{VS}\}$ is

contained in $\mathcal{F}_2 := \{S \in \mathcal{H}^g : S \supseteq \{s_1, \dots, s_{i-1}\}\}$, hence $\bigcap \mathcal{F}_2 \subseteq \bigcap \mathcal{F}_1 \not\ni s_i$. Thus $s_1, s_2, \dots$ is an elastic chain in $\mathrm{St}^g_{h^*}$ for $\mathcal{H}^g$, and there are at most $\mathrm{el}(\mathcal{H}^g, \mathrm{St}^g_{h^*})$ of them. For an elastic chain $s_1, \dots, s_m$, the families $F_i := \{S \in \mathcal{H}^g : S \supseteq \{s_1, \dots, s_i\}\}$ strictly decrease (some member of F_{i-1} lacks s_i) and $F_m \ni \mathrm{St}^g_{h^*}$, so $m \le |\mathcal{H}^g| - 1 \le |\mathcal{H}/{\equiv_g}| - 1$, because $\equiv_g$-equivalent hypotheses have the same g-step relation.

(b) Use the notation of the previous proof. An honest query q escalated at time u has posterior mass $\pi_u := w_{u-1}(\{h : q \notin \mathrm{St}^g_h\}) \ge \delta$. The deterministic answer "yes" removes exactly these hypotheses, so it multiplies Z by $1 - \pi_u \le 1 - \delta$. Other constraints multiply Z by at most 1. A practice datum X multiplies Z by $F := \sum_h w_{\mathrm{prev}}(h)\, p_h(X)/p_{h^*}(X)$, where w_{prev} is the current posterior. Each practice datum is independent of everything that precedes it, including the prover's coins and choices (the prover does not foresee practice data), so $\mathbb{E}[F \mid \mathrm{past}] \le 1$. Let M_n be the product of the factors F over the first n practice data; it is a nonnegative supermartingale with $M_0 = 1$. Since the members of $[h^*]$ keep weight $w(h)$ in Z, $Z_t \ge W^*$. Hence $W^* \le Z_t \le M_{n(t)}(1-\delta)^{\#\mathrm{ESC}_t}$, and on the event $\{\sup_n M_n < 1/\delta''\}$, which has probability at least $1 - \delta''$ by Ville's inequality, $\#\mathrm{ESC}_t \le \ln(1/(W^*\delta''))/\ln(1/(1-\delta)) \le (\ln(1/W^*) + \ln(1/\delta''))/\delta$ for all t.

(c) Let the hypotheses be h_u, $u \in U$, with $\mathrm{St}^g_{h_u} = B \cup (U \setminus \{u\})$, and target h_{u^*}. Any enumeration $s_1, \dots, s_{n-1}$ of $U \setminus \{u^*\}$ is an elastic chain: $\bigcap\{S \in \mathcal{H}^g : S \supseteq \{s_1, \dots, s_{i-1}\}\} = B \cup \{s_1, \dots, s_{i-1}\} \not\ni s_i$. So $\mathrm{el} = n - 1$, the upper bound of (a). It is attained when the practice data lie in B: then VS consists exactly of the h_u with $u \notin \{s_1, \dots, s_{i-1}\}$ when s_i is queried, and s_i is escalated. For the lower bound, take practice distributions supported on B and identical for all hypotheses (shadow-measurable), and the honest prover that queries $s_1, \dots, s_{n-1}$ in order. Under target h_{s_i} the step s_i is invalid, and the answers to $s_1, \dots, s_{i-1}$ are the same ("yes") under h_{u^*} and h_{s_i}. So the law of the transcript up to the verifier's answer in round i is the same under both targets, and δ'-soundness under h_{s_i} gives $\mathbb{P}_{h_{u^*}}[s_i \text{ accepted in round } i] \le \delta'$. Summing, the expected number of these honest queries that are not accepted (escalated or rejected) under h_{u^*} is at least $(1-\delta')(n-1) = (1-\delta')(1/W^* - 1)$ for the uniform prior; a verifier that never rejects, such as $V^{g,\delta}_t$, escalates them. With $\delta = W^*\delta' = \delta'/n$, the bound of (b) is $n(\ln n + \ln(1/\delta''))/\delta'$, and the ratio to the lower bound is at most $2(\ln n + \ln(1/\delta''))/(\delta'(1-\delta')) = O((\ln(1/W^*) + \ln(1/\delta''))/\delta')$ for $n \ge 2$ and $\delta' \le 1/2$. This is the argument of Theorem 4.4 and Corollary 4.17. □

I.2 Identification

Proposition I.1 (No learner sees past the shadow; trivial)**.** *[T4 Prop 3.2, repaired] Suppose the law of the practice data is a function of* $\mathrm{St}^g_{h^*}$ *and the history, escalation and link answers are* $\mathrm{St}^g_{h^*}$*-labels, the law of the object data (which occurrences are shown or queried, and their values) is a function of* $\mathcal{V}_{h^*}$ *and the history,* $\mathcal{A}$ *does not depend on the target, and queries come from the learner or from provers that depend on the target only through its shadow. If* $h \equiv_g h'$*, every data sequence has the same probability under target* h *as under* h'*, so no learner distinguishes them. The closure-level version replaces* St^g *by* $\models$ *and* $\equiv_g$ *by* $\equiv_\infty$*.*

Proof of Proposition I.1. Write the interaction as a sequence of rounds. In round $k+1$ the learner (and any prover whose strategy depends on the target only through its shadow) issues queries Q_{k+1} drawn from a kernel that depends only on the history H_k and its own coins; then the environment issues a datum D_{k+1}. By the assumptions, the conditional law of D_{k+1} given (H_k, Q_{k+1}) is $K(\mathrm{sh}, H_k, Q_{k+1})$ for a kernel K and the shadow sh of the target: practice data depend on $\mathrm{St}^g_{h^*}$ and the history; escalation and link answers are $\mathrm{St}^g_{h^*}$-labels; object data depend

on $\mathcal{V}_{h^*}$ and the history; and paradoxes are computed by the learner from the fixed family $\mathcal{A}$ and the hypotheses themselves, independently of the target. (Since $h \equiv_g h'$ includes agreement on coherence, the same $\mathcal{A}$ satisfies the designation promise under both targets.) Let P_h be the law of the interaction under target h. By induction on k, P_h and $P_{h'}$ agree on (H_k, Q_{k+1}): the step from k to $k+1$ composes the same kernels. Hence they agree on all cylinder sets, and so on the whole sequence. Any learner's output is a function of the history and its coins, so it has the same law under h and h'. The closure-level version is the same argument with $\models$ in place of St^g. □

Proof of Lemma 10.13. Write $D = D_h$ and $T = T_h$.

(i) $\subseteq$: let $v = v_M \circ \rho_h$ with $M \models T$. Since $T \vdash \rho_h(\sim x) \leftrightarrow \neg\rho_h(x)$, $v(\sim x) = 1 - v(x)$. If $\Gamma \subseteq v^{-1}(1)$ is finite and $\Gamma \models_h y$, then $M \models \rho_h\Gamma$ and $T \cup \rho_h\Gamma \models \rho_h y$, so $v(y) = 1$.

$\supseteq$: let v be a coherent valuation and $\Theta := T \cup \{\rho_h(x) : v(x) = 1\}$. Suppose Θ is inconsistent. By compactness some finite $\Gamma \subseteq v^{-1}(1)$ makes $T \cup \rho_h\Gamma$ inconsistent, so $\Gamma \models_h y$ for every $y \in D$. Pick $x \in D$ (here $D \neq \emptyset$ is used); one of $x, \sim x$ has v-value 0, call it y; but $\Gamma \models_h y$ forces $v(y) = 1$, a contradiction. So Θ has a model M. If $v(x) = 1$ then $M \models \rho_h(x)$. If $v(x) = 0$ then $v(\sim x) = 1$, so $M \models \rho_h(\sim x)$ and hence $M \models \neg\rho_h(x)$. Thus $v = v_M \circ \rho_h \in \mathcal{V}_h$.

(ii) ($\Rightarrow$) is (i) $\subseteq$, together with the domain clause in the definition of $\models_h$. ($\Leftarrow$) If $\Gamma \cup \{y\} \subseteq D$ and $\Gamma \not\models_h y$, then $T \cup \rho_h\Gamma \cup \{\neg\rho_h y\}$ is consistent, by completeness; a model M of it gives $v = v_M \circ \rho_h \in \mathcal{V}_h$ with $v(\Gamma) = 1$ and $v(y) = 0$.

(iii) A is h-coherent iff $\bot \notin \mathrm{Cl}_{R_h}(\rho_h A)$ iff $T \cup \rho_h A$ is consistent iff it has a model M, iff $v_M \circ \rho_h$ is a member of $\mathcal{V}_h$ with $v(A) = 1$. For the second equivalence: if A is coherent, pick $x \in D$; in a model of $T \cup \rho_h A$ one of $x, \sim x$ is false, call it y, and then $A \not\models_h y$. If A is incoherent, $T \cup \rho_h A$ is inconsistent and $A \models_h y$ for every $y \in D$.

The converses. If $\mathcal{V}_h \neq \emptyset$, D is the common domain of its members and (ii) recovers $\models_h$. Coherence of finite subsets of D recovers $\models_h$, since for $\Gamma \cup \{y\} \subseteq D$, $\Gamma \models_h y$ iff $T \cup \rho_h\Gamma \cup \{\neg\rho_h y\}$ is inconsistent iff $\Gamma \cup \{\sim y\}$ is incoherent. That neither holds unconditionally: two hypotheses with inconsistent theories and different domains both have $\mathcal{V} = \emptyset$; and two hypotheses that differ only on steps avoiding every member of a fixed $\mathcal{A}$ agree on coherence on $\mathcal{A}$. □

The key step in Theorem 10.14 is the following lemma.

Lemma I.2 (Upper bracket)**.** *Assume the hypotheses of Theorem 10.14, and let $h \in \mathcal{H}$ have $D_h \supseteq D^*$. Call an h^*-*datum *the restriction of some $u \in \mathcal{V}_{h^*}$ to a finite subset of D^*.*

(i) If $\models_h|_{D^*} \not\subseteq \models_{h^*}$, some h^*-datum refutes h.

(ii) If $\models_h|_{D^*} \subseteq \models_{h^*}$, no h^*-datum refutes h, and h is coherent on every finite $A \subseteq D^*$ on which h^* is coherent.

Proof. (i) Take $(\Gamma \rhd y) \in \models_h \setminus \models_{h^*}$ with $\Gamma \cup \{y\} \subseteq D^*$. By Lemma 10.13(ii) for h^*, some $u \in \mathcal{V}_{h^*}$ has $u(\Gamma) = 1$ and $u(y) = 0$. Let $v := u|_{\Gamma \cup \{y\}}$, an h^*-datum with $\mathrm{dom}\, v \subseteq D_h$. If v extended to some $v_M \circ \rho_h \in \mathcal{V}_h$, then $M \models T_h \cup \rho_h\Gamma$ and $M \not\models \rho_h y$, contradicting $\Gamma \models_h y$. So v refutes h.

(ii) Let $v = u|_F$ with $u \in \mathcal{V}_{h^*}$ and $F \subseteq D^*$ finite, and let $\Gamma_v := \{x \in F : v(x) = 1\} \cup \{\sim x : x \in F,\ v(x) = 0\}$, a finite subset of D^* (which is negation-closed). Suppose v extended to no member of $\mathcal{V}_h$. Then $T_h \cup \rho_h\Gamma_v$ is inconsistent, since a model of it would induce an extension of v (using $T_h \vdash \rho_h(\sim x) \leftrightarrow \neg\rho_h(x)$). Pick $x \in D^*$. Then $\Gamma_v \models_h x$ and $\Gamma_v \models_h \sim x$, and both steps have their occurrences in D^*, so $\Gamma_v \models_{h^*} x$ and $\Gamma_v \models_{h^*} \sim x$. But u makes Γ_v true, so by Lemma 10.13(ii) for h^* it makes both x and $\sim x$ true, which is impossible. For coherence: if h were incoherent on $A \subseteq D^*$, then $A \models_h x$ and $A \models_h \sim x$ for $x \in D^*$, hence $A \models_{h^*} x$ and $A \models_{h^*} \sim x$, and $T^* \cup \rho^* A$ would be inconsistent. □

Proof of Theorem 10.14. (a) If $h \approx h'$, then $D_h = D_{h'}$ (both are recovered from the common relation), $\mathcal{V}_h = \mathcal{V}_{h'}$ by Lemma 10.13(i) (both are the coherent valuations of the same relation, for the same negation), and h, h' are coherent on the same sets by Lemma 10.13(iii), applied to $A \cap D_h$ (partial-reading convention). So $h \equiv_\infty h'$. The converse is part of the definition of $\equiv_\infty$. Nonempty domains are needed: with empty readings, a consistent and an inconsistent theory have the same (empty) $\models$ but different $\mathcal{V}$.

(b) By (a), $\equiv_\infty$ and $\approx$ coincide; apply the closure-level version of Proposition I.1.

(c) *Every $h \notin U^*$ is eventually refuted.* If $\models_{h^*} \not\subseteq \models_h$, some $s \in \models_{h^*} \setminus \models_h$ is presented in (P) and refutes h. Otherwise $\models_{h^*} \subseteq \models_h$; then $D_h \supseteq D^*$, because $\{x\} \models_{h^*} x$ for $x \in D^*$; and since all steps of $\models_{h^*}$ have occurrences in D^*, $\models_{h^*} \subseteq \models_h|_{D^*}$, with strict inclusion as $h \notin U^*$. So $\models_h|_{D^*} \not\subseteq \models_{h^*}$, and by Lemma I.2(i) some h^*-datum refutes h; it is presented in (O). *No $h \in U^*$ is ever refuted.* Practice data lie in $\models_{h^*} = \models_h|_{D^*}$; $D_h \supseteq D^*$ as before; and by Lemma I.2(ii) no object datum refutes h and h is coherent on every designated context (which lies in D^* and is h^*-coherent). As $\mathcal{H}$ is finite, after finitely many data the survivors are exactly U^*, and the intersection of their relations, restricted to D^*, is $\models_{h^*}$. Refuting h by a practice datum s requires a certificate that $s \notin \models_h$, i.e. that $T_h \cup \rho_h\Gamma \not\models \rho_h y$; this is co-r.e. and in general not semi-decidable (for $T_h = \mathsf{ZFC}$), so (c) describes the ideal version space.

(d) *Practice.* The data are the elements of $\mathrm{St}^g_{h^*}$, and a datum s refutes h iff $s \notin \mathrm{St}^g_h$. So complete presentation eventually refutes exactly the h with $\mathrm{St}^g_{h^*} \not\subseteq \mathrm{St}^g_h$. Since $\{x\} \rhd x$ needs zero rule applications, $(\{x\} \rhd x) \in \mathrm{St}^g_{h^*}$ for each $x \in D^*$, and practice survivors have $D_h \supseteq D^*$. *Objects and paradoxes.* Among practice survivors, Lemma I.2 shows that some presented object datum refutes h iff $\models_h|_{D^*} \not\subseteq \models_{h^*}$, and that paradoxes refute none of those with $\models_h|_{D^*} \subseteq \models_{h^*}$. *Conclusion.* As $\mathcal{H}$ is finite, the survivors are eventually exactly S^g. $h^* \in S^g$, and every member of S^g contains $\mathrm{St}^g_{h^*}$, so the intersection of the survivors' g-step relations is $\mathrm{St}^g_{h^*}$. The upper bracket holds by definition of S^g. For the lower bracket, let $(\Gamma \rhd y) \in \mathrm{Ch}_{D^*}(\mathrm{St}^g_{h^*})$, witnessed by an argument with lines in $D^* \subseteq D_h$ and inferences and links in $\mathrm{St}^g_{h^*} \subseteq \mathrm{St}^g_h \subseteq \models_h$. Lemma 10.6 under h and monotonicity give $\Gamma \models_h y$. If h^* is step-expressive on D^* at gap g, the two brackets meet, so $\models_h|_{D^*} = \models_{h^*}$ and $h \in U^*$; conversely every $h \in U^*$ with $\mathrm{St}^g_h \supseteq \mathrm{St}^g_{h^*}$ is in S^g. *Effectivity.* Practice refutation is decidable because St^g_h is. An object datum v refutes h iff $T_h \cup \rho_h\Gamma_v$ is inconsistent (notation of Lemma I.2), which is semi-decidable since Cl_{R_h} is r.e.; paradoxes are found by search. Each non-member of S^g is refuted by a finite datum with a finite certificate, and $\mathcal{H}$ is finite, so the budgeted version space equals S^g after finitely many data and a finite budget. Example 10.15 shows that survivors can have strictly weaker meaning. □

Verification of Example 10.15. Both hypotheses are propositional (a fragment of first-order logic) and share the reading; $D^* = D_h$ consists of occurrences read as $p_0, \neg p_0, p_3, \neg p_3$, and $\sim$ maps the occurrence read as p_i to the one read as $\neg p_i$ and back, so D^* is negation-closed. R^* consists of the zero-premise steps $\rhd\tau$ for $\tau \in T^*$ and all steps $\Pi \rhd \varphi$ with Π finite and Π tautologically implying φ; R_h consists of the tautological steps only. Then $\mathrm{Cl}_{R^*}(\Gamma)$ is the set of tautological consequences of $\Gamma \cup T^*$ and $\mathrm{Cl}_{R_h}(\Gamma)$ that of Γ, so both hypotheses are first-order complete, with $T_h = \emptyset$.

Let $1 \le g \le 3$. Since $R_h \subseteq R^*$, $\mathrm{St}^g_h \subseteq \mathrm{St}^g_{h^*}$. Conversely, let $\Gamma \rhd y$ be in $\mathrm{St}^g_{h^*}$, via an R^*-derivation π of $\rho(y)$ from $\rho\Gamma$ with at most g applications. If $\rho(y) \in \rho\Gamma$, the step is in St^g_h with zero applications. Otherwise the last step of π is tautological, because axiom steps conclude members of T^*, which are not readings of occurrences in D^*. So π uses at most $g - 1 \le 2$ axiom steps, and $\rho(y)$ is a tautological consequence of $\rho\Gamma \cup T'$ for some $T' \subsetneq T^*$. We claim $\rho(y)$ is then a tautological consequence of $\rho\Gamma$ alone. Otherwise some valuation v of p_0, p_3 satisfies $\rho\Gamma$ and

falsifies $\rho(y)$ (all four formulas mention only p_0, p_3). Let $p_i \to p_{i+1}$ be a chain axiom missing from T', and extend v by $v(p_k) := v(p_0)$ for $k \le i$ and $v(p_k) := v(p_3)$ for $k \ge i+1$. Every axiom of T' is then of the form $p_k \to p_{k+1}$ with both sides equal, so v satisfies $T' \cup \rho\Gamma$ and falsifies $\rho(y)$, a contradiction. Hence the single tautological step $\rho\Gamma \rhd \rho(y)$ is in R_h, and $\Gamma \rhd y \in \mathrm{St}_h^1 \subseteq \mathrm{St}_h^g$. So $\mathrm{St}_h^g = \mathrm{St}_{h^*}^g$.

$(p_0 \rhd p_3) \in \models_{h^*}$, since T^* implies $p_0 \to p_3$; it is not in $\models_h$, since $p_0 \not\models p_3$. So $\models_h|_{D^*} \subsetneq \models_{h^*}$ and $h \notin U^*$. Every object datum extends to an h-admissible valuation, since $\mathrm{Mod}(T^*) \subseteq \mathrm{Mod}(\emptyset)$ gives $\mathcal{V}_{h^*} \subseteq \mathcal{V}_h$; and every A on which h^* is coherent is h-coherent, since $T_h \subseteq T^*$. So h survives every presentation in the sense of Theorem 10.14(d), although $h \notin U^*$. At $g = 4$, the derivation from p_0 by the three axiom steps and one tautological step puts $(p_0 \rhd p_3)$ in $\mathrm{St}_{h^*}^4 \setminus \mathrm{St}_h^4$, and practice refutes h. □

A random test of Theorem 10.14(d) (computed). On 2604 random (target, rival) pairs over four atoms, the characterization $\{h \in U^* : \mathrm{St}_h^g \supseteq \mathrm{St}_{h^*}^g\}$, which (d) asserts only under step-expressiveness, differs from the true set of survivors 527 times, and S^g never; on the 1545 pairs with a step-expressive target the two characterizations never differ (`verification_checks.py`, check V1).

The nonstandard instance of Proposition 10.16 (modulo the cited transfer principle). Let ρ read informal analysis into *bounded* sentences of a countable sublanguage of the $\in$-language of the superstructure $V(\mathbb{R})$, with constants for the standard objects the practice can name, and let R be first-order logic with bounded axioms true in $V(\mathbb{R})$. With $\mathbb{M} = \{V(\mathbb{R})\}$ and $\mathbb{M}' = \{{}^*V(\mathbb{R})\}$, Robinson's transfer principle (Robinson, 1966), which rests on Łoś's theorem, gives the hypothesis of Proposition 10.16: R is sound for both structures, and they agree on every sentence in $\rho(D_h)$. Boundedness matters: the two structures are not elementarily equivalent in the full $\in$-language ("iterated power sets of $\mathbb{R}$ exist for every finite length" fails at nonstandard lengths).

Proposition I.3 (ε-δ versus internal set theory)**.** *[T4 Prop 3.6, repaired] (modulo cited theorems of Nelson) Let ρ_{IST} read informal analysis literally in Nelson's* IST *(Nelson, 1977), with every contextual constant ("let f be continuous") read as standard ($T_{\mathsf{IST}} := \mathsf{IST} + \{\mathrm{st}(c)\}$) and infinitesimal suppositions folded into bound variables (ξ under "dx is infinitesimal" becomes $\forall dx\,(dx \simeq 0 \wedge dx \neq 0 \to \xi)$). Let D_{st} be the occurrences read as sentences with standard parameters only, and let $h_{\mathrm{W}} = (\mathsf{ZFC} + \text{constants}, \rho_{\mathrm{W}})$, where $\rho_{\mathrm{W}}(x)$ is the internal sentence Nelson's reduction algorithm assigns to $\rho_{\mathsf{IST}}(x)$. Then $h_{\mathrm{W}} \approx_{D_{\mathrm{st}}} h_{\mathsf{IST}}$.*

Outside D_{st} nothing is claimed: $f(x) = Nx$ with N unlimited is ε-δ continuous at 0 but fails the literal reading $\forall x\,(x \simeq 0 \to f(x) \simeq f(0))$ at $x = 1/N$. Nor is agreement of the g-step relations claimed: if infinitesimal proofs are shorter, noise-free bounded-gap practice generated by h_{IST} refutes h_{W} as a granularity hypothesis, leaving meaning unaffected. The proof uses Nelson's reduction algorithm and conservativity of IST over ZFC, with transfer for the standard parameters.

Proof of Proposition I.3. Cited facts (Nelson, 1977). (N1) IST is a conservative extension of ZFC: an internal sentence provable in IST is provable in ZFC. (N2) Reduction: for every IST formula $A(\vec{t})$ there is an internal formula $B(\vec{t})$ with the same free variables such that $\mathsf{IST} \vdash \forall^{\mathrm{st}}\vec{t}\,(A(\vec{t}) \leftrightarrow B(\vec{t}))$. (T) Transfer: for internal $\varphi(\vec{x})$ with no other parameters, $\mathsf{IST} \vdash \forall^{\mathrm{st}}\vec{x}\,\varphi(\vec{x}) \to \forall\vec{x}\,\varphi(\vec{x})$.

Step 1. Let $x \in D_{\mathrm{st}}$, so $\rho_{\mathsf{IST}}(x) = A(\vec{c})$ with $\vec{c}$ contextual constants. By (N2) and $T_{\mathsf{IST}} \vdash \mathrm{st}(c_i)$ for each i, $T_{\mathsf{IST}} \vdash A(\vec{c}) \leftrightarrow B(\vec{c})$, where $B(\vec{c}) = \rho_{\mathrm{W}}(x)$.

Step 2. For internal $\varphi(\vec{c})$: $T_{\mathsf{IST}} \vdash \varphi(\vec{c})$ iff $\mathsf{ZFC} \vdash \varphi(\vec{c})$, where on the right the $\vec{c}$ are fresh constants. ($\Leftarrow$) $\mathsf{IST} \supseteq \mathsf{ZFC}$. ($\Rightarrow$) Finitely many axioms $\mathrm{st}(c_i)$ are used, so $\mathsf{IST} \vdash \bigwedge_i \mathrm{st}(c_i) \to \varphi(\vec{c})$ by the deduction theorem. The c_i do not occur in the axioms of IST, so $\mathsf{IST} \vdash \forall \vec{x}\,(\bigwedge_i \mathrm{st}(x_i) \to \varphi(\vec{x}))$, i.e. $\mathsf{IST} \vdash \forall^{\mathrm{st}} \vec{x}\,\varphi(\vec{x})$. By (T), $\mathsf{IST} \vdash \forall \vec{x}\,\varphi(\vec{x})$; this is an internal sentence, so $\mathsf{ZFC} \vdash \forall \vec{x}\,\varphi(\vec{x})$ by (N1), and hence $\mathsf{ZFC} \vdash \varphi(\vec{c})$.

Step 3. For finite $\Gamma \cup \{y\} \subseteq D_{\mathrm{st}}$: $\Gamma \models_{\mathsf{IST}} y$ iff $T_{\mathsf{IST}} \vdash \bigwedge \rho_{\mathsf{IST}}\Gamma \to \rho_{\mathsf{IST}} y$ iff (Step 1) $T_{\mathsf{IST}} \vdash \bigwedge \rho_{\mathrm{W}}\Gamma \to \rho_{\mathrm{W}} y$ iff (Step 2, the formula being internal) $\mathsf{ZFC} \vdash \bigwedge \rho_{\mathrm{W}}\Gamma \to \rho_{\mathrm{W}} y$ iff $\Gamma \models_{\mathrm{W}} y$. □

Proposition I.4 (What numbers could not be)**.** *[T4 Prop 3.7] (modulo standard facts of* ZF*) Let* D_{ar} *be the occurrences of informal arithmetic sentences, read into the* $\in$*-language via von Neumann numerals* (ρ_{vN}, $n+1 = n \cup \{n\}$) *or Zermelo numerals* (ρ_{Z}, $n+1 = \{n\}$)*, with* $+, \times$ *defined by recursion. Then* $(\mathsf{ZF}, \rho_{\mathrm{vN}}) \approx_{D_{\mathrm{ar}}} (\mathsf{ZF}, \rho_{\mathrm{Z}})$*, although the readings differ on "*$1 \in 3$*", which lies outside* D_{ar}*.*

The proof transports along the recursion-defined isomorphism given by Dedekind's categoricity theorem (Dedekind, 1888). Benacerraf (1965) thus becomes a non-identifiability theorem: practice fixes $\approx$ on its domain and nothing else.

Proof of Proposition I.4. Let ω_{vN} be the set of von Neumann natural numbers, with successor $S_{\mathrm{vN}}(x) = x \cup \{x\}$, and let ω_{Z} be the range of the function z on ω_{vN} defined by recursion, $z(0) = \emptyset$ and $z(S_{\mathrm{vN}} n) = \{z(n)\}$ (Replacement), with successor $S_{\mathrm{Z}}(x) = \{x\}$. ZF proves that both $(\omega_{\mathrm{vN}}, \emptyset, S_{\mathrm{vN}})$ and $(\omega_{\mathrm{Z}}, \emptyset, S_{\mathrm{Z}})$ are Dedekind–Peano systems: the successor is injective, $\emptyset$ is not a successor, and every subset containing $\emptyset$ and closed under successor is everything (for ω_{Z}, by induction along z). Define $+_\bullet, \times_\bullet$ on each by the recursion equations. By Dedekind's categoricity theorem (Dedekind, 1888), provable in ZF, there is a unique $f : \omega_{\mathrm{vN}} \to \omega_{\mathrm{Z}}$ with $f(0) = 0$ and $f(S_{\mathrm{vN}} n) = S_{\mathrm{Z}} f(n)$; it is a bijection, and by induction it preserves $+$ and $\times$.

For an arithmetic formula $\varphi(\vec{v})$ let φ^{vN} and φ^{Z} be its translations: quantifiers relativized to $\omega_\bullet$, and $0, S, +, \times$ interpreted as above. By induction on terms and then formulas, $\mathsf{ZF} \vdash \forall \vec{n} \in \omega_{\mathrm{vN}}\,(\varphi^{\mathrm{vN}}(\vec{n}) \leftrightarrow \varphi^{\mathrm{Z}}(f\vec{n}))$. Atomic formulas use that f commutes with the operations and is injective; connectives are immediate; for $\exists m\,\psi$ use that f is surjective. So for closed arithmetic x, $\mathsf{ZF} \vdash \rho_{\mathrm{vN}}(x) \leftrightarrow \rho_{\mathrm{Z}}(x)$, and for $\Gamma \cup \{y\} \subseteq D_{\mathrm{ar}}$ closed, $\Gamma \models_{\mathrm{vN}} y$ iff $\mathsf{ZF} \vdash \bigwedge \rho_{\mathrm{vN}}\Gamma \to \rho_{\mathrm{vN}} y$ iff $\mathsf{ZF} \vdash \bigwedge \rho_{\mathrm{Z}}\Gamma \to \rho_{\mathrm{Z}} y$ iff $\Gamma \models_{\mathrm{Z}} y$.

Occurrences with contextual constants ("let n be a number") are read with a fresh constant c and the axiom $c \in \omega_{\mathrm{vN}}$, respectively $c \in \omega_{\mathrm{Z}}$. They are not equivalent sentence by sentence, since c denotes different sets under the two readings, but the consequence relations still agree: $\Gamma \models_{\mathrm{vN}} y$ iff $\mathsf{ZF} \vdash \forall \vec{n} \in \omega_{\mathrm{vN}}(\bigwedge \Gamma^{\mathrm{vN}}(\vec{n}) \to y^{\mathrm{vN}}(\vec{n}))$ (deduction theorem and generalization over the fresh constants), iff $\mathsf{ZF} \vdash \forall \vec{m} \in \omega_{\mathrm{Z}}(\bigwedge \Gamma^{\mathrm{Z}}(\vec{m}) \to y^{\mathrm{Z}}(\vec{m}))$ (transport along the bijection f), iff $\Gamma \models_{\mathrm{Z}} y$. Finally, "$1 \in 3$" is not an arithmetic sentence; under ρ_{vN} it is true, and under ρ_{Z} it is false, since $\{\{\{\emptyset\}\}\} \not\ni \{\emptyset\}$. □

I.3 Objects versus paradoxes

Proof of Lemma 10.17. From the false conclusion, repeatedly move to a false antecedent (or link source); stop when there is none. Each move goes one step down a dependency path, so there are at most $\mathrm{depth}(\alpha)$ moves, and only antecedents of visited lines are evaluated. Premises are true, so the walk stops at an inference or link with true antecedents (or source) and false conclusion (or target). If $v = v_M \circ \rho_h$ with $M \in \mathbb{M}_h$, a valid item would have a true conclusion in M. □

In the correction games, VS is the set of surviving hypotheses. A learner is deterministic; the environment may choose the target adaptively, as long as some target is consistent with all its answers, and an item is legal only if it is legal for such a target.

Proof of Theorem 10.19. (a) Announce the set of steps carried by more than half of the surviving weight. Then (+) on $s \notin A_t$ and (−obj) on $s \in A_t$ each keep at most half of it, and h^* always survives; so after M corrections $w(h^*) \le 2^{-M} w(\mathcal{H})$. With the uniform prior this gives $M \le \lfloor \log_2 |\mathcal{H}| \rfloor$.

(b) Fix $r \ge 1$. A (−obj) item $s \in A_t \setminus h^*$ is also a legal ($-\text{bag}_r$) item $B = \{s\}$, and it deletes the same hypotheses ($h \ni s$, i.e. $h \supseteq \{s\}$). (+) items are common to both games. So against any learner, the object environment's moves are a subset of the bag environment's, and every bag-game strategy achieves in the object game at most its bag-game worst case. Hence $M_{\text{obj}} \le M^{(r)}_{\text{bag}}$. At $r = 1$ a legal bag is a set $B \subseteq A_t$ with $|B| \le 1$ and $B \not\subseteq h^*$, i.e. $B = \{s\}$ with $s \in A_t \setminus h^*$, so the games coincide.

(c) Let $\text{Ldim}(\emptyset) = -1$, and for nonempty $\mathcal{C} \subseteq 2^S$ let $\text{Ldim}(\mathcal{C})$ be the largest d such that there is a complete binary tree of depth d whose internal nodes are labelled by elements of S and such that every root-to-leaf path is realized by some $h \in \mathcal{C}$. A path is realized if, at each node labelled s, it goes right iff $s \in h$ (Littlestone, 1988). For $s \in S$ write $\mathcal{C}^1_s := \{h \in \mathcal{C} : s \in h\}$ and $\mathcal{C}^0_s := \mathcal{C} \setminus \mathcal{C}^1_s$. If $\text{Ldim}(\mathcal{C}) = d \ge 0$, then $\min(\text{Ldim}(\mathcal{C}^0_s), \text{Ldim}(\mathcal{C}^1_s)) < d$ for every s. Otherwise both subclasses would have trees of depth d, and a root labelled s above them would give a tree of depth $d+1$ for $\mathcal{C}$.

Upper bound. Announce $A_t := \{s : \text{Ldim}(\text{VS}^1_s) \ge \text{Ldim}(\text{VS}^0_s)\}$. A (−obj) item $s \in A_t$ leaves VS^0_s, and a (+) item $s \notin A_t$ leaves VS^1_s. In both cases the remaining class is the one of smaller Littlestone dimension, which is below $\text{Ldim}(\text{VS})$. The target always survives, so $\text{Ldim}(\text{VS}) \ge 0$, and there are at most $\text{Ldim}(\mathcal{H})$ corrections.

Lower bound. The environment maintains VS with a tree of depth $d = \text{Ldim}(\text{VS})$. While $d \ge 1$, let s label the root. If $s \in A_t$, it answers (−obj) s and commits to VS^0_s. This is legal: any $h \in \text{VS}^0_s$ has $s \in A_t \setminus h$, and $\text{VS}^0_s \ne \emptyset$ because it contains the hypotheses realizing the left subtree. If $s \notin A_t$, it answers (+) s and commits to VS^1_s. Either way the new version space contains a subtree of depth $d-1$. So every learner suffers at least $\text{Ldim}(\mathcal{H})$ corrections, and $M_{\text{obj}}(\mathcal{H}) = \text{Ldim}(\mathcal{H})$. This is Littlestone's theorem, transcribed. Equivalence queries with arbitrary hypotheses (Angluin, 1988) are the same game. □

On 248 random classes the exact minimax value M_{obj} equals the Littlestone dimension (computed) (`verification_checks.py`, check V4).

Proof of Theorem 10.20. (a) The cautious learner announces $A_t = \bigcap \text{VS}_t$. Since $h^* \in \text{VS}_t$, $A_t \subseteq h^*$, so it never receives a negative item; let $s_1, s_2, \dots$ be the (+) items it receives. When s_i arrives, $\text{VS}_t = \{h : h \supseteq \{s_1, \dots, s_{i-1}\}\}$ and $s_i \notin A_t$, so (s_i) is an elastic chain in h^*, and there are at most $\text{el}(\mathcal{H}, h^*) \le \text{el}^*(\mathcal{H})$ corrections. The bound $\text{el}^* \le |\mathcal{H}| - 1$ is proved as in Theorem 10.10(a). Since $M^{(r)}_{\text{bag}}$ is the optimum over learners, $M^{(r)}_{\text{bag}} \le \text{el}^*$.

(b) *Objects.* Announce S. Then $A_t \supseteq h^*$, so the only legal feedback is (−obj) b_{j^*}, which deletes every h_j with $j \ne j^*$; at least one correction is needed since $|\mathcal{H}_n| \ge 2$.

Bags, upper bound. If $n > r$, first announce S. Only bags are legal; a bag B must contain b_{j^*} and deletes every h_j with $b_j \notin B$, leaving at most $|B| \le r$ candidates. Then, while two or more candidates C remain, announce $S \setminus \{b_c\}$ for some $c \in C$. If $h^* = h_c$ there is no feedback. Otherwise a (+) item must be b_c, which deletes h_c; a bag $B \subseteq S \setminus \{b_c\}$ is contained in h_c, so

it also deletes h_c. Each correction removes a candidate, so there are at most $1 + (r-1) = r$ corrections. If $n \le r$, skip the first step: at most $n-1$. Hence $M^{(r)}_{\text{bag}} \le \min(r, n-1)$.

Bags, lower bound. The environment keeps the set C of culprits j consistent with its answers, initially all of $\{1, \dots, n\}$; while $|C| \ge 2$ every announcement differs from some consistent target, so it can always answer. Given A_t:

- if A_t rejects some b_i with $i \notin C$ (a step valid in every survivor), it answers $(+)$ b_i, which deletes nothing;
- otherwise, if A_t rejects some b_i with $i \in C$, it answers $(+)$ b_i, legal for any culprit in $C \setminus \{i\}$, and sets $C \leftarrow C \setminus \{i\}$;
- otherwise $A_t = S$. If $|C| > r$, it answers with a bag B of r candidates' steps and sets $C \leftarrow B$; if $|C| \le r$, it answers $B = \{b_j : j \in C\}$, which deletes only hypotheses already deleted.

Let $f(c)$ be the number of corrections this environment forces from $|C| = c$ until $|C| = 1$. Then $f(1) = 0$, and against the best learner $f(c) = 1 + \min\left(f(c-1),\, [c > r]\, f(r)\right)$, where the second option is unavailable for $c \le r$. So $f(c) = c - 1$ for $c \le r$, and $f(c) = 1 + \min(f(c-1), r-1) = r$ for $c > r$, by induction. Hence $M^{(r)}_{\text{bag}}(\mathcal{H}_n) \ge \min(r, n-1)$.

(c) *Coherence.* Under h_j (axioms $p_{i-1} \to p_i$ for $i \ne j$, with q_i read as p_i and $\sim q_n$ as $\neg p_n$), the valuation with p_i true iff $i < j$ satisfies every axiom together with p_0 and $\neg p_n$. So every h_j is coherent on $A = \{q_0, \sim q_n\}$, and the designation is truthful whatever the target. *Every paradox uses every b_i.* With p_0 true and p_n false, any valuation satisfying the axioms $p_{i-1} \to p_i$ for all $i \ne j$ falsifies the remaining one. So the set of all the b_i together with A is inconsistent, while A together with all b_i but one is consistent. A derivation of $\bot$ from A through suspect steps must therefore use all of them, and every bag is S. Such a bag deletes only hypotheses containing all of S, and there are none. *Lower bound.* As in (b) with bags restricted to S: an announcement that accepts all of S receives the useless bag S; one that rejects a known-valid step receives a useless $(+)$ item; one that rejects a candidate receives $(+)$ on it and loses only it. So $n-1$ corrections are forced. *The object.* Let $u \in \mathcal{V}_{h_{j^*}}$ with $u(q_0) = 1$ and $u(q_n) = 0$. The axioms force $p_0, \dots, p_{j^*-1}$ true (forward along $p_{i-1} \to p_i$, $i < j^*$) and $p_{j^*}, \dots, p_n$ false (backward from $\neg p_n$ along $i > j^*$). So $u(q_i) = 1$ iff $i < j^*$, and descent (Lemma 10.17) along the chain argument $q_0 \rhd q_1 \rhd \cdots \rhd q_n$ stops exactly at b_{j^*}. □

Exact minimax values (computed). For the single-culprit class with $n = 6$, $M_{\text{obj}} = 1$ and $M^{(r)}_{\text{bag}} = 1, 2, 3, 4, 5, 5$ for $r = 1, \dots, 6$; for "two culprits of six" ($|\mathcal{H}| = 15$), $M_{\text{obj}} = 2$ and $M^{(r)}_{\text{bag}} = 2, 4, 4, 4, 4, 4$ (`bag_vs_object_game.py`). On 295 random classes, $M_{\text{obj}} \le M_{\text{bag}} \le \text{el}^* \le |\mathcal{H}| - 1$ always held, with $M_{\text{obj}} < M_{\text{bag}}$ in 43 of them (`verification_checks.py`, check V3).

Proof of Theorem 10.21. Each $s \in A_t$ is missed by less than $\frac{1}{r+1} w(\text{VS})$, so a bag $B \subseteq A_t$ of size $\le r$ leaves (union bound) less than $\frac{r}{r+1} w(\text{VS})$; a $(+)$ item $s \notin A_t$ leaves at most that. So $w(h^*) \le (\frac{r}{r+1})^M$, and $\ln(1 + 1/r) \ge \frac{1}{r+1}$. □

Proof of Proposition 10.22. Let the blocks be $X_1, \dots, X_k$, each of b steps, and let $\mathcal{H}$ consist of the sets that omit exactly one step from each block. Write C_X for the surviving candidate culprits of block X, and call X *unresolved* if $|C_X| \ge 2$.

$M_{\text{obj}} \le k$. Announce S minus the culprits identified so far. Then $A_t \supseteq h^*$, so no $(+)$ item is legal, and each $(-\text{obj})$ item is an unidentified culprit, which resolves its block.

$M_{\text{obj}} \geq k$. The environment always answers inside one unresolved block X: with (+) on some $c \in C_X \setminus A_t$, legal for culprits in $C_X \setminus \{c\}$, or, if $C_X \subseteq A_t$, with (−obj) on some $c \in C_X$, committing to culprit c. Each answer changes only one block and resolves at most one, and while some block is unresolved an answer is available. Each of the k blocks must be resolved, so at least k corrections occur.

$M^{(r)}_{\text{bag}} \geq k \min(r, b-1)$. Let $\Phi := \sum_X \min(r, |C_X| - 1)$, initially $k \min(r, b-1)$. While $\Phi > 0$ some block is unresolved. The environment answers as follows.

- If some unresolved X has $c \in C_X \setminus A_t$: answer (+) c (legal, as the culprit can be another member of C_X). Then C_X loses one element, and Φ drops by at most 1.
- Otherwise every unresolved block has all its candidates accepted. Pick one, X. If $|C_X| > r$, answer with a bag B of r of its candidates (so $B \subseteq A_t$), committing to a culprit in B. Then $C_X \leftarrow B$, and $\min(r, |C_X| - 1)$ drops from r to $r - 1$. If $|C_X| \leq r$, answer $B = C_X$, which is legal and deletes no survivor; Φ is unchanged.

Rejections of known-valid steps and acceptances of identified culprits are ignored. Each correction lowers Φ by at most one, so at least $\Phi_0 = k \min(r, b-1)$ corrections occur. With $b = r + 1$: $|\mathcal{H}| = (r+1)^k$, so $k = \log|\mathcal{H}| / \log(r+1)$, and $M^{(r)}_{\text{bag}} \geq kr$. □

Equality $M^{(r)}_{\text{bag}} = k \min(r, b-1)$ holds for $(b, k) \in \{(2,2), (3,2), (2,3)\}$ and all r (computed) (`bag_product.py`).

Proof of Proposition 10.23. $M_{\text{obj}}(\mathcal{H}) = 1$ means some first announcement A is such that every legal feedback leaves a single hypothesis. A feedback item s against A (either kind) leaves exactly the hypotheses h with $s \in A \triangle h$. The environment may return any $s \in A \triangle h^*$ for any $h^* \in \mathcal{H}$, so the sets $A \triangle h$, $h \in \mathcal{H}$, are pairwise disjoint. In the bag game, announce A. A (+) item $s \notin A$ lies in exactly one $A \triangle h$ and pins the target. A bag $B \subseteq A$ with $|B| \leq r$ leaves the h with $B \not\subseteq h$, i.e. with $B \cap (A \triangle h) \neq \emptyset$. Each element of B lies in at most one such set, so at most r hypotheses survive. Then announce the survivors one at a time. Any feedback against an announced $h \neq h^*$ deletes h: a (+) item $s \in h^* \setminus h$ deletes the hypotheses lacking s, and a bag $B \subseteq h$ with $B \not\subseteq h^*$ deletes the hypotheses containing B. So at most $r - 1$ further corrections occur, at most r in all. (If $|\mathcal{H}| \leq r$, plain elimination costs at most $r - 1$.) □

Evidence for Conjecture 10.24 (computed). The ratio $M^{(r)}_{\text{bag}} / (r \cdot \text{Ldim})$ is at most 1 on all 519 (class, r) pairs of `bag_size_conjecture.py`, and the proved case Proposition 10.23 was also checked on 304 random pairs (check V5). An independent search made during the adversarial verification of this section, exhaustive over all classes with $|S| \leq 4$ and covering about 4.7 million random pairs with $|S| \leq 6$, also found no violation; it is recorded in `research/verification/` and is not among the scripts.

Proposition I.5 (Free barring nullifies objects; trivial)**.** *[T4 Prop 4.7] If barring object data is free, objects constrain nothing: every hypothesis consistent with (P) and (C) can survive forever. If objects are truthful and each barred datum costs $c > 0$ bits, a hypothesis that must bar m data loses to h^* in description length once $m > (\ell(h^*) - \ell(h))/c$.*

Proof. If barring is free, (O) refutes nothing, and only (P) and (C) remain. With cost c, h scores $\ell(h) + mc$ against $\ell(h^*)$, since truthful objects never force h^* to bar. □

I.4 Frege, Russell, Zermelo

Proof of Theorem 10.26. Refutations are sound, so a refuted set is incoherent. (a) I contains every support and is strictly shortest. (b) A coherent c has no refutation; an incoherent c has one of size $k(c)$, found by every search with budget $t \geq k(c)$. (c) For $t \geq t_0$, by (b), the unrefuted members of $\mathcal{R}$ are exactly the coherent ones. (d) Normalizing w_t does not change the arg max. For every $c \in \mathcal{R}$, $|\mathrm{cov}_t(c)/\|w_t\|_1 - \mathrm{cov}_\infty(c)| = \left|\sum_{i \in c}(w_t(i)/\|w_t\|_1 - w_\infty(i))\right| \leq \|w_t/\|w_t\|_1 - w_\infty\|_1 \to 0$, where $\mathrm{cov}_\infty(c) := \sum_{i \in c} w_\infty(i)$. Let c^* be the unique coherent maximizer of cov_∞, and $\gamma > 0$ its margin over the other coherent members ($\mathcal{R}$ is finite). Once $t \geq t_0$ and the ℓ_1 distance is below $\gamma/2$, c^* has strictly larger coverage than every other coherent member, so $\mathrm{REP}_t = c^*$. Pointwise convergence alone would not suffice: the members of $\mathcal{R}$ are typically infinite, and mass escaping along instances inside some c breaks the convergence of coverages. (By Scheffé's lemma, pointwise convergence of probability vectors to a probability vector is enough.) (e) Coh is downward closed, so any superset of the incoherent $c_1 \cup c_2$ is incoherent. If $\mathrm{supp}(w_t) \subseteq c_1 \cap c_2$, then $\mathrm{cov}_t(c_1) = \mathrm{cov}_t(c_2) = \|w_t\|_1$, and the lexicographic comparison reduces to ℓ. □

Proof of Lemma 10.27. Membership of the menu formulas in the classes (Table 13):

- POS: V, S, INT, UNI, PAIR;
- STRAT ∩ Φ: V, EMP, INT, UNI, PAIR, DIFF, CMP;
- SEP: INT, DIFF, ZR;
- Z ∩ Φ: INT, DIFF, ZR, EMP, PAIR, UNI.

The formulas S, R and ZR contain $x \in x$ and so admit no stratification.

(F1) Comp(R) gives y with $\forall x\,(x \in y \leftrightarrow x \notin x)$; instantiating $x := y$ gives $y \in y \leftrightarrow y \notin y$, hence $\bot$.

(F2) Let $\{u\}$ be a one-element structure with $u \in u$. Every atomic formula ($x = y$ or $x \in y$) is true under every assignment, so every positive formula, built from atoms by $\wedge, \vee, \forall, \exists$, is true. Hence $\forall x\,(x \in u \leftrightarrow \varphi)$ holds for every positive φ and all parameters, and $y := u$ witnesses every positive comprehension instance. Extensionality holds trivially.

(F3) The finite and cofinite subsets of $\mathbb{N}$ form a countably infinite Boolean algebra FC; let $e : \mathbb{N} \to \mathrm{FC}$ be a bijection and put $x \in y :\iff x \in e(y)$. For parameters a, b, the extensions of the stratified menu formulas are $\mathbb{N}$ (V), $\emptyset$ (EMP), $e(a) \cap e(b)$ (INT), $e(a) \cup e(b)$ (UNI), $\{a, b\}$ (PAIR), $e(a) \setminus e(b)$ (DIFF) and $\mathbb{N} \setminus e(a)$ (CMP). All are in FC, hence equal $e(y)$ for some y. Injectivity of e gives Extensionality.

(F4) In the hereditarily finite sets with true membership, INT, DIFF and ZR define subsets of $a \in \mathrm{HF}$, which are in HF; EMP, PAIR and UNI define $\emptyset$, $\{a, b\}$ and $a \cup b$, which are in HF. Extensionality and Foundation hold.

(F5) POS ∪ Z and STRAT ∪ Z contain V and ZR. Comp(V) gives v with $\forall x\,(x \in v)$; Comp(ZR) with $a := v$ gives y with $\forall x\,(x \in y \leftrightarrow x \in v \wedge x \notin x)$, i.e. $\forall x\,(x \in y \leftrightarrow x \notin x)$, and (F1) applies. POS ∪ STRAT contains S and CMP. Comp(S) gives s with $\forall x\,(x \in s \leftrightarrow x \in x)$; Comp(CMP) with $a := s$ gives y with $\forall x\,(x \in y \leftrightarrow x \notin s)$, i.e. $\forall x\,(x \in y \leftrightarrow x \notin x)$, and (F1) applies.

(F6) SEP ⊊ Z by the lists. S ∈ POS \ (STRAT ∪ Z), CMP ∈ STRAT \ (POS ∪ Z), V ∈ STRAT \ Z, ZR ∈ Z \ (POS ∪ STRAT), and DIFF ∈ Z \ POS. So the three classes are pairwise incomparable. □

Proof of Theorem 10.28. (i) is Theorem 10.26(a), since NC contains every support and is shortest. (ii) is (F1); NC is the only incoherent member, so t_0 is the size of that refutation. (iii) NC is incoherent; POS, STRAT and Z are coherent (F2–F4); SEP $\subsetneq$ Z is coherent but not maximal; and POS, STRAT and Z are pairwise incomparable (F6), hence all maximal. They are pairwise jointly incoherent by (F5).

(iv) By Theorem 10.26(c), for $t \geq t_0$ the learner REP_t maximizes $(\mathrm{cov}, -\ell)$ over the coherent classes POS, SEP, STRAT and Z, whose description lengths increase in that order. Let W be the total weight and supp the support, with all weights in the support positive; $\mathrm{cov}(c) = W$ iff $\mathrm{supp} \subseteq c$. *Support* $\subseteq$ DC. STRAT and Z contain DC, so both have coverage W, and STRAT beats Z on ℓ. POS has coverage W iff the support avoids DIFF and EMP; SEP iff it avoids UNI, PAIR and EMP. The maximal coverage is W, and the shortest class achieving it is selected: POS if it qualifies, else SEP if it qualifies, else STRAT. So STRAT is selected iff the support meets both $\{\mathrm{DIFF}, \mathrm{EMP}\}$ and $\{\mathrm{UNI}, \mathrm{PAIR}, \mathrm{EMP}\}$, as claimed; the full support DC selects STRAT. $\mathrm{ZR} \in \mathrm{supp} \subseteq \mathrm{DC} \cup \{\mathrm{ZR}\}$. POS and STRAT miss ZR, so their coverage is below W. Z has coverage W, and SEP does iff the support avoids UNI, PAIR and EMP. Since SEP is shorter than Z, SEP is selected in that case and Z otherwise. The full support selects Z. $\{\mathrm{ZR}, \mathrm{V}\} \subseteq \mathrm{supp} \subseteq \mathrm{DC} \cup \{\mathrm{ZR}, \mathrm{V}\}$. No class covers everything. Weights outside the support are 0, and

$$\begin{aligned} \mathrm{cov}(\mathrm{STRAT}) &= W - w(\mathrm{ZR}), \qquad \mathrm{cov}(\mathrm{Z}) = W - w(\mathrm{V}), \\ \mathrm{cov}(\mathrm{POS}) &= \mathrm{cov}(\mathrm{STRAT}) - w(\mathrm{DIFF}) - w(\mathrm{EMP}), \\ \mathrm{cov}(\mathrm{SEP}) &= \mathrm{cov}(\mathrm{Z}) - w(\mathrm{UNI}) - w(\mathrm{PAIR}) - w(\mathrm{EMP}). \end{aligned}$$

So POS ties STRAT iff the support avoids DIFF and EMP, and is below it otherwise; SEP ties Z iff the support avoids UNI, PAIR and EMP, and is below it otherwise. Recall $\ell(\mathrm{POS}) < \ell(\mathrm{SEP}) < \ell(\mathrm{STRAT}) < \ell(\mathrm{Z})$. If $w(\mathrm{ZR}) > w(\mathrm{V})$, the maximal coverage $W - w(\mathrm{V})$ is reached by Z, and by SEP exactly when it ties Z; SEP is shorter, so SEP is selected then and Z otherwise. If $w(\mathrm{V}) > w(\mathrm{ZR})$, the maximal coverage $W - w(\mathrm{ZR})$ is reached by STRAT, and by POS exactly when it ties STRAT, so POS is selected then and STRAT otherwise; for example $\{\mathrm{INT}, \mathrm{UNI}, \mathrm{PAIR}, \mathrm{ZR}, \mathrm{V}\}$ gives POS. If $w(\mathrm{ZR}) = w(\mathrm{V})$, STRAT and Z both reach the common maximum, POS and SEP reach it under their tie conditions, and the shortest class reaching it wins: POS if the support avoids DIFF and EMP, else SEP if it avoids UNI, PAIR and EMP, else STRAT. The full support $\mathrm{DC} \cup \{\mathrm{ZR}, \mathrm{V}\}$ meets all of DIFF, EMP, UNI and PAIR, so Z is selected iff $w(\mathrm{ZR}) > w(\mathrm{V})$, and ties go to STRAT. □

Computation (computed). The script `verification_checks.py` (check V2) compares the conditions of (iv) with the learner of `comprehension_toy.py` on all 93 nonempty sub-supports of DC, alone, with ZR, and with ZR and V, the last family at $w(\mathrm{ZR}) > w(\mathrm{V})$; there is no mismatch. The rule for supports containing ZR and V was also checked against the same learner on all 32 subsets of DC (with ZR and V added) under weight patterns with $w(\mathrm{ZR})$ above, equal to and below $w(\mathrm{V})$, again with no mismatch.

Proposition I.6 (Continuum many maximal repairs)**.** *[T4 Prop 5.4] The instances of naive comprehension have exactly $2^{\aleph_0}$ maximal consistent subsets; even* $\mathrm{Comp}(x \neq x)$ *alone has $2^{\aleph_0}$ pairwise incompatible maximal consistent extensions.*

Proof of Proposition I.6. For $n \geq 1$ let p_n be the sentence "some object has exactly n elements". For a sentence q let $J(q) := \mathrm{Comp}(x \notin x \wedge q)$, an instance of naive comprehension. For $A \subseteq \mathbb{N}_{\geq 1}$

let

$$I_A := \{J(p_n) : n \notin A\} \cup \{J(\neg p_n) : n \in A\} \cup \{\mathrm{Comp}(x \neq x)\}.$$

$J(q) \wedge q \vdash \bot$*:* if q holds, $J(q)$ yields y with $\forall x\,(x \in y \leftrightarrow x \notin x)$, and (F1) applies. So $I_A \vdash \neg p_n$ for $n \notin A$ and $I_A \vdash p_n$ for $n \in A$, and $I_A \cup I_B$ is inconsistent whenever $A \neq B$. *I_A is consistent.* Let M_A have domain $\{e_0, e_1, \dots\} \cup \{a_n : n \in A\}$, where every e_i has no elements, a_n has exactly the elements $e_0, \dots, e_{n-1}$, and there is no other membership. Then p_n holds in M_A iff $n \in A$. Each member of I_A is either $J(q)$ with q false in M_A or $\mathrm{Comp}(x \neq x)$; in both cases the defining formula has empty extension for all parameters, and e_0 witnesses it. *Maximal extensions.* Consistency has finite character, so by Zorn's lemma each I_A extends to a maximal consistent set J_A of instances. If $A \neq B$, then $J_A \cup J_B \supseteq I_A \cup I_B$ is inconsistent, so $J_A \neq J_B$ and the two are incompatible. Every J_A contains $\mathrm{Comp}(x \neq x)$. There are $2^{\aleph_0}$ sets A. There are only countably many instances, so there are at most $2^{\aleph_0}$ maximal consistent sets. The argument uses only pure logic. With Extensionality as background, the same argument works with the Zermelo numerals $z_0 = \emptyset$, $z_{k+1} = \{z_k\}$ in place of the e_i, $a_n = \{z_0, \dots, z_{n-1}\}$ and $A \subseteq \mathbb{N}_{\geq 2}$ (each z_k has at most one element, so p_1 always holds). □

Limit approximation. Enumerate the instances $\iota_0, \iota_1, \dots$, and put $S_0 = \emptyset$, $S_{k+1} = S_k \cup \{\iota_k\}$ if this set is consistent and $S_{k+1} = S_k$ otherwise. Then $S = \bigcup_k S_k$ is consistent by compactness. It is maximal: if $\iota_k \notin S$, then $S_k \cup \{\iota_k\}$ is inconsistent, so $S \cup \{\iota_k\}$ is. Consistency of a finite set of sentences is Π_1, so S is computable from the halting problem, i.e. Δ_2, and by Shoenfield's limit lemma it is the pointwise limit of a computable sequence of guesses. This does not conflict with Incurvati and Murzi (2017): being Δ_2 does not make S recursively axiomatizable.

I.5 The robust core

Proof of Theorem 10.30. (a) Every inference and link of the argument is a non-noise practice item, so by RCH_g it lies in $\mathrm{St}^g_{\mathrm{SV}} \subseteq \mathrm{St}^g_\sigma \subseteq \models_\sigma$, or is a link valid under σ, for every σ. Validity puts every line used in D_{h_σ}, and unused premises are dropped. Lemma 10.6 for h_σ gives the conclusion in $\mathrm{Cl}_{R_\sigma}(\rho_\sigma\mathrm{Prem})$. If T_σ proves the premises, then since Cl_{R_σ} is consequence over T_σ, it proves the conclusion.

(b) Each datum is consistent with each h_σ:

- practice steps lie in $\mathrm{St}^g_{\mathrm{SV}} \subseteq \mathrm{St}^g_\sigma$;
- a "yes" is given only on $\mathrm{St}^g_{\mathrm{SV}} \subseteq \mathrm{St}^g_\sigma$, and a "no" only outside $\bigcup_{\sigma'} \mathrm{St}^g_{\sigma'} \supseteq \mathrm{St}^g_\sigma$, for steps and links alike;
- designated contexts are σ-coherent, so no paradox for h_σ exists;
- an SV-truthful datum at the precise object M is $v(x) = v_{M_\sigma}(\rho_\sigma(x))$ for super-determinate x, the same value for every σ; so v is the restriction of $v_{M_\sigma} \circ \rho_\sigma \in \mathcal{V}_{h_\sigma}$.

Budgeted searches find only certificates, so no h_σ is ever refuted and $\{h_\sigma\} \subseteq \mathrm{VS}_t$. Unanimity then gives acceptance $\subseteq \bigcap_\sigma \mathrm{St}^g_\sigma = \mathrm{St}^g_{\mathrm{SV}} \subseteq \models_\sigma$ for every σ. *Convergence.* Let $\mathcal{H}$ be finite and every element of $\mathrm{St}^g_{\mathrm{SV}}$ eventually presented. Any h lacking some element of $\mathrm{St}^g_{\mathrm{SV}}$ is refuted when that element is presented, so after finitely many data every survivor contains $\mathrm{St}^g_{\mathrm{SV}}$. The h_σ survive, so the intersection of the survivors' relations is contained in $\bigcap_\sigma \mathrm{St}^g_\sigma$, and therefore equals $\mathrm{St}^g_{\mathrm{SV}}$. *Effectivity.* For infinite Σ the robust core can be undecidable. Let the vague sentences be $w_0, w_1, \dots$; let σ_n read w_m as the Δ_0 sentence "machine m does not halt within n steps" over a base theory proving all true Δ_0 sentences; and let each R_σ contain the zero-premise steps $\triangleright\delta$

for true Δ_0 sentences δ, a decidable set. Then $(\emptyset \rhd w_m) \in \mathrm{St}^1_{\sigma_n}$ iff machine m does not halt within n steps. A false Δ_0 sentence is refutable, hence underivable in a sound calculus. So $(\emptyset \rhd w_m) \in \mathrm{St}^1_{\mathrm{SV}}$ iff machine m never halts, which is Π_1-complete. The idealized verifier is an algorithm only for finite Σ or under extra uniformity.

(c) By (b), under these data every h_σ remains a possible target forever. (c1) If an acceptance set A is contained in $\models_\sigma$ for every σ, then $A \subseteq \bigcap_\sigma \models_\sigma = \models_{\mathrm{SV}}$. (c2) The same argument works with St^g_σ. If $\mathcal{H} = \{h_\sigma\}$, all members survive by (b), and the verifier's accepted set is exactly $\bigcap_\sigma \mathrm{St}^g_\sigma = \mathrm{St}^g_{\mathrm{SV}}$ at all times. *The gap.* Let $\Sigma = \{\sigma\}$, let R_σ be natural deduction for classical propositional logic, and let $\rho(x) = q$ and $\rho(y) = q \wedge (q \wedge q)$. Then $(\{x\} \rhd y) \in \models_\sigma$. No single rule application derives $q \wedge (q \wedge q)$ from q. $\wedge$I would need the premise $q \wedge q$; $\to$E would need an implication with that consequent; $\bot$E would need $\bot$; and the classical reductio rule needs a subderivation of at least one further step. So $(\{x\} \rhd y) \notin \mathrm{St}^1_\sigma$, while two $\wedge$I steps suffice.

(d) By (b), the accepted set is contained in $\mathrm{St}^g_{\mathrm{SV}}$, so steps outside it are never accepted. (d1) Let $s \notin \models_\sigma$ for some $\sigma \in \Sigma$. A "yes" on s would be false under σ, so no SV-truthful answer licenses s. Any library extension must keep R_σ sound for $\mathrm{Mod}(T_\sigma)$, so it cannot make s valid under σ. Any expansion into σ-valid steps would make s σ-valid by Lemma 10.6. So while $\sigma \in \Sigma$ and s is unchanged, s stays outside $\mathrm{St}^g_{\mathrm{SV}}$; only shrinking Σ (a definition) or changing the step (lemma-incorporation) removes it. (d2) Let $s = (\Gamma \rhd y) \in \models_{\mathrm{SV}}$. Adding the derived step $\rho_\sigma\Gamma \rhd \rho_\sigma y$ to each library R_σ keeps R_σ sound, because the step is σ-valid. It puts s in $\mathrm{St}^1_\sigma \subseteq \mathrm{St}^g_\sigma$ for every σ, so $s \in \mathrm{St}^g_{\mathrm{SV}}$. Alternatively, an expansion of s into steps of $\mathrm{St}^g_{\mathrm{SV}}$, when one exists, is accepted step by step (Proposition 10.11). □

Theorem I.7 (Steps invalid under a sharpening have monsters; trivial)**.** *[T4 Thm 6.3, repaired]* *If $\Gamma \cup \{y\} \subseteq D_{h_\sigma}$ and $(\Gamma \rhd y) \notin \models_\sigma$ for an admissible σ, some $M \models T_\sigma$ makes $\rho_\sigma\Gamma$ true and $\rho_\sigma y$ false: a σ-*monster*.*

Proof. By definition of $\models_\sigma$, since Cl_{R_σ} is consequence over T_σ; Gödel completeness is what makes such R_σ exist. □

The theorem concerns $\models_\sigma$, not St^g_σ: steps in $\models_{\mathrm{SV}} \setminus \mathrm{St}^g_{\mathrm{SV}}$ have no monster.

Proposition I.8 (Only unanimity chains; the sorites is tight)**.** *[T4 Prop 6.4]* *(known; re-derived) (a) $\models_{\mathrm{SV}}$ is a consequence relation on $\bigcap_\sigma D_{h_\sigma}$. (b) For a probability μ on Σ (with $\{\sigma : s \in \models_\sigma\}$ measurable), if each of the n steps of an argument (links included) is valid with probability $\geq 1-\varepsilon$, the argument is valid with probability $\geq 1 - n\varepsilon$; this is attained for every $\varepsilon \leq 1/n$, and at $\varepsilon = 1/n$ by an argument valid under no sharpening.*

Proof of Proposition I.8. (a) Each $\models_\sigma$, restricted to finite subsets of $D_\sigma := D_{h_\sigma}$, is reflexive ($\{x\} \models_\sigma x$ for $x \in D_\sigma$), monotone, and closed under cut ($\Gamma \models_\sigma x$ and $\Gamma \cup \{x\} \models_\sigma y$ give $\Gamma \models_\sigma y$), because Cl_{R_σ} is a closure operator. These properties are preserved by intersection over σ, on the common domain $\bigcap_\sigma D_\sigma$. (b) Let the argument have steps (inferences and links) $s_1, \ldots, s_n$, and let $E_i := \{\sigma : s_i \in \models_\sigma\}$. Under every $\sigma \in \bigcap_i E_i$, Lemma 10.6 gives $(\mathrm{Prem} \rhd y_m) \in \models_\sigma$. So, by the union bound, the (measurable) event that the argument is valid has probability at least $\mu(\bigcap_i E_i) \geq 1 - \sum_i \mu(E_i^c) \geq 1 - n\varepsilon$. *Tightness.* Take a base theory deciding order between numerals; a vague predicate "heap"; sentences $q_k =$ "k grains make a heap" for $k = 0, \ldots, n$; and sharpenings σ_c, $c = 0, \ldots, n$, defining "heap"(x) as $x \geq c$. The argument is $q_n \rhd q_{n-1} \rhd \cdots \rhd q_0$. Under σ_c, the step $q_k \rhd q_{k-1}$ fails iff $k \geq c > k-1$, i.e. iff $c = k$, and the overall step $q_n \rhd q_0$ fails iff $c \geq 1$. Give $\sigma_1, \ldots, \sigma_n$ mass ε each and σ_0 mass $1 - n\varepsilon$ (for $\varepsilon \leq 1/n$). Each step is valid with probability exactly $1 - \varepsilon$, and the argument exactly with probability $1 - n\varepsilon$. At $\varepsilon = 1/n$, the

argument is valid under no sharpening in the support of μ. `T4-checks/robust_core_and_mdl.py` checks the cases $n = 3, 5, 10$. □

Proof of Proposition 10.31. (a) A step $s = (\Gamma \rhd y) \in A_0$ can be refuted only by a model of T in which $\rho^*\Gamma$ holds and $\rho^* y$ fails; such a model exists iff $s \notin \models_T$. By completeness of monster presentation, s survives iff $s \in \models_T$, so $A_\infty = A_0 \cap \models_T$. By Gödel completeness (Gödel, 1930), each surviving step has a derivation, in any complete calculus C for T, of $\rho^* y$ from $\rho^*\Gamma$. For an argument chained from surviving steps, replacing each step by such a derivation and concatenating yields a C-derivation of the ρ^*-image of the conclusion from the images of the premises. This is Lemma 10.6 with the derivations made explicit.

(b) Here $A_\infty = \{\chi \rhd \theta : \mathsf{PA} \vdash \chi \to \theta\}$, which is r.e. but undecidable: with χ the sentence $0 = 0$, membership decides PA-provability of arithmetic sentences. Let C be a complete calculus for PA satisfying the effectivity assumptions, and suppose $A_\infty \subseteq \mathrm{St}_C^g$ for some g. Since $\mathrm{St}_C^g \subseteq \models_{\mathsf{PA}}$ and $A_0 \cap \models_{\mathsf{PA}} = A_\infty$, we get $A_\infty = A_0 \cap \mathrm{St}_C^g$, which is decidable because A_0 and St_C^g are. Contradiction.

(c) Let $A_0 = \bigcup_{j \le J} \mathrm{inst}(\mathcal{S}_j)$. Under schema-level selection with complete monster presentation, $\mathcal{S}_j$ survives iff none of its instances is refuted, iff all its instances lie in $\models_T$. For each surviving $\mathcal{S}_j$, the template π_j instantiates to a derivation of each instance with at most g rule applications, and its formulas have size at most $|\pi_j|$ times the size of the instance. Choosing the size bound of Definition 10.2 to be at least $\max_j |\pi_j|$ times the input size, every surviving step is in St_C^g. The templates are finitely many derived rules of C, whose instances are the surviving practice. *Pure propositional schemas.* Let $\mathcal{S}$ have metavariables $A_1, \ldots, A_k$ and mention no atoms, with all instances valid. Its generic instance $\mathcal{S}[A_i := p_i]$, for distinct atoms p_i, is valid, so it has a derivation π in a complete propositional calculus with schematic rules. Applying the substitution $p_i \mapsto \varphi_i$ to every formula of π yields a derivation, with the same number of applications, of $\mathcal{S}[A_i := \varphi_i]$, because schematic rules are closed under uniform substitution. So π is a uniform template. *Failure in* PA. Let $\theta(n)$ be $\neg \mathrm{Prf}_{\mathsf{PA}}(n, \ulcorner 0 = 1 \urcorner)$. Each instance $\theta(\bar{n})$ is a true statement about a numeral, decided by a primitive recursive relation, and is therefore provable in PA (by Σ_1-completeness, since PA is consistent). Suppose a template with a schematic letter for the numeral instantiated to a derivation of $\theta(\bar{n})$ for every n. In a standard Hilbert calculus, substituting a fresh variable x for the schematic letter turns it into a derivation of $\theta(x)$, and generalization gives $\mathsf{PA} \vdash \forall x\, \theta(x)$, i.e. $\mathsf{PA} \vdash \mathrm{Con}(\mathsf{PA})$. This contradicts Gödel's second incompleteness theorem. *Calculus relativity.* The argument needs rules closed under substituting a fresh variable for a numeral. A complete calculus whose decidable library contains the zero-premise steps $\rhd \theta(\bar{n})$ for every n (a decidable set of true sentences) makes the schema uniformly 1-step derivable. So uniform derivability, and hence the bound of (c), is relative to the calculus and its library. □

Remark I.9 (Euclid: an analogy). [T4 §6.2] The notes suggest looking at Euclid, who "didn't have a full set of axioms". If a "sharpening" of a diagram is any configuration near it, Manders's *co-exact* properties (Manders, 2008) are the super-determinate ones, his thesis that Euclid reads off only co-exact properties is the analogue of RCH, and the system E of Avigad et al. (2009), sound and complete for a class of sequents relative to ruler-and-compass semantics, plays the role of the robust core of Theorem 10.30(c2). Perturbations are not sharpenings in the sense of Definition 10.29, so this is an analogy only.

I.6 Language invention: what this section adds to the open problem

The hardest open problem of this section is the first problem of Section 14: affordable soundness when the learner must invent language. A universal class (all computable (L, R, ρ) with a description-length prior) restores realizability: Theorem 10.9 is sound with $W^* \geq 2^{-K(h^*)}$, and, information-theoretically, objects need at most $K(h^*)$ corrections (Theorem 10.19(a), weighted halving). But corrections are not soundness, since halving is bold; the sound verifier pays escalations, $\tilde{\Theta}(2^{K(h^*)})$ on unstructured parts of the class (Theorem 10.10(c), Section 4), and paradoxes cannot rescue it (Theorem 10.20). The problem is to find a class that is closed under the definitional extensions history made (say, predicates defined by formulas of bounded quantifier rank over the old language) and has escalation dimension, or correction complexity with objects, polynomial in the target's description length; or to prove that such closure forces exponential antichains like the single-culprit class. Definitions are schemas, so the anti-unification bounds of Section 5 may transfer to definition learning; whether proof-generated concepts can be learned with $O(\text{depth})$ objects each is the concrete sub-question. Further open problems from this section are the bag-price conjecture (Conjecture 10.24), misspecified practice (such as Leibnizian infinitesimals), moving readings, a corpus test of RCH, and identifiability of meaning without step-expressiveness.

J Proofs for Section 11

The following results are proved completely in the main text: Propositions 11.4, 11.17, 11.20 and 11.30, Theorems 11.8, 11.14, 11.23 and 11.28, and Lemma 11.18. Proposition J.14 and the numerical claims marked (computed) are computations, reproduced by the scripts in `research/theory/T3-checks/`; they are not proved here. Everything else is proved below. Secondary results that the main text only summarizes (further export certificates, certification of validity regions, the finite-sun bridge and the prototype's defects) are stated here in full. Throughout, w is a world, $\mathrm{dom} = \mathrm{dom}\,\mathfrak{M}_w$, and $\delta(F \times G) := \{\lambda[D{:=}\mu] : \lambda \in F,\ \mu \in G\}$.

J.1 Preliminaries on the filters of Definition 11.2

Lemma J.1 (Filter bases)**.** (a) *For $c = \mathsf{DEF}(D, \lambda^\circ, \mathcal{G})$ with parent p, the family $\mathcal{B}_c = \{\delta(F \times G) \cap \mathrm{dom} : F \in \mathcal{F}_p(w),\ G \in \mathcal{G}\}$ is a filter base: $X \in \mathcal{F}_c(w)$ iff $X \supseteq B$ for some $B \in \mathcal{B}_c$, and c is proper at w iff every member of $\mathcal{B}_c$ is nonempty.* (b) *For $c = \mathsf{SUP}(A)$ with parent p, $X \in \mathcal{F}_c(w)$ iff $X \supseteq F \cap \|A\|_w$ for some $F \in \mathcal{F}_p(w)$.* (c) *If the parent filter is principal at a point λ, then in (a) one may take $F = \{\lambda\}$: $X \in \mathcal{F}_c(w)$ iff $X \supseteq \{\lambda[D{:=}\mu] : \mu \in G\} \cap \mathrm{dom}$ for some $G \in \mathcal{G}$.*

Proof. (a) $\delta(F_1 \times G_1) \cap \delta(F_2 \times G_2) \supseteq \delta((F_1 \cap F_2) \times (G_1 \cap G_2))$, and $F_1 \cap F_2 \in \mathcal{F}_p(w)$, $G_1 \cap G_2 \in \mathcal{G}$. So $\mathcal{B}_c$ is directed downward, and the filter it generates consists of the supersets (in dom) of its members. It contains $\emptyset$ iff some member is empty. (b) Likewise $\{F \cap \|A\| : F \in \mathcal{F}_p(w)\}$ is closed under finite intersections. (c) If $\mathcal{F}_p(w)$ is principal at λ, every $F \in \mathcal{F}_p(w)$ contains λ, so $\delta(\{\lambda\} \times G) \subseteq \delta(F \times G)$, and $\{\lambda\} \in \mathcal{F}_p(w)$. □

J.2 Proofs for §11.2

Proof of Proposition 11.3. (a) If $\mathrm{Th}_w(c) \models \psi$, compactness gives $\varphi_1, \ldots, \varphi_k \in \mathrm{Th}_w(c)$ entailing ψ, and $\|\psi\| \supseteq \bigcap_i \|\varphi_i\| \in \mathcal{F}_c(w)$, so $\|\psi\| \in \mathcal{F}_c(w)$. (b) If c is proper, every finite subset of $\mathrm{Th}_w(c)$ has a nonempty truth set (the intersection lies in the proper filter), hence a model; by compactness

$\mathrm{Th}_w(c)$ is consistent. If c is improper, $\|\bot\| = \emptyset \in \mathcal{F}_c(w)$. (c) A set lies in a filter iff it lies in every ultrafilter extending it, and Łoś's theorem (Łoś, 1955) gives $\prod_U \mathfrak{M}_w \models \varphi$ iff $\|\varphi\| \in U$. □

Proof of Proposition 11.5. (a) The root filter is principal at λ^*, where $p_{\mathrm{atm}} = 1.0 \times 10^5$. The frame has no phases, and by hypothesis the model exists at $\lambda^*[p_{\mathrm{atm}}{:=}0]$; by Lemma J.1(c) with the principal path the child filter is principal there; hence $c \models p_{\mathrm{atm}} = 0$. Both filters are principal, hence proper, and by Proposition 11.3(b) each context's theory is consistent.

(b) Under limit semantics the only generator is $\{\lambda^*[m_r{:=}0]\} \cap \mathrm{dom} = \emptyset$, so the limit child is improper. Under germ semantics the generators are $\{\lambda^*[m_r{:=}\mu] : 0 < |\mu| < \eta\} \cap \mathrm{dom} = \{\lambda^*[m_r{:=}\mu] : 0 < \mu < \eta\}$, all nonempty, so the germ child is proper (Lemma J.1(a)). In the model at $m_r = \mu > 0$, $a = \Phi/\mu$, so $|a| > K$ whenever $\mu < |\Phi|/\max(K, 1)$; the generator with $\eta = |\Phi|/\max(K, 1)$ lies inside $\||a| > K\|$.

(c) Let $E = \{\mu : \lambda^*[D{:=}\mu] \in \mathrm{dom}\}$. By Lemma J.1(c), $c_{\mathrm{germ}} \models Q \in I$ iff there is $G \in \mathcal{G}$ with $q(\mu) \in I$ for all $\mu \in G \cap E$. "For every open $I \ni q(\lambda^\circ)$ there is $G \in \mathcal{G}$ with $q(G \cap E) \subseteq I$" is the definition of $q(\mu) \to q(\lambda^\circ)$ along the trace of $\mathcal{G}$ on E. (Intervals with endpoints named in L, e.g. rational ones, suffice for convergence.) The limit child is principal at $\lambda^*[D{:=}\lambda^\circ]$, so its value of Q is $q(\lambda^\circ)$. □

Proof of Lemma 11.7. (a) Let $F = \|\varphi\|_w \in \mathcal{F}_p(w)$ and $G \in \mathcal{G}$. Every element of $\delta(F \times G) \cap \mathrm{dom}$ has the form $\lambda[D{:=}\mu] \in \mathrm{dom}$ with $\lambda \in F$; it differs from λ only in coordinates in D, so by D-stability it satisfies φ. Thus this generator lies inside $\|\varphi\|_w$, and $\|\varphi\|_w \in \mathcal{F}_c(w)$. Laws are D-stable because they hold on all of dom. An arithmetic formula in the constants p_i, $i \notin D$, is evaluated in the standard real sort with p_i interpreted as λ_i, so its truth value depends only on $(\lambda_i)_{i \notin D}$, which points differing only on D share.

(b) Let $\lambda, \lambda' \in \mathrm{dom}$ differ only on D, with $\mathfrak{M}_w(\lambda) \models \varphi$. Take the actual point $\lambda^* = \lambda$ and the ideal value $\lambda^\circ = \lambda'_D$. Then $@ \models \varphi$, and the limit child's only generator is $\{\lambda[D{:=}\lambda'_D]\} \cap \mathrm{dom} = \{\lambda'\}$, so its filter is principal at λ' and $c \models \varphi$ says $\mathfrak{M}_w(\lambda') \models \varphi$. Exchanging the roles of λ and λ' gives the converse.

(c) Every generator $\{\lambda[k{:=}0] : \lambda \in F\} \cap \mathrm{dom}$ of the child lies inside $\|k = 0\|$, so $c \models k = 0$ and, by Proposition 11.3(a), $c \models k \leq 1$, for every parent filter and every world (if c is improper it proves everything). So an import of φ into c never leads from a truth to a falsehood. But if dom contains λ and λ' differing only in k, with $k = 0.5$ and $k = 2$ respectively, then $\mathfrak{M}_w(\lambda) \models \varphi$ and $\mathfrak{M}_w(\lambda') \not\models \varphi$. □

J.3 Proofs for §11.3: eternalism and realizability

Proof of Proposition 11.9. (a) Take a classical derivation of $\bot$ from the designated, classically inconsistent A, in a Hilbert or sequent-step format. If every step instance in it were in C, chaining the steps (a consequence relation is monotone and closed under cut) would give $A \vdash_C \bot$, contradicting $\mathcal{A}$-coherence. So some step instance is not in C. It is a substitution instance of the most general instance of its schema, so by structurality that most general instance is not in C either: the schema is not C-valid, and in particular $C \not\supseteq \mathrm{Cn}_{\mathsf{CPC}}$. (b) If $p, \neg p \vdash_C \bot$ for an atom p, substituting α for p gives $\alpha, \neg\alpha \vdash_C \bot$, and weakening gives $A \vdash_C \bot$, again contradicting coherence. □

The eternalist check summarized after Theorem 11.8 is the following.

Corollary J.2 (Eternalist coherence penalizes correct reasoning)**.** *[T3 Cor 2.2, rev.] Call a check* eternalist *if it flags a set J of context-indexed judgments (c, Γ, φ) whenever $\{\bigwedge \Gamma \to \varphi : (c, \Gamma, \varphi) \in J\}$ is unsatisfiable (context indices deleted, suppositions not added). It flags (i) every sound reductio whose internal judgment $(\mathsf{SUP}(A), \emptyset, \bot)$ is recorded; (ii) every sound idealization whose stipulation contradicts a recorded root judgment (Proposition 11.5(a)); (iii) every pair of sound contexts whose different answers to one query are both recorded (IPhO 2025 Theory Problems 2 and 3). If only the discharged $(@, \emptyset, \neg A)$ is recorded, nothing is flagged: what is penalized depends on how much of the reasoning is recorded.*

Proof. The stripped sets are unsatisfiable: in (i) the judgment $(\mathsf{SUP}(A), \emptyset, \bot)$ strips to $\bot$; in (ii) the stipulation and the recorded root judgment contradict each other; in (iii) $Q = 1$ and $Q = 2$ do, by standard names. Each judgment is nevertheless sound in its context (Proposition 11.4, Proposition 11.5(a), and two proper contexts with different declarations). If only $(@, \emptyset, \neg A)$ is recorded for a sound reductio and the other recorded judgments are root judgments true at the actual point, then $\neg A$ also holds there, so the stripped set is satisfiable. □

AUX contexts are identified with their parents throughout. For a set $M \subseteq \mathcal{K}$ and a sentence A, $\|A\|$ in (R1) denotes $\{N \in \mathcal{K} : N \models A\}$. Since SUP contexts have only SUP descendants, the parent of a DEF context is never a SUP context.

Proof of Theorem 11.11(a). Observation 1 (collapse). By (R1) and induction along SUP chains, $M_c = M_b \cap \bigcap_{A \in \mathrm{Asm}(c)} \|A\|$ for every SUP context c with base b. Hence (c, Γ, φ) satisfies (R3) at M_c iff every $N \in M_b$ with $N \models \bigwedge \mathrm{Asm}(c) \wedge \bigwedge \Gamma$ satisfies φ, i.e. iff $(b, \Gamma \cup \mathrm{Asm}(c), \varphi)$ satisfies (R3) at M_b. So, given (R1), the judgments of J hold iff $M_b \subseteq \{N : N \models \mathrm{Th}^*(b)\}$ for every non-SUP context b.

Observation 2 (anchoring lemma). In any realization, $M_c \neq \emptyset$ for every $c \in \mathrm{Anc}$. We induct along the inductive definition of Anc. The root is nonempty by (R0) and certified contexts by (R5). Suppose c enters through an informative use (c, β, t) with $M_{\pi c} \neq \emptyset$, and suppose $M_c = \emptyset$. Then $M_c \models \phi_\beta(v)$ for every $v \in V$ (vacuously), so (R4) gives $M_{\pi c} \models \sigma_\beta \to \varepsilon_\beta(v)$ for all v. Since $(\pi c, \emptyset, \sigma_\beta) \in J$, (R3) gives $M_{\pi c} \models \sigma_\beta$. Any $N \in M_{\pi c}$ then satisfies $\{\sigma_\beta\} \cup \{\varepsilon_\beta(v) : v \in V_0\}$, contradicting informativeness, since $N \in \mathcal{K}$.

($\Rightarrow$) For $c \in \mathrm{Anc}$, $M_c \neq \emptyset$ by Observation 2, and any $N \in M_c$ satisfies $\mathrm{Th}^*(c)$ by Observation 1.

($\Leftarrow$) For each $c \in \mathrm{Anc}$ choose $N_c \in \mathcal{K}$ with $N_c \models \mathrm{Th}^*(c)$, and put $M_c = \{N_c\}$. Put $M_c = \emptyset$ for every non-SUP context outside Anc (these are DEF contexts, since $@ \in \mathrm{Anc}$), and define SUP contexts by (R1). Then (R0), (R1), (R2) and (R5) hold by construction, since $@ \in \mathrm{Anc}$ and $C_{\mathrm{cert}} \subseteq \mathrm{Anc}$. (R3) holds by Observation 1: at anchored b because $N_b \models \mathrm{Th}^*(b)$, and at non-anchored b vacuously. For (R4), take a use (c, β, t). If $\pi c \notin \mathrm{Anc}$, then $M_{\pi c} = \emptyset$ and (R4) is vacuous. If $\pi c \in \mathrm{Anc}$, then $c \in \mathrm{Anc}$ because β is informative, so $M_c = \{N_c\}$. The judgment $(c, \emptyset, Q = \bar{t})$ lies in J^* unchanged (c is not a SUP context), so $Q^{N_c} = t$; by standard names, $M_c \models Q = \bar{v}$ only for $v = t$. Finally $(\pi c, \emptyset, \varepsilon_\beta(t)) \in J$ gives $N_{\pi c} \models \varepsilon_\beta(t)$, so $M_{\pi c} \models \sigma_\beta \to \varepsilon_\beta(t)$. □

Proof of Theorem 11.11(b). Fix a frame realization at w.

Filter anchoring lemma: every $c \in \mathrm{Anc}$ is proper at w. The root is proper, and certified contexts are proper by assumption. Suppose c enters through an informative use (c, β, t) with πc proper, and suppose c is improper. Then $c \models_w \phi_\beta(v)$ for every v, so bridge soundness gives $\pi c \models_w \sigma_\beta \to \varepsilon_\beta(v)$ for all v. The judgment $(\pi c, \emptyset, \sigma_\beta)$ holds, so $\pi c \models_w \varepsilon_\beta(v)$ for all v (Proposition 11.3(a)). By informativeness, $\{\sigma_\beta\} \cup \{\varepsilon_\beta(v) : v \in V_0\}$ has no model in $\mathcal{K}$ for some

finite V_0; every $\mathfrak{M}_w(\lambda)$ is in $\mathcal{K}$, so the truth set of the finite conjunction is $\emptyset$. That set lies in $\mathcal{F}_{\pi c}(w)$, contradicting properness.

Conclusion. Let $c \in \mathrm{Anc}$. For a SUP descendant $c' = \mathsf{SUP}(A_k)$ of ... of $\mathsf{SUP}(A_1)$ of c, iterating Proposition 11.4 and using Proposition 11.3(a) shows that $c' \models_w \chi$ iff $c \models_w \bigwedge_i A_i \to \chi$. Hence each judgment (c', Γ, φ) holds iff $c \models_w \bigwedge \mathrm{Asm}(c') \wedge \bigwedge \Gamma \to \varphi$, and so $\mathrm{Th}^*(c) \subseteq \mathrm{Th}_w(c)$. Since $\mathrm{Th}^*(c)$ is finite, the intersection of the truth sets of its members lies in $\mathcal{F}_c(w)$, and it is nonempty because c is proper. Any λ in it gives a model $\mathfrak{M}_w(\lambda) \in \mathcal{K}$ of $\mathrm{Th}^*(c)$. No compactness is used. The argument does not depend on whether paths are principal or germs. □

Proof of Theorem 11.11(c). CE1. $C_{\mathrm{cert}} = \{c\}$, so $\mathrm{Anc} = \{@, c\}$; $\mathrm{Th}^*(@) = \{p_1 = 1\}$ and $\mathrm{Th}^*(c) = \{p_1 = 0\}$ are each satisfiable in $\mathcal{K}$ (the constant p_1 may denote any real), so the criterion holds. In a frame realization, the root judgment forces $\lambda_1^* = 1$. The certified c is proper, so $\lambda^*[2{:=}0] \in \mathrm{dom}$ and $\mathcal{F}_c$ is principal at that point, whose first coordinate is 1; hence $c \models p_1 = 1$, and $c \not\models p_1 = 0$ since $\mathrm{Th}_w(c)$ is consistent. *CE2.* Two siblings with the same declaration have the same generators, hence the same filter. Both are certified, hence proper; the judgments $Q = 1$ and $Q = 2$ would put $\|Q = 1\| \cap \|Q = 2\| = \emptyset$ in this proper filter. The criterion holds because each of $\{Q = 1\}$, $\{Q = 2\}$ is satisfiable. Both counterexamples are also brute-forced over all frames on $\Lambda = \{0, 1\}^2$ (`realizability_frame.py`). □

Brute-force checks of Theorem 11.11. Brute force agrees with (a) on 6,500 random instances with 0 mismatches: 1,500 from `realizability.py`, 3,000 for the SUP-collapse from `realizability_sup.py`, and 2,000 from a deeper stress test. With uninformative bridges (a) fails on 117/600 instances, so the hypothesis is needed. Over all frames on $\Lambda = \{0, 1\}^2$, 8,000 instances show 0 violations of (b) and 598 instances in the gap of (c) (computed).[16] Informativeness is defined in finite form because $\mathcal{K}$ is not compact: $\{Q \neq v : v \in \mathbb{R}\}$ is finitely satisfiable but has no model with a standard real sort.

The exact frame criterion under limit semantics, summarized in the main text after Theorem 11.11, is the following.

Proposition J.3 (Frame realizability under limit semantics)**.** *[T3 Prop 2.7] (brute-force checked) Assume every* DEF *context in J uses limit semantics, every used bridge is an informative value-export bridge, and* SUP *contexts have only* SUP *descendants. For a candidate actual point λ, define the points of the non-*SUP *contexts by $\lambda_@ = \lambda$ and $\lambda_c = \lambda_{\pi c}[D_c{:=}\lambda_c^\circ]$. Let $S(\lambda)$ be the least set of non-*SUP *contexts that contains @ and C_{cert} and is closed under (i)* parents*: $c \in S \Rightarrow \pi c \in S$; (ii)* informative bridges*: $(c, \beta, t) \in U$ and $\pi c \in S \Rightarrow c \in S$; (iii)* coincidence*: $\pi c \in S$ and $\lambda_c = \lambda_{c'}$ for some $c' \in S \Rightarrow c \in S$. Then J is frame-realizable iff there is $\lambda \in \Lambda$ such that every point $x \in \{\lambda_c : c \in S(\lambda)\}$ lies in Λ and $\bigcup\{\mathrm{Th}^*(c) : c \in S(\lambda),\ \lambda_c = x\}$ has a model $N \in \mathcal{K}$ with $p_i^N = x_i$ for all i.*

The criterion is non-local in two ways: it quantifies over the actual parameter values, and it conjoins the theories of all forced-proper contexts that sit at one parameter point. CE1 fails it because c's point shares p_1 with the root, and CE2 because the two siblings share a point. `realizability_frame.py` compares it with brute-force frame realizability over all frames on $\Lambda = \{0, 1\}^2$ (up to three DEF and two SUP contexts, 0–2 bridge uses per context): 8,000 instances, 0 mismatches.

[16] The deeper and the frame counts need non-default arguments: `realizability_deep.py 11 2000`; `realizability_frame.py 7 3000` and `realizability_frame.py 8 5000`. The defaults run 400 and 3,000 instances.

Proof of Proposition J.3. Shape of the filters. Under limit semantics every non-SUP context's filter is improper or principal at its point. Indeed $\mathcal{F}_@$ is principal at $\lambda^* = \lambda_@$. If $\mathcal{F}_{\pi c}$ is principal at $\lambda_{\pi c}$, Lemma J.1(c) with $G = \{\lambda_c^\circ\}$ shows that $\mathcal{F}_c$ is generated by $\{\lambda_c\} \cap \mathrm{dom}$: principal at λ_c if $\lambda_c \in \mathrm{dom}$, improper otherwise. If $\mathcal{F}_{\pi c}$ is improper, so is $\mathcal{F}_c$, since $\delta(\emptyset \times G) = \emptyset$. A SUP context below a base principal at x is principal at x or improper (Lemma J.1(b)), and its judgments collapse into the base by Proposition 11.4.

($\Rightarrow$) Let $(\mathfrak{M}, \lambda^*)$ be a frame realization and let P be the set of proper non-SUP contexts. P contains @ and C_{cert}. It is closed under (i), since an improper parent has an improper child; under (ii), by the filter anchoring lemma in the proof of Theorem 11.11(b); and under (iii): if $\pi c \in P$ and $\lambda_c = \lambda_{c'}$ with $c' \in P$, then $\lambda_{c'} \in \mathrm{dom}$ (a proper context is principal at its point), so $\mathcal{F}_c$ is principal at $\lambda_c \in \mathrm{dom}$ and c is proper. Hence $S(\lambda^*) \subseteq P$. For each point $x = \lambda_c$ with $c \in S(\lambda^*)$ we have $x \in \mathrm{dom} \subseteq \Lambda$; every $c' \in S(\lambda^*)$ at x is principal at x, so $N = \mathfrak{M}(x)$ satisfies all their collapsed theories, and $p_i^N = x_i$.

($\Leftarrow$) Fix such a λ and put $S = S(\lambda)$, $\lambda^* = \lambda$, $\mathrm{dom} = \{\lambda_c : c \in S\} \subseteq \Lambda$, and let $\mathfrak{M}(x)$ be the given model at x; this is a frame, since $p_i^{\mathfrak{M}(x)} = x_i$, and $\lambda^* = \lambda_@ \in \mathrm{dom}$. *Contexts in S are proper:* all their ancestors are in S by (i), so all the ancestors' points are in dom, and by induction each is principal at its point. *Non-SUP contexts outside S are improper:* by induction on depth; if $\pi c \notin S$, then πc is improper and so is c; if $\pi c \in S$, then (iii) gives $\lambda_c \neq \lambda_{c'}$ for all $c' \in S$, i.e. $\lambda_c \notin \mathrm{dom}$. *Judgments* hold at a proper $c \in S$, because $\mathfrak{M}(\lambda_c)$ satisfies $\mathrm{Th}^*(c)$, which by the collapse covers the judgments of c and of its SUP descendants; at improper contexts they hold vacuously. *Certified contexts* lie in S, hence are proper. *Bridge uses* (c, β, t): if πc is improper, soundness is vacuous. Otherwise $\pi c \in S$, so $c \in S$ by (ii), and c is principal at λ_c. Then $c \models Q = \bar{v}$ iff $Q^{\mathfrak{M}(\lambda_c)} = v$, which by the judgment $(c, \emptyset, Q = \bar{t})$ and standard names happens only for $v = t$; and $\pi c \models \varepsilon_\beta(t)$ by the judgment $(\pi c, \emptyset, \varepsilon_\beta(t))$. □

Corollary J.4 (Reductio immunity; well-posedness where it matters)**.** *[T3 §2.2 Cors, rev.] Under the hypotheses of Theorem 11.11: (i) a SUP context deriving $\bot$ contributes exactly $\neg \bigwedge \mathrm{Asm}$ to its base, in both semantics; (ii) if an idealized context deriving $\bot$ is anchored, J is frame-free unrealizable, and if it is not, that judgment imposes no constraint on frame-free realizability; in the frame semantics the first half holds, the second can fail; (iii) eternalist coherence is strictly stronger than frame-free realizability.*

Proof of Corollary J.4. (i) In the frame-free sense, by the collapse a judgment $(c, \Gamma, \bot)$ at a SUP context c with base b becomes $\neg(\bigwedge \mathrm{Asm}(c) \wedge \bigwedge \Gamma) \in \mathrm{Th}^*(b)$. In the frame semantics the same follows from Proposition 11.4, iterated as in the proof of Theorem 11.11(b). Adding this sentence to $\mathrm{Th}^*(b)$ makes a satisfiable theory unsatisfiable only if the theory already entails $\bigwedge \mathrm{Asm}(c) \wedge \bigwedge \Gamma$.

(ii) Let c be a DEF context with $(c, \emptyset, \bot) \in J$; then $\bot \in \mathrm{Th}^*(c)$. If $c \in \mathrm{Anc}$, $\mathrm{Th}^*(c)$ is unsatisfiable, so J is not frame-free realizable by Theorem 11.11(a), and not frame-realizable by (b). If $c \notin \mathrm{Anc}$: Anc depends only on C_{cert} and U, and removing $(c, \emptyset, \bot)$ does not affect U (bridge uses involve only judgments of the forms $Q = \bar{t}$, σ_β, $\varepsilon_\beta(t)$; we assume $\bot$ is none of these). So by (a), J is frame-free realizable iff $J \setminus \{(c, \emptyset, \bot)\}$ is: the judgment imposes no constraint. In the frame semantics this second half can fail: a non-anchored context may still be forced proper. Example: $J = \{(@, \emptyset, p_1 = 0), (c, \emptyset, \bot)\}$ with c an uncertified limit child deforming $D = \{1\}$ toward 0 and no bridges. Then $c \notin \mathrm{Anc}$ and J is frame-free realizable, but the root forces $\lambda_1^* = 0$, so c sits at $\lambda^* \in \mathrm{dom}$, is proper, and cannot satisfy $\bot$.

(iii) Let $N \in \mathcal{K}$ satisfy the stripped set. Put $M_c = \{N\}$ at every non-SUP context and define

SUP contexts by (R1). (R0), (R2), (R5) hold. (R3): each judgment (c, Γ, φ) has $N \models \bigwedge \Gamma \to \varphi$, and $M_c \subseteq \{N\}$. (R4): for a use (c, β, t), $N \models Q = \bar{t}$, so $M_c \models Q = \bar{v}$ only for $v = t$, and $N \models \varepsilon_\beta(t)$; here πc is not a SUP context, so $M_{\pi c} = \{N\}$. The same construction works if N satisfies $\bigcup_c \mathrm{Th}^*(c)$ (the collapsed variant), using Observation 1. Conversely, the three judgment sets of Corollary J.2 are realizable but flagged: (i) $\{(\mathsf{SUP}(A), \emptyset, \bot)\}$ is realized by $M_@ = \{N\}$ with $N \models \neg A$; (ii) $\{(@, \emptyset, p_{\mathrm{atm}} = 10^5), (c, \emptyset, p_{\mathrm{atm}} = 0)\}$ and (iii) $\{(c_1, \emptyset, Q = 1), (c_2, \emptyset, Q = 2)\}$ by one structure per context. Each stripped set is unsatisfiable. The collapsed variant does not flag (i), but flags (ii) and (iii). □

J.4 Proofs for §11.3: blame localization and hygiene

Proof of Corollary 11.15. (a) Let w be a world at which d is proper. The root is proper at every $w \in W_\rho$, and $W_\rho \neq \emptyset$. If every Ax in the cone were true at w and every Step, Imp and Exp instance in it sound at w, Theorem 11.14 (which uses only the items of the derivation tree, i.e. the cone) would give $d \models_w \bot$, i.e. $\emptyset \in \mathcal{F}_d(w)$. (b) A derivation of $d \Vdash \bot$ contains only the items in its tree, and no rule of Definition 11.13 takes ist-claims, or a SUP context's undischarged conclusions, as premises. (c) Let πd be proper at w with $\pi d \models_w \omega_d$. Take a generator $\delta(F \times G) \cap \mathrm{dom}$ of $\mathcal{F}_d(w)$. Since $F \cap \|\omega_d\| \in \mathcal{F}_{\pi d}(w)$ and πd is proper, there is $\lambda \in F \cap \|\omega_d\|$. Then $E_\lambda = \{\mu : \lambda[D{:=}\mu] \in \mathrm{dom}\} \in \mathcal{G}$, and since $\mathcal{G}$ is proper, $G \cap E_\lambda \neq \emptyset$. For μ in it, $\lambda[D{:=}\mu] \in \delta(F \times G) \cap \mathrm{dom}$. So every generator is nonempty, and d is proper by Lemma J.1(a). □

Proposition J.5 (Propagation of $\bot$ along informative bridges)**.** *[T3 Cor 2.6, propagation clause] Let c be a* DEF *context with an informative value-export bridge β cited for c, and suppose $c \Vdash \bot$, $\pi c \Vdash \sigma_\beta$ and $\pi c \Vdash \omega_c$ are derived. Suppose $\hat{R}$ contains ex falso ($\bot \rhd \psi$) and the kernel of πc is refutation-complete for finite sets (it derives $\bot$ from any finite set of derived sentences that has no model in $\mathcal{K}$). Then $\pi c \Vdash \bot$ is derivable by adding finitely many synthetic Exp instances. Iterating along an anchoring chain reaches the root or a certified context. Moreover, fix a world w at which πc is proper:* (i) *if c is proper at w, some item of c's own derivation of $\bot$ is unsound at w;* (ii) *if c is improper at w, all of c's own items are sound at w, and some item of the derivation of $\pi c \Vdash \omega_c$ is unsound at w;* (iii) *without the certificate premise in Exp, in case (ii), if the derivation of $\pi c \Vdash \sigma_\beta$ is sound at w, some synthetic Exp instance is unsound at w: the bridge applied to an ill-posed context.*

Proof. Informativeness gives $t_1, \ldots, t_k$ with $\{\sigma_\beta, \varepsilon_\beta(t_1), \ldots, \varepsilon_\beta(t_k)\}$ unsatisfiable in $\mathcal{K}$. Ex falso gives $c \Vdash Q = \bar{t}_i$ for each i, and k Exp instances give $\pi c \Vdash \varepsilon_\beta(t_i)$. Refutation completeness gives $\pi c \Vdash \bot$. Iterate for πc along the chain by which c entered Anc.

(i) is the contrapositive of Theorem 11.14 in c. (ii) If c is improper at w, so is every context below it (SUP filters refine their parent's, and a DEF child of an improper context has the empty generator $\delta(\emptyset \times G) \cap \mathrm{dom}$). Every set lies in an improper filter, so every Ax in c's subtree is true at w, and every Step, Imp and Exp instance concluding in that subtree is sound at w. By Corollary 11.15(c), $\pi c \not\models_w \omega_c$, so by Theorem 11.14 the derivation of $\pi c \Vdash \omega_c$ contains an item unsound at w. (iii) Without ω_c, suppose the derivation of σ_β is sound at w, so $\pi c \models_w \sigma_\beta$. If every synthetic instance were sound at w, then, as $c \models_w Q = \bar{t}_i$ trivially, $\pi c \models_w \varepsilon_\beta(t_i)$ for all i, and the finitely many truth sets would have empty intersection in the proper filter $\mathcal{F}_{\pi c}(w)$. □

Proof of Theorem 11.16. (a) Let $c' = \mathsf{SUP}(A)$ be the child of c in which the reductio is carried out. By Lemma J.1(b), $\mathcal{F}_{c'}(w) \supseteq \mathcal{F}_c(w)$, and $\|A\| \in \mathcal{F}_{c'}(w)$. So the premises K_c (true in c) are

true in c', A is true in c', and every c-sound instance has its truth set in $\mathcal{F}_c(w) \subseteq \mathcal{F}_{c'}(w)$, i.e. is c'-sound. By induction along the derivation (filter closure, Proposition 11.3(a)) every line is true in c', so $c' \models_w \bot$, and Proposition 11.4 gives $c \models_w A \to \bot$, i.e. $c \models_w \neg A$.

(b) A derivation of $\bot$ from K_c is also a derivation of $\bot$ from $K_c \cup \{B\}$ for every B, in particular for $B = A$ and $B = \neg A$. The set of refuted sentences is then all sentences, whatever the world.

(c) From $A \to q$ and $\neg A \to q$, case analysis gives q, contradicting $\neg q$; so K is unsatisfiable. The valuation $A = q = 0$ satisfies $\{A \to q, \neg q\}$, and $A = 1, q = 0$ satisfies $\{\neg A \to q, \neg q\}$. D_A derives q from A and $A \to q$ (modus ponens) and then $\bot$ from $q, \neg q$; $D_{\neg A}$ is the same with $\neg A$. As a derivation (used premises, steps, conclusion), D_A is identical to the reductio of A from the consistent set $K' = \{A \to q, \neg q\}$, and that reductio is sound because $K' \cup \{A\}$ is unsatisfiable. So a test that depends only on the derivation and accepts every sound reductio from a consistent premise set accepts D_A; likewise $D_{\neg A}$ with $K'' = \{\neg A \to q, \neg q\}$. The accepted conclusions $\neg A$ and $\neg\neg A$ are jointly unsatisfiable. (`hygiene.py` checks these facts by truth tables.)

(d) Suppose $N \models A$. Every leaf of the reductio is in $K_c \cup \{A\}$, hence true in N; every step has true premises and an instance true in N, hence a true conclusion. So $N \models \bot$, which is impossible; thus $N \models \neg A$. If both A and $\neg A$ were refuted by reductios whose instances N satisfies, then $N \models \neg A$ and $N \models \neg\neg A$. For the counterexample: if $c \models q$, the zero-premise instance $\rhd q$ is c-sound; with $K_c = \{\neg q\}$, every $\mathsf{SUP}(B)$ derives q and $\neg q$, hence $\bot$ by the sound instance $q, \neg q \rhd \bot$, so every B is refuted; a model of $\neg q$ exists but violates the instance $\rhd q$. For existence: if c is proper at w, $c \models_w K_c$ and the finitely many instances used are c-sound at w, then the truth sets of the members of K_c and of the instance conditionals $\bigwedge \varphi_i \to \psi$ all lie in $\mathcal{F}_c(w)$; their intersection lies in $\mathcal{F}_c(w)$ and is nonempty, and any λ in it gives $N = \mathfrak{M}_w(\lambda)$.

(e) By (b), each A yields both labels; they are jointly unsatisfiable, a fixed function of K_c, and the same in every world. If K_c is consistent and $\hat{R}$ logically sound, a model of K_c satisfies every instance, so by (d) no A and $\neg A$ are both refuted. □

J.5 Export certificates and proofs for §11.4

Proof of Theorem 11.19. In each case we exhibit f meeting the hypotheses of Lemma 11.18.

(a) Let $f(\lambda) = e^{-1/\lambda^2}$ for $\lambda > 0$ and $f(0) = 0$. It is C^∞ on $[0, \Lambda]$ with all derivatives 0 at 0, and $f(\lambda^*) > 0$. For $Q \in \mathcal{C}$ and $A \in \mathbb{R}$, $Af \in C^\infty$, so $Q + Af \in \mathcal{C}$ by the closure hypothesis, and $Q + Af$ has the same jet at 0 as Q.

(b) Since $\lambda^* \notin \overline{S}$, some interval $(\lambda^* - r, \lambda^* + r)$ misses S. Let f be a C^∞ bump supported in it with $f(\lambda^*) = 1$. Then $Af \in C^\infty$, $Q + Af \in \mathcal{C}$, and $(Q + Af)|_S = Q|_S$. The germ at 0 is the special case $S = [0, \lambda^*/2)$, since two functions that agree on $[0, \lambda^*/2)$ have the same germ.

(c) Let $f(\lambda) = \lambda^{N+1}$. It is a polynomial, so $Q + Af \in \mathcal{C}$ by the closure hypothesis; its derivatives of order $\leq N$ at 0 vanish, so $Q + Af$ has the same N-jet as Q; and $f(\lambda^*) \neq 0$. (Mere inclusion in the entire functions would not do: a singleton class admits an exact export rule.)

(d) is (b) with the finite, hence closed, set $S = \{\lambda_1, \ldots, \lambda_m\}$.

Weaker hypothesis. If only $\mathcal{C} \supseteq C^\infty$, a rule sound on $\mathcal{C}$ is sound on C^∞, which satisfies $C^\infty + C^\infty \subseteq C^\infty$, so the rule abstains on every $Q \in C^\infty$. In (d) it abstains on every $Q \in \mathcal{C}$: let p be the polynomial interpolating Q at the λ_j and $f(\lambda) = \prod_j (\lambda - \lambda_j)$; then $p + Af \in C^\infty \subseteq \mathcal{C}$ has the information $I(Q)$ for every A, and $f(\lambda^*) \neq 0$, so a non-abstaining output on $I(Q)$ would have to contain $p(\lambda^*) + Af(\lambda^*)$ for every A. In (b) the conclusion can fail outside C^∞: let $\mathcal{C} = C^\infty \cup \{\sqrt{\lambda}\}$, I the germ at 0, and let the rule output $[\sqrt{\lambda^*}, \sqrt{\lambda^*}]$ when shown the germ of

$\sqrt{\lambda}$ and abstain otherwise. No smooth function has the germ of $\sqrt{\lambda}$ (it is not differentiable at 0 from the right), so the rule is sound on $\mathcal{C}$, and it does not abstain on $\sqrt{\lambda}$. □

Proof of Theorem 11.21. (a) At every world compatible with the parent's derivation, $\lambda^* \in U$, so $|Q(\mathfrak{M}(\lambda^*)) - q| \le \epsilon$ by the frame theorem. (b) The negative half is Theorem 11.19(b) (the germ is information on a set whose closure omits λ^*); the positive half is (a) with U and the stated rate. (c) The root filter is the declared germ, so "$Q \sim q$" is a root claim; the decidability statement is Gruntz's algorithm for limits of exp-log functions, modulo zero-equivalence of constants (Gruntz, 1996), which is undecidable for richer classes (Richardson, 1968). □

In IPhO 2025 Theory Problem 2, part C.3, the official premise "X is a few cm" is false, but its conclusion is robust over the whole apparatus range: what justifies it is a uniform bound over that range, the form of Theorem 11.21(a), not the value at the stated point.

Further certificates. The main text summarizes three certificates of the form of Theorem 11.21.

Theorem J.6 (Gronwall finite-horizon export)**.** *[T3 Thm 3.6, rev.: (b) new; Gronwall 1919]* *Let $\dot{x} = f(x)$ and $\dot{y} = f(y)+g(t,y)$ have solutions on $[0,T]$, $L > 0$, and $B(t) = |x_0-y_0|e^{Lt}+\frac{\eta}{L}(e^{Lt}-1)$. (a) If f is L-Lipschitz on a set K containing both trajectories and $|g(t,y(t))| \le \eta$, then $|x(t)-y(t)| \le B(t)$. (b)* Checkable form. *Let K_r be the closed tube of radius r around $x([0,T])$. If f is L-Lipschitz on K_r, $|g| \le \eta$ on $[0,T]\times K_r$, and $B(T) < r$, then y stays in K_r and the bound of (a) holds.*

Every hypothesis of (b) concerns the computed idealized trajectory and explicit bounds, so the parent can certify it; those of (a) concern the unknown trajectory. The bound grows with T: qualitative long-time claims from structurally unstable idealizations do not export, finite-horizon quantitative ones do.

Proof of Theorem J.6. (a) Let $u(t) = |x(t)-y(t)|$. From $x(t)-y(t) = x_0-y_0+\int_0^t (f(x)-f(y)-g(s,y))ds$, the Lipschitz bound on K and $|g(s,y(s))| \le \eta$ give $u(t) \le u(0)+\int_0^t (Lu+\eta)\,ds =: v(t)$. Then v is C^1 with $v' = Lu+\eta \le Lv+\eta$, so $\frac{d}{dt}(e^{-Lt}(v+\eta/L)) = e^{-Lt}(v'-Lv-\eta) \le 0$, and $u \le v \le (u(0)+\eta/L)e^{Lt}-\eta/L = B(t)$.

(b) Let $t_1 = \sup\{t \in [0,T] : |x(s)-y(s)| \le r \text{ for all } s \in [0,t]\}$. The set contains 0, because $|x_0-y_0| = B(0) \le B(T) < r$, and it is closed by continuity, so the supremum is attained. On $[0,t_1]$ both $x(s)$ and $y(s)$ lie in K_r, and $|g(s,y(s))| \le \eta$ there; so (a) applies on $[0,t_1]$ with $K = K_r$ and gives $|x-y| \le B(t) \le B(T) < r$. If $t_1 < T$, continuity gives $|x-y| < r$ on a neighbourhood of t_1, contradicting the definition of t_1. Hence $t_1 = T$. □

Theorem J.7 (Rigorous projectile certificate)**.** *[T3 Thm 3.7]* *A point mass is launched with speed v_0 at angle $\theta \in (0,\pi/2)$ under gravity g, with neglected force per unit mass $a_n = -\kappa(t)v$, $0 \le \kappa(t) \le k|v(t)|$ (quadratic drag with any $C_d \le C_{d,\max}$, or any antiparallel force dominated by it, time-varying ones included; lift is not covered). With $x := kv_0^2/g < 1$ and $R_0 = v_0^2 \sin 2\theta/g$,*

$$R_0\Big[\frac{1}{1+x} - \frac{x\tan\theta}{(1+x)^2}\Big] \;\le\; R \;\le\; \frac{R_0}{1-x}.$$

Numbers (computed). Steel ball ($r = 1\,\text{cm}$, $m = 32.7\,\text{g}$, $C_d = 0.47$, $v_0 = 10\,\text{m/s}$, 45°): $x = 0.0276$, certified $R \in [9.653, 10.483]\,\text{m}$, simulated $9.978\,\text{m}$. Ping-pong ball: $x = 1.42$; the certificate is silent and the export blocked, while the simulated range is 49% short of R_0, although the in-context derivation is word for word the same. Random ($x \in [0, 0.9], \theta$): 0/300

violations; an independent test with bang-bang $\kappa(t)$: 0/400. At 45° the 5% validity region is $x \le (1/0.95)^{1/2} - 1 = 0.0260$ certified, against $x \le 0.0679$ true: sound and about 2.6× conservative. As a schema in the sense of Theorem 11.22(c), σ is $x \le \eta$ with $\eta < 1$, $\omega = \top$, and $\epsilon(v, \theta) = v \max\left(\frac{\eta}{1-\eta}, \frac{\eta}{1+\eta} + \frac{\eta \tan\theta}{(1+\eta)^2}\right)$.

Proof of Theorem J.7. Write $v = (v_x, v_y)$, $v(0) = v_0(\cos\theta, \sin\theta) = (v_{0x}, v_{0y})$, and $\dot{v} = -g\hat{y} - \kappa v$; existence of the solution up to landing is the root model's well-posedness. Let $T = \inf\{t > 0 : y(t) = 0\}$; since $\dot{y}(0) = v_{0y} > 0$, $y > 0$ on $(0, T)$.

Speed bound. On $[0, T]$, $\frac{d}{dt}(\frac{1}{2}|v|^2 + gy) = v \cdot \dot{v} + gv_y = -\kappa|v|^2 \le 0$, and $y \ge 0$; so $|v| \le v_0$ and $|a_n| = \kappa|v| \le k|v|^2 \le kv_0^2 = xg$.

Flight time. On $[0, T]$, $\ddot{y} = -g + a_{n,y} \in [-g(1+x), -g(1-x)]$, so $v_{0y}t - \frac{1}{2}g(1+x)t^2 \le y(t) \le v_{0y}t - \frac{1}{2}g(1-x)t^2$. The upper bound vanishes at $T_{hi} = 2v_{0y}/(g(1-x))$ and is negative after it; if $T > T_{hi}$ (or $T = \infty$), then $y(T_{hi}) \le 0$ with $T_{hi} \in (0, T)$, a contradiction. So $T \le T_{hi}$. The lower bound is positive on $(0, T_{lo})$ with $T_{lo} = 2v_{0y}/(g(1+x))$; if $T < T_{lo}$ then $y(T) > 0$, a contradiction. So $T \ge T_{lo}$.

Horizontal motion. $\dot{v}_x = -\kappa v_x$ gives $v_x = v_{0x}e^{-\int_0^t \kappa} > 0$, so X is increasing, and $\ddot{X} = -\kappa v_x \in [-\kappa|v|, 0] \subseteq [-xg, 0]$. Hence $R = X(T) \le v_{0x}T \le v_{0x}T_{hi} = 2v_{0x}v_{0y}/(g(1-x)) = R_0/(1-x)$, using $R_0 = 2v_{0x}v_{0y}/g$. And $R \ge X(T_{lo}) \ge v_{0x}T_{lo} - \frac{1}{2}xgT_{lo}^2 = \frac{R_0}{1+x} - \frac{2xv_{0y}^2}{g(1+x)^2}$; since $R_0 \tan\theta = 2v_{0y}^2/g$, this is the stated lower bound. □

Theorem J.8 (Error propagation through chains). *[T3 Thm 3.8, rev.] (a) For a chain $\lambda^\circ = \lambda^{(0)}, \ldots, \lambda^{(k)} = \lambda^*$ with certified $|Q(\lambda^{(i)}) - Q(\lambda^{(i-1)})| \le \delta_i$, $|Q(\lambda^*) - Q(\lambda^\circ)| \le \sum \delta_i$. (b) If the answer is $F(q_1, \ldots, q_m)$ from exports with $|Q_j - q_j| \le \delta_j$, and F is differentiable on the box $\prod_j [q_j \pm \delta_j]$ with $L_j = \sup|\partial_j F|$ there, then $|F(Q) - F(q)| \le \sum_j L_j\delta_j$. (c) If $\sum_j L_j\delta_j \ge |F(q)|$, not even the sign of $F(Q)$ is certified by this budget. For "$100 - 99.9$" with tolerances 0.2 the budget is $0.4 > 0.1$, and indeed $Q_1 - Q_2 \in [-0.3, 0.5]$.*

A direct range enclosure can still certify a sign the budget cannot: $F(q) = q^2$ at $q = 1$ with $\delta = 0.6$ has budget 1.92, yet $F(Q) \in [0.16, 2.56]$.

Proof of Theorem J.8. (a) Triangle inequality. (b) The segment from q to Q lies in the convex box, on which F is differentiable; the mean value theorem gives $F(Q) - F(q) = \nabla F(\xi) \cdot (Q - q)$ for some ξ on the segment, and $|\nabla F(\xi) \cdot (Q - q)| \le \sum_j L_j\delta_j$. (c) If the budget is at least $|F(q)|$, the certified interval $[F(q) - \sum L_j\delta_j, F(q) + \sum L_j\delta_j]$ contains 0. With $Q_1 \in [99.8, 100.2]$ and $Q_2 \in [99.7, 100.1]$, $Q_1 - Q_2 \in [-0.3, 0.5]$. For $F(q) = q^2$ on $[0.4, 1.6]$, $L = 3.2$ and the budget is 1.92, while $F([0.4, 1.6]) = [0.16, 2.56]$. □

Proof of Theorem 11.22. (a) In a child with canonical stipulation $p_D = 0$ (e.g. $k = 0$), the smallness parameter (e.g. $s = kv_0^2/g$) is 0, so $c \Vdash s \le \eta$ is derivable soundly in the child, whatever the world. An Exp rule that accepts it therefore exports $Q(\lambda^*) \in [Q(0) \pm \epsilon]$ from the value of Q at the ideal point alone. This is an export rule whose information is a single sample at $0 \ne \lambda^*$, so by Theorem 11.19(d) it is unsound on any class closed under adding smooth functions, for every ϵ. The ping-pong numbers are those of Theorem J.7 (computed).

(b) With k and g both stipulated in the child, the child's range is $R_0(g') = v_0^2 \sin 2\theta/g'$, which takes any positive value as g' varies, while the parent condition $kv_0^2/g \le \eta$ is evaluated at the actual g and holds for the steel ball (0.0276). The export $R \in [R_0(g') \pm \epsilon]$ is false for g' far from g.

(c) By Theorem 11.14 and Lemma 11.7(a), it suffices that each Exp instance is sound at every w. Let $p = \pi c$ and suppose $c \models_w Q = \bar{v}$, $p \models_w \sigma_\beta$ and $p \models_w \omega_c$. By Lemma J.1(a) there are $F_1 \in \mathcal{F}_p(w)$ and $G \in \mathcal{G}$ with $\delta(F_1 \times G) \cap \mathrm{dom} \subseteq \|Q = \bar{v}\|$. Let $F = F_1 \cap \|\sigma_\beta\| \cap \|\omega_c\| \in \mathcal{F}_p(w)$ and fix $\lambda \in F$. Since $\lambda \in \|\omega_c\|$, $E_\lambda = \{\mu : \lambda[D{:=}\mu] \in \mathrm{dom}\} \in \mathcal{G}$. For $\mu \in G \cap E_\lambda$, $\lambda[D{:=}\mu] \in \delta(F_1 \times G) \cap \mathrm{dom}$, so $Q(\mathfrak{M}_w(\lambda[D{:=}\mu])) = v$. Hence $\{\mu : \lambda[D{:=}\mu] \in \mathrm{dom},\ Q(\mathfrak{M}_w(\lambda[D{:=}\mu])) = v\} \supseteq G \cap E_\lambda \in \mathcal{G}$, and the schema's antecedent holds at λ; as $\mathfrak{M}_w(\lambda) \models \sigma_\beta$, the schema gives $\mathfrak{M}_w(\lambda) \models \varepsilon_\beta(v)$. So $F \subseteq \|\varepsilon_\beta(v)\|$, i.e. $p \models_w \varepsilon_\beta(v)$.

The rope. $\mathrm{dom} = \{m_r > 0\}$, $D = \{m_r\}$, limit semantics. The child's generator $\{\lambda^*[m_r{:=}0]\} \cap \mathrm{dom}$ is empty, so the child is improper and proves $a = 0$ by (vacuously) sound steps. The schema's antecedent requires $\lambda[m_r{:=}0] \in \mathrm{dom}$, which never holds, so the schema is a theorem of the frame. The parent proves $m_r \le \eta$ when the actual $m_r \le \eta$, and Exp without ω_c would export $|a| \le \epsilon$, whereas $|a| = |\Phi|/m_r \ge |\Phi|/\eta > \epsilon$ for $\eta < |\Phi|/\epsilon$. With the certificate requirement, $\|\omega_c\|$ must be empty (no λ has $\lambda[m_r{:=}0] \in \mathrm{dom}$), so the proper root cannot soundly derive ω_c. □

Lemma J.9 (Asymptotic schemas below a principal parent)**.** *[T3 Remark after Thm 3.9] Read the child judgment "$Q \simeq v$" as: $c \models_w |Q - \bar{v}| < \bar{r}$ for every rational $r > 0$. Call β an* asymptotic schema *for the declaration $(D, \lambda^\circ, \mathcal{G})$ if, for all w, all $\lambda \in \mathrm{dom}\,\mathfrak{M}_w$ with $\mathfrak{M}_w(\lambda) \models \sigma_\beta$ and $E_\lambda = \{\mu : \lambda[D{:=}\mu] \in \mathrm{dom}\,\mathfrak{M}_w\} \in \mathcal{G}$, and all v: if $Q(\mathfrak{M}_w(\lambda[D{:=}\mu])) \to v$ along the trace of $\mathcal{G}$ on E_λ, then $\mathfrak{M}_w(\lambda) \models \varepsilon_\beta(v)$. If πc is principal at w (the root, or a limit-semantics context below principal ancestors), the Exp instance from $c \Vdash Q \simeq v$, $\pi c \Vdash \sigma_\beta$ and $\pi c \Vdash \omega_c$ to $\pi c \Vdash \varepsilon_\beta(v)$ is sound at w.*

Proof. Let πc be principal at λ (if it is improper, soundness is vacuous). Then $\mathfrak{M}_w(\lambda) \models \sigma_\beta$ and $\lambda \in \|\omega_c\|$, so $E_\lambda \in \mathcal{G}$. By Lemma J.1(c), $c \models_w |Q - \bar{v}| < \bar{r}$ iff there is $G \in \mathcal{G}$ with $|Q(\mathfrak{M}_w(\lambda[D{:=}\mu])) - v| < r$ for all $\mu \in G \cap E_\lambda$. Holding for every rational $r > 0$, this is convergence to v along the trace of $\mathcal{G}$ on E_λ. The schema gives $\mathfrak{M}_w(\lambda) \models \varepsilon_\beta(v)$, i.e. $\pi c \models_w \varepsilon_\beta(v)$. Without ω_c the argument fails at $E_\lambda \in \mathcal{G}$: an improper child proves $Q \simeq v$ for every v. Under a non-principal parent the convergence would have to be uniform on a parent filter set, which is an additional side condition. □

J.6 Validity regions and proofs for §11.5

Minimal repair and common-mode error. Coherent tolerance vectors form an up-set: enlarging a tolerance enlarges an interval, so coherence is preserved upward, and a minimal coherent repair only ever widens tolerances. Exporting a pendulum's T/T_0 by the small-angle schema and the exact one at amplitudes up to 179.9° drives the small-angle tolerance from 0.0077 to 4.4, and coherence never restricts the schema's regime (computed). Two small-angle-based schemas never disagree, so both export ratio 1 at 120°, where the truth is 1.373.

Theorem J.10 (Split conformal, and its failure under selection)**.** *[T3 Thm 4.3] (a) If $e_1, \ldots, e_{n+1}$ are exchangeable and $\hat{\epsilon}$ is the k-th smallest of $e_1, \ldots, e_n$, $k = \lceil (n+1)(1-\alpha) \rceil$ ($\hat{\epsilon} = \infty$ if $k > n$), then $\mathbb{P}(e_{n+1} \le \hat{\epsilon}) \ge 1 - \alpha$ (Vovk et al., 2005; Lei et al., 2018). (b) If the solver picks the instance after $\hat{\epsilon}$ is fixed, from a regime where $\sup e > \hat{\epsilon}$, coverage on it is 0.*

Calibrated on $\theta_0 \sim U[0°, 30°]$ ($n = 200$, $\alpha = 0.1$), the small-angle schema gets coverage 0.913; the adversary's 60° has error 7.3×10^{-2} against $\hat{\epsilon} = 1.44 \times 10^{-2}$ (computed). Mondrian (per-regime) calibration narrows the gap but cannot close it: this is the commitment problem of Section 3 again.

Proof of Theorem J.10. (a) If $k > n$ then $\hat{\epsilon} = \infty$ and there is nothing to prove; note $k \geq 1$ for $\alpha < 1$. Call index $i \in \{1, \ldots, n+1\}$ *high* if e_i is strictly greater than at least k of the other n values. If $e_{n+1} > \hat{\epsilon}$, the k-th smallest of $e_1, \ldots, e_n$, then e_{n+1} exceeds at least k of them, so $n+1$ is high. In any realization at most $n+1-k$ indices are high (in sorted order, only positions $k+1, \ldots, n+1$ can have k strictly smaller values). By exchangeability every index is high with the same probability p, so $(n+1)p = \mathbb{E}[\#\text{high}] \leq n+1-k$, and $\mathbb{P}(e_{n+1} > \hat{\epsilon}) \leq p \leq (n+1-k)/(n+1) \leq \alpha$ because $k \geq (n+1)(1-\alpha)$. Ties are handled by the strict inequality. (b) If the chosen instance has $e > \hat{\epsilon}$, it is not covered, and the solver can choose such an instance whenever one exists. □

Theorem J.11 (No certification without regularity)**.** *[T3 Thm 4.4] Let $\mathcal{C}$ be closed under $e \mapsto e + Af$ for every continuous bump $f \geq 0$ and $A > 0$. A deterministic certifier that makes finitely many oracle queries and outputs a region V with $e \leq \tau$ on V, soundly on $\mathcal{C}$, has V inside the set of queried points. A randomized certifier that is sound almost surely satisfies this inclusion almost surely.* Lean: Export.no_certification_without_regularity *(deterministic clause)*

No free export and no free certification rest on one argument: simulation is an oracle for the top model, but without a regularity guarantee it certifies only the points queried.

Proof of Theorem J.11. We take an exact oracle (or any oracle whose answer is a fixed function of the value at the queried point); this only strengthens the certifier.

Deterministic. Fix $e \in \mathcal{C}$, let $\pi_1, \ldots, \pi_q$ be the queries made on e and V the output. Suppose $\pi \in V$ is not queried. Let $f \geq 0$ be a continuous bump with $f(\pi) = 1$ whose support avoids the finite set $\{\pi_1, \ldots, \pi_q\}$, and let $A > 0$. On $e' = e + Af \in \mathcal{C}$ the first query is π_1, with the same answer since $f(\pi_1) = 0$; by induction the whole adaptive query sequence and the output coincide, so $\pi \in V(e')$. Soundness on e' requires $e(\pi) + A \leq \tau$ for all $A > 0$, which is impossible.

Randomized. Suppose that on some $e \in \mathcal{C}$ the event that V contains an unqueried point has positive probability (we assume the events below are measurable). For each outcome in this event there is an unqueried $\pi \in V$ and a ball B from a fixed countable base (rational centres and radii) with $\pi \in B$ and $\overline{B}$ disjoint from the finitely many queried points. Some fixed B occurs with positive probability. Let $f(\pi') = \operatorname{dist}(\pi', \Pi \setminus B)$, a continuous bump, positive exactly on B. For outcomes in the event for B, the transcript on $e + Af$ equals that on e, so V still contains a point $\pi \in B$ with $f(\pi) > 0$, and $e(\pi) + Af(\pi) > \tau$ once A is large enough for that outcome. By monotone convergence, some fixed A makes the certifier unsound on $e + Af \in \mathcal{C}$ with positive probability. □

With regularity, sound certification is possible at a tight price.

Theorem J.12 (Conservative Lipschitz certifier)**.** *[T3 Thm 4.5] Let $\Pi = [0,1]^d$, e be L-Lipschitz in $\|\cdot\|_\infty$, and a* validated oracle *return $e_{hi}(\pi) \in [e(\pi), e(\pi) + \eta]$ (interval ODE enclosures; Tucker 2011; Immler 2018). With G_h the centres of the $\lceil 1/h \rceil^d$ cubes of side $\leq h$ tiling Π, output $V = \Pi \cap \bigcup_{\pi_j \in G_h} \{\pi : \|\pi - \pi_j\|_\infty \leq (\tau - e_{hi}(\pi_j))/L\}$. (a)* Soundness: *$e \leq \tau$ on V for every L-Lipschitz e, deterministically, against any solver. (b)* Completeness: *if $\gamma > \eta$ and $h \leq (\gamma - \eta)/L$, then $V \supseteq \{e \leq \tau - \gamma\}$, with $\lceil L/(\gamma - \eta) \rceil^d$ calls. (c)* Lower bound: *every deterministic certifier sound on the L-Lipschitz class and complete at margin γ makes at least $\lfloor L/(2\gamma') \rfloor^d$ queries for every $\gamma' > \gamma$, even with an exact oracle.* Lean: Export.lip_cert_sound Lean: lip_cert_complete Lean: lip_lower_bound_cube

The bounds match up to 2^d and the oracle slack: the standard $\Omega((L/\gamma)^d)$ barrier for Lipschitz problems (Nemirovsky and Yudin, 1983). The constant L must be *proved* in the top model; an

L estimated from simulations makes the projectile's region $x \in [0, 0.066]$ (true: $x \le 0.0679$) a calibrated guess, not a certificate.

Proof of Theorem J.12. (a) If $\pi \in V$, then $\|\pi - \pi_j\|_\infty \le (\tau - e_{hi}(\pi_j))/L$ for some centre π_j, so $e(\pi) \le e(\pi_j) + L\|\pi - \pi_j\|_\infty \le e_{hi}(\pi_j) + (\tau - e_{hi}(\pi_j)) = \tau$.

(b) Let $e(\pi) \le \tau - \gamma$, and let π_j be the centre of a cell containing π, so $\|\pi - \pi_j\|_\infty \le h/2$. Then $e_{hi}(\pi_j) \le e(\pi_j) + \eta \le e(\pi) + Lh/2 + \eta \le \tau - \gamma + Lh/2 + \eta$, so the radius at π_j is at least $(\gamma - \eta)/L - h/2 \ge h - h/2 = h/2 \ge \|\pi - \pi_j\|_\infty$, and $\pi \in V$. With $h = (\gamma - \eta)/L$ there are $\lceil 1/h \rceil^d = \lceil L/(\gamma - \eta) \rceil^d$ centres.

(c) Let $e_0 \equiv \tau - \gamma$; completeness forces $V(e_0) = \Pi$. Put $\rho = \gamma'/L$ and pack $M = \lfloor 1/(2\rho) \rfloor^d$ disjoint open $\|\cdot\|_\infty$-balls of radius ρ in Π. With fewer than M queries on e_0, some ball $B(\pi_0, \rho)$ contains no query. Define $e_1 = e_0 + \max(0, \gamma' - L\|\pi - \pi_0\|_\infty)$. It is L-Lipschitz, equals e_0 outside the open ball, and $e_1(\pi_0) = \tau - \gamma + \gamma' > \tau$. All answers on e_1 agree with those on e_0, so the deterministic certifier outputs $V(e_1) = \Pi \ni \pi_0$, which is unsound. □

Theorem J.13 (Monotone certifier)**.** *[T3 Thm 4.6, rev.: query count, adversary] Let e be non-decreasing on $[0,1]$ (no continuity needed), with a validated upper oracle.* Lazy bisection *starts with $lo = 0$, $hi = 1$ unqueried, queries $m = (lo + hi)/2$, sets $lo = m$ ("accepted") if $e_{hi}(m) \le \tau$ and $hi = m$ otherwise, k times, and outputs $V = [0, lo]$ if some query was accepted, $\emptyset$ otherwise. (a) It is sound. (b) With an exact oracle, V misses at most an interval of length 2^{-k} of $\{e \le \tau\}$ (a variant that first queries both endpoints guarantees only $2^{-(k-2)}$). (c) Missing at most length h on every non-decreasing e needs $\lceil \log_2(1/h) \rceil$ queries, even with an exact oracle. (d) For down-sets in dimension $d \ge 2$, $\Theta(h^{1-d})$ queries are necessary and sufficient (proof sketch).* Lean: Export.lazy_bisection_sound Lean: lazy_bisection_miss Lean: mono_lower_bound *((a)–(c))*

For the small-angle schema, $e(\theta_0) = \frac{2}{\pi}K(\sin^2 \frac{\theta_0}{2}) - 1$ is provably increasing (a power series in $k^2 = \sin^2 \frac{\theta_0}{2}$ with positive coefficients), with rigorous enclosure $[k^2/4,\ k^2/4 + \frac{9}{64}k^4/(1 - k^2)]$. Twenty queries on $[0, 0.9\pi]$ certify $\theta_0 \le 7.243°, 16.168°, 22.810°, 49.946°$ for $\tau = 10^{-3}, 5 \times 10^{-3}, 10^{-2}, 5 \times 10^{-2}$ (exact thresholds 7.2441°, 16.1687°, 22.8138°, 50.1040°), each truncated downward and re-checked in interval arithmetic (computed).

Proposition J.14 (Laymon monotonicity can fail)**.** *[T3 Prop 4.7] (computed) For a pendulum with quadratic drag β at amplitude 6.75°, the signed relative period shift ($\times 10^{-6}$) is -3.0, -5.3, -7.7, -3.9, $+10.9$, $+36.2$ at $\beta = 0.0125$, 0.025, 0.05, 0.1, 0.15, 0.2: amplitude decay shortens the period, damping lengthens it at second order. With $\tau = 6 \times 10^{-6}$, a monotone certifier querying $\beta = 0.1$ ($|e| = 3.9 \times 10^{-6}$) certifies $[0, 0.1]$, which contains $\beta = 0.05$ with $|e| = 7.7 \times 10^{-6} > \tau$.*

Proof of Theorem J.13. (a) lo changes only to accepted points, so if $V = [0, lo]$ then $e_{hi}(lo) \le \tau$ was observed, and for $\pi \le lo$, $e(\pi) \le e(lo) \le e_{hi}(lo) \le \tau$.

(b) Each query halves $hi - lo$, so $hi - lo = 2^{-k}$ at the end. With an exact oracle a rejection at hi means $e(hi) > \tau$, so by monotonicity $\{e \le \tau\} \subseteq [0, hi)$; if nothing was rejected, $hi = 1$ and $\{e \le \tau\} \subseteq [0, 1]$. If $V \ne \emptyset$, the missed part lies in $(lo, hi]$, of length 2^{-k}. If $V = \emptyset$, every query was rejected, so $hi = 2^{-k}$ and the missed part lies in $[0, hi)$. The endpoint variant spends two queries before bisecting, leaving $k - 2$ halvings.

(c) Let $\tau > 0$ and $e_t = 2\tau \mathbf{1}[\pi \ge t]$, non-decreasing with $\{e_t \le \tau\} = [0, t)$. Exact answers on e_t take only the values 0 and 2τ, so a deterministic certifier with q queries has at most 2^q transcripts and hence at most 2^q outputs. Choose $M = \lceil 1/h \rceil$ thresholds in $(0, 1]$ pairwise more than h apart (possible since $(M - 1)h < 1$). If $2^q < M$, two thresholds $t < t'$ share a transcript

and hence an output V. Soundness on e_t gives $V \subseteq [0,t)$, so on $e_{t'}$ the missed part contains $[t,t')$, of length more than h. Hence $2^q \geq M \geq 1/h$, i.e. $q \geq \lceil \log_2(1/h) \rceil$.

(d) *Sketch.* On a grid of side h, the cells meeting a hyperplane $\pi_1 + \cdots + \pi_d = \text{const}$ form an antichain of size $\Theta(h^{1-d})$ for the coordinatewise order; down-sets can include or exclude each of them independently, so each must be probed. A staircase walk along the boundary of the down-set achieves the bound. □

J.7 Proofs for §11.6

Proof of Theorem 11.27. Fix $w \in W_\rho$. We check that every Ax in the accepted derivation is true at w and every Step, Imp and Exp instance is sound at w; then Theorem 11.14 gives $@ \models_w Q \in [q-\delta, q+\delta]$, and since $\mathcal{F}_@(w)$ is principal at λ^*_w, $Q^{\mathfrak{M}_w(\lambda^*_w)} \in [q-\delta, q+\delta]$.

- *Step.* By (TB1) the instance conditional $\bigwedge \varphi_i \to \psi$ (and every cited law) is valid in every $\mathfrak{M}_w(\lambda)$, $\lambda \in \mathrm{dom}$; its truth set is dom, which lies in every filter on dom. So the instance is c-sound for every context c.
- *Root axioms* lie in Ax_ρ, which is true at $\mathfrak{M}_w(\lambda^*_w)$ because ρ is a reading (a hypothesis of the theorem, secured by the Reading check when a formal specification is supplied), hence in the principal root filter.
- SUP *axioms:* $\|A\| \in \mathcal{F}_{\mathsf{SUP}(A)}(w)$ by definition.
- DEF *axioms* are canonical (Ax legality). Under limit semantics $\{\lambda^\circ\} \in \mathcal{G}$, and the generator $\delta(F \times \{\lambda^\circ\}) \cap \mathrm{dom}$ lies in $\|p_D = \lambda^\circ\|$. Under germ semantics, for definable $U \in \mathcal{G}$ the generator $\delta(F \times U) \cap \mathrm{dom}$ lies in $\|p_D \in U\|$. This check is what blocks Theorem 11.22(b).
- *Imports into* DEF *contexts* are catalogue laws (valid on dom by (TB1)) or undeformed data, hence D-stable at w, hence sound by Lemma 11.7(a). Imports into SUP children are sound because $\mathcal{F}_{\mathsf{SUP}(A)}(w) \supseteq \mathcal{F}_p(w)$. AUX bridges are kernel-checked theorems, sound by (TB1).
- *Discharges* are sound by Proposition 11.4.
- *Exports.* Schema indexing ensures that the cited β is, by (TB2), a theorem for the context's actual declaration. In the pointwise form, the core of the proof of Theorem 11.22(c) shows that the instance is sound at w; in the asymptotic form, cited only below a principal parent, Lemma J.9 does. Validated enclosures used in side conditions are correct by (TB3). The premises σ_β and ω_c are derived in the parent, so in the induction of Theorem 11.14 they are true there at w. By schema indexing, ω_c is the certificate ω_β of the catalogue entry, which by (TB2) has the semantic property required in Definition 11.13; a solver-chosen sentence such as $\top$ could not be cited, which blocks the rope counterexample of Theorem 11.22(c).

Nothing in this argument depends on how the derivation was produced, so it holds for every solver. For a germ root d with trusted declaration, the same argument yields $d \models_w \chi$. If $W_{\rho^*} \subseteq W_\rho$, the conclusion holds at every $w \in W_{\rho^*}$, i.e. it is supertrue for the intended problem; if the actual system is a world in W_{ρ^*}, the conclusion is true of it. □

Proof of Theorem 11.29. (a) Supertruth of $Q \in [q \pm \delta]$ means $|Q^w - q| \leq \delta$ for every $w \in W_\rho$, i.e. $\sup_w |Q^w - q| \leq \delta$. If some q satisfies this, all Q^w lie in an interval of length 2δ, so $\operatorname{osc} Q \leq 2\delta$. Conversely, if $\operatorname{osc} Q \leq 2\delta$ (and $W_\rho \neq \emptyset$), the midrange $q = \frac{1}{2}(\sup Q^w + \inf Q^w)$ has $|Q^w - q| \leq \frac{1}{2} \operatorname{osc} Q \leq \delta$. (b) is Theorem 11.27 applied to root derivations. (c) By Theorem 11.28(a),

a checker sound for every $\rho^* \in \mathcal{R}$ accepts only claims holding on $\bigcup_{\rho\in\mathcal{R}} W_\rho$; and accepting exactly the claims supertrue on that union is sound for every $\rho^* \in \mathcal{R}$, since $W_{\rho^*} \subseteq \bigcup_{\rho\in\mathcal{R}} W_\rho$. □

J.8 The worked example: further bridges, defects, and proofs

Exact formula. A 40-million-ray Monte Carlo gives MC/exact = 0.9997–1.0099 on five test bins of Proposition 11.31, consistent with ≈0.6% noise (computed).

Thin-leg bridge. Without the lit-domain hypothesis the lower bound of Proposition 11.32 fails ($L = 60a$, $\alpha = 30°$, $r = 40a$, $\varphi = \pi$ gives $E = 0$). The bound holds on 2.5×10^6 test points and is rigorous but conservative (computed): it certifies 5% from $r = 21a, 31a, 44a, 111a$ for $\sigma = 1, 0.5, 0.3, 0.1$, while the exact error exceeds 5% only for $r \le 10.5a, 10.4a, 27.6a, 98a$ (root-finding on exact σ level sets); near $\sigma = 1/\sqrt{2}$ the gap reaches 23×, and it is unbounded as the tolerance shrinks, because the exact error $\approx \epsilon(2\sigma^2 - 1)/(2\sigma)$ vanishes to first order there. The export domain is an annulus $a \ll r < L \tan\alpha$ away from the shadow-side kink.

Finite sun. The sun is not a point; the next bridge accounts for its angular radius.

Proposition J.15 (Finite-sun bridge)**.** *[T3 Prop 6.3, rev.: hypotheses] Let the admissible zenith angles be $[\alpha_{\min}, \alpha_{\max}] \subseteq (0, \pi/2]$, the sun's angular radius $\delta_s < \alpha_{\min}$ with $\alpha_{\max} + \delta_s \le \pi/2$ (the whole disc above the horizon), and $\Delta = \arcsin(\sin\delta_s / \sin\alpha_{\min})$. If $r \le L\tan(\alpha_{\min} - \delta_s)$, $\epsilon < 2/\sqrt{5}$ and $\sigma - \Delta/2 > m$, then for any radiance profile over the solar disc, $\frac{E_\odot}{E_0} \in \big[(1 - \frac{\Delta}{2\sigma})\,\mathrm{lo}(\epsilon, \sigma - \frac{\Delta}{2}),\ (1 + \frac{\Delta}{2\sigma})\,\mathrm{hi}(\epsilon, \sigma - \frac{\Delta}{2})\big]$, with* lo, hi *the bounds of Proposition 11.32.*

The proof writes $E_\odot / I_0$ as a convex combination of the α-free geometric factor over the sun's azimuth spread, bounded by Δ via Napier's rule. An independent test with a limb-darkened disc found 0/859 violations (computed).

Residual defects of the prototype. A purely syntactic version of the checks accepts V6–V9 of Table 16: if C1 allows α in the answer, C3 checks only the oscillation over α of answer over certified form, never its offset from 1, and C4 does not enforce $\epsilon < 2/\sqrt{5}$ or $\sigma_{\min} - \Delta/2 > m$ (beyond $2/\sqrt{5}$ the upper bound turns negative and is silently dropped). The semantic version is still a toy. (i) It hard-codes one top model and two bridge schemas. (ii) SymPy and floating-point evaluation of the budget at a corner stand in for a kernel and interval arithmetic; that the corner is worst rests on monotonicity of the bounds, argued, not machine-checked. (iii) C3 accepts only answers *symbolically identical* to the certified form, so it is incomplete; a fuller checker would add a rigorously enclosed offset $|\text{answer}/\text{certified} - 1|$ to the budget. (iv) V2 and V5 are rejected because the SPS *declares* the violation; the semantic protection is that C4 recomputes side conditions from root data and C3 catches answer-changing undeclared imports (V11). (v) V0 is a completeness gap, and V4 is caught only because the specification is formal. (vi) Numeric inputs are not validated: a NaN tolerance is accepted.

A germ-root example. IPhO 2025 Theory Problem 2, part C.4, is a germ-root problem whose leading-order $W^* = 19.8\,\mathrm{mJ}$ differs from the exact optimum 20.1 mJ by about 1.5% (1.3% before rounding), inside the marking interval $[19, 21]\,\mathrm{mJ}$, so the tolerance-aware check of Theorem J.8 passes (computed).

Proofs.

Proof of Proposition 11.31. (a) The incoming direction is $u = (\sin\alpha\, e(0), -\cos\alpha)$. Reflection in the vertical surface with outward normal $n = (e(\vartheta), 0)$ keeps the vertical component and maps the horizontal one to $\sin\alpha\,(e(0) - 2\cos\vartheta\, e(\vartheta)) = \sin\alpha\, e(2\vartheta - \pi)$; the surface is lit where $u \cdot n < 0$, i.e. $\cos\vartheta < 0$. Descending a height h takes horizontal travel $h\tan\alpha = s$, so $P = a\, e(\vartheta) + s\, e(\psi)$. For the second form, $e(\vartheta)\cdot e(\psi) = \cos(\pi - \vartheta) = -\cos\vartheta = \sin\frac{\psi}{2}$ and $e(\vartheta)\cdot e(\psi + \frac{\pi}{2}) = \sin\vartheta = \cos\frac{\psi}{2}$.

(b) A direct computation (checked in SymPy) gives $\det \partial P/\partial(\vartheta, s) = a\cos\vartheta - 2s$, which is negative on the lit side, so $|\det| = 2s - a\cos\vartheta = 2s + ac$. For injectivity, note first that along one reflected ray the coefficient $s + ac$ of $e(\psi)$ is strictly increasing in s, and distinct ϑ give distinct ψ. So it suffices that two distinct reflected rays never meet. By (a) the ray with parameter ψ is $\{t\, e(\psi) + a\cos\frac{\psi}{2}\, e(\psi + \frac{\pi}{2}) : t \geq a\sin\frac{\psi}{2}\}$. Take $\psi_1 = 2x < \psi_2 = 2y$ with $x, y \in (0, \pi)$ and put $d = y - x \in (0, \pi)$. If $\sin 2d \neq 0$, solving the 2×2 linear system for the intersection of the two lines gives

$$t_1 = \frac{a(\cos x \cos 2d - \cos y)}{\sin 2d}, \qquad t_2 = \frac{a(\cos x - \cos y \cos 2d)}{\sin 2d}$$

(checked in SymPy), and the rays meet only if $t_1 \geq a\sin x$ and $t_2 \geq a\sin y$. *Case* $\sin 2d > 0$, *i.e.* $d < \pi/2$. $t_2 \geq a\sin y$ is $\cos x \geq \cos(y - 2d) = \cos(d - x)$; as $x, |d - x| \in [0, \pi)$ and cos decreases there, $x \leq |d - x|$, which forces $d \geq 2x$. $t_1 \geq a\sin x$ is $\cos(x + 2d) \geq \cos(x + d)$; with $u = x + d \in (0, \pi)$ and $u < x + 2d < 2\pi$, this forces $x + 2d > \pi$ and $2\pi - (x + 2d) \leq u$, i.e. $2x + 3d \geq 2\pi$. Then $4d \geq 2x + 3d \geq 2\pi$, contradicting $d < \pi/2$. *Case* $\sin 2d < 0$, *i.e.* $d > \pi/2$. The inequalities reverse: $\cos x \leq \cos(d - x)$ gives $x \geq |d - x|$, hence $d \leq 2x$; and $\cos(x + 2d) \leq \cos(x + d)$, where $u = x + d \in (\pi/2, \pi)$ and $x + 2d \in (\pi, 2\pi)$, gives $2\pi - (x + 2d) \geq u$, i.e. $2x + 3d \leq 2\pi$. Then $4d \leq 2x + 3d \leq 2\pi$, contradicting $d > \pi/2$. *Case* $d = \pi/2$. Then $e(2y) = -e(2x)$, the lines are parallel, and they coincide only if $a\cos x = -a\cos y = a\sin x$, i.e. $x = \pi/4$; then $t_1 e(2x)$ and $-t_2 e(2x)$ with $t_1, t_2 > 0$ point apart, so the rays do not meet. (A numerical search over 2×10^6 random pairs also found no forward intersection.)

(c) The irradiance on a surface element is $I\sin\alpha\,|\cos\vartheta|$ and the element has area $a\, d\vartheta\, dh$ with $dh = ds/\tan\alpha$. So the reflected flux through $d\vartheta\, ds$ is $I\cos\alpha\, a|\cos\vartheta|\, d\vartheta\, ds$; dividing by the floor area $(2s + ac)\, d\vartheta\, ds$ and using $|\cos\vartheta| = c$, $I\cos\alpha = I_0$ gives $E = I_0\, ac/(2s + ac)$. Floor points outside the image receive no reflected light, and points whose preimage has $s > L\tan\alpha$ would need reflection above the lit height. The total flux is $\int_0^L \int_{\pi/2}^{3\pi/2} I\sin\alpha\,|\cos\vartheta|\, a\, d\vartheta\, dh = 2aLI\sin\alpha$, the flux intercepted by the lit side.

(d) Neither $P(\vartheta, s)$ nor E as a function of (ϑ, s, I_0) contains α; only the constraint $s \leq L\tan\alpha$ does. □

Proof of Proposition 11.32. Since $\epsilon < 2/\sqrt{5} < 1$, $r > a$.

Step 0: the point is in the image and lit. On the ray with parameter ψ, the point at distance r from the axis is $u\, e(\psi) + a\cos\frac{\psi}{2}\, e(\psi + \frac{\pi}{2})$ with $u = \sqrt{r^2 - a^2\cos^2\frac{\psi}{2}} > 0$, and it lies on the forward ray ($s = u - ac \geq 0$) because $u^2 - a^2c^2 = r^2 - a^2 \geq 0$. Its polar angle is $\varphi(\psi) = \psi + \delta(\psi)$ with $\tan\delta = a\cos\frac{\psi}{2}/u$, continuous in ψ, tending to $\arcsin\epsilon$ as $\psi \to 0^+$ and to $2\pi - \arcsin\epsilon$ as $\psi \to 2\pi^-$. By the intermediate value theorem every $\varphi \in (\arcsin\epsilon, 2\pi - \arcsin\epsilon)$ is attained. And $\sigma > m$ forces φ into this interval: for $\varphi \in [0, \arcsin\epsilon]$, $\sigma = \sin\frac{\varphi}{2} \leq \frac{\varphi}{2} \leq m$, and symmetrically near 2π. So the point has a preimage, unique by Proposition 11.31(b), with $s \leq u \leq r \leq L\tan\alpha$, so it is lit.

Step 1. With $u = s + ac \geq 0$, Proposition 11.31(a) gives $r^2 = u^2 + a^2\cos^2\frac{\psi}{2}$, so $u \in [r\sqrt{1 - \epsilon^2}, r]$.

Step 2. $\sin\delta = a\cos\frac{\psi}{2}/r$, so $|\sin\delta| \le \epsilon$ and, as $\delta \in (-\pi/2, \pi/2)$, $|\delta| \le \arcsin\epsilon$. Since $c = \sin\frac{\psi}{2} \ge 0$ and sin is 1-Lipschitz, $|\sigma - c| \le |\sin\frac{\varphi}{2} - \sin\frac{\psi}{2}| \le |\delta|/2 \le m$.

Step 3. $E/E_0 = \frac{ac}{2s+ac} \cdot \frac{2r}{a\sigma} = \frac{c}{\sigma} \cdot \frac{2r}{2u-ac}$. The second factor is at least 1 because $2u - ac \le 2r$, and at most $1/(\sqrt{1-\epsilon^2} - \epsilon/2)$ because $2u - ac \ge 2r\sqrt{1-\epsilon^2} - a$; the denominator is positive exactly when $\epsilon < 2/\sqrt{5}$.

Step 4. By Step 2, $c/\sigma \in [1 - m/\sigma, 1 + m/\sigma]$, and $1 - m/\sigma > 0$. Multiplying the two ranges of nonnegative factors gives the claim. □

Proof of Proposition J.15. Domain. Each direction d in the solar disc has zenith angle $\alpha_d \in [\alpha - \delta_s, \alpha + \delta_s] \subseteq [\alpha_{\min} - \delta_s, \pi/2]$, so $\cos\alpha_d \ge 0$ and tan is increasing there. The preimage of a floor point at distance r has $s \le u \le r$ (Step 0 above), and $r \le L\tan(\alpha_{\min} - \delta_s) \le L\tan\alpha_d$, so the point is lit by every direction, for every admissible α.

Convex combination. By Proposition 11.31(d) and rotational symmetry, direction d with azimuth offset β_d contributes $G(r, \varphi - \beta_d)\cos\alpha_d\, dI(d)$, where $G = E/I_0$ is the α-free geometric factor. The direct illuminance is $I_0 = \int \cos\alpha_d\, dI(d)$, so $E_\odot/I_0$ is a convex combination of the values $G(r, \varphi - \beta)$ over the azimuth spread, for any radiance profile over the disc.

Azimuth spread. In the right spherical triangle formed by the zenith, the sun's centre and a tangent point of the disc, Napier's rule gives $\sin\beta_{\max} = \sin\delta_s/\sin\alpha$. Since $\alpha \in [\alpha_{\min}, \pi/2]$ and $\delta_s < \alpha_{\min}$, this is at most $\sin\delta_s/\sin\alpha_{\min}$, so $|\beta| \le \Delta$.

Combination. Let $\sigma' = |\sin\frac{\varphi-\beta}{2}|$. As sin is 1-Lipschitz, $|\sigma' - \sigma| \le |\beta|/2 \le \Delta/2$, so $\sigma' \ge \sigma - \Delta/2 > m$ and Proposition 11.32 applies at $\varphi - \beta$: $G(r, \varphi-\beta) \in [\mathrm{lo}(\epsilon, \sigma'), \mathrm{hi}(\epsilon, \sigma')] \cdot E_0(\varphi-\beta)/I_0$. Since lo increases and hi decreases in σ, these factors lie in $[\mathrm{lo}(\epsilon, \sigma - \frac{\Delta}{2}), \mathrm{hi}(\epsilon, \sigma - \frac{\Delta}{2})]$, and $E_0(\varphi - \beta)/E_0(\varphi) = \sigma'/\sigma \in [1 - \frac{\Delta}{2\sigma}, 1 + \frac{\Delta}{2\sigma}]$. All factors are nonnegative ($\sigma > \Delta/2 + m$), so multiplying and averaging over the convex combination gives the claim. □

J.9 The physicist's checks in detail

This expands the paragraph after Table 17. *Dimensions* form a free abelian group, and dimension inference is unification modulo abelian groups (Kennedy, 1994). The Buckingham Π theorem (Buckingham, 1914) turns "express in terms of" into a type signature: in IPhO 2025 Theory Problem 2, part A.1, the force must be $mg\,\Phi(\rho Sh/m, P_0S/(mg))$, and the official $(m + \rho Sh)g$ is $\Phi = 1 + \pi_1$. Dropping a small Π group assumes complete similarity, which fails in intermediate asymptotics (Barenblatt, 1996), so "$\Pi_2 \ll 1$, ignore it" is a bridge (role 4), not a type fact. *Context symmetries* are context-indexed (time reversal holds without drag), and a declared symmetry gives free negative data: a non-equivariant candidate rule is refuted by a symmetry image of one of its own accepted instances (dimension inference as learning meaning from positive examples is discussed in Section 5). *Limiting cases* are cross-context regression tests whose validity is Norton's continuity (Proposition 11.5(c)), and *order-of-magnitude* checks are export side conditions (Theorem J.8). Qualitative checks are one-sided refutations, as in qualitative simulation, which is sound but incomplete (Kuipers, 1986). The closest precedent for the in-model layer is the Andes tutor (VanLehn et al., 2005), which checks each equation for units and against an author-supplied model, and never checks bridges, because the author fixes the idealizations.

K Lean formalization

A subset of the discrete results is formalized in Lean 4 (de Moura and Ullrich, 2021) with Mathlib (The mathlib Community, 2020) (tag `v4.32.0`). The formalization is in the directory `lean/`, in

the library `InfLearn`. It has 17 modules and about 14,600 lines.

- There is no `sorry`, no `axiom` declaration and no `native_decide`.
- `lake build` reports no errors and no warnings.
- `lake build Audit` runs `#print axioms` on 483 main declarations. Each depends only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`, or on a subset of them. The output is in `lean/audit-output.txt`.

Every result listed in Table 24 is formalized in the sense given there. In the body, the tag $^{\text{Lean:}\ldots}$ marks some of them with a representative declaration. The column "fidelity" takes one of four values:

- *exact*: the paper statement, possibly in a more general form;
- *special case*: a strict special case, as described in `lean/README.md`;
- *weaker*: a weaker conclusion than the paper states;
- *variant*: a closely related statement, neither stronger nor weaker.

Modelling conventions. The following conventions apply throughout.

- **Language.** The object language is propositional, with atoms $p_0, p_1, \ldots$ and the connectives $\bot, \top, \neg, \wedge, \vee, \to$. Results stated for arbitrary Boolean signatures are proved for every fragment of this set. As a consequence, the truth-table tell-tale has $13 + 1$ data rather than $11 + 1$, because $\bot$ and $\top$ are constants of the language. The bilateral completeness theorems are proved for sequents over an arbitrary type of formulas.
- **Consequence operators** are arbitrary closure operators. Finitarity is not assumed.
- **Learners, verifiers and certifiers** are arbitrary, non-computable functions. Interactive lower bounds are proved against deterministic agents only.
- **Probability** appears in two forms only. One is non-negative summable weights on a countable step set. The other is finite distributions, on the 2^k worlds or on a finite class of labellers.
- **Real analysis.** Smooth bumps are replaced by Lipschitz tents, because the compiled Mathlib subset does not include derivatives.

Not formalized. The following are outside the formalization:

- all probabilistic and anytime results, including Ville's inequality and the Bayes-conservative verifiers (Section 4.4);
- anti-unification and the schema-specific escalation bounds;
- the noise results of Section 5;
- arithmetic (Section 6.3);
- the two-tier main theorem (Theorem 7.12);
- the kink theorem and the other Kolmogorov-complexity results of Section 8;

- first-order and real-parameter context semantics, with the exception of the export results listed below;
- the Structured Physics Solution checker theorem;
- the identification results of Section 10.

These rest on the paper proofs and on the adversarial verification described in Section 1.

Table 24: Map from paper results to Lean declarations. The namespace prefix `InfLearn.` is omitted; the second component names the module (`Post` is `PostCompleteness`).

Paper	Lean declaration(s)	Fidelity
Lemma 2.3	`Cl_eq_sInter`, `ClOp`, `Cl_subset_Cl_iff_subset_Sound`	exact
Lemma 3.1	`StepSoundness.lemma_1_1`	exact
Theorem 3.2	`StepSoundness.thm_2_1_i`, `thm_2_1_ii`, `thm_2_1_error_and_trivial`	exact
Corollary 3.3	`StepSoundness.cor_2_2_trivializes`	weaker (deterministic core)
Theorem 3.12	`CoherenceGames.thm_2_4`, `thm_2_4_environment`	exact
Theorem 4.2 (a)	`StepSoundness.thm_3_1_a`	exact
Theorem 4.2 (c)	`StepSoundness.thm_3_1_b`, `vsVerifier_optimal`	exact (deterministic part; the randomized bound (b) is not formalized)
Lemma 5.1 (a),(b)	`StepSoundness.thm_3_1_a_static`, `thm_3_1_b_static`, `subset_sInter_VS_iff`	exact for (a),(b)
Theorem 4.4	`Unstructured.thm_3_2`, `vsVerifier_attains`	special case (deterministic verifiers)
Theorem 4.12	`Unstructured.thm_3_9_Esc_le_card`, `thm_3_9_coSingleton`	special case (deterministic verifiers)
Lemma 6.2 (a),(b)	`CoherenceGames.negative_bag`, `refuted_of_superset`	exact for (a),(b)
Theorem 6.5	`CoherenceGames.thm_2_2`, `thm_2_2_iii_iv`	special case (finite $\mathcal{H}$)
Proposition 6.6	`CoherenceGames.prop_2_3`	special case (finite)
Theorem 6.7	`CoherenceGames.thm_2_5`	special case (finite)
Theorem 6.13	`Post.eq_Cn2_or_eq_trivial`, `structural_extensions_Cn2`, `frag_eq_Cn2F_or_eq_trivial`, `impFrag_postComplete`	exact (connectives in $\{\bot, \top, \neg, \wedge, \vee, \to\}$)
Theorem 6.15	`Post.eq_Cn2_iff_exists_coherent`, `learner_identifies`, `learner_mindChanges_le`, `versionSpace_eq_singleton_iff`	exact, except the finite-basis clause of (a)
Proposition E.1 (b), first clause	`Post.le_Cn2_of_empty_eq_Taut`	special case (C_{adm} not defined)
Lemma 6.23	`Carnap.Val_mrel`, `lemma_4_1_b`, `intClosure_BV_eq`	exact
Theorem 6.24	`Carnap.carnap_problem`, `carnap_single`, `carnap_not_learnable_single`	exact
Theorem 6.25	`Carnap.denialRank_eq_zero_iff` (and `_one_iff`, `_two_iff`, `_top_iff`)	exact
Theorem 6.26	`Carnap.structural_eq_BV`, `ttLearner_finite_identification`, `no_finite_telltale`, `BV_not_BC_learnable`	exact (13+1 data)
Lemma 7.7	`Blame.isMaxClean_iff`, `iInter_maxClean_eq`, `existsUnique_minTransversal_iff`	exact
Lemma 7.22	`Post.schema_invalid_iff_closedRefutation`, `card_candidates_le`	exact

Paper	Lean declaration(s)	Fidelity
Theorem 8.2 (product part)	`NoAdaptation.thm_2_1_i_ennreal_maxF`, `unifHyp_loss`, `minimax_of_transitive`, `uniform_marginals_of_transitive`	exact for the lower bound and for the value $\sum_i \log \lvert\mathcal{C}(x_i)\rvert$ under the group condition; the general value $\max_Q \sum_i H(Q_{x_i})$ is not formalized
Lemma 8.6	`RateThreshold.onLowerHull_iff`, `lemma_3_1_ii`, `lemma_3_1_iii_hyp`	special case (finite class)
Theorem 8.7	`RateThreshold.prod_isMin_iff`, `rate_threshold`, `thm_3_2`	exact
Corollary 8.8	`RateThreshold.selectable_iff`	exact
Theorem 8.9	`RateThreshold.rate_blindness`	special case (separable setting)
Proposition 8.10	`RateThreshold.valid_not_minimizer`, `pareto_dominated`	exact
Theorem 9.8	`Bilateral.thm_2_2_a`, `thm_2_2_b`, `coherent_iff_realizable`	exact
Corollary 9.9	`Bilateral.cor_2_3`	exact
Theorem 9.10	`Bilateral.scottClosedIso`, `isScott_iff_Th_Val_eq`	exact (without the filter formulation)
Theorem 9.11	`Bilateral.thm_2_6_a`, `structuralClosedIso`, `thm_2_6_b_realize`	special case (fixed signature)
Theorem 9.14	`Specker.thm_3_4`, `specker_parable`	exact
Proposition 9.23 (probabilistic part)	`Specker.frustrated_triangle`	exact
Lemma 10.17	`Blame.LineArg.descent_depth`, `ArgTree.descent_tree_not_mem`	exact
Theorem 10.19 (a),(b)	`CoherenceGames.thm_4_3_a`, `ParadoxLowerBound.thm_4_3_b`	exact
Theorem 10.20 (b),(c)	`ParadoxLowerBound.thm_4_4_b`, `thm_4_4_c`	special case (deterministic learners; one concrete chain)
Theorem 10.21	`CoherenceGames.thm_4_5`	exact
Proposition 11.3 (a),(b)	`Contexts.cn2_closed_ThIn`, `consistent_ThIn_iff`	special case (propositional)
Proposition 11.4	`Contexts.holdsIn_sup_iff`	special case (propositional)
Lemma 11.7	`Contexts.import_sound`	special case (propositional)
Theorem 11.8	`Contexts.not_exists_truthFunctional_reading`	special case
Proposition 11.9	`Contexts.misdesignation_not_Cn2_le`, `atomic_paraconsistent`	exact
Theorem 11.16	`Contexts.refutes_of_inconsistent`, `derivation_local_test_accepts_both`, `certificate_not_both`	exact for (b)–(d)
Lemma 11.18, Theorem 11.19 (b)–(d)	`Export.invisible_perturbation`, `no_free_export_samples`, `no_free_export_finite_jet`, `no_free_export_germ`	exact for (d); variant for (b) and (c)
Proposition 11.20	`Export.exists_interval_iff_radius_le`	exact
Theorem 11.23	`Export.coherence_cannot_calibrate`	exact
Theorem J.11	`Export.no_certification_without_regularity`	weaker (deterministic part)
Theorem J.12	`Export.lip_cert_sound`, `lip_cert_complete`, `lip_lower_bound_cube`	exact
Theorem J.13	`Export.lazy_bisection_sound`, `lazy_bisection_miss`, `mono_lower_bound`	exact